» Grid Incident in Spain and Portugal on 28 April 2025

ICS Investigation Expert Panel
Final Report

20 March 2026

Preamble

This final report is prepared by the Expert Panel established to perform this technical investigation, in accordance with Article 15 of Commission Regulation (EU) 2017/1485 of 2 August 2017 establishing a guideline on electricity transmission system operation (SO GL) and the Incident Classification Scale (ICS) methodology.

The final report provides a factual, technical, and objective account of the incident to transparently inform stakeholders and governance bodies. Its purpose is to identify the causes of the incident and to recommend measures that strengthen the resilience of the European power system. It aims to support transparency, learning and continuous improvement in system operation across Europe.

The analyses, findings, and recommendations contained in this final report are based on data, documents, and other materials provided by the impacted TSOs and DSOs, as well as some generators and other third parties. While the Expert Panel has exercised due care and applied its best effort in reviewing and evaluating the information received, it does not independently verify, audit, or validate the completeness, accuracy, or reliability of any external data sources. Importantly, the report is not intended to assign liability or responsibility to any party and should not be interpreted as doing so in any way.

This report has been prepared and agreed upon by the Expert Panel. Any analyses, findings, and recommendations set out in this report reflect the Expert Panel's technical assessment at the time of writing and are without prejudice to any investigation or enforcement action that may be taken by the competent authorities.

The recommendations outlined in this report represent a comprehensive set of measures designed to enhance operational robustness, improve cross-stakeholder information exchange, and contribute to maintaining a high level of security of supply across the European power system. The monitoring of the recommendations' implementation provided in this report does not fall within the mandate or remit of the Expert Panel. While voluntary in nature, the recommendations are intended to prompt and support concrete action to help prevent the recurrence of similar events in the future. Any responsibility for the consideration, follow-up and implementation of the recommendations lies with each addressee.

Finally, the recommendations may serve as a basis for future regulatory development, inviting competent authorities to consider, within their respective mandates, possible regulatory or implementation measures, with a view to ensuring that the lessons learned from this incident are duly reflected in the applicable requirements and to establishing a framework aimed at preventing similar incidents in the future.

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1 MANAGEMENT SUMMARY

1.1 Introduction

On 28 April 2025, at 12:33 Central European Summer Time (CEST), the power systems of Spain and Portugal experienced the most severe and unprecedented blackout that had occurred in Europe in the past 20 years, with a major impact on citizens and society as a whole in both countries. A small area in France near the Spanish border also experienced disruptions for a limited duration. The remainder of the Continental Europe (CE) power system did not experience any significant disturbance.

The blackout occurred after an uncontrolled, rapid rise in system voltage and loss of voltage control on a day with multiple concurrent phenomena, accelerated by rapid generation output reductions and disconnections, leading to voltage instability and cascading generation disconnections in Spain.

Following the incident, the affected transmission system operators (TSOs) REN (Portugal) and Red Eléctrica (Spain) immediately activated their respective system restoration plans. In addition, RTE (France) initiated all necessary procedures and protocols to restore voltage levels in the French electricity system. System restoration in Portugal was completed by 00:22 on 29 April 2025, and in Spain, the transmission system was fully restored by 04:00 on the same day.

1.1.1 Expert Panel and Structure of Factual Report

The incident has been classified as a scale 3 event – the highest level in terms of severity in accordance with the **Incident Classification Scale (ICS) Methodology**[1]). Consequently, pursuant to the same legal framework, an Expert Panel was set up, which began investigating the incident on 12 May 2025 to deliver the factual report first and the final report thereafter. The reports were prepared in line with the ICS methodology applicable at the time of the incident.

In accordance with the ICS methodology, the Expert Panel comprises representatives from both affected and non-affected TSOs, the Agency for the Cooperation of Energy Regulators (ACER), national regulatory authorities (NRAs), regional coordination centres (RCCs), and convenors from relevant ENTSO-E bodies. The Panel is led by two experts from TSOs not directly affected by the incident: Klaus Kaschnitz from APG (Austria) and Richard Balog from Mavir (Hungary). Overall, the Expert Panel comprises 49 experts: 30 from ACER and NRAs and 19 from TSOs, RCCs, and ENTSO-E bodies across Europe (the complete list of Expert Panel members is available on the final page of this report).

One member from each affected TSO participated and actively contributed to the plenary Expert Panel. In line with the terms of reference, their contribution involved providing input and suggestions but not acting as the primary authors of either the factual or the final reports.

Given the technical complexity of this incident and the numerous aspects to be investigated, the Expert Panel organised its work of analysing the data and drafting this report across several areas (voltage control, oscillations, etc.). Each area was assigned to a specific subgroup of experts reporting to the Expert Panel. Each subgroup was led by a member of the Expert Panel who was not affiliated with the affected TSOs.

In these subgroups, additional experts from affected TSOs participated and contributed via text suggestions, figures, analyses, and/ or other input. Their contribution was assessed by the relevant subgroup of experts before being submitted to the Expert Panel. These measures aim to ensure the neutrality of the final report.

1 The ICS Methodology was developed pursuant to Regulation (EC) No 714/2009 of the European Parliament and of the Council of 13 July 2009, as repealed by the Regulation (EU) 2019/943, and updated to fulfil the objectives under Article 15(5) of the Commission Regulation (EU) 2017/1485 of 2 August 2017 establishing a guideline on electricity transmission system operation (SO GL)

This final report builds on the **factual report** published by the Expert Panel on 3 October 2025, as foreseen by the ICS methodology. For the sake of completeness, almost all data presented in the factual report is also included in the final report, so that the reader does not have to consult both reports in parallel. A limited number of changes were made to the data presented in the factual report, reflecting a deeper understanding of the matter. Furthermore, the final report also includes specific data that were unavailable at the time the factual report was published. These data are included to ensure the highest possible quality and correctness of the final report.

The structure of the final report is similar to the factual report, with two major differences: (i) a new Chapter 4 is added, presenting the detailed analyses performed by the Expert Panel and identifying the root causes of the incident; and (ii) Chapter 9 presents the conclusions of the investigation and the recommendations of the Expert Panel aimed at improving the performance and resilience of the power system across Europe.

This first chapter provides a high-level summary of the report, following the same structure as the report itself. It also includes a specific sub-chapter on data collection, given that collecting complete and high-quality data proved very challenging for this investigation.

In this report, all times are in CEST, which corresponds to UTC+02:00. Furthermore, when referring to Spain and Portugal as relevant areas, this refers to continental Spain and continental Portugal, respectively.

1.1.2 Data Collection

The ICS methodology provides that an Expert Panel established to investigate a scale 2 or 3 incident carries out its investigation based on data reported by the affected TSOs, as listed in Annexes 1 ("*Common data for reporting*"), 2 ("*Specific data reported for depending on the ICS criterion*") and 3 ("*Additional data for the investigation of scale 2 and scale 3 incidents*"). The methodology also indicates that the Expert Panel shall request additional data and information deemed necessary for the investigation, including data owned by third parties not represented in the Expert Panel (e.g., distribution system operators (DSOs) and generators), to be provided by the affected and other relevant TSOs.

In accordance with this provision, the affected TSOs delivered the required data under the ICS methodology. The Expert Panel also considered it necessary to request and collect specific data from DSOs and significant grid users – especially generation facilities – in the affected countries. Furthermore, given that DSOs did not seem to have access to actual production data from small embedded generators with a production capacity of less than 1 MW (typically rooftop PV), the Expert Panel contacted several manufacturers of PV inverters to obtain such data in an aggregated manner. Two of these companies delivered data in time for the Panel to consider. Finally, the Expert Panel received several voluntary contributions from various third parties (i.e., companies and a trade association).

Data collection from DSOs and significant grid users

Data and information requested from DSOs comprised information on the following, among others:

- any disconnection of generators before the incident;
- the generation mix in the DSO grid before the incident;
- any voltage alarms or exceeded voltage thresholds;
- load shedding;
- the restoration process.

Data and information requested from significant grid users (generators) comprised, among others:

- a list of generation facilities in operation on 28 April and any related power system stabilisers;
- information on generation tripping, including fault recorder files, protection relay settings, and schematic drawings;
- information on any identified malfunctions related to oscillations;
- communications with the TSO before the incident;
- information on the black-start process related to the restoration phase.

In line with the ICS methodology and practices applied in previous investigations, the task of collecting data from third parties was entrusted to the affected TSOs.

In Portugal and France, the TSOs collected the necessary data and transferred it to the Expert Panel in a timely manner. The following parties delivered data:

- DSO in Portugal: E-Redes.
- Generators in Portugal: Akuo, Aquila, Axpo, CWPOWER, Dos Grados, EDP-Produção, EML, Energi Innovation, Engie, EXUS, Galp, Iberdrola, Powersun Solutions, Neoen, Prosolia, Vector Renewables, Voltalia, Welink and WiseEnergy.
- DSO in France: ENEDIS.
- Generators in France: EDF, EDF Renouvelable and SHEM.

In Spain, collecting data from third parties proved more challenging. To facilitate the provision of third-party data to the Expert Panel, two initial letters were sent at the end of May 2025: one from the Expert Panel leaders to the Spanish TSO, Red Eléctrica, on 26 May, and the other from ENTSO-E to the Spanish authorities on 28 May. Following these letters, Red Eléctrica obtained consent from 33 generation companies[2] and DSOs to share all relevant data at its disposal with the Expert Panel, while eight others delivered their data directly to the Expert Panel after being requested by the Panel leaders.

More than 240 emails and letters were exchanged between the Expert Panel leaders and the third parties concerned between 28 June 2025 and 17 February 2026 to finalise the data collection process. Additionally several meetings and workshops were organised with third parties impacted by the incident (DSOs, generators), with the aim to clarify the information submitted to the Expert Panel.

Overall, the following Spanish parties have delivered data:

- DSOs: E-Distribucion, Electra Caldense, Electrica Bermejales, Electricas Pitarch, E-Redes, Estabanell, i-DE, Medina Garvey, UFD and Viesgo.
- Generators (including control centres): Acciona, Alpiq, Axpo, Cecovi, Cepsa, Cogen-Energia, EDP España, EDP Renewables, Elawan, Endesa, Energya-VM, Engie, Galp, Gamesa, Gesternova, Gnera, Holaluz, Iberdrola, Ibereolica, Ignis, Magnon, Mercuria Sostenible, Naturgy, Nexus, Norvento, Repsol, RWE, Saica, Samca, SEC Hueneja, Total Energies and Wind to Market.

Consequently, the Expert Panel collected a significant amount of data from DSOs and significant grid users. However, some data remain missing, particularly for generation trips that occurred before the blackout, which prevented the Expert Panel from identifying the cause of tripping for generation units due to missing or no fault recordings. The concerned parties (namely, the owners of those facilities) all cited the lack of fault record data as the reason they were unable to provide this information to the investigation.

Other contributions

In addition to the raw data collected from TSOs and other parties, the Expert Panel also considered the following additional material:

- Ex-post evaluation reports submitted by Spain and Portugal to the European Commission, prepared pursuant to Article 17 of Regulation (EU) 2019/941 of the European Parliament and of the Council of 5 June 2019 on risk preparedness in the electricity sector.
- Several spontaneous submissions, including letters, reports, presentations and other technical documents from various parties: AELEC, E.DSO, Endesa, Eurelectric, Iberdrola, Red Eléctrica and UFD.
- Spanish government report published on 17 June 2025.
- Red Eléctrica report required by the Spanish regulation (Operational Procedure PO 9) published on 18 June 2025.
- Presentations given by E.DSO, EU DSO Entity and Eurelectric during the joint workshops of the System Operations European Stakeholder Committee (SO ESC) and of the Grid Connection European Stakeholder Committee (GC ESC), chaired by ACER, which took place on 18 July and 13 October 2025.

While the investigation is primarily based on the raw data and information collected at the Expert Panel's request, the Expert Panel also considered these additional external documents in preparing the final report.

2 Generation control centre: in Spain, all generation facilities with power equal or greater than 1 MW must send real-time information to the TSO. Considering that a very high number of facilities meet this criterion (around 4,200), an additional role has been established based on Operational Procedure 8.2 and RD 413/2014, namely the generation control centre. The generation control centre acts as an intermediary between generation facilities and the TSO control centre to send real-time measurements to the TSO and in the other direction to send to generation facilities the reference power values required by the TSO.

Treatment of confidential information

The collection of data required for the purposes of this investigation and the drafting and elaboration of the factual and final reports by the Expert Panel have been performed in line with the ICS methodology and the confidentiality requirements established in both the European and relevant national regulatory frameworks. To safeguard confidentiality while upholding transparency, the Expert Panel has adopted a balanced approach, under which any data provided by the affected TSOs linked to individual generators and/or by third parties and classified as confidential has been anonymised.

Such classification is based on the due justification of the sensitive nature of the specific data by the providing party to the Expert Panel. The anonymisation measures applied are without prejudice to the integrity of the data analysis conducted by the Expert Panel, ensuring confidentiality while preserving the evidential value and robustness of the data provided.

In light of this, this report mentions the technology of individual power plants and, if applicable, the region where they are located instead of their names. Furthermore, absolute values of their active or reactive power infeed are replaced by per-unit values.

1.2 System and Market Conditions Before the Incident

During its investigation, the Panel has determined that the incident began at 12:32:00 on 28 April 2025. Chapter 2 of the report therefore covers the period prior to that time, starting at 9:00.

The day of the incident was a typical spring day in Spain, with mild temperatures and mostly sunny weather. The system's solar photovoltaic generation was similar to previous days, while wind generation was more variable but within the ranges observed in previous days.

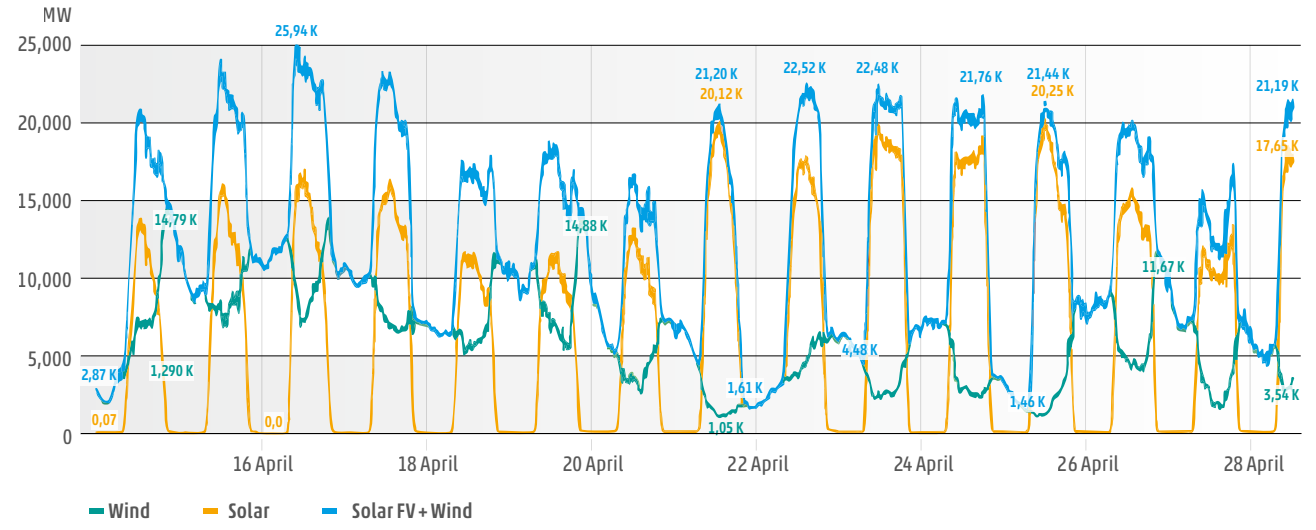


Figure 1-1: Solar and wind production (in MW) in Spain on previous days and the day of the incident, not including small PV units (<1 MW)

The morning hours of 28 April 2025 were characterised by an increasing generation of renewables, which led to lower prices on the day-ahead market and to Spain's exports reaching 5 GW in total. From approximately 09:00, the voltage variability in Spain began to increase, albeit without significant fluctuations, until shortly after 10:30, when the voltage in a part of the 400 kV transmission network briefly approached – but did not exceed – 435 kV.

With the exception of one node (Olmedilla), the voltages in the 400 kV network remained below 435 kV during the period before the incident. Oscillatory behaviour was noted in the morning, but no significant oscillations with amplitudes above 20 mHz were detected until 12:03.

Figure 1- 2 illustrates voltage evolution in the main 400 kV transmission substations (pilot nodes) in Spain from 9:00.

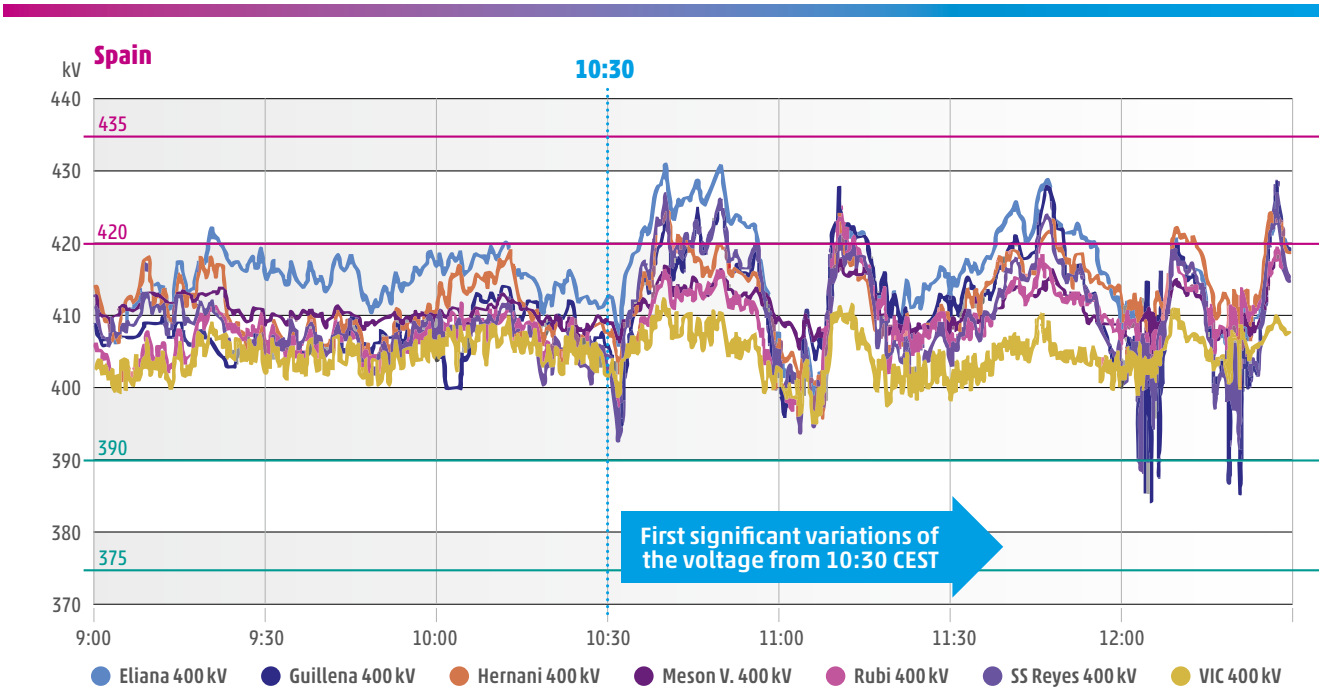


Figure 1-2: Voltage evolution in the main 400 kV transmission substations (pilot nodes) in Spain

During the observed time period, the voltage levels on the 400 kV network in Portugal remained within the normal range of operation (380 kV to 420 kV), which differs from the ranges allowed in Spain.

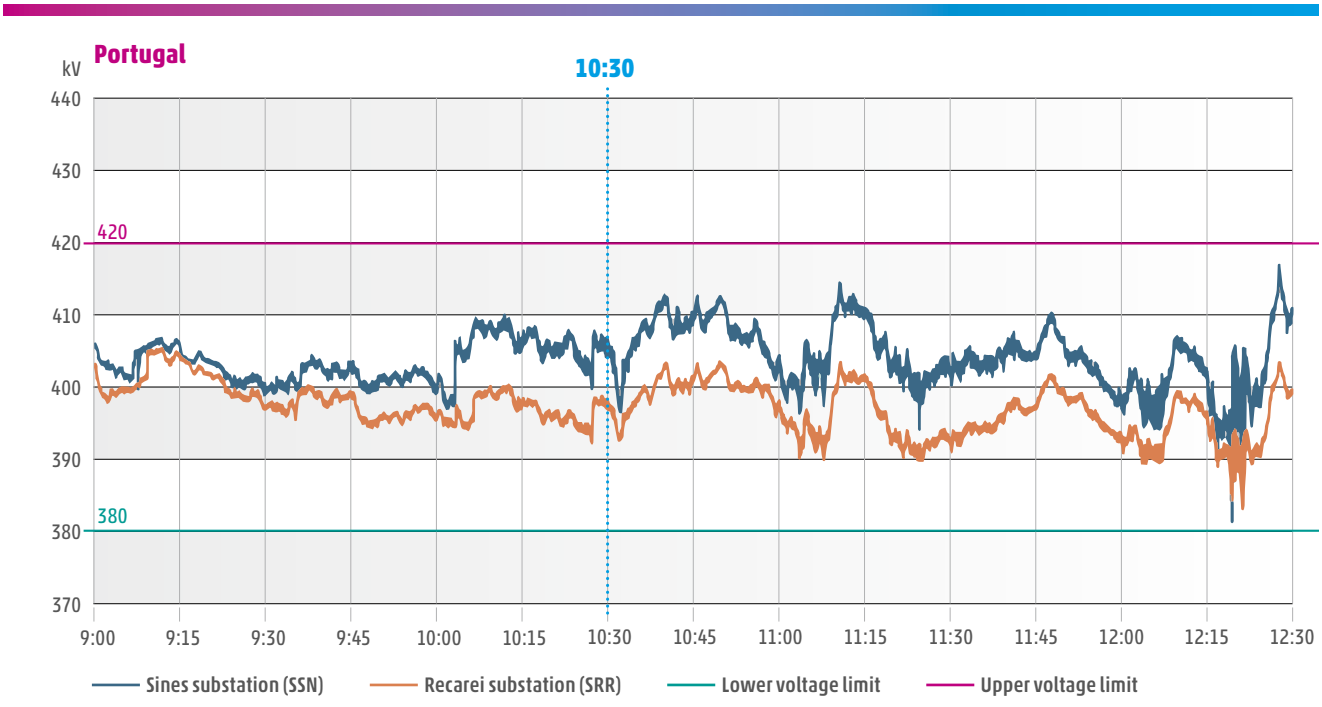


Figure 1-3: Voltage evolution in the main 400 kV transmission substations (pilot nodes) in Portugal

An analysis of the geographical distribution of generation and consumption in Spain reveals that electricity production in the south-west of the country was very high relative to consumption in that region. This results in high electricity transit flows from the southwest of Spain to the surrounding areas.

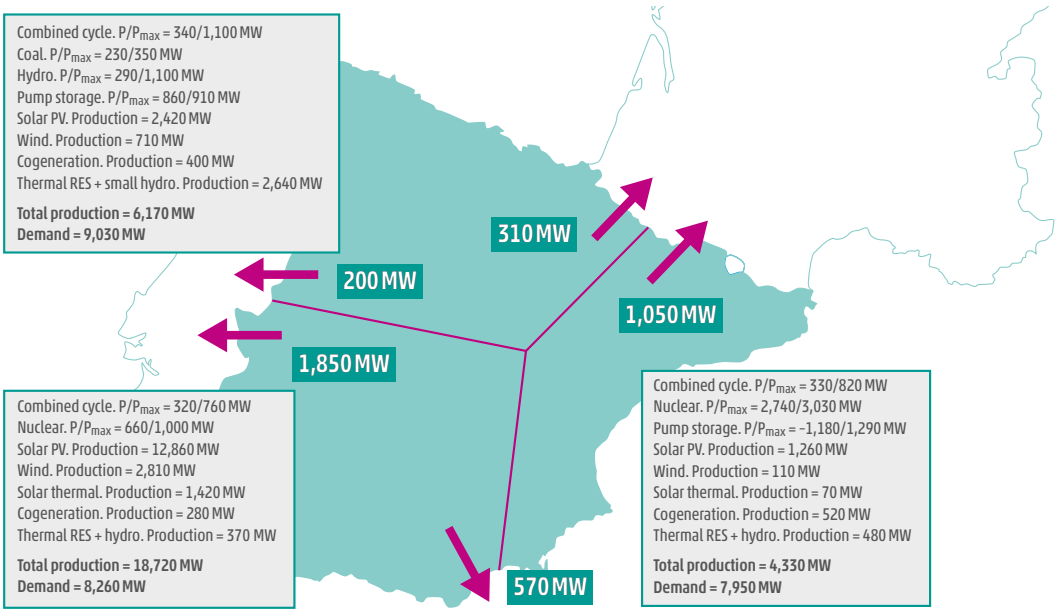


Figure 1-4: Geographical distribution of the generation and consumption in Spain at 12:32

During the half hour preceding the blackout, two main periods of oscillations – power, voltage and frequency swings – were observed in the Continental Europe Synchronous Area (CE SA), the first of which took place from 12:03 to 12:08. The analysis indicates that this oscillation had a local character classified as converter-driven instability phenomenon primarily affecting the Spanish and Portuguese power systems with a dominant frequency of 0.63 Hz.

The second oscillation occurred between 12:19 and 12:22 as an inter-area oscillation, with a dominant frequency of 0.2 Hz, corresponding to the East-Centre-West continental mode.

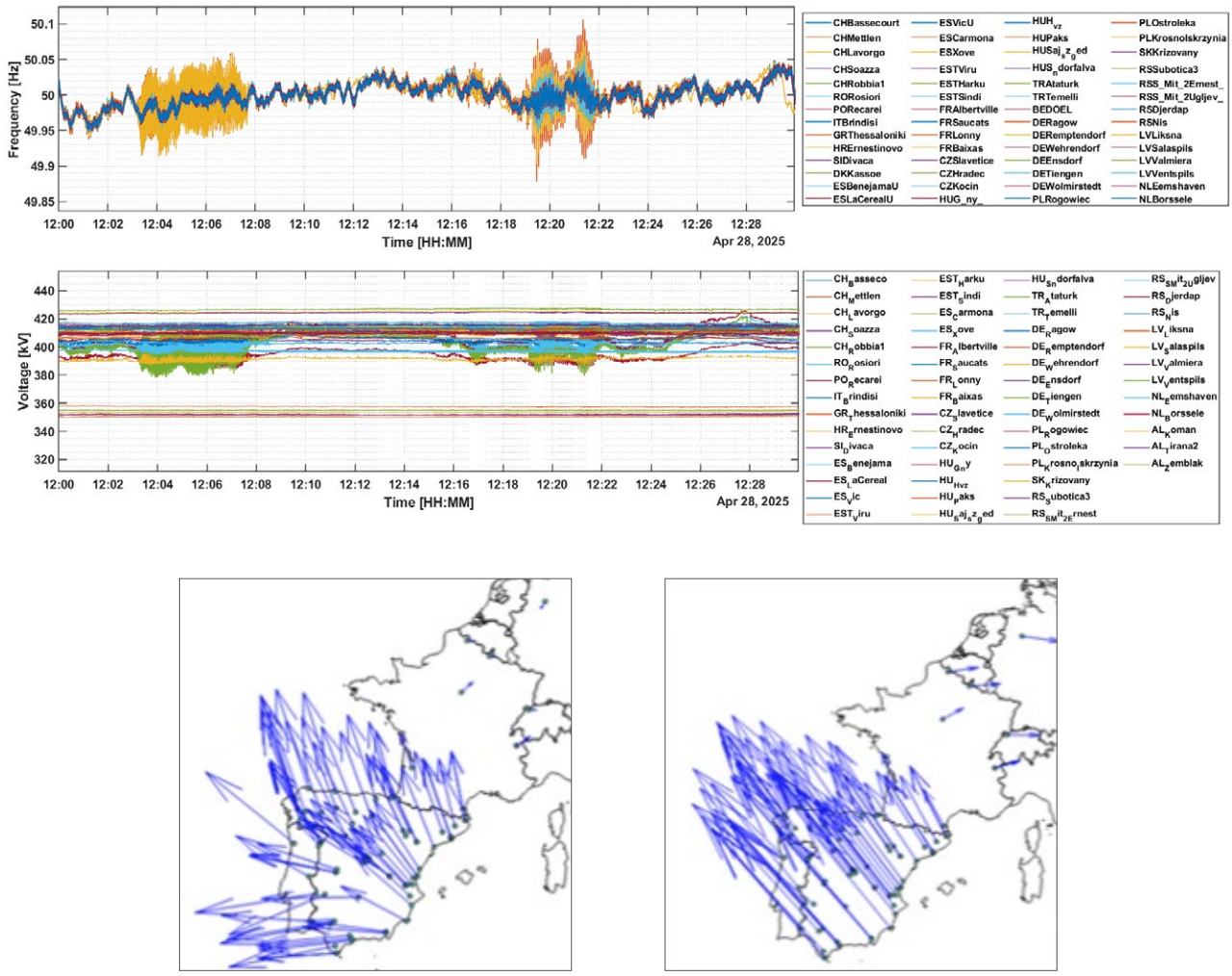


Figure 1-5: Oscillatory episodes during the half hour preceding the blackout: Frequency and voltage phasor magnitude measurements from European PMUs (top); mode shapes of the two oscillations (bottom)

To damp these oscillations, the operators in the control rooms of the relevant TSOs took several mitigating measures, such as reducing exports from Spain to France, coupling internal power lines in the South of Spain, and changing the operating mode of the high-voltage direct current (HVDC) link between France and Spain. While these measures effectively mitigated the oscillations, their nature led to an increase in voltage in the Iberian power system.

At 12:32:00 the voltage of the Iberian power system at the 400 kV level was below 420 kV and no notable oscillation with amplitude higher than 20 mHz could be observed. The evolution of the system and market conditions up to this point is explained in further detail in Chapter 2.

1.3 System Conditions during the Incident

After 12:32:00, the voltages at numerous nodes rose, as shown in the Figure 1-6.

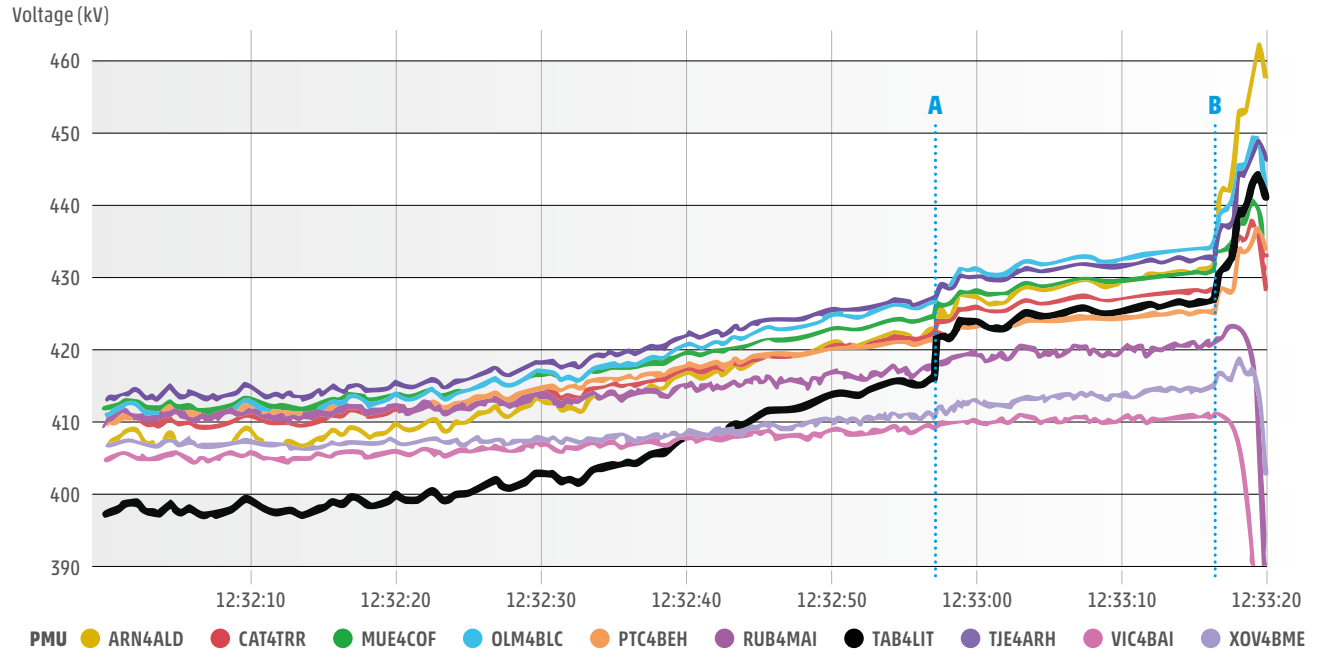


Figure 1-6 : Voltage (positive symmetrical component) evolution in the Spanish power system after 12:32:00 (PMU data). Phase asymmetry (not visible in this chart) was observed in some nodes.

Around 12:30, several generators changed their active power output. From 12:32:00 to 12:32:48, concurrent with the voltage rise shown in Figure 1-6, the active power output of large Renewable Energy Sources (RES) generators (higher than 5 MW) in Spain decreased by approximately 500 MW. The RES power plants maintained a fixed power factor, meaning that changes in active power also affected the generators' reactive power output.

Additional events (mainly generation-related) occurred from 12:32:00 onwards. Between 12:32:00 and 12:32:57, 208 MW of identified distributed wind and solar generating units in northern and southern Spain experienced either a fast downwards change of operating point of a unit without a ramping restriction, or a disconnection of the unit due to unknown reasons.

In the same period, there was an increase in net load in the distribution grids of approximately 317 MW, which can be attributed partly to disconnections of small embedded generators <1 MW (mainly rooftop PV[3]) and partly to an increase in voltage-dependent load as a result of a voltage rise. The exact reasons for some of these events are not known.

3 See Section 4.3 for further explanations.

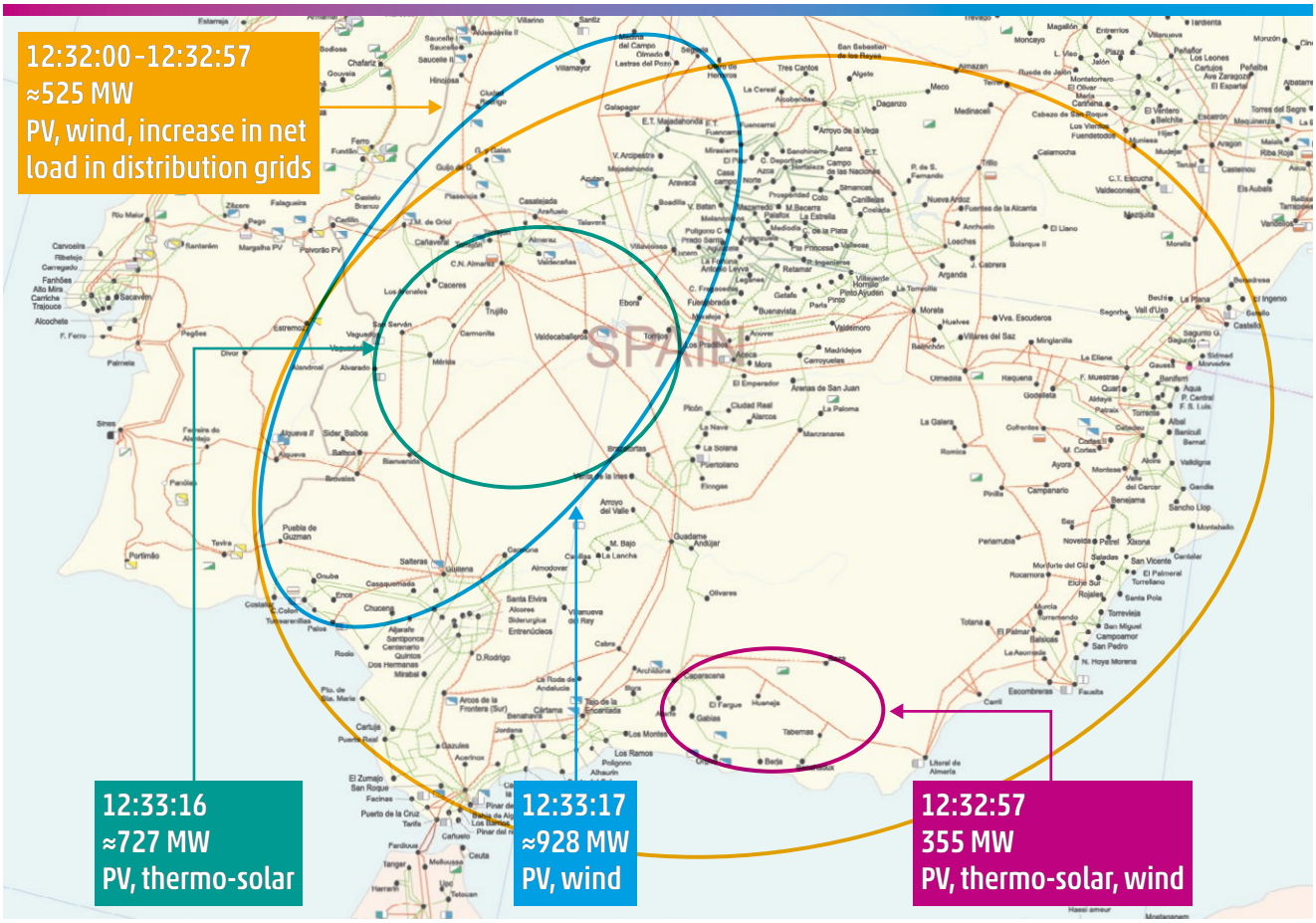


Figure 1-7: Areas of major generation disconnection events in Spain until 12:33:18

From around 12:32:20, a particularly steep rise in local voltage occurred at a substation in the area of Granada (see the nearby PMU coloured in black in Figure 1-6). Some milliseconds after 12:32:57, a 400/220 kV transformer in the same substation tripped due to the activation of an overvoltage protection in the 220 kV side of the transformer, which connects several generation facilities (photovoltaic, wind and thermo-solar) to the transmission grid. Just before the disconnection, the transformer was injecting 355 MW into the grid and absorbing 165 Mvar of reactive energy, and the voltage at the 400 kV level was 417.9 kV. This event created an additional step change in voltage (see Figure 1-6).

The next event consisted of two sets of trips, resulting in an additional loss of 727 MW of PV and thermo-solar facilities connected to two 400 kV transmission substations in the area of Badajoz. At the first substation, an evacuation line tripped at 12:33:16.443 due to the activation of the overvoltage protection. The positive-sequence voltage level at the time of this trip is estimated at 432.4 kV via indirect calculations due to the lack of high-resolution voltage data at the connection point. In the second substation, the trip occurred at 12:33:16.820 due to overvoltage protection.

After that, several trips between 12:33:17 and 12:33:18.020 occurred, leading to the disconnection of wind and solar generation in Segovia, Huelva, Badajoz, Sevilla and Caceres, for a total of 928 MW. Some of these trips occurred due to overvoltage protection, but the Expert Panel was not able to establish the cause of most of these trips. The analysis carried out by the Expert Panel indicates that the overvoltage protection settings at some generation units were set below the voltage limits established in accordance with the applicable requirements.

Figure 1-8 illustrates an increase in voltage beyond 435 kV during this sequence, excluding the 0.5 GW generation schedule changes. The figure therefore shows 317 MW net load increase assumed at 12:32:00, 208 MW fast downwards change of operating point or tripping due to an unknown reason between 12:32:00 and 12:32:57, and identified trips from 12:32:57 onwards until 12:33:18.020, adding up to more than 2.5 GW in total).

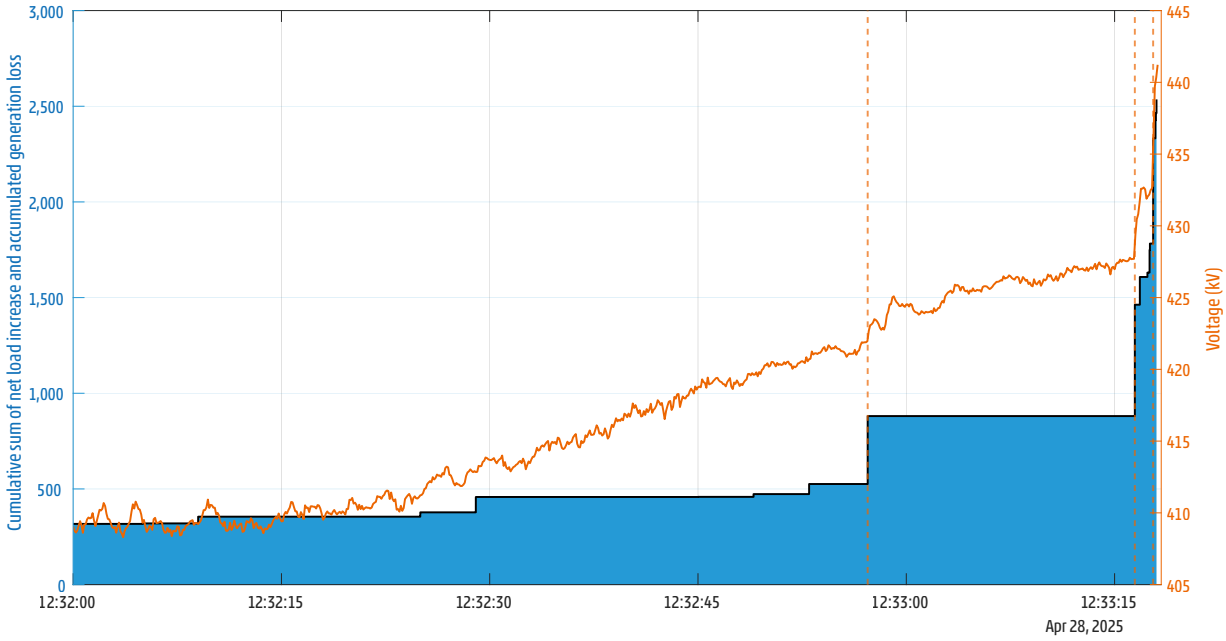


Figure 1-8: Accumulated generation loss combined with net load increase in the Iberian Peninsula system vs voltage at the 400 kV substation in Carmona (Spain). Dotted lines represent the first three main generation disconnection events.

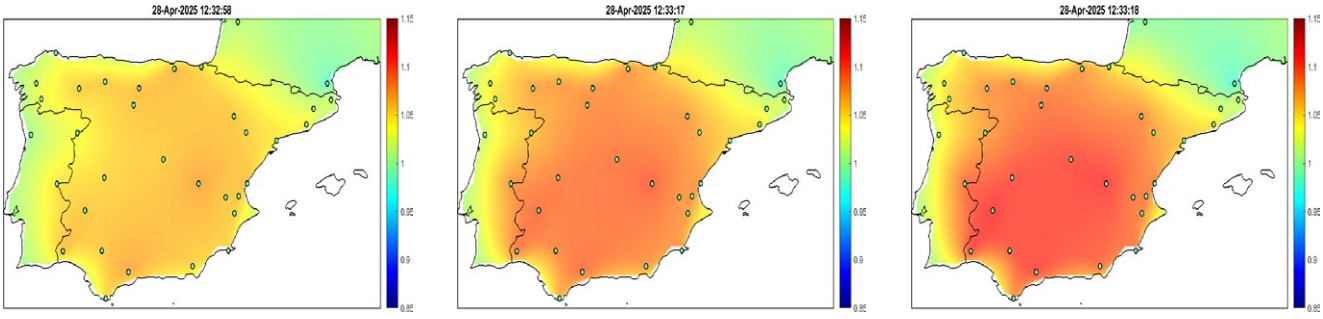


Figure 1-9: Heat map of the Iberian Peninsula for the temporal instants at (a) 12:32:58, (b) 12:33:17, and (c) 12:33:18, showing voltage magnitudes at different substations.

No generation trips were observed in Portugal and France within the 12:32:00 – 12:33:18 timeframe.

Some generation units were consuming reactive power, which reduced the voltage. However, the disconnections of these units without adequate compensation for the loss of reactive power absorption by other resources capable of controlling reactive power led to increased voltages not only in Spain but also in Portugal. Furthermore, the frequency decreased.

Between 12:33:18 and 12:33:21, the voltage in the South of Spain increased sharply, and consequently also in Portugal. The overvoltage triggered a cascade of generation losses, causing the frequency of the Spanish and Portuguese power systems to decline.

At 12:33:19, the power systems of Spain and Portugal started losing synchronism with the rest of the European system.

Between 12:33:19 and 12:33:22, the automatic load shedding and system defence plans of Spain and Portugal – implemented in accordance with Commission Regulation 2017/2196 of 24 November 2017 establishing a network code on electricity emergency and restoration (NC ER) – were activated, but were unable to prevent the collapse of the Iberian power system.

At 12:33:20.473, the AC interconnection to Morocco tripped due to underfrequency. At 12:33:21.535, the AC overhead lines between France and Spain were disconnected by protection devices due to a loss of synchronism, preventing the propagation of the perturbation into the CE power system.

From that moment on, the Spanish and Portuguese grids operated in island mode. After this AC separation of the Iberian Peninsula, the power imbalance continued to grow, leading to a further decline in frequency.

Finally, at 12:33:23.960, the electrical separation of the Iberian system was completed by tripping the HVDC lines that were still transmitting power from Spain to France (due to the constant power mode setting at the time). All system parameters of the Spanish and Portuguese electricity systems collapsed.

Figure 1-10 illustrates the evolution of the frequency and voltage in Spain (substation of Carmona) and the frequency in the rest of CE (substation of Bassencourt, Switzerland) during the incident.

The evolution of the Rate of Change of Frequency (RoCoF) in the moments before the blackout indicates that RoCoF in the area remained within the absolute range of 1 Hz/s up until 12:33:20.560, when the frequency was already around 49 Hz. After that, the absolute value of RoCoF exceeded 1 Hz/s, indicating that the system conditions had already degraded.

Compared with the blackouts in Spain and Portugal, France was marginally affected by the incident. In addition to a loss of approximately 7 MW of load, one nuclear power plant tripped due to the incident.

The evolution of system conditions from 12:32:57 until the blackout is explained in further detail in Chapter 3.

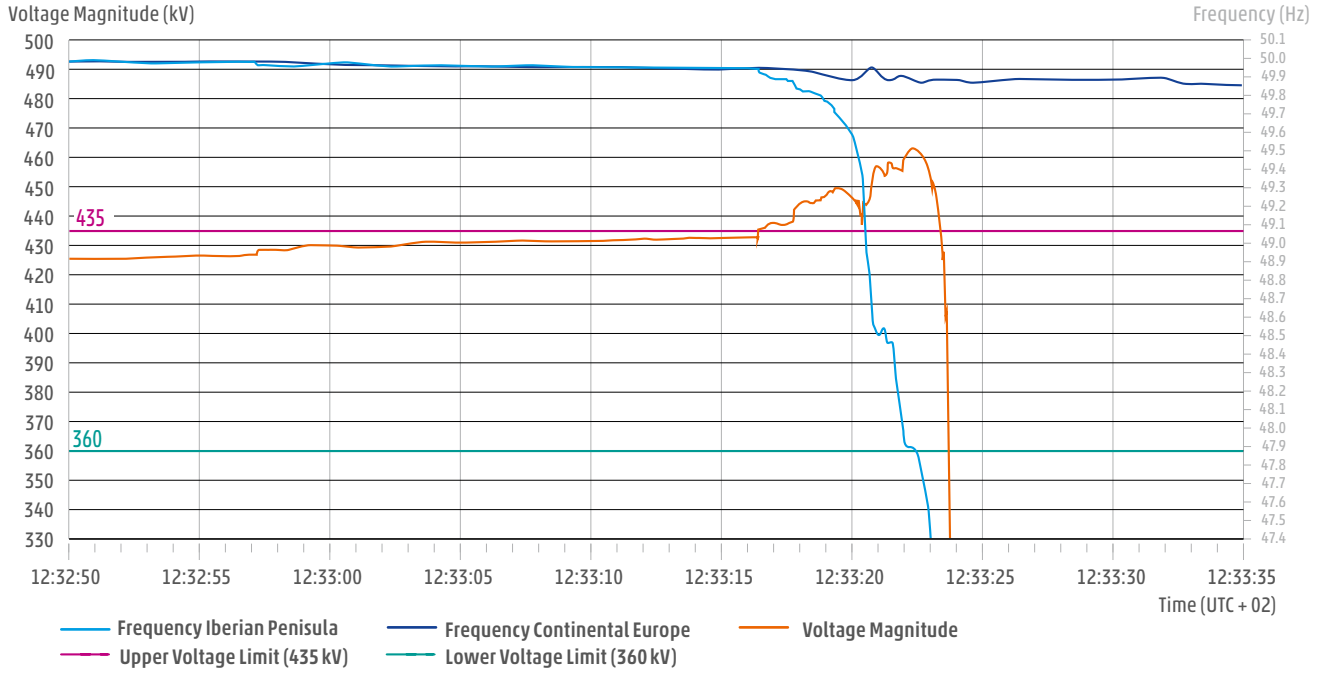


Figure 1-10: Evolution of the frequency and the voltage in the substation of Carmona (Spain) and of the frequency in the rest of Continental Europe (substation of Bassencourt, Switzerland) during the incident.

A comprehensive description of the system conditions during the incident is available in Chapter 3.

1.4 Technical analysis

Chapter 4 of the report presents detailed technical analyses of the phenomena and actions that preceded the blackout, to understand the precise causes of the incident. All these analyses are based on the factual data collected and presented in the relevant chapter of the report. Based on the results of the technical analyses, the Expert Panel established a root cause tree describing the key factors that ultimately led to the blackout.

While the root cause tree analysis describes the role of each element in the tree and the relationship between them, the detailed explanation of the elements is provided in the technical analysis.

It should be noted that these technical analyses focus on the Spanish system, as the facts described in chapters 2 and 3 indicate that the blackout originated in Spain, not in Portugal.

1.4.1 Voltage control

The first area of technical analysis concerns voltage control, which was an important aspect of this incident. The voltage control capability of a system depends on many factors, including the reactive power capacity of the system and the design of the voltage control scheme at all voltage levels (including TSO-operated integrated network components and the requirements applicable to the reactive-power-related behaviour of third parties, particularly generators).

The analysis identifies several factors that contributed to the deterioration of system conditions on 28 April 2025, particularly in the sudden voltage increase observed from around 12:32:00 onwards. They included the following, in no specific order of importance:

- Switching of network components for voltage control, such as shunt reactors, was undertaken manually,[4] which required a certain amount of decision-making and processing time.
  - Some shunt reactors had been previously disconnected due to low voltages during the oscillation episodes up to 12:24.
- According to the requirements applicable on the day of the incident, the RES generators followed a fixed power factor for reactive power provision, thereby not contributing effectively to the voltage control of the system.
  - The fixed power factor mode means that generators absorb reactive power proportionally to active power, and do not respond to voltage fluctuations.
- Regarding conventional synchronous power plants, the analysis shows that the reactive power output of several generators reached Q-reference less than 75 % of the samples (per hour). The reactive power output was not aligned with the TSO expectations based on the requirements under the applicable Operational Procedure 7.4[5].

4 The Expert Panel has not comprehensively assessed the approach across all TSOs. Nevertheless, several TSOs participating in the Panel indicated that they only have manual control of shunt reactors, while RTE indicated they have a hybrid approach (combination of manual and automatic control).

5 This is without prejudice to any potential derogations from the requirements, which the Expert Panel is not competent to assess. The scope of this investigation and the mandate of the Expert Panel do not provide it with any authority or intention to interpret the applicable legislation or to determine liabilities for non-compliance with legal requirements. The analysis is based on RE's SCADA data.

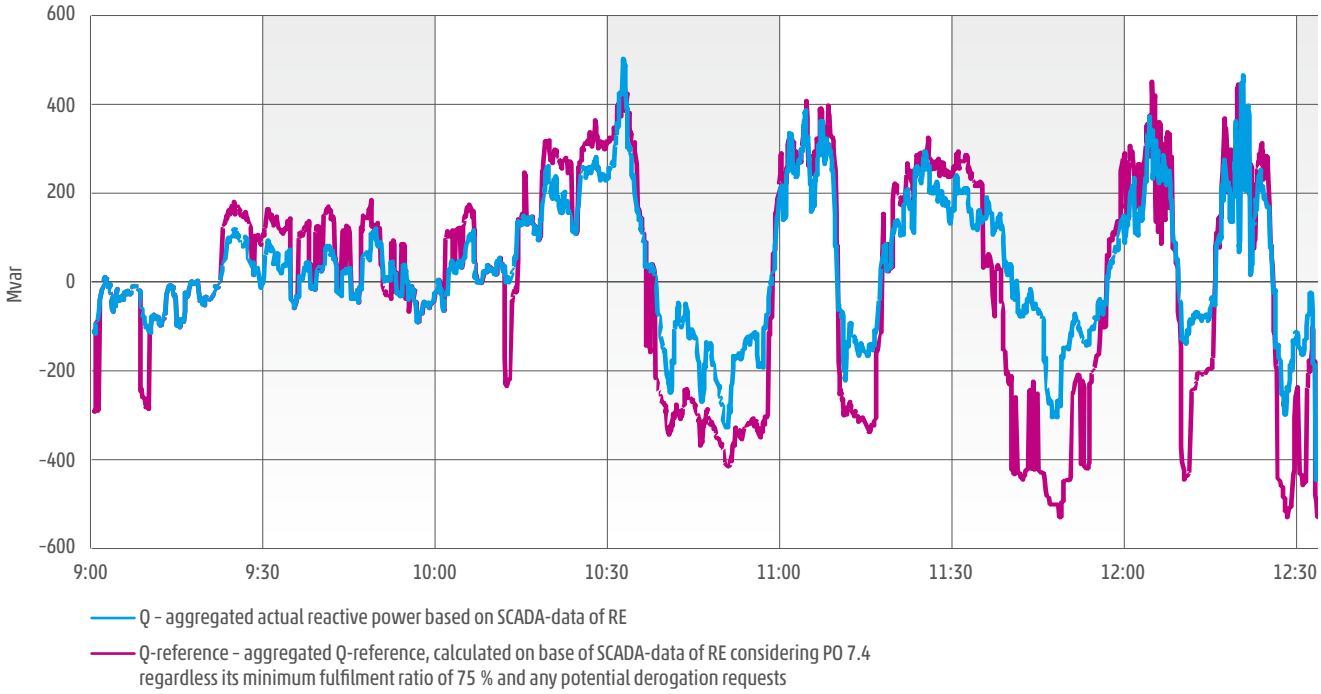


Figure 1-11: Reactive power provided (Q) and the reference reactive power (Q-reference) aggregated for conventional generation units larger than 100 MW of power installed capacity in the centre/southwest area of Spain, based on SCADA data from RE.

- The applicable Operational Procedure 7.4 concerning reactive power provision by synchronous generators did not include explicit specifications concerning dynamic behaviour, and there were no economic consequences if the requirements regarding voltage control were not met.
- The design of voltage control of some local generation networks (extensive grids behind the transmission grid connection point, shared by several generation portfolios) was not aligned with system needs.
  - This also contributed to some of these facilities disconnecting when the voltage at the connection point was within limits.
- Due to Spain's wider operating voltage range of the 400 kV grid compared to the rest of Europe, the margin between the allowed operating voltage limit and the voltage at which generators are allowed to disconnect was very small or non-existent.

Together, the reactive power assets in the system were not able to address the sudden voltage rise. The reactive power absorption status of three asset groups is shown in Figure 1-12. The figure aggregates, by asset group, the total reactive power capacity in the Spanish system on the day of the incident, as well as the actual reactive power absorption at the moment of the incident. However, it should be noted that these categories of assets have different static and dynamic reactive power characteristics: on the day of the incident, conventional synchronous machines (here: PO 7.4 generators) dynamically controlled voltage locally, inverter-based assets (here: a large share of the column RES) contributed with reactive power according to a fixed power factor, and fixed reactive power assets (here: shunt reactors) were manually switched by the TSO to maintain static voltage.

The Expert Panel also assessed the voltage fluctuations that occurred on certain days prior to 28 April 2025, in particular on 22 April. On this day, the voltage reached very high values shortly after 19:00, leading to some disconnections but not to a cascade of overvoltage disconnections, as the system conditions and configuration on that day differed from those on 28 April. In particular, the system did not experience any significant episode of oscillation, either local or inter-area, on 22 April.

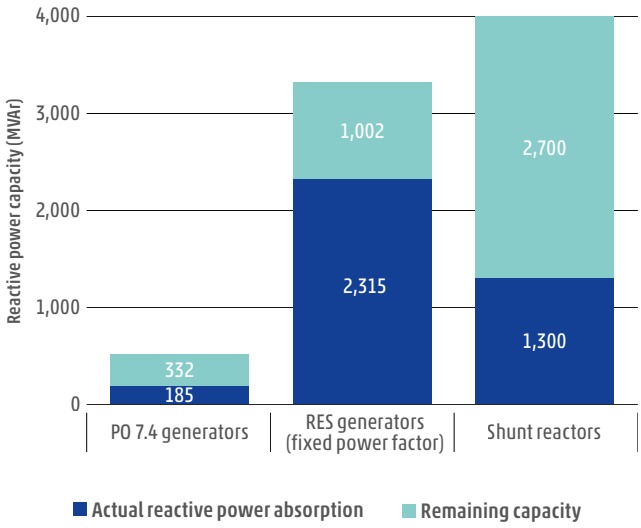


Figure 1-12: Capacity [6] of the different categories of reactive power assets in South Spain at 12:33:00

1.4.2 Oscillations

The second area of technical analysis concerns the two episodes of oscillation that occurred during the half-hour preceding the blackout and are described in Chapter 2. To perform the analysis of these phenomena, the Expert Panel (i) developed a simulation model of the CE power system that can represent the oscillatory behaviour of the system and (ii) used available data from a large number of PMUs located in the Iberian Peninsula and in the rest of the CE synchronous area.

Regarding the first oscillatory episode at 0.63 Hz, the data analysis shows that this mode is mainly located in the Iberian Peninsula, rather than in the rest of the European system, with higher activity in a specific area of Spain between Carmona and Almaraz. It also reveals that this mode is not naturally present in the system. The Expert Panel notes that, in the immediate vicinity of the node where the highest amplitude was observed (jointly with the highest power oscillations), an inverter-based power plant is connected, classified as an existing generator under the applicable requirements in Spain. The analysis concludes that this oscillation is a converter-driven forced oscillation.

The second oscillatory episode at 0.2 Hz was a classic inter-area oscillation of the East-Centre-West mode in the Continental Europe Synchronous Area. The modal analysis of the frequency measurements shows a significant participation of the Iberian Peninsula oscillation, together with the centre of Europe, and the eastern part. The causes of this second oscillation are the combination of different elements that enable the phenomenon. Specific conditions prevailing in the Iberian peninsula created the preconditions to the triggering of the oscillations: the high transmission angle (around 80° between Carmona and S. Llogaia), some lines out of service, the absence of power system stabilisers at some large units, and the insufficient damping of the devices already installed. The simulations carried out also show that the 0.63 Hz oscillation mode can contribute to exciting the 0.2 Hz inter-area oscillation mode, along with other factors.

The Expert Panel also assessed the effectiveness of the measures taken to dampen the oscillations. It concludes that the measures applied during the first oscillatory episode (0.63 Hz), which were not designed to damp oscillations in this frequency range, had a minor, but nevertheless positive impact on the oscillation. On the other hand, the analysis confirms the effectiveness of the operational measures applied during the inter-area oscillation (0.2 Hz).

Finally, the analysis identified several measures to improve the damping of forced and inter-area oscillations, respectively and showed that the measures taken had a positive impact on damping oscillations.

6 For explanation of the asterisk, see full Figure note in Section 4.1.2.

1.4.3 Analysis of the flows between TSO and DSO grids in Spain

The Expert Panel conducted a specific analysis to identify the main influencing factors explaining changes in flows between the transmission and distribution grids (hereafter TSO-DSO flows), particularly after the oscillations and during the critical time period from 12:32:00.

Changes in TSO–DSO flows and load result from complex interactions among various factors within the distribution and transmission grids. Changes in distribution grids (e.g., activation of distribution grid-connected units providing balancing services, including automatic frequency restoration reserves (aFRRs) and manual frequency restoration reserves (mFRRs)) can influence TSO–DSO flows, which in turn can impact voltage and regional flows across the transmission system.

There is also feedback from the transmission grid to the distribution grids, e.g. voltage-dependent loads or possible disconnections of small embedded generators due to voltage variations on medium- or low-voltage DSO feeders.

The analysis indicates that:

- the changes in the total TSO–DSO exchange can only be slightly linked to aFRR or mFRR activations;
- based on the data provided by two PV-inverter manufacturers, the changes in total TSO–DSO flows can be partially linked to disconnections of small PV installations (<1 MW) connected to low-voltage grids due to activation of over-voltage inverter protection.

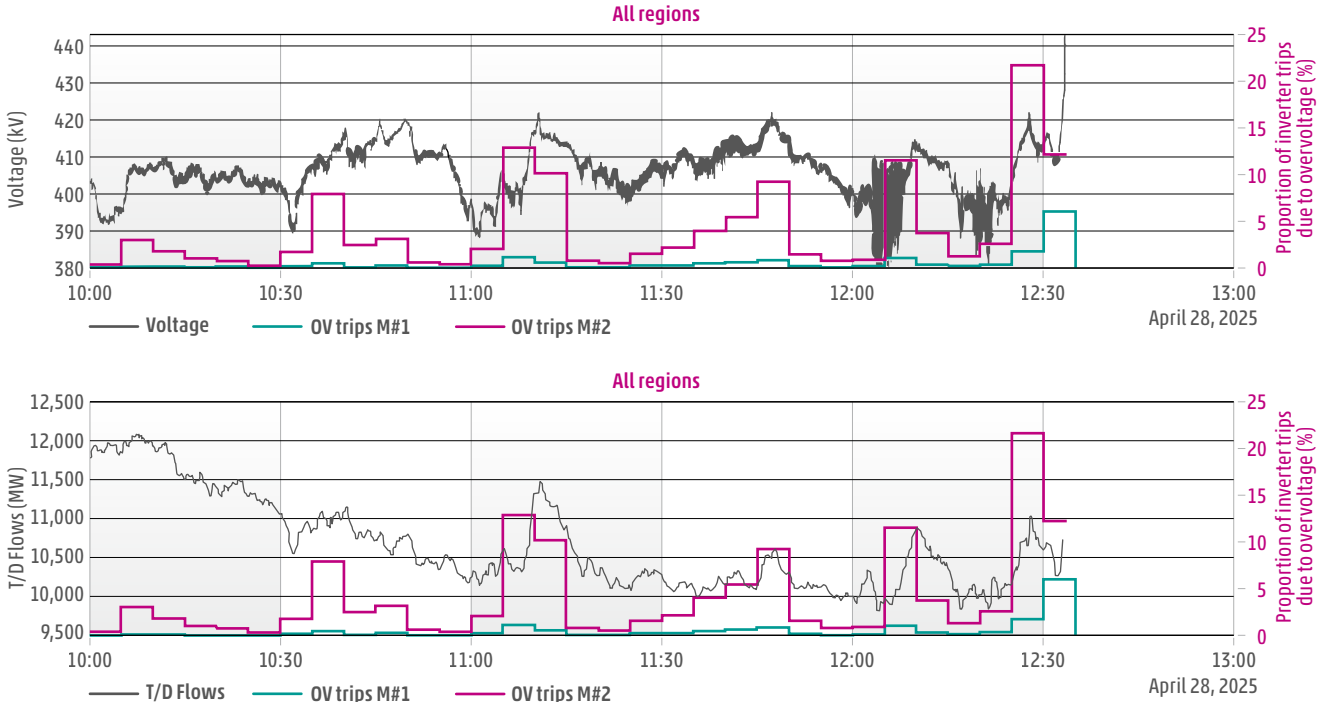


Figure 1-13: Correlation between voltage at the 400 kV substation Carmona , sum of TSO-DSO flows and proportion of inverter trips in Spain by automatic activation of overvoltage protections, aggregated in 5-minute intervals (Source: Red Eléctrica, PV-inverter manufacturers M#1 and M#2)

The data shows a correlation between the proportion of inverter trips and periods when the voltage rises on the transmission grid and when TSO–DSO flows increase. Furthermore, the data provided by one of the two PV-inverter manufacturers also allowed to estimate the total amount of disconnected generation[7]. This amount is highly dependent on the actual number of trips and on the total time between the disconnection and the reconnection of the disconnected inverters.

7 The methodology used, described in chapter 4.3, is based on several assumptions and simplifications in the absence of further information. These factors should therefore be taken into account when interpreting the results

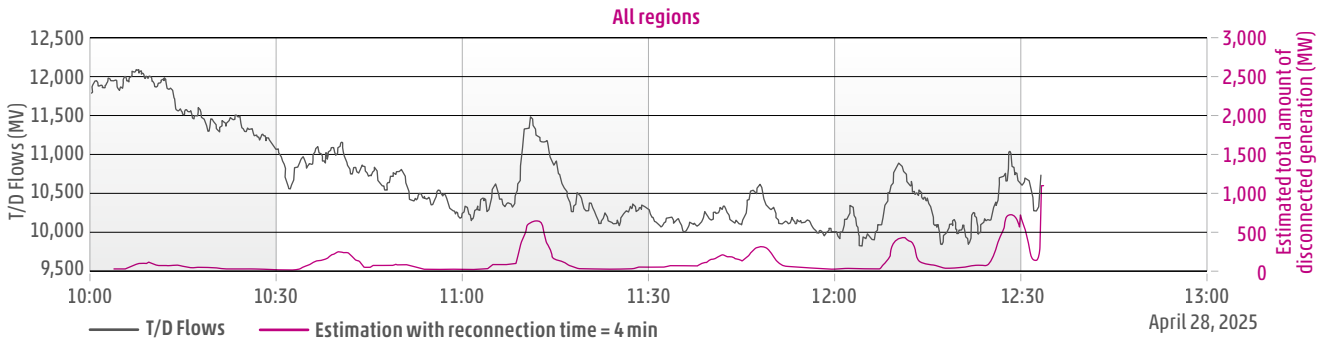


Figure 1-14: Correlation between the TSO–DSO flows and the estimated total amount of disconnected generation in Spain (units <1 MW connected to low-voltage grids)

» Because the voltage dependency of loads is an inherent property of power systems, it can also be linked to a certain extent to the load increase in the distribution grids (and hence to the changes in TSO–DSO flows). Indeed, a correlation is observed between a simplified load model and DSO measurements. However, detailed models, more comprehensive data on load composition, and additional representative measurements from DSOs would be needed to accurately model the behaviour of TSO/DSO interfacing points.

1.4.4 System defence plans

The system defence plans in Spain and Portugal were automatically triggered by the drop in frequency in the final stages leading up to the blackout. They were unable to prevent the system's collapse, despite the fact that their activation proceeded as designed.

The technical analysis of the system defence plan assesses several areas and factors using models suitable for analysing the frequency and voltage stability phenomena, respectively.

The frequency stability analysis includes:

» simulation of a perfectly executed load shedding and hydro-pump plant disconnection;

» the effect of inertia on the frequency variation;

» the role of HVDC power reversal mode.

In particular, the analysis of the impact of inertia indicates that, even with significantly higher inertia values, the loss of system synchronism would not have been avoided, considering the sequence of events. This can be explained by the rapid reduction in synchronising torque in the Iberian Peninsula (due to cascading generator tripping) that led the system to rapidly reach the point of no return.

Furthermore, analysis shows that the rapid intervention on the HVDC interconnection between France and Spain aimed at restoring the Iberian Peninsula's system balance could have worsened system behaviour due to fast transient phenomena.

The second part of the analysis focuses on voltage stability. After the validation of the simulation model, the analysis assesses the sensitivity regarding the following aspects:

» the effect of different parameters (cross-border exchanges, TSO/DSO exchanges, generation changes, connecting transmission lines, connecting/disconnecting shunt reactors) on voltage variations;

» the role of voltage regulation from conventional generators;

» the role of voltage regulation from additional synchronous condensers;

» the role of the defence system.

The simulations demonstrated that, at the time of the incident, the voltage of the Spanish power system was sensitive to any variations – whether related to exchanges with Portugal, France or Morocco, TSO-to-DSO flows, or generation– resulting in significant voltage fluctuations.

The analyses above clearly indicate that the key phenomenon in the incident was the non-effectiveness of voltage control within the Spanish power system. Simulations show that increased reactive power margins could have prevented system collapse, enabling it to operate at lower voltage levels and maintain overall system stability.

Finally, the Expert Panel conducted a detailed analysis of the performance of the Low-Frequency Demand Disconnection (LFDD) scheme.

The goal was to verify the load shed at each step, and to identify the exact instant of the trips, and the frequency value at the moment of pickup and tripping of the load shedding relays. Overall, this analysis concludes that the performance of the LFDD scheme at transmission- and distribution-level was in line with the applicable requirements and in accordance with the design.

1.4.5 Management of voltage-related alarms

Given the numerous voltage fluctuations that preceded the blackout, as detailed in Chapter 2, the Expert Panel considered it useful to analyse the voltage-related alarms recorded by the TSOs (in Spain, Portugal and France), DSOs (in Spain and Portugal) and generation control centres (in Spain) in the half-hour preceding the incident, and how they were handled.

The analysis indicates that the distribution of alarms recorded by Red Eléctrica exhibited zonal clustering, with the southern and eastern zones repeatedly prominent for undervoltage alarms, while the interior, southern and northern zones recorded the majority of overvoltage alarms.

The remedial actions taken within the control rooms (coupling and decoupling of reactors, opening and closing of lines, setting HVDC voltage setpoints) were effective in bringing the voltage back under the alarm threshold within minutes in most cases (typically less than 5 minutes). In the last overvoltage episode that preceded the system collapse, Red Eléctrica's dispatchers were unable to implement manual remedial actions due to the rapid development of the incident.

1.4.6 Root cause tree

The analyses summarised above enabled the Expert Panel to develop a root cause tree of the incident. Its purpose is to explain, in schematic, synthetic and simplified form, how the incident unfolded as a result of a combination of key factors, as well as the causal relationships between the various elements.

The root cause tree is presented in Figure 1-15.

Root cause tree

The blue boxes correspond to factors that contributed to the incident and on which the Expert Panel is making recommendations.

The white boxes correspond to additional elements or explanations that allow to understand how the incident developed.

(x) Link to recommendation number

Abbreviations

aFRR	automatic Frequency Restoration Reserve
BRP	Balancing Responsible Party
CESA	Continental Europe Synchronous Area
DRS	loss of synchronism protection (Débouclage suite à Rupture de Synchronisme)
PSS	Power System Stabiliser
Q-reference	reference reactive power (as defined in § 2.6.6)
RES	Renewable Energy Sources
RfG	Requirements for Generators Network Code
SPS	Special Protection Scheme

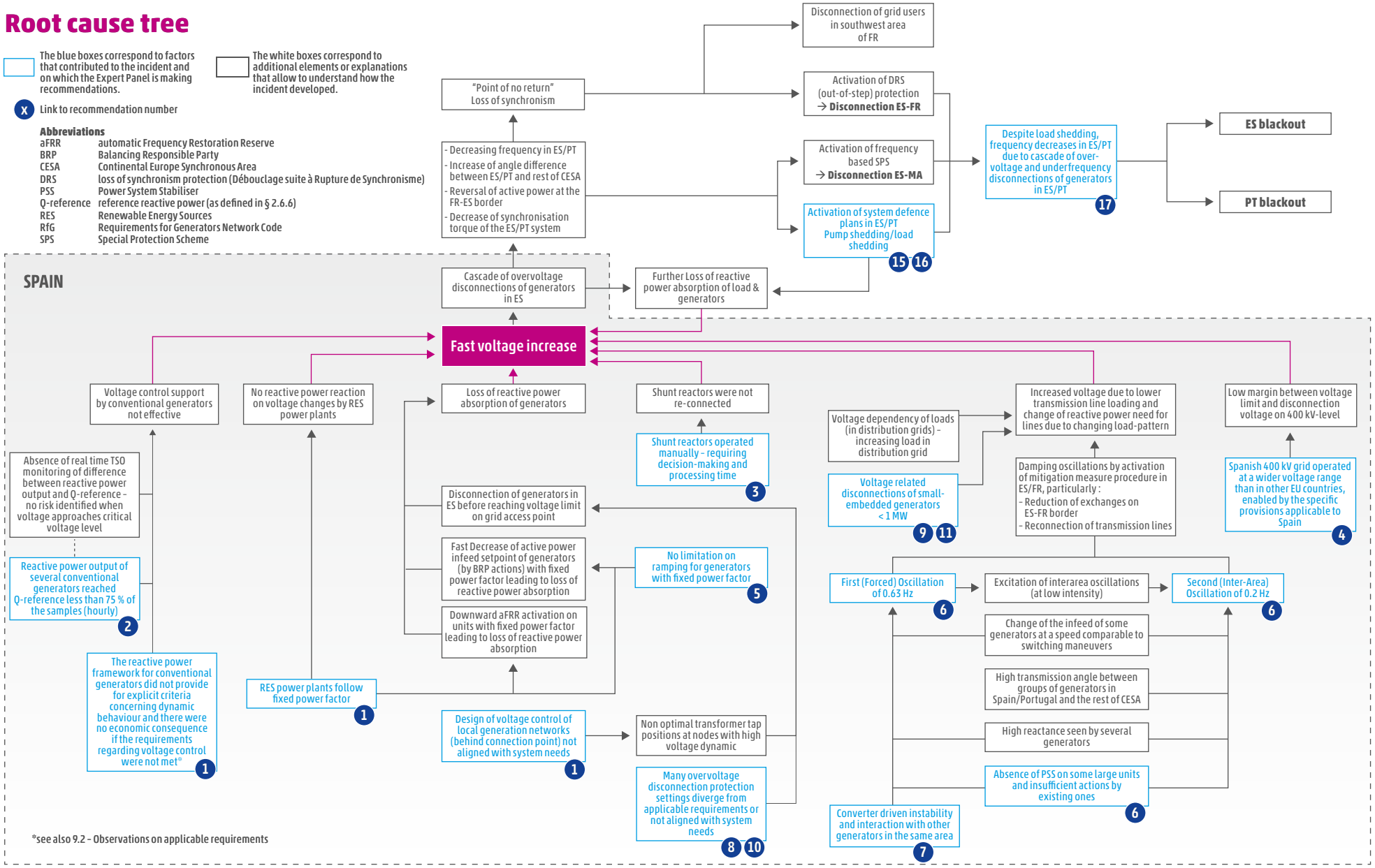


Figure 1-15: Root cause tree

From the tree, it can be seen that the fast voltage increase, which led to the cascade of overvoltage disconnections in Spain and ultimately to the blackout, occurred due to the combination of numerous factors, and particularly (the order in this list does not reflect any particular ranking in terms of criticality):

- the reactive power output of several conventional generators reached Q-reference less than 75 % of the samples (hourly);
- the framework for conventional generators' reactive power output did not include specifications for dynamic behaviour and there were no economic consequences if the reactive power output was not aligned with the 75 % rule;
- the RES power plants followed a fixed power factor;
- the design of voltage control of local generation networks (behind connection point) is not aligned with the system needs;
- there was no limitation on ramping for generators with fixed power factor;
- shunt reactors are operated manually, requiring decision-making and processing time;
- according to data provided and estimations performed, many overvoltage disconnection protection settings diverged from applicable requirements or were not aligned with system needs;
- two oscillatory episodes occurred in the half-hour preceding the blackout: first, a (forced) oscillation of 0.63 Hz, and second, an (inter-area) oscillation of 0.2 Hz:
  - The power system experienced converter-driven instability that interacted with other generators in the same area;
  - There was an absence of PSS at some large units and insufficient damping by the existing ones;
- A certain proportion of small PV units (<1 MW) experienced voltage-related disconnections by activation of inverter protection;
- Spanish 400 kV grid is operated at a wider voltage range than in other EU countries, enabled by specific provisions applicable to Spain;
- the system entered an operating point where, by design, its defence plan was unable to interrupt the cascade of overvoltage disconnections and prevent the total collapse of the Spanish and Portuguese power systems.

More explanation on the root cause tree is provided in Chapter 4.6.

1.5 Restoration Process

Following the incident, each affected TSO immediately activated their respective system restoration plans, implemented in accordance with the applicable requirements, as well as any other relevant procedures and protocols for restoring the electricity system's voltage.

Power system restoration in some regions of the Portuguese and Spanish systems was facilitated, among other measures, by activating power system resources, such as black-start processes in certain power plants, and by leveraging existing interconnections with France and Morocco, as schematically illustrated in Figure 1-16.

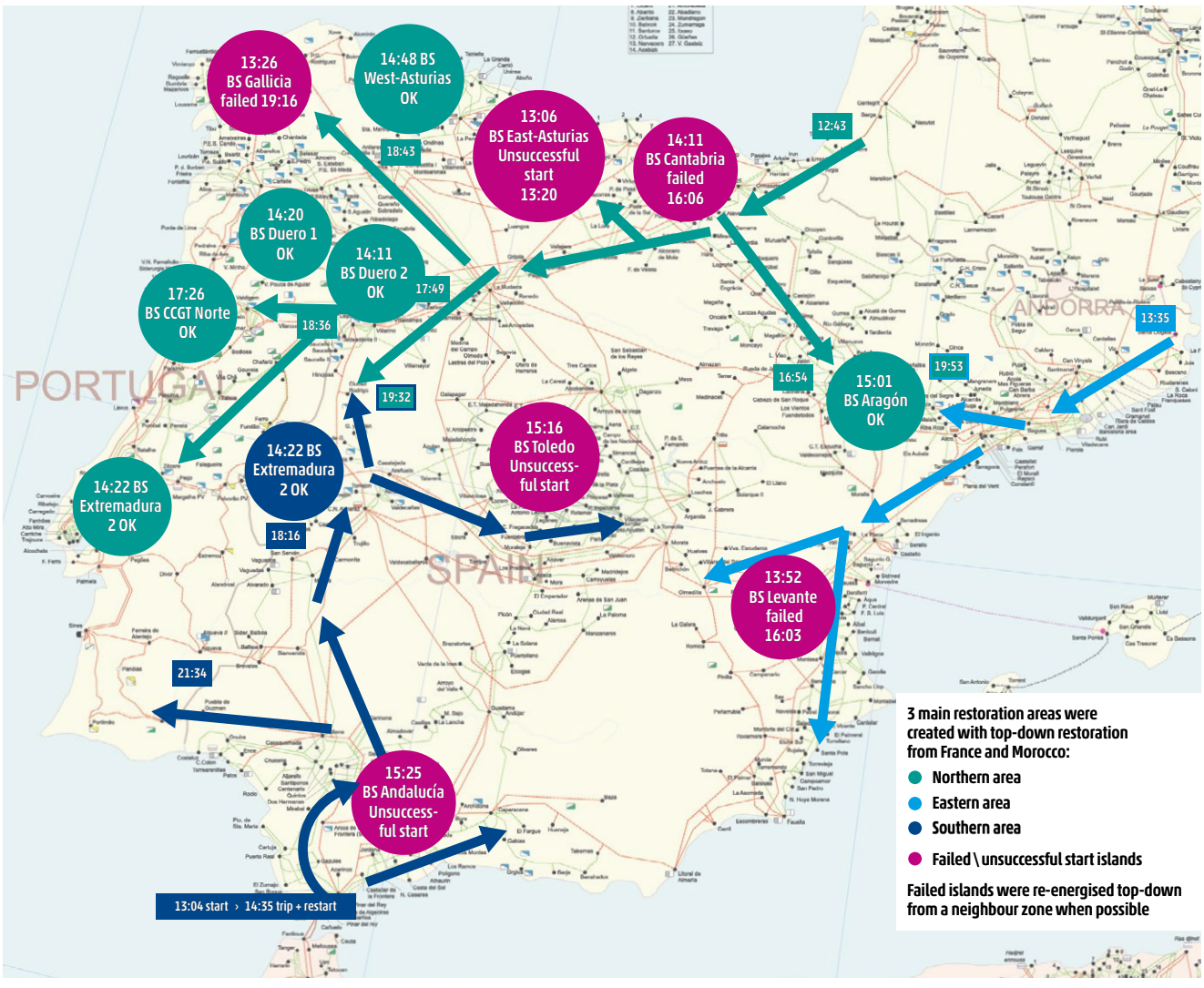


Figure 1-16: Mapping of the main milestones of the restoration process in Spain and Portugal

In Figure 1-16, the green and blue circles represent islands energised "bottom-up" from black-start facilities (some after multiple attempts) and eventually synchronised with a "top-down" re-energised part of the grid (arrows in magenta, green or light blue). The magenta circles indicate islands where bottom-up restoration failed. These areas were re-energised later from one of the "top-down" re-energised grids, from either France (L-400 kV Argia—Hernani energised at 12:43 and L-400 kV Baixas—Vic energised at 13:35) or Morocco.

At 13:04, the interconnection between Spain and Morocco was energised, but synchronisation had to be repeated at 14:34 after tripping at 14:27.

At 15:07, all Spanish nuclear power plants confirmed that they had external supply for their auxiliary services.

At 18:36, the L-220 kV Aldeadavila—Pocinho 1 interconnector between Spain and Portugal was energised, and the Portuguese transmission grid received voltage with continental frequency again. In the meantime, two black-start islands could be built up in Portugal.

At 19:32, by switching on the L-400 kV Almaraz—C. Rodrigo line, the southern zone connected with the Moroccan system could be synchronised with the northern zone connected to CE SA.

At 00:22 and around 04:00 on 29 April 2025, the transmission grid restoration was completed in Portugal and Spain, respectively. Therefore, thanks to comprehensive restoration procedures, fallback strategies, and the full commitment of operators from TSO, DSOs, producers and other parties, the Spanish system was fully restored within 16 hours, while the Portuguese system was back online within 12 hours after the blackout.

Nevertheless, the Expert Panel identified several issues that impacted the restoration process, including difficulties in starting black-start units or maintaining stable islands, problems with the voice communication systems of several DSOs and significant grid users, and insufficient observability of facilities involved in bottom-up restoration scenarios of distribution grids. Chapter 9 of the report includes several recommendations to address these issues.

More details on the restoration process are described in Chapter 5.

1.6 Operational Planning

The results of the various tasks performed by the RCCs before the incident indicate that the grid was considered secure and no major issues were detected in the affected area during the operational planning phase. The outage planning coordination (OPC) task conducted by SEleNe CC and Coreso to assess outage incompatibilities for transmission elements revealed no violations of N-1 criteria (no flow congestions and no voltage out of range) for the Iberian Peninsula transmission network.

The short-term adequacy (STA) analysis, led by Nordic RCC, confirmed that the available production capacity could meet the expected consumption. The coordinated security analysis (CSA) conducted by Coreso revealed no significant operational security risks, and the grid was deemed N-1 secure. The common grid model (CGM) and coordinated capacity calculation (CCC) processes did not reveal any unsafe grid situation.

The RCC analysis conducted before the incident and the operational planning procedures applied in Spain are described in further detail in Chapter 6.

1.7 Communication of Synchronous Area Monitors and Between TSOs

During and after the blackout, communication between TSOs and synchronous area monitors (Swissgrid and Amprion) supported crisis management and restoration coordination. Exchanges via direct calls, ENTSO-E Awareness System (EAS) notifications, and email enabled the situational awareness, alignment on cross-border actions and prioritisation of restoration steps. Key operational measures included adjustments in cross-border exchanges and HVDC interconnection modes to support grid stability.

Unaffected TSOs actively shared information and offered support throughout the incident. The synchronous area monitors streamlined information flow, enabling the affected TSOs to focus on urgent operational needs.

This structured, timely communication framework helped manage the incident and restore system stability across the affected regions, as described in further detail in Chapter 7.

1.8 Classification of the Incident Based on the ICS Methodology

The Incident Classification Scale (ICS) Methodology has been developed in accordance with Regulation (EC) No 714/2009 of the European Parliament and of the Council of 13 July 2009, as repealed by the Regulation (EU) 2019/943, and updated to fulfil the objectives under Article 15(5) of the Commission Regulation (EU) 2017/1485 of 2 August 2017 establishing a guideline on electricity transmission system operation (SO GL). The ICS methodology aims to provide a realistic view of system states during incidents. The incident classification criteria are ranked by priority, with the highest-priority criterion determining the incident scale. The methodology stipulates that every incident classified as scale 2 or 3 must be investigated by an Expert Panel.

On 28 April 2025, the scale 3 of incident severity was met in Spain and Portugal as the result of both loss of demand in the Iberian Peninsula and the total absence of voltage. In France, the highest scale of incident severity reached was scale 1 – violation of standards on voltage – due to the high voltage experienced due to the incident. Consequently, the Expert Panel was established based on the requirements of the applicable legislation.

Additionally, the incident met the RCC investigation threshold described in the RCC Post-Operation and Post-Disturbances Analysis and Reporting methodology. This prompted the RCC investigation subgroup to initiate an RCC investigation. Further explanations can be found in Chapter 8.

1.9 Conclusions and Recommendations

The blackout that affected Spain and Portugal on 28 April 2025 was the most severe and unprecedented incident in the European power system for more than 20 years. The investigation conducted by the Expert Panel to establish the most accurate sequence of events possible, to understand the root causes of this incident, and to identify all the factors that contributed to it was also exceptional in terms of its complexity, scope and the nature of the events that led to the blackout.

This final report of the Expert Panel's investigation highlights that the blackout is the result of multiple interacting factors. The incident evolved through a sequence involving a combination of voltage fluctuations and oscillatory phenomena, leading to widespread generation disconnections in Spain, particularly inverter-based resources, followed by a cascade of overvoltage disconnections and culminating in the loss of synchronism of the Iberian system with the Continental Europe Synchronous Area.

The root cause tree provides a comprehensive picture of the causes, their relation and the sequence of events. Despite the correct activation of the system defence plans, the nature and magnitude of the cascading events led to a full collapse of the Spanish and Portuguese systems within seconds. After the blackout, the restoration process began immediately and was completed in 12 hours in Portugal and 16 hours in Spain, thanks to comprehensive restoration procedures, fallback strategies and the full commitment of operators from TSOs, DSOs, producers and other parties.

After analysing all the data at its disposal, establishing the sequence of events, and identifying the causes and other factors that contributed to the incident, the Expert Panel's work has focused primarily on deriving clear, actionable recommendations to reduce the likelihood of similar events and strengthen the overall security and resilience of the European grid.The recommendations are grouped into two categories:

(i) those directly linked to root causes of the incident;

(ii) those not directly linked to the root causes of the incident.

In addition, the report contains several other proposals identified as "areas for improvement" at the end of certain chapters.

These proposals are not included among the recommendations, as they do not have a direct or indirect link to the incident. Nevertheless, they have been identified as potentially useful for further consideration. The report also revisits past recommendations from previous ICS investigations that remain relevant to this event, highlighting the importance of implementation and long-term follow-up. In addition, the Expert Panel notes several broader areas where further work by ENTSO-E and third parties could enhance system security.

Finally, the report contains a list of observations concerning the applicable requirements. In line with the scope of the investigation and the mandate of the Expert Panel, these observations do not aim to interpret the applicable legislation or determine liabilities for non-compliance with legal requirements. These observations have been included without prejudice to any inquiries and/or follow-up actions carried out by the competent national authorities.

Ultimately, the recommendations outlined in this chapter represent a comprehensive set of measures designed to enhance operational robustness, improve cross-stakeholder information exchange and help maintain a high level of security of supply across the European power system. As such, they aim to ensure that the lessons learned from this exceptional incident are duly reflected in the applicable requirements on both European and national levels.

The monitoring of the implementation of the recommendations provided in this chapter does not fall within the mandate or remit of the Expert Panel. Any responsibility for the consideration, follow-up and implementation of the recommendations lies solely with each addressee.

Strengthened coordination among TSOs, DSOs, SGUs and other stakeholders is essential to effectively managing complex system events like this one. Sustained efforts to enhance operational practices and information sharing will help maintain the security of supply for all consumers.

All recommendations are detailed in Chapter 9 of the report.

2 SYSTEM AND MARKET CONDITIONS BEFORE THE INCIDENT

During the night spanning from 27 to 28 April, the Iberian power system operated normally with voltages in Spain in the range 399 – 426 kV. This chapter presents the system and market conditions on Monday, 28 April 2025 for the period between 9:00:00 and 12:32:00. For the purpose of this report, it is considered that the incident started at 12:32:00.

2.1 Information on Topology

2.1.1 Planned and Unplanned Outages

In this section, all planned outages with an impact on neighbouring systems are presented by the TSO as defined in methodologies for coordinated operations, in accordance with Commission Regulation (EU) 2017/1485 (see Section 2.8.1 for the detailed list of RTE, RE, and REN outages).

There were no unplanned outages with an impact on neighbours' systems on either RE, REN, or RTE's grid. Therefore, the figure below presents the location of planned outages with an impact on neighbours' systems. Planned outages without an impact on neighbouring systems are listed in Tables 2-9, 2-10 and 2-11 in Section 2.8.1.

Figure 2-1: Location of planned unavailable elements with an impact on neighbouring systems

Note: The figure shows (in blue colour) all transmission elements with an impact on neighbouring systems that were in planned maintenance on 28 April between 12:00 and 13:00.

From the planned outages with an impact on neighbouring systems, two notable border tie lines were out of service:

- Brovales–Alqueva 400 kV (Spain–Portugal border)
- Biescas–Pragnères 220 kV (France–Spain border)

The Brovales–Alqueva 400 kV (Spain–Portugal) tie line was put into service on 29 April, following the interruption of this outage, initially planned to terminate on 3 May.

Table 2-1 below provides a list of works and manoeuvres from RE between 9:00 and 12:32.

HOUR	ELEMENT	ZONE	COMMENTS
09:00	SE 220 kV SERRALLO	EAST	
09:03	SE 220 kV STA. ELVIRA	SOUTH	
09:16	SE 220 kV ACECA pos PRADILLOS	CENTRE	
09:21	SE 220 kV TORRELLANO	EAST	
09:37	SE 400 kV ALDEADAVILA: JBP2	NORTHWEST	
09:37	SE 400 kV FAUSITA	EAST	
09:52	L-220 kV PRADO SANTO DOMINGO–VILLAVICIOSA	CENTER	
09:52	SE 220 kV VILLAVICIOSA pos ACJ	CENTER	Operation in 2 nodes
09:53	SE 400 kV GUILLENA: L/COLLECTOR 1	SOUTH	
10:46	SE 400 kV PALOS: AT-2 and TM-2	SOUTH	
11:15	L-220 kV VILLAVICIOSA–LUCERO–LEGANES	CENTER	
11:36	SE 220 kV ACECA: 522-1 Switch	CENTER	
12:16	SE 220 kV SS. REYES: L/ PS. FERNANDO split with L/ ALCOBENDAS	CENTER	Topological manoeuvre at SS REYES 220 kV to avoid overload on L-220 kV LOECHES–PS. FERNANDO post contingency at DC-400 kV ALMARAZ–C. RODRIGO/ALDEADAVILA–ARAÑUELO

Table 2-1: List of works and topological manoeuvres from RE

Throughout this document, the following geographic zones are defined:

- South: Andalucía (except Almería) and Extremadura
- Centre: Madrid and Castilla–La Mancha (except Albacete)
- East: Comunidad Valenciana, Murcia, Albacete, and Almería
- North: País Vasco, Navarra, La Rioja, and Aragón
- North East: Cataluña
- North West: Galicia, Asturias, and Cantabria y Castilla León

2.1.2 Topological actions for voltage control

This section lists all manual topological actions taken for voltage control by TSOs (see section 2.8.2 for the detailed list of open lines and RE zonal SCADA images at 12:25). The situation from 9:00 to 12:32 is depicted in Figure 2-2:

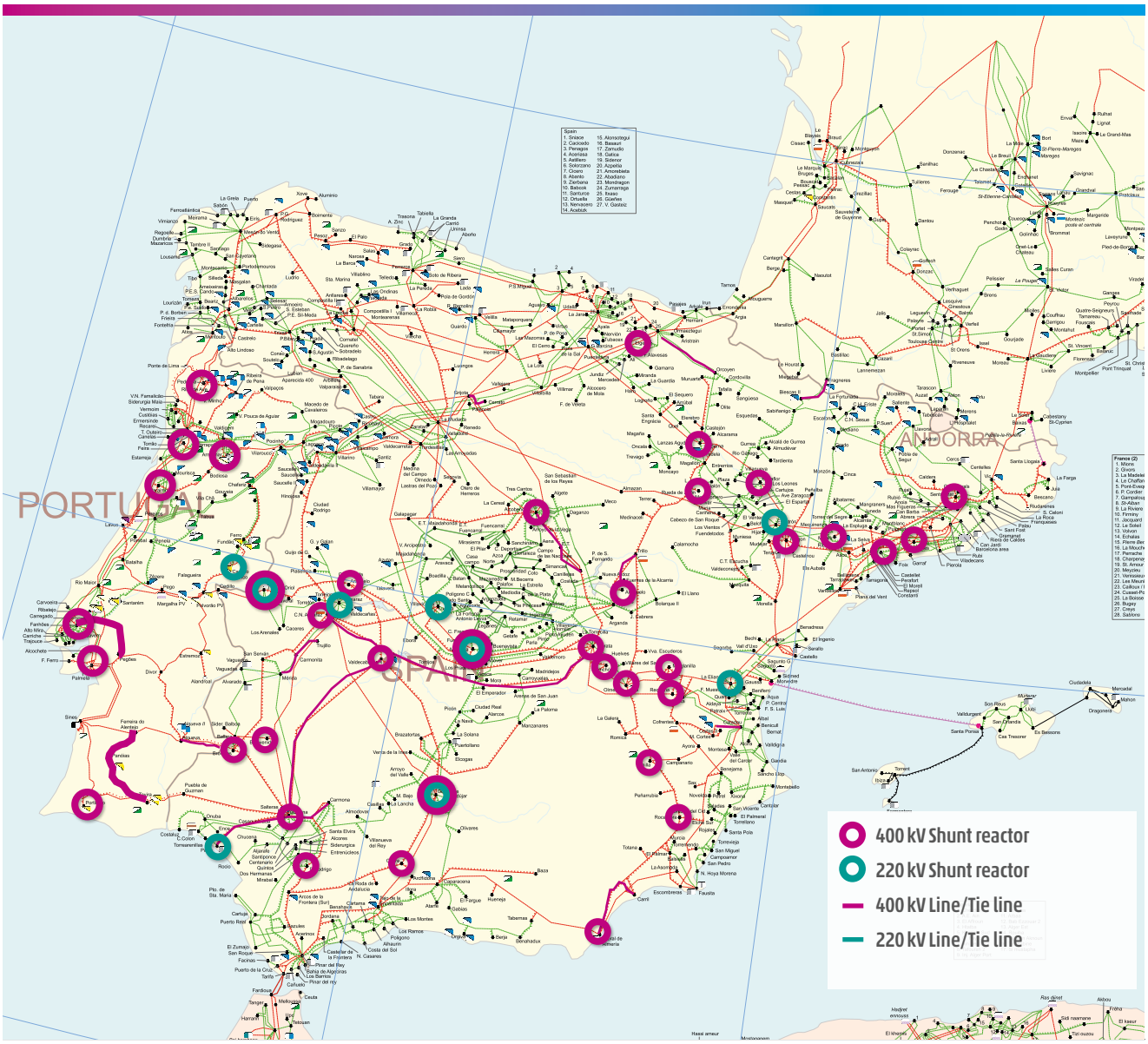


Figure 2-2: Iberian scenario: Open lines and reactances in Portugal and in Spain from 9:00 until 12:32

For context, shunt reactors and lines are connected or disconnected manually based on decisions of the operators in the RE and REN control rooms. Connecting lines decreases system impedance. This action increases damping, and on the other hand increases the reactive power production in the system and hence increases voltage. Disconnecting lines has the opposite effect. Connecting shunt reactors increases reactive power consumption in the system and thus decreases voltage. Disconnecting shunt reactors has the opposite effect.

Additionally, in the table below a list of manoeuvres from Red Eléctrica before 12:32 is provided. For every manoeuvre, or group of manoeuvres, that it is done in the control room, the decision to perform the manoeuvre is taken either based on the experience of the operator or based on a static power flow simulation performed right before, to check the expected impact of the action on voltages and loads of elements.

HOUR	ELEMENT	NAME	MOVEMENT	ZONE
09:02	LINE	L-400 kV ALMARAZ–SAN SERVÁN 1	SWITCH ON	SOUTH
09:02	SHUNT REACTOR	VALDECABALLEROS 400 REA 2	SWITCH OFF	SOUTH
09:02	SHUNT REACTOR	ANCHUELO REA 1	SWITCH OFF	CENTRE
09:05	SHUNT REACTOR	MINGLANILLA 400 REA 1	SWITCH OFF	EAST
09:08	SHUNT REACTOR	LITORAL 400 REA 1	SWITCH OFF	EAST
09:13	LINE	L-400 kV BRAZATORTAS–MANZANARES 1	SWITCH ON	CENTRE
09:13	LINE	L-220 kV GURREA–VILLANUEVA 1	SWITCH ON	NORTH
09:17	LINE	L-400 kV SALLENTE–CALDERS	SWITCH ON	NORTHEAST
09:13	SHUNT REACTOR	RUEDA 400 REA 2	SWITCH OFF	NORTH
09:14	SHUNT REACTOR	BELINCHON 400 REA 1	SWITCH OFF	EAST
09:22	SHUNT REACTOR	VITORIA 400 REA 2	SWITCH ON	NORTH
09:23	SHUNT REACTOR	GUADAME 400 REA 2	SWITCH OFF	SOUTH
09:24	SHUNT REACTOR	DRODRIGO 400 REA 1	SWITCH OFF	SOUTH
09:25	SHUNT REACTOR	ARANUELO 400 REA 1	SWITCH OFF	SOUTH
09:26	SHUNT REACTOR	BIENVENIDA 400 REA 1	SWITCH OFF	SOUTH
09:27	SHUNT REACTOR	MORALEJA 400 REA 1	SWITCH OFF	CENTRE
09:31	SHUNT REACTOR	JM. ORIOL 400 REA 2	SWITCH OFF	SOUTH
09:32	SHUNT REACTOR	MORALEJA 220 REA 12	SWITCH OFF	CENTRE
09:34	SHUNT REACTOR	ALMARAZ 400 REA 3	SWITCH OFF	SOUTH
09:41	SHUNT REACTOR	VALDECABALLEROS 400 REA 1	SWITCH OFF	SOUTH
09:44	SHUNT REACTOR	BROVALES 400 REA 1	SWITCH OFF	SOUTH
09:49	SHUNT REACTOR	EALMARAZ 220 REA 1	SWITCH OFF	SOUTH
09:52	SHUNT REACTOR	MAGALLON 400 REA 2	SWITCH OFF	NORTH
09:54	LINE	L-400 kV ALMARAZ–MORATA 2	SWITCH ON	CENTRE
10:02	LINE	L-400 kV BROVALES–SAN SERVAN 1	SWITCH OFF	SOUTH
10:04	SHUNT REACTOR	GUILLENA 400 REA 2	SWITCH OFF	SOUTH
10:05	SHUNT REACTOR	CABRA 400 REA 1	SWITCH OFF	SOUTH
10:05	LINE	L-400 kV ARCOS–D. RODRIGO 2	SWITCH ON	SOUTH
10:18	SHUNT REACTOR	JM. ORIOL 220 REA 1	SWITCH OFF	SOUTH
10:19	SHUNT REACTOR	MORALEJA 220 REA 13	SWITCH OFF	CENTRE
10:20	SHUNT REACTOR	OLMEDILLA 400 REA 1	SWITCH OFF	EAST
10:22	SHUNT REACTOR	VILLAVICIOSA 220 REA 2	SWITCH OFF	CENTRE
10:29	SHUNT REACTOR	ROCAMORA 400 REA 1	SWITCH OFF	EAST
10:32	LINE	L-220 kV ACECA–PICON	SWITCH ON	CENTRE
10:32	SHUNT REACTOR	MAGALLON 400 REA 1	SWITCH OFF	NORTH
10:32	SHUNT REACTOR	PINILLA 400 REA 1	SWITCH OFF	EAST
10:32	SHUNT REACTOR	SS REYES 400 REA 3	SWITCH OFF	CENTRE
10:33	LINE	L-400 kV BROVALES–GUILLENA 1	SWITCH ON	SOUTH
10:33	HVDC 320 kV STA. LLOGAIA–BAIXAS	RAISE THE SETPOINT TO 413 kV	SETPOINT	NORTHEAST
10:35	LINE	L-400 kV GUADAME–VALDECABALLEROS	SWITCH ON	SOUTH
10:39	SHUNT REACTOR	MAGALLON 400 REA 2	SWITCH ON	NORTH
10:40	SHUNT REACTOR	GUADAME 400 REA 2	SWITCH ON	SOUTH
10:40	SHUNT REACTOR	PINILLA 400 REA 1	SWITCH ON	EAST
10:40	SHUNT REACTOR	MORALEJA 220 REA 12	SWITCH ON	CENTRE
10:43	HVDC 320 kV STA. LLOGAIA–BAIXAS	REDUCE THE SETPOINT TO 409 kV	SETPOINT	NORTHEAST
10:44	SHUNT REACTOR	RUEDA 400 REA 2	SWITCH ON	NORTH

HOUR	ELEMENT	NAME	MOVEMENT	ZONE
10:44	CONDENSER	JUIA 220 CONDEN1	SWITCH OFF	NORTHEAST
10:45	SHUNT REACTOR	VALDECABALLEROS 400 REA 2	SWITCH ON	SOUTH
10:50	SHUNT REACTOR	CABRA 400 REA 1	SWITCH ON	SOUTH
10:50	SHUNT REACTOR	REQUENA 400 REA 1	SWITCH ON	EAST
10:51	HVDC 320 kV STA. LLOGAIA–BAIXAS	REDUCE THE SETPOINT TO 404 kV	SETPOINT	NORTHEAST
10:59	SHUNT REACTOR	VALDECABALLEROS 400 REA 2	SWITCH OFF	SOUTH
10:59	SHUNT REACTOR	SENTMENAT 400 REA 1	SWITCH OFF	NORTHEAST
10:59	HVDC 320 kV STA. LLOGAIA–BAIXAS	RAISE THE SETPOINT TO 410 kV	SETPOINT	NORTHEAST
11:00	SHUNT REACTOR	GUADAME 400 REA 2	SWITCH OFF	SOUTH
11:01	SHUNT REACTOR	CABRA 400 REA 1	SWITCH OFF	SOUTH
11:02	SHUNT REACTOR	LA SERNA 400 REA 2	SWITCH OFF	NORTH
11:03	LINE	L-400 kV OLMEDILLA–ROMICA 2	SWITCH ON	EAST
11:03	SHUNT REACTOR	BEGUES 400 REA 1	SWITCH OFF	NORTHEAST
11:03	SHUNT REACTOR	REQUENA 400 REA 1	SWITCH OFF	EAST
11:03	SHUNT REACTOR	VITORIA 400 REA 2	SWITCH OFF	NORTH
11:03	SHUNT REACTOR	GUADAME 220 REA 3	SWITCH OFF	SOUTH
11:04	SHUNT REACTOR	ESCATRON 220 REA 1	SWITCH OFF	NORTH
11:04	SHUNT REACTOR	MORALEJA 220 REA 12	SWITCH OFF	CENTRE
11:04	SHUNT REACTOR	PALOS 220 REA 1	SWITCH OFF	SOUTH
11:04	SHUNT REACTOR	RUEDA 400 REA 2	SWITCH OFF	NORTH
11:04	SHUNT REACTOR	MAIALS 400 REA 1	SWITCH OFF	NORTHEAST
11:07	SHUNT REACTOR	MAGALLON 400 REA 2	SWITCH OFF	NORTH
11:07	SHUNT REACTOR	RUBI 400 REA 1	SWITCH OFF	NORTHEAST
11:07	LINE	L-400 kV AGUAYO–ABANTO	SWITCH ON	NORTHWEST
11:07	LINE	L-400 kV GUADAME–CABRA 1	SWITCH ON	SOUTH
11:08	LINE	L-400 kV PINAR–TAJO	SWITCH ON	SOUTH
11:08	HVDC 320 kV STA. LLOGAIA–BAIXAS	RAISE THE SETPOINT TO 413 kV	SETPOINT	NORTHEAST
11:08	SHUNT REACTOR	PINILLA 400 REA 1	SWITCH OFF	EAST
11:09	LINE	L-400 kV MONTEARENAS–MUDARRA 2	SWITCH ON	NORTHWEST
11:10	SHUNT REACTOR	CABRA 400 REA 1	SWITCH ON	SOUTH
11:10	SHUNT REACTOR	GUADAME 220 REA 3	SWITCH ON	SOUTH
11:11	HVDC 320 kV STA. LLOGAIA–BAIXAS	REDUCE THE SETPOINT TO 409 kV	SETPOINT	NORTHEAST
11:11	SHUNT REACTOR	LA SERNA 400 REA 2	SWITCH ON	NORTH
11:12	SHUNT REACTOR	MAIALS 400 REA 1	SWITCH ON	NORTHEAST
11:14	HVDC 320 kV STA. LLOGAIA–BAIXAS	REDUCE THE SETPOINT TO 405 kV	SETPOINT	NORTHEAST
11:17	LINE	L-400 kV ARCOS–CABRA	SWITCH ON	SOUTH
11:18	SHUNT REACTOR	RUBI 400 REA 1	SWITCH ON	NORTHEAST
11:20	LINE	L-400 kV PIEROLA–VANDELLÓS	SWITCH ON	NORTHEAST
11:22	SHUNT REACTOR	ELIANA 220 REA 1	SWITCH OFF	EAST
11:43	SHUNT REACTOR	ELIANA 220 REA 1	SWITCH ON	EAST
11:43	SHUNT REACTOR	ESCATRON 220 REA 1	SWITCH ON	NORTH
11:46	SHUNT REACTOR	SENTMENAT 400 REA 1	SWITCH ON	NORTHEAST
11:47	SHUNT REACTOR	GUADAME 400 REA 2	SWITCH ON	SOUTH
11:47	SHUNT REACTOR	MINGLANILLA 400 REA 1	SWITCH ON	EAST
11:48	SHUNT REACTOR	RUEDA400 REA 2	SWITCH ON	NORTH
11:48	HVDC 320 kV STA. LLOGAIA–BAIXAS	REDUCE THE SETPOINT TO 401 kV	SETPOINT	NORTHEAST

HOUR	ELEMENT	NAME	MOVEMENT	ZONE
11:48	SHUNT REACTOR	EALMARAZ 220 REA 1	SWITCH ON	SOUTH
11:48	SHUNT REACTOR	MORALEJA 220 REA 12	SWITCH ON	CENTRE
11:50	SHUNT REACTOR	PALOS 220 REA 1	SWITCH ON	SOUTH
11:59	HVDC 320 kV STA. LLOGAIA–BAIXAS	RAISE THE SETPOINT TO 406 kV	SETPOINT	NORTHEAST
11:59	SHUNT REACTOR	GUADAME 400 REA 2	SWITCH OFF	SOUTH
12:01	SHUNT REACTOR	EALMARAZ 220 REA 1	SWITCH OFF	SOUTH
12:02	LINE	L-220 kV C.PLATA–VILLAVERDE BAJO 2	SWITCH ON	CENTRE
12:04	SHUNT REACTOR	VILLAVICIOSA 400 REA 1	SWITCH OFF	CENTRE
12:04	SHUNT REACTOR	GUADAME 220 REA 3	SWITCH OFF	SOUTH
12:05	HVDC 320 kV STA. LLOGAIA–BAIXAS	RAISE THE SETPOINT TO 412 kV	SETPOINT	NORTHEAST
12:05	SHUNT REACTOR	RUEDA 400 REA 2	SWITCH OFF	NORTH
12:07	LINE	L-400 kV GRIJOTA–VILLARINO 2	SWITCH ON	NORTHWEST
12:07	LINE	L-400 kV P. GUZMAN–GUILLENA 1	SWITCH ON	SOUTH
12:07	LINE	L-400 kV PALMAR–CARRIL	SWITCH ON	EAST
12:07	SHUNT REACTOR	ARAGON 400 REA 1	SWITCH OFF	NORTH
12:08	LINE	L-400 kV LA ROBLA–MUDARRA	SWITCH ON	NORTHWEST
12:08	LINE	L-400 kV PALMAR–ROCAMORA 2	SWITCH ON	EAST
12:15	LINE	L-400 kV MORATA–VILLAVICIOSA	SWITCH ON	CENTRE
12:17	SHUNT REACTOR	CABRA 400 REA 1	SWITCH OFF	SOUTH
12:21	SHUNT REACTOR	PEÑAFLOR 400 REA 1	SWITCH OFF	NORTH
12:21	LINE	L-400 kV PINILLA–ROMICA 2	SWITCH ON	EAST
12:22	LINE	L-400 kV PINILLA–ROCAMORA 1	SWITCH ON	EAST
12:24	SHUNT REACTOR	PALOS 220 REA 1	SWITCH OFF	SOUTH
12:24	SHUNT REACTOR	MORATA 400 REA 4	SWITCH OFF	CENTRE
12:25	LINE	L-400 kV GUADAME–CABRA 3	SWITCH ON	SOUTH
12:25	LINE	L-400 KV TORDESILLAS–GALAPAGAR	SWITCH ON	CENTRE
12:26	SHUNT REACTOR	VITORIA 400 REA 2	SWITCH ON	NORTH
12:27	SHUNT REACTOR	PEÑAFLOR 400 REA 1	SWITCH ON	NORTH
12:27	SHUNT REACTOR	GUADAME 220 REA 3	SWITCH ON	SOUTH
12:27	SHUNT REACTOR	GUADAME 400 REA 2	SWITCH ON	SOUTH
12:28	SHUNT REACTOR	MORATA 400 REA 4	SWITCH ON	CENTRE
12:32	HVDC 320 kV STA. LLOGAIA–BAIXAS	REDUCE THE SETPOINT TO 409 kV	SETPOINT	NORTHEAST

Table 2-2: List of works and manoeuvres from RE for voltage control

Table 2-3 provides a list of topological actions – shown in Figure 2-2 – taken for voltage control by REN:

Element type	Switched-off element name	Start date and time	End date and time	Reason
Line	Fanhoes–Pegoes 400	26/04 19:46	30/04 06:23	Manual voltage control
Line	Panoias–Tavira 400	27/04 02:18	28/04 09:07	Manual voltage control
Line	Ferreira do Alentejo–Panoias 400	27/04 02:18	28/04 09:07	Manual voltage control
Shunt Reactor	RS1–S. Feira 180 Mvar	28/04 09:09	29/04 05:24	Manual voltage control
Shunt Reactor	RS1–S. Castelo Branco 70 Mvar	28/04 09:09	29/04 00:12	Manual voltage control
Shunt Reactor	RS1–S. Portimão 180 Mvar	28/04 10:03	28/04 23:33	Manual voltage control
Shunt Reactor	RS1–S. Pedralva 180 Mvar	28/04 10:03	29/04 02:41:	Manual voltage control
Shunt Reactor	RS1–S. Paraimo 180 Mvar	28/04 10:06	29/04 00:37	Manual voltage control
Shunt Reactor	RS1–S. Armamar 180 Mvar	28/04 10:27	29/04 02:39	Manual voltage control
Shunt Reactor	RS1–S. Fanhões 180 Mvar	28/04 10:27	28/04 22:51	Manual voltage control
Shunt Reactor	RS2–S. Palmela 180 Mvar	28/04 12:19	28/04 23:56	Trip due to low voltage protection

Table 2-3: List of manual topological actions taken for voltage control by REN

2.1.3 Grid Topology Snapshots at 12:00:00 and 12:32:00

The following two figures provide a view of the grid topology in Spain at the time instants of 12:00:00 (Figure 2-3) and 12:32:00 (Figure 2-4). The figure shows all the lines that are connected in that time instance.

The lines that are disconnected (either due to planned maintenance or due to voltage control) are shown on grey colour in the figure.

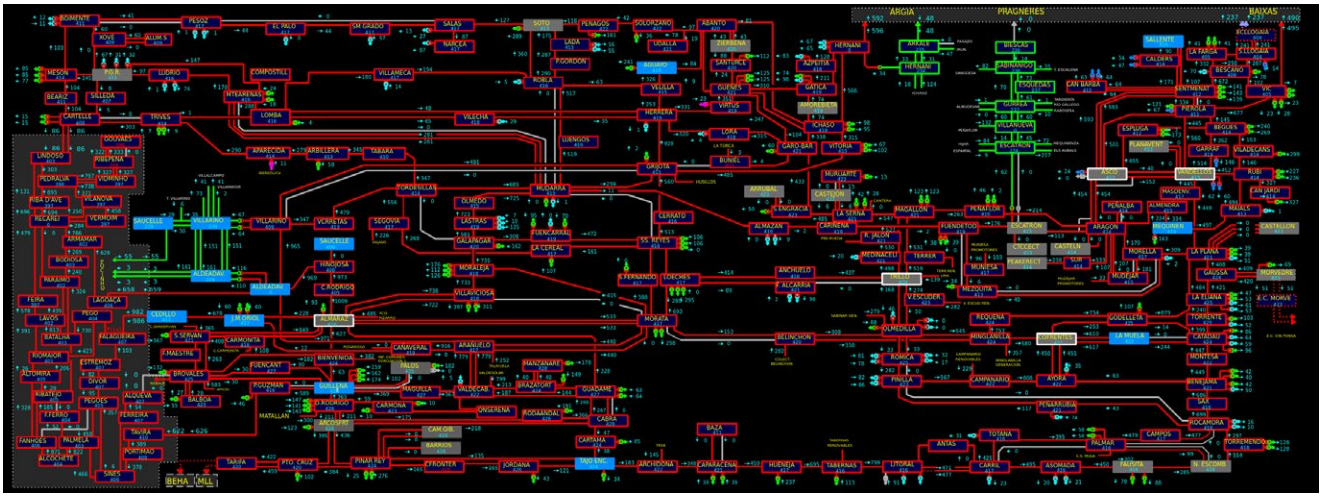


Figure 2-3: Grid topology in Spain at 12:00:00

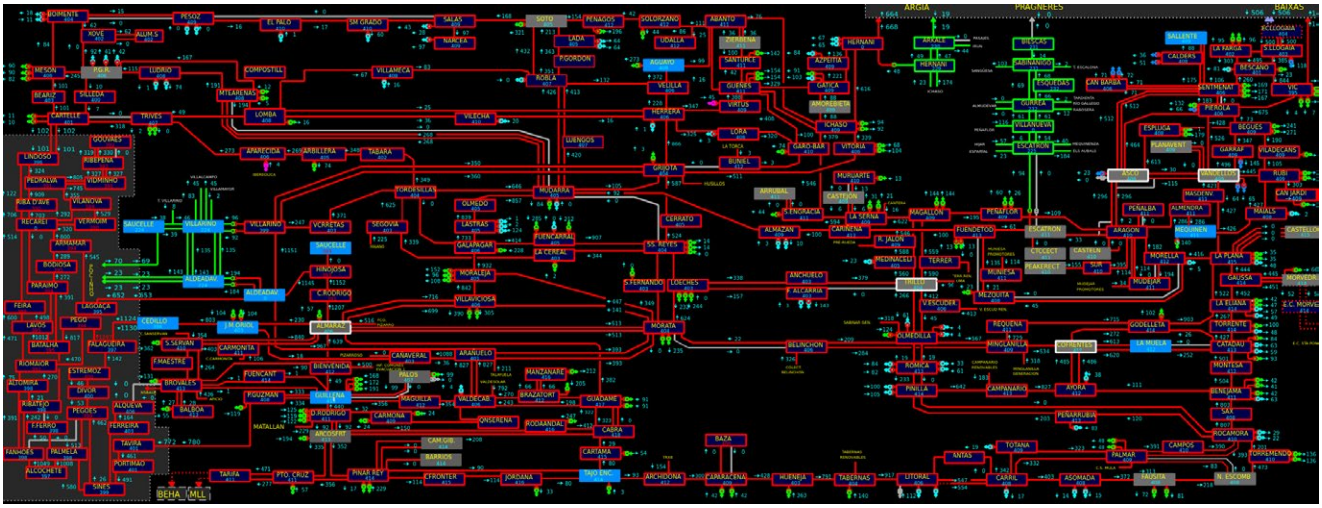


Figure 2-4: Grid topology in Spain at 12:32:00

2.1.4 Demand and Generation Forecasts

The purpose of demand and generation forecasts is to provide the most accurate information possible to support essential security analyses – such as adequacy assessments and N-1 contingency evaluations – and ensure the secure operation of the power system.

Affected TSOs publish renewable generation and electricity demand forecasts, both are carried out in-house, incorporating external weather forecasts. In the case of renewable generation, RE further enhances internal forecasting models with data provided by external suppliers.

Concerning the STA process, forecasts on demand, available generation, and all necessary inputs are provided each D-1 for adequacy analysis. In order to ensure correct delivery, they are sent at 3:00 and again at 8:00. Moreover, a manual check of the correct delivery is performed daily. Concerning the forecasts for the STA process, a time lag – as mentioned in chapter 5 – was detected in the files sent to STA, starting from the date of the hour shift from CET to CEST (30 March).

This bug was detected at the beginning of April and was corrected in May. It only affected STA forecast input files and had no operational impact related to the real-time incident of 28 April.

Concerning N-1 contingency evaluations, the time lag had no impact on 28 April.

The following figures show forecasts estimated by TSOs two days in advance, one day in advance, and the latest available on the same day at 10:00, 11:00, and 12:00:

» 1) Demand forecast, estimated as generation minus pumping, minus exchanges with other systems

» 2) Wind forecast

» 3) PV forecast

2.1.4.1 Spanish Demand Forecast

Demand forecast is generated by combining several proprietary models developed by RE, which use temperature, solar radiation, and labour days demand patterns as exogenous variables, along with real-time demand data.

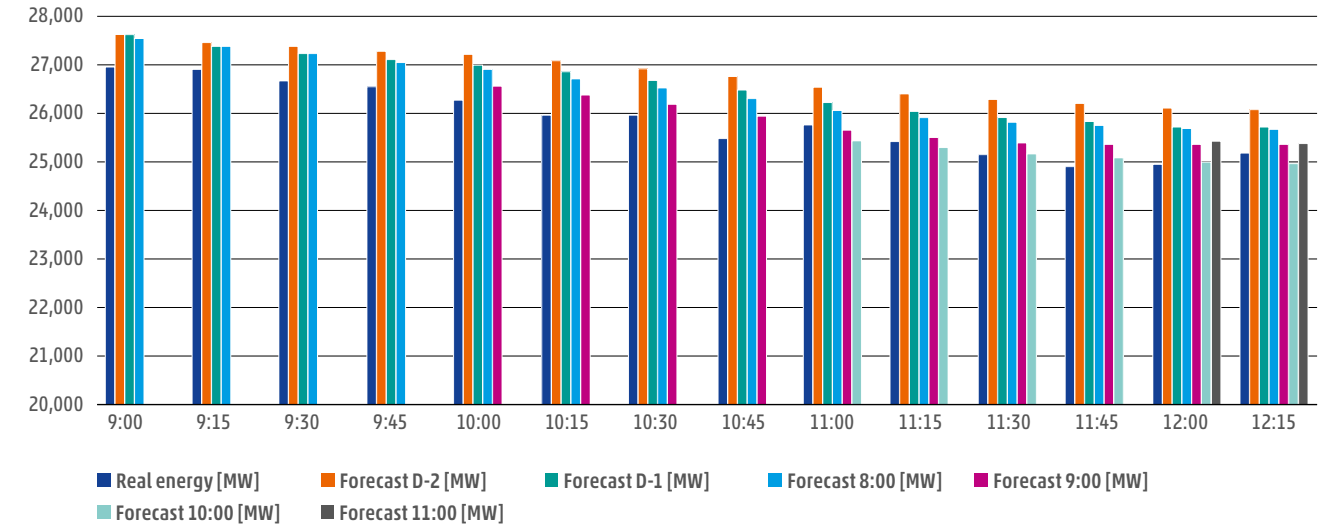


Figure 2-5: Spanish demand forecast

2.1.4.2 Portuguese Demand Forecast

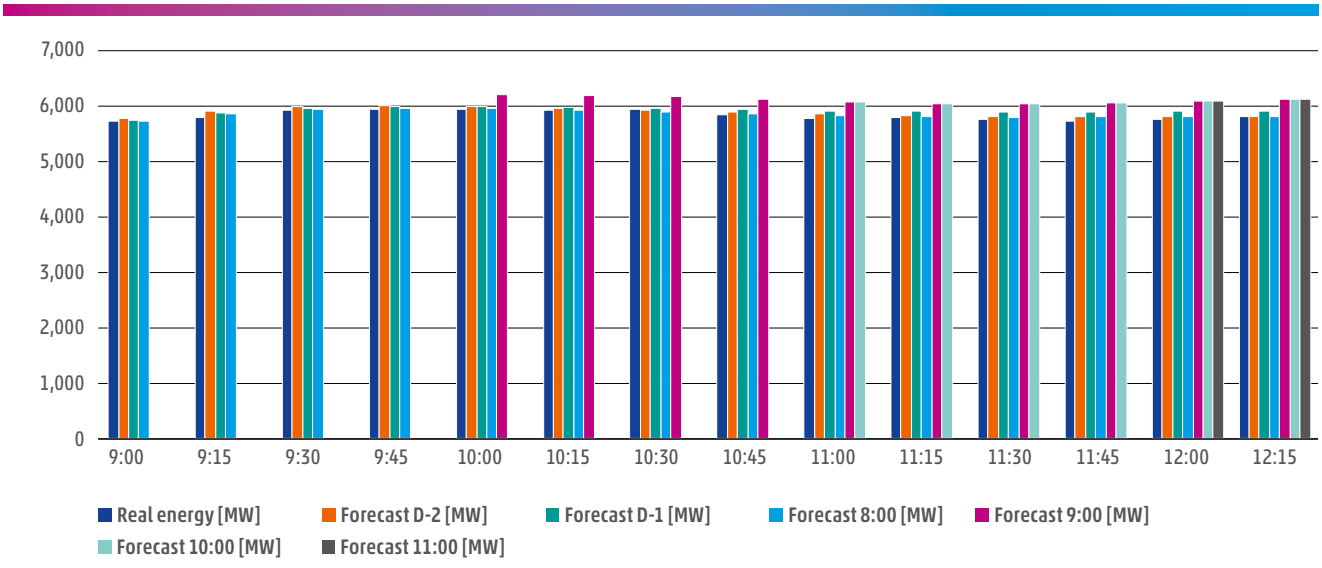


Figure 2-6: Portuguese demand forecast

2.1.4.3 Spanish Wind Forecast

The renewable generation potential forecast – defined as the maximum expected renewable generation based on weather conditions – produced by RE combines its proprietary forecasting models with data from various external providers. For wind generation, forecasting models use wind speed and direction at a height of 100 metres as exogenous variables.

Subsequently, RE refines the abovementioned forecast by applying constraints such as the unavailability of renewable installations and network limitations, which help approximate the potential forecast to the expected actual production. Finally, the forecast is adjusted for the initial time horizons using real-time renewable generation data received via telemetry by RE.

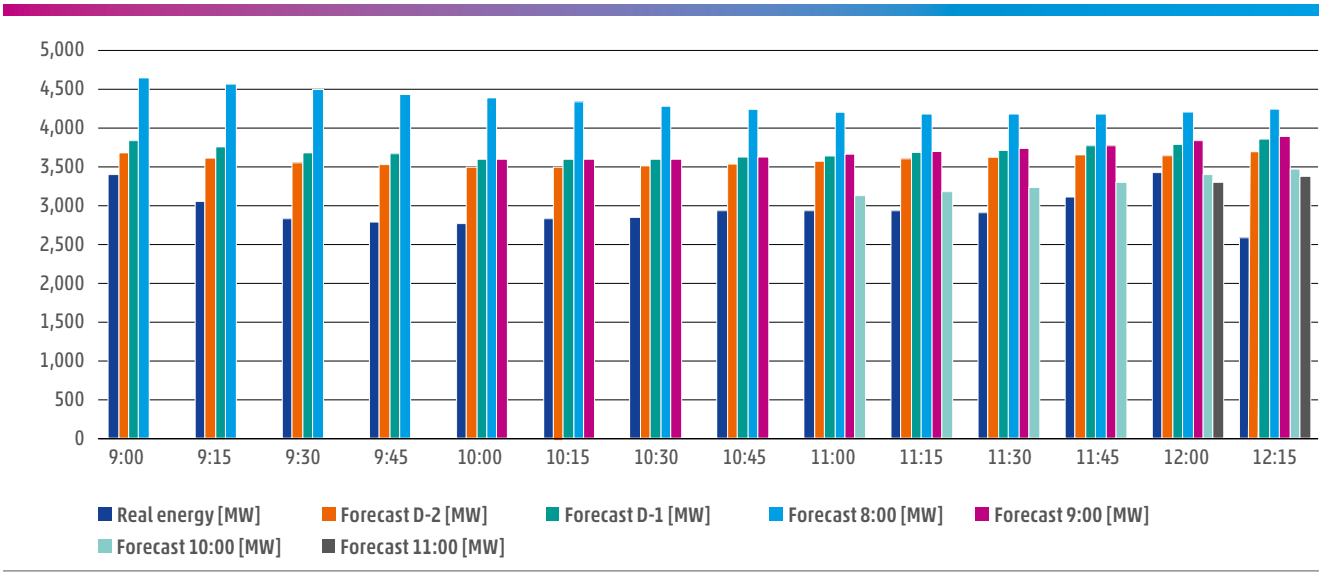


Figure 2-7: Spanish wind forecast

RE corrects these forecasts in their initial horizons using the latest actual production value received. At 08:00, production increased unexpectedly, prompting an upward correction of the forecast. However, at 09:00, production dropped again, whereby the upward correction made at 08:00 resulted in an error.

2.1.4.4 Portuguese Wind Forecast

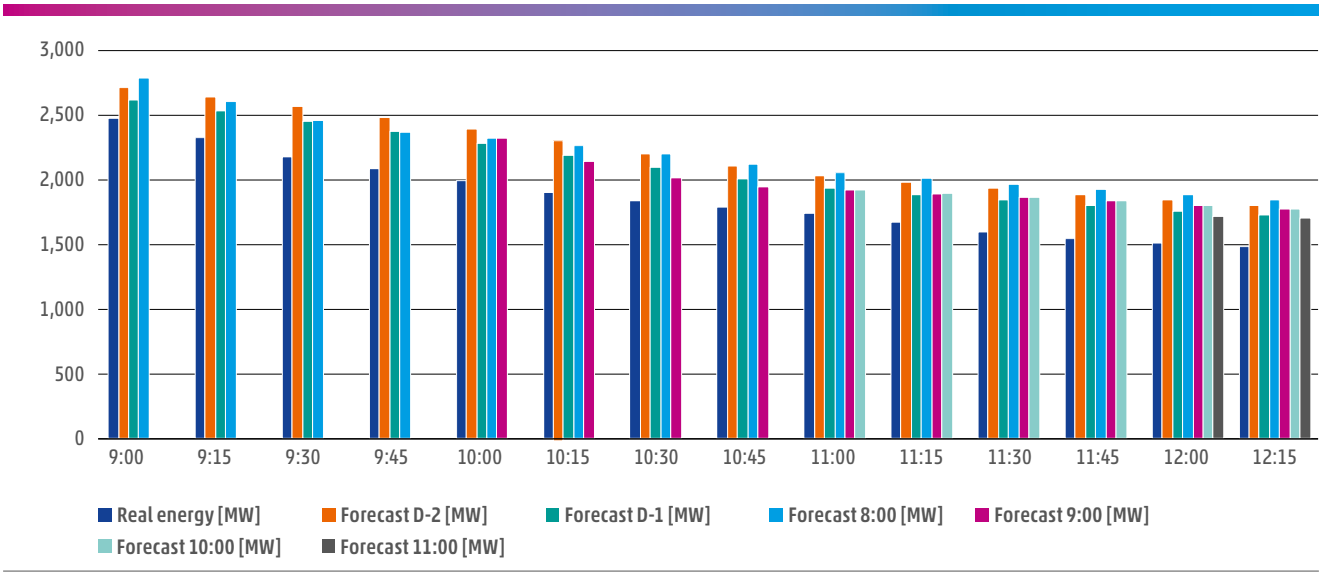


Figure 2-8: Portuguese wind forecast

2.1.4.5 Spanish PV Forecast

The aforementioned renewable generation potential forecast also applies to photovoltaic (PV) generation. In this case, models incorporate global radiation, cloud cover, and temperature.

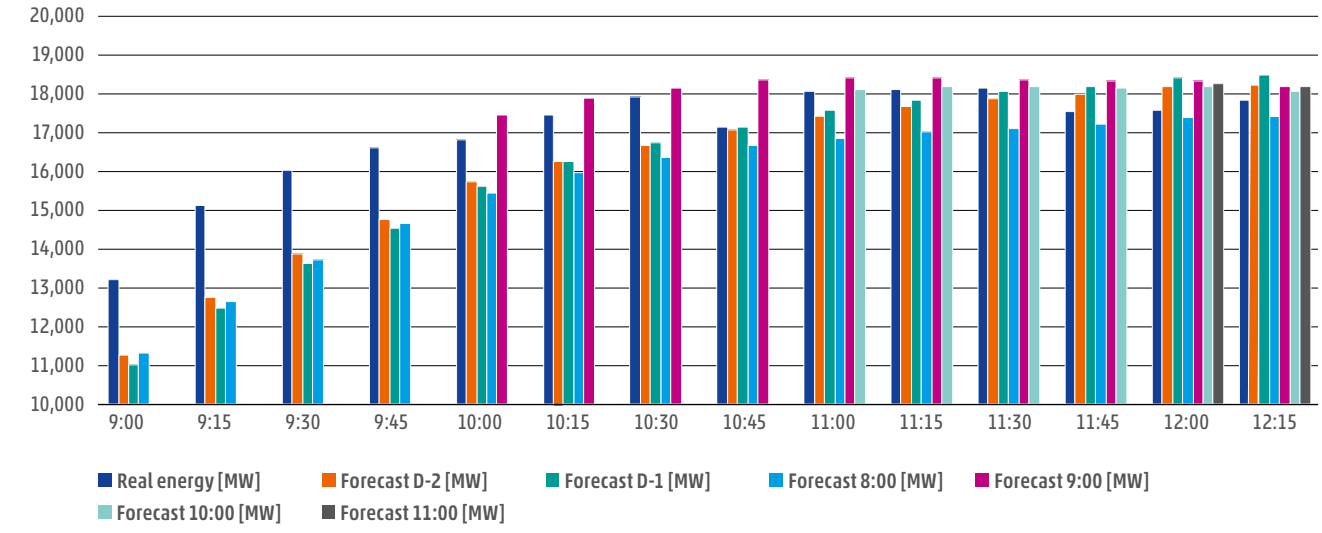


Figure 2-9: Spanish PV forecast (PV installations < 1 MW not considered)

2.1.4.6 Portuguese PV Forecast

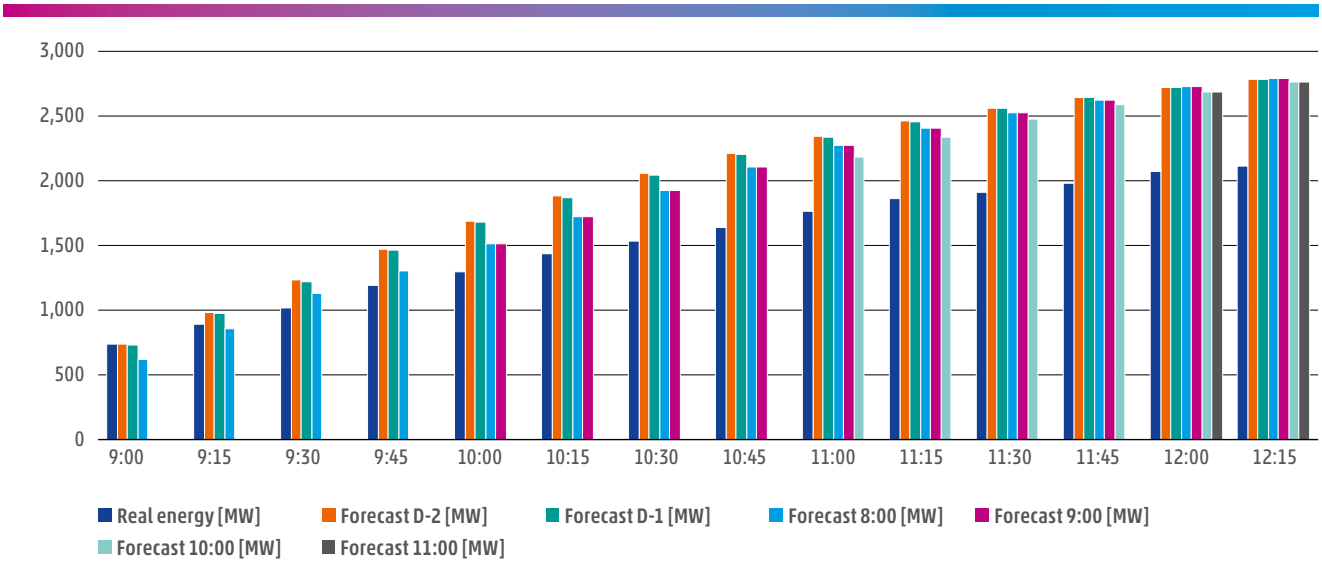


Figure 2-10: Portuguese PV forecast

2.2 Market Conditions

2.2.1 Day Ahead Prices

After the day-ahead market, the prices for delivery before 9:00 settled at low values in Spain and Portugal, in comparison to France. After 9:00, the prices dropped to near 0 €/MWh in the RE price zone.

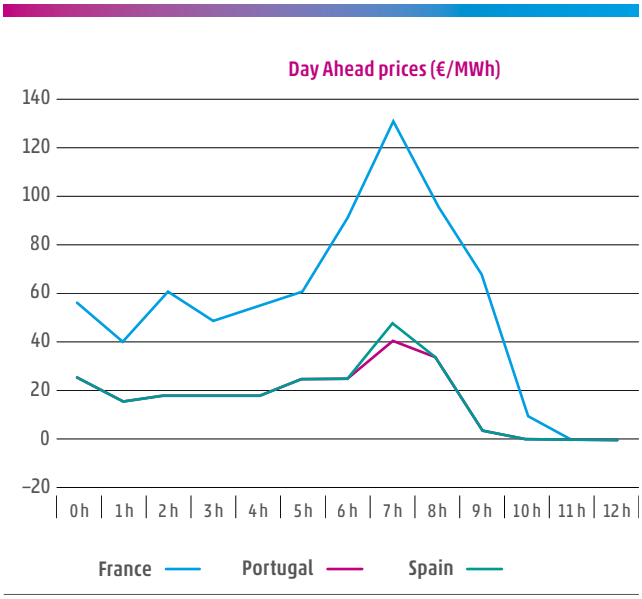


Figure 2-11: Day-ahead prices

2.3 Active Power Flows Before the Incident

2.3.1 Load Patterns

2.3.1.1 Spanish Total Load

Figure 2-12 below shows the evolution of the total load on the Spanish network from 9:00 to 12:32 on 28 April, as well as the total load on a similar day (24 April). The total load is calculated as the sum of the output of all generation units with a capacity greater than 1 MW connected to the transmission or distribution network, adjusted for net electricity exchanges with the neighbouring TSOs. Hence, it does not include the demand directly supplied by smaller generation units connected to the distribution network. Effectively, if small generation units disconnect, the total load as defined here will increase.

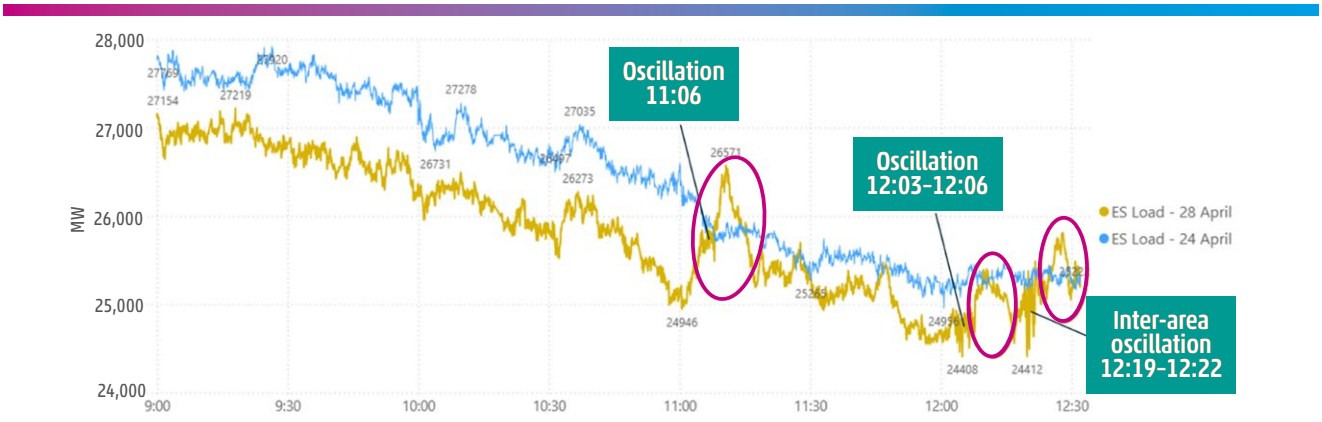


Figure 2-12: Total load in Spain on 24 and 28 April between 9:00 and 12:32

This figure shows that three significant temporary total load increases occurred on 28 April but not on 24 April, namely from 11:07 to 11:10, from 12:07 to 12:15, and from 12:25 to 12:29. There is no distinguishable direct cause explaining these patterns, although it can be observed that these three increases in load occurred after three oscillations that took place at 11:06, between 12:03 and 12:06, and between 12:19 and 12:22, respectively, as described in details in Section 2.5.6.

Figure 2-13 shows the power flows between transmission and distribution networks aggregated across continental Spain on the same days (24 and 28 April). It is calculated as the sum of the active power of all transformers 400/132 kV and lower and 220/132 kV and lower connected to the distribution.

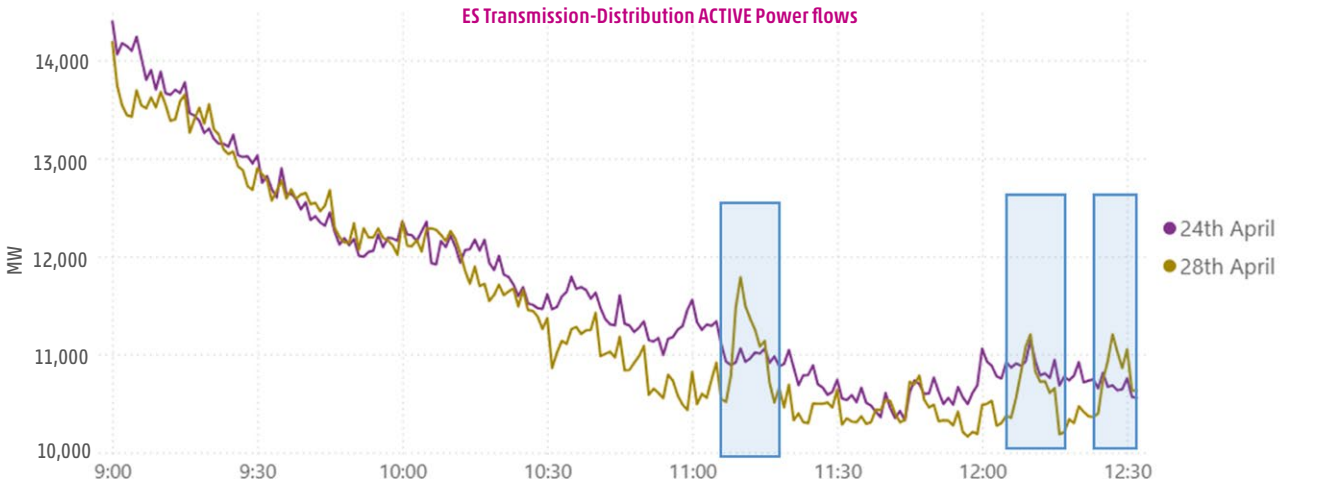


Figure 2-13: Spanish peninsula's transmission-distribution flows on 24 and 28 April between 9:00 and 12:32 (Source: RE SCADA)

The patterns observed in the previous graphs are also visible in this figure, possibly indicating that the total load increases are due to temporary disconnections of small generation units (<1 MW) connected to the distribution networks.

Chapter 4.3 provides a more detailed analysis of the TSO-DSO flows and disconnection of small PV units connected to low voltage grids.

As a regional example, Figure 2-14 presents the power flow in the Madrid area.

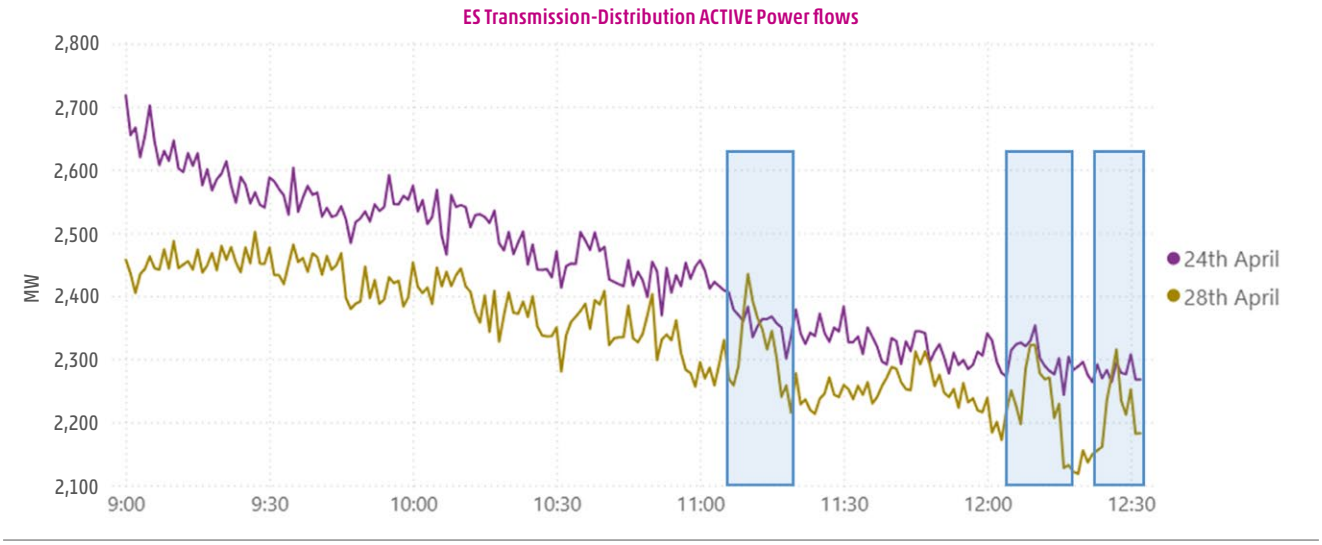


Figure 2-14: Madrid's transmission-distribution flows on 24 and 28 April between 9:00 and 12:32 (Source: RE SCADA)

At the substation level, the active power flow through two transformers in parallel 220/132 kV at "TS 1–Madrid" (located in Madrid) is shown in Figure 2-15 below as an example, compared to the same period on Thursday, 24 April.

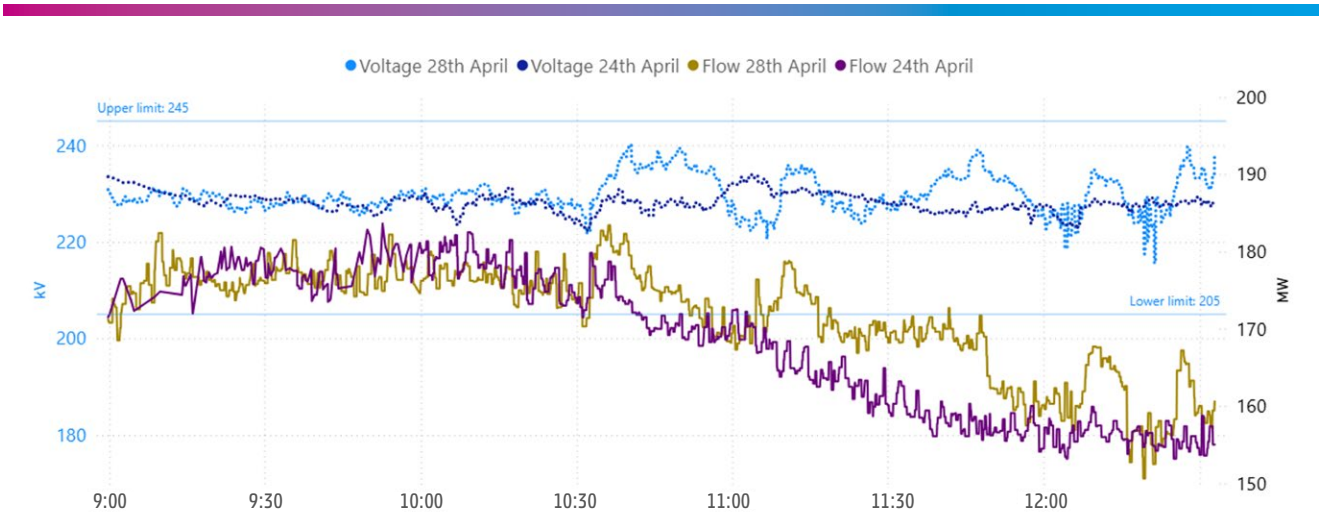


Figure 2-15: Voltage and Active power through the two 220/132 kV transformers at "TS 1–Madrid" between 9:00 and 12:32

2.3.1.2 Portuguese Load

Concerning the Portuguese system, Figure 2-16 provides a comparison of the demand with a similar day (24 April).

No unusual behaviour regarding Portuguese demand at the transmission network was observed on 28 April prior to the incident.

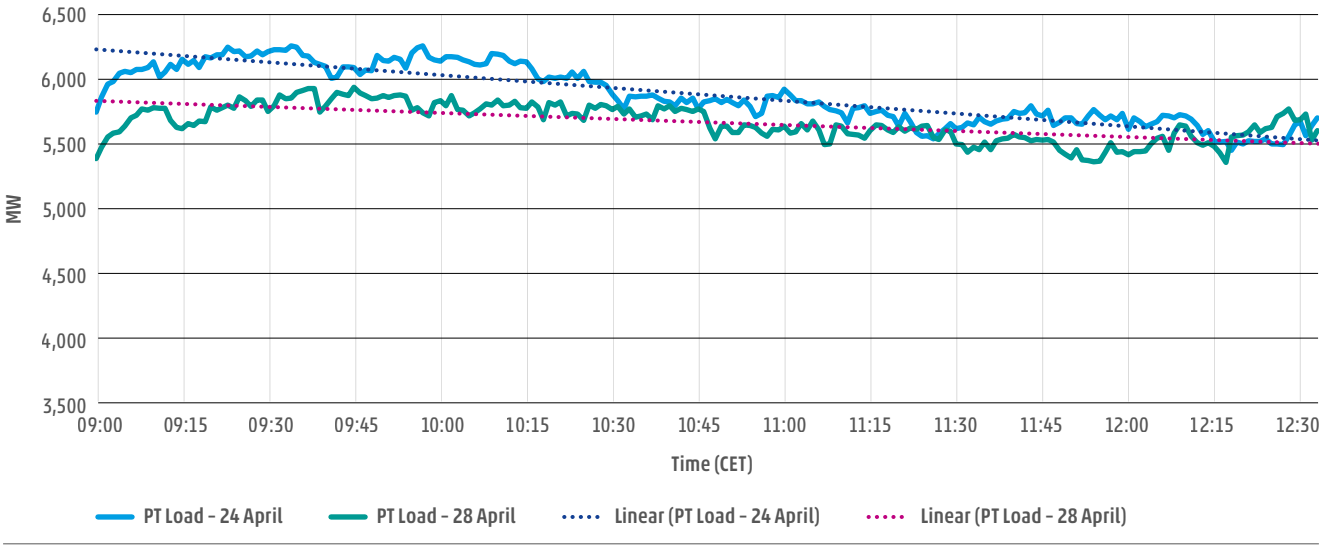


Figure 2-16: Comparison of Portuguese demand evolution on 28 April compared to 24 April between 9:00 and 12:30 (Source: REN's real-time SCADA measurements)

2.3.2 Production Patterns

2.3.2.1 Spanish Generation Mix

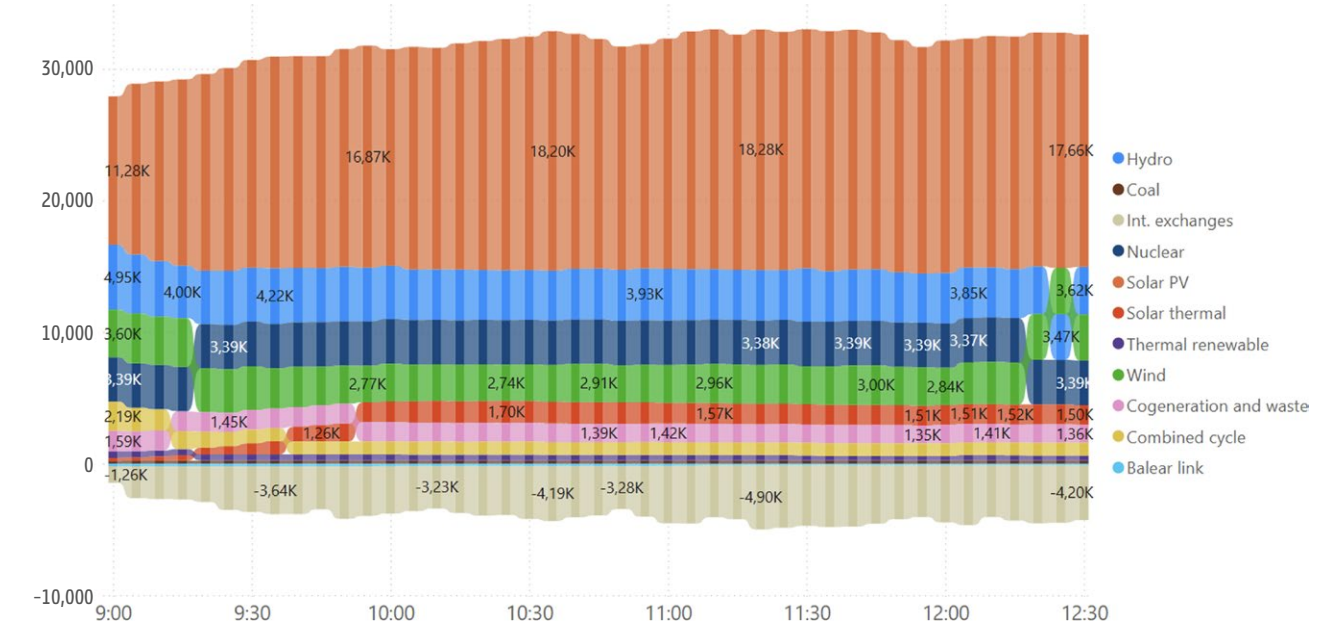


Figure 2-17: Spanish generation mix on 28 April from 9:00 to 12:30 (PV installations < 1 MW not considered)

28 April was a typical spring day in Spain, with mild temperatures and sunshine. According to Figure 2-18, the system's solar photovoltaic generation was similar to previous days, while wind generation was more variable but within the ranges observed in previous days.

This behaviour of renewable energies is characteristic of this time of year, where favourable weather maintains consistent solar energy production, while wind generation can fluctuate due to changes in wind speed and direction.

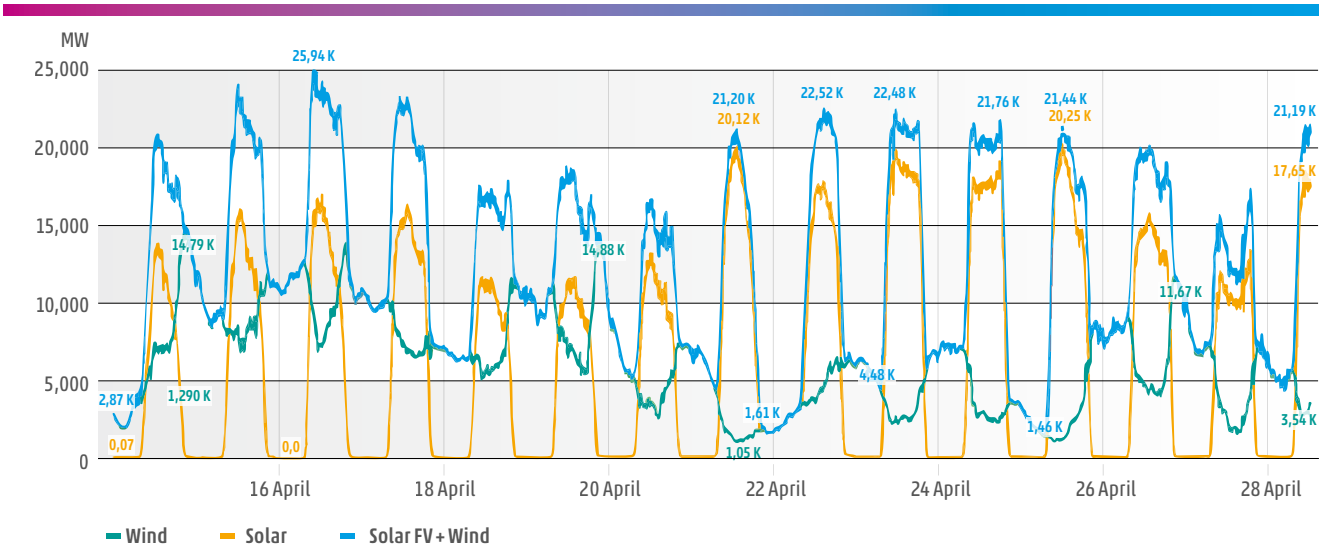


Figure 2-18: Solar PV and wind production in the two weeks in Spain prior to the incident (PV installations < 1 MW not considered)

2.3.2.2 Portuguese Generation Mix

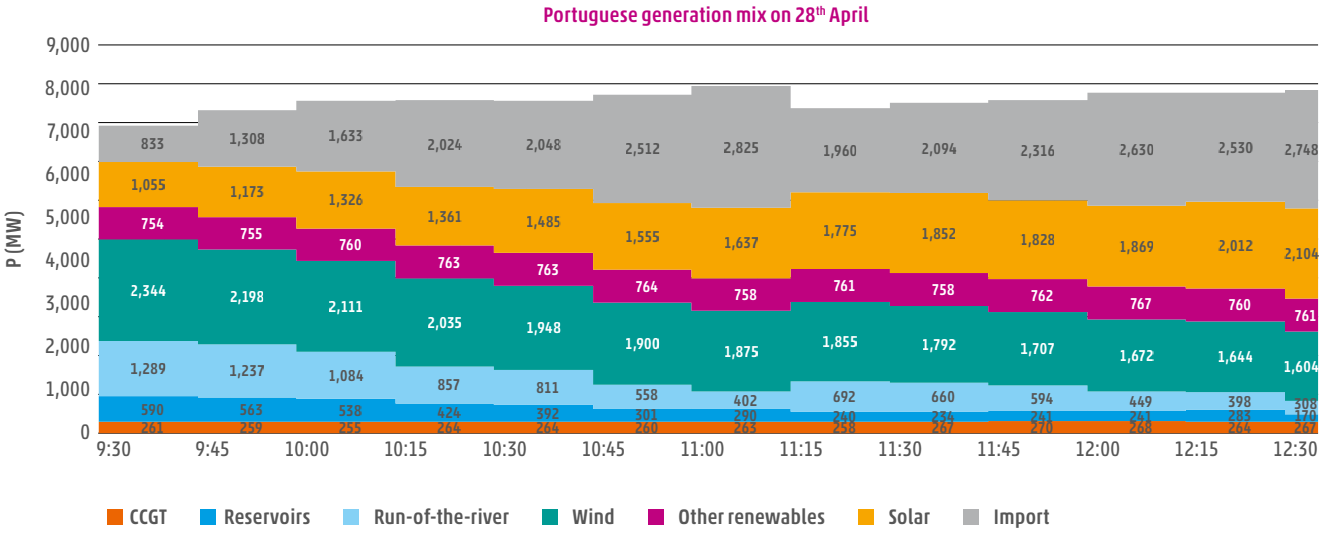


Figure 2-19: Portuguese generation mix on 28 April from 9:30 to 12:30

In Portugal, the temperature and wind speed were relatively moderate on 28 April, with little cloud cover and no precipitation observed. According to Figure 2-20, the photovoltaic generation of the Portuguese system remained stable and showed minimal variation during the days leading up to 28 April.

Wind generation increased in the preceding days due to periods of higher wind speeds, causing slightly more variability in this type of generation. However, these fluctuations remained within the typical range for this time of year.

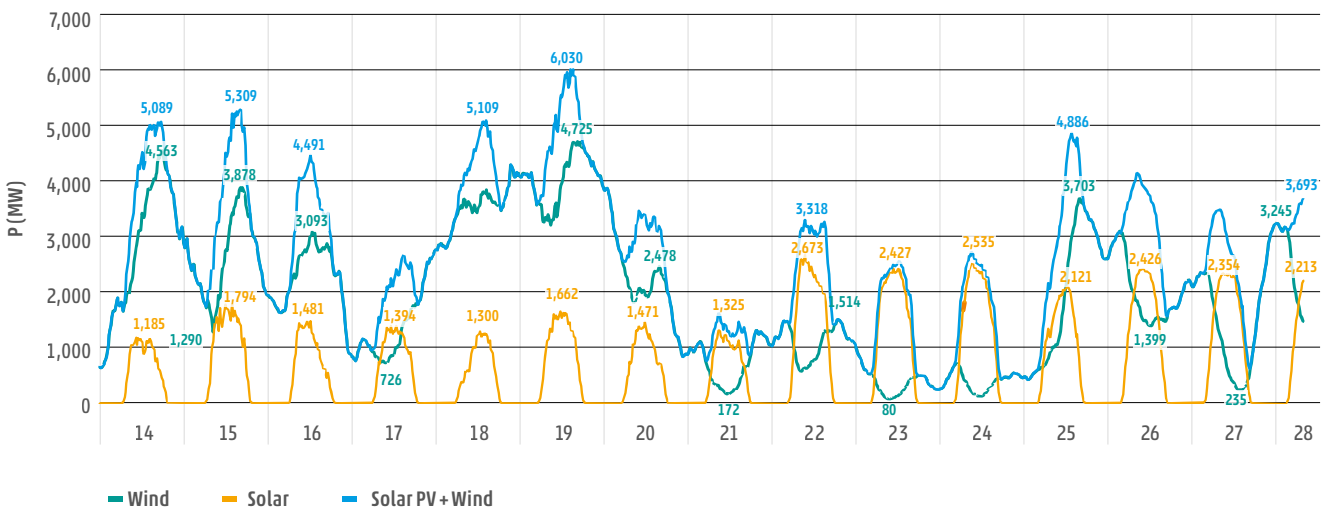


Figure 2-20: Solar PV and wind production in Portugal in the two weeks prior to the incident

2.3.3 Cross-Border Flows

Figure 2-21 shows physical flows through the Spain's interconnections with France, Portugal, and Morocco from 9:00 to 12:32, as well as the total amount.

Furthermore, the load curve is presented for reference. The source is Red Eléctrica SCADA with a four-second resolution.

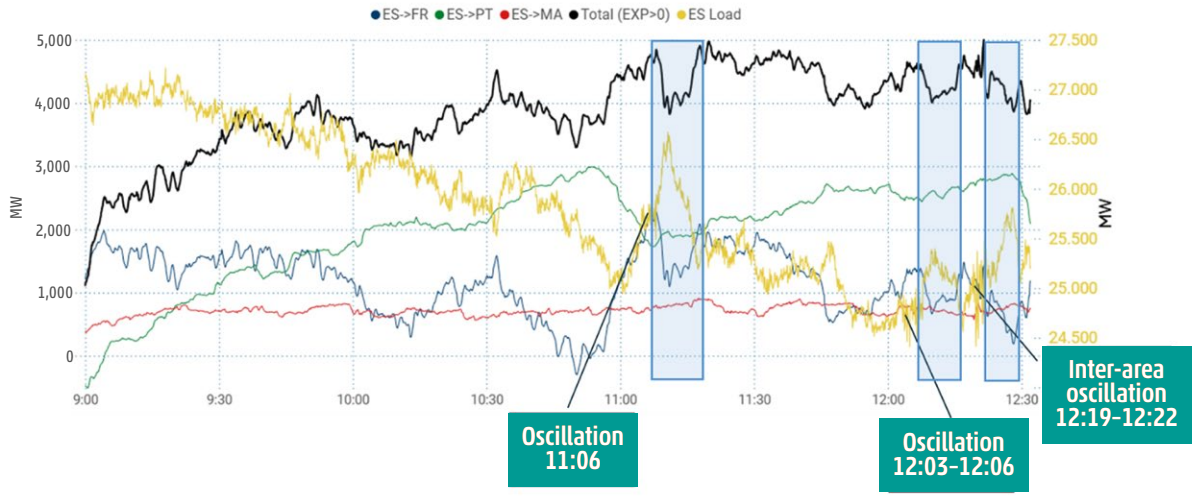


Figure 2-21: Spanish load and physical flows through ES–FR, ES–PT, and ES–MA between 9:00 and 12:32 (source: RE SCADA)

Figure 2-21 above shows the behaviour of the total load and the exchanges across the three Spanish borders, as well as the sum of these exchanges. Coinciding with the total load increases described above, a reduction of approximately 1,000 MW in the exchange through interconnections is observed during the period between 11:07 and 11:10, declining from 4,800 MW exporting to 3,800 MW exporting (approximate values), and the deviation continued for several minutes. This delay can be explained by the correction value of the International Grid Control Cooperation (IGCC), a real-time process of imbalance netting between TSOs that aims to avoid simultaneous activation of automatic frequency restoration reserves (aFRR) in opposite directions. This process corrects the input of the involved frequency restoration processes accordingly. It can be seen as a transfer of imbalance between TSOs when they are in opposite sign (upward/downward direction).

The IGCC flow during this period increases up to the maximum value of 1,000 MW (the available cross-border capacity limit).

The same behaviour is also observed just after the two subsequent total load increases described above. At 12:07, the value of the export programme with France is 2,000 MW, while the actual exchange with France is around 1,300 MW (see figures in Section 2.4.4). At 12:10, the programme does not change, and the value of the export exchange with France is around 670 MW. Including the interconnections with Portugal and Morocco, this means that the peninsular system reduces its exports by 600 MW (from 4,600 to 4,000 MW).

At 12:24, the value of the export programme with France is 1,000 MW, while the actual exchange with France is 866 MW. At 12:28, the programme does not change, and the value of the export exchange with France is 190 MW (see Figure 2-23). Including the interconnections with Portugal and Morocco, this means that the peninsular system reduces its exports by 513 MW.

Figure 2-22 below focuses on the window from 12:00 to 12:32.

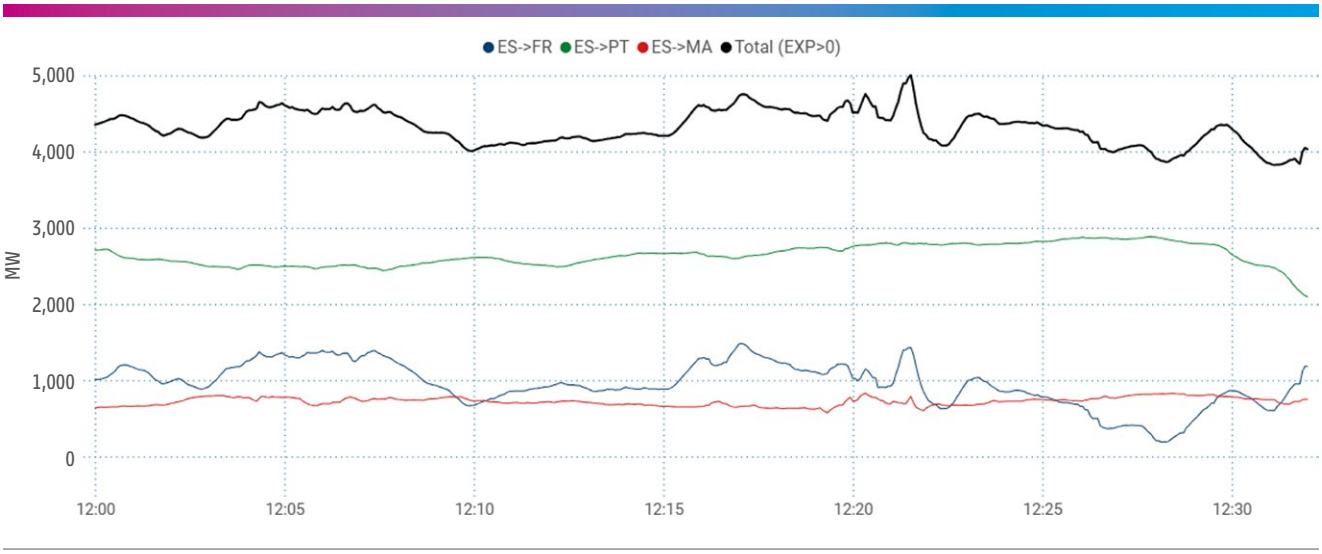


Figure 2-22: Physical flows through ES–FR, ES–PT, and ES–MA between 12:00 and 12:32 (source: RE SCADA)

2.3.4 Scheduled Commercial Exchanges

The cross-border programs are implemented with 10-minute ramps (starting 5 minutes before and ending 5 minutes after the start of the 15 minutes period), except when an urgent measure is required, such as in the case of a counter-trading for security reasons.

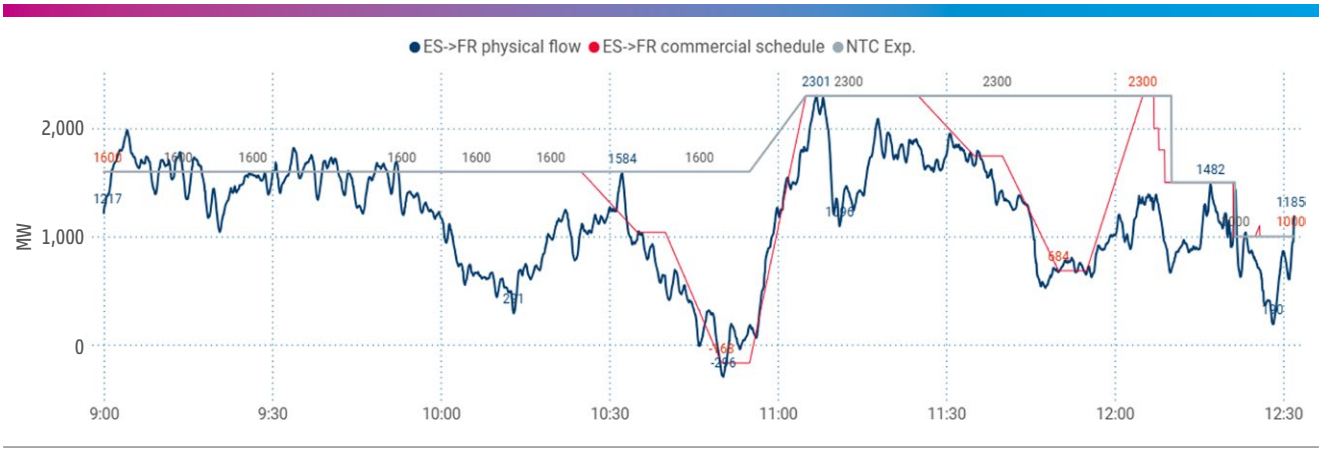


Figure 2-23: Commercial schedule and physical flow through the ES–FR border (source: RE SCADA)

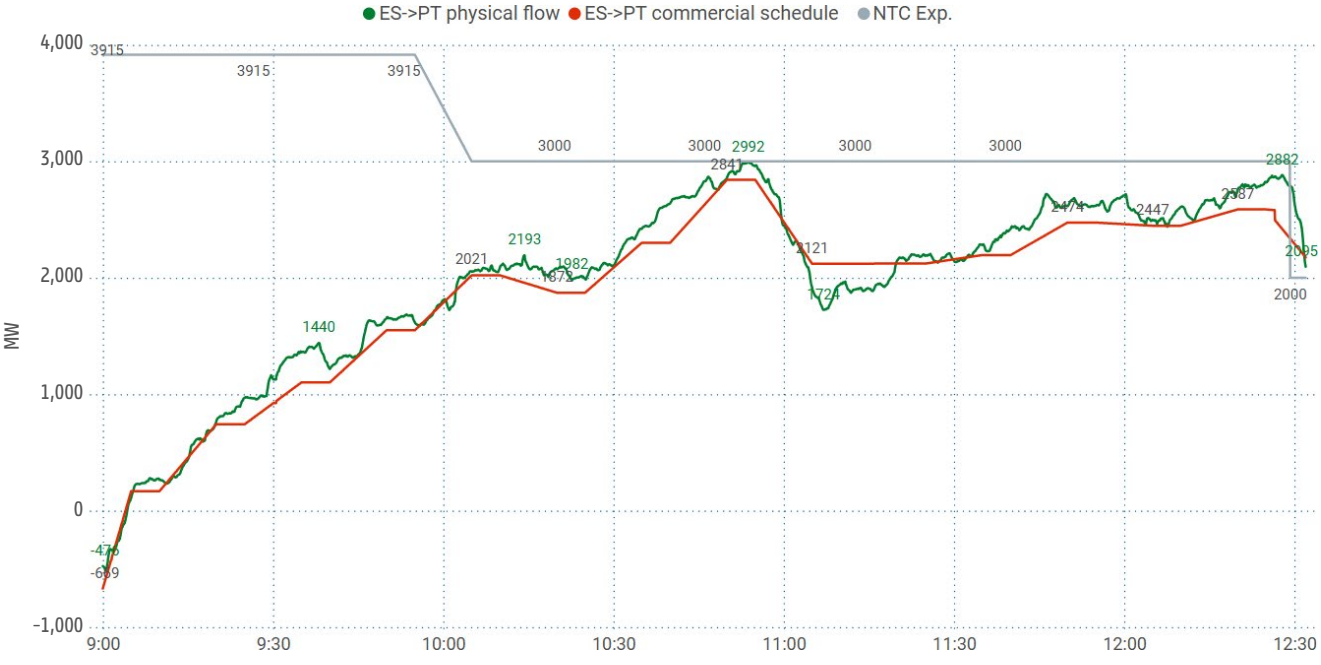


Figure 2-24: Commercial schedule and physical flow through the ES–PT border (source: RE SCADA)

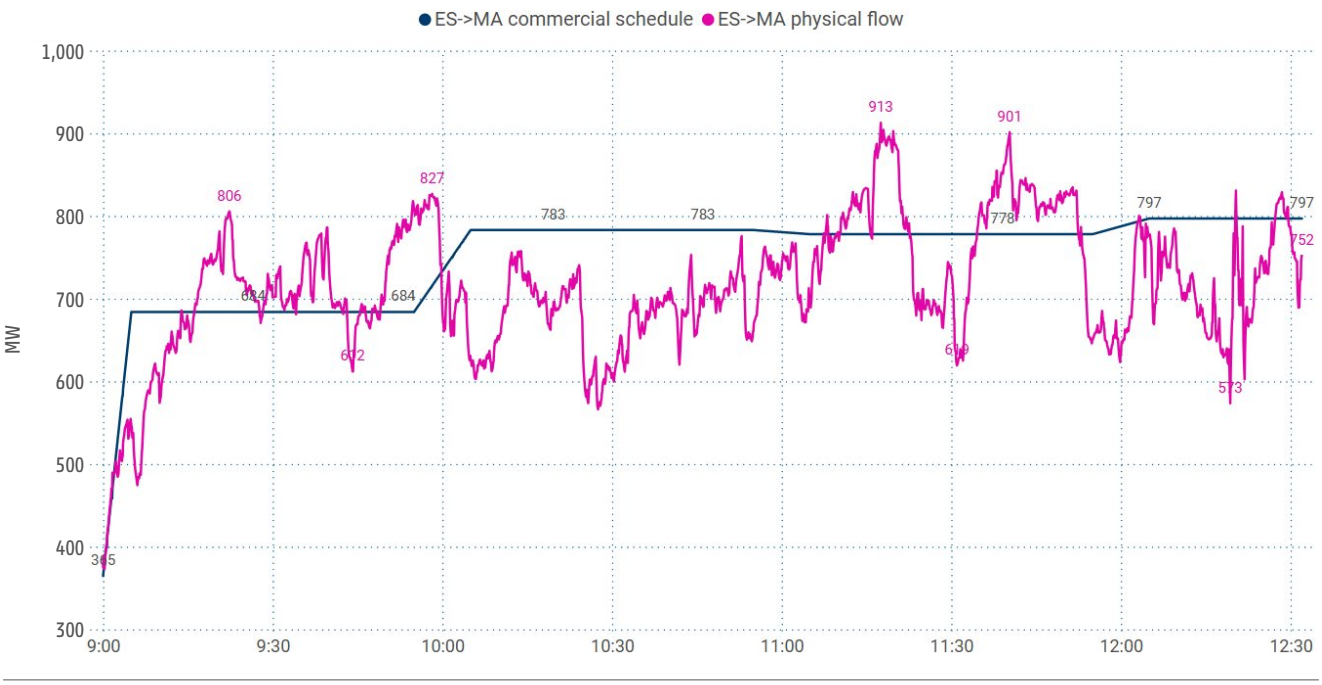


Figure 2-25a: Commercial schedule and physical flow through the ES–MA border (source: RE SCADA)

Figure 2-25b shows the aFRR demand in Spain throughout the timeframe.

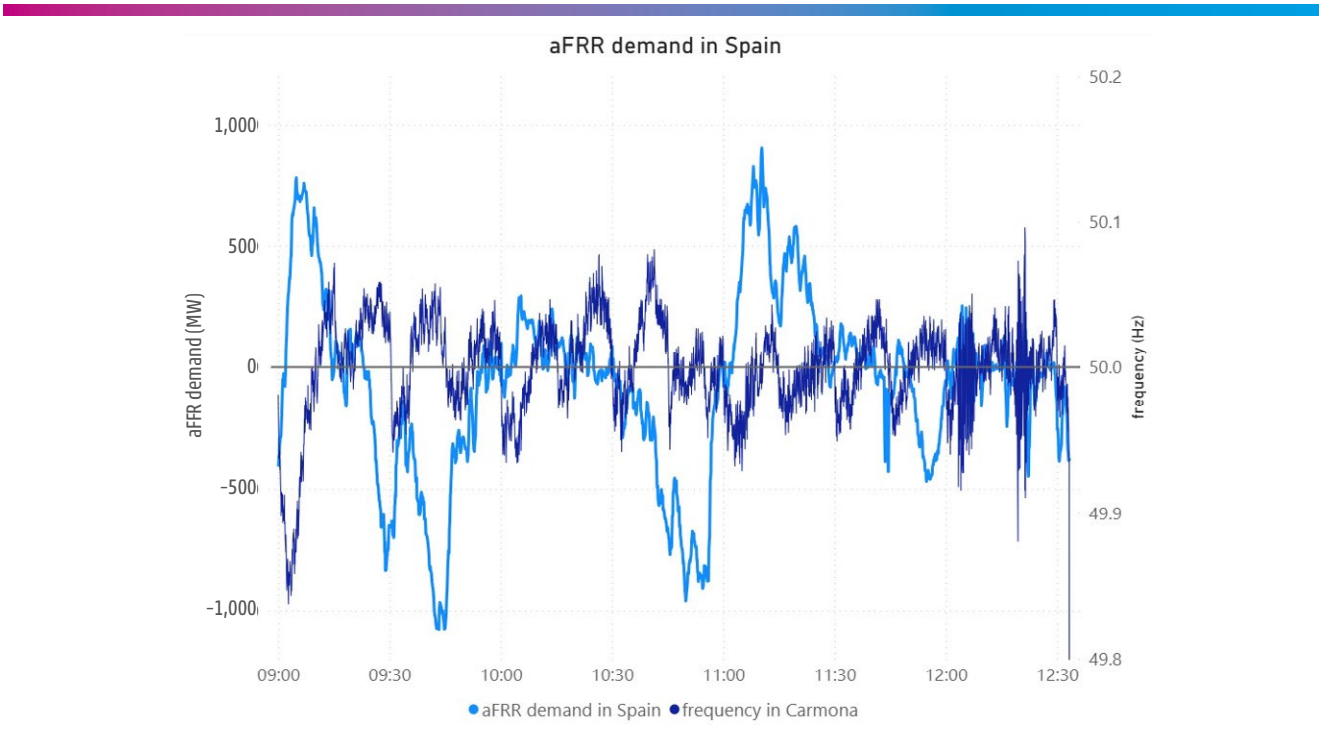


Figure 2-25b: aFRR demand in Spain (source: RE)

2.4 Inertia

In order to estimate the total inertia of the Iberian Peninsula power system, the following procedure was applied.

For a system comprising N rotating generators, the total stored kinetic energy (KE) is:

$$E_{\text{kin}} = \sum_{i}^{N} H_{\text{gen,i}} S_{\text{gen,i}}$$

Where:

- H is the inertia constant of the i-th generator;
- S is the rated apparent power of the i-th generator.

The equivalent inertia constant is given by

$$H_{\text{eq}} = \frac{\sum_{i}^{N} H_{\text{gen,i}} S_{\text{gen,i}}}{\sum_{i}^{N} S_{\text{gen,i}}}$$

Inverter-based resources such as batteries, wind, and photovoltaic systems do not contribute to system inertia. Thermal solar and certain other renewable sources have a non-zero inertia constant.

With reference to the rated apparent power of inverter-based resources, it is assumed that their energy contribution is approximately equal to the active power P injected into the system. We can assume that the rated power of static generation is approximately equal to the actual delivered active power.

Under these hypothesis, the equivalent inertia of a system where M inverter-based resources are connected and operating is:

$$H_{\text{eq}} = \frac{\sum_{i}^{N} H_{\text{gen,i}} \times S_{\text{gen,i}} + \sum_{j}^{N} 0 \times S_{\text{inverters},j}}{\sum_{i}^{N} S_{\text{gen,i}} + \sum_{j}^{M} S_{\text{inverters},j}} = \frac{\sum_{i}^{N} H_{\text{gen,i}} \times S_{\text{gen,i}}}{\sum_{i}^{N} S_{\text{gen,i}} + \sum_{j}^{M} S_{\text{inverters},j}}$$

The scientific literature [1] [2] [3] [4] [5] [6] underlines that the total inertia of the grid must consider the load contribution, namely the effect of rotating machines on industrial and domestic loads. It is worth underlining that inertia calculation is affected by significant uncertainty, mainly for the following reasons:

- The equivalent total inertia is a linearisation of a process affected by several non-linearities.
- The rated inertia of generators, turbines, and other rotating parts is estimated and not precisely calculated in several cases.
- The load's contribution can vary in a wide range depending on the type of loads, aggregations, etc.

Accordingly, the final value for the total inertia will be expressed as a plausible range, rather than a single number.

The total system inertia can thus be expressed as:

$$H_{\text{tot}} = H_{\text{eq}} + H_{\text{loads}}$$

Where the inertial contribution by loads is expressed in the same reference basis of the system.

Table 2-4 summarises the main calculated parameters for each electrical system (Spain, Portugal, and the total Iberian Peninsula) immediately prior to the incident at 12:30.

Spain		Portugal		Iberian Peninsula	
KE (MWs)	H_{tot} (s)	KE (MWs)	H_{tot} (s)	KE (MWs)	H_{tot} (s)
97,590	2.17 - 2.67	21,884	2.45 - 2.95	119,474	2.21 - 2.71

Table: 2-4

The calculation of inertia for days preceding the incident will be implemented in the final report.

The following figures focus on the Spanish network and show for each day from 1 January until 28 April 2025:

- the minimum number of conventional units coupled to the network (>30 MW);
- the number of conventional units coupled to the network (>30 MW) in the time periods 10:00 – 11:00, 11:00 – 12:00 and 12:00 – 13:00;
- the total installed power of the connected conventional units, each time at the hour with the lowest number of coupled units; and
- the total installed power of the connected conventional units, in the time periods 10:00 – 11:00, 11:00 – 12:00 and 12:00 – 13:00.

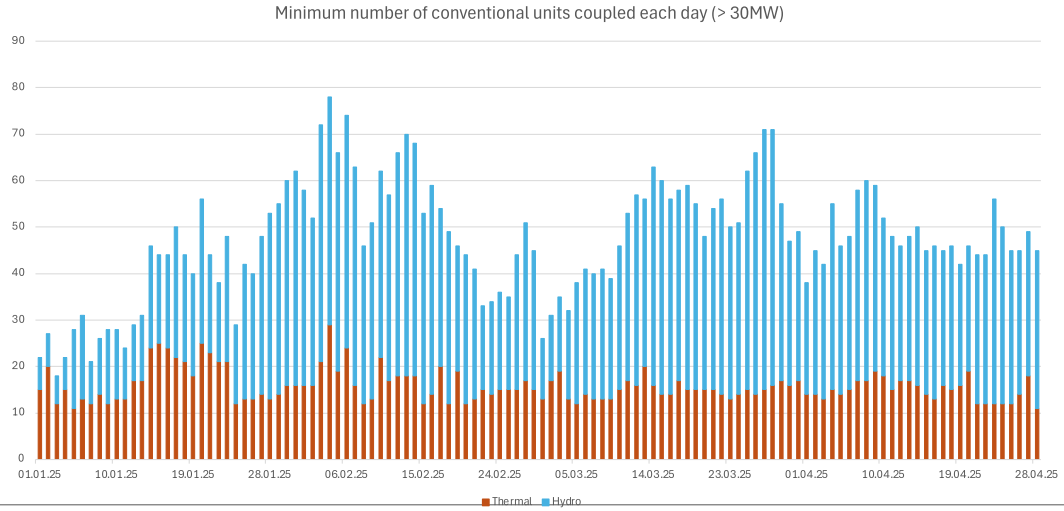


Figure 2-26: Minimum number of conventional units (>30 MW) coupled to the Spanish network each day from 1 January to 28 April

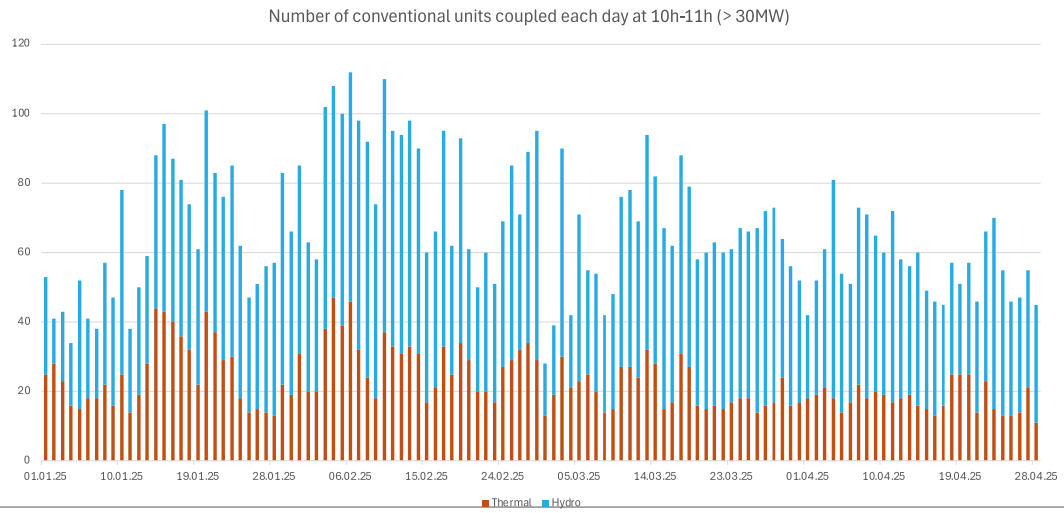


Figure 2-27: Number of conventional units (>30 MW) coupled to the Spanish network each day in the period from 10:00 to 11:00 (from 1 January to 28 April)

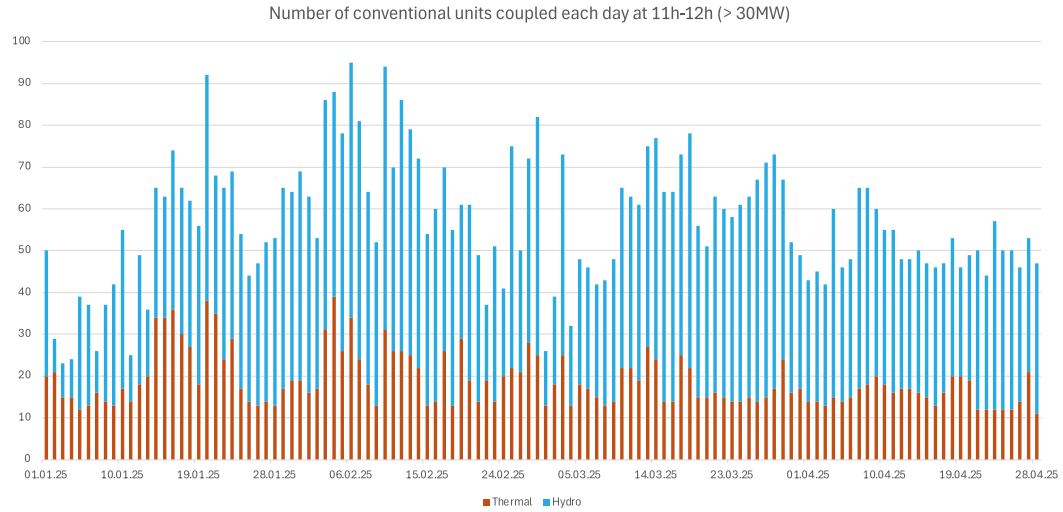


Figure 2-28: Number of conventional units (>30 MW) coupled to the Spanish network each day in the period from 11:00 to 12:00 (from 1 January to 28 April)

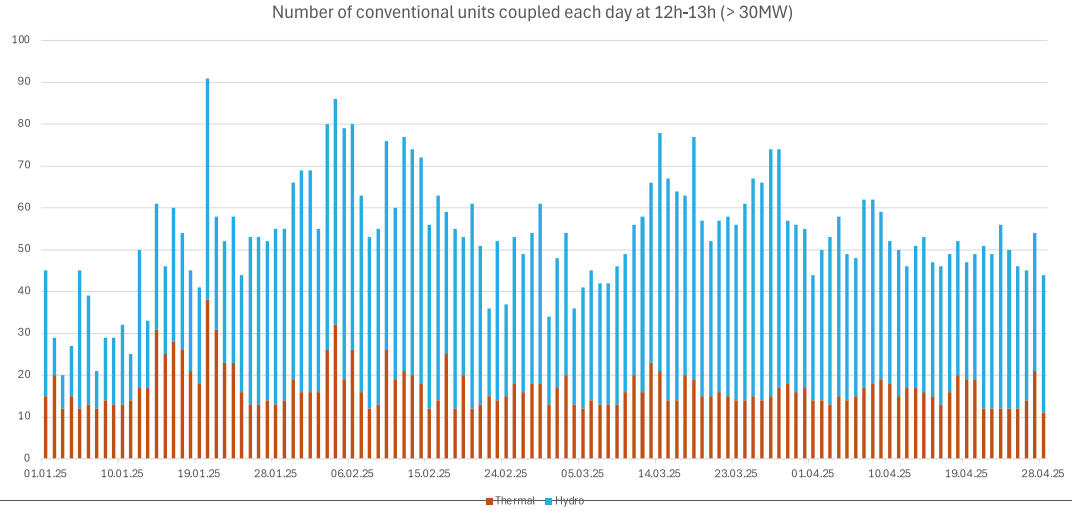


Figure 2-29: Number of conventional units (>30 MW) coupled to the Spanish network each day in the period from 12:00 to 13:00 (from 1 January to 28 April)

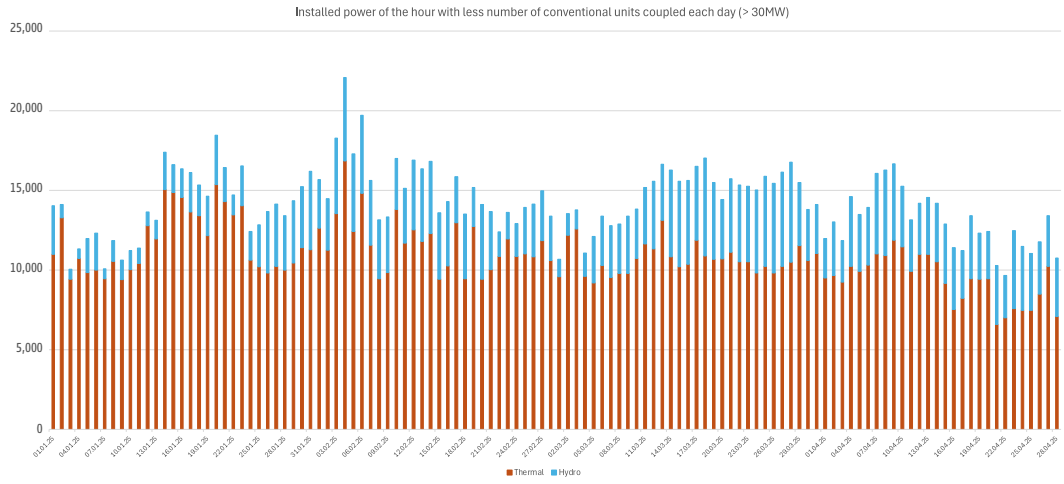


Figure 2-30: Installed power of the conventional units connected to the Spanish network, each time at the hour with the lowest number of coupled units (from 1 January to 28 April)

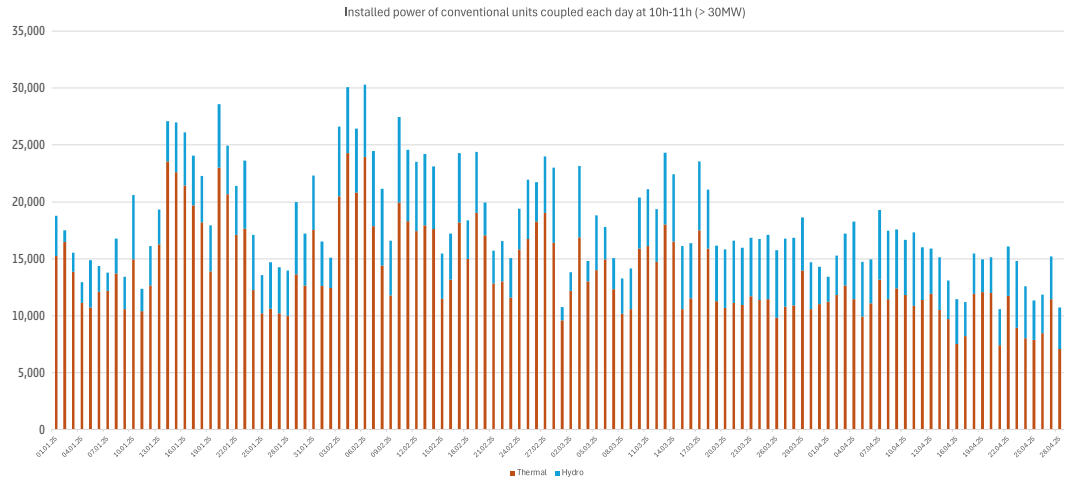


Figure 2-31: Installed power of the conventional units connected to the Spanish network during the period from 10:00 to 11:00 (from 1 January to 28 April)

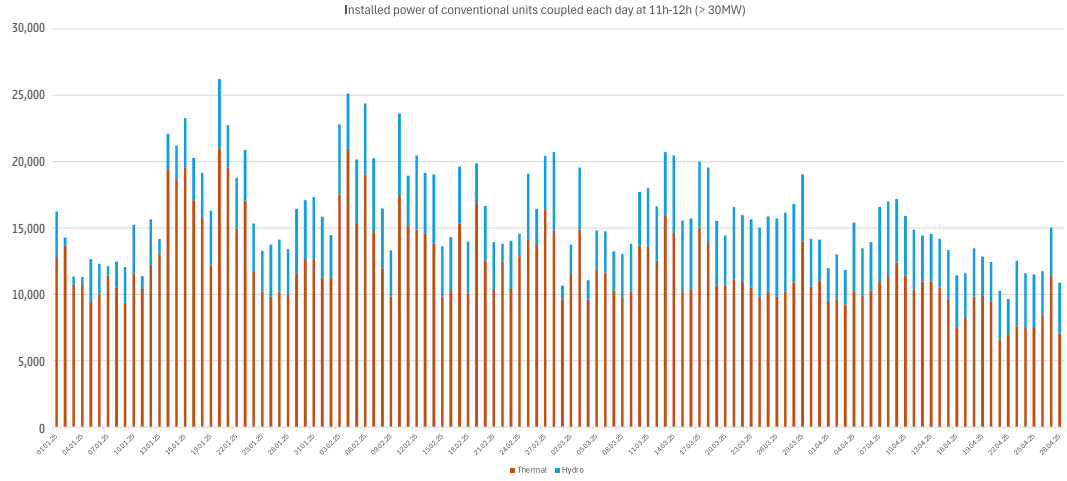


Figure 2-32: Installed power of the conventional units connected to the Spanish network during the period from 11:00 to 12:00 (from 1 January to 28 April)

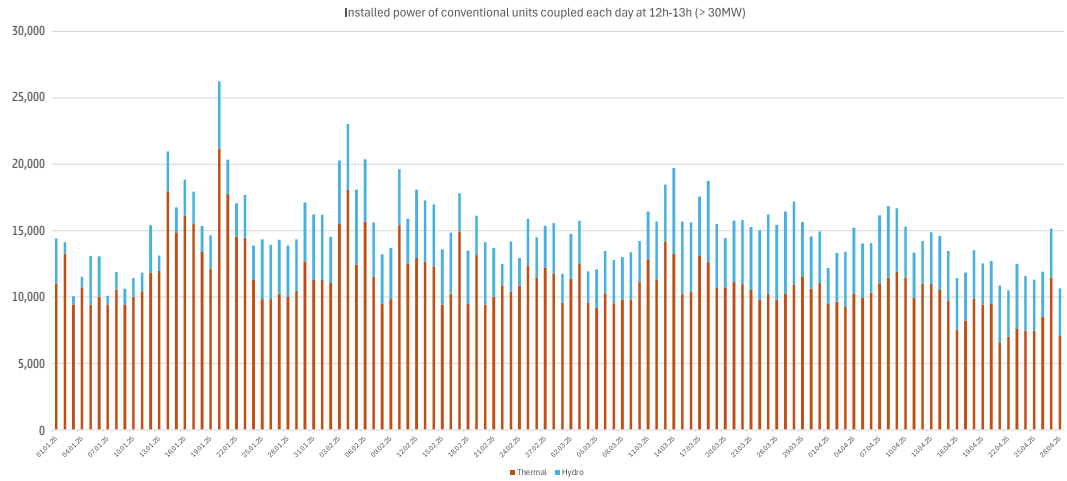


Figure 2-33: Installed power of the conventional units connected to the Spanish network during the period from 12:00 to 13:00 (from 1 January to 28 April)

2.5 Oscillations

2.5.1 Stability Main Concepts

Power system stability phenomena classification involves different branches:

- Frequency stability
- Voltage stability
- Rotor angle stability

The rise in the number and capacity of inverter-based resources (i.a., generators, batteries, HVDC links) recently required the introduction of two new families of dynamic phenomena:

- Resonance stability
- Converter-driven stability

Figure 2-34[1] schematically depicts the different classifications of power system stability.

A subcategory of rotor angle stability is small-disturbance angle stability, defined as the ability to maintain synchronism under minor disturbances. Small-disturbance angle stability can be further divided into non-oscillatory instability (a consequence of a lack of sufficient synchronising torque) and oscillatory instability (resulting from a lack of sufficient damping torque). Minor disturbance rotor angle stability problems (including oscillatory stability) can be either local or global.

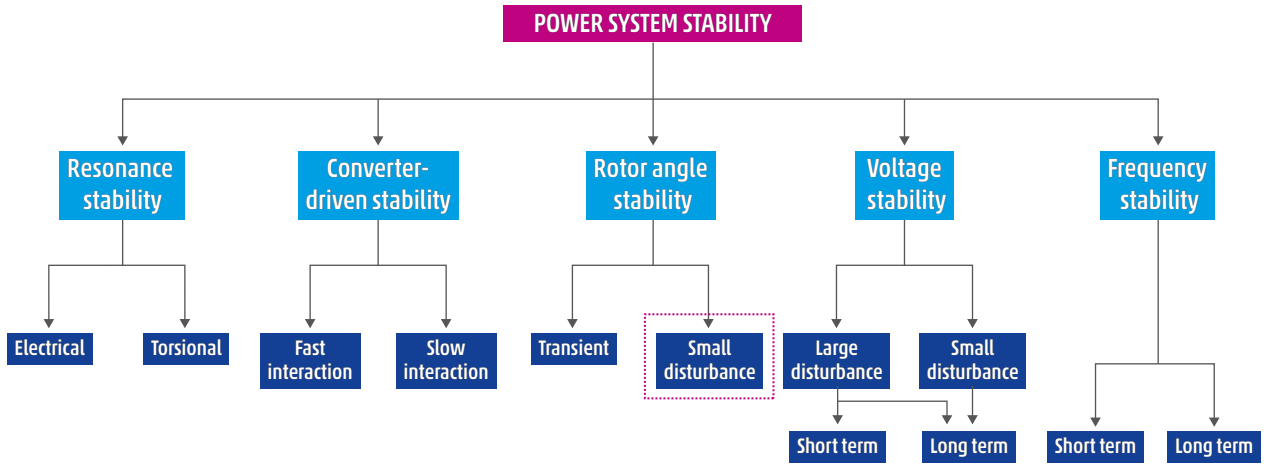


Figure 2-34: Classification of power system stability

Focusing on the oscillatory stability, we can distinguish between different cases. First, depending on the geographical and electrical topological scope, oscillations can be categorised into:

- Inter-plant oscillations: In the same power plant, units oscillate against each other.
- Local oscillations: In a power system, a single generation unit oscillates against the rest of the system.
- Inter-area oscillations: In a power system, a cluster of generation units in one part of the system oscillates against a cluster of generation units in another part, covering a wide extension of the grid.

1 Stability definitions and characterisation of dynamic behavior in systems with high penetration of power electronic interfaced technologies, IEEE Power and Energy Society, Tech. Rep. PES-TR77, May 2020.

Second, depending on the origin of oscillatory behaviour, oscillations can be:

- Forced:[2] Where the oscillation signal is introduced into the power system by a certain source. For example, a controller in a power plant might malfunction or be improperly tuned. The frequency of such oscillation will depend on the driving source. The forced oscillation could be also generated by the interaction between inverter based generators and conventional generators/inverter generators.
- Natural: Where the power system is susceptible to various intrinsic oscillatory modes due to its characteristics, such as topology, generators etc. (such as the 0.15 – 0.30 Hz modes in the Continental Europe grid explained below, or 0.3 – 0.9 Hz modes in the Nordic grid[3]). These oscillation modes are usually sufficiently damped and characterised by low amplitude and energy. They are permanently visible in the system due to the continuous presence of small perturbations owing to normal operational events (manoeuvres, change of topology, load or generation change of path, etc.)

2.5.2 Inter-Area Oscillations

Inter-area oscillatory stability problems are caused by the interaction among large groups of generators and have widespread effects. They involve oscillations between a group of generators in one area and a group of generators in another area. In general, the frequency range depends on the system topology and size. The inter-area oscillations that commonly occur in the Continental Europe (CE) system are characterised by three different modes that physically represent the continuous conversion of kinetic energy of rotating machines into potential energy and vice versa. Figure 2-35 shows the geographical displacement of modes with an equivalent spring mass mechanical analogy to better explain the physics of oscillation. The typical range of inter-area oscillatory modes in the CE system is 0.15 – 0.30 Hz.

It is worth underlining that in a large power system, additional oscillatory frequencies are present with characteristics that are specific to a certain CE system geographic area and not dominant in the dynamics of large parts of the system. Consequently, they are not the object of the present investigation.

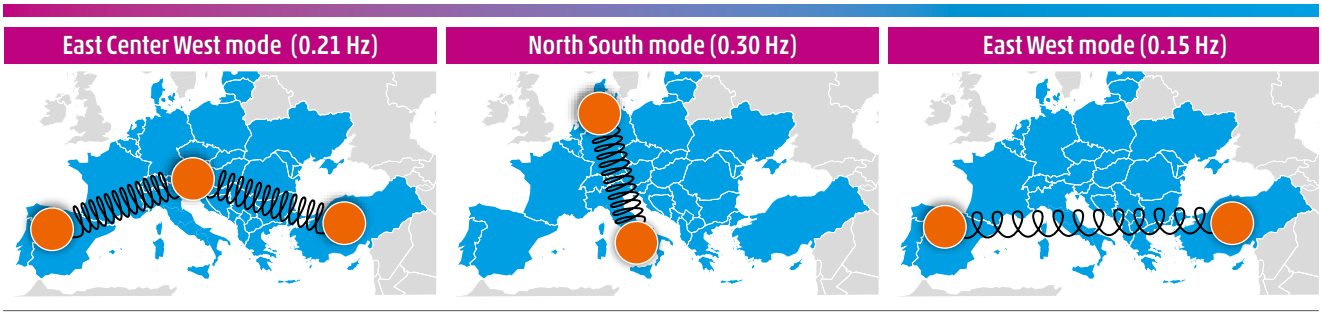


Figure 2-35: Inter-area modes representation with equivalent spring mass mechanical analogy

From an operational perspective, oscillatory instability is detected when the main system variables (frequency, voltages, angle differences, etc.) exceed a limit amplitude jointly with a weak damping for a minimum reference time window.

2 IEEE PES PSDP/PSSC Task Force on Forced Oscillations

3 https://www.sciencedirect.com/science/article/pii/S0378779624012525

As mentioned above, intrinsic oscillation modes are always active in the grids, but stable in terms of amplitude and damping. Regarding the inter-area oscillatory stability of the CE SA, worldwide technical literature and continuous monitoring and studies performed by ENTSO-E experts have identified the following factors as important:

- High power flows between two areas of the system from the peripheral parts of the system directed to the centre (and consequently significant transmission angles spread)
- Increased system impedance due to lines open (i.e. maintenance, voltage control, etc.).
- Underexcitation of generators.
- Power oscillation stabilisers not effective to damp the oscillations.
- Loads as "natural dampers" not sufficient to smooth the oscillation.

In addition, weak control schemes or failures in generator control systems can also have an impact, since forced oscillations may couple with inter-area modes in different ways:

- forcing oscillatory frequency is very near to the interarea natural frequency, the interarea mode can be excited and amplified
- forcing oscillatory frequency is not coincident with interarea natural frequency, so the interarea mode is excited transiently or quasi permanently but with a smaller amplitude

Historically, the Iberian Peninsula participated to East Central West mode and East West mode in a frequency range between 0.15 Hz and 0.2 Hz. When higher oscillatory frequencies are detected, it is in general possible – based on signal analysis techniques – to evaluate whether these oscillations have local or inter-area characteristics.

2.5.2.1 Oscillatory Stability Protocol Adopted Between RTE and RE

With reference to the Iberian Peninsula, a significant oscillatory activity was experienced on 1st December 2016, when a 400 kV line was unexpectedly opened at 11:18, triggering a 0.15 Hz oscillation with 0.140 Hz as the maximum amplitude of recorded frequency. The 2016 report[4] concluded that the control rooms (RTE and RE) reacted promptly (two minutes after the oscillation detection), reducing the export flow from Spain to France. The report also concluded that as dynamic stability limits get closer to static limits, N-1 power flow analyses might need to be complemented with dynamic assessments close to real time.

Focusing on HVDC behaviour, after 1 December 2016 event, some studies were carried out to investigate the maximum improvement of stability that could be given by the Llogaia–Baixas HVDC, which led to the following conclusions:[5]

- It was concluded that the Power Oscillation Damping (POD) functionality of the Llogaia–Baixas HVDC (the functionality affects both the active and reactive power of the HVDC) should always be operative. However, the POD functionality of this HVDC alone is insufficient to stabilise inter-area oscillations between the Iberian Peninsula and CE, due to the grid configuration (meshing and load distribution) near the terminals of the link.
- It was also concluded that when stability needs to be improved (i.e. damping of inter-area oscillations needs to be increased), the active power control of the link should be changed. Accordingly, the active power control mode of the link should be changed from the default hybrid mode[6] (which emulates the AC line dynamics and is typically used in voltage source converter HVDCs) to constant power mode.

At present, a specific protocol exists between RTE and RE that outlines all actions to be performed by the control room Operators to increase damping of inter-area oscillations. The relevant part of the protocol regarding logic, conditions and remedial actions is shown in Figure 2-36. Examples of remedial actions including setting the HVDC link to constant power mode, increasing the active power flow through the HVDC, increasing grid meshing, and reducing the active power export from Spain to France. It is important to underline that the protocol is focused on oscillatory phenomena in the inter-area frequency band 0.15 ... 0.3 Hz.

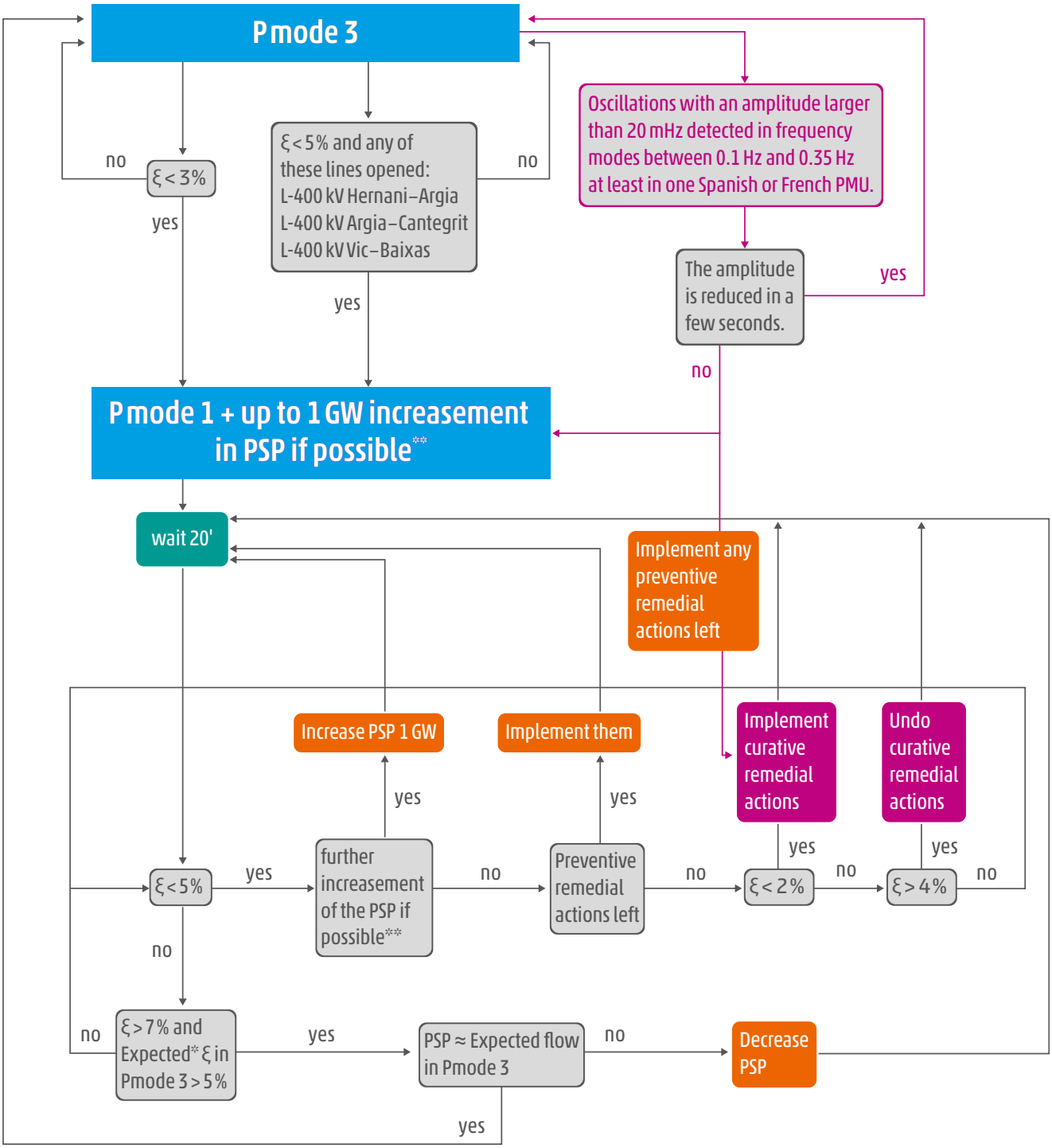
4 https://eepublicdownloads.entsoe.eu/clean-documents/SOC%20documents/Regional_Groups_Continental_Europe/2017/CE_inter-area_oscillations_Dec_1st_2016_PUBLIC_V7.pdf

5 "Improvement of the oscillatory behavior of the HVDC link between Spain and France": 2020 CIGRE Paris session.

6 Hybrid operation implies that active power flowing is $P = P_0 + K \cdot \Delta\vartheta$ with typically $P_0 = 0$. K is a constant and ϑ is angle between the terminals of the link.

Inter-area oscillations Risk management

RE or RTE identifies a constraint on inter-area oscillations when the damping measurement is lower than 3%, when it is lower than 5% and any of the following lines are open: Hernani–Argia, Argia–Cantegrit or Vic–Baixas 400 kV or when real time oscillations with an amplitude larger than 20 mHz in oscillation modes between 0.1 Hz and 0.35 Hz are detected and the amplitude is not reduced in a few seconds. The next diagram summarises the decision making.



* Based on machine learning software.
** It reduces power flow through the AC interconnector with increase damping, we consider it is possible when:
It does not imply static violations such as overloads or voltages out of ranges in the Spanish or French networks.

Figure 2-36: Flow chart of RTE-RE protocol for inter-area oscillation management

2.5.3 Local Oscillations

Local oscillations concern a small part of the power system and are usually associated with rotor oscillations of a single power plant against the rest of the power system. Such oscillations are generally characterised by a frequency in the range of 0.8 – 2 Hz and observed in a smaller geographical area.

As described in Section 2.5.6.3, detailing the 0.63 Hz oscillation at 12:03 – 12:08 on 28 April, RE explained to the Expert Panel that the control room recognised the presence of an 0.63 Hz oscillation with a low damping (<1 %), and determined at the moment that applying the countermeasures described in the common protocol (described in Section 2.5.2.1) to be applied in the event of inter-area oscillations with low damping was the most effective action to improve the system stability. The intention was to gain a positive effect by changing the operating point of the system and creating again an increase of intrinsic system damping.

2.5.4 Forced Oscillations

Some oscillation might be forced; for example, caused by inverter-based resources (IBRs). The frequency range for this type of oscillation is more challenging to define, as they are generally driven by their control systems rather than by dynamics based on the physical characteristics of generators, as is mainly the case with synchronous generators. An inverter-based controller that malfunctions and sustains this malfunction could force this effect onto the grid. This might also exacerbate inter-area oscillation modes and induce other generators into oscillation.

2.5.5 Focus on Automatic Stabilising Countermeasures in the Iberian Grid

The aim of this section is to describe the behaviour and status of POD functions over STATCOM and HVDC and power system stabilisers.

2.5.5.1 STATCOMs

The only STATCOM that was in service in the Spanish grid on 28 April is Vitoria 220 kV. The nominal reactive power of this STATCOM is ±150 Mvar (inductive and capacitive), for V = 1 pu in the point of connection. Another 150 Mvar STATCOM has been commissioned in Tabernas 220 kV after 28 April 2025 and two additional 150 Mvar STATCOMs are planned[7] to be commissioned in 2025 (Lousame 220 kV and Moraleja 400 kV).

The steady-state voltage control modes available at STATCOM Vitoria are Q-mode and V-mode. The V-mode was active during the incident and means that STATCOM injects a current proportional to the voltage deviation from a reference value. The reference voltage set by the RE Control Centre at the time of the incident was 222 kV.

Additionally, the POD and Fast Current Injection Controller (FCIC) controls were enabled. The POD modulates the reactive power supplied by the STATCOM based on the frequency deviation to damp electromechanical oscillations in the range of [0.05 – 1.00] Hz, while the FCIC acts in case of large voltage deviations by generating a supplementary reactive current proportional to the difference between the instantaneous voltage and the voltage one second prior to its activation. It acts if phase voltages are outside the range of 0.85 pu to 1.15 pu or voltage variations greater than 0.1 pu in 1 s.

7 Pages 203 and 205 of the **Spanish Transmission network development plan for 2021 – 2026**.

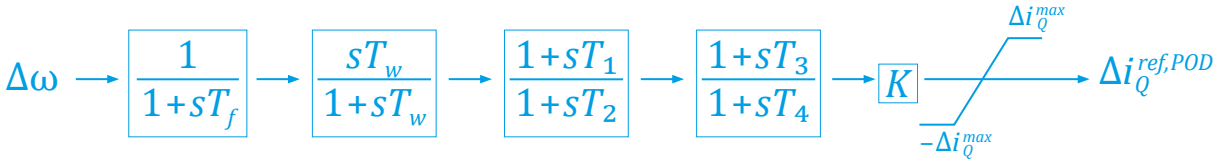


Figure 2-37: Vitoria STATCOM POD block scheme

2.5.5.2 HVDC

There is a HVDC link on the border between France and Spain, named INELFE-1. It has a rated power of 2 × 1,000 MW and is a VSC-type HVDC. Station A of the HVDC is connected to Santa Llogaia 400 kV substation (Spain) and Station B is connected to Baixas 400 kV substation (France). The HVDC is further described in Section 2.7.

With reference to the ability of the link to damp the inter-area oscillations, the HVDC is equipped with a flexible POD function, as detailed in Figure 2-38 below.

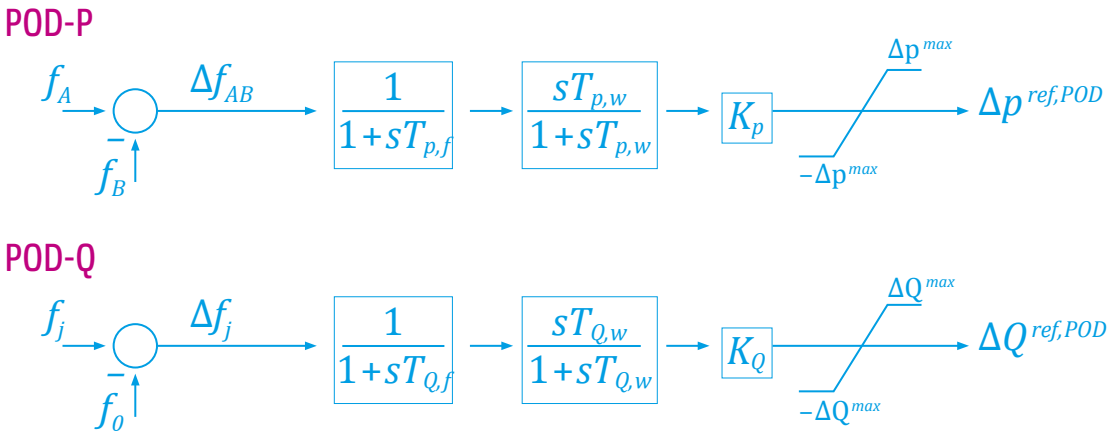


Figure 2-38: INELFE-1 POD block scheme

The POD-P logic processes as input the frequency at terminals A and B, calculating the difference Δf_{AB} for the POD-P or the difference between the frequency in the terminal and the nominal frequency Δf_j for the POD-Q, which is filtered by a cascade of low pass and high pass filters and multiplied by a gain K. POD-Q uses the local frequency deviation from a reference value f0, namely the frequency of terminal A (B) for POD-Q in terminal A (B). The final signal is summed to the HVDC active power and reactive power reference value, respectively. These logics were active during the incident of 28 April 2025. It is important to remark that for safety reasons and to prevent malfunction of the POD, a selective control implemented by the manufacturer prevents operation of POD-Q when the instantaneous output of the controller excessively frequently reaches its limits (±100 Mvar per HVDC pole).

If this occurs, POD-Q is disabled, although the total reactive power capacity of the HVDC is not affected (+400/-600 Mvar per HVDC pole).

The POD-Q functionality on the Spanish side remained active from 4 March at 20:03 on Link 1, and from 8 March at 14:34 on Link 2 – the dates on which it was disabled due to works on the links – until 28 April at 12:03:51. After the beginning of the 12:03 episode of the 0.6 Hz oscillation, the POD-Q of HVDC on the Spanish side was automatically blocked., according to manufactured design. This POD-Q remained blocked until the blackout on the Spanish side. The remote reactivation of POD-Q on this side was not applied due to the need to carry out a detailed verification of the cause of the automatic deactivation prior to re-enabling it to preserve the integrity of the HVDC. On the French side, the POD-Q remained active until the blackout.

2.5.5.3 Power System Stabilisers

The Power System Stabilisers (PSSs) are the first line of the system defence against inter-area and local oscillations. The installed PSSs are listed in the following Tables 2-5 and 2-6, for Spain and Portugal, respectively.

Technology	PSS installed	Type	Note
Nuclear	NO		
Hydro	NO		
Pumping storage	YES	PSS2A	One power plant (with five units)
CCGT	YES	PSS2A, PSS2B, PSS3B, IEEEST	Six CCGT plants (eight units in total) One CCGT has a single input PSS
Coal	NO		

Table 2-5: Availability of power system stabilisers on the main rotating generators in the Spain grid in operation on 28 April 2025

Technology	PSS installed	Type	Note
Thermal	YES	PSS2A	Only one thermal plant in service
Hydro	YES	PSS2A, PSS2B	Three power plants (two, the largest) out of fourteen with PSSs active
Pumping storage	YES	PSS2A, PSS2B	Nine power plants (seven, the largest) out of fourteen with PSSs active. The PSS is also active in pumping mode

Table 2-6: Availability of power system stabilisers on the main rotating generators in the Portugal grid in operation on 28 April 2025

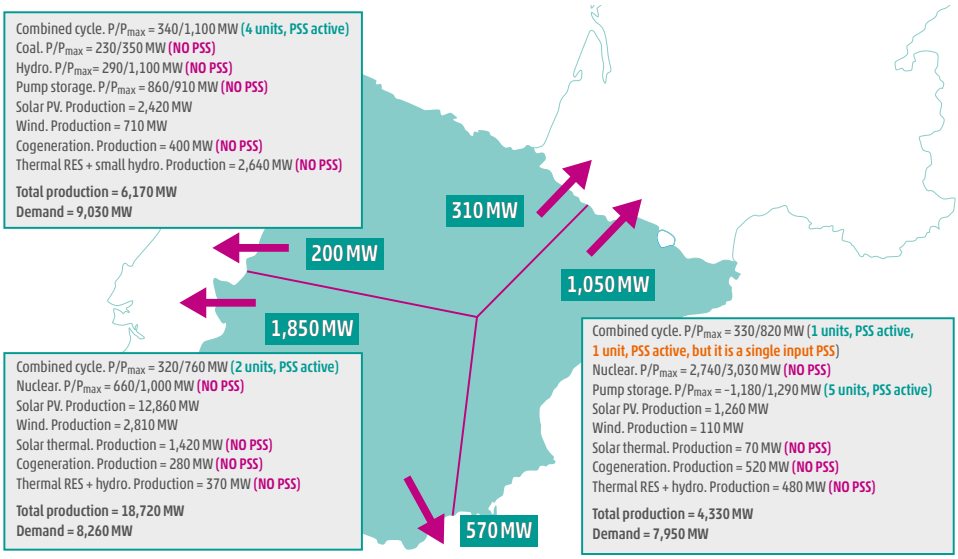


Figure 2-39: Geographical displacement of PSSs

Figure 2-39 represents the geographical displacement of PSSs in Spain. The effectiveness of PSSs in RE and REN grids is assessed in the analysis phase of the investigation.

2.5.6 Oscillatory Stability on 28 April

2.5.6.1 Angular Displacement Before the Oscillations

The heatmaps (Figure 2-40) report the angular displacement between different locations in Spain and Portugal, taking as a reference a PMU in southeast of the interconnection between Spain and France (marked as REF in the figure). It can be observed that maximum angle spread is in the southwest part of Spain.

There were 76 PMUs operational in Spain on 28 April. The extensive PMU coverage in Spain enabled the Expert Panel to establish several relevant facts and enabled producing detailed heatmaps.

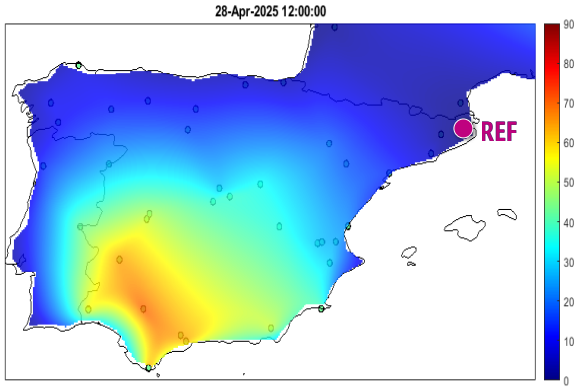


Figure 2-40: Heatmap of angular displacement at 12:00 (source: PMU)

Figure 2-41 helps to interpret the heatmap of the previous figure, aiming to divide the Spanish grid into three equivalent geographic areas in terms of demand and highlight the heterogeneous distribution of generation in Spain before the incident, as well as the generation mix per area. From the numbers indicated in the figure, it can be observed that within the Spanish system, the southwest region is "pushing" active power in the direction of the centre-north and east.

The geographic spread of dominant production capacity can be summarised as follows:

- Renewables (mainly photovoltaic and wind) mainly located in the southwest
- Nuclear in the east and southwest
- CCGT spread across the three regions

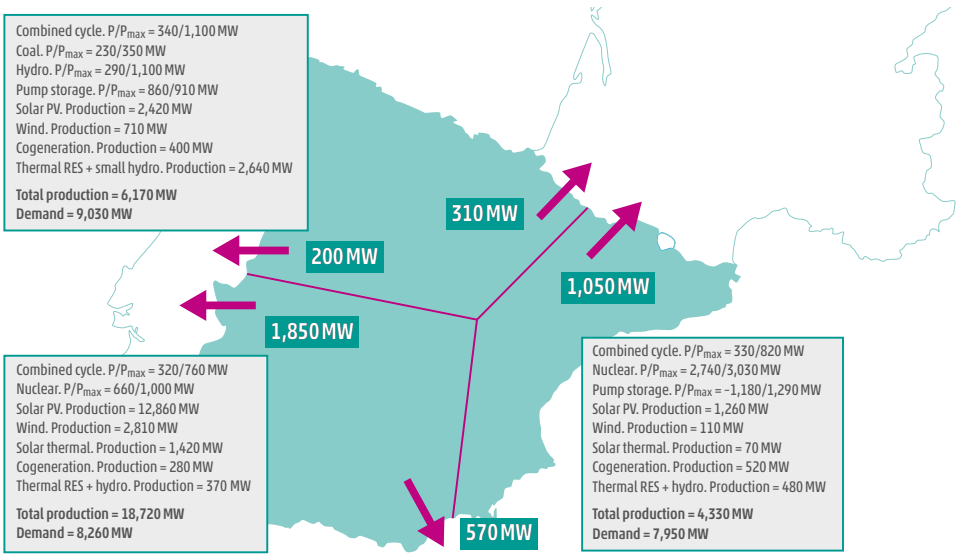


Figure 2-41: Geographic distribution of generation in Spain at 12:32

2.5.6.2 Oscillations

Between 09:00 and 12:00, Figure 2-42 shows some oscillations with small amplitude. In accordance with the common protocol between RTE and RE, no remedial actions were taken to damp these oscillations, including the one at 11:06, because the criteria of the procedure RTE-RE were not met (the damping was higher than 3 % and the amplitude of the oscillations was below 20 mHz). It is also worth specifying that the amplitude in the graphs is not exactly the amplitude of the actual system frequency (expressed in mHz) but rather the output of the modal analysis tool available to the RE control room staff, which provides a value of the estimate mode in a certain range (the actual amplitude in the grid might be higher).

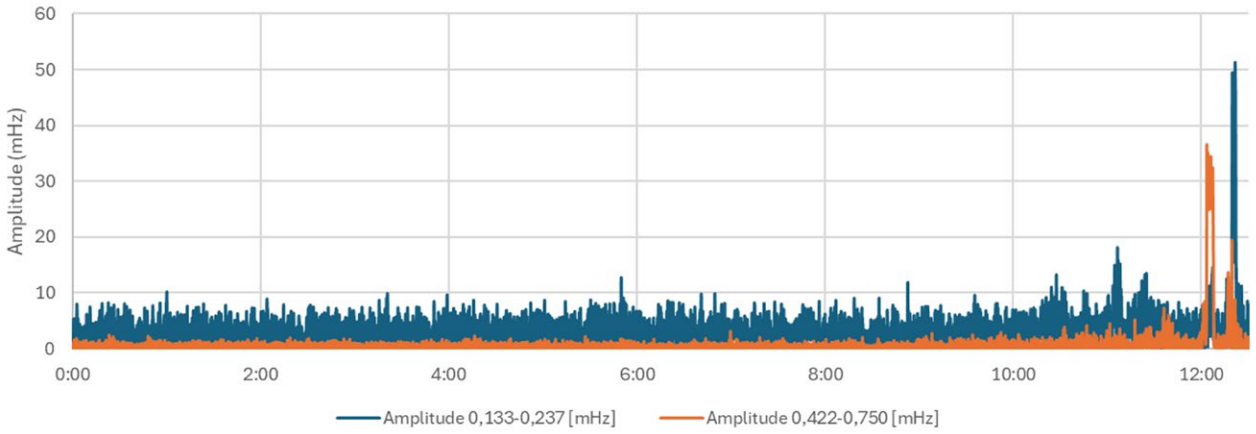


Figure 2-42: Amplitude of modes identified by the modal analysis tool in the RE control room in real time

In the 30 minutes before the blackout event, the Iberian Peninsula was affected by two prominent oscillation phenomena, i.e. periodic fluctuation of all electrical quantities such as frequency, voltage magnitude, active and reactive power. The analyses of these oscillations – conducted by the Expert Panel – are based on two main datasets:

- Frequency PMU measurements in some of the main buses of the CE SA.
- Frequency and voltage PMU recorded in the main buses of Spain, Portugal and South France along the Spanish border grids.

Figure 2-43 depicts frequency and voltage magnitudes from a selection of CESA PMUs, showing the dynamic behaviour of the CE SA between 12:00 and 12:23. These measurements reveal two distinct oscillatory events that affected both frequency and voltage signals, which will be analysed in detail in the following sections. As preliminary remark, it is possible to observe that in terms of amplitude, the first oscillation mainly involves voltages and the second one frequencies.

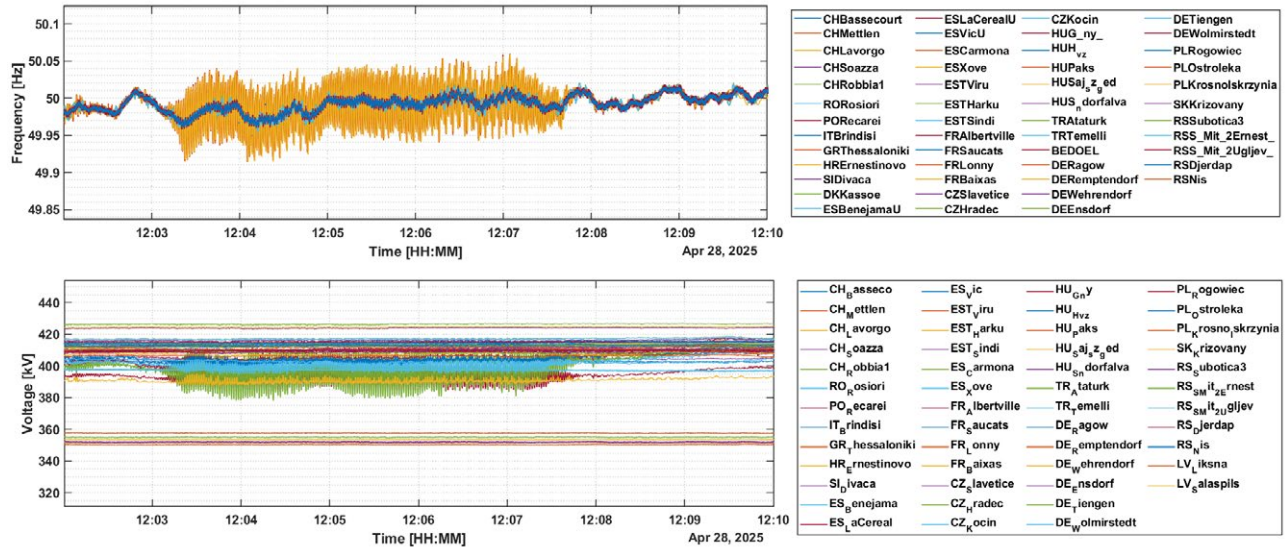


Figure 2-43: Frequency and voltage phasor magnitude measurements from European PMUs

2.5.6.3 Oscillation at 12:03 – 12:08

Oscillatory behaviour with a frequency around 0.63 Hz appears visually discernible in a time series plot at around 12:00:30 in the voltage plot below from PMU in Carmona (Figure 2-44).

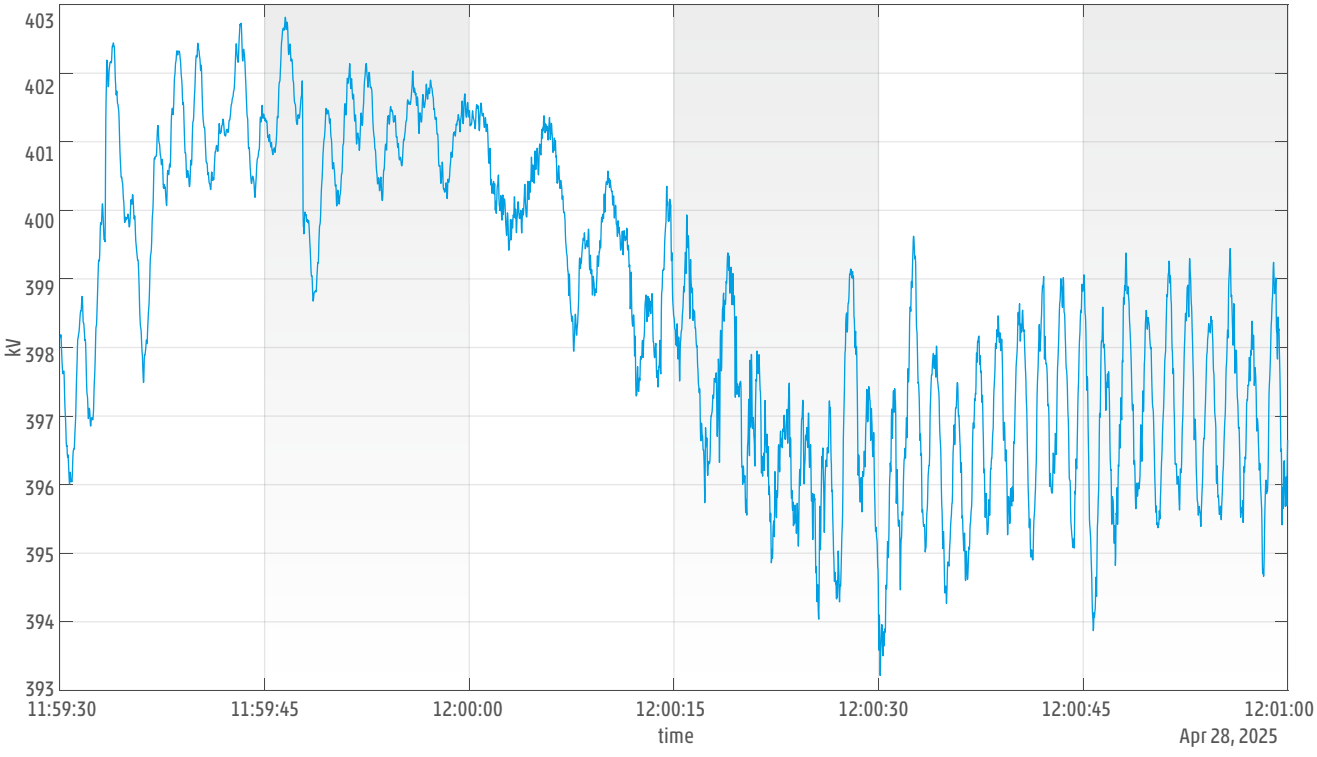


Figure 2-44: Voltage at PMU ONA4DRO, 11:59:30 – 12:01:00

The 0.63 Hz sinusoidal pattern is also visually discernible in the time series plot from around 12:00:30 in the PMU in Santa Llogaia in both the voltage and active power signals, i.e. also in the cross-border power flow on the HVDC. As previously explained, the HVDC link connected at Santa Llogaia is equipped with a POD-Q control system designed to contribute to the damping of oscillations.[8]

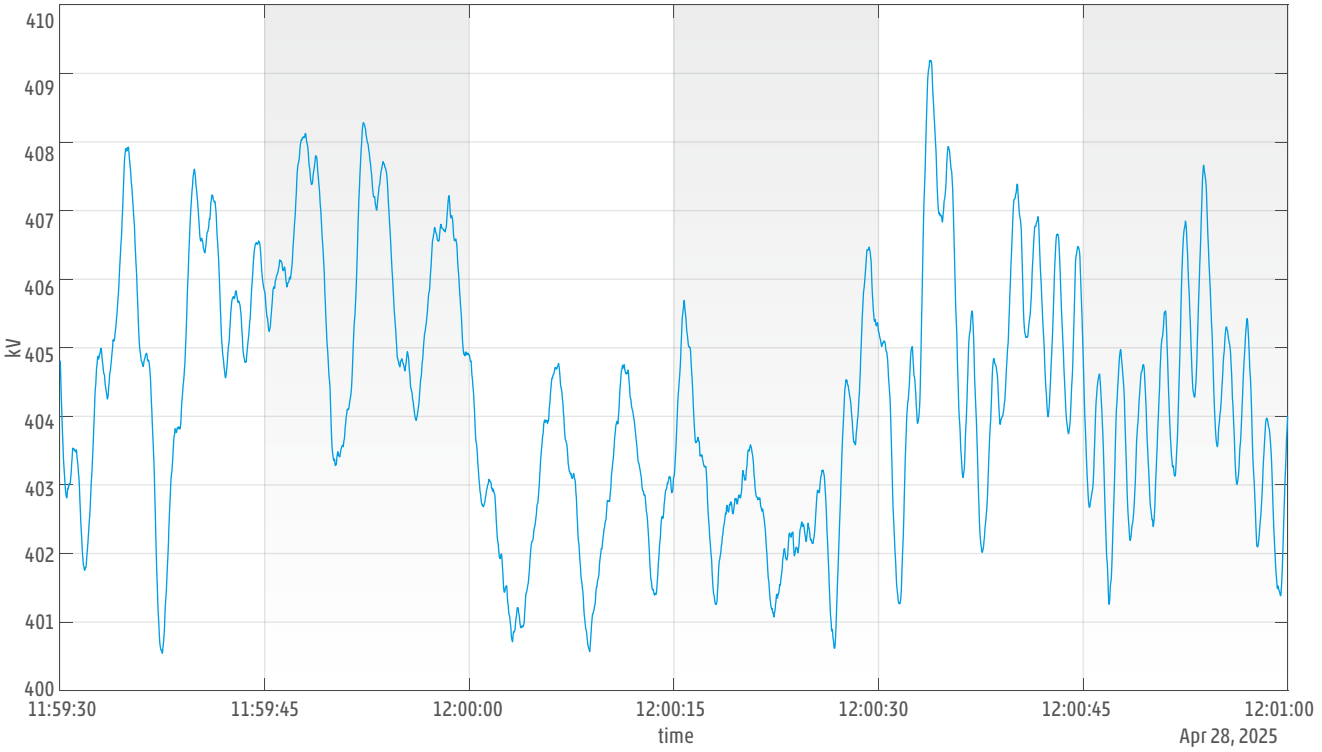


Figure 2-45: Voltage at PMU LLG4ECSL bay 1, 11:59:40 – 12:01:00

8 This control modulates reactive power with the aim of adjusting the voltage in the area, thereby influencing local demand and electrical active power of generators nearby to counteract the oscillation. The reactive power modulation performed by the HVDC is what causes the voltage oscillation observed at Santa Llogaia. RE explained that the oscillation in active power through the link is caused by the angle variation that was inducing at the HVDC terminals the 0.6 Hz oscillation. When operating in AC emulation mode, the HVDC transmits power proportionally to the angular difference between Santa Llogaia and Baixas; therefore, any variation in angle results in a corresponding variation in transmitted power. This oscillation in active power is also influenced by the POD-P control, which modulates the active power to mitigate the oscillation currently present in the system.

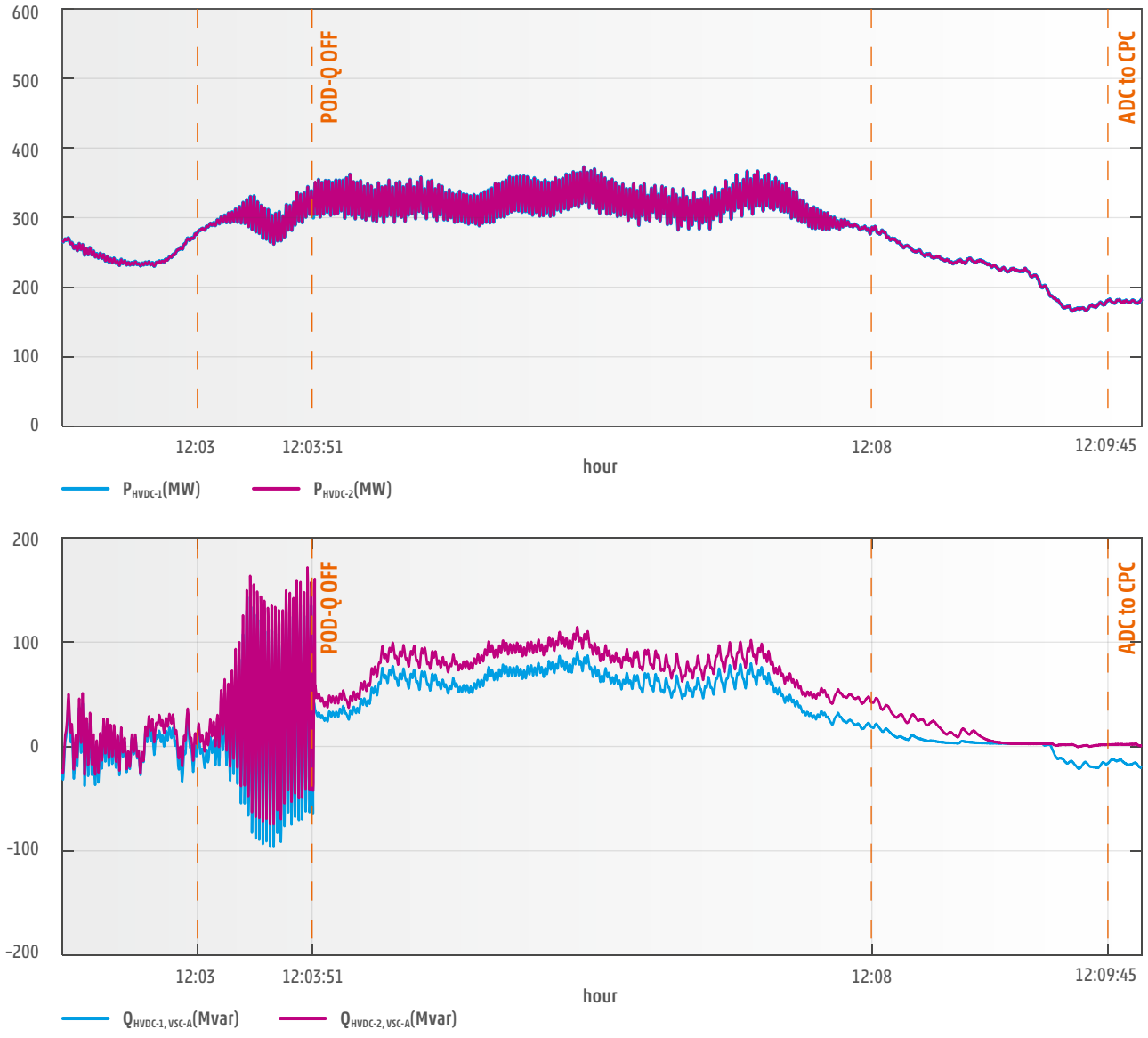


Figure 2-46: Active and reactive power flow at PMU LLG4ECSL bay 1, 12:00:00 - 12:10:00

The oscillation then appeared in all nodes of the system (sustained amplitude higher than 20 mHz) at 12:03 and lasted approximately five minutes, with a maximum peak-to-peak amplitude of 100 mHz in the frequency.

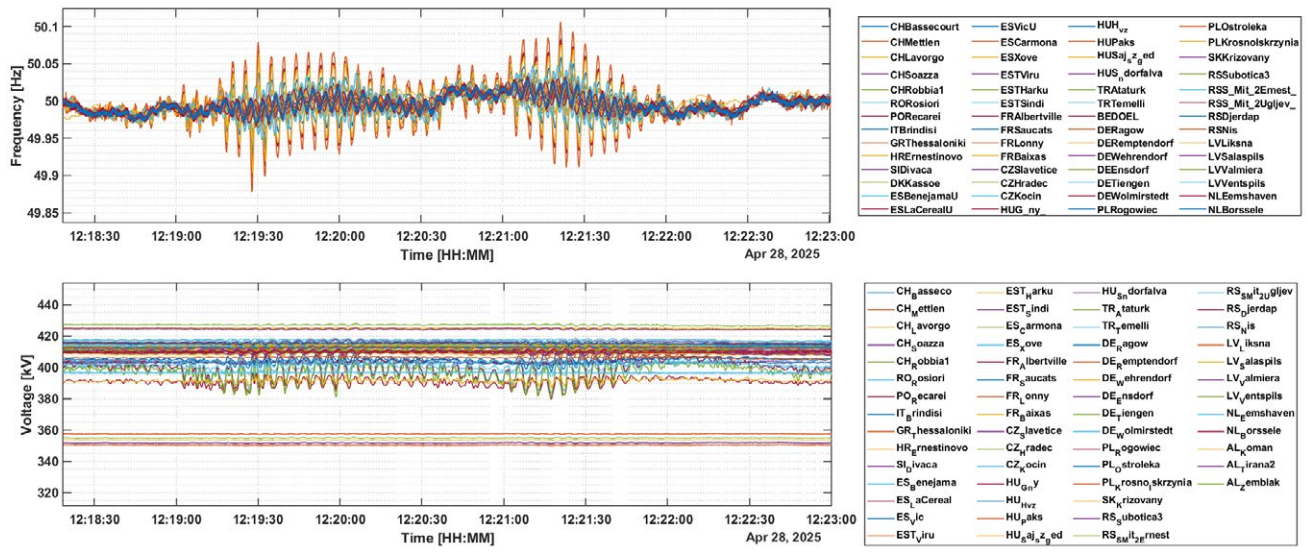


Figure 2-47: Frequency and voltage phasor magnitude measurements from European PMUs

A focused view on Spanish PMU measurements is provided in Figure 2-48, highlighting the system behaviour during the first oscillation and the moments immediately after. The figure displays the evolution of frequency, oscillation amplitude, voltage, and cross-border power exchanges.

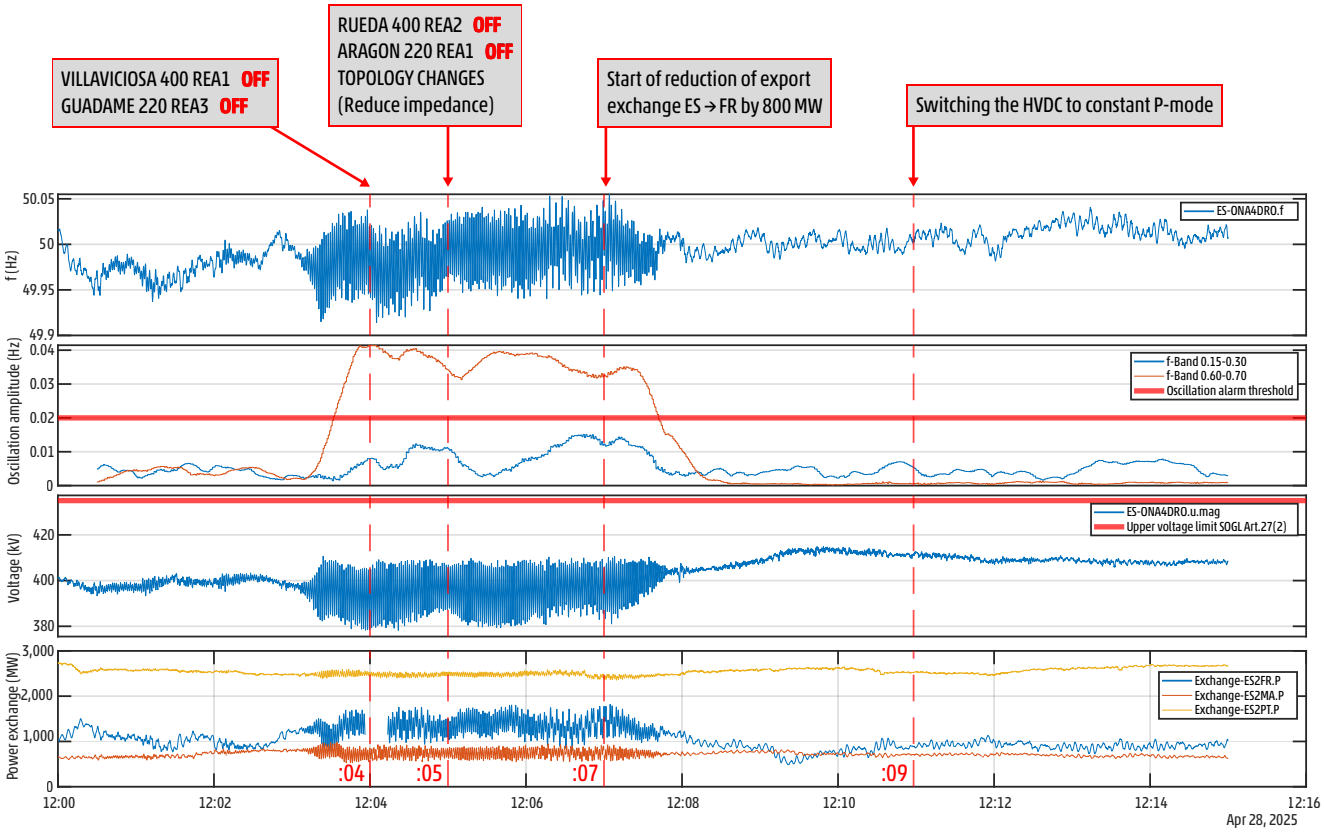


Figure 2-48: Characteristics data of the first oscillations (source: WAMS 100 ms sampling rate in the 6 kV Carmona (Spain) substation) and countermeasures applied

The second box of Figure 2-48 shows the modal analysis result, demonstrating the dominant presence of a 0.6 Hz component (in orange) and a less prominent component at 0.2 Hz, namely East Central West inter-area oscillation (in blue).

Voltage oscillations were also observed, primarily in the southwestern area. In Figure 2-49, voltages from south-west of Spain (Almaraz and Carmona), north-west (Xove), east (Benejama) and north-east (Vic) are shown. During the oscillation, in certain substations, voltage levels approach the lower threshold established by Spanish regulations of 375 kV, although the highest recorded voltage values barely exceed 410 kV.

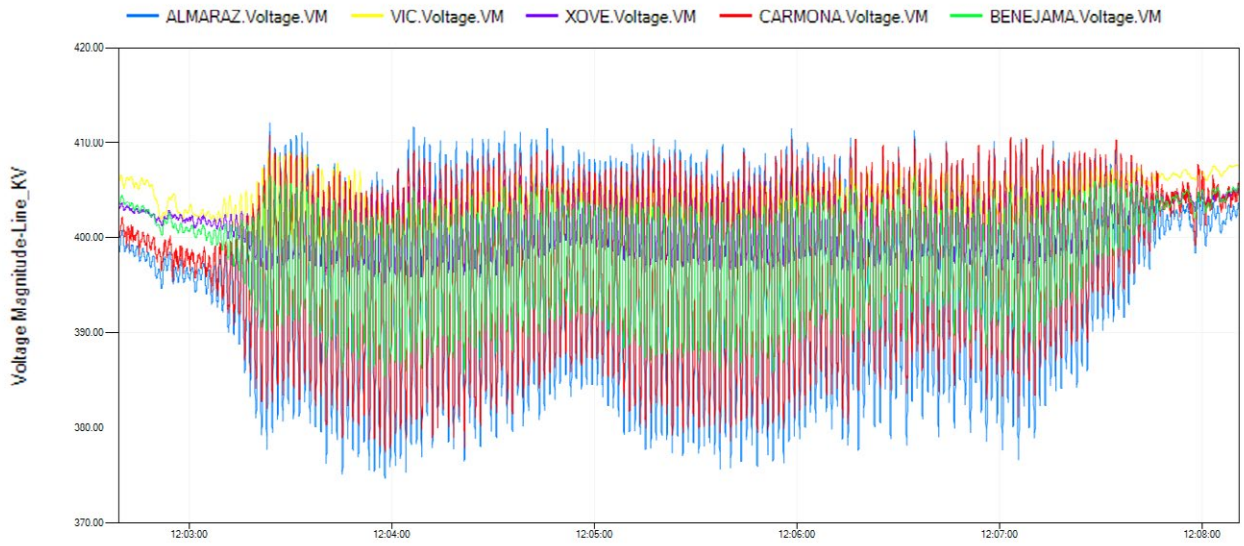


Figure 2-49: Voltage magnitudes in several Spanish substations

Figure 2-50 displays the same voltage values, but after applying a band-pass filter (0.55 to 0.70 Hz), which enhances the visibility of the oscillation amplitude at each substation. It can be seen that the voltage oscillation amplitude reaches 30 kV peak to peak at the Almaraz 400 kV substation.

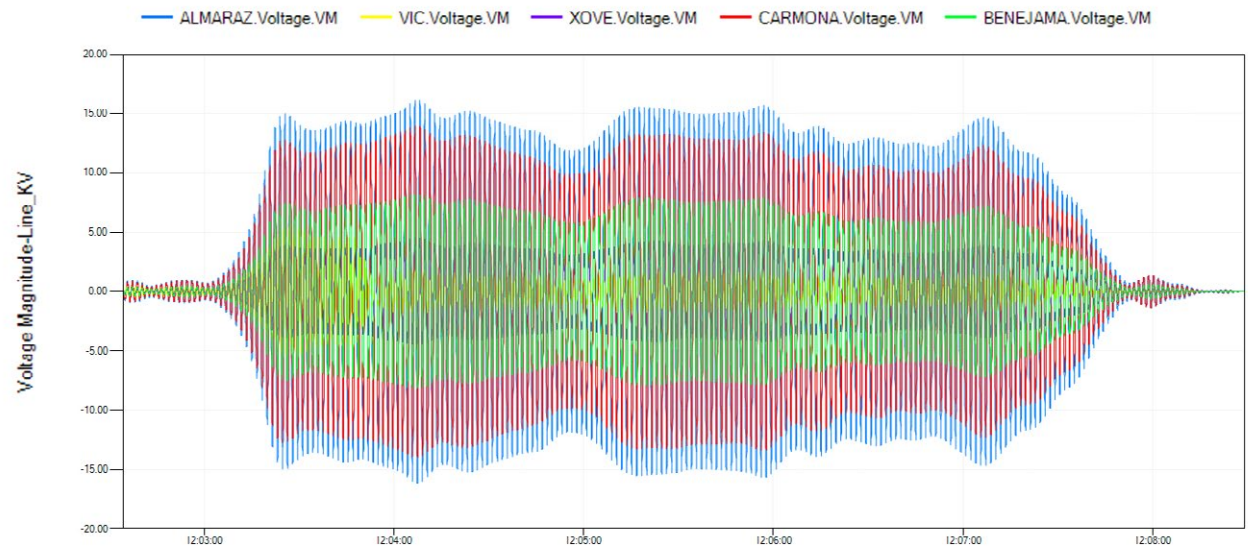


Figure 2-50: Voltage oscillation alter apply a pass-band filter [0.55 - 0.70 Hz] in several Spanish substations

After applying a complex principal component analysis algorithm[9] to all 400 kV PMUs, it is possible to recreate the finding that the dominant mode is 0.63 Hz. In fact, processing a set of several time series (i.e. voltage, frequency, etc.), the algorithm is able to find a dominant characteristic mode like reported below in Figure 2-51. Observing the sinusoidal path, it is possible to estimate the oscillatory frequency (i.e. measuring time between two adjacent peaks).

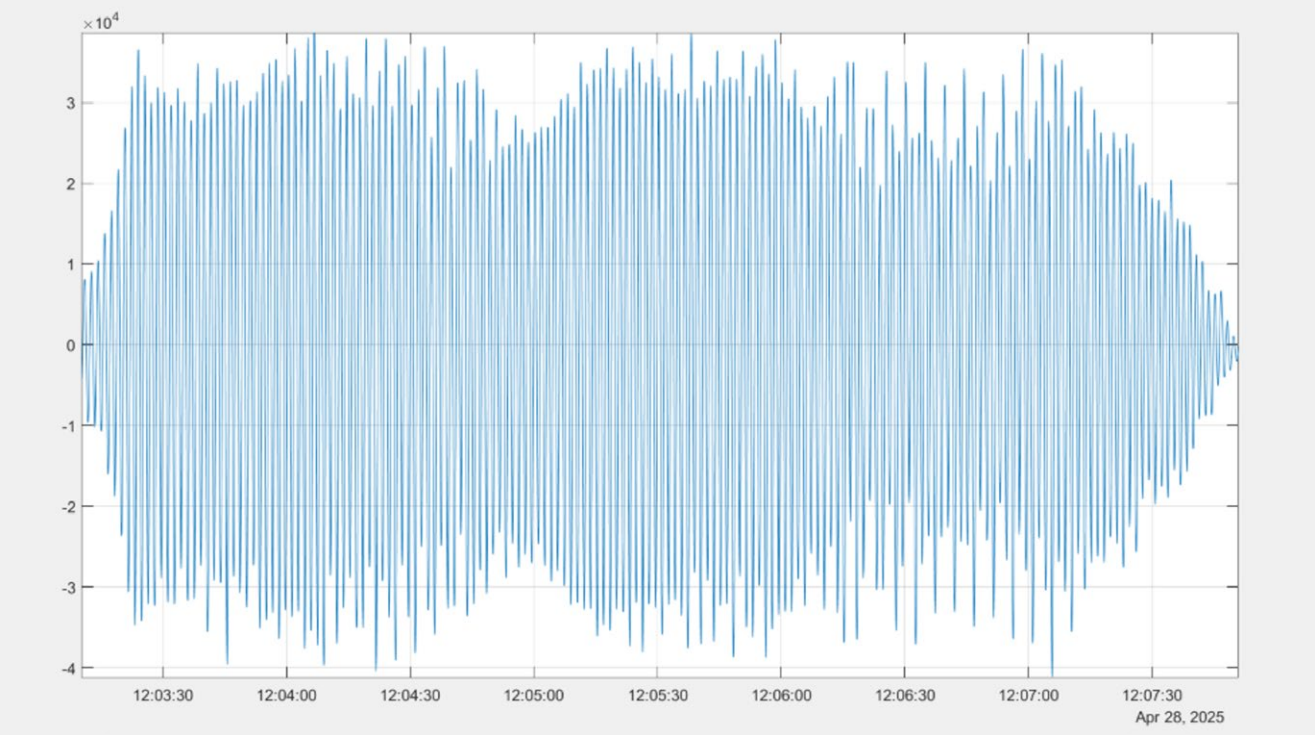


Figure 2-51: Voltage oscillation principal component analysis (PCA)

9 Complex principal component analysis is a technique that processes several time series and extracts the dominant components versus time, which enables identifying the frequencies, damping, amplitude and mode shape of the dominant modes. Ref. "Complex Principal Component Analysis: Theory and Examples" by J. D. Horel.

In order to evaluate the 400 kV nodes that are the most active in the 0.63 Hz oscillation, a complex principal component analysis was performed, showing the maximum activity (in this case, voltage amplitudes) in the nodes of Almaraz and Puebla de Guzman.

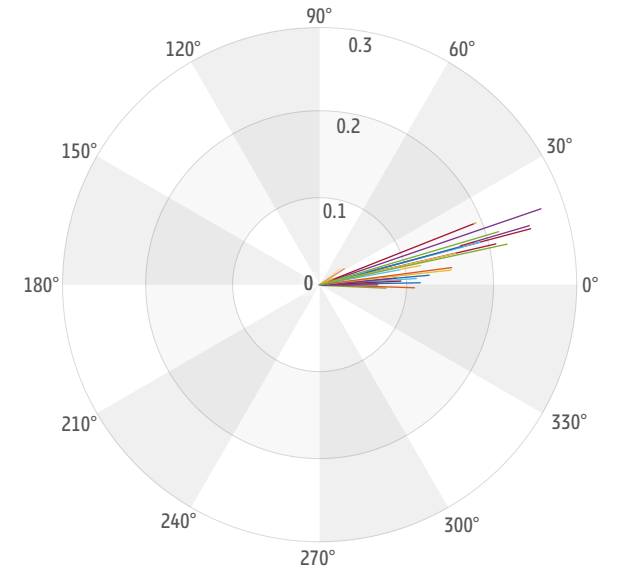


Figure 2-52: Mode shape by complex principal component analysis on PMU voltages normalised (each colour in the polar plot represent a different location)

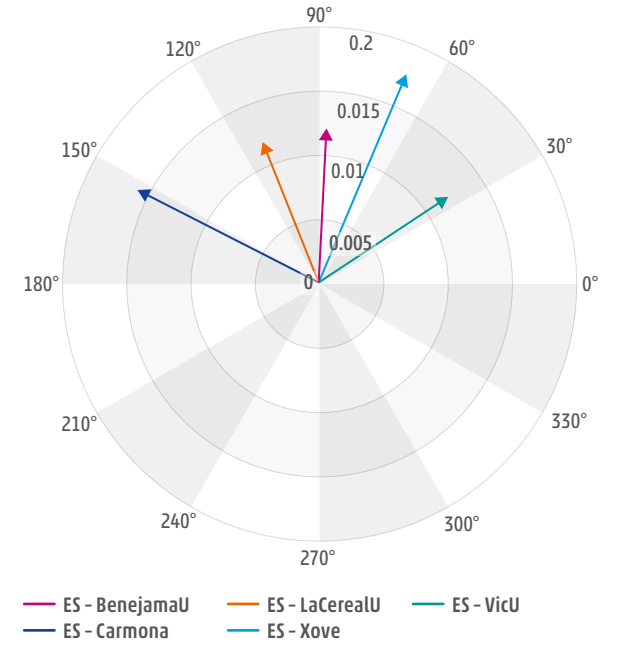


Figure 2-53: Mode shape on PMU frequencies normalised (each colour in the polar plot represents a different location)

Figure 2-53 clearly shows how the significant amplitudes of mode shape are located in Spain and the angle spread is around 120°, with no phase opposition oscillations.

One minute after detecting the oscillation, the control rooms of RE and RTE activated the common protocol described in Section 2.5.2. In terms of event sequence, the RE control room called the RTE control room to initiate the activation of the protocol. The protocol is intended to be applied depending on the damping measurement or when oscillations in the range of 0.1–0.35 Hz and an amplitude higher than 20 mHz are detected.

RE and RTE initiated a change of the operating mode of the HVDC (switching from hybrid mode to constant power mode) and a countertrading procedure involving 800 MW across the France—Spain exchange borders. The countertrading procedure was performed in the following way: as the total imbalance in Spain was negative (i.e. less production in comparison with consumption), there were no need for generators to compensate for countertrading through RR and mFRR allocations.

The imbalance generated by countertrading was compensated with aFRR or reduced the need for mFRR. On the RTE side, countertrading was used to increase French production. This production adjustment was made at the national level, based on electricity market prices using the merit order method. While not part of the oscillation control protocol, four shunt reactors were switched off by RE due to the lower voltages seen in the system during the oscillation:

- 12:04 Villaviciosa 400 kV REA 1
- 12:04 Guadame 220 kV REA 3
- 12:05 Rueda 400 kV REA 2
- 12:05 Aragón 400 kV REA 1

Furthermore, several topological actions were undertaken, aiming to reduce the impedance of the grid.

Regarding the behaviour of the POD controllers of INELFE HVDC, the POD-P controller remained active during the whole period, while the POD-Q controller was active at the start of the 0.63 Hz oscillation, but disabled at 12:03:51, i.e. approximately 50 seconds after the start of the oscillation. The disabling of the POD-Q was caused by the logic implemented by the manufacturer to prevent its operation when the output of the controller saturates in ±100 Mvar (reaching the upper and lower limit too often).

Figure 2-54 reports the frequency and voltage measured at the Spanish terminal of the HVDC. In the first chart, it can be observed that when the POD-Q is disabled, the amplitude of the frequency oscillation does not change. In the second chart, it can be observed that after the disabling the POD-Q, the amplitude of the local oscillation of voltage at the Llogaia substation decreases. The increase in amplitude of voltage during the POD-Q action is correct, due to the action of the HVDC control aimed at increasing damping. The voltage oscillation amplitude did not change in other substations.

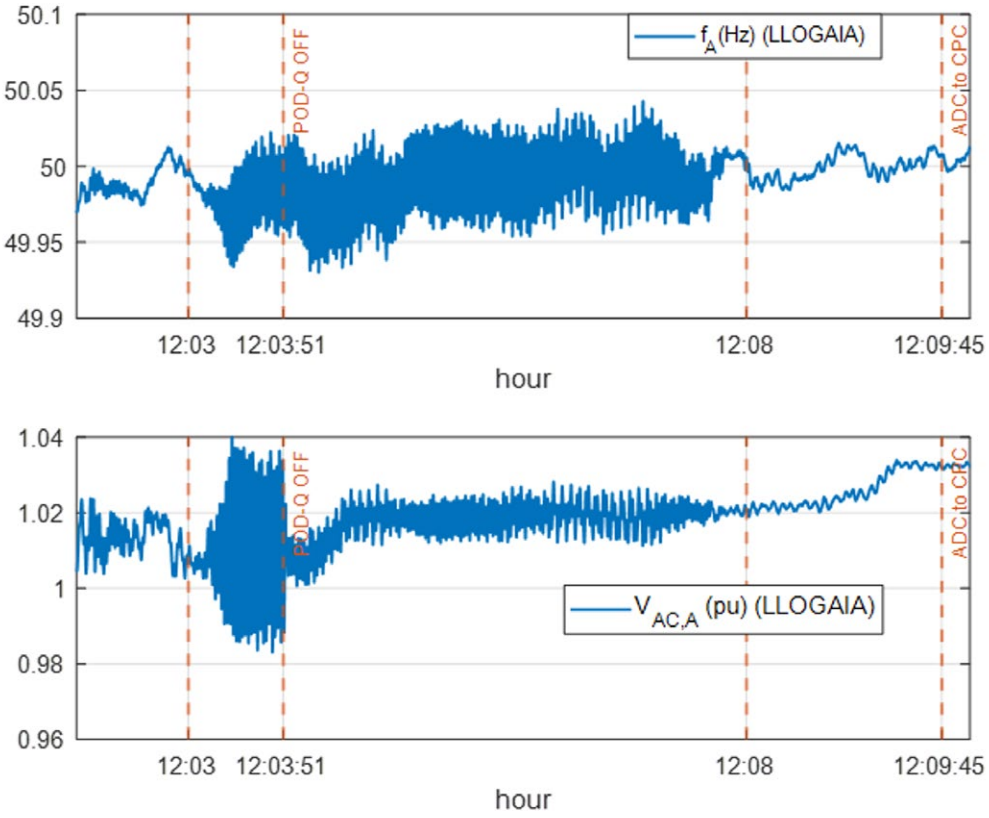


Figure 2-54: First oscillation of 0.63 Hz frequency and voltage at Sta. Llogaia 400 kV substation

The subsequent switch to constant power mode was applied by the control room at 12:09, when the oscillation amplitude was just decreasing. This decision was made by RE/RTE with the aim to further improve the damping. After the power mode was switched to constant power, an additional action was the change of the P set point by increasing it to 1,000 MW in the export direction from the Spanish perspective.

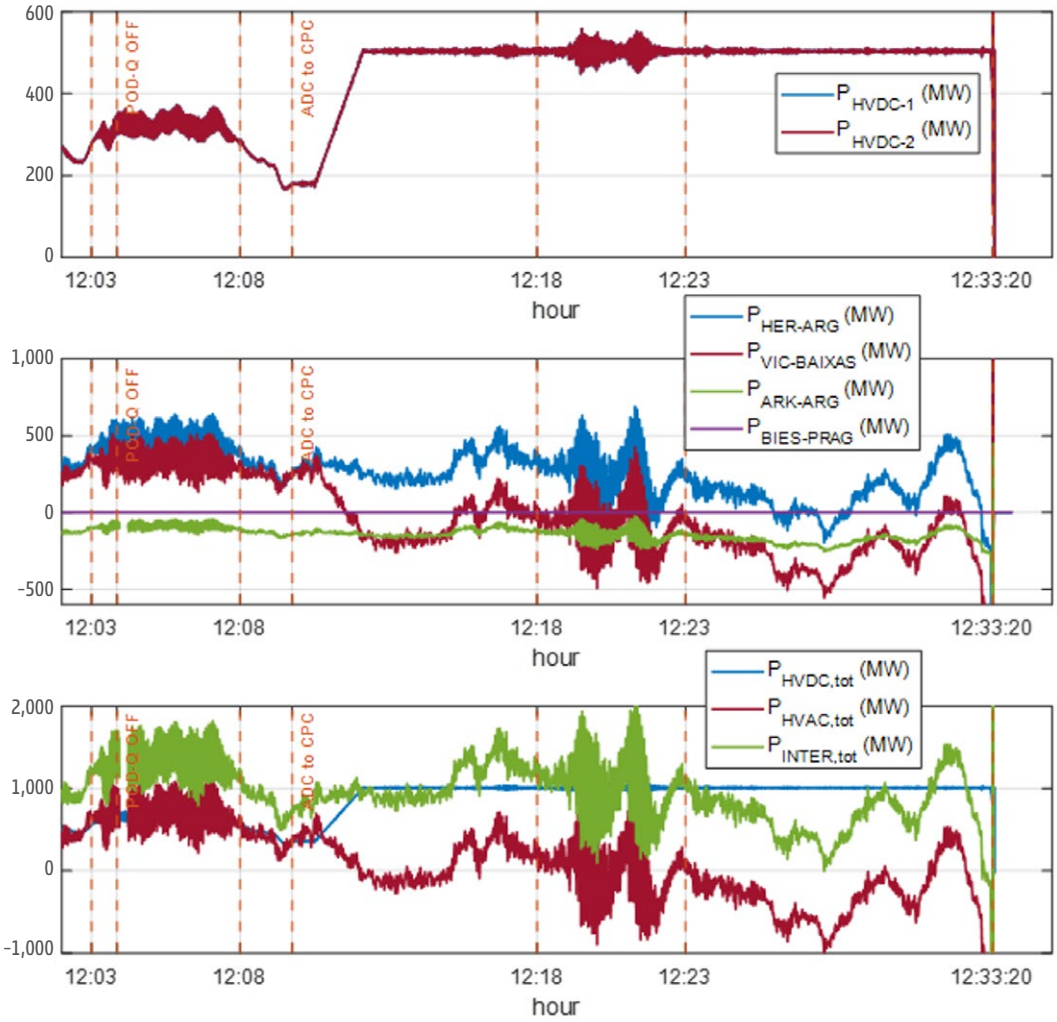


Figure 2-55: Power flows in the Spain–France interconnection (sign criterion: positive = exported from Spain to France)

The frequency domain analysis – performed using a sliding-window fast Fourier transform (FFT) on all PMUs – shows the emergence of a distinct oscillatory mode at approximately 0.63 Hz during the first oscillation.

The FFT results clearly indicate that this mode has a predominant oscillation frequency of 0.63 Hz, most likely corresponding to a local mode.

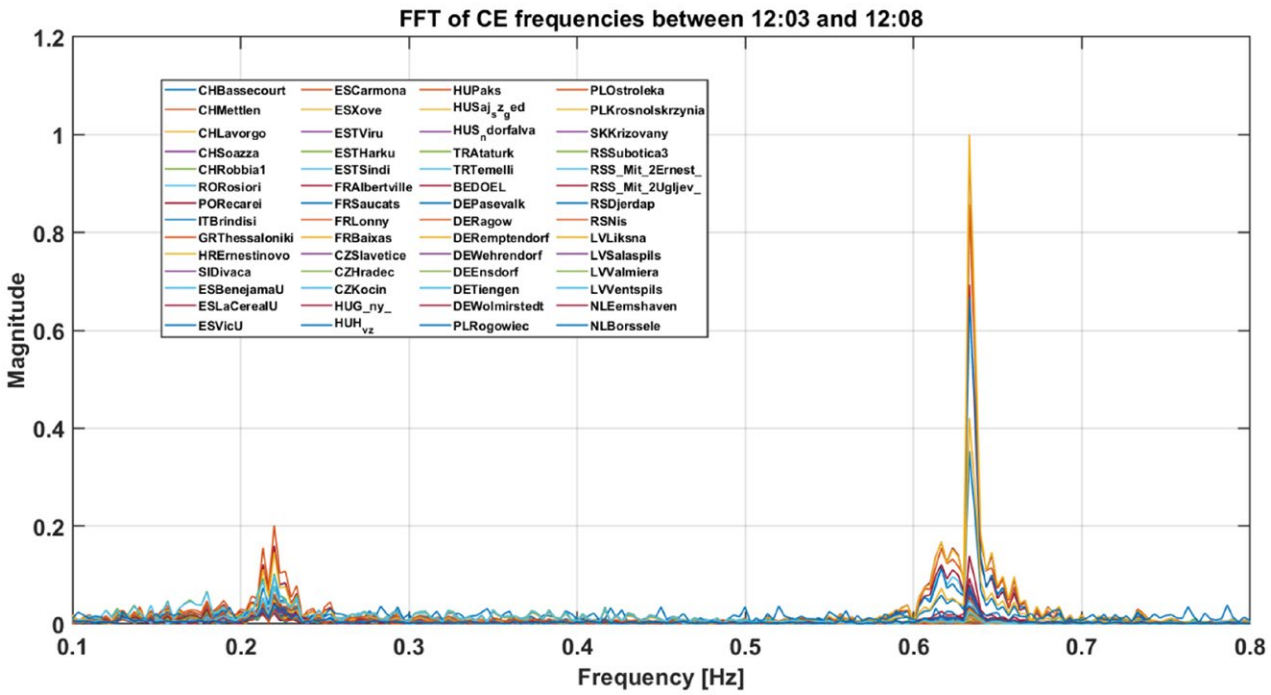


Figure 2-56: Normalised fast Fourier transform of the frequency from various PMUs across Continental Europe between 12:03 and 12:08

To classify this 0.63 Hz oscillation geographical distribution, a mode shape analysis was performed, and the results are showed in Figure 2-57 below.

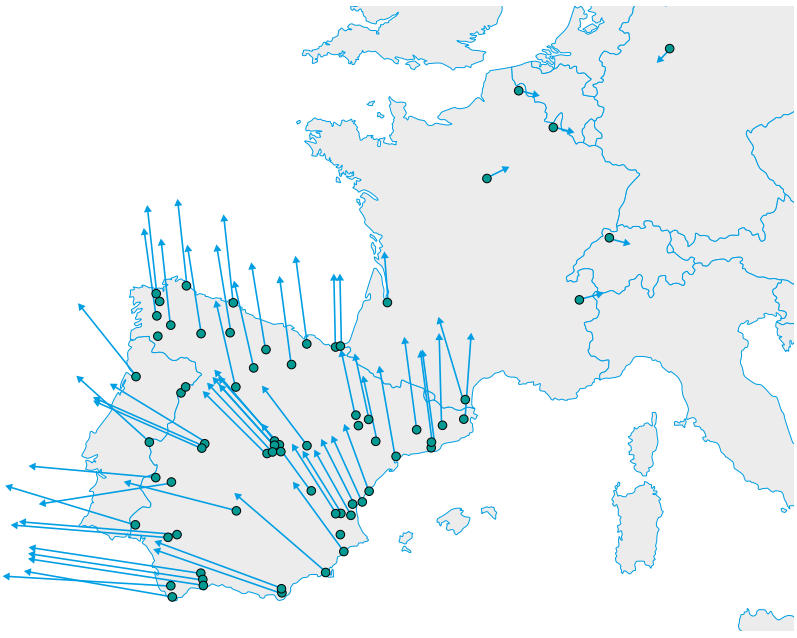


Figure 2-57: Mode shape of the 0.63 Hz oscillation (frequencies)

The arrow in Figure 2-57 above indicates the direction of the oscillation and amplitude. It can be noted that the angular spread of frequencies between north and south of Iberia is around 90° to 100°, with a local geographic displacement.

Figure 2-58 displays a zoom of voltage and frequency at various locations to estimate the angle between the two quantities, which is approximately 147°, confirming unstable behaviour. In fact, the more voltage in a node that is in phase with the frequency, the more the loads can act as a stabilising action.

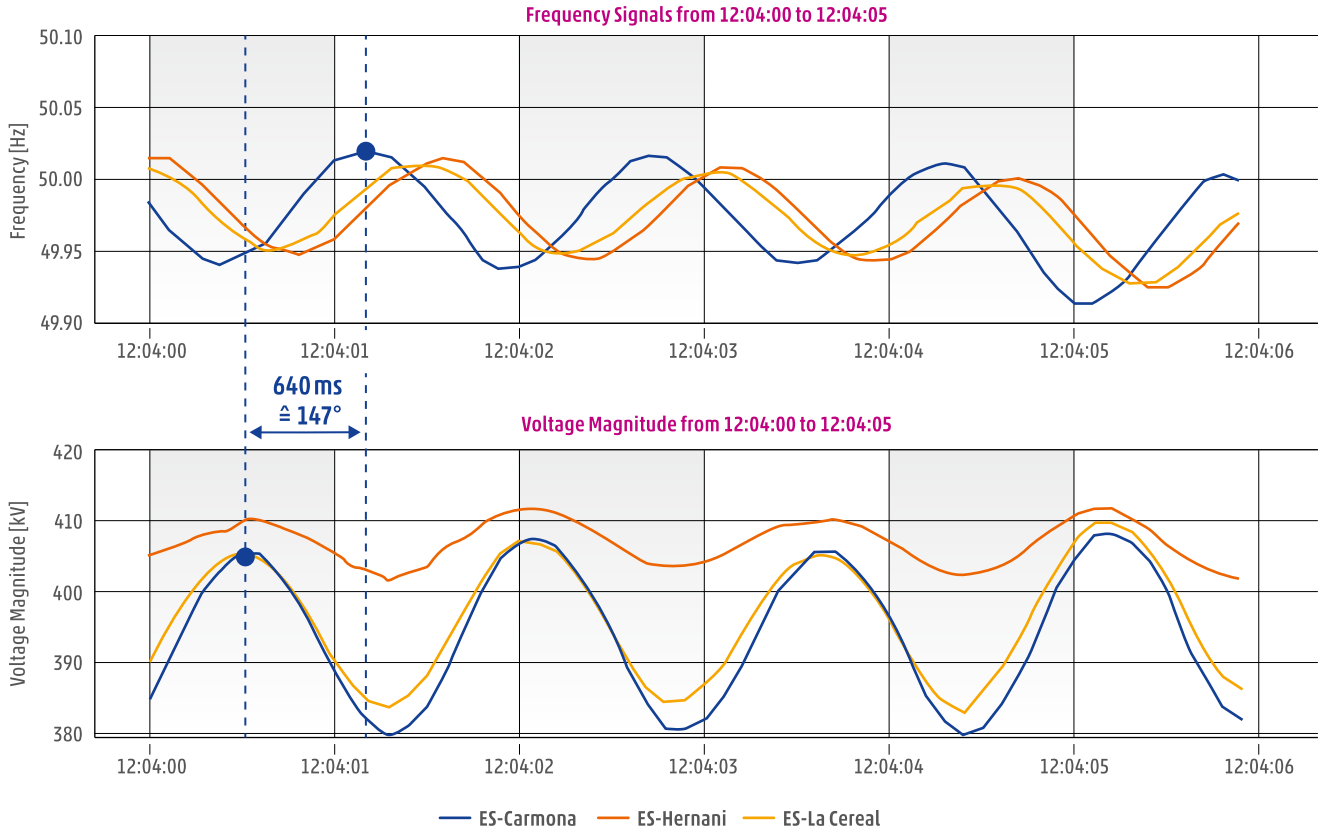


Figure 2-58: Angle between voltages and frequencies

Regarding generation behaviour during oscillations, based on the available data, the maximum resolution available is insufficient to display a clear trend of active, reactive power, and voltage fluctuations for particular generators.

Nonetheless, based on PMU data, it is possible to understand that the dominant content of energy is associated with the reactive power and voltage. In addition, large power fluctuations were detected in the SCADA data of generation in the area of the nodes of Almaraz and Puebla de Guzman.

Figure 2-59 shows fluctuations of active power with a peak-to-peak amplitude of around 0.5 pu and reactive power with a peak-to-peak amplitude of around 0.25 pu, occurring between 12:03 and 12:08.

Chapter 4.2 includes a detailed analysis of the 0.63 Hz oscillations.

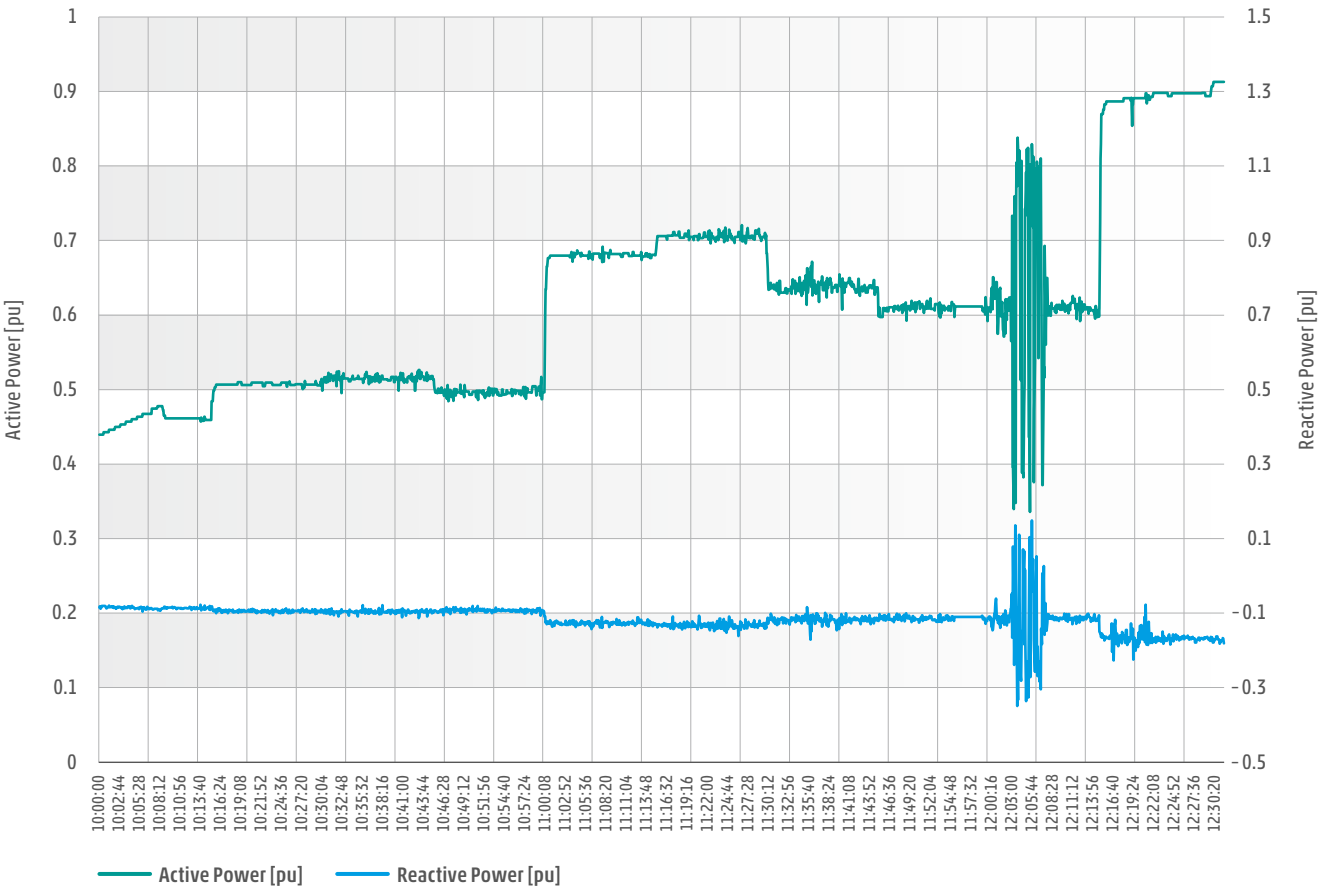


Figure 2-59: Active and reactive power generated by a power plant connected in the province of Badajoz

There have been smaller episodes of oscillatory behaviour registered on some PMUs after the first major oscillation described above and before the second one, which started at 12:19, as shown in Figure 2-60 below, specifically on voltage in PMU ALD4HIN.

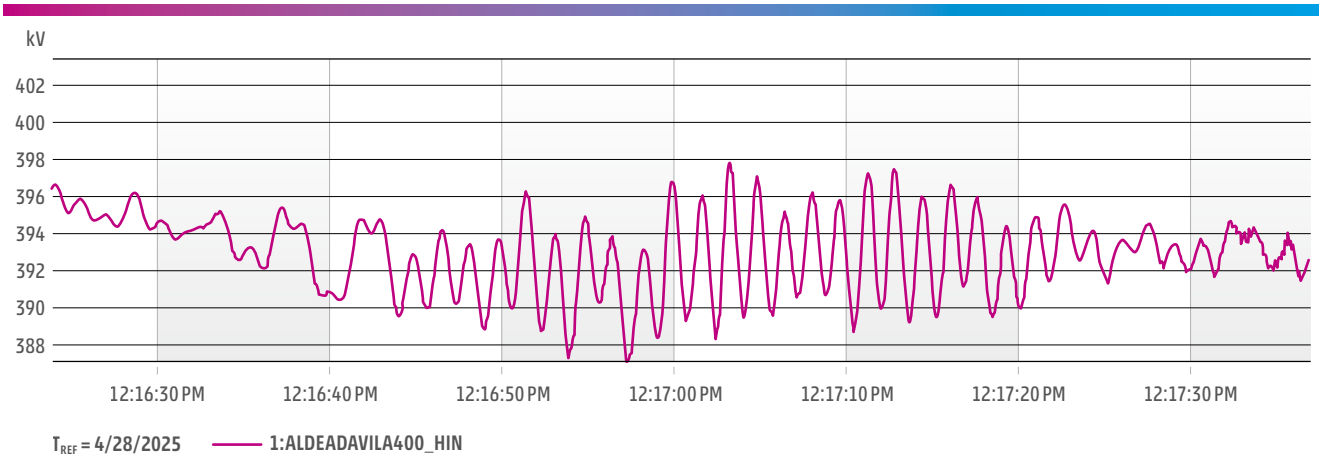


Figure 2-60: Voltage at PMU ALD4HIN between 12:16:30 and 12:17:30

Figure 2-61 shows the period from 12:16:30 to 12:20:00, showing the minor oscillations at the beginning of the interval, as well as the beginning of the 12:19:00 oscillation.

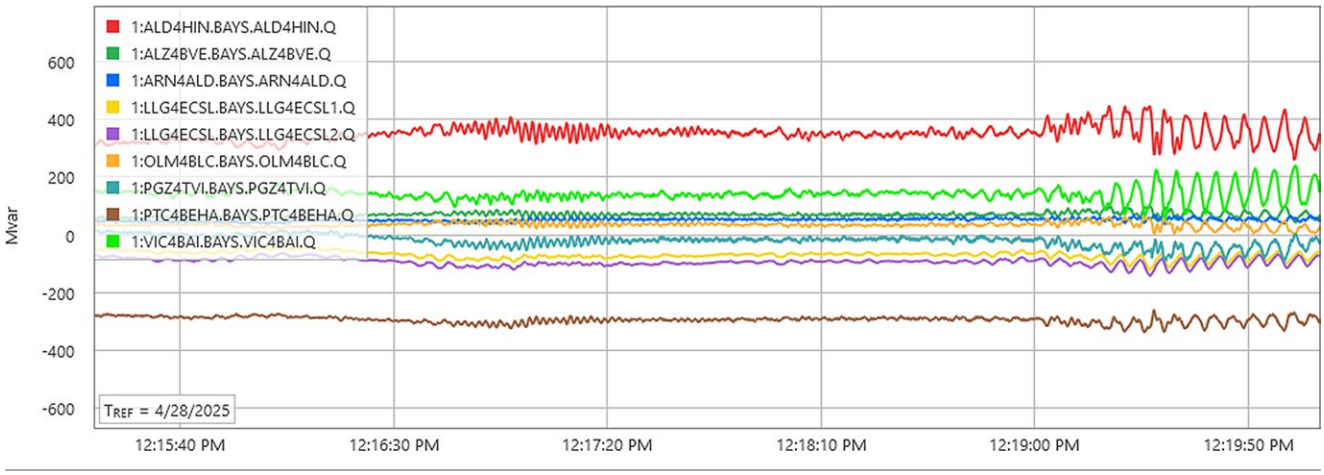


Figure 2-61: Reactive power at various PMUs between 12:15:30 and 12:20:00

2.5.6.4 Oscillation at 12:19 – 12:22

The second major oscillation is considered to have begun (an amplitude higher than 20 mHz) on all PMUs at 12:19.

The 12:19 oscillation is considered to have lasted for approximately three minutes, with a maximum frequency amplitude of around 200 mHz, significantly larger than the 12:03 – 12:08 event.

Following this event, RE and RTE initiated another countertrading procedure involving 500 MW across the France – Spain exchange borders. The countertrading procedure was performed in the following way: as the total imbalance in Spain was negative (i.e., less production in comparison with consumption), there was no need for generators to compensate for countertrading through RR and mFRR allocations.

The imbalance generated by countertrading was compensated with aFRR or reduced the need for mFRR. On the RTE side, countertrading was used to increase French production This production adjustment was made at the national level, based on electricity market prices (mFRR and RR) using the merit order method.

In addition, further actions were adopted on shunt reactors:

- 12:17 Cabra 400 kV REA 1 was disconnected before the second oscillation

Two minutes after the beginning of the second oscillation, the following shunt reactors were disconnected by RE:

- 12:21 Peñaflor 400 kV REA 1
- 12:24 Palos 220 kV REA 1
- 12:24 Morata 400 kV REA 4

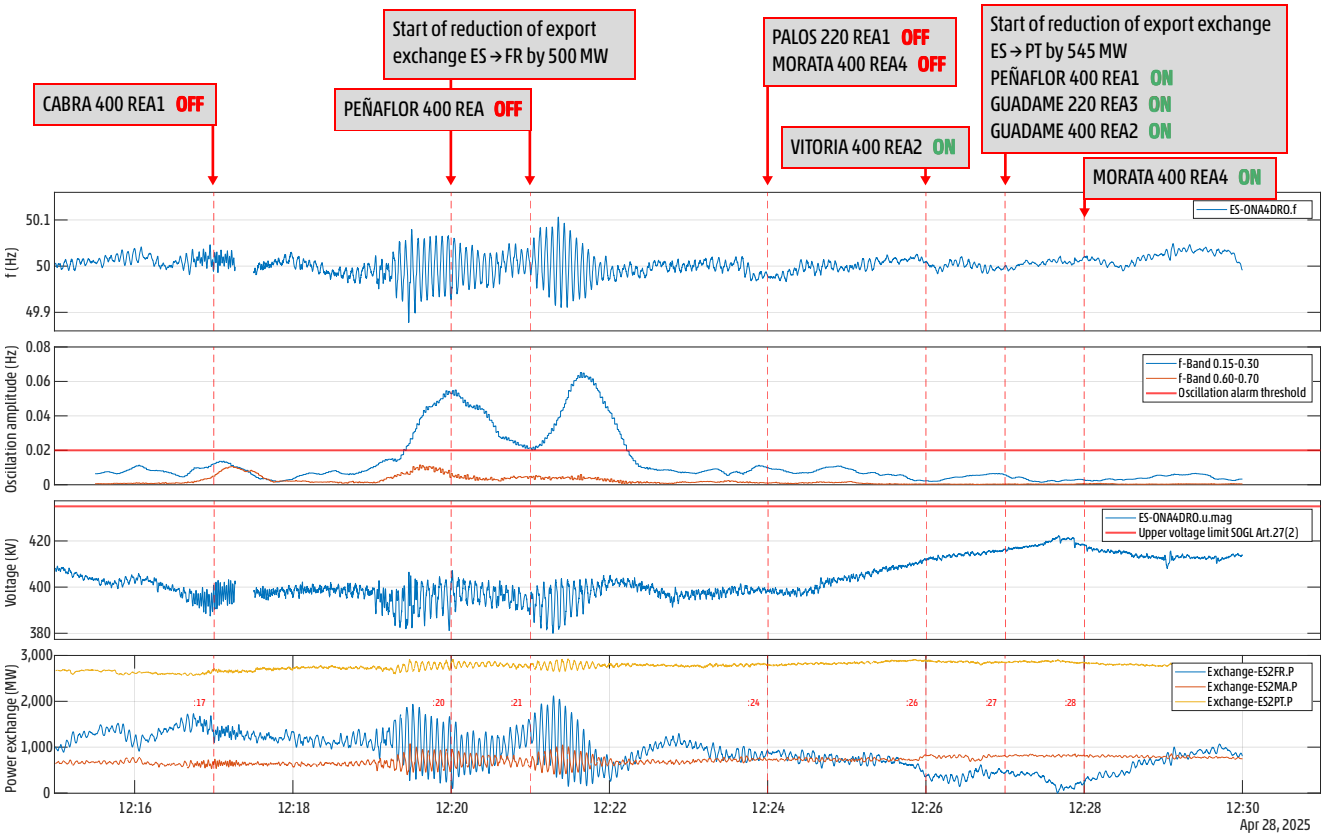


Figure 2-62: Characteristics data of the second oscillations and increasing voltage (source: WAMS 100 ms sampling rate at the 400 kV Carmona substation) and countermeasures

From Figure 2-62, it is possible to observe the frequency behaviour on the first graph. The second graph shows the amplitudes of the two oscillatory frequency components, namely the 0.2 Hz inter-area mode in blue (dominant) and the 0.63 Hz local mode in orange (with smaller amplitude).

Figure 2-63 presents the system-wide trends of frequency and voltage across CE:

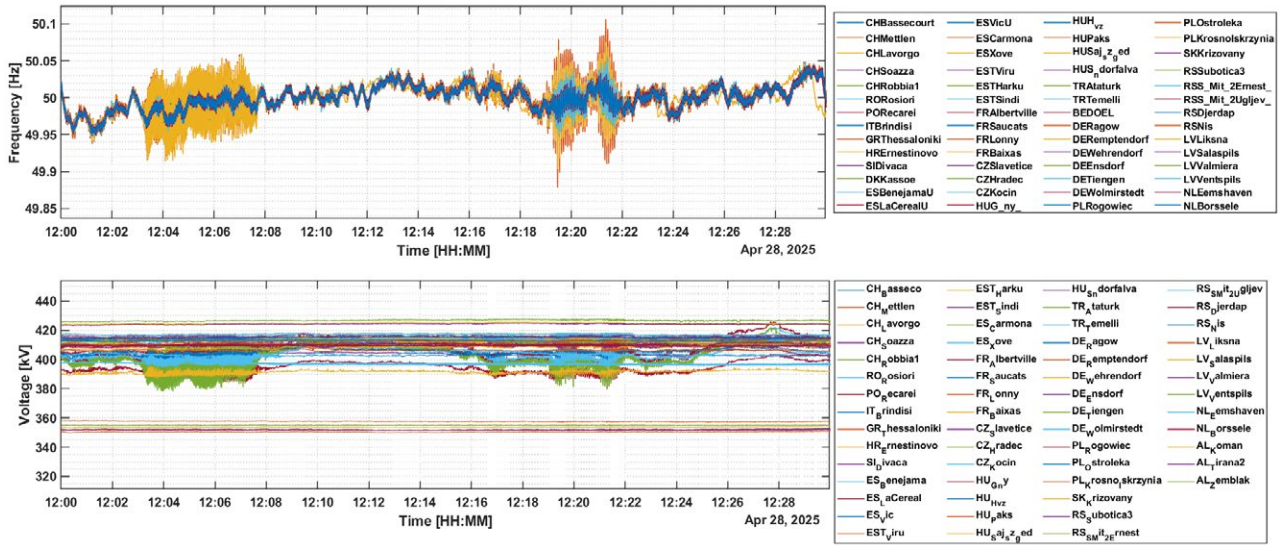


Figure 2-63: Frequency and voltage phasor magnitude measurements from European PMUs

Unlike the first oscillation – which was confined to the Iberian Peninsula in terms of very low amplitude outside the Iberian system – this second event exhibited a clear inter-area character. This is confirmed by both the modal analysis estimates and the polar plot, which indicate that the oscillation corresponds to the ECW mode. Figure 2-64 shows a spectrogram over all the PMUs of Spain and Portugal available (frequency measurement sampled at 100 ms); it is clearly visible the fingerprint of interarea mode at 0.2 Hz.

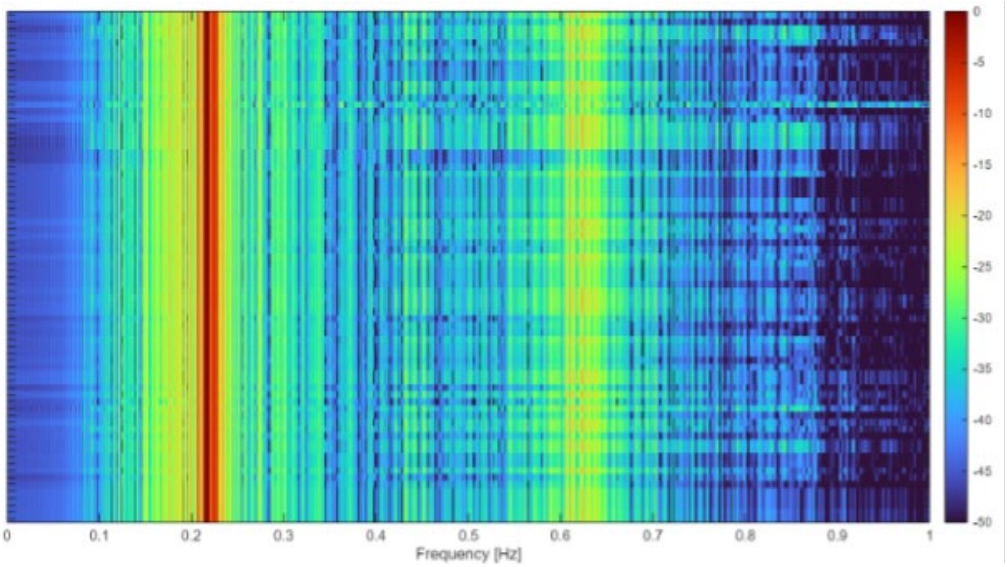


Figure 2-64: Spectrogram of all the Spain/Portugal PMUs during the timeframe of 12:18 to 12:25

The polar plot in Figure 2-65 was obtained processing 128 measurands spreading several European locations; to better readability of the picture, the results of mode shape vectors were aggregated in three main equivalent vectors with a weighting procedure that aggregates vectors with reciprocal phase degrees less than 30°.

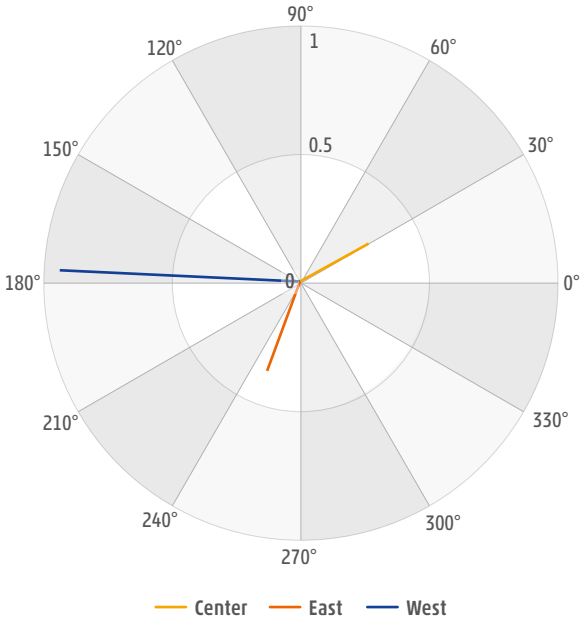


Figure 2-65: Polar plot of the ECW mode at 12:20:00

The energy distribution across the clusters confirms that the Western area contributed more strongly to this oscillatory mode.

The modal estimates produced using a FFT algorithm[10] (Figure 2-66) show a progressive degradation in the calculated damping of the ECW mode.

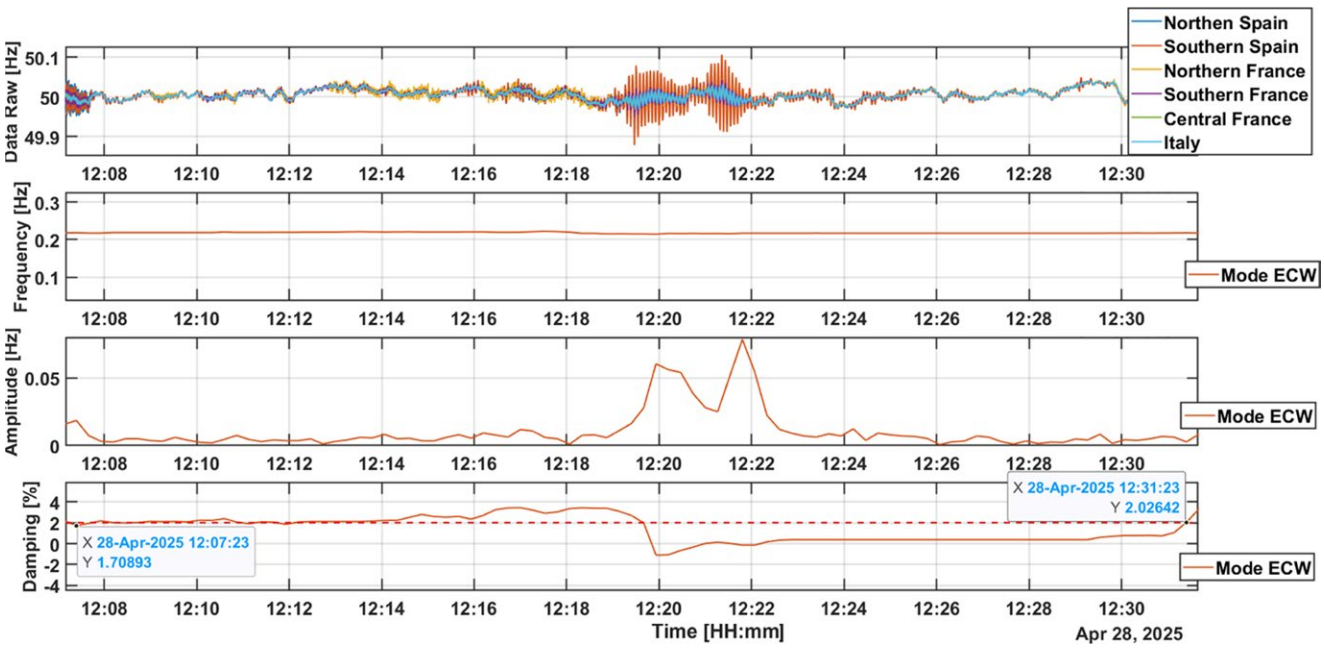


Figure 2-66: Estimated frequency, amplitude, and damping ratio of the dominant oscillation mode over time, obtained using a mode estimation algorithm

Damping remained consistently low for several minutes before becoming negative around 12:19.

This deterioration in damping is accompanied by a marked increase in modal energy, as indicated by the growing amplitude of the oscillation frequency.

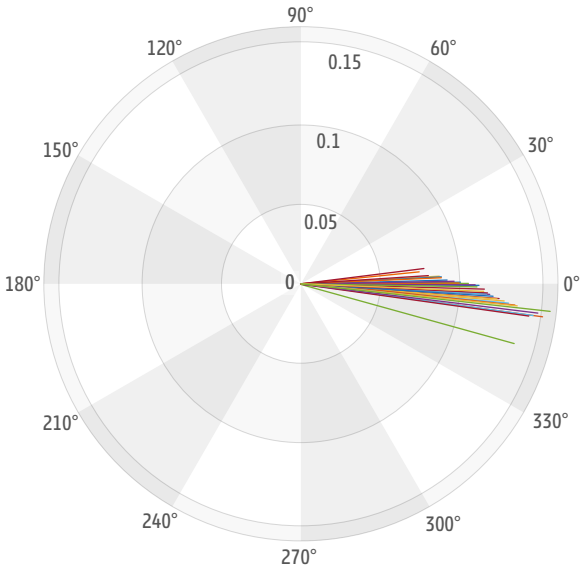


Figure 2-67: Mode shape by complex principal component analysis on PMU normalised voltages (colours in the polar plot represent different locations)

10 The adopted technique was based on a fast Fourier analysis on the selected PMUs time series and frequency/damping estimation with the Tufts-Kumaresan algorithm.

In this case, voltage oscillations are also observed, with an amplitude of the same order as those recorded during the 12:03 oscillation, and even slightly lower. Once again, a decrease in the average voltage value is observed.

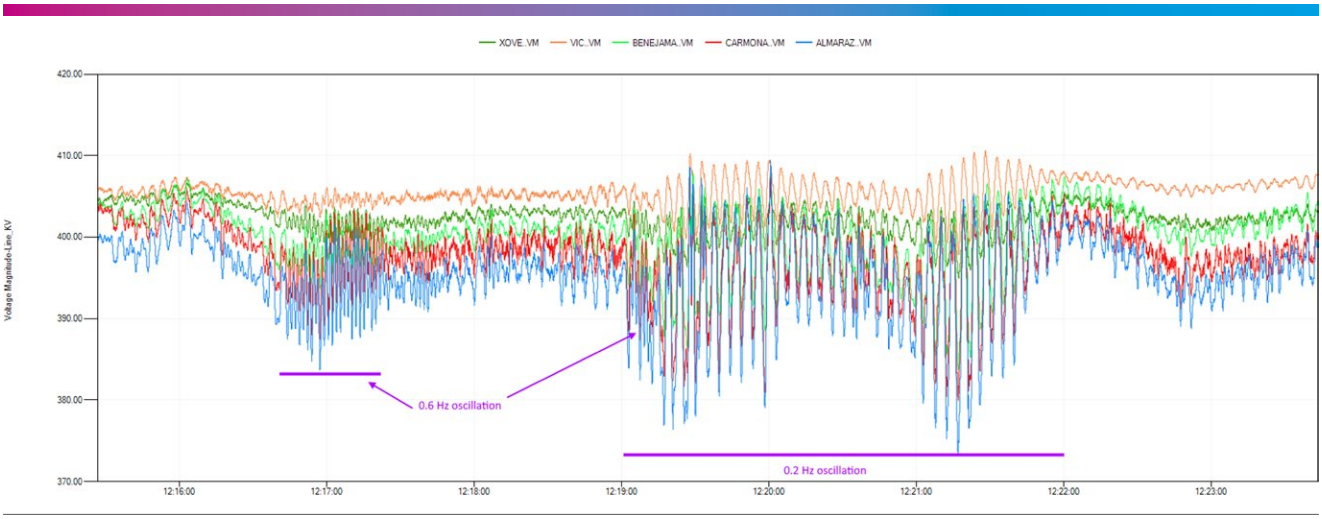


Figure 2-68: Voltage magnitudes at several Spanish substations

Figure 2-69 displays the same voltage values, but after applying a band-pass filter (0.18 to 0.25 Hz), which enhances the visibility of the oscillation amplitude at each substation. It can be seen that the voltage oscillation amplitude reaches 28 kV peak to peak at the Almaraz 400 kV substation.

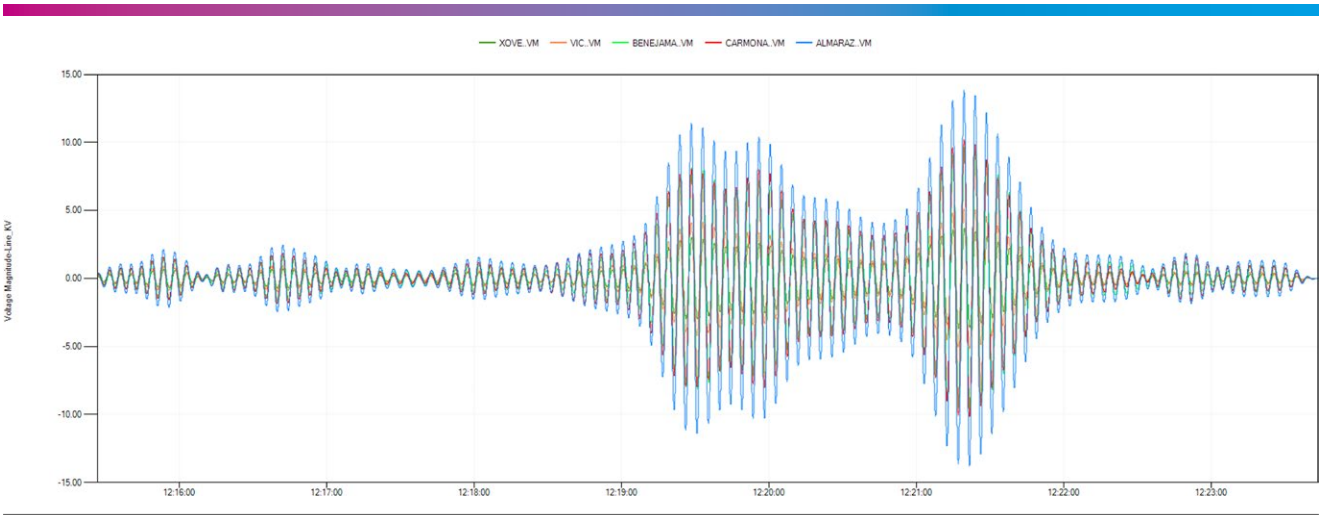


Figure 2-69: Voltage oscillations alter applying a pass-band filter [0.18 – 0.25 Hz] at several Spanish substations

2.6 Reactive Power and Voltages

2.6.1 Voltage Standards in Spain and Portugal

Voltage standards in Spain are governed by Operational Procedures 1.1, 1.2, and 1.4 from 1998. The Operational Procedure 1.4 mentions two upper voltage limits on the 400 kV grid (420 kV and 435 kV).[11]

Following letters received from two stakeholders, the Expert Panel has formally asked CNMC – the Spanish National Regulatory Authority – to clarify whether the 420 – 435 kV range is only applicable in case of a contingency situation. CNMC responded as follows:

> P.O.1.1 "Performance and safety criteria for the operation of the electrical system" sets the security criteria that must be applied in the operation of the Spanish peninsular electrical system and establish the limits for contingency situations. This procedure does not set specific voltage limits in normal operation. It refers to local (zonal) procedures that will collect nodal values determined according to the criteria established by PO1.3 "Establishment of permissible voltages at nodes in the network managed by the System Operator".
>
> The operational procedure P.O. 1.4 "Energy delivery conditions at the border points of the network managed by the System Operator" was passed simultaneously as the P.O. 1.3. This procedure P.O. 1.4 sets that under normal operating conditions, the voltage at the 400 kV level at the border points will range from 390 and 420 kV but eventually maximum levels of 435 kV may be reached. Any installation directly connected to the transmission network must be capable of withstanding these values without damage or disconnection.
>
> Therefore, although the 435 kV limit is consistent with the value regulated in operational procedure P.O.1.1 "Performance and safety criteria for the operation of the electrical system" for contingency situations, the P.O.1.4 does not only link the voltage level of 435 kV to these contingency situations, but to normal operating conditions. Hence 435 kV have to be considered in all type of operation and not only limited to contingency situations.
>
> In this regard, references in European regulations to the possibility that the Spanish TSO may operate the transmission grid at 435 kV (article 27.2 of Regulation (EU) 2017/1485, notwithstanding provisions under article 28) and may oblige generators subject to article 16.2.iii of Regulation (EU) 2016/631 to remain connected at the voltage level of 435 kV for an unlimited time, do not condition it to a contingency either."

In Portugal, voltage standards are defined in Procedure 5 of the "Manual de Procedimentos da Gestão Global do Sistema," where the upper voltage limit on the 400 kV grid is 420 kV. Note that ERSE has published a new version of the code after the 28 April 2025, with new internal numbering.

In terms of deviations from voltage standards reported in the past, RE reported thirteen scale 1 and thirteen scale 0 voltage violations in the 2024 ICS report[12] (related to 2023) and no voltage violations in the 2025 ICS report (related to 2024). REN did not report any voltage violations in 2023 and 2024.

11 In Spain, the following limits apply in accordance with the official Operational Procedure 1.4, § 3.2 (fRE translation): *"Voltages at the nodes – Under normal operating conditions, the voltage at the 400 kV level at the connection nodes will be between 390 and 420 kV. At the 220 kV level, the voltage will be between 205 and 245 kV. Eventually, maximum values of up to 435 kV and minimum values of up to 375 kV may occur at the 400 kV level. At the 220 kV level, voltages may eventually drop to 200 kV. Any installation directly connected to the transmission network must be able to withstand the above values without damage or disconnection."*

12 **https://www.entsoe.eu/network_codes/sys-ops/annual-reports/**

2.6.2 Voltage Profiles

2.6.2.1 Voltage Profiles in Spain

In this section, voltage evolution is analysed from 09:00 to 12:32. From 09:00, with the appearance of PV generation in the system, voltage variability increases, albeit without significant excursions. From 10:30, greater excursions are observed. The higher (red lines, 420/245 kV and 435 kV) and lower (green line, 390/205 kV and 375/200 kV) voltage limits mentioned in the Spanish Operational Procedure 1.4[13] from 1998 are shown in Figures 2-70 and 2-71 (based on data from the Red Eléctrica SCADA system).

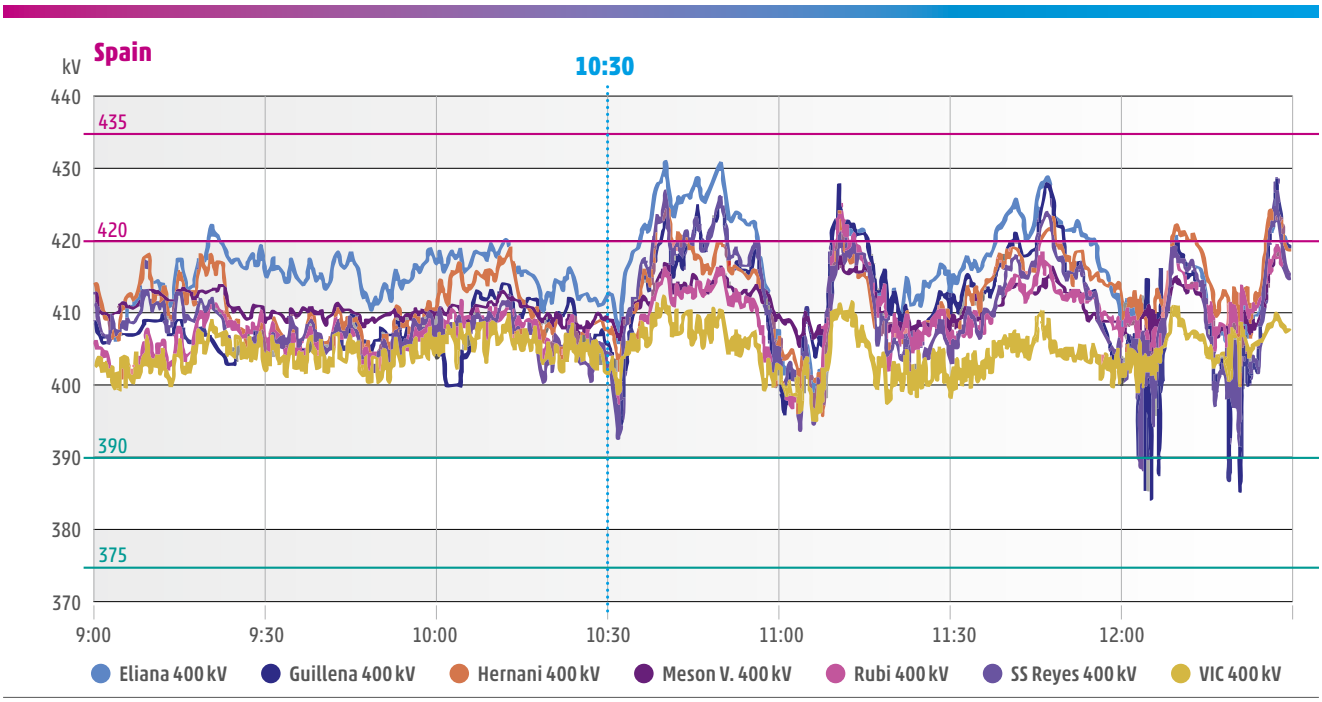


Figure 2-70: Voltage evolution at the main 400 kV transmission substations (pilot nodes) in Spain

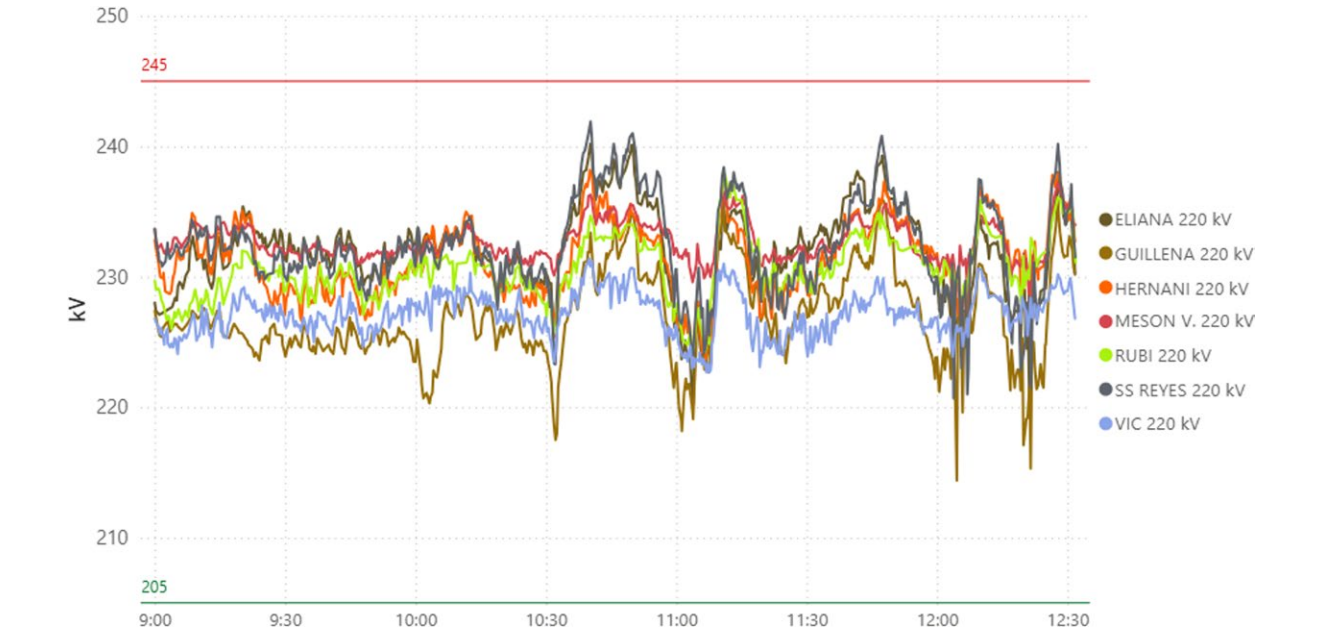


Figure 2-71: Voltage evolution at the main 220 kV transmission substations (pilot nodes) in Spain

13 https://www.boe.es/buscar/doc.php?id=BOE-A-1998-20053

In the following, the voltage profile evolution of the main transmission substations are aggregated by area.

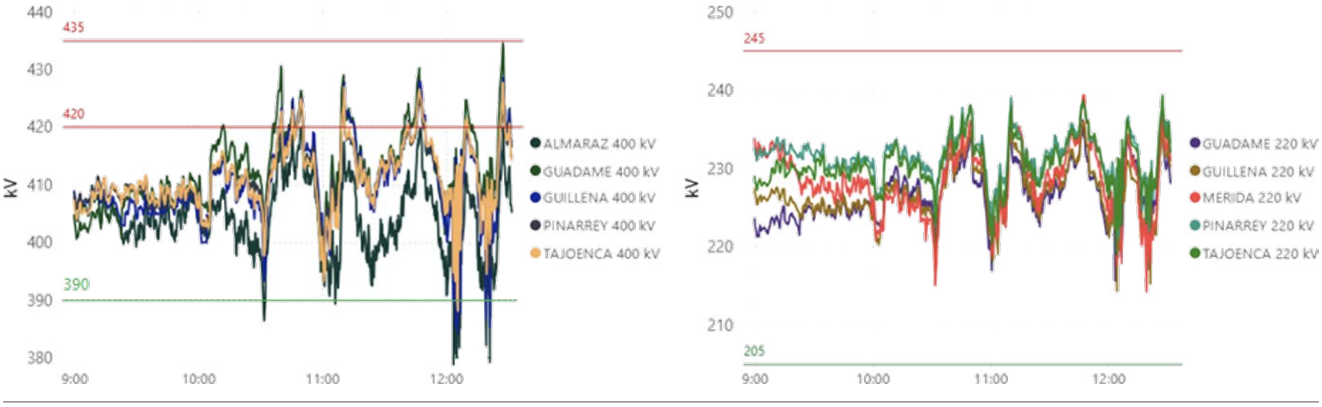


Figure 2-72: Voltage evolution at the main transmission substations (pilot nodes) in the south of Spain

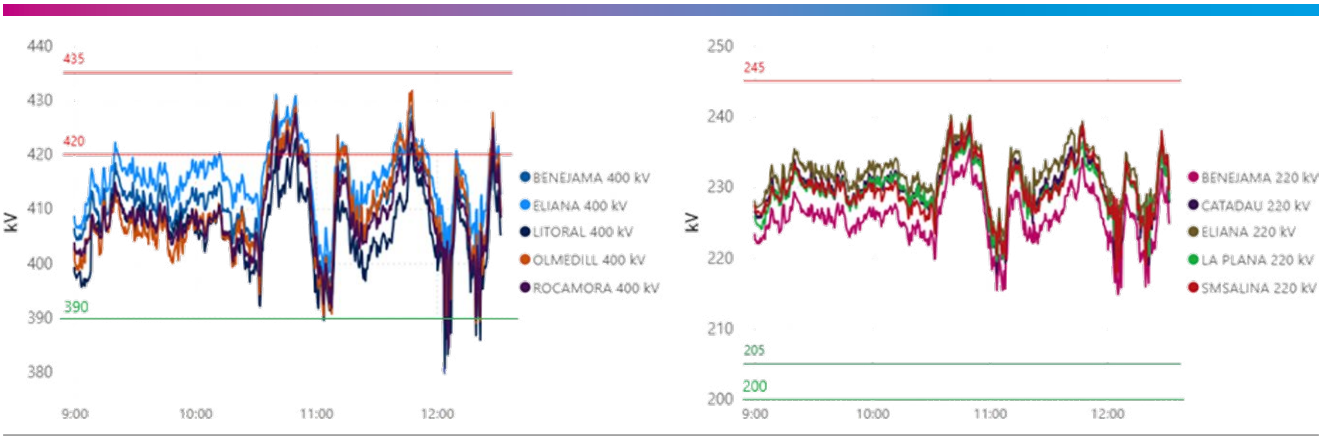


Figure 2-73: Voltage evolution at the main transmission substations (pilot nodes) in the east of Spain

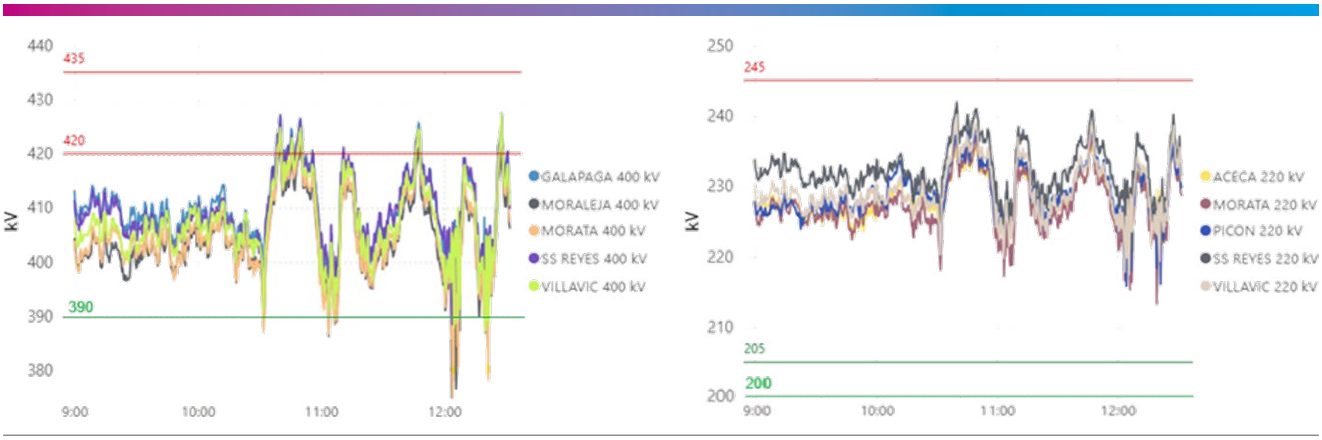


Figure 2-74: Voltage evolution at the main transmission substations (pilot nodes) in the centre of Spain

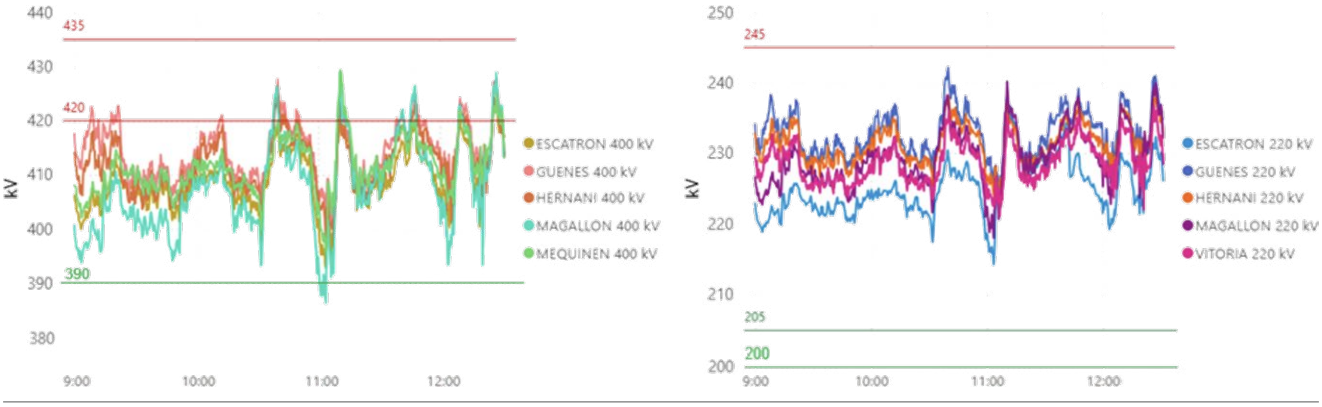


Figure 2-75: Voltage evolution at the main transmission substations (pilot nodes) in the north of Spain

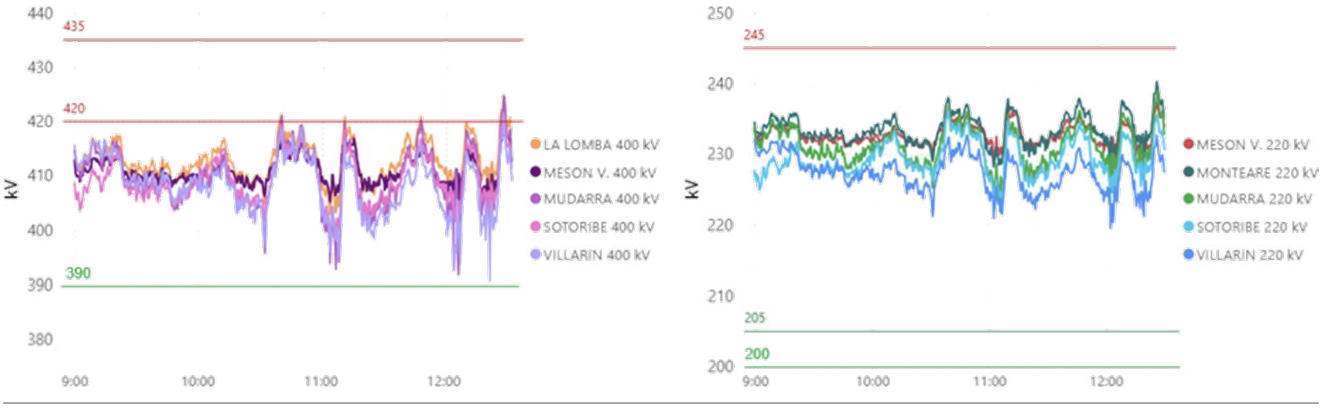


Figure 2-76: Voltage evolution at the main transmission substations (pilot nodes) in the northwest of Spain

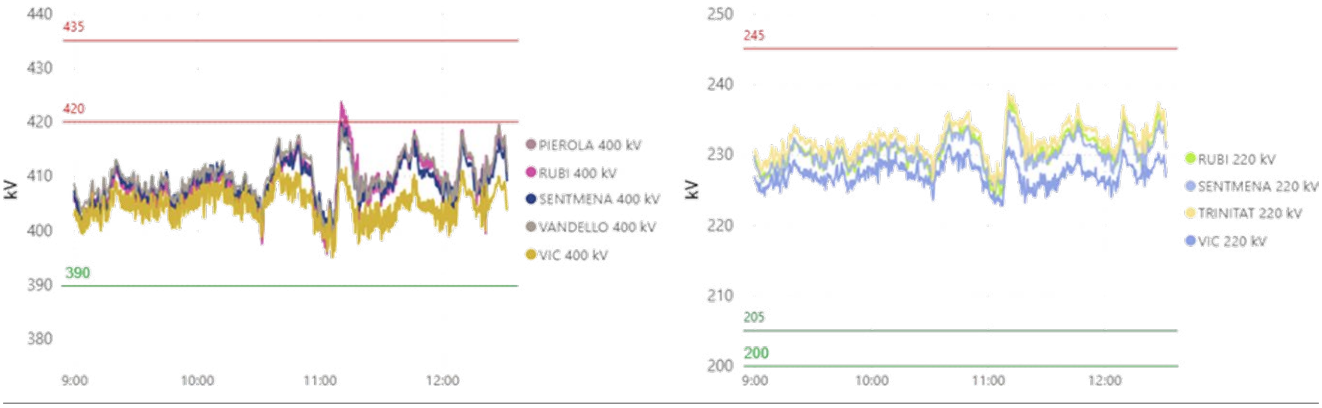


Figure 2-77: Voltage evolution at the main transmission substations (pilot nodes) in the northeast of Spain

Chapter 4.1 includes an analysis of these voltage fluctuations and considers also the voltage behaviour observed in Spain on 22 April 2025.

2.6.2.2 Voltage Profiles in Portugal

In this section, the evolution of voltage in the REN grid is analysed from 09:00 to 12:30 in the substations equipped with available PMU, namely Sines Substation (SSN) and Recarei Substation (SRR). The voltage level is within the nominal values for the observed timeframe.

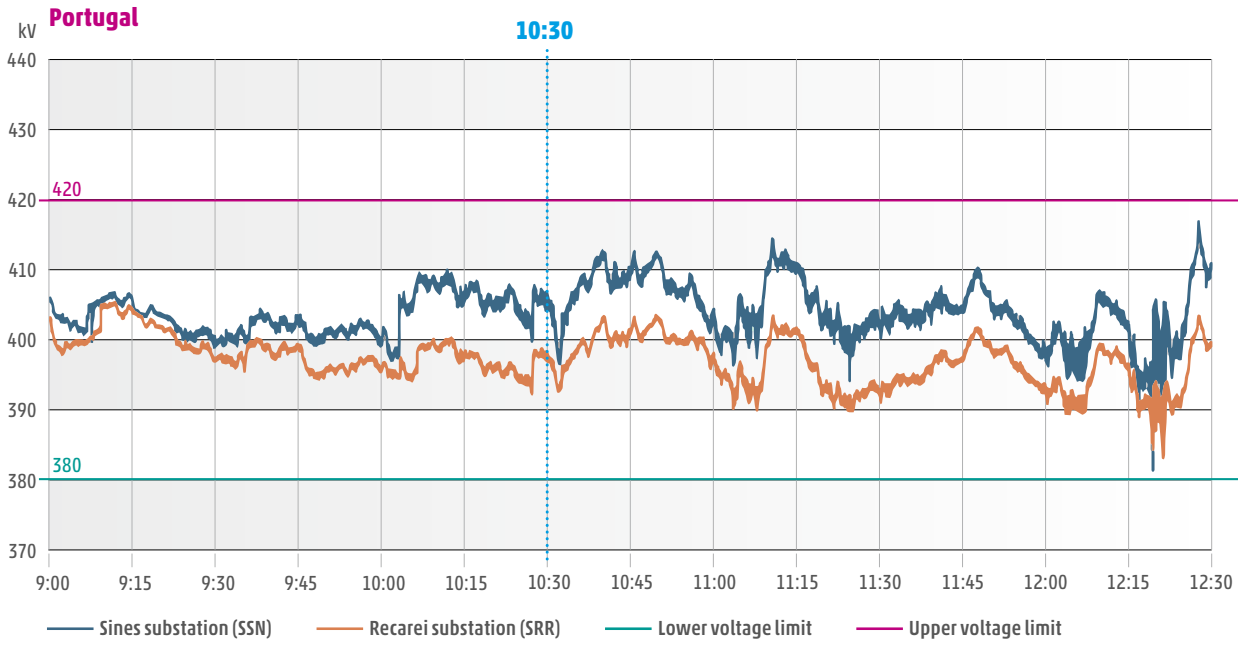


Figure 2-78: Voltage measurements at Sines and Recarei substations (Portugal)

2.6.2.3 Voltage Profiles in France

In this section, the evolution of voltage in the RTE grid is analysed from 09:00 to 12:30 at the Baixas and Saucats substations (southern part of France). The voltage level is within the nominal values for the observed timeframe.

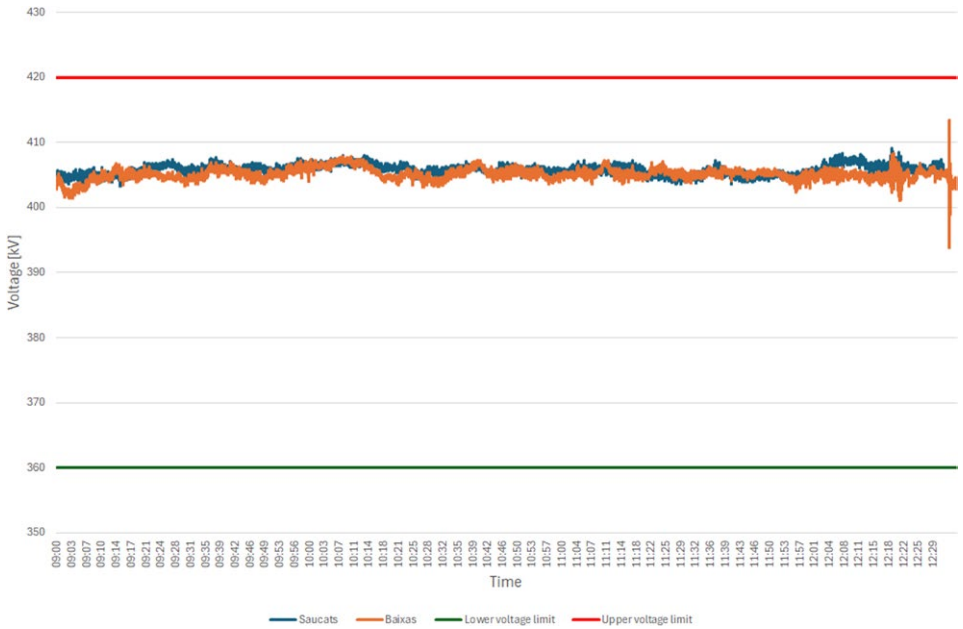


Figure 2-79: Voltage measurements at Baixas and Saucats substations (France)

2.6.3 Voltage Heatmaps of the Iberian Peninsula

In this section, the voltage heatmaps of the Iberian Peninsula are depicted every fifteen minutes from 9:00 to 12:30.

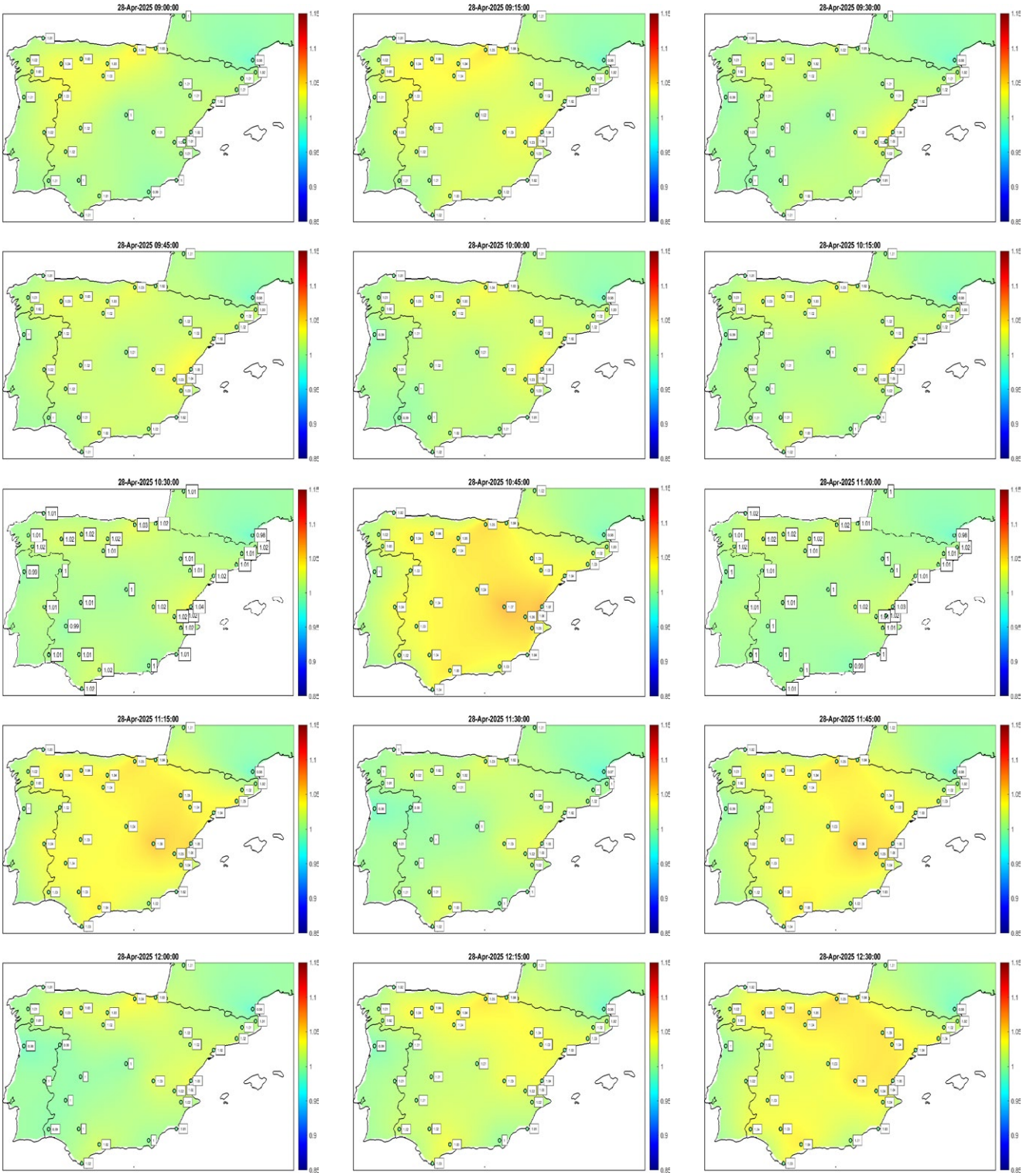


Figure 2-80: Voltage heatmaps of the Iberian Peninsula

2.6.4 Chronology of Voltage Control Actions

2.6.4.1 RE Shunt Reactors, Capacitors, STATCOMs, and HVDC Patterns

The shunt reactors connected to the 400 kV network of Red Eléctrica have 150 Mvar of reactive power capacity, and the shunt reactors/condensers connected to the 220 kV network have 100 Mvar of reactive power capacity. The shunt reactors are three-phase.

Section 2.1.2 details the effect of connecting shunt reactors and the chronology of the voltage control remedial actions. The example in Figure 2-81 shows the impact of connecting the shunt reactor in the Morata 400 kV substation at 12:28:01, showing the reactive power flow and voltage.

A summary of the shunt reactors and condensers operated by RE is provided in Table 2-7 below, together with the nominal power, location, and position at 12:32:00.

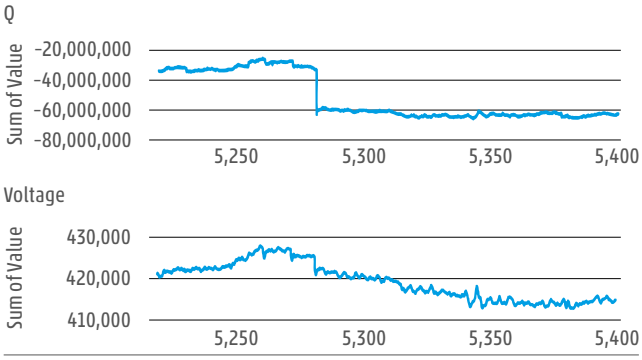


Figure 2-81: Reactive power flow and voltage at PMU MOR4MOT from 12:27:00 to 12:30:00 (x-axis is in seconds in the reference timeframe of the PMU)

SHUNT REACTORS and CAPACITORS	Nominal power (Mvar)	Location	Position at 12:32:00
ALMARAZ 400 REA 3	150	SOUTH	Disconnected
ANCHUELO REA 1	150	CENTRE	Disconnected
ARAGON 400 REA 1	150	NORTH	Disconnected
ARANUELO 400 REA 1	150	SOUTH	Disconnected
BEGUES 400 REA 1	150	NORTH-EAST	Disconnected
BELINCHON 400 REA 1	150	EAST	Disconnected
BIENVENIDA 400 REA 1	150	SOUTH	Disconnected
BROVALES 400 REA 1	150	SOUTH	Disconnected
CABRA 400 REA 1	150	SOUTH	Disconnected
DRODRIGO400 REA 1	150	SOUTH	Disconnected
EALMARAZ 220 REA 1	100	SOUTH	Disconnected
ELIANA 220 REA 1	100	EAST	Connected
ESCATRON 220 REA 1	100	NORTH	Connected
GUADAME 220 REA 3	100	SOUTH	Connected
GUADAME 400 REA 2	150	SOUTH	Connected
GUILLENA 400 REA 2	150	SOUTH	Disconnected
JM.ORIOL 220 REA 1	100	SOUTH	Disconnected
JM.ORIOL 400 REA 2	150	SOUTH	Disconnected
JUIA 220 CONDEN1	150	NORTH-EAST	Disconnected
LA SERNA 400 REA 2	150	NORTH	Connected
LITORAL 400 REA 1	150	EAST	Disconnected
MAGALLON 400 REA 1	150	NORTH	Disconnected
MAGALLON 400 REA 2	150	NORTH	Disconnected
MAIALS 400 REA 1	150	NORTH-EAST	Connected
MINGLANILLA 400 REA 1	150	EAST	Connected
MORALEJA 220 REA 12	100	CENTRE	Connected
MORALEJA 220 REA 13	100	CENTRE	Disconnected
MORALEJA 400 REA 1	150	CENTRE	Disconnected
MORATA 400 REA 4	150	CENTRE	Connected
OLMEDILLA 400 REA 1	150	EAST	Disconnected
PALOS 220 REA 1	100	SOUTH	Disconnected
PEÑAFLOR 400 REA 1	150	NORTH	Connected
PINILLA 400 REA 1	150	EAST	Disconnected
REQUENA 400 REA 1	150	EAST	Disconnected
ROCAMORA 400 REA 1	150	EAST	Disconnected
RUBI 400 REA 1	150	NORTH-EAST	Connected
RUEDA 400 REA 2	150	NORTH	Disconnected
SENTMENAT 400 REA 1	150	NORTH-EAST	Connected
SS REYES 400 REA 3	150	CENTRE	Disconnected
VALDECAB 400 REA 1	150	SOUTH	Disconnected
VALDECABALLEROS 400 REA 2	150	SOUTH	Disconnected
VILLAVICIOSA 220 REA 2	100	CENTRE	Disconnected
VILLAVICIOSA 400 REA 1	150	CENTRE	Disconnected
VITORIA 400 REA 2	150	NORTH	Connected

Table 2-7: Main characteristics and status of shunt reactors in RE network

2.6.4.2 REN Shunt Reactor Patterns

In this section, the main characteristics related to shunt reactors of REN are described. The shunt reactors are three-phase.

Substation	Site ID	SR ID	Voltage Level kV	Nominal Power Mvar	Position at 12:32:00
ARMAMAR	SAMM	R1	400 kV	150	Disconnected
CASTELO BRANCO	SCC	R1	220 kV	70	Disconnected
FANHÕES	SFN	R1	400 kV	150	Disconnected
FEIRA	SFRA	R1	400 kV	150	Disconnected
PEDRALVA	SPDV	R1	400 kV	150	Disconnected
PARAIMO	SPI	R1	400 kV	150	Disconnected
PALMELA	SPM	R2	400 kV	150	Disconnected
PORTIMÃO	SPO	R1	400 kV	150	Disconnected
RIO MAIOR	SRM	R1	400 kV	150	Connected
TÁBUA	STBA	R1	220 kV	70	Connected
TAVIRA	STVR	R1	150 kV	75	Connected

Table 2-8: Main characteristics of the shunt reactors in REN data

Section 2.1.2 details the chronology of manoeuvres of shunt reactors by REN. The shunt reactors were gradually disconnected by the control room operator as a normal operation during the observed timeframe, and at 12:19, the Palmela shunt reactor tripped due to undervoltage (the registered voltage value was 379.80 kV, and the protection setting was U = 380 kV and t = 2 s). After the system collapse at 12:33:30, the three remaining shunt reactors of Tabua, Tavira, and Rio Maior were also disconnected. Figure 2-82 depicts the gradual decrease in the amount of inductive reactive power provided by REN's shunt reactors.

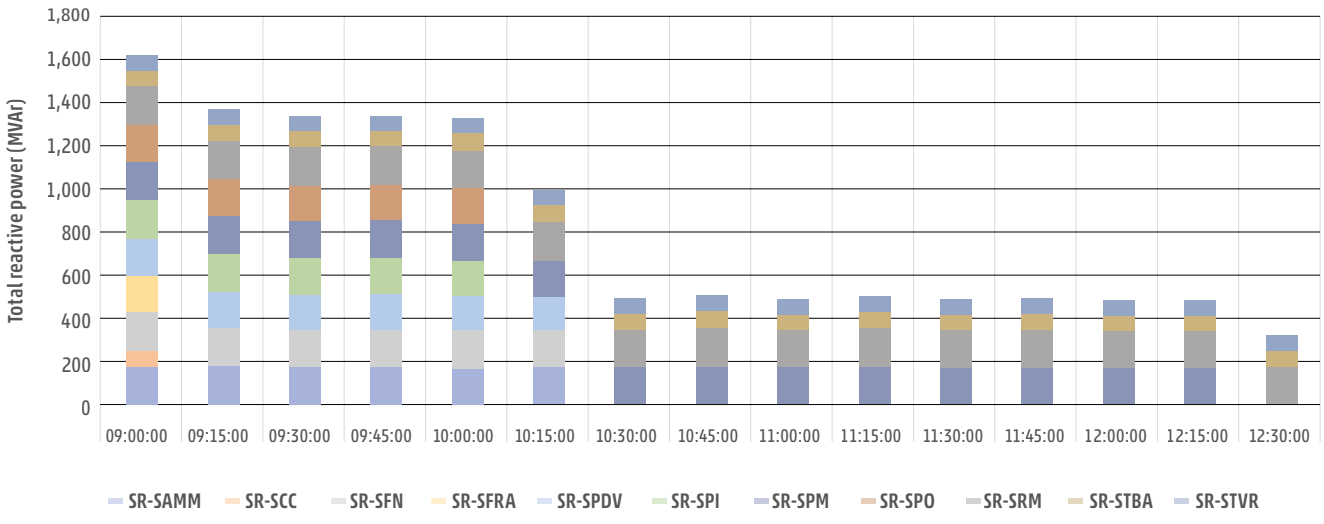


Figure 2-82: Trend of the total reactive power of the shunt reactors in the REN grid

2.6.5 Voltage Regulation Schemes

2.6.5.1 Voltage Regulation Scheme in Spain

In Spain, the voltage regulation at the time of the blackout was governed by the Operating Procedure 7.4 from March 2000, which was subsequently **amended** in June 2025.

Operating Procedure (P.O.) 7.4[14] **"Ancillary Voltage Control Service of the Transmission Network"** applies to conventional generators with an installed capacity equal to or greater than 30 MW connected to buses of the transmission network, transmission operator, consumers connected to the transmission grid with a contracted power equal to or greater than 15 MW, and DSOs.

Below, some literal extracts from Operational Procedure 7.4 are included:

*"**3. Definition***

Voltage control consists of a set of actions involving resources for the generation and absorption of reactive power (generators, reactors, capacitors, etc.) and other voltage control elements, such as transformers with tap changers. These actions are aimed at maintaining voltage levels at the nodes of the transmission network within specified margins to ensure compliance with safety and quality criteria for electricity supply."

*"**4. Service providers***

The service providers shall be:

a) All generating units operating under the ordinary regime, with a registered net capacity equal to or greater than 30 MW and directly connected, or connected through a dedicated evacuation line, to nodes of the transmission network. [...]

b) Transmission companies.

c) Qualified consumers not covered by a tariff (1), directly connected or connected through a dedicated line to nodes of the transmission network (hereinafter referred to as "service-providing consumers"), with a contracted power equal to or greater than 15 MW.

d) Distribution System Operators [...]"

*"**6. Service provision***

[...]

6.1 Mandatory Requirements

As a technical condition for connection to the transmission network, and to ensure proper operation and system security, providers of this ancillary service must deliver the following minimum services:

6.1.1 Generators

Generators must have a mandatory minimum margin of reactive power capacity, both for generation and absorption, to provide the service. They must adjust their reactive power production and absorption within these limits to help maintain voltage levels at the plant busbars within the variation margins defined by the voltage setpoint and the acceptable variation band established by the System Operator.

For generators, the required minimum reactive power margin at plant busbars at nominal transmission network voltage is defined based on the installed net active power, as recorded in the Administrative Register of Electricity Production Facilities, and the following power factor values:

a) Capacitive power factor (cos φ) of 0.989 (reactive power generation equivalent to 15 % of maximum net active power).

b) Inductive power factor (cos φ) of 0.989 (reactive power absorption equivalent to 15 % of maximum net active power).

14 https://www.boe.es/buscar/doc.php?id=BOE-A-2000-5204

This reactive power generation/absorption margin must be deliverable by the unit across the entire range of active power variation, from its technical minimum to its maximum net active power.

These requirements will vary depending on the voltage value at the corresponding node of the transmission network, according to the linear function graphically shown in Annex 6 […]."

"***10. Measurement and Monitoring of Service Compliance***

To monitor compliance with the service, the System Operator will use telemetry data received through the CECOEL real-time energy control system. […]

10.1 Generators

The System Operator will perform sampling every five minutes of the voltage values at the control node and the active and reactive power generated/absorbed by the unit at the plant busbars.

To assess service compliance, an acceptable deviation band of ±2.5 kV around the voltage setpoint established by the System Operator for the control node is defined.

The service will be considered properly delivered when at least 75 % of the sampled values within each hour meet one of the following two conditions:

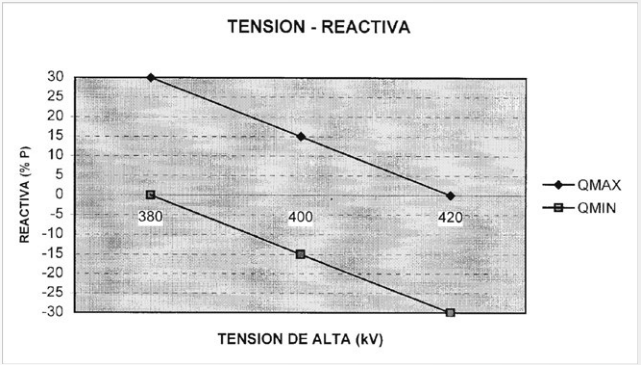
a) *The voltage at the control node assigned to the unit remains within the acceptable variation margins.*
b) *The unit has reached the mandatory reactive power limit or, where applicable, the mandatory limit plus the additional assigned resources, in the appropriate direction.*

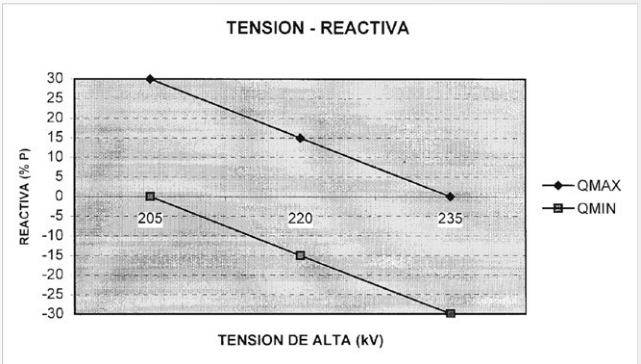
To verify this, it will be checked that the voltage telemetry from the control node – or, if unavailable, the values resulting from state estimation – are within the acceptable band (±2.5 kV around the voltage setpoint established by the System Operator for the control node) in at least 75 % of the sampled values within each hour. If so, the service will be considered properly delivered.

If the voltage has been outside the acceptable band in more than 25 % of the sampled values during the hour, the active and reactive power values at the plant busbars will be analysed. For each set of active power and busbar voltage values, the reactive power limit that the unit should have delivered or absorbed in that situation will be determined, taking into account both the mandatory minimum requirements and, where applicable, the additional assigned resources.

In this latter case, even if the voltage setpoint for the corresponding control node (plant busbars) was not met, as long as the unit has reached the corresponding reactive power limit (mandatory requirements + assigned additional resources) in that situation, in at least 75 % of the samples taken during each hour in which the voltage was out of limits, the service will be considered properly delivered. […]"

ANNEX 6: Variation of Mandatory Requirements for Units Based on the Voltage of the Node of the Transmission Network





Following a presentation shared by a stakeholder, the Expert Panel formally asked CNMC – the Spanish National Regulatory Authority – to explain the figure above ("Annex 6"). More specifically, the Expert Panel asked which of the following two options is correct:

» Option 1
- <405 kV: Q has to be at least Qmax (Qmax as minimum generation of reactive power).
- 405 – 410 kV: No requirement.
- >410 kV: Q has to be at least Qmin (Qmin as minimum absorption of reactive power).

» Option 2
- Q has to be within the range of Qmin and Qmax.

CNMC responded as follows:

"OP 7.4 defines a minimum mandatory capability of reactive power control expressed as a margin both in generation and absorption for the provision of the service. Generators must be able to modify their production and absorption of reactive power within these limits (Qmax, Qmin), so that they contribute to maintain the voltage within the variation margins defined by the voltage setpoint value (400 kV) and the permissible variation band around it established by the System Operator (405 – 410 kV).

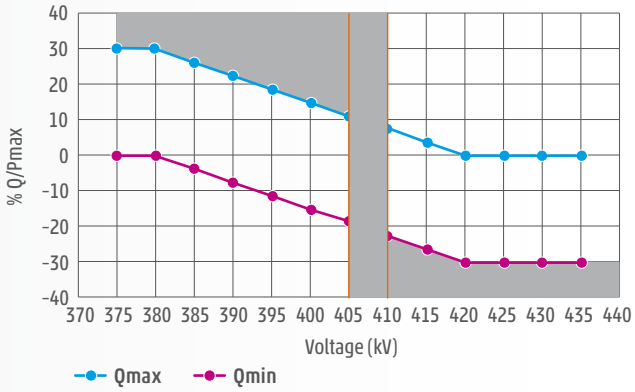
The terms of the service provision are complemented in article 10 of P.O. 7.4. According to article 10.1 of the P.O. 7.4, the service is considered to have been provided adequately if at least 75 % of the values sampled each hour meet one of these two conditions:

a) The voltage is maintained within the permissible variation margins (405 – 410 kV for the voltage level of 400 kV).

b) The group has reached the mandatory reactive power limit or, where applicable, the mandatory limit plus the additional limit assigned, in the appropriate direction.

Based on the above, for an adequate compliance with the requirements of the service, when the voltage level is under 405 kV generators must generate a reactive power of at least the generation limit of its reactive capacity (the appropriate direction in this case), when the voltage level is over 410 kV they must absorb at least the absortion limit of its reactive capacity (the appropriate direction in this case), and there is no special requirement of reactive power when the level voltage is between 405 – 410 kV.

Grey area in the figure below shows the right area, that is, Option 1 in your question, without prejudice that compliance allows a time margin of 25 % outside that area."



Document: "Voltage Setpoints in the Transmission Network" (PCT-0-006: submitted to voltage control service providers in December 2011):

Below some literal extracts from PCT-0-006 are included:

"ANNEX 2:

The purpose of this document is to publish the voltage setpoints applicable for the provision of the Voltage Control Service on the transmission network, which will be in effect from January 1, 2012, until further notice from Red Eléctrica.

GENERAL SETPOINTS

The voltage setpoints to be used as reference[15] during the different peak, flat, and off-peak periods are as follows:"

400 kV		
PEAK	**FLAT**	**OFF-PEAK**
405 - 410	405 - 410	405 - 410

220 kV		
PEAK	**FLAT**	**OFF-PEAK**
225 - 230	225 - 230	225 - 230

[...]"

Royal Decree 413/2014 *applies to electricity generation facilities using renewable energy sources, cogeneration, and waste-to-energy (RCR). Section 7 of the regulation establishes the following mandatory voltage control requirements:*

1. *Facilities must maintain a power factor within the range of 0.98 inductive to 0.98 capacitive on an hourly basis. Accordingly, they must inject or withdraw reactive power depending on their active power output within this range, which can be modified annually by resolution of the Secretary of State for Energy, upon proposal by the TSO according to system needs[16].*
2. *Facilities with an installed capacity equal to or greater than 5 MW must follow the instructions issued by the TSO to adjust their power factor within the established range, based on system requirements. The TSO gives inductive power factor instructions by email to these facilities so that they absorb as much reactive power as possible, respecting the maximum inductive limit of 0.98 set by the Royal Decree, which corresponds to 20 % reactive power relative to the active power generated. The TSO can update its instructions when necessary and requests the facility owner to implement the change within a few days.*
3. *In cases where the facility is connected to the distribution network, any modification to the power factor range must take into account the limitations that might be established by the distribution system operator for the safety of its network. For this purpose, the distribution network operator may propose specific instructions to the TSO that must be considered.*

It is worth noting that Royal Decree 413/2014 includes a penalty of 0.261 c€/kWh (2.61 €/MWh) for non-compliance with the hourly obligations established for RCR generators.

Article 9, Section 5 of **CNMC Circular 3/2020** *– which establishes the methodology for calculating electricity transmission and distribution tariffs – includes a billing term for reactive energy applicable to all consumers, except those connected at low voltage with a contracted power of 15 kW or less.*

The reactive power billing term applies to all time periods except period 6, provided that reactive energy consumption exceeds 33 % of active energy consumption during the billing period (i.e., power factor ≤ 0.95) and only affects such excesses. During period 6, consumers must maintain a power factor greater than 0.98 capacitive, although no penalty is currently associated with non-compliance for this period.

15 There is a specific setpoint applicable to Sabón and Meirama.

16 This has never happened

2.6.5.2 Voltage Regulation Scheme in Portugal

The voltage regulation in the Portuguese transmission system is nodal.

Synchronous power plants (hydro and thermal) can provide reactive power compensation and/or voltage control. Hydro power plants provide this service not only in generation mode but also in pumping/synchronous compensator operation mode, when available. The (voltage and reactive) setpoints are communicated by phone by the TSO. The synchronous generators are obliged to deliver reactive power in order to keep voltage/reactive power at the setpoint defined by the TSO, within the restrictions defined by a P-Q area and as long as the power factor remains higher than 0.90 inductive or 0.95 capacitive. As an exception, one hydro power plant is contracted to provide reactive power control even when not producing active power – thus, it functions as a synchronous compensator. This service uses a reactive power setpoint.

The old wind power plants provide reactive power compensation (predefined tan(φ) between –0.2 and 0.2, which that means the ratio between reactive and active power outputs is predefined).

Solar power plants with an installed capacity larger than 1 MW (as do new wind power plants) provide automatic voltage/reactive power control, receiving real-time setpoints sent by REN's SCADA for voltage control or reactive control (the voltage control setpoint is the most commonly used). Note that service is provided during the daylight period and in night operation mode.

REN uses the measure of the connection point of the power plant as a reference. The power plants have a closed-loop voltage regulation on the high voltage side.

2.6.5.3 Voltage Regulation Scheme in France

Figure 2-83 provides an overview of the voltage regulation system in France. The reactive power requirement of each control zone – calculated by the secondary voltage regulator (SVR) – is expressed as a per-unit value K, ranging from –1 to +1. This value is transmitted from the control centre to the local regulating units. The communication delay of approximately ten seconds is determined by the SCADA system's sampling rate.

In accordance with French grid connection requirements, all generation units with an installed capacity above 50 MW are mandated (with remuneration) to participate in the SVR scheme. This ensures a broad and distributed contribution to voltage regulation across the transmission network.

At the unit level, a reactive power control loop (RPCL) receives the K value from the SVR and computes a voltage reference Usref (t), which is then sent to the automatic voltage regulator (AVR). The RPCL can be modelled as a proportional-integral (PI) controller, which regulates the generator's terminal reactive power output Qs(t) to follow a time-varying reference Qcs(t), defined as:

$$Q_{cs}(t) = K(t) \times Q_r(t)$$

Here, Qr(t) represents the unit's maximum reactive power capability (either injection or absorption) at a given time, acting as a participation factor.

For thermal units, Qr is typically considered constant and proportional to the nominal reactive power of the alternator (e.g., Qr = 1.4 × Qnom). However, in practice, the actual reactive power capability can vary depending on the unit's operating conditions, such as terminal voltage and active power output. This discrepancy can lead to situations where the reactive power output Qs(t) does not reach the theoretical maximum even when K = ±1, due to physical limitations such as rotor current or internal angle constraints.

In terms of dynamic response, each control block in Figure 2-83 operates on a different time scale:

- The AVR is the fastest, with a response time in the order of seconds.
- The RPCL follows, with a typical response time of approximately 10 seconds.
- The SVR is the slowest, with a response time on the order of 100 seconds.

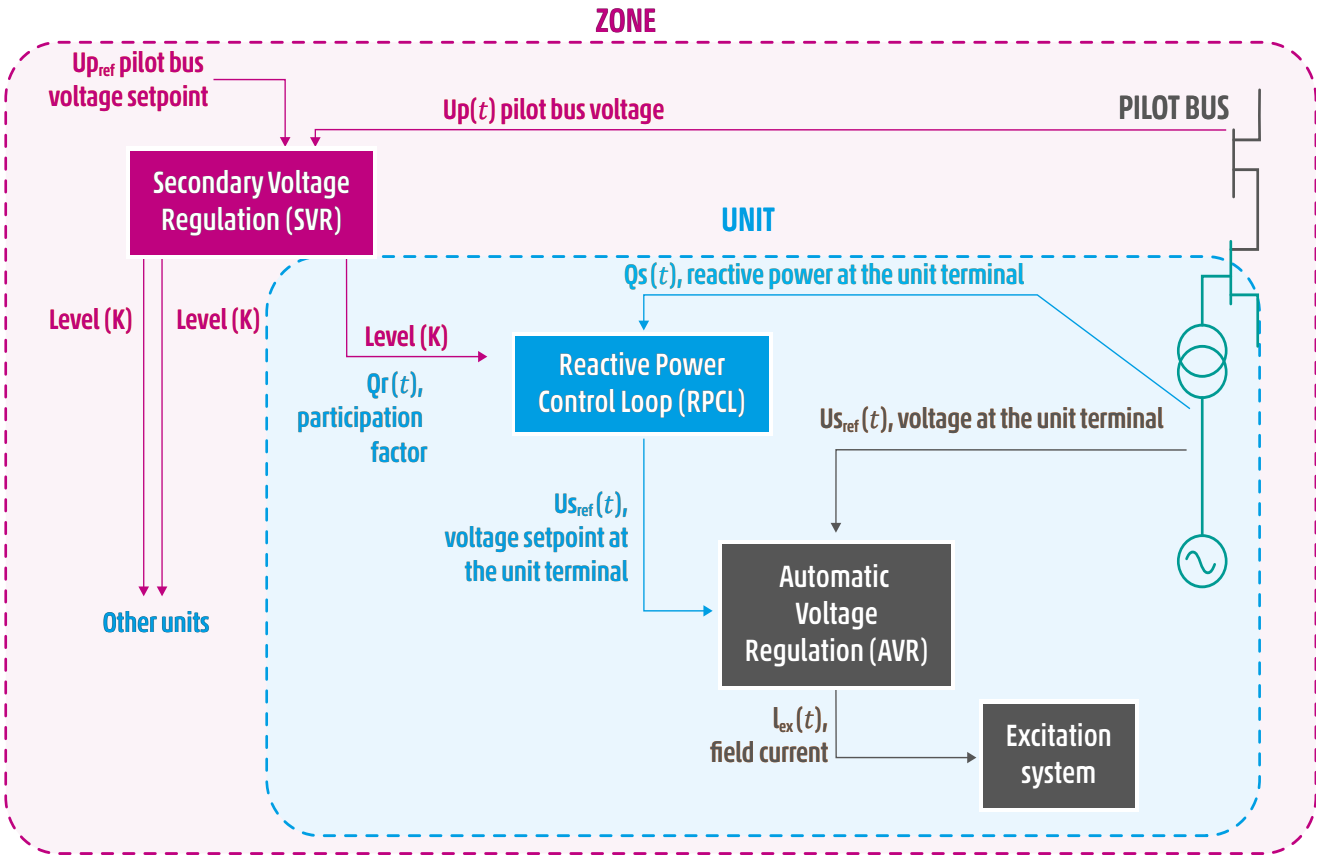


Figure 2-83: SVR principle diagram

2.6.5.4 Voltage Regulation by the HVDC Baixas Santa Llogaia

See Section 2.7.

2.6.6 Aggregated Data from Power Plants Connected to the Spanish Transmission Network

Figures 2-84 – 2-86 show per region, the aggregation of the reactive power provided by all conventional synchronous generation units with an installed capacity larger than 100 MW (Q), as well as the aggregation of their reference reactive power (Q-reference), where Q-reference corresponds to the minimum reactive power that a generation unit must generate or absorb, based on CNMC's explanation of Operating Procedure 7.4 applicable at the time of the incident (see Section 2.6.5.1). This is without prejudice to the fact that it is permissible, under the Operating Procedure applicable in Spain at the time of the incident, for each unit not to comply with this requirement up to 25 % of the time per hour, and without prejudice to any potential derogations from the requirements, which the Expert Panel is not competent to assess.

For calculation of the aggregated values of Q-reference the individual Q-reference value of a generator was set equal to its actual Q value in periods when voltage was within the permissible variation band (405 – 410 kV) or the generator was in a direct voltage setpoint control mode (pilot project). When the measured voltage crosses the edge of the variation band in either direction, this is reflected in step changes in Q-reference.

Further observations and analysis on the behaviour of conventional synchronous generators regarding reactive power output is described in Chapter 4.1.

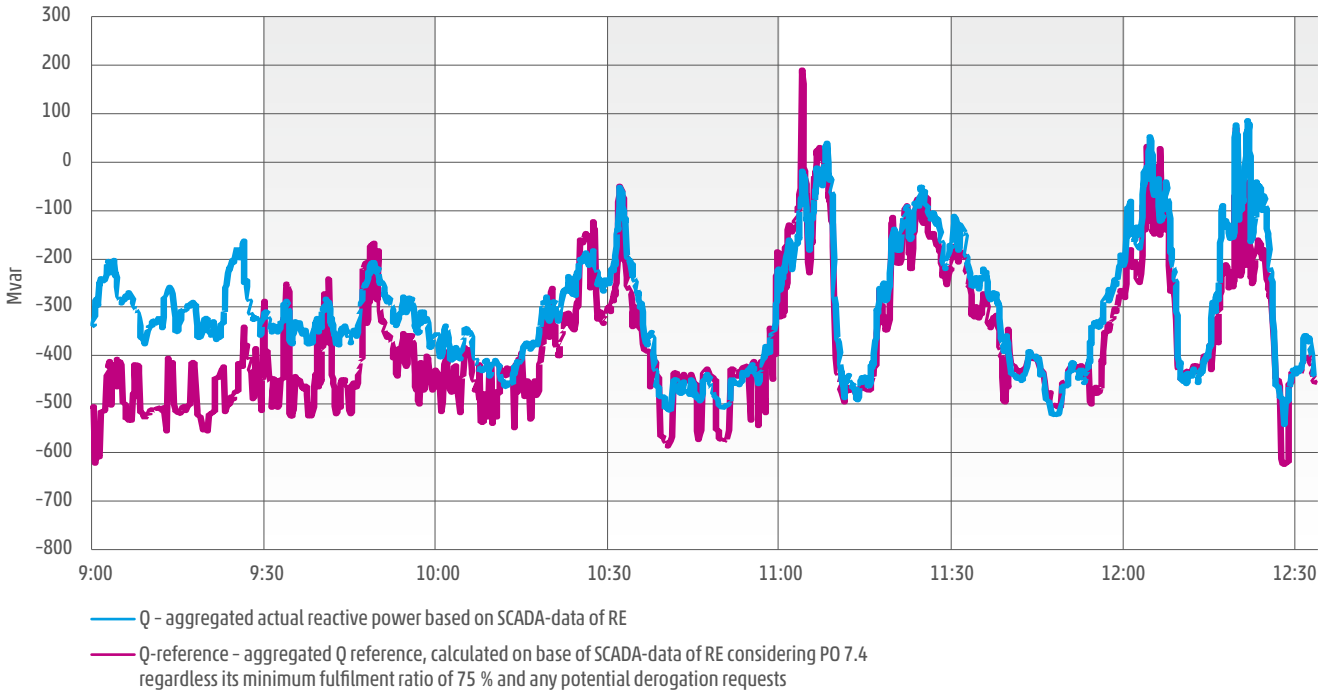


Figure 2-84: Reactive power provided (Q) and the reference reactive power (Q-reference) aggregated for conventional generation units larger than 100 MW of power installed capacity in the north/north-west area of Spain, based on SCADA data of RE

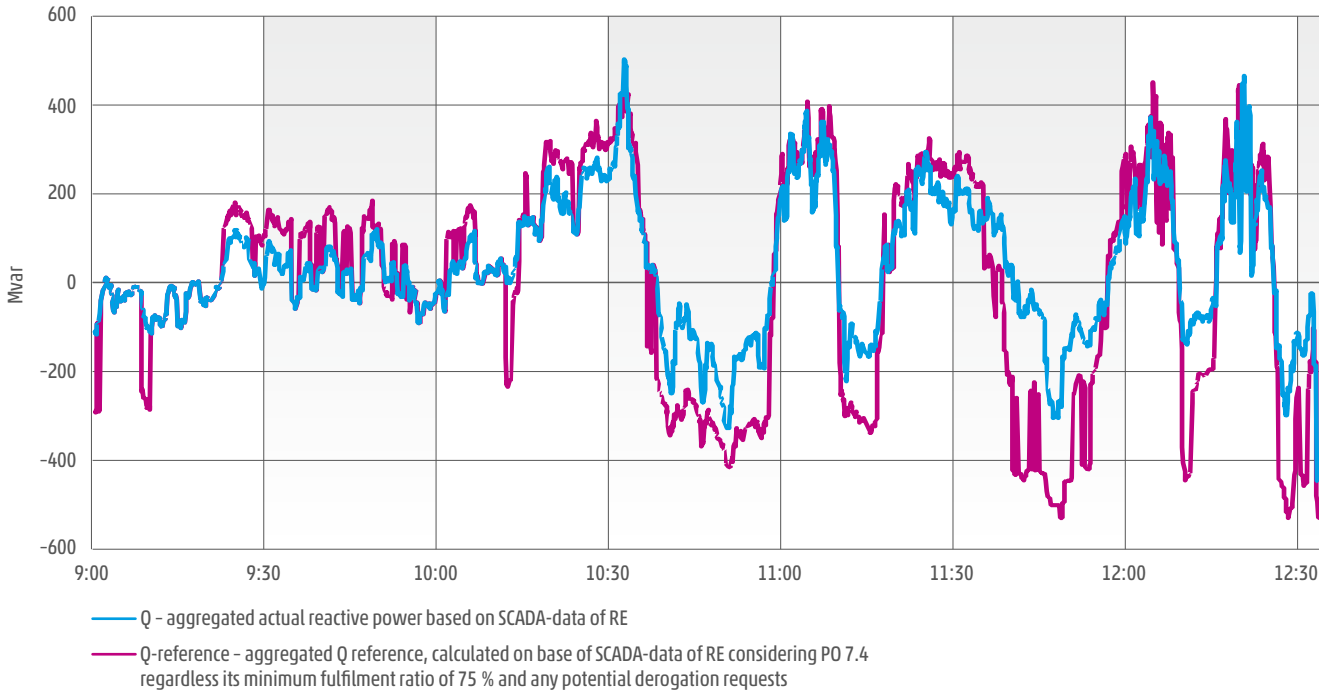


Figure 2-85: Reactive power provided (Q) and the reference reactive power (Q-reference) aggregated for conventional generation units larger than 100 MW of power installed capacity in the centre/south-west area of Spain, based on SCADA data of RE

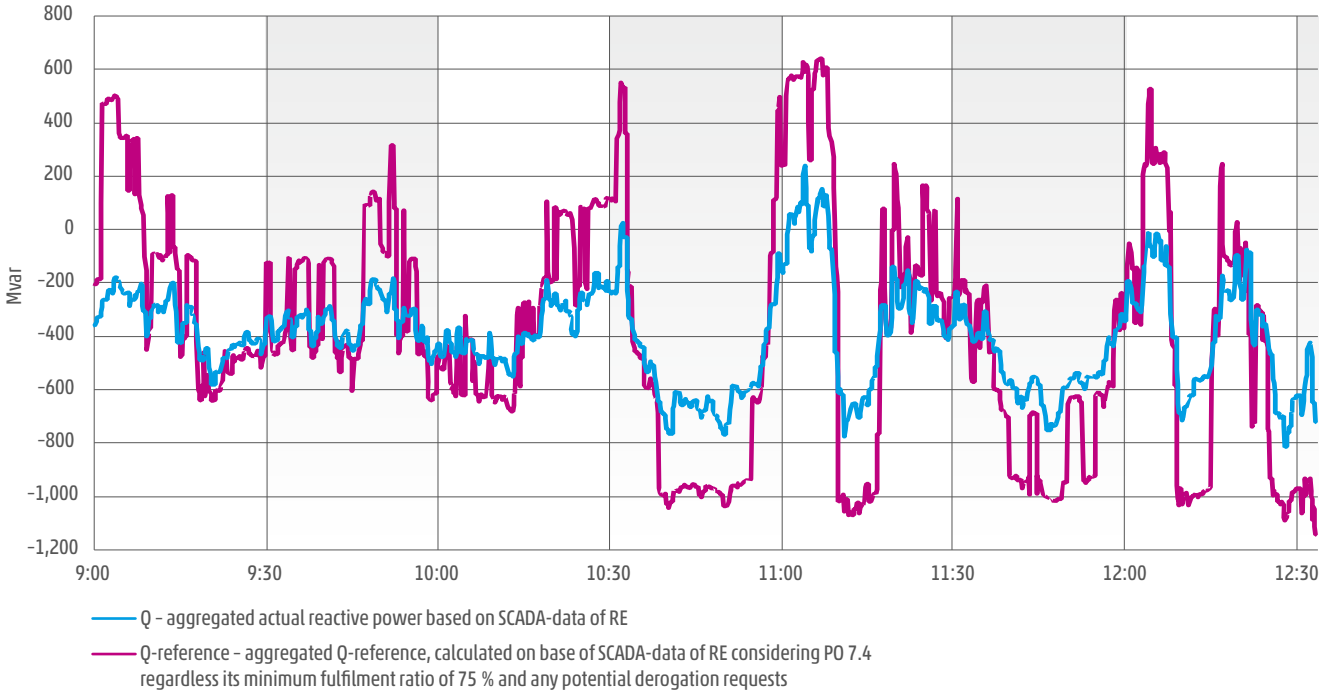


Figure 2-86: Reactive power provided (Q) and the reference reactive power (Q-reference) aggregated for conventional generation units larger than 100 MW of power installed capacity in the east/north-east area of Spain, based on SCADA data of RE

Figures 2-87 and 2-88 show the aggregation of the active and reactive power provided respectively by all PV and wind units >1 MW in Spain.

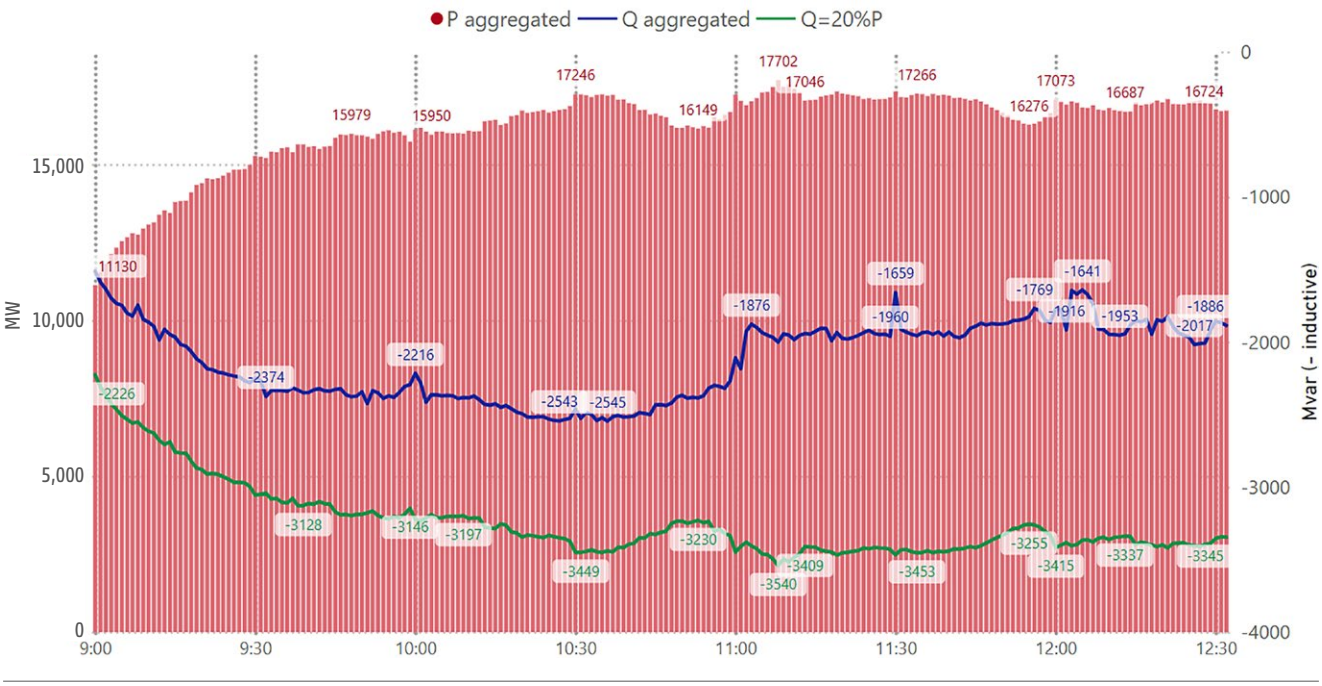


Figure 2-87: Aggregated P and Q of PV generation (facilities or aggregations with installed capacity >1 MW) , based on SCADA data of RE. The green line corresponds to a 0.98 inductive power factor, which is the lowest limit of the generic range defined by the RD 413/2014.

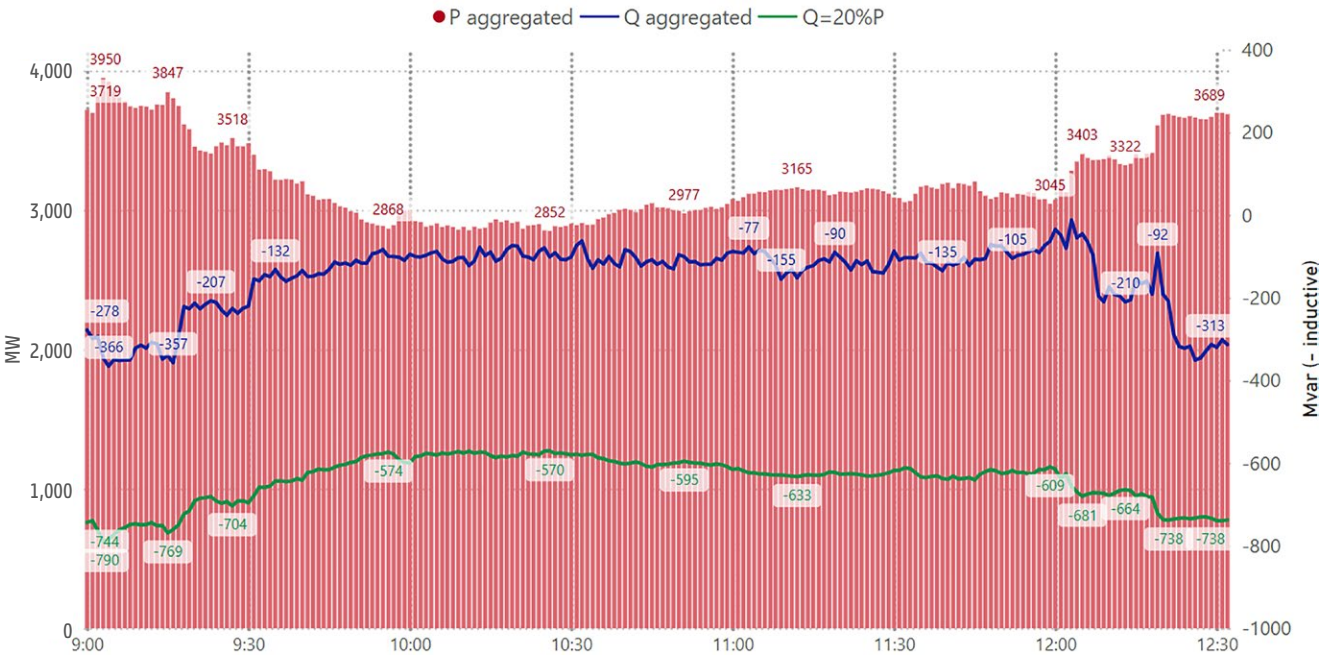


Figure 2-88: Aggregated P and Q of wind generation (facilities or aggregations with installed capacity >1 MW) , based on SCADA data of RE. The green line corresponds to a 0.98 inductive power factor, which is the lowest limit of the generic range defined by the RD 413/2014.

2.6.7 Data from Relevant Power Plants Connected to the Portuguese Transmission Network

Figure 2-89 shows the aggregation of the reactive power provided by generating power plants connected to the 400 kV voltage level with a power installed capacity of more than 100 MW in the south area of Portugal, as well as the aggregation of their reference reactive power. Q-reference is calculated taking into account the RfG requirements related to voltage control, complemented by the Portuguese framework "Portaria n.º 73/2020", which defines the reactive power envelope for each generator type. For the Q-reference, REN considers the voltage setpoint that was active at the time, and computes it assuming a 2 % slope and no deadband for the voltage characteristic.

If the voltage at the connection point is above or below 2 % of the voltage setpoint, the unit should provide the maximum reactive power. If voltage on connection point is ±2 % of voltage setpoint, Q-reference is set equal to the actual reactive power (Q), as long as it follows the defined slope

Due to different approaches to voltage control by synchronous generators in Spain and Portugal, these implementation aspects should be considered when comparing Figure 2-89 to Figures 2-84, 2-85 and 2-86.

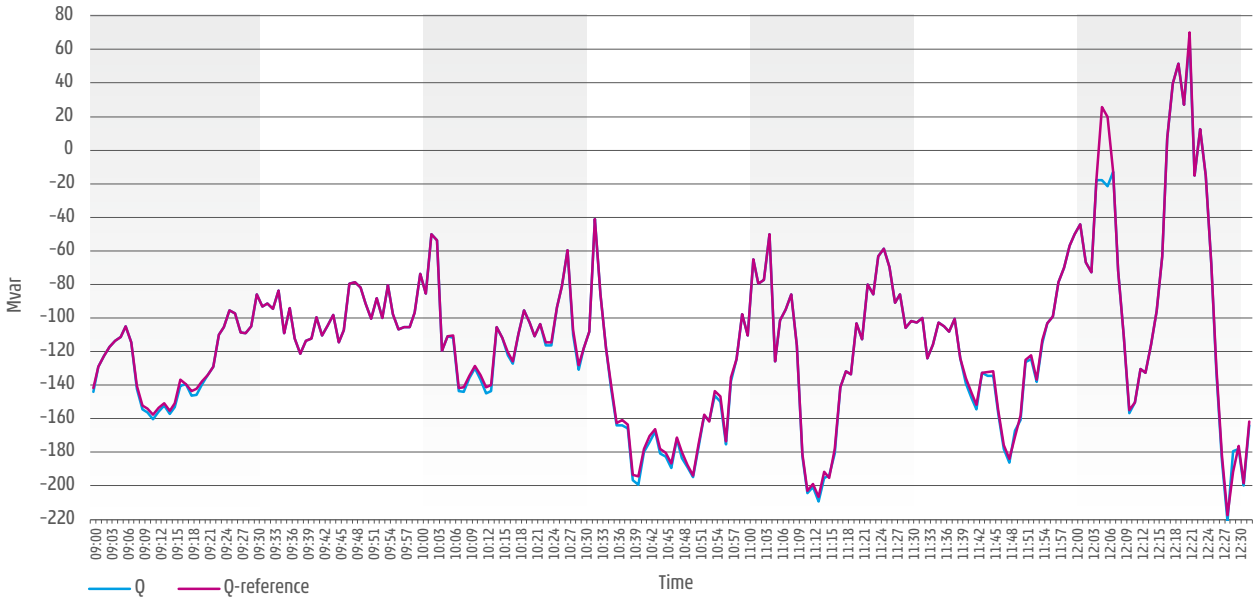


Figure 2-89: Provided reactive power (Q) and reference reactive power (Q-reference) for generating relevant power plants with more than 100 MW of power installed capacity in the south area of Portugal (REN's SCADA measurements)

2.6.8 Technical Constraints in Spain

The technical constraints solving process in Spain is described in Operational Procedure 3.2 "Technical restrictions" (the latest amendment was published on 17 March 2025 and includes adjustments for a fifteen-minute market timeframe). This process has two differentiated parts, one of which is undertaken after the day-ahead market gate closure time, and another is performed continuously in real time to solve remaining issues. The objective is to guarantee security with the minimum and more economical number of changes to the results that come out of the daily market. This process allows the Spanish TSO to introduce any change that it considers justified to guarantee the security of supply in the scheduling of the generation and storage units (and consumers on voluntary bases), connected to both transmission and distribution networks, including the dispatch of any power plant not scheduled by the market but needed by the system to provide services such as voltage control or balancing. Needs for maintaining security in distribution are assessed and communicated to the TSO by the DSO. The process has two differentiated parts: one of them taken care after the day ahead market, with the aim of guaranteeing the physical feasibility of the economic dispatch of the market, and another one under continuous bases to solve any constraint found in real-time operation. The TSO must choose the group of actions that solves the security issues (mainly congestion and voltages out of range) at the lowest possible cost.

The cost of this process is daily transferred to the consumers (directly to consumers for those who buy directly in the wholesale market and through suppliers for the rest), in proportion to their consumption, as part of the cost of energy. However, the TSO is financially neutral and has no budget limit, so it has no direct economic impact if it reduces the costs.

After the D-1 results are published, situations such as unexpected unavailability or changes in the forecasts can occur that need to be solved in the real-time constraint resolution process.

In general, this process implies continuous monitoring and adjustment, and it offers the advantage of less uncertainty due to being closer to real time.

Even if there was not a remunerated voltage control service in place, power plants scheduled by the TSO under PO3.2 to solve situations of lack of dynamic voltage control receive the technical constraints remuneration (pay as bid) for their active power redispatch. In security studies conducted on 27 April, for 28 April, the combined cycle "Thermal 4–Centre /South-West" was scheduled for the entire day to regulate voltage in Western Andalusia. At 19:52 on 27 April, the unit was declared unavailable due to an internal problem, initially until 22:00 on 27 April and later extended to 00:00 on 30 April.

The connection of "Thermal 5–Centre/South-West" was extended during the night to secure voltages. During the morning of 28 April, RE considered that the "Thermal 5–Centre/South-West" plant was not needed.

There is no operational procedure approved in Spain where a minimum number of generation units coupled is required, and there is also no maximum limit. The criterion to decide the coupling of an additional generation unit is the fulfilment of Operational Procedure 1.1 with foreseen scenarios (generation, demand, and network).

At 12:20 on 28 April, RE ordered the connection of an additional thermal power plant equipped with PSS, following the detection of system oscillations. The selected group was a combined-cycle gas plant in centre/south-west, which indicated that it could be connected in 90 minutes. At 12:26, the confirmation was issued to the power plant to connect at 14:00. Due to the blackout occurring before 14:00, this connection never occurred. In general, RE is aware of the start-up times of the combined-cycle gas plants in its control area.

2.6.9 Short Circuit Power

Figures 2-90 and 2-91 plot the calculated short circuit power over time, demonstrating that most trends appear to be uniform during the observed timeframe.

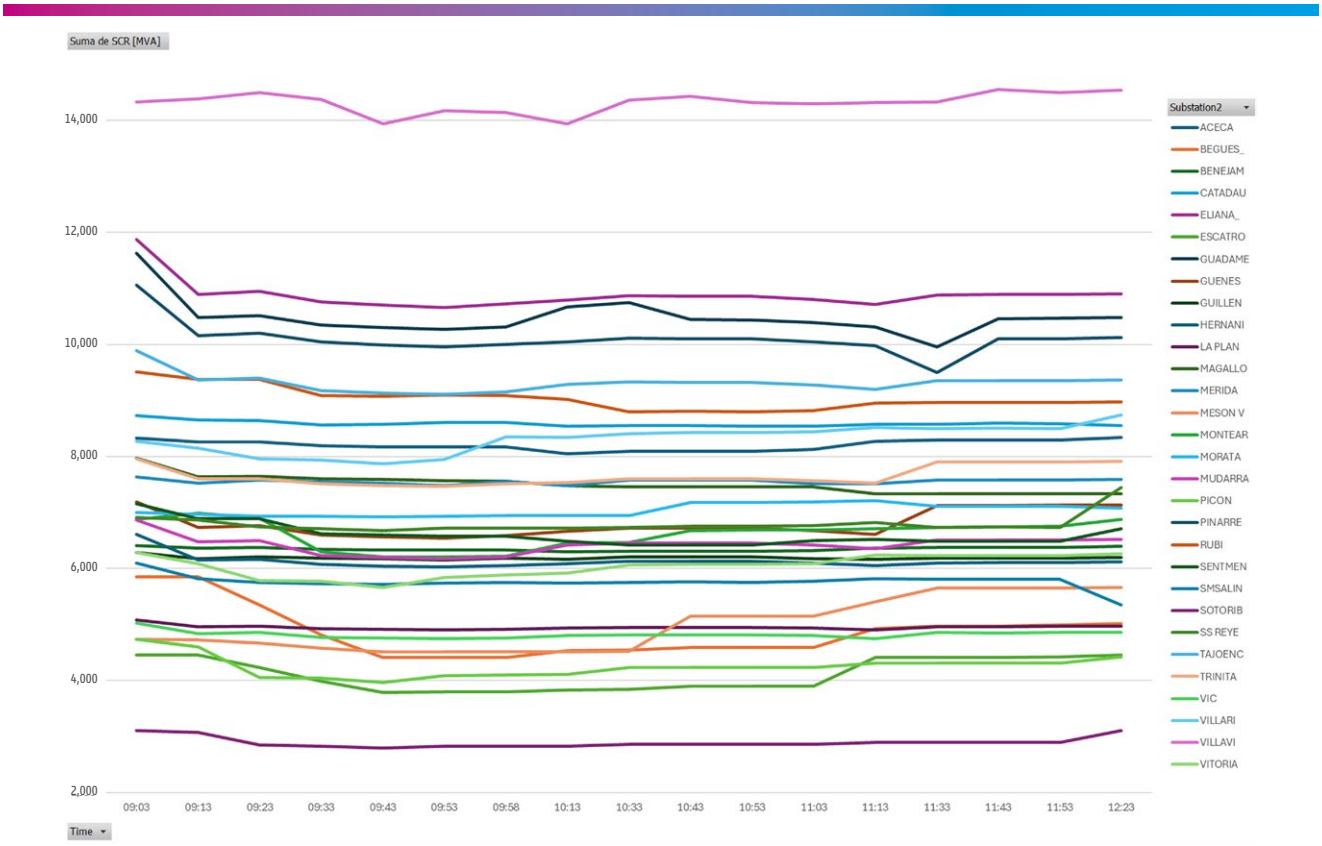


Figure 2-90: Short circuit power evolution in the Spanish 220 kV network (pilot nodes)

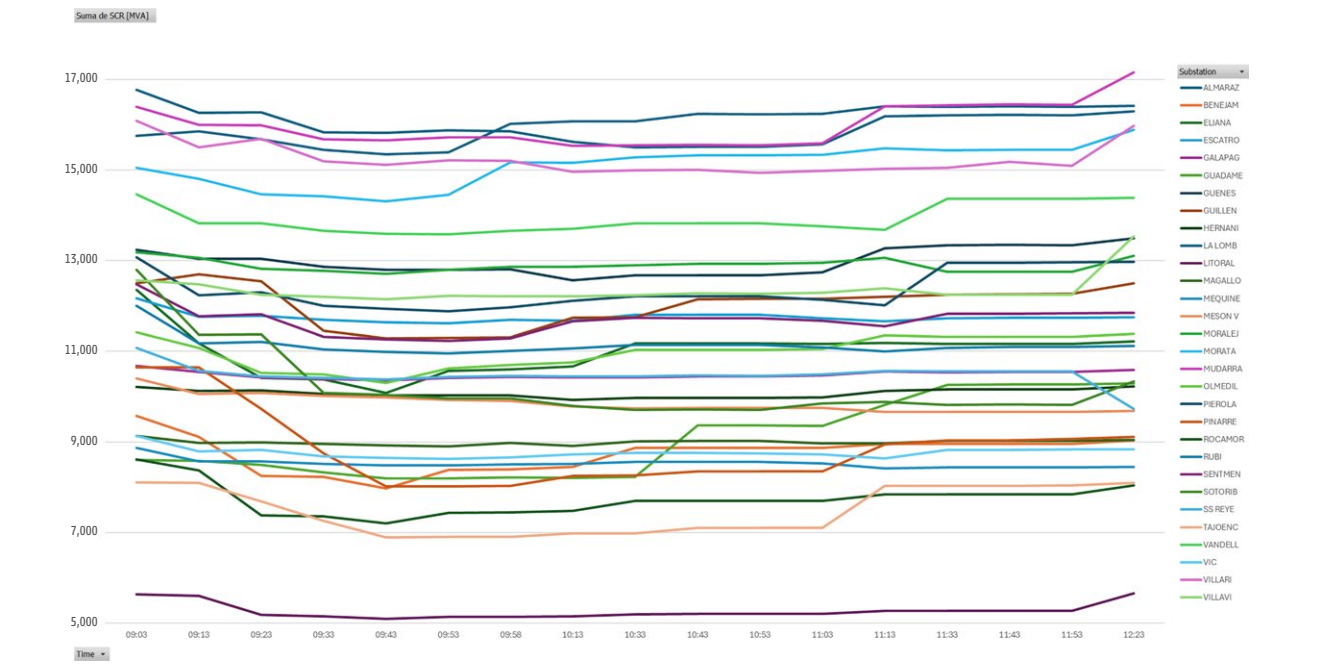


Figure 2-91: Short circuit power evolution in the Spanish 400 kV network (pilot nodes)

2.6.10 Reactive Power Flows with Neighbouring TSOs

Figures 2-92 – 2-94 plot profiles of the reactive power flows with the neighbouring TSOs.

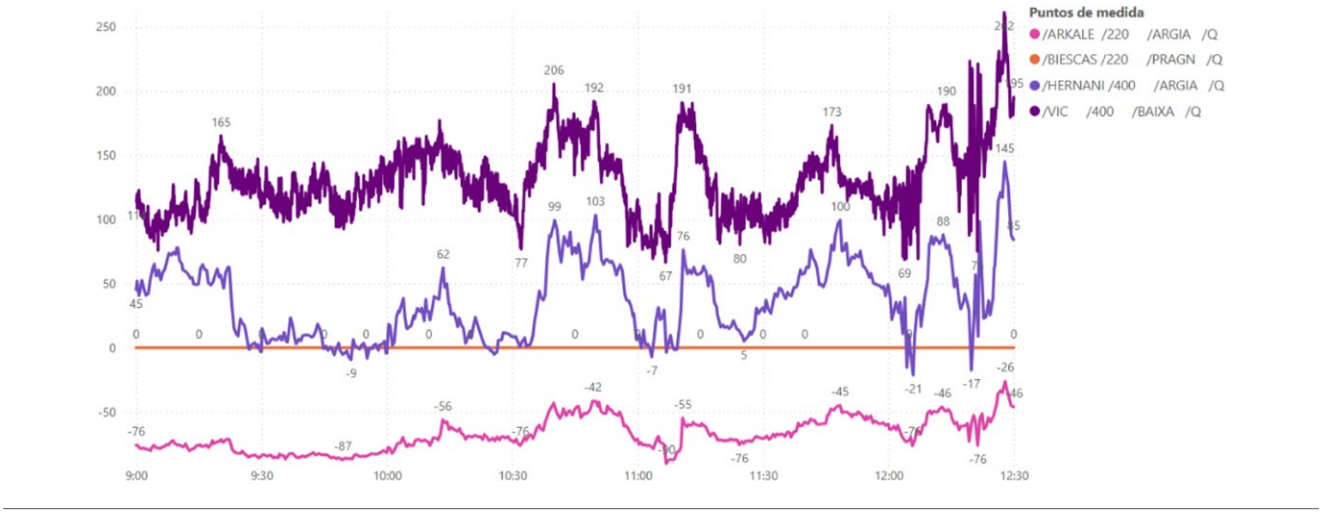


Figure 2-92: AC interconnectors reactive power flow from Spain to France [Mvar]

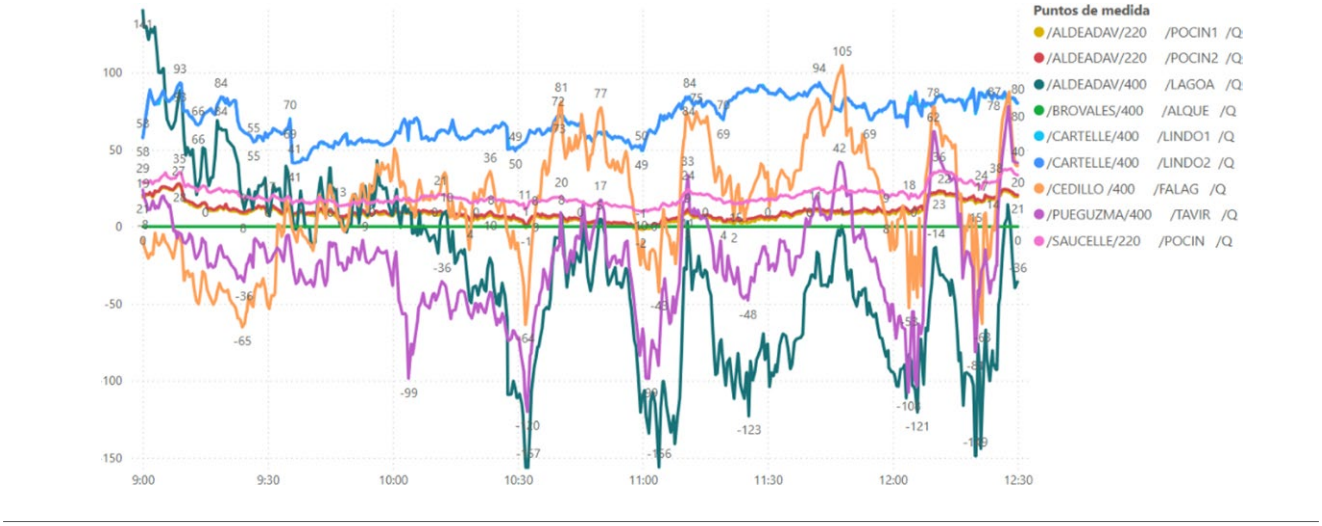


Figure 2-93: AC interconnectors reactive power flow from Spain to Portugal [Mvar]

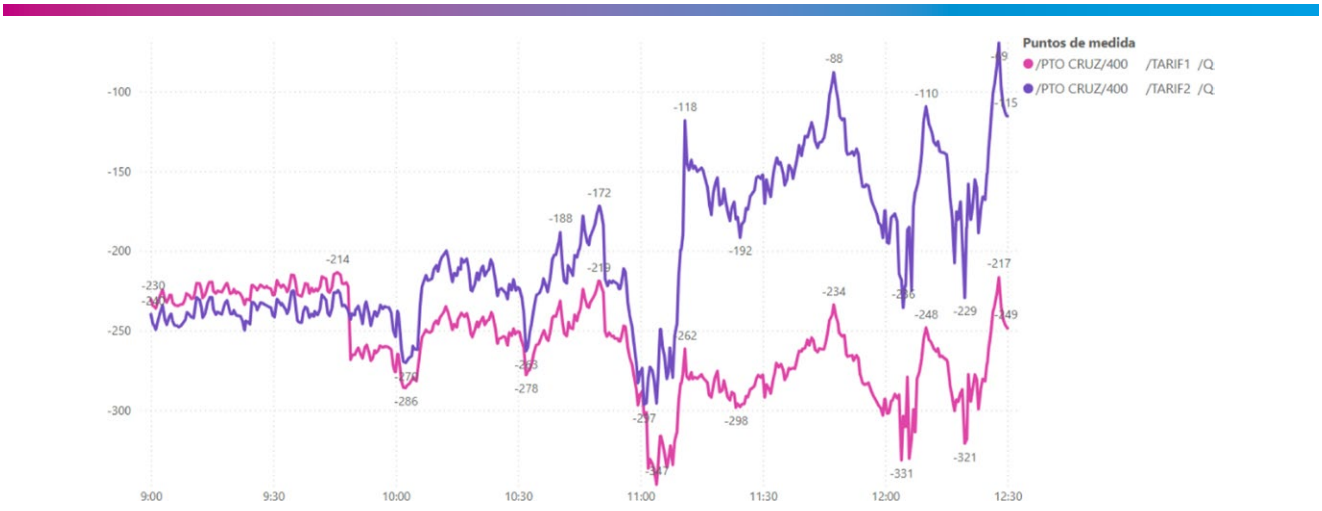


Figure 2-94: AC interconnectors reactive power flow from Spain to Morocco [Mvar]

2.7 Behaviour of the HVDC Link

The Santa Llogaia–Baixas HVDC link is a VSC-type HVDC with transmission capacity of 2 × 1,000 MW commissioned in 2015. In 2019, it was the VSC HVDC with the highest transmission capability.[17]

A VSC-type HVDC is a relatively new HVDC technology, as opposed to the classic LCC. VSC can control active and reactive power independently, provide black start capability, and work as a STATCOM.

This HVDC connects the substations of Santa Llogaia 400 kV (in Spain) and Baixas 380 kV (in France). It runs almost in parallel with the Vic–Baixas 400 kV AC interconnection line, forming an AC-DC corridor.

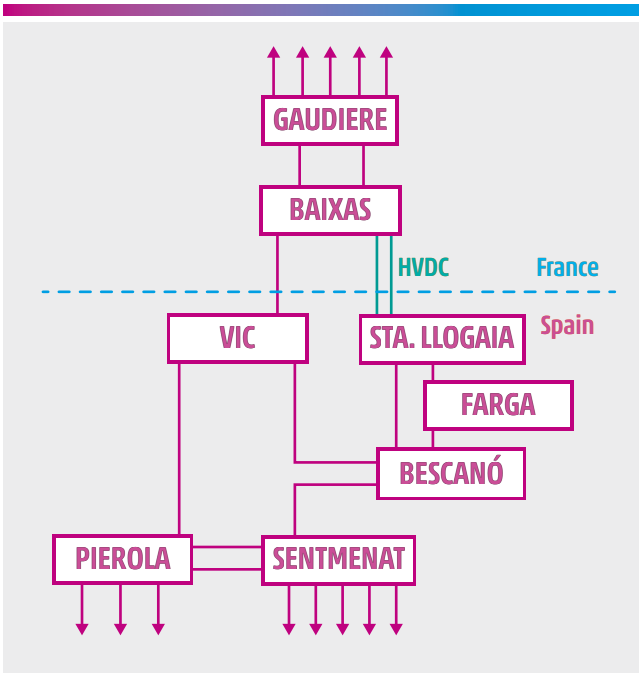


Figure 2-95: 400 kV grid in the area of the HVDC Baixas–Santa Llogaia

2.7.1 HVDC Control

The operator of a VSC-based HVDC independently controls the active and reactive power at both HVDC terminals, with a priority assigned to one of these controls if needed in extreme operating points.

In terms of active power control, the operator of a VSC HVDC sets the active power reference point. This active power setpoint is achieved based on either the desired real power flow, desired DC voltage, or desired frequency. For each moment in time, in one of the two terminals, Psetpoint will be determined. In the other terminal, the DC line voltage depends on the Psetpoint and will be set in accordance. The maximum active power capability is 2 × 1,000 MW.

On the Santa Llogaia–Baxias HVDC, the active power setpoint is determined for the Spanish and French ends and based on one of the three modes. In PMODE1, a fixed active power transmission value and its direction are determined. In PMODE2, the HVDC active power setpoint depends on an external analogue signal, which is intended to be the active power flow on several lines. This mode is not used in operation. The third mode – PMODE3 – is an AC emulation mode, where the active power setpoint is calculated based on the difference in angles between the HVDC terminals. Effectively, in this mode, the flow on the HVDC will resemble the flow on an AC line.

17 See ENTSO-E's 2019 report " HVDC links in system operations" available at https://eepublicdownloads.entsoe.eu/clean-documents/SOC documents/20191203_HVDC links in system operations.pdf.

The function for PMODE3 on the Santa Llogaia–Baixas HVDC is as follows:

$$P_{setpoint} = P_0 + (K / (1 + sT) * (\delta_A - \delta_B)$$

The values for P_0, K and T are as follows: $P0 = 0$, $K = 360\ MW/º$, $T = 50\ s$. The angles δ_A and δ_B are measured at substations 400 kV Baixas in France and 400 kV Santa Llogaia in Spain.

In terms of **reactive power control**, the operator of a VSC HVDC sets the reactive power reference point. This reactive power setpoint is achieved based on either the desired reactive power flow or desired AC voltage.

For each moment in time, in each of the terminals independently, the reactive power setpoints Qsetpoint will be determined.

Each HVDC link is capable of generating up to 400 Mvar and absorbing up to 600 Mvar in each of the terminals. Working both links together can generate up to 800 Mvar and absorb up to 1,200 Mvar in each of the terminals. These maximum capabilities can be limited by the active power transmitted by the link. The maximum capability of Q depending on the transmitted P is shown in the guaranteed operation area curve in Figure 2-96. This implies that in terms of control priority, the Santa Llogaia–Baxias HVDC prioritises active over reactive power.

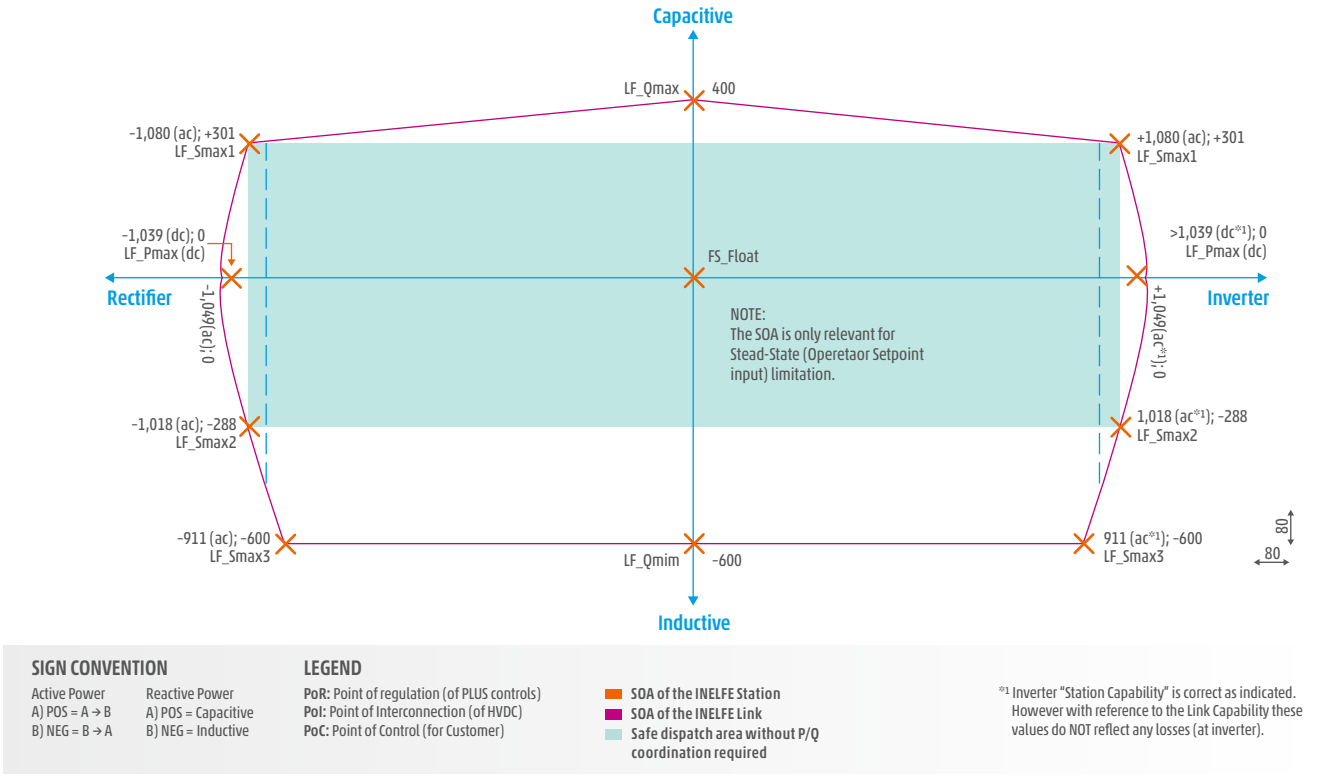


Figure 2-96: Guaranteed operation area curve

On the Santa Llogaia–Baxias HVDC, the reactive power setpoints on the French and Spanish sides vary. On both sides, the reactive power setpoint is determined based on a $V_{setpoint}$.

On the French side, the $V_{setpoint}$, is controlled by the secondary voltage control. This means that measurements from several substations in the French power systems are collected, and a voltage setpoint at the HVDC terminal calculated based on a pre-determined approach.

On the Spanish side, the Vsetpoint is manually inserted by the control room. The management of the HVDC voltage setpoint is integrated within the northeast area to ensure coordination with other voltage control devices operating in that region. On 28 April, the setpoint was set as outlined in Figure 2-97.

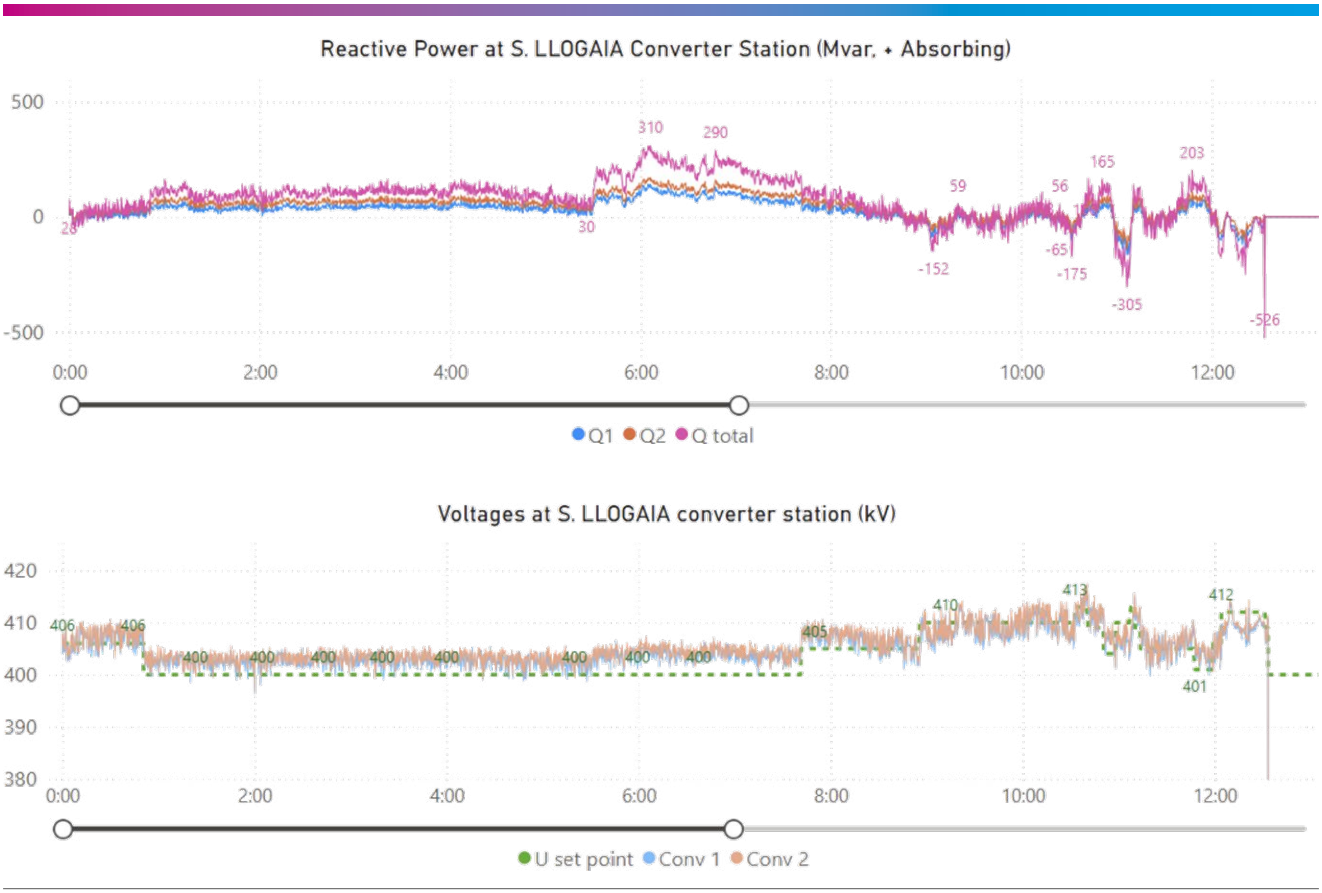


Figure 2-97: Reactive power and voltages at the Santa Llogaia converter station on 28 April (source: RE SCADA)

2.7.1.1 Voltage and Reactive Power Flows at HVDC Terminals

This section describes the behaviour of the HVDC link, with a focus on voltage and reactive power at the terminals on both the French and Spanish sides.

The observed timeframe is 11:30:00 – 12:00:00, with a focus on 11:35:20 – 11:36:20 (a minor oscillation with amplitude around 0.6 Hz can also be observed in this latter timeframe).

2.7.1.1.1 French side

It is observed that in the period 11:30:00–12:00:30, the reactive power at a HVDC node (Baixas 1) varies between +340 and +443 MVar (both generating). The voltage is a minimum of 388 kV and a maximum of 394 kV.

It is observed that in the period from 11:35:20 to 11:36:20, the reactive power at a HVDC node (Baixas 1) varies between +372 and +395 MVar (both generating). The voltage is a minimum of 389 kV and a maximum of 392 kV.

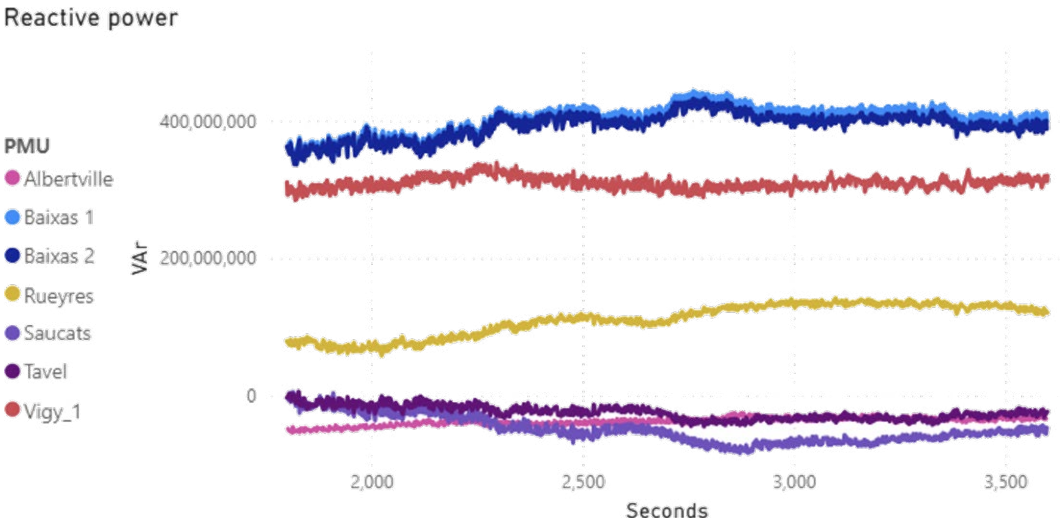


Figure 2-98: Reactive power flow measured by a set of PMUs in France, 11:30:00 –12:00:00 (x-axis is in seconds in the reference timeframe of the PMUs)

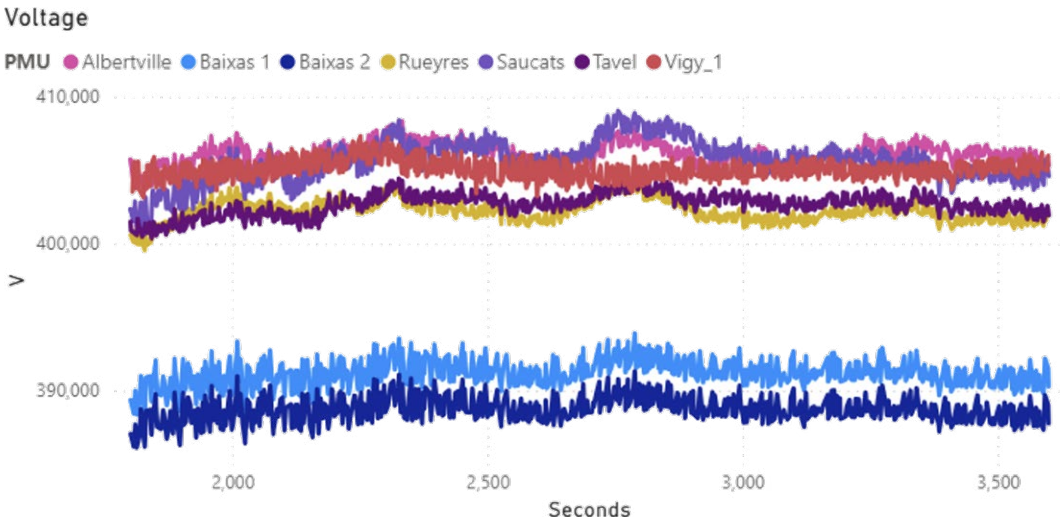


Figure 2-99: Voltage measured by a set of PMUs in France, 11:30:00 –12:00:00 (x-axis is in seconds in the reference timeframe of the PMUs)

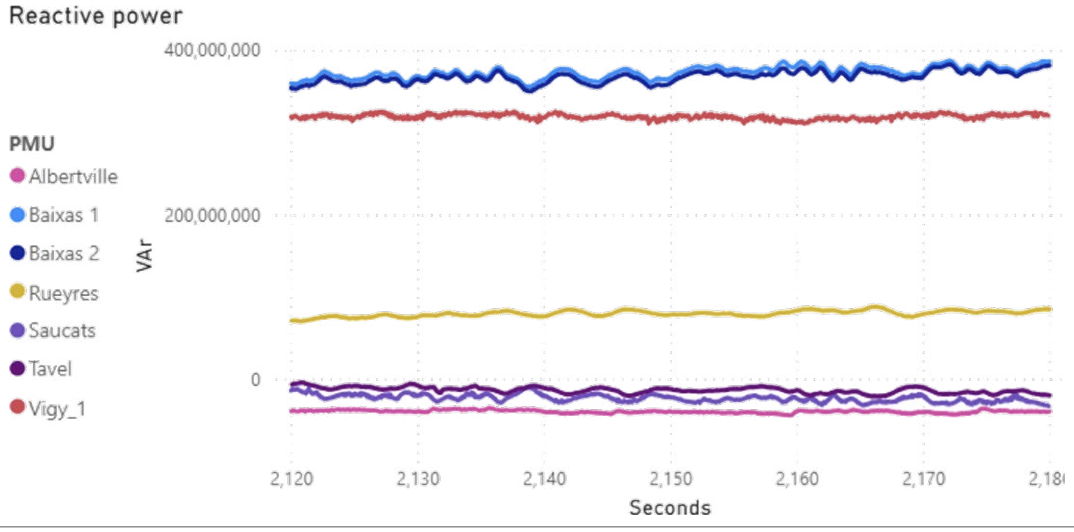


Figure 2-100: Reactive power flow measured by a set of PMUs in France, 11:35:20 -11:36:20 (x-axis is in seconds in the reference timeframe of the PMUs)

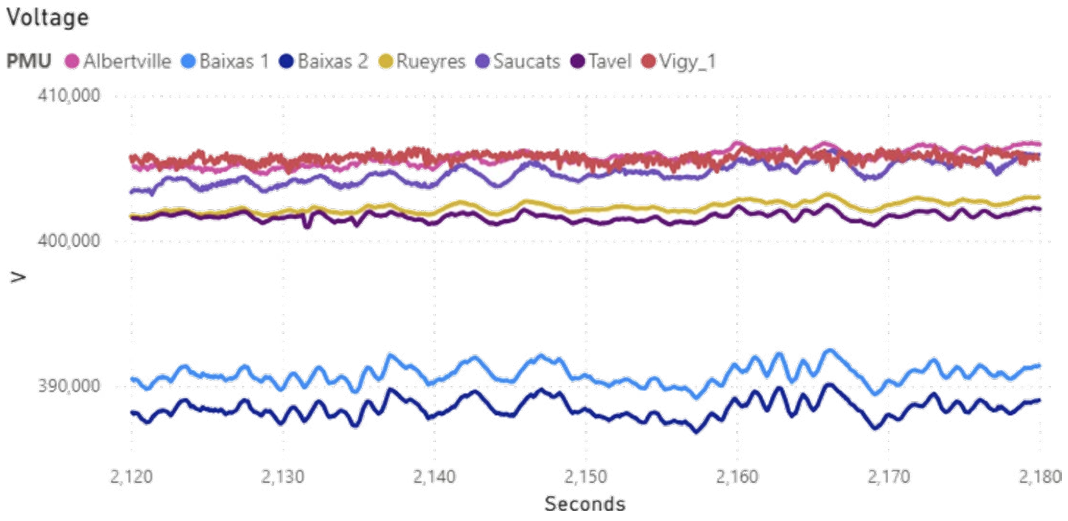


Figure 2-101: Voltage measured by a set of PMUs in France, 11:35:20 -11:36:20 (x-axis is in seconds in the reference timeframe of the PMUs)

2.7.1.1.2 Spanish side

It is observed that in the period from 11:30:00 to 12:00:00, the reactive power at a HVDC node (Santa Llogaia 1) varies between –51.6 MVAr (absorbing) and +128.4 MVar (generating). The voltage is minimum 398 kV and maximum 411.0 kV.

It is observed that in the period from 11:35:20 to 11:36:20, the reactive power at a HVDC node (Santa Llogaia 1) varies between –43.6 MVAr (absorbing) and +35.7 MVar (generating). The voltage is a minimum 400.5 kV and maximum 408.6 kV.

It is observed from Figures 2-98 and 2-100 that the variation of reactive power is the highest at the PMU corresponding to the HVDC node. Figure 2-102 shows that in the Spanish HVDC node, reactive power fluctuates between positive and negative values. It is also observed from Figure 2-105 that the variation of voltage in the PMU corresponding to the Spanish HVDC node is higher than the variation of voltage in adjacent PMUs (Vic and Rubi).

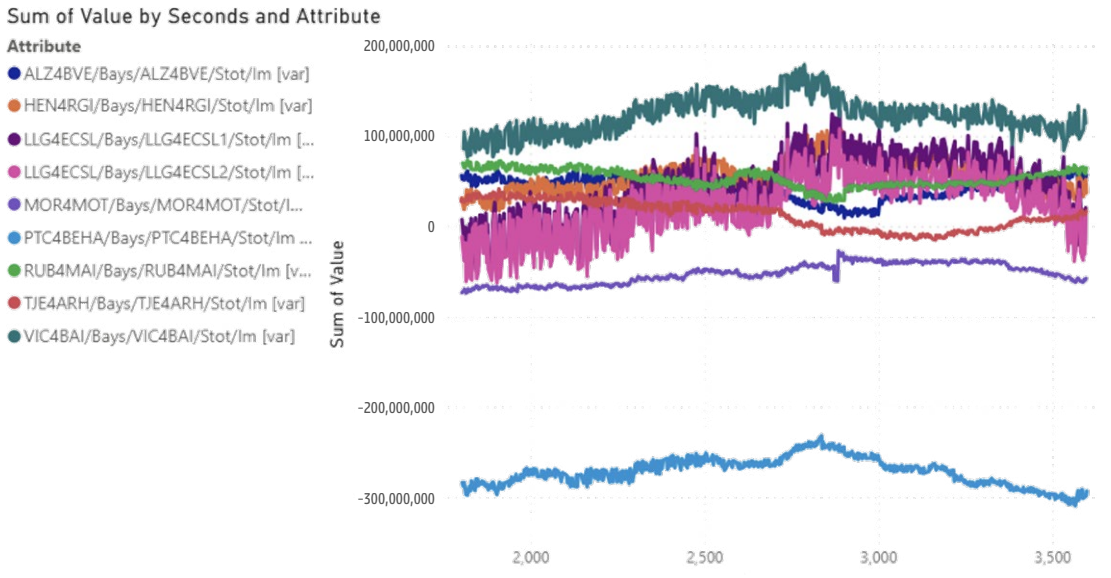


Figure 2-102: Reactive power measured by a set of PMUs in Spain, 11:30:00 –12:00:00 (x-axis is in seconds in the reference timeframe of the PMUs)

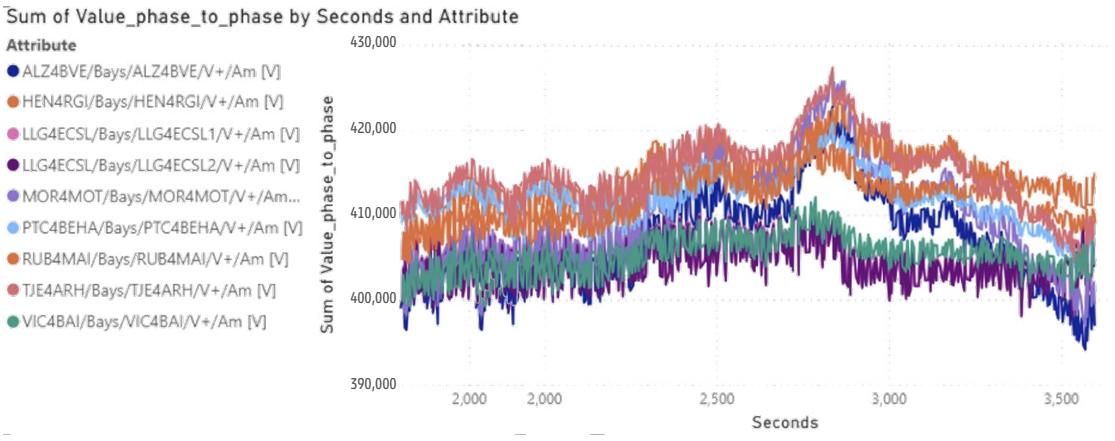


Figure 2-103: Voltage measured by a set of PMUs in Spain, 11:30:00 –12:00:00 (x-axis is in seconds in the reference timeframe of the PMUs)

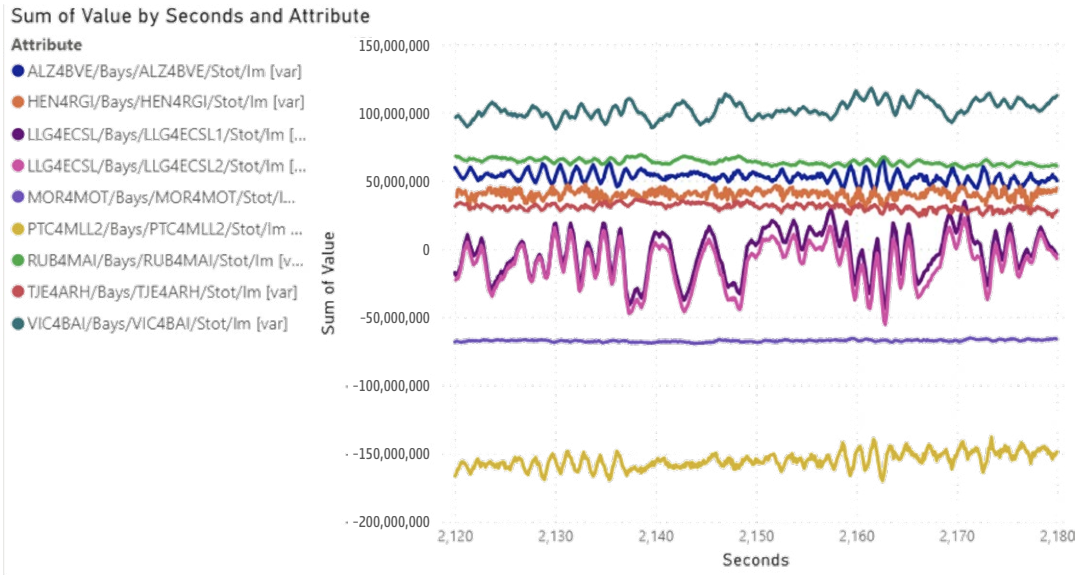


Figure 2-104: Reactive power measured by a set of PMUs in Spain, 11:35:20 -11:36:20 (x-axis is in seconds in the reference timeframe of the PMUs)

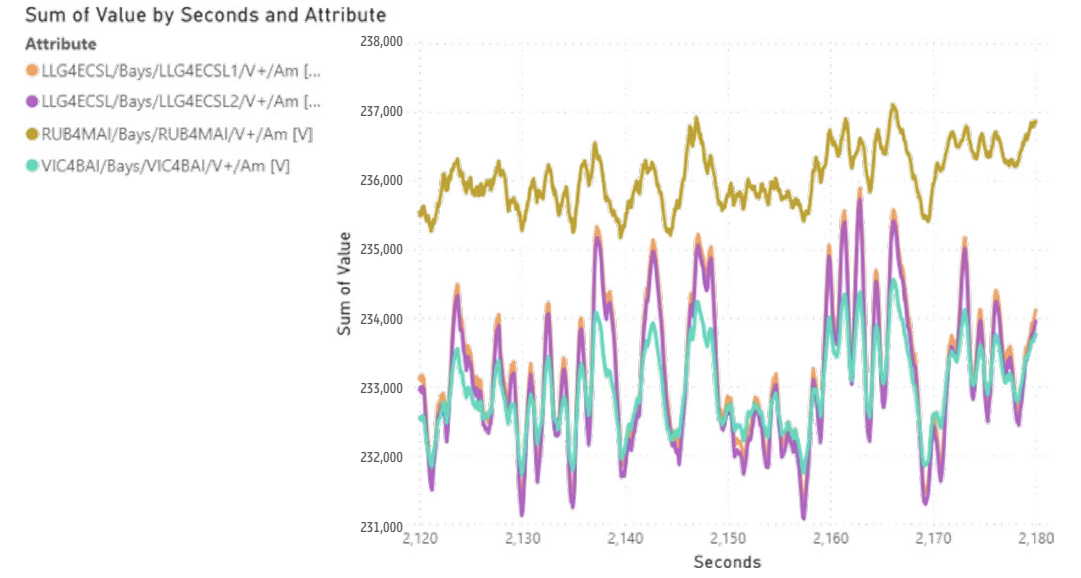


Figure 2-105: Voltage measured by a set of PMUs in Spain (the HVDC node and two nodes in the geographical vicinity), 11:35:20 -11:36:20 (x-axis is in seconds in the reference timeframe of the PMUs)

2.7.1.2 Active and Reactive Power Flows and Voltage at HVDC Terminals

This section describes the behaviour of the HVDC link, with a focus on active power, reactive power, and the AC voltage at the Spanish end. Figures 2-106 to 2-108 show the time series from 12:00:00 to 12:16:00.

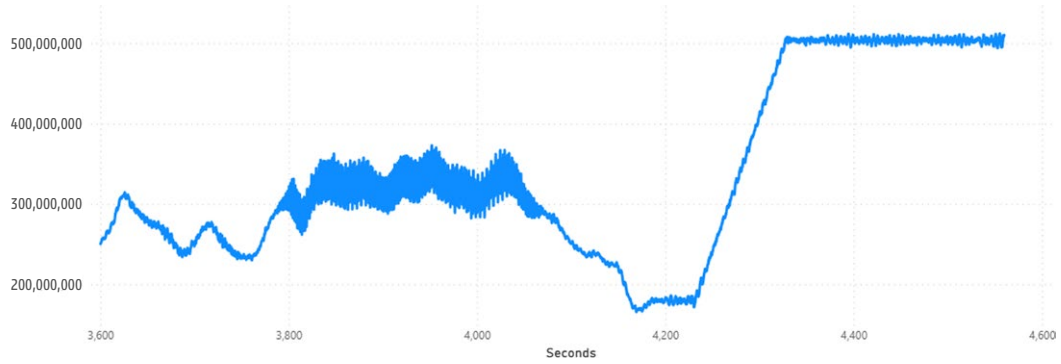


Figure 2-106: Active power flow, Llogaia bay 1, 12:00 – 12:16 (x-axis is in seconds in the reference timeframe of the PMU)

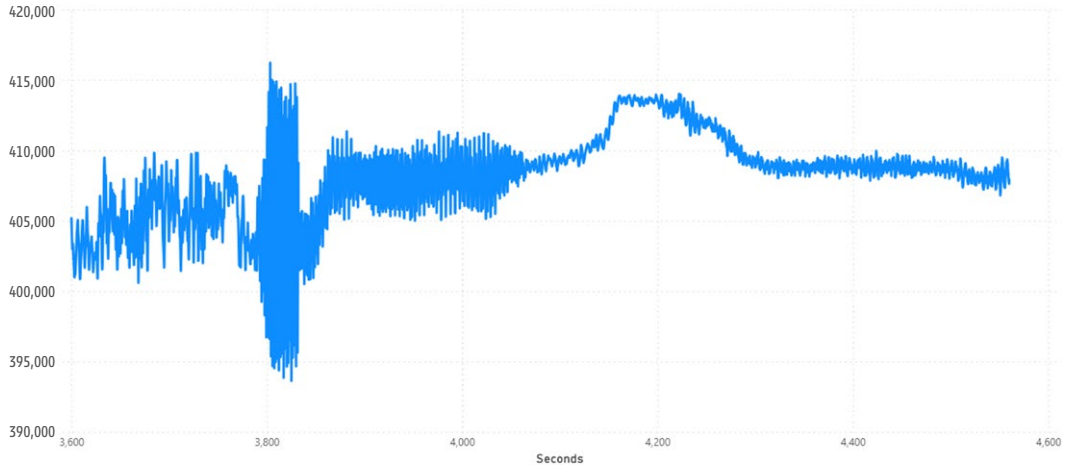


Figure 2-107: AC voltage, Llogaia bay 1, 12:00 – 12:16 (x-axis is in seconds in the reference timeframe of the PMU)

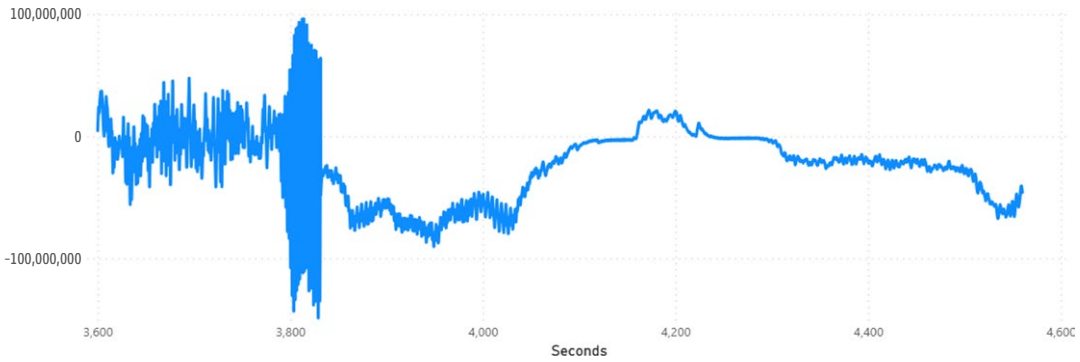


Figure 2-108: Reactive power flow, Llogaia bay 1, 12:00 – 12:16 (x-axis is in seconds in the reference timeframe of the PMU)

Figures 2-109 to 2-111 show the data from 12:16:00 to 12:32:00.

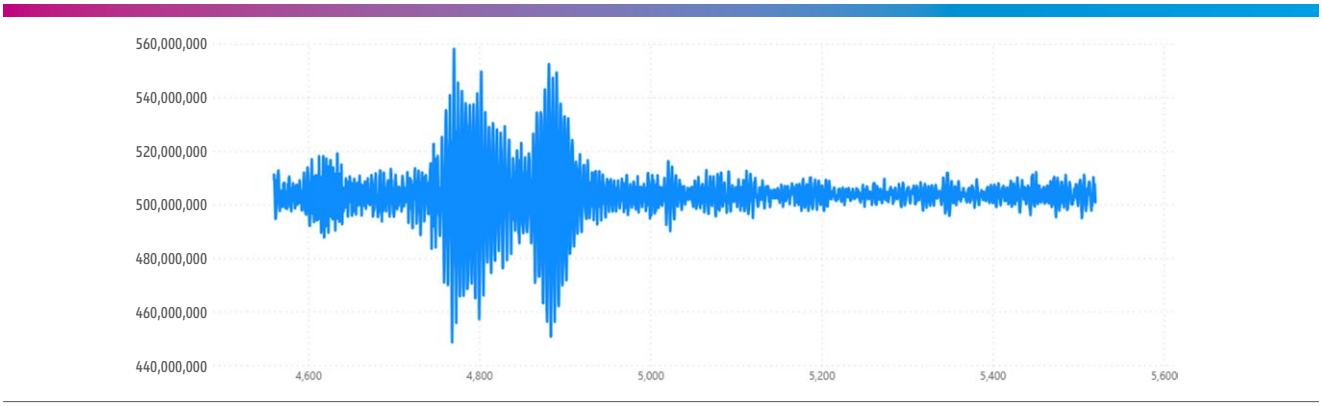


Figure 2-109: Active power flow, Llogaia bay 1, 12:16 -12:32 (x-axis is in seconds in the reference timeframe of the PMU)

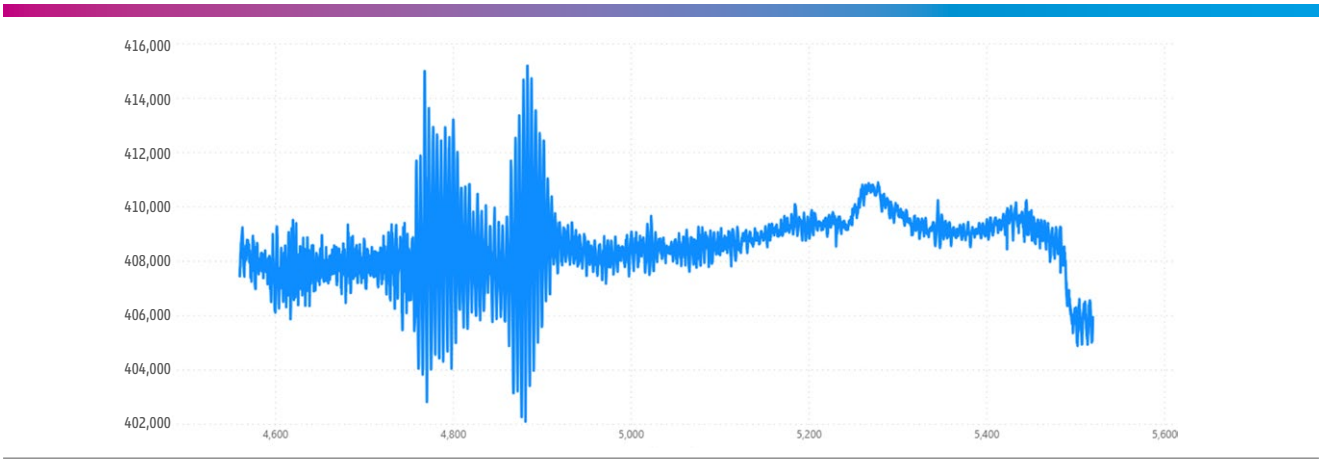


Figure 2-110: AC voltage, Llogaia bay 1, 12:16 -12:32 (x-axis is in seconds in the reference timeframe of the PMU)

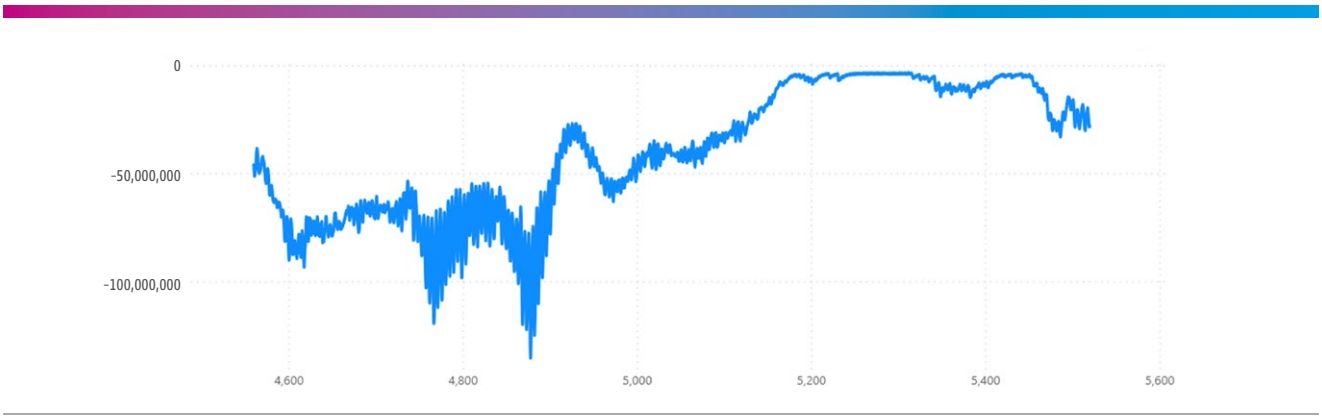


Figure 2-111: Reactive power flow, Llogaia bay 1, 12:16 -12:32 (x-axis is in seconds in the reference timeframe of the PMU)

2.7.2 HVDC Oscillation Damping

The output of active and reactive power of the HVDC can be altered by the output of POD control loops. The final voltage and current signals produced by the HVDC depend on all inner control loops and how they are prioritised. Section 2.5.5.2 describes the POD functionalities of the HVDC.

2.7.3 Additional functionalities

Besides the functionalities mentioned earlier in the chapter, the Santa Llogaia–Baxias HVDC is also capable of the following functions:[18]

- special protections scheme run-back/run-up (meaning a rapid decrease or increase in power flow to stabilise grids during emergencies);
- providing frequency containment reserves in case of a system split;
- island detection.

The HVDC has the capability to automatically adjust the active power flow or active power parameters in response to an external signal. This functionality is referred to as modulation modules (power run-up/run-down). The modulation modules (MMs) are a station control functionality, meaning that they are independently defined in each converter station (Santa Llogaia and Baixas).

In normal operation, MMs are only used on the Spanish side, and the external signal is associated with predefined contingencies. The MMs considered at Santa Llogaia end have the following features:

- **MM1:** Switch to Pmode1 under the contingency of double circuit 400 kV Santa Llogaia–Bescanó/Santa Llogaia–La Farga
- **MM2:** Switch to Pmode1 under the contingency N-1 400 kV Bescanó–Sentmenat
- **MM3:** Switch to Pmode1 under the contingency of double circuit 400 kV Bescanó–Sentmenat/Vic–Pierola
- **MM4:** Switch Pmode1 under the contingency N-1 400 kV Bescanó–Vic
- **MM5:** Switch to Pmode1 under the contingency of double circuit 400 kV Bescanó–Sentmenat/Bescanó–Vic
- **MM6:** Limiting Operating Power Range (OPR) under the contingency of double circuit 400 kV Bescanó–La Farga/Bescanó–Santa Llogaia
- **MM7:** Limiting OPR under the contingency of double circuit 400 kV La Farga–Santa Llogaia/Bescanó–Santa Llogaia

The HVDC control system automatically activates the relevant MM and adjusts the active power transmitted through the link when both of the following conditions are satisfied:

- The relevant MM is enabled.
- The line(s) for this MM tripped.

In real-time operation, MMs are enabled depending on network conditions. Because enabling or disabling them affects power levels, these actions are coordinated with RTE.

At 12:32, the MMs enabled at the Santa Llogaia end were:

- MM1
- MM3
- MM5
- MM6

None of the aforementioned MMs were activated during the day.

18 Per Table 3 of https://eepublicdownloads.entsoe.eu/clean-documents/SOC documents/20191203_HVDC links in system operations.pdf

2.8 Annex

2.8.1 Detailed List of All Planned Outages

Element type	Hour (from – to)	Switched off element name
Tie-line	All Day	400 kV BROVALES–ALQUEVA
Line	All Day	400 kV BUNIEL GRIJOTA 1
Line	09 to 18	400 kV ALMARAZ CN MORATA 2
Line	09 to 24	400 kV CATADAU MUELA 1
Line	All Day	400 kV CARRIL LITORAL 1
Line	All Day	400 kV FUENTES DE LA ALCARRIA TRILLO
Line	All Day	400 kV LOECHES SS.REYES
Line	All Day	400 kV SAN FERNANDO SS.REYES 1
Line	All Day	400 kV BROVALES ALQUEVA
Line	09 to 24	400 kV PALOS GUILLENA 2
Line	All Day	400 kV CARMONA VALDECABALLEROS
Line	09 to 18	220 kV ATIOS PAZOS DE BORBEN
Line	09 to 24	220 kV HARO LAGUARDIA
Line	All Day	220 kV ITXASO ORCOYEN 2
Line	09 to 18	220 kV ORTUELLA SANTURCE
Line	09 to 24	220 kV VALLADOLID NUEVO ZARATAN 2
Line	All Day	220 kV BADALONA GUIXERES 1
Line	09 to 18	220 kV ABRERA GIS RUBI
Line	All Day	220 kV BEGUES CAN JARDI
Line	All Day	220 kV BIESCAS PRAGNERES
Line	09 to 24	220 kV EIXAMPLE VILANOVA
Line	All Day	220 kV LA ESPLUGA JUNEDA
Line	09 to 24	220 kV LA FARGA JUIA 1
Line	09 to 24	220 kV CAN JARDI CERVELLO
Line	09 to 18	220 kV ACECA PRADILLOS
Line	All Day	220 kV ALDAIA QUART DE POBLET
Line	All Day	220 kV CASA DE CAMPO NORTE
Line	All Day	220 kV ANTONIO LEYVA PARQUE INGENIEROS
Line	09 to 18	220 kV LA PLANA EL SERRALLO 1
Line	09 to 24	220 kV LEGANES T DE LEGANES
Line	09 to 24	220 kV LUCERO T DE LEGANES
Line	09 to 24	220 kV PRADO VILLAVICIOSA
Line	All Day	220 kV LA TORRECILLA VILLAVERDE BAJO
Line	09 to 24	220 kV VILLAVERDE BAJO T DE LEGANES
Line	09 to 24	220 kV ALCORES SANTA ELVIRA 2
Line	16 to18	220 kV CARTUJA DON RODRIGO
Line	09 to 18	220 kV DON RODRIGO ALJARAFE
Line	09 to 13	220 kV DON RODRIGO DOS HERMANAS 2
Line	All Day	220 kV MERIDA TRUJILLO
Line	16 to 18	220 kV ALJARAFE CHUCENA 1
Line	09 to 15	220 kV ALJARAFE SANTIPONCE
Line	All Day	132 kV IRUN ERRONDENIA
Transformer	09 to 24	AT ESCATRÓN 400/220 kV
Transformer	09 to 24	AT PALOS 400/220 kV
Transformer	All Day	AT LITORAL 400/132 kV

Table 2-9: RE planned outages

Element type	Switched off element name	Start date/time CET	End date/time CET	Reason
Line	Paraimo–Pereiros 1 220 kV	06/01 09:10	14/08 18:35	Upgrade line capacity
Line	Fernao Ferro–Ribatejo 400 kV	07/03 08:53	11/07 18:00	Upgrade line capacity
Line	Palmela–Fernão Ferro 2 150	31/03 16:02	24/05 14:18	Palmela substation remodulation
Transformer	Auto Transformador 4 400/150 kV Subestação Ferreira do Alentejo	07/04 09:39	31/10 18:00	Ferreira Alentejo substation remodulation
Line	Recarei–Vermoim 3 400 kV	14/04 09:12	22/09 18:00	Recarei substation remodulation
Line	Tunes–Estoi 150 kV	16/04 18:01	22/09 18:00	Tunes substation remodulation and line maintenance
Tie-line	Brovales–Alqueva 400 kV	24/04 09:04	29/04 13:40	Corrective maintenance, installing bird diverters after a protected species collision
Line	Fanhões–Carriche2 220	28/04 07:59	02/05 18:12	Creation of safety conditions for other company work near the line
Line	Alto Mira–Carriche/Trajouce 220	28/04 08:02	29/04 18:58	Substation maintenance
Line	Chafariz–Vila Cha 1/Ramal Gouveia 220 kV	28/04 09:52	29/04 09:15	Other company works

Table 2-10: REN planned outages

Element type	Switched off Element name	Start date	End date	Reason
Tie-line	225 kV Biescas Pragnères	22/04	02/05	PST maintenance
Line	400 kV Eguzon Valdivienne 2	08/04	06/06	Switch replacement
Line	400 kV Braud Cubnezais 4	28/04	05/05	Preventive and curative maintenance
Line	440 KV Breuil Marmagne	28/04	30/04	Curative maintenance
Busbar	Valdivienne 2 A	08/04	02/05	Development of the grid
Busbar	Granzay 1B	22/04	28/04	Preventive maintenance
Busbar	Marsillon 1	14/04	28/04	Development of the grid

Table 2-11: RTE planned outages

2.8.2 Detailed List of Open Lines for Voltage Control

Element name	Date start Time	Hour start time	Date end time	Hour End Time	Zone
L-400 kV MORELLA–MUDEJAR 1	15/03	23:49	29/04	20:32	NORTH
L-400 kV ARAGÓN–PEÑALBA 2	31/03	23:12	29/04	20:37	NORTH
L-400 kV CASTEJON–MURUARTE 1	07/04	22:27	29/04	19:38	NORTH
L-400 kV PALMAR–ROCAMORA 2	13/04	00:20	28/04	12:08	EAST
L-400 kV PALMAR–CARRIL	13/04	02:59	28/04	12:07	EAST
L-400 kV CARTELLE–TRIVES 1	14/04	23:21	28/04	21:30	NORTH WEST
L-400 kV LA ROBLA–MUDARRA 1	14/04	23:24	28/04	12:08	NORTH WEST
L-400 kV BAZA–CAPARACENA 2	16/04	01:53	29/04	21:04	EAST
L-220 kV TALAVERA–VILLAVERDE	16/04	15:55	28/04	16:54	CENTRE
L-400 kV N.ESCOMBRERAS–PALMAR 2	17/04	01:55	28/04	18:55	EAST
L-400 kV PINILLA–ROMICA 2	17/04	23:44	28/04	12:21	EAST
L-400 kV MONTEARENAS–MUDARRA 1	19/04	03:07	30/04	06:52	NORTH WEST
L-400 kV PINILLA–ROCAMORA 1	19/04	23:59	28/04	12:22	EAST
L-400 kV MONTEARENAS–MUDARRA 3	20/04	00:38	28/04	05:20	NORTH WEST
L-400 kV TORDESILLAS–GALAPAGAR	20/04	01:37	28/04	12:25	CENTRE
L-220 kV RUBI–GRAMANET 2	20/04	13:55	28/04	15:38	NORTH-EAST
L-220 kV MARAGALL–TRINITAT1	20/04	13:55	02/05	11:36	NORTH-EAST
L-400 kV BOIMENTE–PESOZ 2	21/04	16:09	29/04	01:58	NORTH WEST
L-400 kV SALAS–PESOZ	21/04	16:12	29/04	01:59	NORTH WEST
L-400 kV BROVALES–GUILLENA1	21/04	18:43	28/04	10:33	SOUTH
L-400 kV GUADAME–CABRA 1	23/04	00:21	28/04	11:07	SOUTH
L-220 kV GURREA–VILLANUEVA 1	23/04	03:57	28/04	09:13	NORTH
L-400 kV ARCOS–CABRA	23/04	19:23	28/04	11:17	SOUTH
L-400 kV MUDARRA–S. SEBASTIÁN	23/04	23:00	30/04	06:30	NORTH WEST
L-400 kV PINAR–TAJO	23/04	23:48	28/04	11:08	SOUTH

Element name	Date start Time	Hour start time	Date end time	Hour End Time	Zone
L-400 kV GUADAME–CABRA 3	24/04	19:47	28/04	12:25	SOUTH
L-400 kV GUADAME–VALDECABALLEROS 2	24/04	19:52	28/04	10:35	SOUTH
L-400 kV BELINCHON–MORATA 1	24/04	23:55	29/04	20:35	CENTRE
L-400 kV AGUAYO–ABANTO	25/04	16:16	28/04	11:07	NORTH WEST
L-400 kV P.GUZMAN–GUILLENA 1	25/04	20:09	28/04	12:07	SOUTH
L-400 kV OLMEDILLA–ROMICA 2	25/04	23:57	28/04	11:03	EAST
L-400 kV MORATA–VILLAVICIOSA	26/04	00:52	28/04	12:15	CENTRE
L-400 kV DON RODRIGO–ARCOFRONT 2	26/04	01:01	28/04	10:06	SOUTH
L-400 kV MONTEARENAS–MUDARRA 2	26/04	15:59	28/04	11:09	NORTH WEST
L-400 kV HERRERA–LOMBA	26/04	18:06	28/04	06:14	NORTH WEST
L-400 kV GRIJOTA–VILLARINO 2	26/04	19:51	28/04	12:07	NORTH WEST
L-400 kV ALMARAZ–MORATA 2	26/04	19:56	28/04	09:54	CENTRE
L-400 kV GUILLENA–PALOS 2	27/04	00:15	28/04	08:19	SOUTH
L-220 kV AENA–SS REYES 2	27/04	00:26	14/05	15:08	CENTRE
L-400 kV RUBÍ–VANDELLÓS	27/04	01:14	28/04	07:37	NORTH-EAST
L-400 kV COFRENTES–LA MUELA 1	27/04	03:06	28/04	14:01	EAST
L-400 kV CATADAU–LA MUELA 1	27/04	03:10	28/04	09:38	EAST
L-400 kV PIEROLA–SENTMENAT 2	27/04	11:28	28/04	15:20	NORTH-EAST
L-400 kV PIEROLA–VANDELLÓS	27/04	12:40	28/04	11:20	NORTH-EAST
L-400 kV SALLENTE–CALDERS	27/04	12:47	28/04	09:17	NORTH-EAST
L-400 kV ABANTO–ICHASO 1	27/04	15:14	28/04	06:20	NORTH
L-400 kV BRAZATORTAS–MANZANARES 1	27/04	15:17	28/04	09:13	CENTRE
L-220 kV BESOS NUEVO–VILANOVA	27/04	17:09	28/04	06:39	NORTH-EAST
L-400 kV BROVALES–SAN SERVAN1	27/04	19:55	28/04	10:02	SOUTH
L-400 kV ALMARAZ–S.SERVAN 1	27/04	19:56	28/04	09:02	SOUTH
L-400 kV ARAÑUELO–MORATA 2	27/04	20:02	28/04	07:29	CENTRE

Element name	Date start Time	Hour start time	Date end time	Hour End Time	Zone
L-400 kV ARAÑUELO–VALDECABALLEROS 2	27/04	20:07	28/04	07:23	SOUTH
L-400 kV FUENDETODOS–MEZQUITA	27/04	20:09	28/04	20:29	NORTH
L-400 kV MEZQUITA–MORELLA 1	27/04	20:09	29/04	20:33	NORTH
L-400 kV ALMARAZ–VILLAVICIOSA 2	27/04	22:48	28/04	07:17	CENTRE
L-400 kV BIENVENIDA–GUILLENA	27/04	23:01	28/04	07:48	SOUTH
L-400 kV ARCOS–RODA	27/04	23:03	28/04	06:14	SOUTH
L-400 kV ALMARAZ–GUADAME	27/04	23:29	28/04	07:09	SOUTH
L-220 kV CERRO PLATA–VILLAVERDE BAJO 2	28/04	00:45	28/04	12:02	CENTRE
L-220 kV ACECA–PICON	28/04	00:46	28/04	10:32	CENTRE
L-220 kV PICON–PUERTOLLANO	28/04	00:46	28/04	22:00	CENTRE
L-400 kV LA CEREAL–SEGOVIA	28/04	00:47	28/04	06:59	CENTRE
L-400 kV SALLENTE–SENTMENAT	28/04	01:14	28/04	06:47	NORTH-EAST
L-400 kV ALMARAZ–C. RODRIGO	28/04	03:39	28/04	05:28	SOUTH
L-400 kV ALDEADAVILA–HINOJOSA	28/04	03:40	28/04	05:27	NORTH WEST

Table 2-12: RE list of open lines for voltage control

3 SYSTEM CONDITIONS DURING THE INCIDENT

This chapter details the events that took place during the incident, namely from 12:32:00 until the blackout.

Subchapter 3.1 presents the dynamic behaviour of the system during the incident. In Section 3.1.1, a sequence of events is provided in both tabular and graphical form. They are grouped in so-called "event clusters" for easier geographical identification on maps. The next section (3.1.2) presents the evolution of the main electrical quantities in the affected TSO networks that characterised the event, including voltage magnitude, frequency, rate of change of frequency (RoCoF), and active power.

Subchapter 3.2 covers protection behaviour, including an analysis of generation trips, and is preceded by a summary of general grid code requirements for generators.

Subchapter 3.3 provides information on system defence plans for the affected TSOs. Where PMU measurements are considered in this chapter, it should be noted that PMUs have a temporal accuracy of approximately ±30 ms.

3.1 Dynamic Behaviour of the System During the Incident

This subchapter provides a factual representation of the evolution of the system during the event. As described above, this section deals with all the events that occurred between 12:32:00 and 12:34:00. The time before this period is covered in the previous chapter.

3.1.1 Sequence of Events

This section describes the sequence of events that led to the separation of the Iberian Peninsula power system from the Continental Europe Synchronous Area and Morocco, and to the subsequent blackout.

Table 3-1 lists the following types of events: generation loss (including type of generation), line trip, hydro Pump trip, and load shedding, along with the estimated value of lost power (country abbreviations used in Table 3-1: ES = Spain, PT = Portugal, IP = Iberian Peninsula). Causes are not included in the table as they are unknown in most cases. The known causes are reviewed in Section 3.2. Causes are estimated in cases where no factual information is available.

The events in Table 3-1 are grouped according to geographical, temporal, event-type, and energy-source criteria.

The resulting clusters are displayed on the network map (Figure 3-1), showing either the specific location of the affected plant or substation, or the centroid when events within the same cluster occurred across different geographical areas.

It is worth noting that many events occurred within a very short time frame, making it difficult to determine their exact sequence, as not all data is time-synchronised with sufficient accuracy. In addition, some of these events occurred in parts of the system that are not fully observable[19] from the TSO perspective, requiring estimation of the imbalance caused by the disconnection of certain power plants. Therefore, the Table 3-1 reports the sequence of events based on the information available to the Expert Panel[20], noting that some additional generation or load trips may be missing.

19 In Spain and Portugal, all generation facilities greater than 1 MW must send production in real time to the TSO. Smaller facilities do not have this obligation.

20 In case of lack of direct and punctual measurement of single generators (e.g. distributed generation) the SCADA information allowed to calculate the electrical power imbalance.

Event Cluster	Country	Time (CEST)	Event Type	Description	Electrical Power Imbalance (MW)
1	ES	12:32:00 - 12:32:57	Net load increase	Disconnection of small embedded generators and/or actual increase of load at the distribution level	317.30
2a/b	ES	12:32:00 - 12:32:57	Fast downwards change[21] of operating point of generation unit without ramping limitation OR disconnection of generation unit due to unknown reason	Wind/PV	Total 208
2a	ES	12:32:05	-	Wind	2.57
2a	ES	12:32:09	-	Wind	21.62
2a	ES	12:32:09	-	Wind	5.89
2a	ES	12:32:25	-	Wind	19.21
2a	ES	12:32:29	-	Wind	22.42
2a	ES	12:32:29	-	Wind	55.63
2a	ES	12:32:45	-	PV	0.59
2a	ES	12:32:49	-	Wind	13.35
2a	ES	12:32:49	-	PV	1.14
2a	ES	12:32:53	-	PV	11.94
2a	ES	12:32:53	-	Wind	19.98
2b	ES	12:32:09	-	Wind	4.79
2b	ES	12:32:09	-	Wind	2.86
2b	ES	12:32:25	-	Wind	3.43
2b	ES	12:32:29	-	Wind	2.26
2b	ES	12:32:53	-	Wind	4.78
2b	ES	12:32:53	-	Wind	10.01
2b	ES	12:32:53	-	Wind	5.99
3	ES	12:32:57.220	Generation loss due to power transformer trip	PV, wind, and thermo-solar	355.00
4a	ES	12:33:16.443	Generation loss	PV and thermo-solar	582.00
4b	ES	12:33:16.820	Generation loss	PV	145.00
5a1	ES	12:33:17.368	Generation loss	Wind	22.87
5a2	ES	12:33:17.780	Generation loss	PV	550.00
5a3	ES	12:33:17.940	Generation loss	Wind	94.00
5b1	ES	12:33:17.520	Generation loss	PV	118.00
5b2	ES	12:33:17.547	Generation loss	Wind and PV	33.73
5b3	ES	12:33:17.975	Generation loss	PV	37.50
5b4	ES	12:33:18.020	Generation loss	PV	71.90
6a1	ES	12:33:18.102	Generation loss	PV	3.00
6a2	ES	12:33:18.220	Generation loss	PV	20.00
6a3	ES	12:33:18.360	Generation loss	PV	16.00
6a4	ES	12:33:18.410	Generation loss	PV	41.00
6a5	ES	12:33:18.540	Generation loss	PV	63.00
6a6	ES	12:33:18.630	Generation loss	PV	25.70
6a7	ES	12:33:18.680	Generation loss	PV	127.50
6a8	ES	12:33:18.846	Generation loss	PV	154.00
6a9	ES	12:33:19.000	Generation loss	PV	16.00
6b	ES	12:33:18.951	Generation loss	PV	530.00
7a	PT	12:33:19.000	Load shedding	Pump	–221.00
7b	PT	12:33:19.090	Load shedding	Pump	–124

21 Most of downwards change in production was not reflected in the communicated program.

Event Cluster	Country	Time (CEST)	Event Type	Description	Electrical Power Imbalance (MW)
7c	PT	12:33:19.332	Load shedding	Pump	–113.00
8a	ES	12:33:19.095	Generation loss	PV	150.00
8a	ES	12:33:19.320	Generation loss	PV	0.71
8a	ES	12:33:19.339	Generation loss	Wind	11.17
8a	ES	12:33:19.380	Generation loss	Wind	6.21
8b	ES	12:33:19.252	Generation loss	Wind	364.00
8b	ES	12:33:19.260	Generation loss	PV	70.00
8b	ES	12:33:19.396	Generation loss	PV	509.00
8b	ES	12:33:19.407	Generation loss	PV	63.30
8b	ES	12:33:19.920	Generation loss	PV	72.00
8b	ES	12:33:19.951	Generation loss	PV	12.00
8b	ES	12:33:19.960	Generation loss	PV	22.30
8b	ES	12:33:19.973	Generation loss	PV	24.60
9	ES	12:33:19.971	Line trip	Loeches (RE)–Arganda (RE) 220 kV	-
10	ES	12:33:20.000	Pump disconnection	Pump	–102.00
11	PT	12:33:20.000	Load shedding	Pump	–219.00
11	PT	12:33:20.000	Load shedding	Pump	–219.00
11	PT	12:33:20.102	Load shedding	Pump	–207.00
11	PT	12:33:19.520	Load shedding	Pump	–18
11	PT	13:33:19.780	Load shedding	Pump	–18
11	PT	12:33:20.540	Load shedding	Pump	–74
11	PT	12:33:20.660	Load shedding	Pump	–75
12a	ES	12:33:20.020	Generation loss	PV	313.00
12a	ES	12:33:20.040	Generation loss	PV and thermo-solar	50.30
12a	ES	12:33:20.100	Generation loss	Wind	48.90
12a	ES	12:33:20.100	Generation loss	Combined cycle	165.00
12b	ES	12:33:20.175	Generation loss	PV	75.80
12b	ES	12:33:20.200	Generation loss	PV	305.00
12b	ES	12:33:20.235	Generation loss	PV	139.50
12b	ES	12:33:20.300	Generation loss	PV and thermo-solar	35.00
12b	ES	12:33:20.420	Generation loss	PV	131.00
12b	ES	12:33:20.476	Generation loss	PV	212.40
13a	ES	12:33:20.133	Load shedding	Pump	–69.00
13a	ES	12:33:20.180	Load shedding	Pump	–129.00
13a	ES	12:33:20.193	Load shedding	Pump	–139.00
13a	ES	12:33:20.320	Load shedding	Pump	–26.00
13a	ES	12:33:20.367	Load shedding	Pump	–25.00
13a	ES	12:33:20.405	Load shedding	Pump	–201.00
13a	ES	12:33:20.416	Load shedding	Pump	–131.00
13a	ES	12:33:20.418	Load shedding	Pump	–130.00
13a	ES	12:33:20.423	Load shedding	Pump	–130.00
13a	ES	12:33:20.506	Load shedding	Pump	–69.00
13b	ES	12:33:20.224	Load shedding	Pump	–89.00
13b	ES	12:33:20.243	Load shedding	Pump	–414.00
13b	ES	12:33:20.254	Load shedding	Pump	–79.00
13b	ES	12:33:20.259	Load shedding	Pump	–192.00

Event Cluster	Country	Time (CEST)	Event Type	Description	Electrical Power Imbalance (MW)
13b	ES	12:33:20.285	Load shedding	Pump	–51.00
13b	ES	12:33:20.306	Load shedding	Pump	–192.00
13b	ES	12:33:20.480	Load shedding	Pump	–78.00
14	ES	12:33:20.353	Load shedding	Pump	–103.00
14	ES	12:33:20.476	Load shedding	Pump	–53.00
15	ES	12:33:20.370	Load shedding	Pump	–75.00
16a	ES	12:33:20.229	Tie-line trip	Puerto de la Cruz (RE)–Beni Harchane (ONEE) 400 kV	–
16b	ES	12:33:20.473	Tie-line trip	Puerto de la Cruz (RE)–Mellousa (ONEE) 400 kV	–
17	ES	12:33:20.561	Load shedding	Pump	–181.00
17	ES	12:33:20.605	Pump disconnection	Pump	–30.00
18	IP	12:33:20.574	Load shedding 49.0 Hz	Distribution	–448.8
19	ES	12:33:20.650	Generation loss	PV	167.70
20	ES	12:33:20.650	Load shedding	Demand	–970.10
20	ES	12:33:20.810	Load shedding	Demand	–181.70
21	ES	12:33:20.740	Generation loss	PV	89.00
21	ES	12:33:21.080	Generation loss	PV	170.67
21	ES	12:33:21.254	Generation loss	PV	129.00
22	IP	12:33:20.788	Load shedding 48.8 Hz	Distribution	–1,733.39
23	ES	12:33:20.800	Load shedding	Pump	–68.00
24	PT	12:33:21.000	Generation loss	Distributed generation on DSO side (estimate)	1,073.00
25	PT	12:33:21.048	Net load disconnection	Unexpected trip due to improper activation of load shedding function	110.00
26	ES	12:33:21.060	Load shedding 48.7 Hz	Distribution	–71.03
27	PT	12:33:21.155	Load shedding	Pump	–337.00
28	ES	12:33:21.168	Load shedding	Demand	–250.70
29	IP	12:33:21.300	Load shedding 48.6 Hz	Distribution	–1,495.9
30	PT	12:33:21.390	Net load disconnection	Unexpected trip due to improper activation of load shedding function	17.00
31a	ES	12:33:21.407	Tie-line trip	Baixas (RTE)–Vic (RE) 400 kV	–
31b	ES	12:33:21.437	Tie-line trip	Arkale (RE)–Argia (RTE) 220 kV	–
31b	ES	12:33:21.535	Tie-line trip	Argia (RTE)–Hernani (RE) 400 kV	–
32	ES	12:33:21.440	Generation loss	PV	26.00
32	ES	12:33:21.503	Generation loss	PV	140.00
33	IP	12:33:21.537	Load shedding 48.4 Hz	Distribution	–1,809.09
34	PT	12:33:21.562	Load shedding	Pump	–30.00
34	PT	12:33:21.782	Load shedding	Pump	–329.00
35	IP	12:33:21.838	Load shedding 48.2 Hz	Distribution	–1,582.3
36a	ES	12:33:21.872	Generation evacuation line trip	Pinar del Rey (RE)–San Roque 1 220 kV	–
36b	ES	12:33:21.980	Generation evacuation line trip	Olmedilla (RE)–Sabinar 400 kV[22]	–
37	IP	12:33:22.080	Load shedding 48.0 Hz	Distribution	–1,566.1
38a	ES	12:33:22.160	Generation loss	PV	55.00
38a	ES	12:33:22.460	Generation loss	PV	52.00
38a	ES	12:33:22.470	Generation loss	Wind and PV	114.00
38a	ES	12:33:22.860	Generation loss	PV	57.00
38b	ES	12:33:22.600	Generation loss	PV	117.50

22 At the time of the trip, the line's active power flow was null.

Event Cluster	Country	Time (CEST)	Event Type	Description	Electrical Power Imbalance (MW)
39	PT	12:33:22.213	Load shedding	Pump	–114.00
40	ES	12:33:22.330	Generation loss	Wind and PV	54.00
40	ES	12:33:22.900	Generation loss	PV	170.50
41a	ES	12:33:22.702	Generation loss	Nuclear	1,045.00
41b	ES	12:33:23.460	Generation loss	Combined cycle	166.00
41c	ES	12:33:23.515	Generation loss	Nuclear	653.00
42	ES	12:33:23.076	Line trip	Maguilla (RE)–Valdecaballeros (RE) 400 kV	–
43a	ES	12:33:23.140	Generation loss	PV	125.00
43a	ES	12:33:23.360	Generation loss	PV	117.50
43a	ES	12:33:23.360	Generation loss	PV	167.50
43b	ES	12:33:23.400	Generation loss	PV	380.00
43a	ES	12:33:23.260	Generation loss	PV	57.00
44	PT	12:33:24.519	Generation loss	Hydro	86.19
45	ES	12:33:23.520	HVDC block	HVDC Iberian Peninsula (RE)–Majorca (RE)	–
46	ES	12:33:23.590	Generation loss	Nuclear	1,692.00
47	ES	12:33:23.648	Generation loss	Combined cycle	87.00
48	PT	12:33:23.644	Generation loss	Combined cycle	330.69
48	PT	12:33:23.790	Generation loss	Hydro	64.76
49	PT	12:33:23.837	Generation loss	Hydro	40.12
49	PT	12:33:23.845	Generation loss	Hydro	39.62
50	ES	12:33:23.883	Pump disconnection[23]	Pump	–78.00
50	ES	12:33:23.888	Pump disconnection[24]	Pump	–78.00
51	ES	12:33:23.960	HVDC block	HVDC Iberian Peninsula (RE)–France (RTE)	–
52	PT	12:33:23.954	Line trip	Mogadouro (REN)–Central da Valeira (REN) 220 kV	
53	PT	12:33:24.195	Generation loss	Hydro	65.34
54	ES	12:33:24.230	Generation loss	Combined cycle	78.00
54	ES	12:33:24.725	Generation loss	Combined cycle	83.00
54	ES	12:33:24.800	Generation loss	Combined cycle	140.00
55a	ES	12:33:25.060	Tie-line trip	Aldeadávila (RE)–Lagoaça (REN) 400 kV	
55b	ES	12:33:25.144	Tie-line trip	Cartelle (RE)–Lindoso (REN) (ckt 2) 400 kV	
55c	ES	12:33:25.183	Tie-line trip	Cartelle (RE)–Lindoso (REN) (ckt 1) 400 kV	
55d	ES	12:33:25.325	Tie-line trip	Cedillo (RE)–Falagueira (REN) 400 kV	
55e	ES	12:33:25.410	Tie-line trip	Saucelle (RE)–Pocinho (REN) 220 kV	
55f	ES	12:33:26.165	Tie-line trip	Aldeadávila (RE)–Pocinho (REN) (ckt 1) 220 kV	
56	PT	12:33:24.434	Line trip	Lagoaça (REN)–Armamar (REN)	
57	PT	12:33:26.078	Generation loss	Hydro	45.81
57	PT	12:33:26.080	Generation loss	Hydro	45.82
57	PT	12:33:26.081	Generation loss	Hydro	45.96
58	PT	12:33:27.154	Generation loss	Hydro	24.10
59	PT	12:33:29.225	Generation loss	Hydro	39.36
59	PT	12:33:29.317	Generation loss	Hydro	61.40
60	PT	12:33:37.800	Generation loss	Hydro	49.04

Table 3-1: Sequence of events (imbalances shown in red for >500 MW, in orange for >50 MW, and in yellow for the remainder)

23 This pumping unit was tripped by the undervoltage protection function, without the underfrequency protection function having been triggered beforehand.

24 This pumping unit was tripped by the loss of synchronism protection function, without the underfrequency protection function having been triggered beforehand.

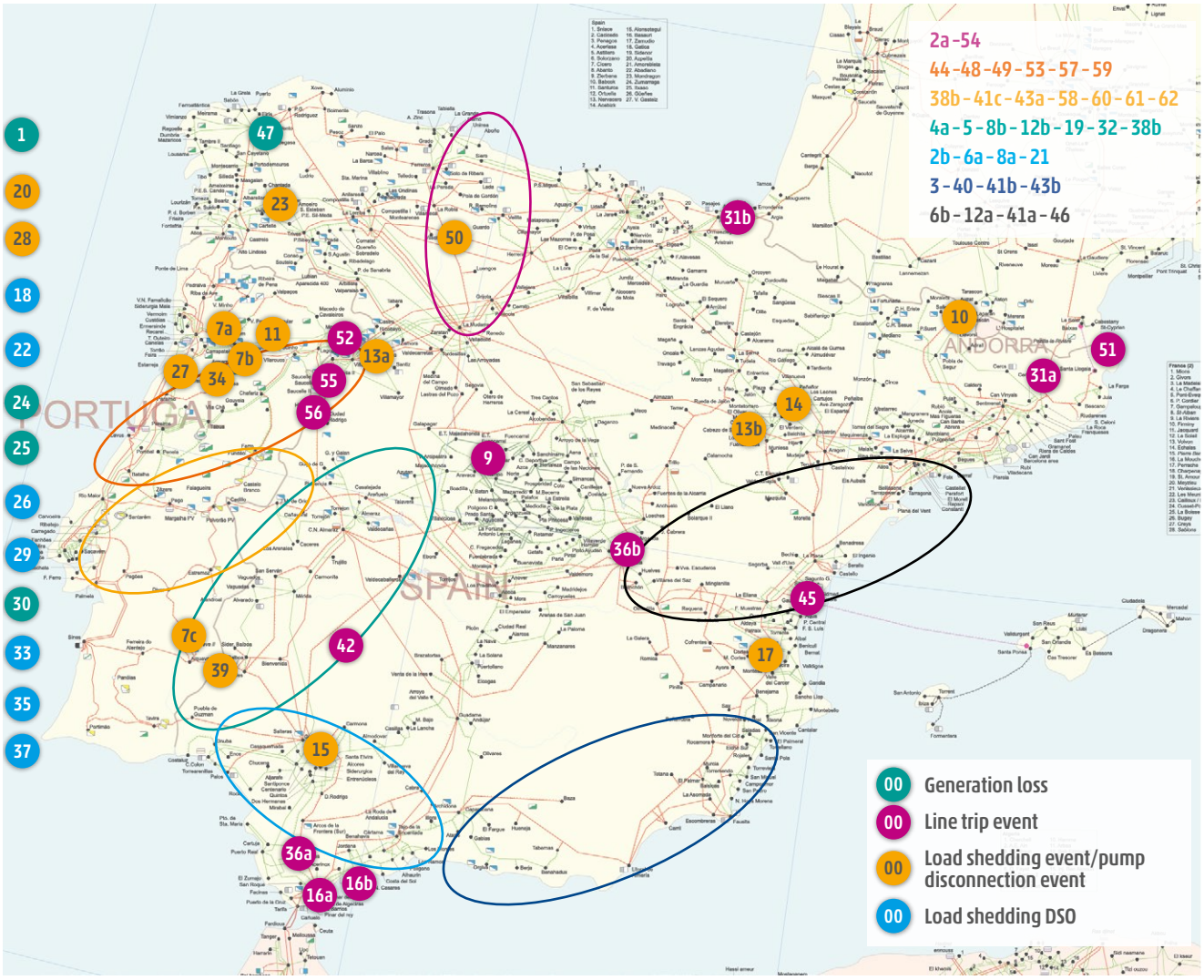


Figure 3-1: Temporal and spatial representation of the events in the Spanish and Portuguese power system. The events noted on the left side of the map refer to distributed resources. The events mentioned on the top-right corner of the map are not precisely located on the map for confidentiality reasons: their approximate location is indicated by the circled area in the corresponding colour (e.g. event 2b is located in the area outlined in blue)

The sequence of the main events can be summarised as follows:

- From 12:32:00.000 to 12:32:57.000, there was a net load increase of approximately 317 MW on the distribution grids **(event 1)** in Spain. A detailed analysis regarding the potential influencing factors of event 1 is included in Chapter 4 in the Analysis on TSO-DSO Flows and Load Changes subchapter.
- From 12:32:00.000 to 12:32:57.000 there was a fast downwards change of operating point of generation units without ramping limitation (not corresponding with communicated program) or disconnection of these units due to unknown reasons representing approximately 208 MW from wind and solar generation facilities >1 MW, mostly connected to the TSO grid in both northern **(event 2a)** and southern **(event 2b)** Spain.

  For the events classified as 2a and 2b, the Expert Panel notes that the limited quality of the available data[25] does not allow for a precise determination of facilities behavior between 12:32:00 and 12:32:57. The absence of oscillography data, inconsistencies in setpoint information, and the limited detail provided by the available SCADA measurements prevent any definitive identification of unit tripping or exact operational changes.

  Given these constraints, the Expert Panel concludes that the most plausible explanation is that an unquantified share of the 208 MW portfolio experienced either disconnection or a fast operational change during the interval of interest, recognising that the existing data do not support a more detailed assessment.
- Some milliseconds (ms) after 12:32:57, a generation evacuation power transformer owned by a third party A and operated by third party B in the 400 kV transmission substation (TS) 1 – Granada tripped (**event 3**). This transformer transfers power from several wind and solar power plants to the Spanish transmission system. At the moment of the trip, the active power flow on this transformer was approximately **355 MW**, from the 220 kV side to the 400 kV side.

  The owner reported the cause of the trip as the overvoltage function on the 220 kV side of the transformer. As reported by the owner, at 12:32:57.155, the protection equipment on the 220 kV side recorded a voltage of 242.9 kV on phase B and tripped (the protection setting for the overvoltage protection relay was Ut = 110 % Un (242 kV), t = 0 s). At 12:32:57.215, coinciding with the transient associated with the circuit breaker opening, phases A and C tripped (at registered voltage 244.3 kV).

  The owner provided no voltage measurement for the 400 kV side of the transformer. A COMTRADE recording is available at the 400 kV TS 1 – Granada, coinciding with the tripping of the 220 kV transformer circuit breaker. This recording was captured by a transmission line protection system at the aforementioned substation: the voltage value observed in this record at the moment of the transformer trip was 417.9 kV.

  In a facility connected downstream of the 400/220 kV transformer that tripped, and linked via a 500-metre line to the 220 kV Generation substation 1 – Granada, an oscillography record is available with voltage measurements at the 220 kV level, showing that the voltage was 237.5 kV at the moment of the transformer trip. According to SCADA data, the reading at 12:32:56 was 233.4 kV (1.061 pu) at the transformer on the 220 kV voltage side.
- In the 12:33:16–12:33:17 time frame, an additional **727 MW** of thermo-solar and PV generation tripped in 400 kV TS 1 – Badajoz and 400 kV TS 2 – Badajoz in central-southern Spain (**event 4**).

  In the 400 kV generation substation (GS) 1 – Badajoz, an evacuation line tripped at 12:33:16.443 for unknown reasons. GS 1 – Badajoz is a collector substation for generation evacuation, with its connection point located at the 400 kV TS 1 – Badajoz. At the time, the disconnected generation plants were injecting 582 MW into the grid. Immediately afterwards, the generation connected downstream of the transformer disconnected due to overfrequency.

  The transmission line monitored by the PMU at 400 kV TS 1 – Badajoz was out of service during the event; therefore, no PMU measurements from this substation are available. The last SCADA voltage reading in substation 400 kV TS 1 – Badajoz (at an unknown phase) at 12:33:10 was 434.0 kV at evacuation bay, 430.5 kV at busbar 1, and 429.9 kV at busbar 2.

  At 12:33:16.420, the positive sequence voltage in the PMU at 400 kV TS 3 – Cáceres was 431.72 kV, and the voltage measure in phases A, B, and C in the same PMU were 428.1, 436.8, and 430.2 kV, respectively. A significant difference is observed between the phase B voltage measurement and the other two phases, ranging between 6 and 8 kV. This may be partly due to system imbalance but also to measurement errors. To assess this, COMTRADE records from protection relays dated 22 April were analysed.

25 At the request of the Expert Panel, data were provided by the generation facility owners. Upon comparison with the information supplied by the TSO, discrepancies were identified. For some of these units, the Expert Panel received unit tripping data showing protection activations occurring after 12:33:16. In addition, for some units, the Expert Panel received voltage data showing that voltage in 12:32:00 – 12:32:57 did not exceed operational limits, at those units.

These records show that, in four bays of the same substation, the difference between phase B and the other two phases is approximately 2.15 kV, whereas in the bay from which the PMU measures, the difference is 5.34 kV. Therefore, it is possible that the voltage transformer for phase B is measuring about 3 kV higher than the actual voltage of that phase in the substation.

At 12:33:16:443, the positive sequence voltage at 400 kV TS 3—Cáceres was 43,278 kV, while the power flow was 907.21 MW towards a substation in the north. These values already reflected the changes caused by the generation loss. At 12:33:16.680, the positive sequence voltage at this substation reached 440.0 kV. The voltage measured by the PMU then remained above 440 kV until 12:33:23.600.

A PV power plant connected to it, 400 kV TS 2—Badajoz, tripped at 12:33:16.820 for unknown reasons. At the time, the plant was injecting 145 MW into the grid.

The PMU located in the substation was out of service at the time of the event. The last SCADA voltage reading in the substation was 436.1 kV at 12:33:16.

- In the 12:33:17–12:33:18 time frame, approximately **928 MW**[26] of wind and solar generation disconnected in the 400 kV TS 1—Segovia, 400 kV TS 1—Seville (event 5a), 400 kV TS 1—Badajoz, 220 kV TS 1—Huelva, 220 kV TS 1—Cáceres, and 220 kV TS 3—Badajoz (**event 5b**).
- After event 5b, the high voltage and low frequency profiles in the Spanish transmission system led to cascading tripping of many other generators, both conventional and renewable.
- At 12:33:20.473, the AC interconnection to Morocco tripped due to underfrequency (the frequency threshold of the protection is 49.5 Hz). At the same time, pump storages disconnected both in Spain and Portugal in line with the underfrequency defence plan schemes.
- At 12:33:21.535, the AC interconnection to France was lost, leading to a system split between the Iberian Peninsula and the CE SA due to a loss of synchronism relay intervention.
- After the AC separation of the Iberian Peninsula caused by the AC synchronism loss, the power imbalance continued to increase, causing the frequency to drop further. The HVDC link between Santa Llogaia and Baixas tripped at 12:33:23.960, completing the electrical separation of the Iberian Peninsula from France.

3.1.2 Evolution of Electrical Quantities During the Incident

This subchapter describes the evolution of the main quantities that characterise the event in the main high voltage (HV) buses of the affected TSO networks (i.e. voltage magnitudes and phases, frequency, and active power in the minute before the blackout event).

The data used for these factual analyses are collected from PMUs installed across the affected TSOs grid.

Figure 3-2 shows voltage magnitude heat maps at various moments in the Iberian Peninsula and southern France. The figure clearly shows the increase of voltage magnitudes starting in the Spanish network. At 12:32:58, the positive sequence voltage magnitudes were over 1.05 pu. At 12:33:17, maximum voltage magnitudes reached 442.5 kV (1.106 pu) in the Spanish grid. One second later, at 12:33:18, voltage magnitudes reached 451.4 kV (1.128 pu), with higher values in the southern part of Spain.

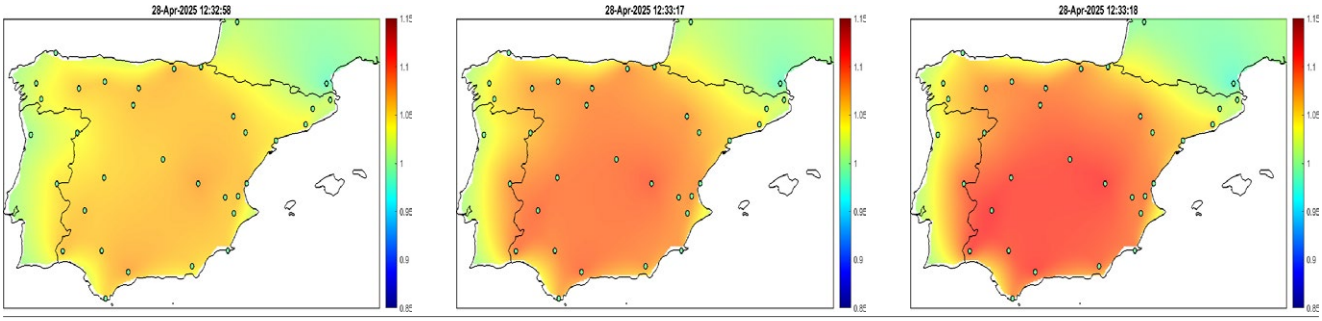


Figure 3-2: Heat map of the Iberian Peninsula for the temporal instant (a) 12:32:58, (b) 12:33:17, and (c) 12:33:18 showing voltage magnitudes at different substations

26 This is the total amount of power that could be estimated based on measurements. However, the effects on frequency deviation suggest that this trip could have been up to 1,100 MW.

3.1.2.1 Spain

The dataset used for the analysis reported in this section comes mainly from Spanish PMU devices located as shown in Figure 3-3.

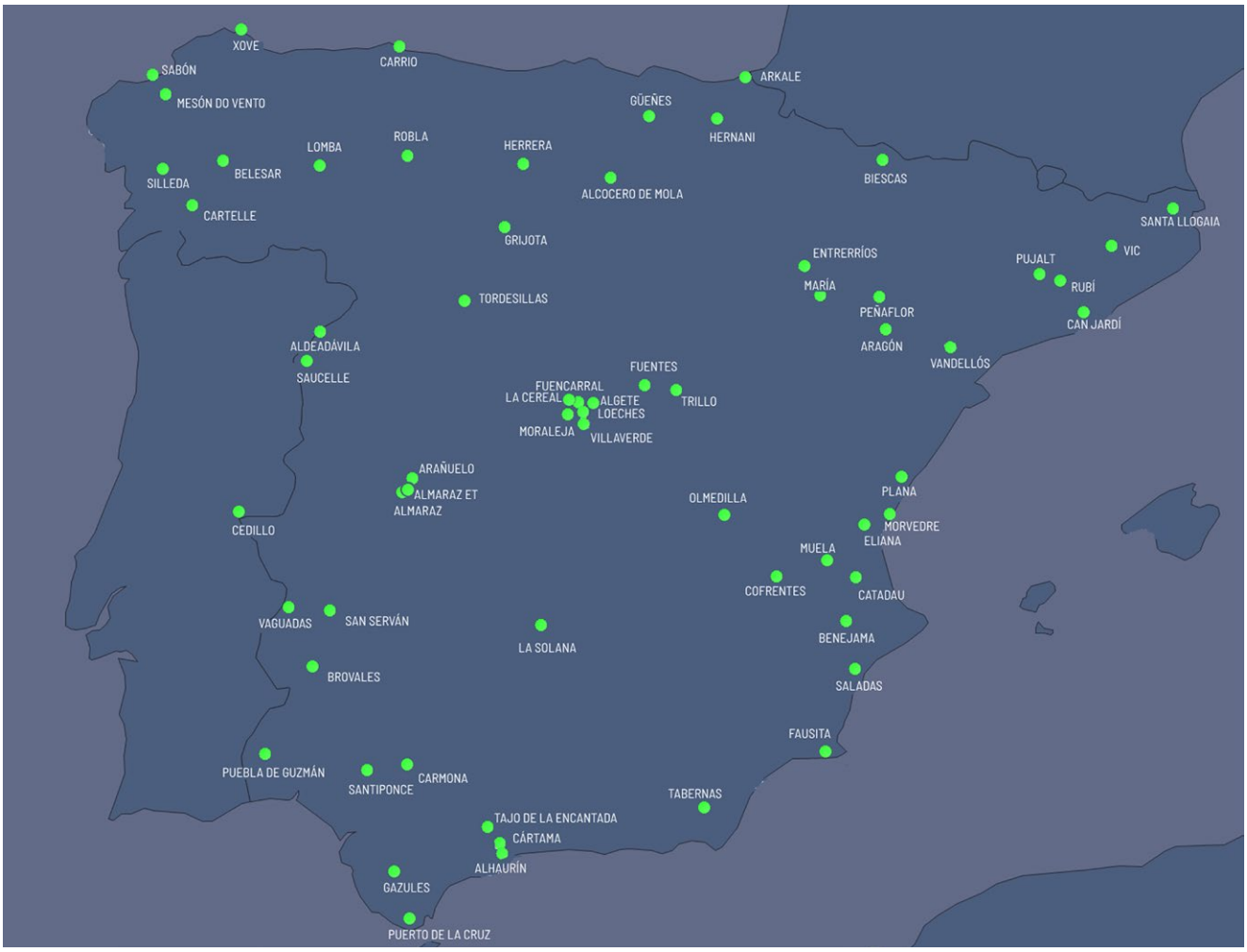


Figure 3-3: Location of PMU devices in the Spanish power system

Figure 3-4 shows the evolution of the electrical quantities in the main buses of the Spanish power system from 12:32:00 to 12:33:25 on 28 April 2025.

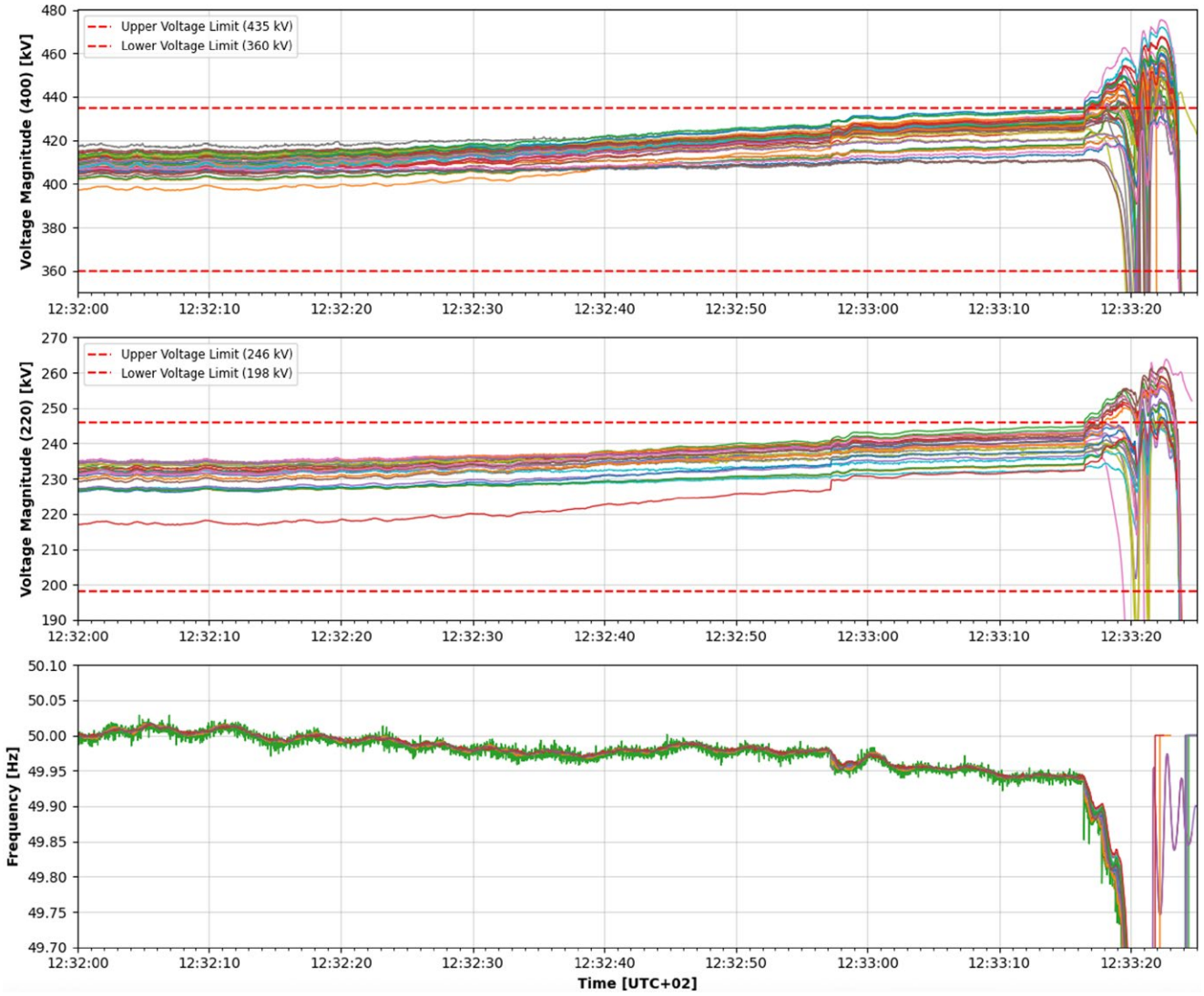


Figure 3-4: Evolution of the electrical quantities in the Spanish power system during the event

3.1.2.1.1 Voltage

Admissible voltage ranges in the Spanish transmission system

More detailed explanation of voltage limits in Spain is included in Chapter 2, Section 2.6. According to these prescriptions, voltage in Spain can range:

- from 375 to 435 kV for the 400 kV network
- from 200 to 245 kV for the 220 kV network

Voltage profiles in the Spanish transmission system

Unless otherwise specified, references to PMU voltage values in this section refer to the positive-sequence voltage component, defined as $V_+ = {}^1/_3 \times (V_a + a \times V_b + a^2 \times V_c$, where $a = e^{j120°}$.

Voltage protection of transmission lines is based on the measurement of phase-to-ground voltage. Voltage magnitude profiles, shown in Figure 3-5, indicate that voltages at the measuring points on the transmission system remained within operational limits until 12:33:16 (event 4) at all buses in the observed perimeter, both on the 400 kV and 220 kV voltage levels. At 12:32:00, voltage began to increase rapidly, possibly also due to the progressive disconnection between 12:32:00 and 12:32:55 of 208 MW of wind and solar generation connected to the distribution grids, along with an increase of 317 MW in net load that cannot be directly identified but possibly involve rooftop PV generation (total disconnection of roughly 525 MW). The Expert Panel made this preliminary assessment based on the flows measured by PMUs at the borders with neighbouring countries.

At 12:32:57, the trip of a power transformer in the "400 kV TS 1—Granada" (event 3) caused the disconnection of 355 MW generation from PV, wind, and thermo-solar power plants. The reactive power being absorbed through this transformer was 165 Mvar. At that time, the voltage in the HV network further increased almost instantaneously, while remaining below 435 kV (i.e. within operational limits).

At 12:33:16.680, after the disconnection of generation connected to "400 kV TS 1—Badajoz", the positive sequence voltage in at least one bus in the Spanish system exceeded 440 kV (1.10 pu). The disconnections in the "400 kV TS 2—Badajoz" substation at 12:33:16.820 (completing event 4) led to voltage beyond operational limits at several buses in the Spanish transmission system. The loss of reactive power absorption at those two substations was 165 and 38 Mvar, respectively.

At 12:33:17, the disconnection of approximately 930 MW of PV, and wind power plants connected to several 400 kV and 220 kV substations led to voltages exceeding 440 kV in several substations.

Figure 3-5: Evolution of the voltage in the Carmona substation and of the net active power exchange position between Spain and neighbouring countries between approximately 12:32:00 and 12:33:00 (source: PMU data from Red Eléctrica)

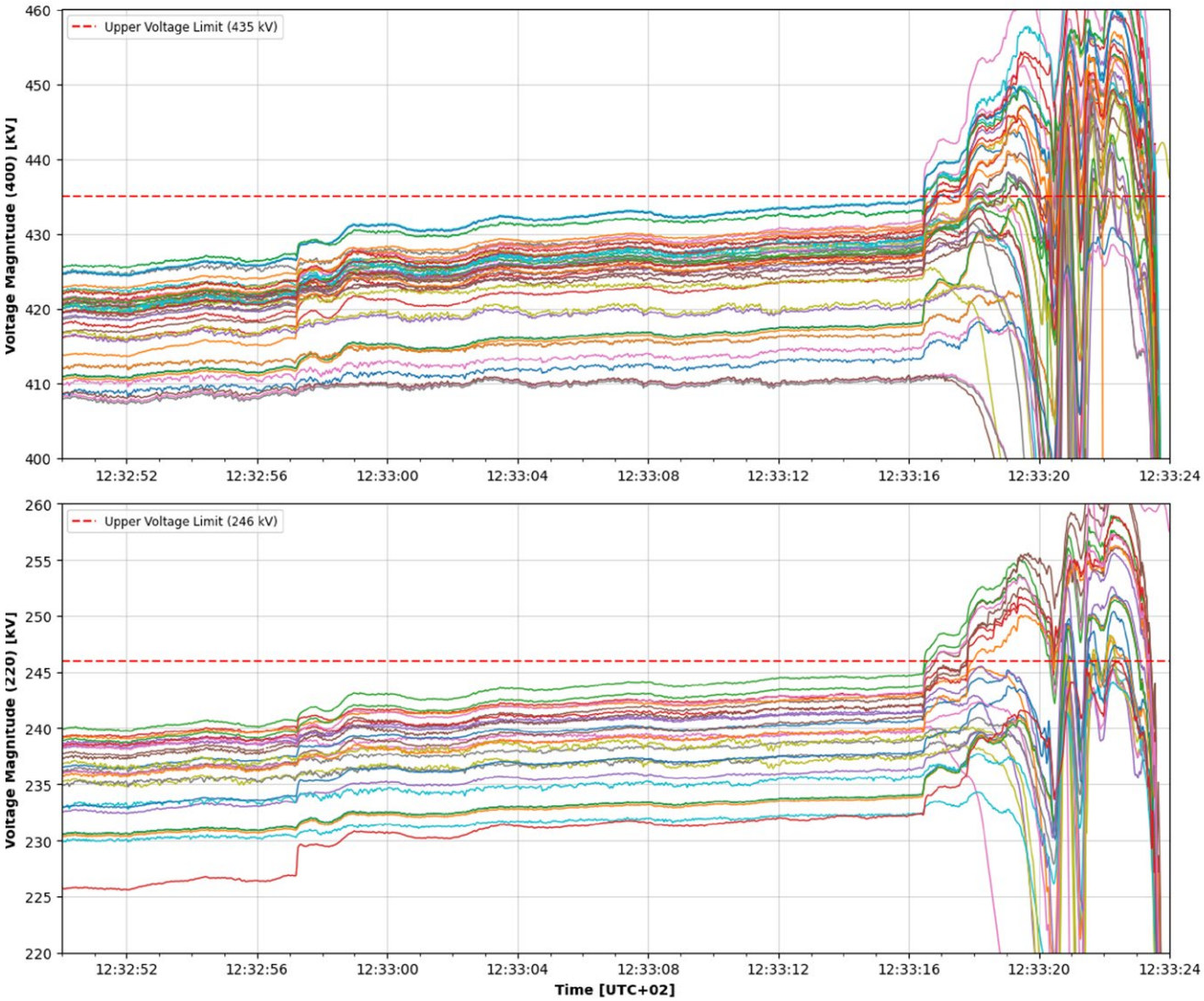


Figure 3-6: Voltage in the Spanish power system during the incident

At this stage, voltage control was fully lost, with cascading disconnection of further power generators leading to the loss of synchronism of the Iberian Peninsula with continental Europe. The low frequency (49.5 Hz) triggered underfrequency protection in Morrocco, and the border was opened at 12:33:20.4.

The loss of synchronism triggered DRS (*Débouclage suite à Rupture de Synchronisme, translated in this document as loss of synchronism protection* – see Section 3.2.2.4) on the France–Spain border, leading to the AC separation of the Iberian Peninsula from the French power transmission system at 12:33:21.5, disconnection of the ES–FR HVDC at 12:33:23.960, and blackout at 12:33:27.

Figure 3-7 depicts, in blue, the sum of net load increase (event 1) and accumulated generation loss in the Iberian Peninsula system[27] and, in orange, the maximum voltage profile on the 400 kV network. Event 1 is displayed as one step on this figure at 12:32:00, as insufficient information is available regarding the distribution of this event between 12:32:00 and 12:32:57.

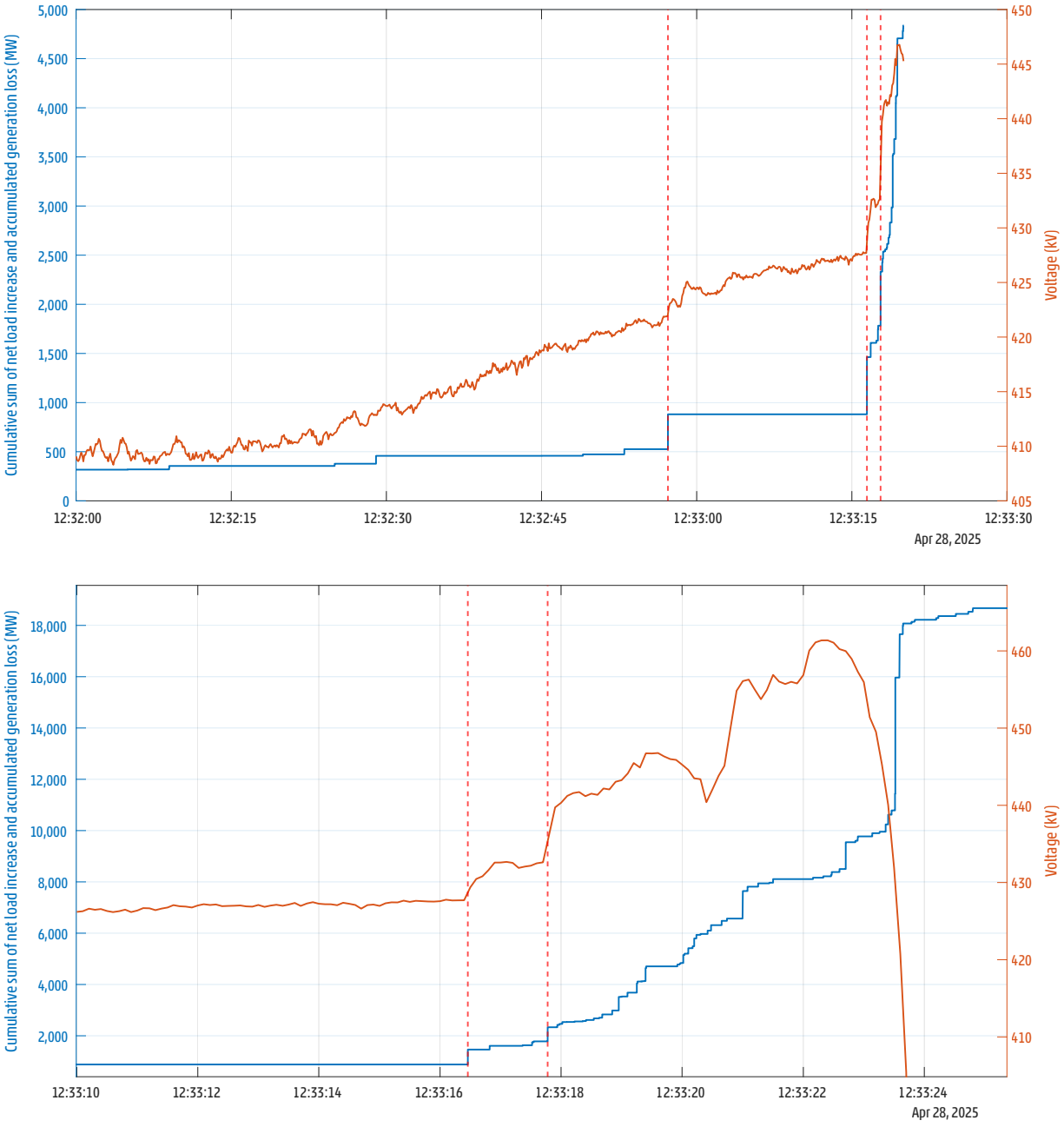


Figure 3-7: Accumulated generation loss combined with net load increase (event 1) in the Iberian Peninsula system vs 400 kV maximum voltage. Dotted lines represent the three main generation disconnection events (event 3, event 4, and event 5).

27 Net load increase and accumulated generation loss is based on the events reported in Table 3-1.

3.1.2.1.2 Frequency and Rocof

The sequence of events caused the frequency in Spain to rapidly deteriorate after the loss of generation. Figure 3-8 shows the frequency measurements and RoCoF in the last minute before the blackout. The RoCoF was calculated using a 500 ms sliding window accordingly to the best practices suggested by ENTSO-E.

Figure 3-8: Frequency and RoCoF in the Spanish power system during the incident

In the minutes before the event, the frequency in Spain was stable, with typical fluctuations for a large, interconnected system. Before 12:32:00, the frequency was around the nominal value of 50 Hz. Following the progressive increase in net load (event 1) and loss of generation (events 2a and 2b) at distribution level, the frequency began to decrease, dropping to 49.98 Hz from 12:32:00 to 12:32:55.

At 12:32:57, the sudden disconnection of 355 MW of generation (event 3) caused the frequency to drop further, to 49.94 Hz.

At 12:33:16, the loss of an additional 725 MW of generation (event 4) led the frequency to further decrease, to around 49.9 Hz. From that point onwards, the frequency in Spain and Portugal began deviating from that of continental Europe and did not return to synchronism.

At 12:33:17, the disconnection of at least 930 MW of generation (event 5) led to frequency values of around 49.8 Hz in Spain and Portugal.

Before 12:33:20, at least 2,600 MW more generation was disconnected, and the frequency in the Iberian Peninsula dropped to 48.5 Hz. At the same time, the frequency in continental Europe reaccelerated towards 50 Hz. The AC interconnection lines between France and Spain were still in service at that time, but the Iberian Peninsula lost synchronism with continental Europe at around 12:33:19.62. The loss of synchronism caused large amplitude swings in electric power across the AC interconnection lines between Spain and France, as shown in Figure 3-10 (middle).

In the following five seconds, cascading events caused a further ~–8,000 MW imbalance within the Iberian Peninsula transmission system, caused by the disconnection of ~7,000 MW of generation and the disconnection of ~15,000 MW of loads and pumping hydro from the defence system. Table 3-2 shows a breakdown of the disconnected power by type of energy source.

Type of energy source	Aggregated disconnected power (MW)
Combined Cycle	719.0
Load	-8,504.3
Nuclear Power Plant	3,390.0
Pumped Hydro	-2,756.0
PV	3,198.1
PV and Thermo-Solar	85.3
Wind	48.9
Wind and PV	168.0
Total generation	7,609.3
Total demand + Pump	-11,260.3

Table 3-2: Focus of power disconnection from 12:33:20 to 12:33:25

Figure 3-9 depicts the overall generation loss in the Iberian Peninsula system and the frequency trajectory during the incident.

Figure 3-9: Cumulative power imbalance in the Iberian Peninsula system vs average frequency.

Figure 3-10 shows frequency and RoCoF in the Spanish transmission system estimated over a 500 ms sliding window, as suggested by the most relevant scientific literature and ENTSO-E best practices on the topic.

This approach is due to the need to calculate a rocof related to the average system frequency, a representative indicator of the overall dynamic response of the grid[28]. It can be observed that immediately after 12:33:20, the absolute values of RoCoF were well beyond 1 Hz/s.

28 https://eepublicdownloads.entsoe.eu/clean-documents/SOC%20documents/Regional_Groups_Continental_Europe/2018/TF_Freq_Meas_v7.pdf

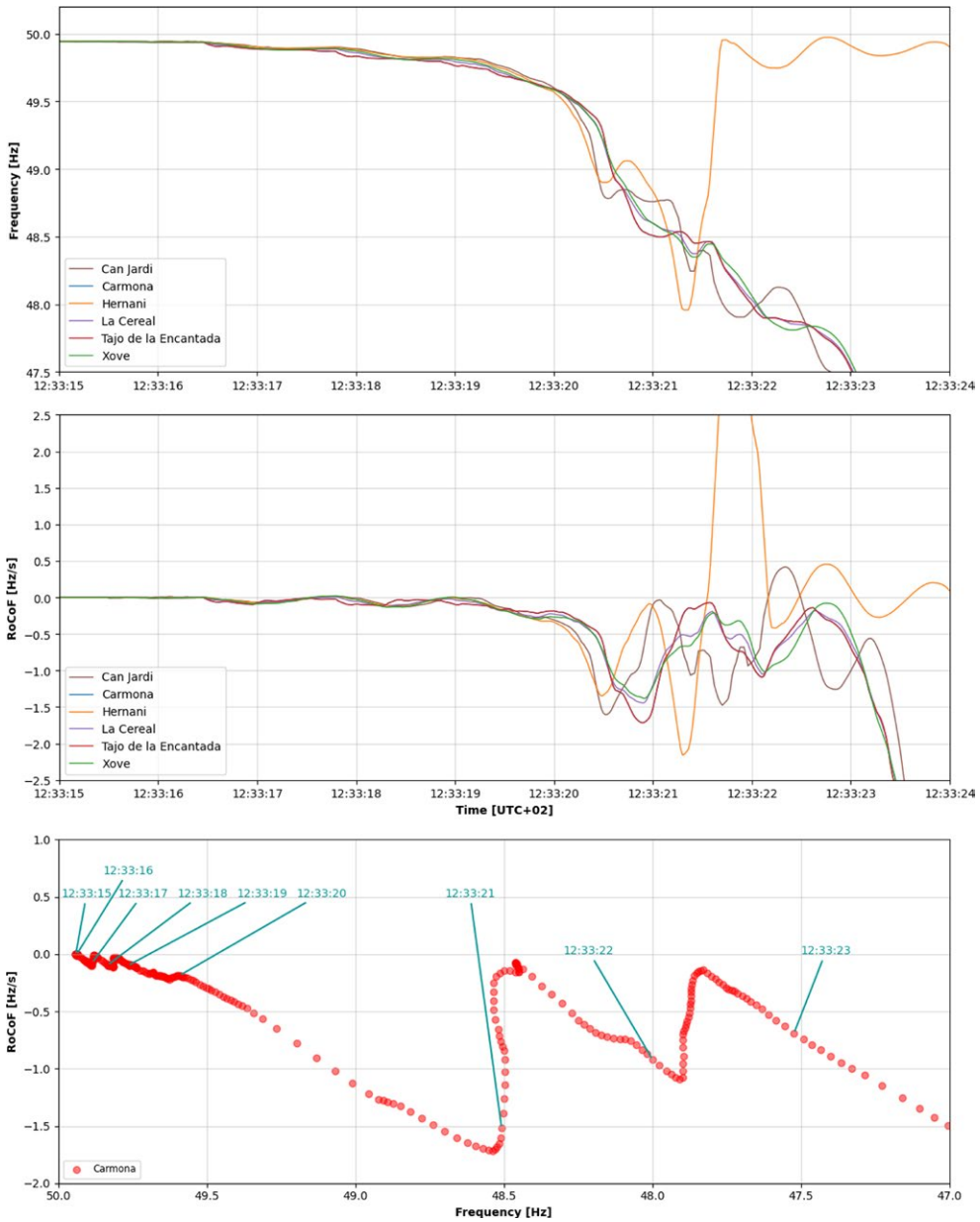


Figure 3-10: Frequency and RoCoF in the Spanish transmission system

Figure 3-10 shows how in certain substations near the France–Spain border, high RoCoFs were observed during the transient caused by the loss of synchronism. During the transient associated with the loss of synchronism, significant frequency drops were recorded in the border area when power was flowing from Spain to France; conversely, when power was flowing from France to Spain, sharp frequency increases were observed. Therefore, the RoCoFs observed near the France–Spain border were not only caused by the demand generation imbalance within the Iberian Peninsula but were also strongly influenced by the transient associated with the loss of synchronism. This occurred due to the loss of a massive amount of generation in the Iberian Peninsula, which led to such a high level of active power import from France that the stability angle (90°) in the border was exceeded. Beyond this angle difference, any further increase results in a decrease in imported power rather than an increase, thereby worsening the generation-demand imbalance in the system. When the angle difference reaches 270°, at which point electricity exports from Spain to France reach their maximum level, power exports from Spain to France begin to decrease. This leads to a reversal in the direction of cross-border flows, with power starting to be imported from France into Spain. This process continues to repeat with increasing speed due to the growing frequency difference between systems, until the systems are fully separated.

To more accurately assess the impact of the demand generation imbalance on frequency during the early stages, Figure 3-11 shows the frequency and RoCoF at the La Cereal substation, located in the north of Madrid – therefore near the centre of the Iberian Peninsula – along with the total power exchange between the Iberian Peninsula and France and Morocco.

Figure 3-11 shows that the RoCoF of –1 Hz/s was reached at 12:33:20.560. At that moment, Spain had experienced the disconnection of at least 5,750 MW of generation and the load shedding of nearly 2,500 MW from pumped-storage facilities. In Portugal, 2,100 MW of pumped-storage load had also been shed. Additionally, due to the loss of synchronism, the Iberian Peninsula was exporting 5,440 MW – representing a difference of approximately 5,000 MW compared to the value at 12:33:16.443 – thus further increasing the existing demand generation imbalance in the Iberian Peninsula. Therefore, at the moment when the rate of change of –1 Hz/s was first reached, the demand generation imbalance in the Iberian Peninsula amounted to at least 6,150 MW.

The actual imbalance at that moment may have been even greater, as the precise disconnection times for most of the generation could not be determined, and it is also possible that rooftop PV were disconnected at that time Figure 3-11 also shows that the RoCoF threshold of –2 Hz/s – established in the Spanish TED 749/2020 Order and Portuguese regulation Portaria 73/2020 as the value that generation units should withstand – is reached at approximately 12:33:23.360, nearly seven seconds after event 4a occurred (it should be noted that only power generation modules (PGMs) connected after the regulation's entry into force are required to maintain their performance within this threshold). Following the disconnection of a large amount of generation triggered by the overvoltage event in the system, the estimated RoCoF value during the transient exceeded the withstand capabilities for which the large new generators were designed.

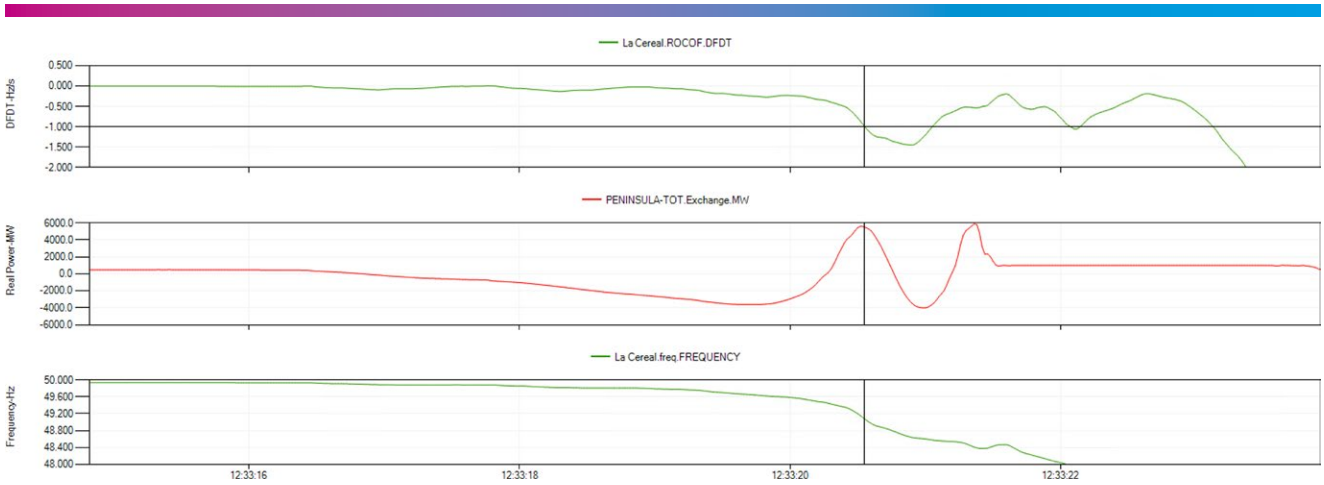


Figure 3-11: RoCoF and frequency in the La Cereal 400 kV substation and total exchange between the Iberian Peninsula and France and Morocco.

In Figure 3-11b it is possible to show detection of the point of no return w by compiling the available WAMS measurements of two PMUs, one from each of the separation areas.The space trajectory representation clearly shows how the no return boundary between the two equivalent systems results in asynchronism.

More in detail it is visible the angular swing that leads to the intervention of asynchronism protections on the french border; it is worth to recall that the protection action avoid the propagation of the perturbation to the whole central europe grid.

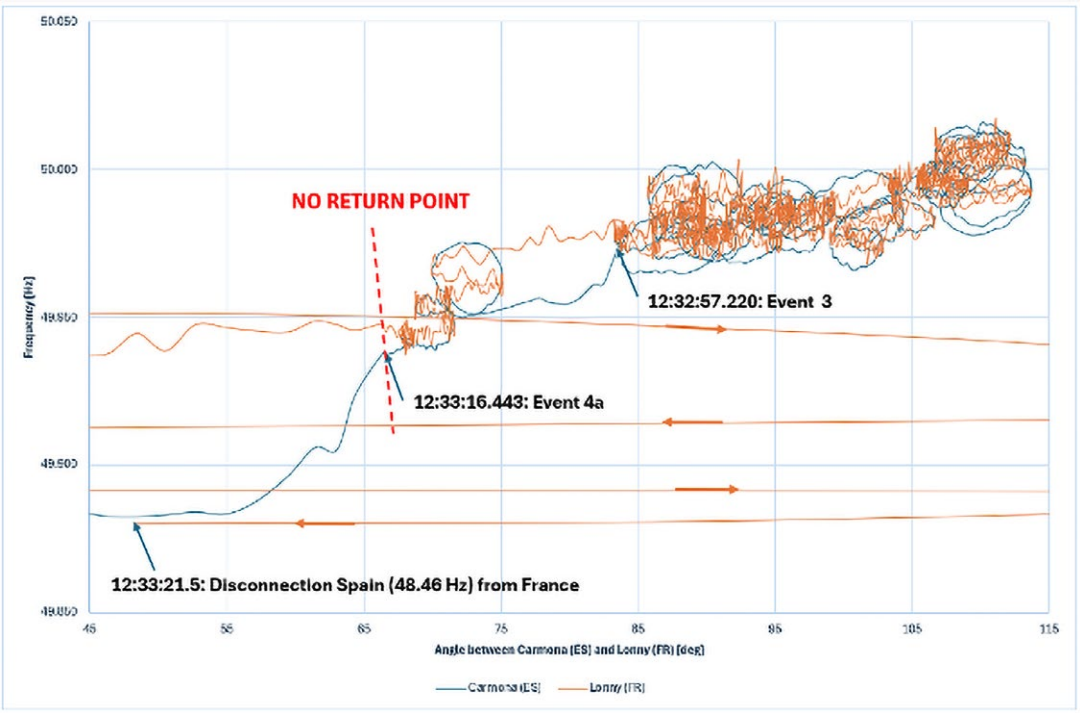


Figure 3-11b: Illustration of the Point of no Return by using WAMS measurement

3.1.2.1.3 Active Power Flow on the Border

Figure 3-12 depicts the evolution of active power flows on the Spanish power system borders to France, Portugal, and Morocco. Positive values are exports from Spain.

The connections with Morocco and Portugal did not experience any changes – the flows remained constant throughout the pre-event period at a stable value of 750 MW and 2,100 MW, respectively.

On the border with France, active power flows exhibited a visible change (a decrease of 1,500 MW) between 12:32:00 and 12:33:00.

After the 4a event, Spain began importing until the lines with France and Morocco tripped due to the intervention of loss of synchronism relays.

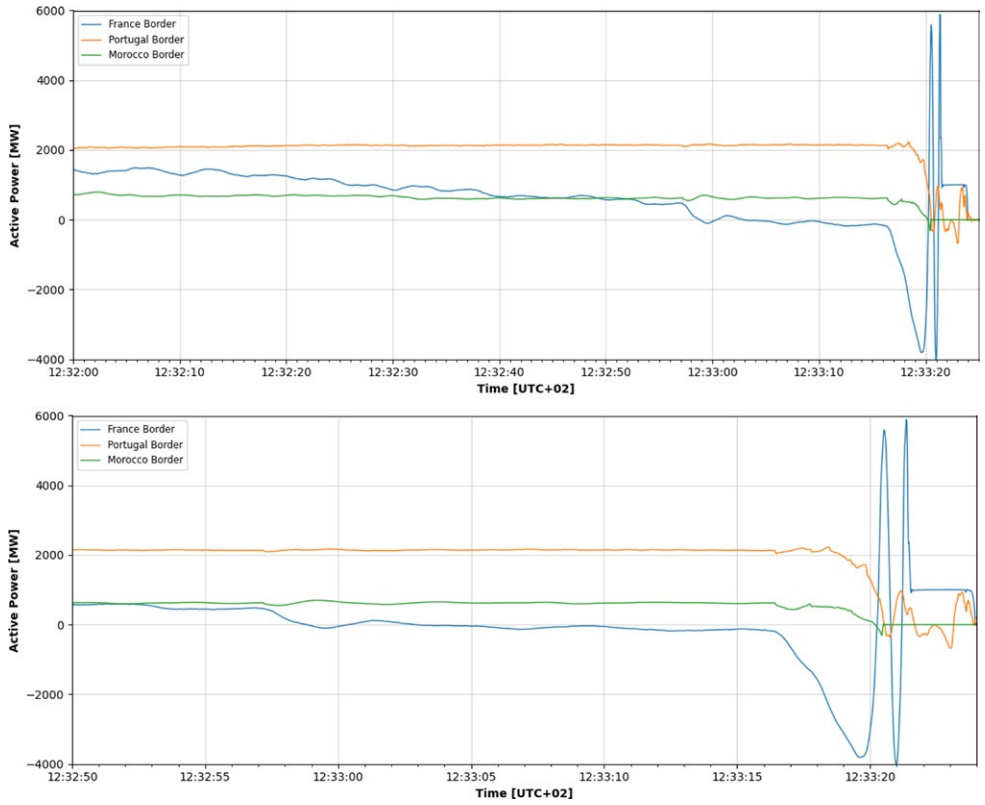


Figure 3-12: Active power on the Spanish power system borders.

Figure 3-13 shows the correlation between voltage magnitude and angle with respect to the net position of the Spanish grid (which is mainly influenced by the exchange with France).

The graphs show that as soon as exchange began to decrease – due to the events 1, 2a and 2b – the voltage increased noticeably.

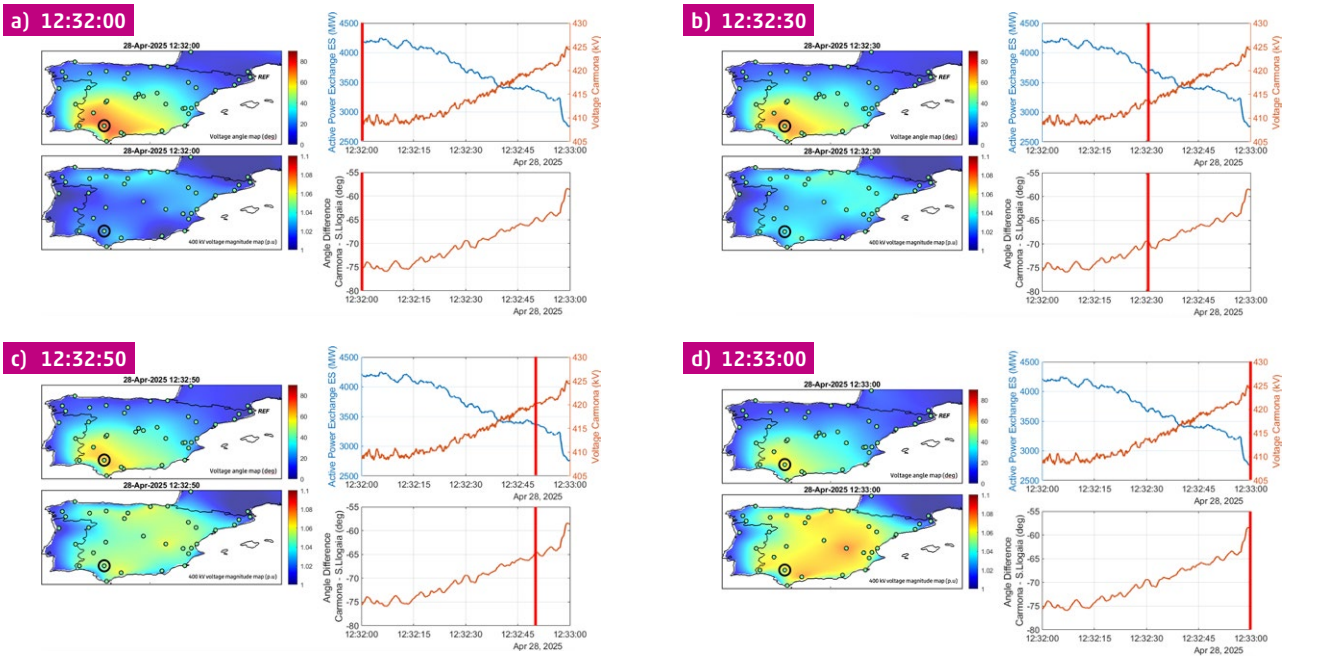


Figure 3-13: Voltage phasors vs physical net position of Spain (the reference angle is indicated in the maps as REF)

3.1.2.2 Behaviour of the HVDC link

This section presents the behaviour of the HVDC link Santa Llogaia–Baixas. At the time the incident began (12:32:00), the HVDC was set to the following setpoints:

- Active power: 2 × 500 MW in the Spain to France direction
- AC voltage setpoint at the Spanish end: 405 kV
- AC voltage setpoint at the French end: 395 kV

Figure 3-14 shows the power flow and AC voltage on the PMU in Santa Llogaia from 12:32:00 to 12:33:25.

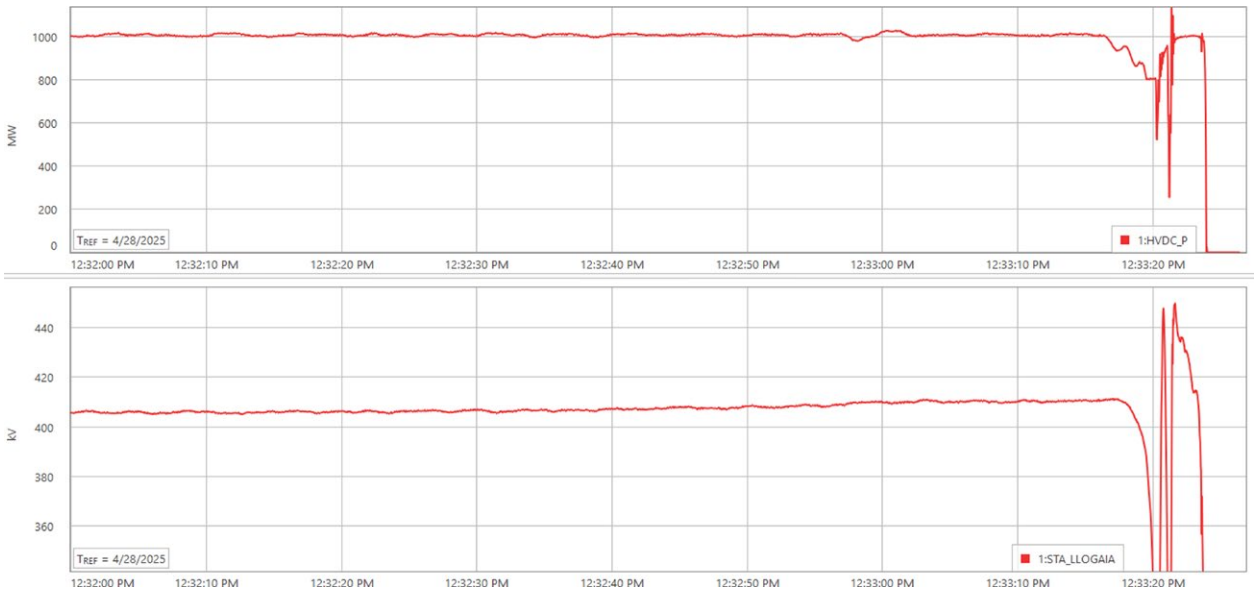


Figure 3-14: Active power through both HVDC links and voltage at Santa Llogaia 400 kV

With regard to the power flow on the France–Spain border, Figure 3-15 complements Figure 3-12, showing the active power flow on the two HVDC pole and in the 3 AC lines in service (2 of 400 kV and 1 of 220 kV) for the time frame 12:33:15 –12:33:25. The chart shows the flow on the AC lines dropping to 0 MW at 12:33:21 following the line trips triggered by the loss of synchronism protections. The HVDC continued exporting energy from Spain to France until 12:33:24, when the flow on the HVDC line dropped to zero as well, after the voltage collapse in Santa Llogaia.

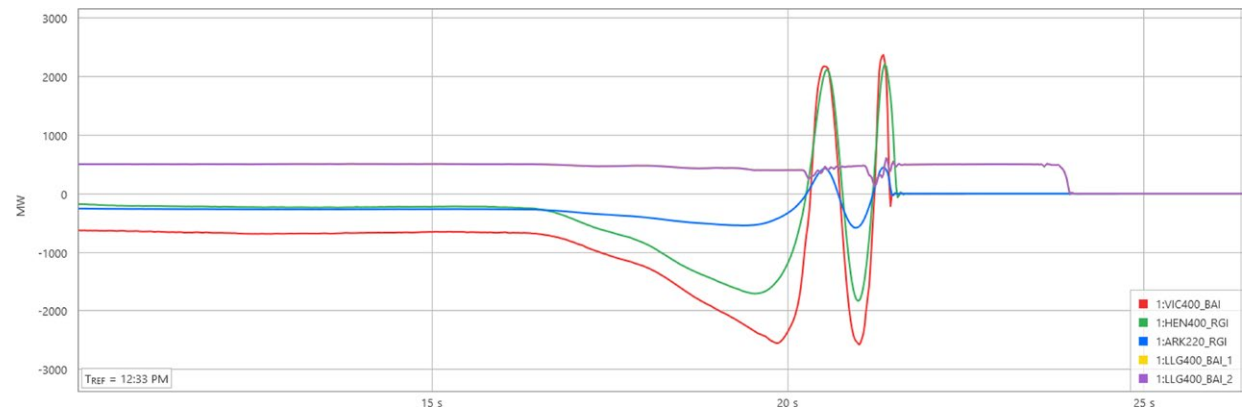


Figure 3-15: Active power through France–Spain interconnections (positive indicates power flow from Spain to France)

At 12:33:00, the voltage at the HVDC nodes was 393 kV in Baixas and 410 kV in Santa Llogaia. The HVDC was absorbing 2 × 440 Mvar of reactive power on the French side and injecting 7.7 Mvar of reactive power on the Spanish side.

3.1.2.3 Portugal

The dataset used for the analysis reported in this section comes from five PMU devices located in the Portuguese power system. Figure 3-16 depicts the evolution of the electrical quantities in the main buses of the Portuguese power system during the event.

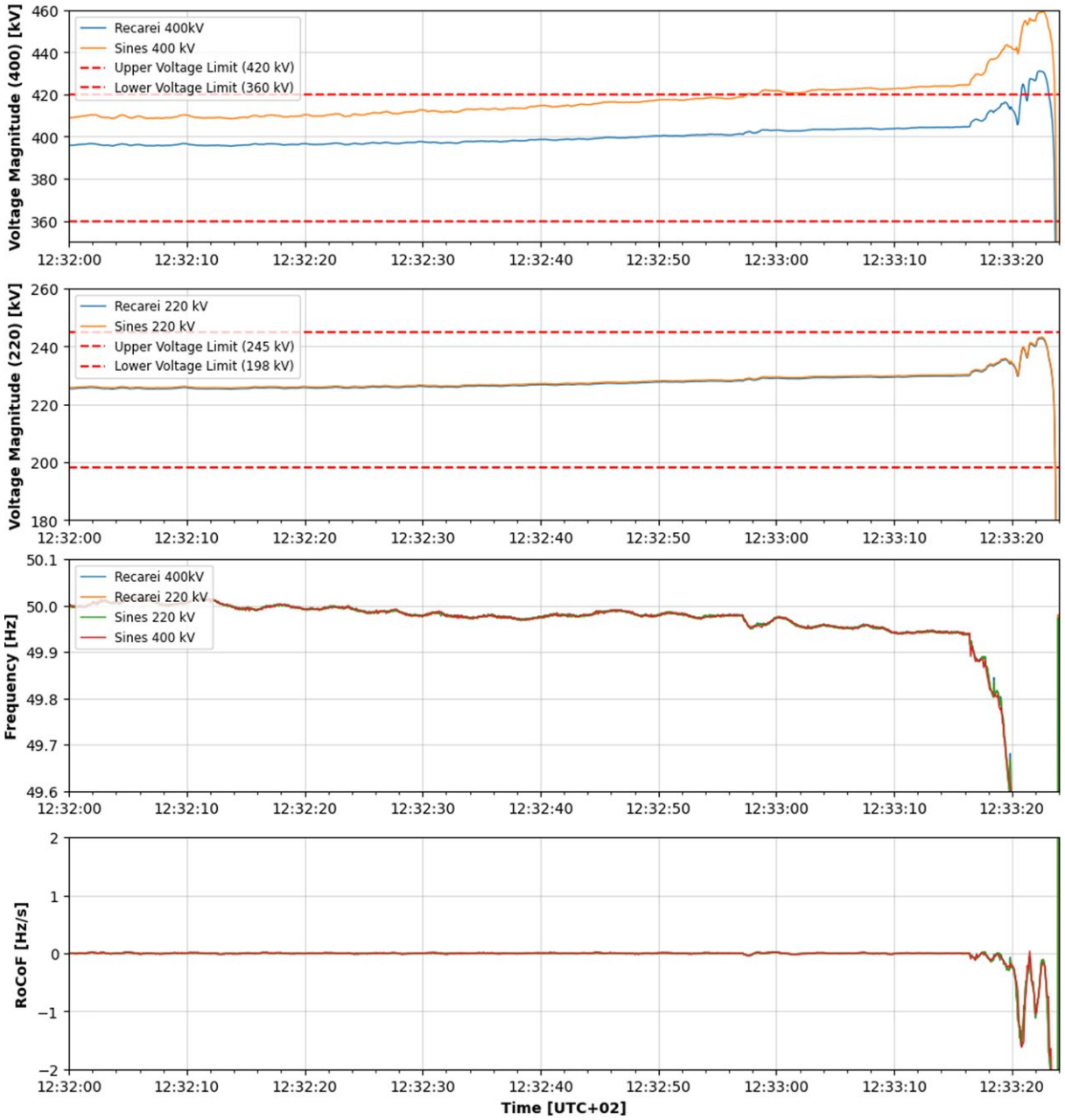


Figure 3-16: Evolution of the electrical quantities in the Portuguese power system during the event

Looking at the voltage magnitude profiles on the transmission system, it can be observed that from 12:32:00 to 12:32:57, voltages rose on average by 3 % at 400 kV buses and 5.5 % at 220 kV buses within the observed perimeter, while remaining within normal operating limits.

Regarding frequency and RoCoF, the same considerations for Spain apply. Portuguese regulation Portaria 73/2020 of 16 March, which corresponds to the national implementation of the EU Requirements for Generators (RfG 2016/631), establishes the same RoCoF limit of 2 Hz/s.

3.1.2.4 France

The dataset used for the analysis reported in this section comes from PMU devices located as shown in Figure 3-17.

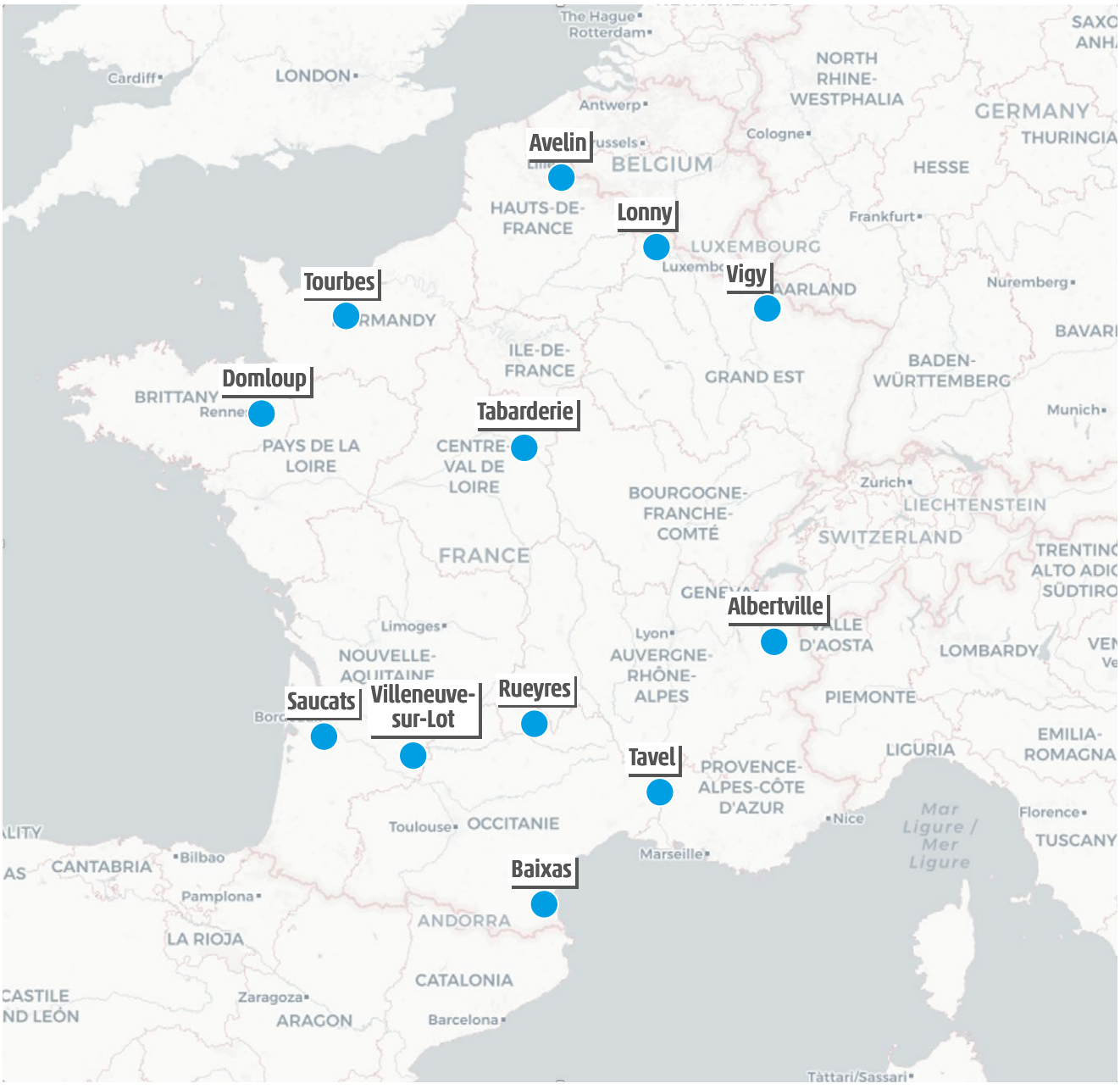


Figure 3-17: Location of PMU devices in the French power systems

Figure 3-18 depicts the evolution of the electrical quantities in the main nodes of the French power system during the incident.

Figure 3-18: Evolution of the electrical quantities in the French power system during the incident

At 12:35, at least six 400 kV substations were above 430 kV (the upper voltage limit on the RTE grid is 420 kV), and one substation was above 435 kV (Baixas 400 kV).

More than 10 225 kV substations were above 260 kV (the upper voltage limit on the RTE grid is 245 kV) on the southwest part of the RTE network.

Loss of voltage control capacities (value)

- Nuclear 1 South-West was absorbing 426 Mvar at 12:33.
- Hydro 6 South-West hydraulic power pump was absorbing 27 Mvar at 12:33.
- HVDC Baixas Santa Llogaia was absorbing 870 Mvar at 12:33:29 (based on measurements from French PMU).

3.2 Performance of the Protection System During the Incident

The objective of designing, setting up, and maintaining a protection system is to prevent dangerous situations for people, limit damage to electrical system components when an anomaly or failure occurs, minimise the consequences of service discontinuities in any network situation, and reduce the risk of transient instability in the transmission network.

Reconstructing system operation actions, comparing them with operational planning results, and assessing the utilisation of system elements are essential for analysis. Events such as the blackout on 28 April 2025 must be thoroughly analysed and investigated to understand their causes and the sequence of events, and to identify corrective actions that could help prevent similar incidents in the future.

This section analyses the protection's operation (tripping) using the available data: mainly the SCADA event lists and the disturbance fault recordings (DFRs).

3.2.1 Generation Protection Behavior During the Incident

3.2.1.1 Grid Code Requirements for Generators in Spain

In Spain, the technical requirements for generation facilities regarding voltage and frequency depend on their commissioning date, with the following main categories:

- Facilities commissioned before the application date of the EU RfG (i.e. before 27 April 2019), which are subject to the technical requirements published in 1998 in Operating Procedure 1.4 for voltage requirements and to the System Operation Guideline.
- Facilities commissioned after the application of the EU RfG (27 April 2019) and before the entry into force of Order TED 749/2020 (8 January 2021), which are subject to the exhaustive requirements of the RfG.
- Facilities commissioned after the entry into force of Order TED 749/2020 (8 January 2021) which are subject to all the requirements of the RfG and Order TED 749/2020.
- Renewable energy production facilities commissioned after the entry into force of Royal Decree 413/2014 (11 June 2014).

Accordingly, the following operating voltage ranges are distinguished (labelled for the purpose of this report with V1–V5). For voltage values that are not listed in any of the ranges listed for each category, generators are allowed to disconnect:

- **Category V1:** Facilities commissioned prior to 27 April 2019 (Operating Procedure 1.4 and System Operation Guideline apply):
  - Generator connected to a voltage of 400 kV:
    - 390 – 420 kV: Normal operation, generators must stay connected for an unlimited time
    - 375 – 390 kV and 420 – 435 kV: Occurs eventually, generators must stay connected for an unlimited time
  - Generator connected to a voltage of 220 kV:
    - 205 – 245 kV: Normal operation, generators must stay connected for an unlimited time
    - 200 – 205 kV: Occurs eventually, generators must stay connected for an unlimited time
- **Category V2:** Facilities commissioned after 27 April 2019 (exhaustive RfG criteria apply) that are type D electricity generation modules must be able to remain connected to the grid and operate within the voltage ranges (at the connection point and expressed in per unit values relative to the base) for the minimum time periods listed below:
  - Type D electricity generation module connected to 110 kV ≤ V < 300 kV:
    - 0.85 – 0.90 pu: 60 minutes
    - 0.90–1.118 pu: Unlimited time
    - 1.118–1.15 pu: Minimum 20 minutes, maximum 60 minutes

- Type D electricity generation module connected to 300 kV ≤ V ≤ 400 kV:
  - 0.85 – 0.90 pu: 60 minutes
  - 0.90–1.0875 pu: Unlimited time (including the Spanish exception)
  - 1.0875–1.10 pu: Minimum 20 minutes, maximum 60 minutes

» **Category V3:** Facilities commissioned after 8 January 2021 (Order TED 749/2020 and RfG applies) that are type D electricity generation modules must be able to remain connected to the grid and operate within the voltage ranges (at the connection point and expressed in per unit values relative to the base) for the minimum time periods listed below:

- Type D electricity generation module connected to 110 kV ≤ V < 300 kV:
  - 0.85 – 0.90 pu: 60 minutes
  - 0.90–1.118 pu: Unlimited time
  - 1.118–1.15 pu: 60 minutes
- Type D electricity generation module connected to 300 kV ≤ V ≤ 400 kV:
  - 0.85 – 0.90 pu: 60 minutes
  - 0.90–1.0875 pu: Unlimited time
  - 1.0875–1.10 pu: 60 minutes

» **Category V4:** Facilities commissioned after 8 January 2021 that are type B, C, and D electricity generation module connected to a radial distribution network with a voltage below 110 kV (note: the operator of the distribution grid has the right to define different disconnection times in accordance with the TSO):

- < 0.85 pu: 1.5 seconds
- 0.85–1.10 pu: Unlimited time
- 1.10–1.15 pu: 1 second
- > 1.15 pu: 0.2 seconds

» **Category V5:** Facilities commissioned prior to 27 April 2019 that are connected to voltages lower than 220 kV (hence out of scope of category V1). There is no national voltage withstand requirement for generators in this category.

The required operating frequency ranges are (labelled for the purpose of this report with F1–F4):

» **Category F1:** Facilities prior to 11 June 2014: Protection settings are based on the characteristics of each facility, considering their technical capabilities.

» **Category F2:** Renewable energy facilities commissioned after 11 June 2014 (Royal Decree 413/2014 applies):

- < 48 Hz: 3 seconds
- 48 – 51 Hz: Unlimited time
- 51 Hz: 0.5 seconds

» **Category F3:** Facilities after 27 April 2019 (exhaustive RfG criteria apply) that are type A, B, C, and D electricity generation modules must be able to remain connected to the grid and operate within the frequency ranges referred to below:

- 47.5 – 48.5 Hz: Non-exhaustive time (minimum 30 minutes)
- 48.5 – 49.0 Hz: Non-exhaustive time (minimum 30 minutes)
- 49.0 – 51.0 Hz: Unlimited time
- 51.0 – 51.5 Hz: 30 minutes

» **Category F4:** Facilities after 8 January 2021 (Order TED 749/2020 applies) that are type A, B, C, and D electricity generation modules must be able to remain connected to the grid and operate within the frequency ranges referred to below:

- 47.5 – 48.5 Hz: 30 minutes
- 48.5 – 49.0 Hz: Unlimited time
- 49.0 – 51.0 Hz: Unlimited time
- 51.0 – 51.5 Hz: 30 minutes

Additionally, Order TED (749/2020) establishes that for type A, B, C, and D electricity generation modules commissioned after 8 January 2021, the requirements for combined frequency and voltage variations are defined such that the frequency ranges indicated in the previous point are modified according to the voltage. The following figures show, for each combined frequency–voltage range, the minimum time that an electricity generation module must remain connected to the grid.

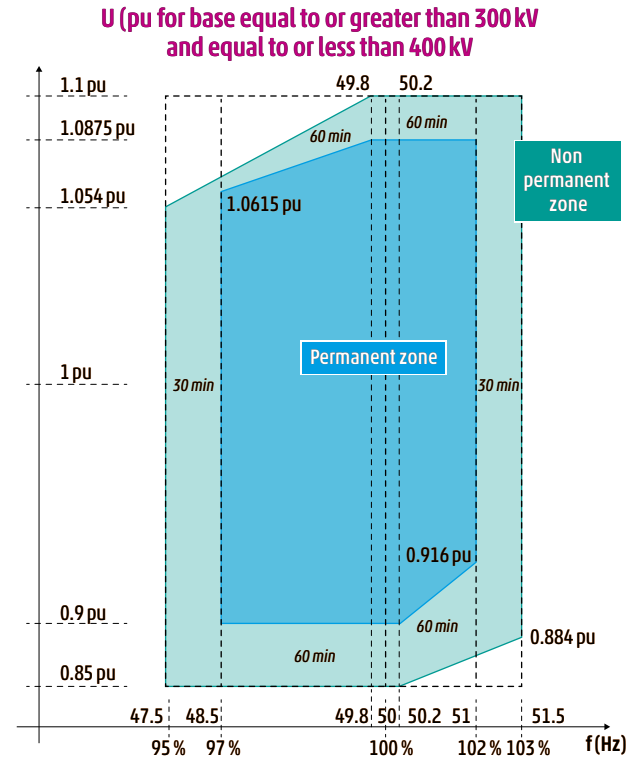


Figure 3-19: Minimum time periods during which an electricity generation module must be able to operate without disconnecting from the grid, for different combined values of frequency and voltage, when the nominal voltage of the connection point is equal to or greater than 300 kV and equal to or less than 400 kV

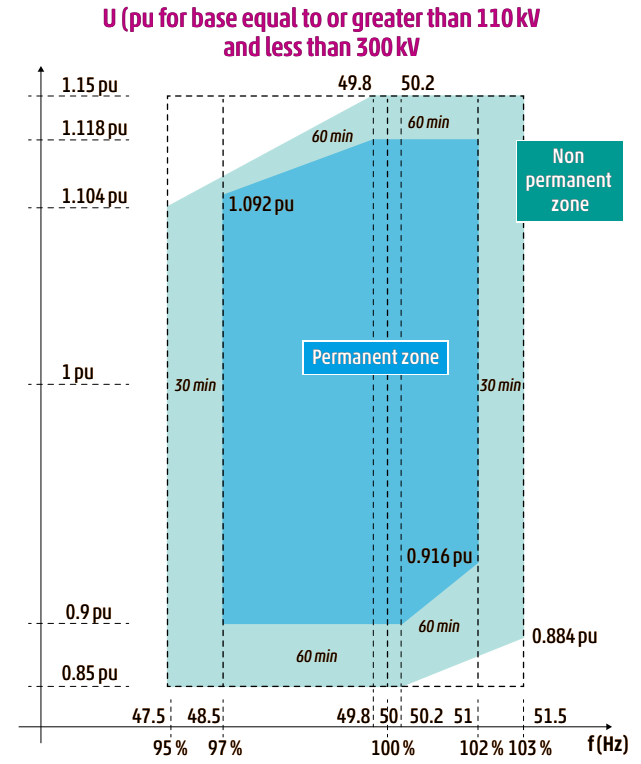
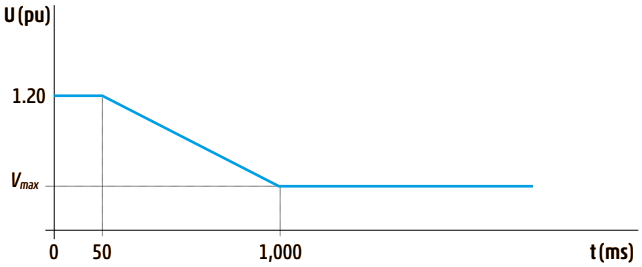


Figure 3-20: Minimum time periods during which an electricity generation module must be able to operate without disconnecting from the grid, for different combined values of frequency and voltage, when the nominal voltage of the connection point is equal to or greater than 110 kV and less than 300 kV

Likewise, Order TED/749/2020 established that electricity generation modules of types B, C, and D, with a commissioning date later than 8 January 2021, must maintain their performance in accordance with the requirements to withstand temporary overvoltages. They must remain connected to the grid and continue operating stably during temporary overvoltage events, in accordance with Figure 3-21 and Figure 3-22.



V_{max}: Highest admissible voltage considered within the voltage ranges and minimum durations that must be withstood without disconnection, in accordance with Regulation (EU) 2016/631 of 14 April 2016, or, if this regulation is not applicable, the highest admissible voltage considered within the normal operating ranges.

Figure 3-21: Minimum overvoltage durations at the point of connection (RMS voltage to ground in one or all phases, expressed in per unit of the voltage base at the point of connection) that a type D power park module connected to the transmission network must be capable of withstanding without disconnection

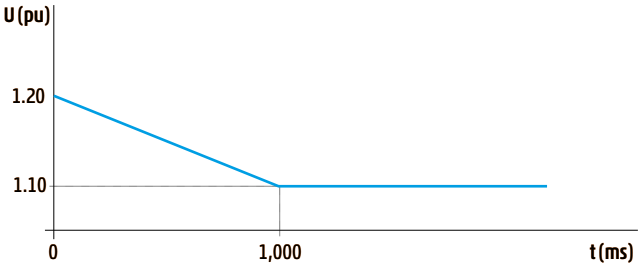


Figure 3-22: Minimum overvoltage durations at the point of connection (RMS voltage between any pair of phases, expressed in per unit of the voltage base at the point of connection) that a type B, C, or D power park module not connected to the transmission network must be capable of withstanding without disconnection

In addition to these requirements for generators, the Operation Procedure 11.1 includes the General Protection Criteria document[29]. This document states that:

"In power plants, overvoltage relays are used, featuring two operating stages. The first stage is instantaneous and is set to approximately 1.4 times the nominal voltage. The second stage is adjusted between 1.10 and 1.20 times the maximum service voltage and may operate with either a fixed-time characteristic or an inverse-time characteristic.".

It should be noted the difference between nominal voltage and maximum service voltage, which is the voltage below which the generation plant must remain connected.

Regarding the requirement to maintain and provide records enabling the analysis of generation dynamics and the operation of protection systems, Spanish regulations establish the following:

- Spanish Operating Procedure 9 (P.O. 9) stipulates that when an incident occurs, as defined in Section 10.3 thereof, the owner of the facilities or the party responsible for the affected supply must provide TSO with a preliminary report of the event based on the best available information. Additionally, the SO may request further clarification regarding the contents of the report and gather the information outlined in Annex V that has not yet been received and is deemed necessary.
- Annex V of P.O. 9 specifies the requirement to provide oscillographic records, protection event logs, and chronological records from the control systems of substations affected by the incident, as well as the complete configuration files of the protection devices that were triggered. The same procedure establishes that oscillographic data must be submitted in electronic format, preferably in COMTRADE format (as defined in IEEE Standard C37.111).
- Furthermore, Regulation EU RfG 2016/631 states: "power-generating facilities shall be equipped with a facility to provide fault recording and monitoring of dynamic system behaviour. This facility shall record the following parameters: voltage, active power, reactive power, and frequency".
- Spanish Order TED 749/2020 establishes: "In the case of type C and D power-generating modules, the activation of any protection relay must be recorded along with the oscillographic data. The owner of the power-generating module shall be obliged to provide to the relevant system operator the fault record and oscillographic data upon request".

29 Criterios Generales de Protección del Sistema Eléctrico Peninsular Español

3.2.1.2 Grid Code Requirements for Generators in Portugal

In Portugal, generator requirements are aligned with the EU RfG 2016/631, complemented by national legislation, specifically Portaria No. 73/2020 from 16 March covering the non-exhaustive requirements for grid connection.

Regarding frequency range withstand capability requirements, these two regulations define the operational frequency ranges that generating modules must be capable of supporting:

- 47.5 Hz–48.5 Hz: 30 minutes
- 48.5 Hz–49.0 Hz: Unlimited time
- 49.0 Hz–51.0 Hz: Unlimited time
- 51.0 Hz–51.5 Hz: 30 minutes

Similarly, regarding voltage range withstand capability requirements, the operational voltage ranges that generating modules must be capable of supporting are as follows:

For voltage base values from 110 kV to 300 kV

- 0.85 pu–0.90 pu: 60 minutes
- 0.90 pu–1.118 pu: Unlimited time
- 1.118 pu–1.15 pu: 20 minutes

For voltage base values from 300 kV to 400 kV

- 0.85 pu–0.90 pu: 60 minutes
- 0.90 pu–1.05 pu: Unlimited time
- 1.05 pu–1.10 pu: 20 minutes

Without prejudice to the above voltage levels required to be supported, type D power park modules (PPMs) shall be capable of withstanding transient overvoltages and must remain connected to the grid for at least the following values of transient overvoltage amplitude and duration:

- 1.25 pu during 100 ms and 1.20 pu during 5 s

It is the understanding of the Expert Panel that the facilities commissioned before the application date of the EU RfG and Portaria No. 73/2020 must comply with requirements established in Portaria No. 596/2010 from 30 July, which approved and published the national Transmission Grid Regulation and Distribution Grid Regulation. In those cases, the applicable requirements for the frequency and voltage withstand capability ranges are similar to the current requirements.

3.2.1.3 Generation Disconnection in Spain During the Incident

The generation disconnections in Spain until 12:33:19.000 are listed in Table 3-3. The identification of the events is in line with the sequence of events provided in Table 3-1.

These events will be analysed in the following subchapters.

Event ID	Time (CEST)	Substation	Voltage Level (kV)	Technology	Production disconnected (MW)	Measured/ *estimated* voltage at the time of the disconnection (kV)	Alignment with the applicable requirements
3	12:32:57.220	400kVTS1-Granada	400	PV, wind and thermo-solar	355	417.9	Diverges
4a	12:33:16.443	400kVTS1-Badajoz	400	PV, thermo-solar	582	*432.6 ≤ V ≤ 437.3*	Diverges
4b	12:33:16.820	400kVTS2-Badajoz	400	PV	145	*438.7*	Diverges
5a1	12:33:17.368	400kVTS1-Segovia	400	wind	22.87	436.5	Aligns
5b1	12:33:17.520	400kVTS1-Badajoz	400	PV	118	*443*	Possibly aligns *
5b2	12:33:17.547	220kVTS1-Huelva	220	Wind and PV	33.73	NA	Unknown
5a2	12:33:17.780	400kVTS1-Sevilla	400	PV	550	437.7	Diverges
5a3	12.33:17.940	400kVTS1-Segovia	400	Wind	94	443.4	Aligns
5b3	12:33:17.975	220kVTS1-Cáceres	220	PV	37.5	250.4	Diverges
5b4	12:33:18.020	220kVTS3-Badajoz	220	PV	71.9	NA	Unknown
6a1	12:33:18.102	220kVTS2-Sevilla	220	PV	3	246.5	Diverges
6a2	12:33:18.220	220kVTS3-Badajoz	220	PV	20	NA	Unknown

Event ID	Time (CEST)	Substation	Voltage Level (kV)	Technology	Production disconnected (MW)	Measured/ *estimated* voltage at the time of the disconnection (kV)	Alignment with the applicable requirements
6a3	12:33:18.360	400kVTS3-Badajoz	400	PV	16	447.7	Aligns
6a4	12:33:18.410	400kVTS2-Cáceres	400	PV	41	444.3 (440 kV threshold was exceeded only 570 ms before the trip)	Diverges
6a5	12:33:18.540	400kVTS1-Murcia	400	PV	63	NA	No requirements to this facility
6a6	12:33:18.630	220kVTS1-Cádiz	220	PV	25.7	247.5	Diverges
6a7	12:33:18.680	220kVTS1-Cádiz	220	PV	127.5	248.2	Diverges
6a8	12:33:18.846	400kVTS1-Málaga	400	PV	154	448.6	Possibly aligns *
6b	12:33:18.951	400kVTS1-Cuenca	400	PV	530	448.6	Possibly aligns *

* The available information does not allow verification of the transient overvoltage requirements

Table 3-3: Generation disconnection in Spain

According to the requirements set out in Order TED 749/2020, generation modules, that are required to maintain their performance in accordance with the applicable requirements, must be able to withstand overvoltage values higher than the maximum operating voltage defined in said regulation for a temporary period equal to or less than 1 second, and in accordance with what is shown in Figures 3-21 and 3-22. In order to assess the conditions during which a disconnection has occured, it is necessary to have records that include at least 1 second prior to the trip event. In some cases, such a record is not available, and only a few milliseconds prior to the trip are recorded. In these situations, if the voltage at the instant of tripping was above the maximum service voltage, the event has been considered as possibly aligns with the applicable requirements".

3.2.1.3.1 Event ID 3: "400 kV TS 1—Granada" 355 MW disconnection

At 12:32:57.220, in GS 1—Granada (Spain), the generation evacuation transformer 400/220 kV tripped at the 220 kV level. This caused the loss of 355 MW of active power generation and 165 Mvar of reactive absorption.

During the investigation, the loss of this generation was identified based on the alarms available to Red Eléctrica. This identification was confirmed to be consistent with observations from PMU measurements, and an oscillographic record on the 400 kV substation coinciding with the tripping was also available. However, this oscillography is from a line within the substation and therefore does not include direct measurements from the transformer itself. Nonetheless, it is possible to observe changes in the substation voltage, and in the current flowing through the line, resulting from the generation loss.

The tripped 400/220 kV evacuation transformer does not belong to the transmission grid, as it is owned by a consortium of companies comprising several entities that inject power through the transformer. There are various facilities downstream of this transformer. Some facilities belong to Category V1 (523 MW) and thus fall outside of the requirements specified in either the TED Order or the RfG.

However, based on the best understanding of the Expert Panel during the investigation, there are also several facilities (total: 127 MW) downstream that do fall under the application of the requirements TED Order 749/2020 (they belong to category V3). This is outlined in a following paragraph of this subsection.

No oscillographic records from the protection system that operated were delivered to the Expert Panel. However, the owner of this transformer reported that the cause of the trip was overvoltage and that the trip occurs when the 220 kV side of the 400/220 kV transformer reaches 242 kV (1.10 pu), with no delay (t = 0 s). This trip setting was set by the owner. The trip setting was not reviewed by the TSO because it was not needed to coordinate with transmission protection (short-circuit protections).

The owner of the transformer provided an event record showing that the voltage exceeded the setting threshold (70 secondary volts or 140 kV primary phase-to-ground or 242 kV primary phase-to-phase). Five ms later, the activation of the output contact was issued, which means that the trip signal was initiated. At that moment, the voltage was already below the tripping threshold. Fifty-five ms after this activation, new operations of the overvoltage function were observed, which may coincide with the breaker opening transient.

LISTADO DE SUCESOS						
HORA	SUCESO	Va	Vb	Vc	Vn	Frec
11:32:05.716	Arraq temp Fase A tensión #2	48.64	52.61	58.96	0.61	49.22
02:51:14.924	Arraq temp Fase C tensión #2	62.06	58.96	49.33	1.07	51.03
07:40:06.328	Arraq temp Fase A tensión #2	50.80	62.06	58.96	0.57	50.85
12:32:57.155	Arraq temp Fase B tensión #1	68.60	70.14	68.21	0.69	49.99
12:32:57.160	Act sal temp Fase B tensión #1	68.60	69.77	68.21	0.54	49.99
12:32:57.215	Arraq temp Fase A tensión #1	70.53	69.39	70.53	0.17	49.80
12:32:57.215	Arraq temp Fase C tensión #1	70.53	69.39	70.53	0.17	49.80
12:32:57.220	Act sal temp Fase A tensión #1	70.53	71.66	70.53	0.57	49.80
12:32:57.220	Act sal temp Fase C tensión #1	70.53	71.66	70.53	0.57	49.80
12:33:21.929	Arraq temp (Un. Frec #1)	74.59	76.00	74.59	0.44	47.98
12:33:23.616	Arraq temp Fase A tensión #1	70.53	68.21	66.23	0.21	44.34
12:33:23.621	Arraq temp Fase C tensión #1	65.83	68.60	70.14	0.41	44.11
12:33:23.627	Arraq temp Fase A tensión #1	70.14	68.60	65.41	0.29	44.11
12:33:23.632	Arraq temp Fase C tensión #1	66.63	66.63	70.53	0.41	44.08
12:33:23.632	Act sal temp Fase A tensión #1	66.63	66.63	70.53	0.41	44.08
12:33:23.642	Arraq temp Fase C tensión #1	66.63	65.83	70.14	0.47	43.97

Figure 3-23: Event record of GS 1—Granada transformer protection

An oscillographic record corresponding to this trip is available, captured by one of the line protection relays at the 400 kV TS 1—Granada, the point of connection. This record (see Figure 3-24) shows that the maximum phase-to-ground voltage prior to the trip was 241.3 kV, which corresponds to a phase-to-phase voltage of 417.9 kV (1.045 pu). The same voltage was measured in phases A and B.

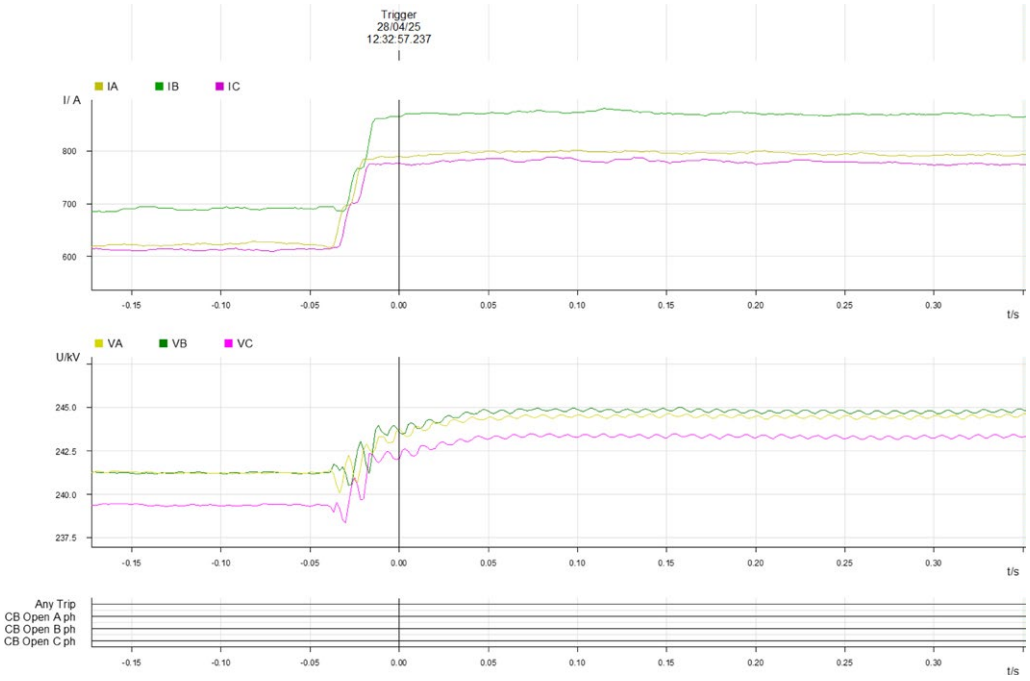


Figure 3-24: Oscillographic record from a line protection in 400 kV TS 1—Granada

Although no oscillographic records are available from the relay that caused the trip at the 220 kV level, a record is available from a plant located downstream from this substation, approximately 500 meters from 220 kV GS 1—Granada, where the voltage at the 220 kV level can be observed. Figure 3-25 shows this record, indicating that the phase-to-ground voltage was 137.1 kV, which corresponds to a phase-to-phase voltage of 237.5 kV (1.079 pu).

The secondary voltage values of phases A and C in this COMTRADE record (68.55 V secondary in phase A and 68.05 V secondary in phase C) are consistent with those observed in the event log shown in Figure 3-23. However, the 5 ms transient in phase B that triggered the transformer overvoltage protection is not observed. Following the upstream circuit breaker trip, an increase in voltage is observed at the register due to the loss of the evacuation path.

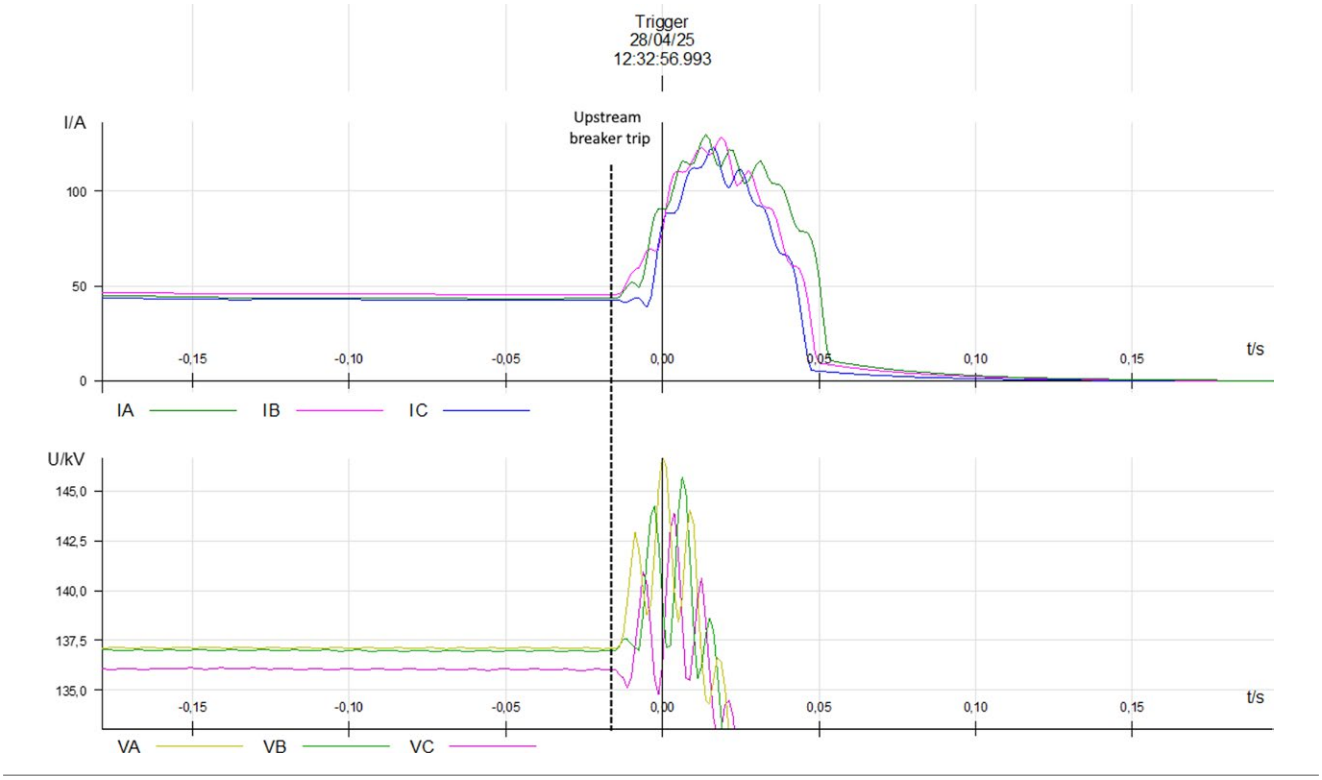


Figure 3-25: Oscillographic record of a generation evacuation line connected downstream of 220 kV GS 1 – Granada

The breakdown of the installed capacity of the generation connected downstream of the tripped element, in terms of the voltage withstand capabilities (described in Section 3.2.1.1) is as follows

- Category V1: 523 MW
- Category V2: 0 MW
- Category V3: 127 MW
- Category V4: 0 MW
- Category V5: 0 MW

This disconnection occurred when the voltage at the transmission network connection point was at a value (418 kV) significantly lower than the threshold established in the applicable regulation (Order TED 749/2020), which permits the disconnection of this generation evacuation facility only from 440 kV onwards. Therefore, it appears that this trip diverges from the applicable requirements.

The absence of oscillographic records from the relay that operated prevents a definitive determination of why the relay measured a value above the trip setting, as such values were not observed in an adjacent facility. However, it is possible that this was caused by transients in the measurement secondary circuit or by an erroneous measurement by the relay itself, which, having zero time delay, resulted in an instantaneous trip. It is concluded that the overvoltage protection functions have been enabled without any actuation time delay.

Likewise, in accordance with the applicable requirements for V3 category plants connected downstream of the facility that tripped, these units must remain connected indefinitely up to a voltage of 1.1 pu, and may disconnect at higher voltages only if the transient voltage curve requirements are met. Downstream from this plant there are facilities that must withstand transient overvoltages of up to 1.2 pu, in accordance with the applicable requirementys. Since the protection is set with zero delay, this transient overvoltage requirement is not met.

3.2.1.3.2 Event ID 4a: 400 kV TS 1—Badajoz 582 MW Disconnection

At 12:33:16.443, at 400 kV GS 1—Badajoz, the evacuation line to 400 kV GS 2—Badajoz tripped. This caused the loss of 582 MW of active power generation and 165 Mvar of reactive absorption.

Several oscillographic records from one downstream operator indicate that, at this moment, an upstream collector installation was lost, as overfrequency was observed immediately afterwards. Additionally, a chronological log provided by another downstream operator shows the opening of circuit breakers on a line at the 400 kV GS 1—Badajoz, which connects to the substation where the generation loss occurred.

Although there are facilities that were commissioned prior to the entry into force of the RfG, there are two facilities whose commissioning took place afterwards, and which also obtained certification as compliant with TED Order 749/2020.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel. It is unclear whether the relay that tripped was the one located at 400 kV GS 1—Badajoz or the one at the remote end in GS 2—Badajoz, which could have sent a transfer trip signal to GS 1—Badajoz. However, what is available is an event log showing the opening of circuit breakers 521-1 and 520-2 at 400 kV GS 1—Badajoz at that moment.

There is no high-resolution voltage data available at the connection point, 400 kV TS 1—Badajoz. The only data available is from the SCADA system. The SCADA voltage value registered at 12:33:10 at the TS 1—Badajoz generation evacuation bay was 434.0 kV, 430.5 kV at busbar 1, and 429.9 kV at busbar 2. An examination of PMU measurements from four nearby substations shows that from 12:33:04 – corresponding to the moment of lowest voltage around 12:33:10 – the maximum voltage increase up until the moment of the trip was 3.3 kV.

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 556 MW
- V2: 0 MW
- V3: 69 MW
- V4: 0 MW
- V5: 0 MW

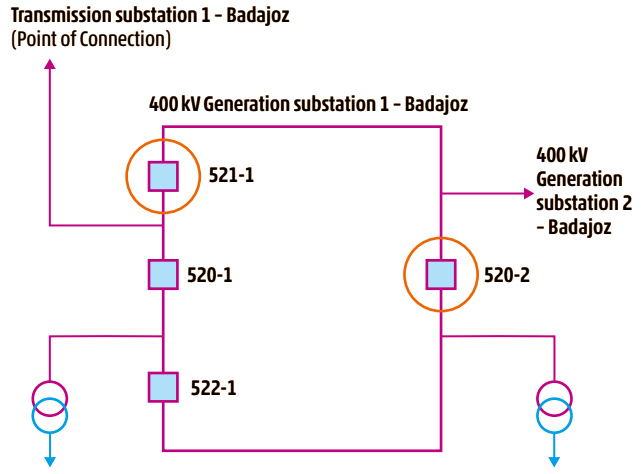


Figure 3-26: Single line diagram of generation evacuation substation 400 kV GS 1—Badajoz

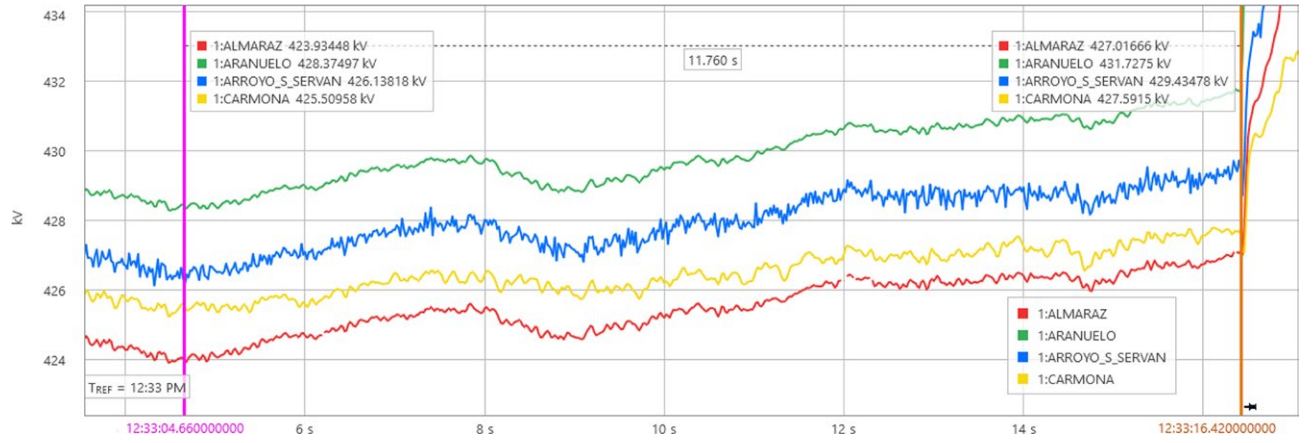


Figure 3-27: Voltage measurements from PMUs at 400 kV substations in the vicinity of TS 1–Badajoz

The 5b1 event also occurred on the 400-kV GS 1–Badajoz bay, and it can be observed that the voltage at the moment of the 5b1 event was 443 kV. Based on PMU data from nearby substations, it is possible to estimate the voltage variation that took place between event 4a and event 5b1. From the analysis of the PMU measurements, it is observed that, among the near substations where PMU data are available, the largest voltage step occurs at Arañuelo 400 kV, where the measured jump is 10.6 kV. In addition, it must be considered that the expected voltage rise after event 4a should be higher at the 400 kV GS 1–Badajoz substation, and therefore at the 400-kV TS 1–Badajoz substation, than at Arañuelo 400 kV, since it is at the 400 kV TS 1–Badajoz node where reactive power absorption is lost.

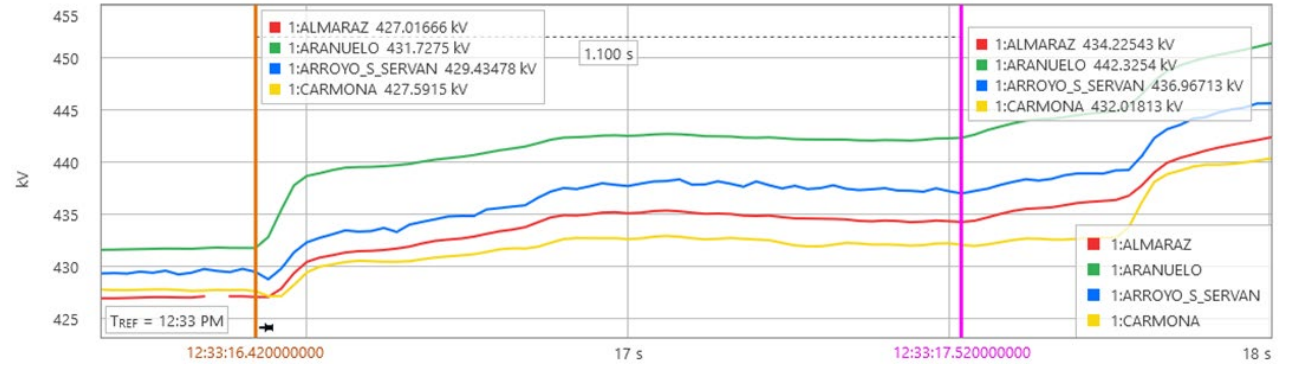


Figure 3-28: Voltage measurements from PMUs at 400 kV substations in the vicinity of TS 1–Badajoz between events 4a and 5b1

Therefore, using the 443 kV measured during event 5b1 and the 10.6 kV voltage variation measured at the Arañuelo 400-kV substation between instants 4a and 5b1, it is estimated that the voltage at said 400 kV GS 1–Badajoz node was approximately 432.4 kV when the event 4a happened. Using the pre-trip SCADA measurements of active and reactive power and the voltage measured at the connection line with the transmission network, as well as the parameters of this line, it has been calculated that the voltage difference between 400 kV GS 1-Badajoz and 400 kV TS 1–Badajoz is less than 0.2 kV.

So the estimation of the voltage at 400 kV GS 1–Badajoz yields results consistent with the estimation obtained from SCADA data at the 400-kV TS 1–Badajoz node and with the voltage variation measured by PMUs in nearby substations.

Therefore, all evidence indicates that the overvoltage protection operated when the voltage at 400 kV GS 1–Badajoz and 400 kV TS1–Badajoz was below 440 kV, and most likely even below 435 kV. Therefore, it appears that this trip diverges from the applicable requirements.

3.2.1.3.3 Event ID 4b: 400 kV TS 2—Badajoz 145 MW Disconnection

At 12:33:16.820, a PV plant connected to 400 kV TS 2 – Badajoz tripped. This PV plant was generating 145 MW and absorbing 37 Mvar. This PV plant was certified as compliant with Order TED 749/2020.

The last voltage values registered by the SCADA system were 427.7 kV at 12:32:52 and 436.1 kV at 12:33:16.

During the preparation of the final report, the Expert Panel received the settings values for the overvoltage protection function at the generation plant's end. According to the owner, the voltage threshold was set at 435.6 kV (1.089 pu) with a time delay of 1.5 seconds and 455,6 kV (1.139 pu) with a time delay of 0.2 seconds. Additionally, the COMTRADE record from the generator plant terminal that tripped has also been received. Also the settings received from overvoltage function protection at the collector substation end were received. These settings indicate that two thresholds are configured for the overvoltage protection function. The first threshold is set at 435.6 kV phase-to-phase (1.089 pu), with a time delay of 1.5 seconds, and the second threshold is set at 455.4 kV phase-to-phase (1.1385 pu), with a time delay of 0.2 seconds.

The COMTRADE record shows that, at the moment immediately prior to the circuit breaker opening, the maximum voltage measured was 439.4 kV (1.098 pu). Furthermore, considering the timing configuration of this protection, it is known that the overvoltage protection function picked up 1.5 seconds before the trip, and according to the oscillographic record received, the maximum voltage measured at that moment was 433.9 kV (1.08 pu), which is 0.5 % below the configured threshold.

Based on the P, Q, and V data measured by a PMU installed at the opposite end of a line connected to the 400 kV TS 2-Badajoz substation, and using the parameters of this line, it has been calculated that the voltage when the overvoltage protection function picked up, 1.5 seconds before the trip, at the 400 kV TS 2-Badajoz was approximately 432.8 kV and at 12:33:16.800, just before the plant trip, the voltage would have been approximately 438.7 kV at the point of connection to transmission grid.

Figure 3-29 shows the oscillographic record of the line protection that tripped at the power plant substation end.

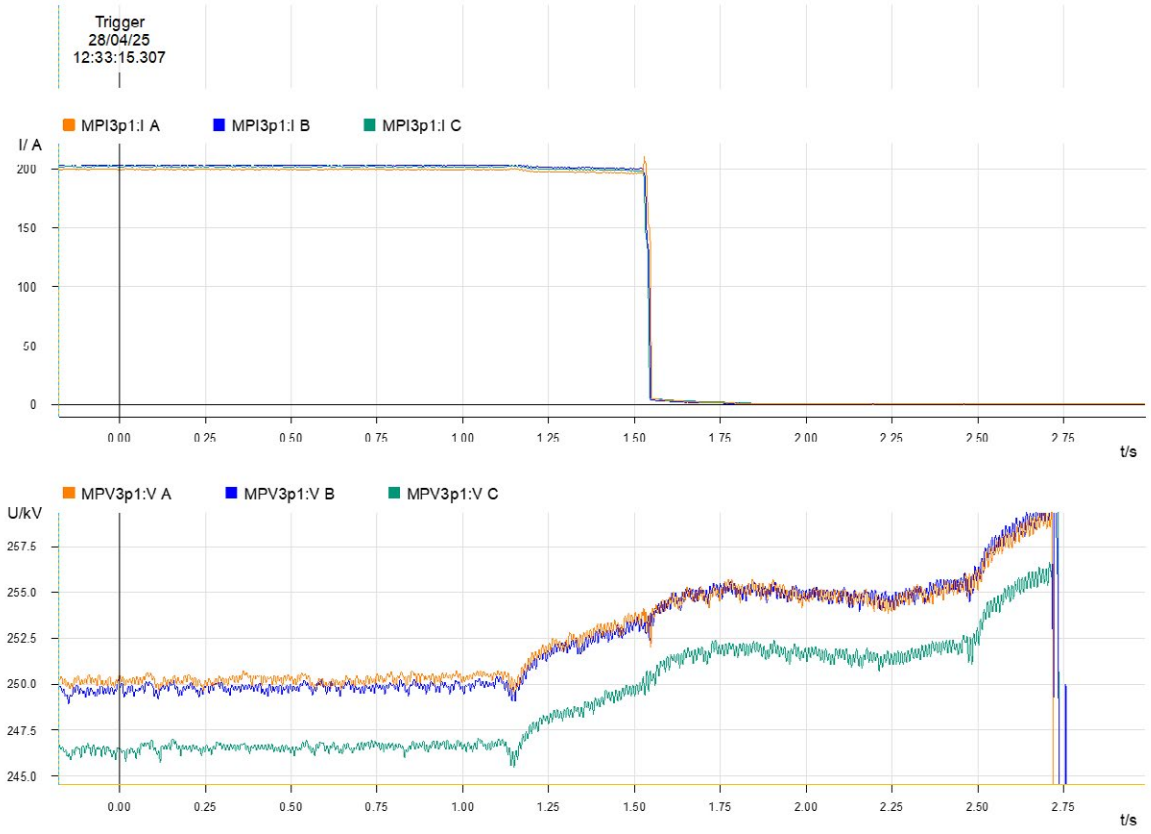


Figure 3-29: Oscillographic record of the protection system for the line connecting the PV plant to the collector substation at the generation substation end

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 150 MW
- V4: 0 MW
- V5: 0 MW

It appears that the settings reported by the owners for each end of the evacuation line (435.6 kV, 1.089 pu, and a time delay of either 1.5 seconds) diverge from the applicable requirements. The voltage measured at the generator plant terminal when it tripped was below the applicable required thresholds.

Likewise, the estimations based on the available PMU data and the parameters of a connecting line indicate that the voltage was below 435 kV at the moment when the overvoltage protection function picked up, which subsequently tripped after 1.5 seconds. Furthermore, at the instant of tripping, the voltage remained below the threshold, under which the plant should have remained permanently connected.

Therefore, it appears that this disconnection diverges from the applicable requirements.The owner of the power plant that tripped has informed the Expert Panel that the overvoltage protection settings at the power plant end have been modified. The current settings are as follows: 444.4 kV (1.111 pu) with a time delay of 1.5 seconds as first stage and 480 kV (1.2 pu) with a time delay of 0.2 seconds as second stage.

3.2.1.3.4 Event ID 5a1: 400 kV TS 1—Segovia 22.9 MW Disconnection

At 12:33:17.368, three wind farms connected to a 132 kV evacuation grid that is connected to 400 kV TS 1—Segovia were disconnected due to the tripping of an element in the upstream section of the generation collector network. This caused the loss of 22.9 MW of active power generation and the loss of 12.3 Mvar in reactive power absorption. These three wind farms were commissioned before the RfG entered into force.

The generation loss was identified through the oscillographic record of a collector transformer. The synchronisation of the recording was verified by analysing the frequency of the voltage signal.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel. However, an oscillographic record was received from the 400/132 kV generation transformer at TS 1—Segovia, in which the voltage levels at both 400 kV (connection point) and 132 kV can be observed at the time of the trip. The phase-to-ground voltage measured at the 400 kV level at the time of the trip is 252.0 kV, which corresponds to a phase-to-phase voltage of 436.5 kV. The phase-to-ground voltage measured at the 132 kV level was 83.3 kV, corresponding to a phase-to-phase voltage of 144.3 kV.

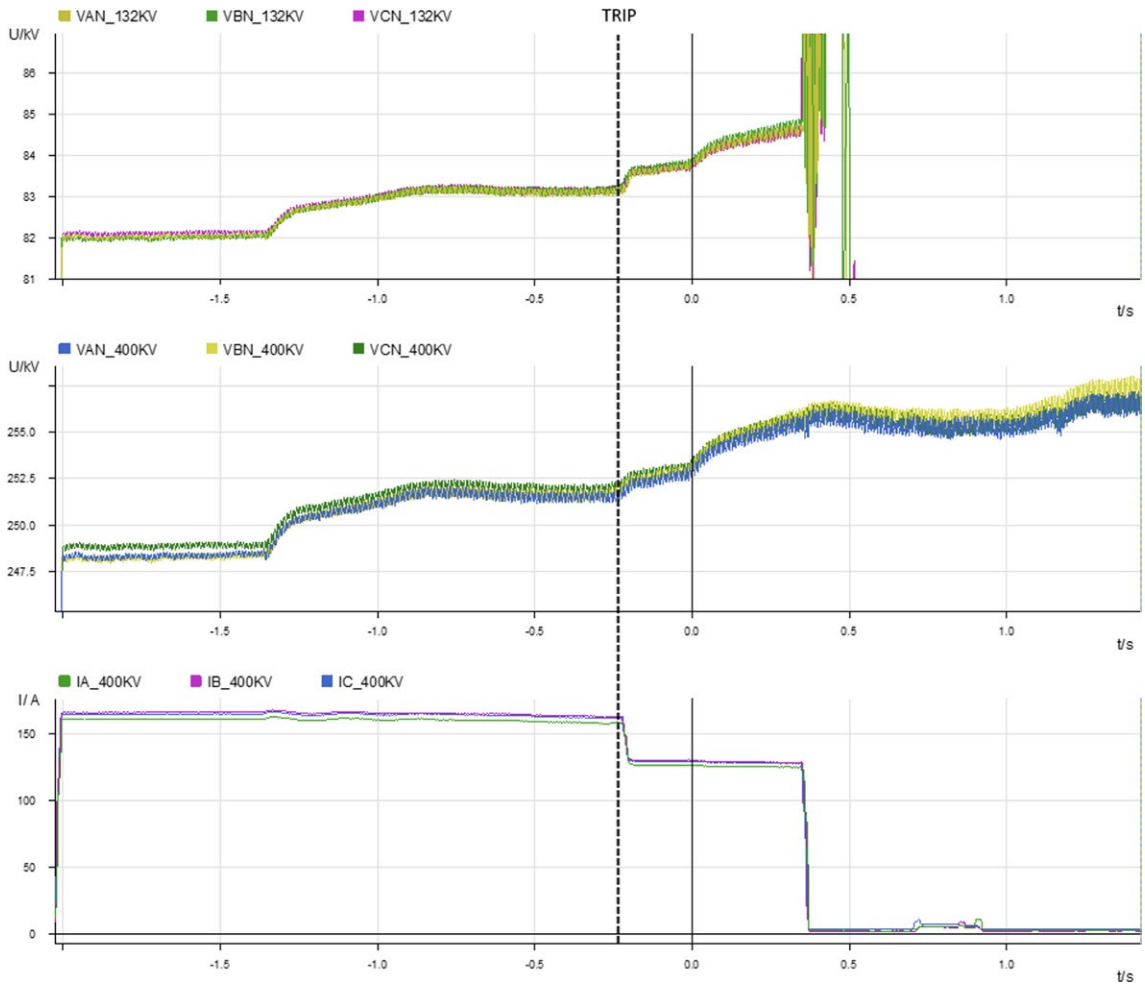


Figure 3-30: Voltage at the 400 kV and 132 kV voltage levels of the TS 1 – Segovia generation transformer and transformer current at the 400 kV level

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 49 MW
- V2: 0 MW
- V3: 0 MW
- V4: 0 MW
- V5: 0 MW

The disconnection of this generation collector installation occurs when the voltage at the connection point was slightly above the applicable voltage threshold (435 kV). Hence, it appears that this trip is aligned with the applicable requirements. However, the trip was issued when the voltage exceeded that 435 kV threshold by only 0.3 %, meaning that in a different scenario it is possible that a trip in this 132 kV evacuation installation could occur with the 400 kV point of connection voltage remaining below the 435 kV threshold. The setting threshold of the protection that operated is not known, but such setting should incorporate a sufficient safety margin to avoid these potential situations.

3.2.1.3.5 Event ID 5b1: 400 kV TS 1—Badajoz 118 MW Disconnection

At 12:33:17.520, the evacuation line to 400 kV TS 1—Badajoz tripped in 400 kV GS 1—Badajoz. This generation evacuation network has its point of connection at 400 kV TS 1—Badajoz. This caused the loss of 118 MW of active power generation and the loss of 22 Mvar in reactive power absorption.

The generation loss was identified through an oscillographic record from the transformers at the GS 1—Badajoz and an event log.

It is inferred that the relay that operated was the line relay, as the circuit breaker that opened is the one shared between the line and Transformer 1 (520-1, see Figure 3-26). However, the circuit breaker shared between Transformer 1 and Transformer 2 is 522-1 (see Figure 3-26), which was recorded to have opened approximately 1.5 seconds later due to the undervoltage function.

It is the understanding of the Expert Panel that the disconnected facility must comply with Order TED 749/2020.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel. The owner of the transformers sent oscillographic records from the transformers, which show that the overvoltage function only initiates during the trip transient, and that it is the undervoltage function that triggers 1.5 seconds after the line circuit breaker opens. In this record, there is a voltage channel that, although identified as phase-to-ground voltage, appears to correspond to phase-to-phase voltage, which measured 443 kV at the moment of the trip. This voltage was measured at the 400 kV GS 1—Badajoz, not the 400 kV TS 1–Badajoz, which is the connection point to the transmission grid.

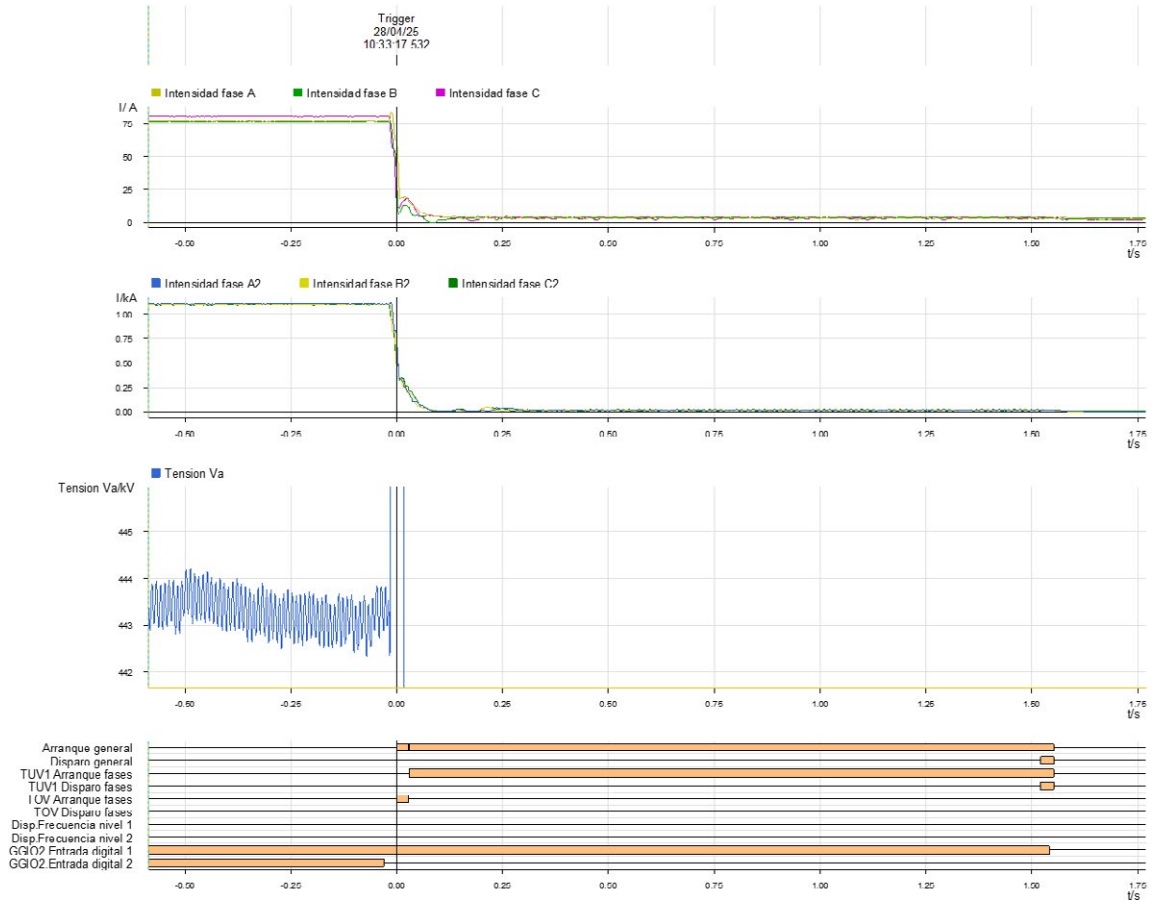


Figure 3-31: Oscillographic record from Transformer 1 at GS 1—Badajoz

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is the following:

- V1: 0 MW
- V2: 0 MW
- V3: 206 MW
- V4: 0 MW
- V5: 0 MW

The applicable requirements specify that the installation must withstand up to 440 kV for at least 60 minutes, with the system frequency above 49.8 Hz. The voltage measured at the generation substation at the instant of the trip was above this value. This assessment has to be done at the point of connection but there are not high-resolution voltage measurements that would allow to do this. In addition, it is not possible to assess alignment with the transient overvoltage curve, as no recording is available that allows identifying the moment at which 440 kV was exceeded. Based on the voltage measured at the generation substation at the trip instant, 443 kV, the protection should not have operated until at least 929 ms had elapsed after the 440 kV threshold was surpassed.

The trip was triggered when the voltage exceeded that 440 kV threshold by only 0.6 %. The setting threshold of the protection that operated is not known, but such setting should incorporate a sufficient safety margin to avoid trips when the voltage at the connection point is below the required threshold taking into account the measurement errors and that the voltage is measured in the generation substation and not at the connection point which is where this should be assessed.

3.2.1.3.6 Event ID 5b2: 220 kV TS 1—Huelva 33.7 MW Disconnection

At 12:33:17.547, a PV plant and a wind farm connected to 220 kV TS 1—Huelva were disconnected.

The wind farm was commissioned before the RfG entered into force. It is the understanding of the Expert Panel that the PV plant must comply with Order TED 749/2020. To determine the timing of the generation loss, event records provided by the control centre were used. It was verified that, at that moment, a voltage step is observable in the PMU measurements from 400 kV TS 1–Huelva.

During the preparation of the final report, the overvoltage settings for one of the two plants were received, along with the fault record of an overvoltage protection from that plant. However, the record shows that the interrupted current at the moment of opening is 0 A, indicating that the plant must have tripped beforehand. From this, it is concluded that the provided record does not correspond to the actual instant of disconnection of this plant. Therefore, no oscillographic records from the protection system that tripped the plant were delivered to the Expert Panel. For the second plant no oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel.

The control centre associated with this connection point indicated that the cause of the trip was the overvoltage protection function, and that the voltage reached was 267 kV at the 220 kV level and 76 kV in the 66 kV evacuation network. However, since no oscillographic data were provided, it is not possible to determine whether this voltage was reached before the trip, during the trip transient, or after the trip.

For instance, the 220 kV level remained connected for several seconds after the event, and significant overvoltage was recorded in this voltage level following that moment.

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 17 MW
- V2: 0 MW
- V3: 42 MW
- V4: 0 MW
- V5: 0 MW

The overvoltage protection settings received for one of the two plants are: a first stage at 1.1 pu with a time delay of 1.5 seconds, and a second stage at 1.15 pu with a time delay of 0.2 seconds. The applicable requirements specify that this plant must withstand 1.114 pu (245 kV) for an unlimited time, given that its grid connection point is at the 220 kV transmission network it appears that the settings are not aligned with the applicable requirements. The owner of this plant has also provided the settings of a third plant connected to the same evacuation network. These settings are identical to those of this plant.

As no valid records are available at the point of connection, it is not possible to assess what was the voltage level at the instant of the trip.

3.2.1.3.7 Event ID 5a2: 400 kV TS 1—Seville 550 MW Disconnection

At 12:33:17.780, the generation–transmission interface line tripped at the end corresponding to the generation evacuation network and additionally sent a remote trip signal to the transmission network substation end (400 kV TS 1—Seville). This caused the disconnection of several PV plants, resulting in the loss of 550 MW of active power injection and absorption of 195 Mvar of reactive power.

It is the understanding of the Expert Panel that all PV plants connected to this point of connection must comply with Order TED 749/2020.

An oscillographic record was available from the evacuation facility at the transmission network end, which was used to identify the generation loss.

Figure 3-32 shows the oscillographic record captured by the line protection system at the transmission substation (400 kV TS 1—Seville), which received the remote trip signal from the opposite end. It shows that the maximum phase-to-ground voltage recorded at the time of the trip was 252.7 kV, which corresponds to 437.7 kV phase-to-phase.

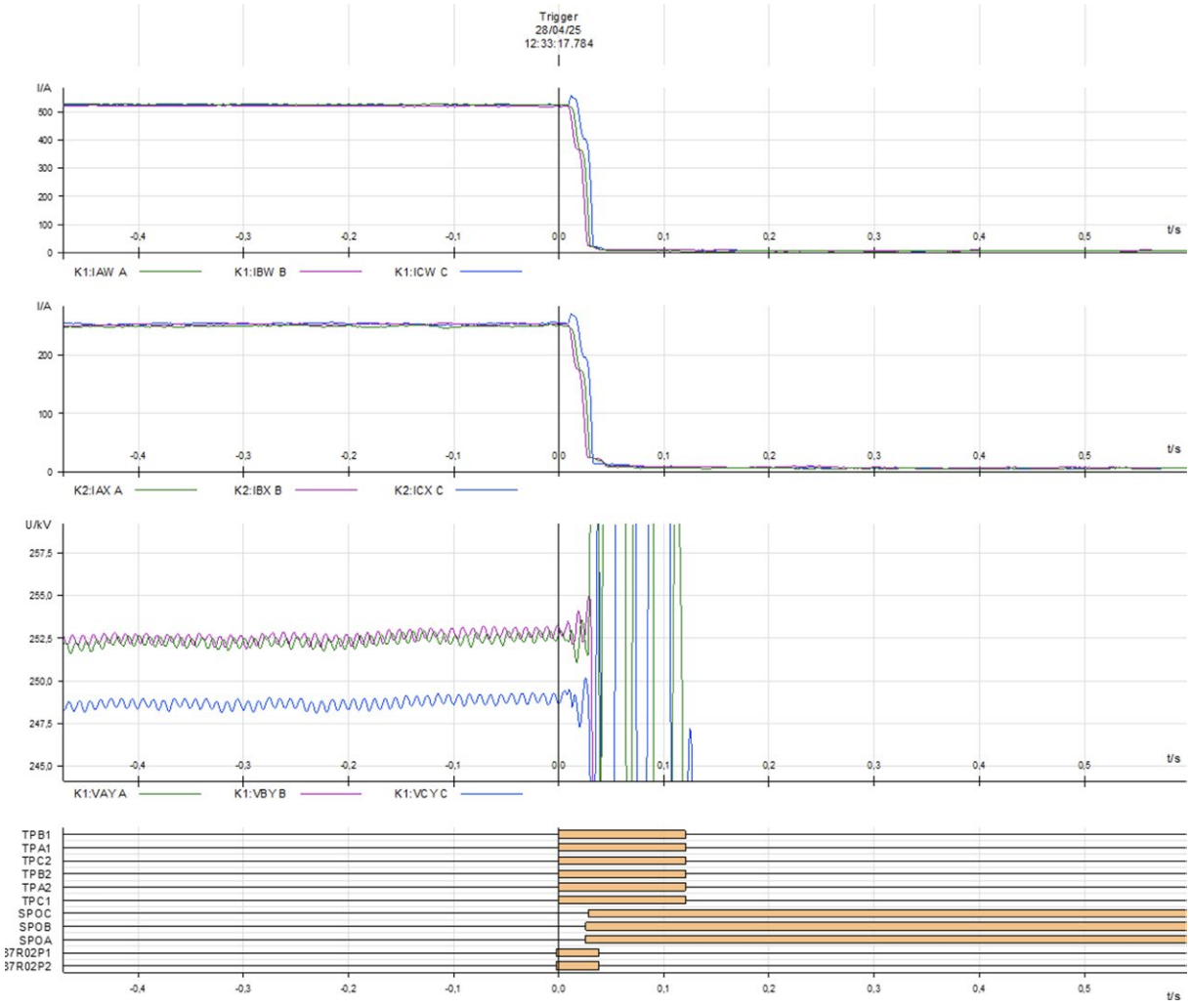


Figure 3-32: Oscillographic record from 400 kV S 1 Seville substation, 400 kV GIS 1 Seville bay (TPX1: breaker 1 trip of each phase, TPX2: breaker 1 trip of each phase, SPOX: Single pole open of each phase, 87R02P1: transfer trip received through communication channel 1, 87R02P2: transfer trip received through communication channel 2

An oscillographic record from the protection system that operated at the generation end was provided. It has also been indicated that the overvoltage protection function was the cause of this trip. The over-voltage settings of this protection function, as reported, are 436.6 kV (1.09 pu) with one second time delay.

Figure 3-33 shows the oscillographic record from the line protection that tripped at the 400 kV GS 1—Seville end. It can be observed that the maximum phase-to- ground voltage measured at the time of the trip was 251.6 kV, which corresponds to 435.8 kV phase-to-phase.

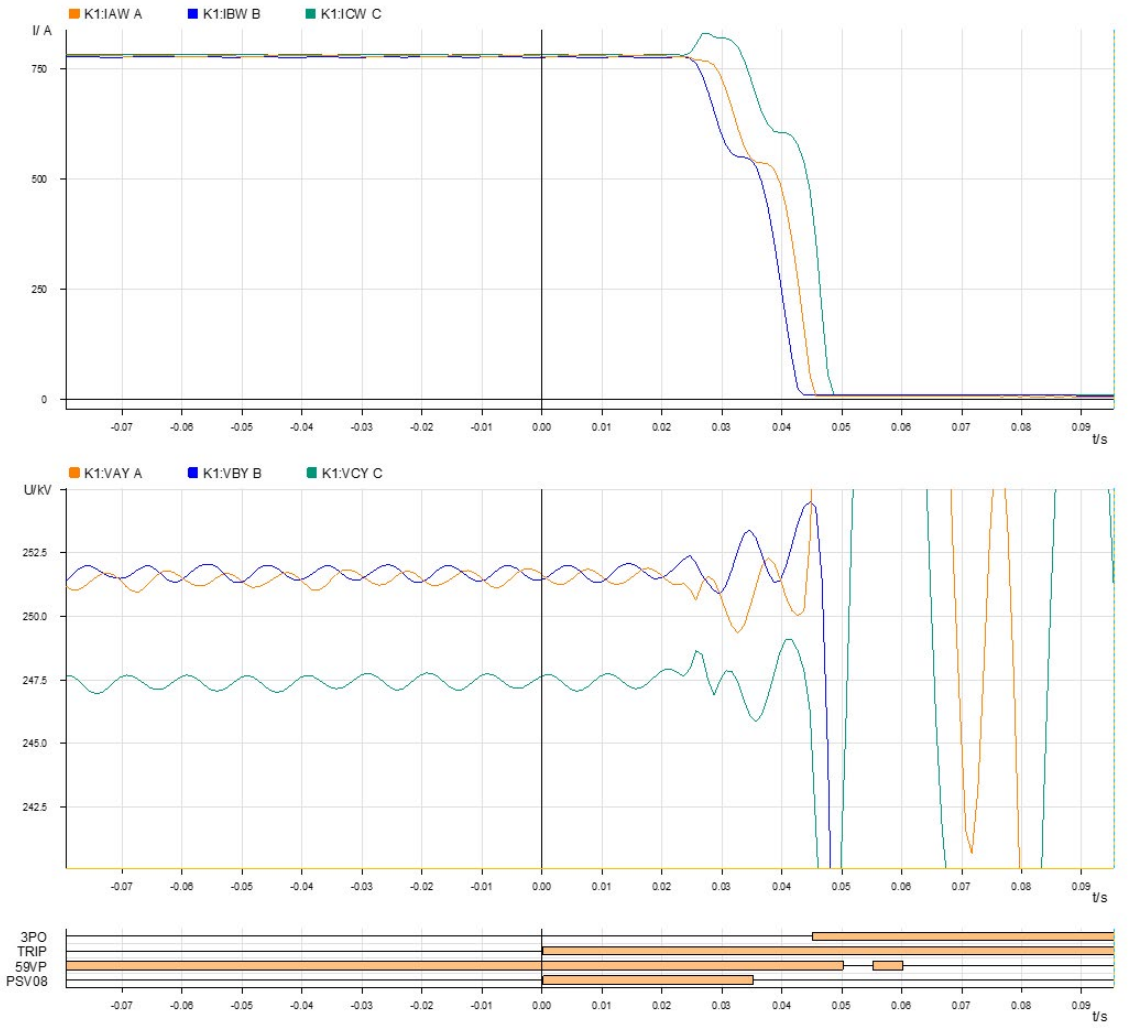


Figure 3-33: Oscillographic record from 400 kV GS 1—Seville substation bay to TS 1—Seville (3PO: 3 breaker poles open, PSV08: overvoltage trip)

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 698 MW
- V4: 0 MW
- V5: 0 MW

Therefore, based on the available information, it is concluded that the trip occurred when the voltage at the connection point with the transmission grid, and even at the generator collector substation terminal, was below the 440 kV threshold defined according to the applicable requirements. Therefore, it appears that this trip diverges from the applicable requirements.

The settings provided by the facility owner show that the implemented thresholds (1.09 pu) diverge from the applicable requirements. It can also be observed that the trip occurred when the voltage was slightly below the configured threshold.

The possible reason for this is the relay hysteresis: the threshold may have been exceeded at the instant of pickup, but subsequently the voltage dropped below that value and the function was not reset because the voltage did not decrease sufficiently to allow the function reset.

In protection relays, once the threshold defined in the overvoltage function is exceeded, will only be reset if the measured voltage not only falls below that pick-up threshold but decreases somewhat further in order to prevent the protection element from repeatedly starting and stopping when the voltage is close to the pickup level. This characteristic is known as hysteresis. Depending on the relay model, this hysteresis may be fixed or adjustable, with default adjustable values around 5 % of the threshold being common in many devices.

In other words, for the function to reset once it has picked up, the measured voltage should fall below 0.95 × "pickup threshold". If the settings applied do not take this circumstance into account, it is possible that, if the function picks up for a few milliseconds, it may remain active even though the voltage has already dropped below the applicable pickup threshold. Likewise, if the pickup threshold is set very close to the limit, it is possible that tripping may occur when the voltage was already within the zone in which the generator should remain connected indefinitely, or that tripping may occur after 1 or 2 seconds (depending on the delay setting) at a voltage level for which the unit should remain connected for several minutes.

The following figure shows, as an example, the voltage thresholds and time requirements that a generation module must withstand without disconnecting, in accordance with the applicable requirements under Order TED 749/2020, when its point of connection is a 220 kV transmission-grid substation. In this illustrative case, the tripping thresholds have been set above the limits established in accordance with the applicable requirements; however, the safety margin left above the required values is too small.

Even though the thresholds are set above the required values – and assuming there are no measurement errors – the hysteresis associated with the deactivation of the overvoltage function (0.95 × voltage threshold) prevents the requirements from being fulfilled. In both cases represented in the figure, the installation should have remained connected indefinitely (blue curve) but trips after 60 minutes, or it should have remained connected for at least 60 minutes (red curve) but instead will trip after 1.5 seconds.

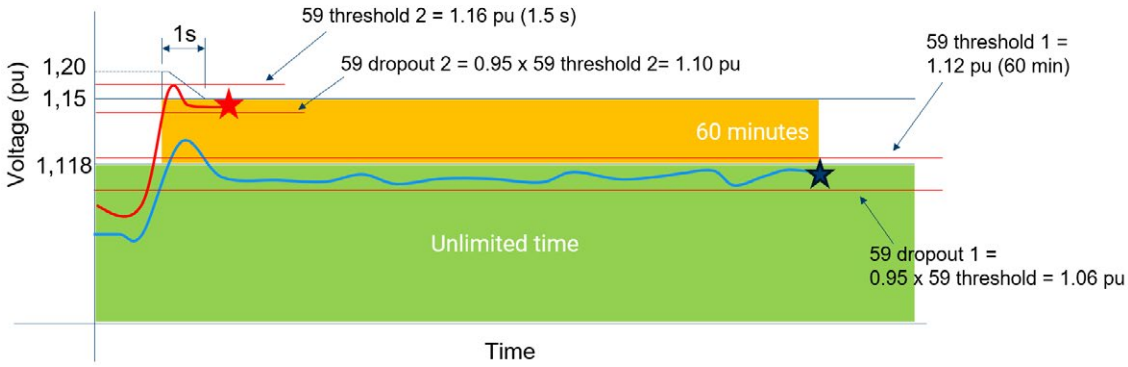


Figure 3-34: overvoltage threshold settings and relays hysteresis

Therefore, the pickup threshold must take into account the hysteresis implemented in the protection device, ensuring that the voltage level at which the overvoltage function resets remains above the corresponding limits established in accordance with the applicable requirements.

An analysis of the settings provided shows that the secondary-side setting was 1.101 pu of the nominal secondary voltage of the voltage transformer, which is slightly above the 1.1 value defined in accordance with the applicable requirements, with a too small safety margin. However, it seems that the nominal voltage of the voltage transformer, 396 kV rather than 400 kV, was not taken into account. Consequently, the transformation ratio is 396,000/110, which results in the primary-side threshold being 1.09 pu, as previously indicated.

3.2.1.3.8 Event ID 5a3: 400 kV TS 1—Segovia 94 MW Disconnection

At 12:33:17.940, at 400 kV TS 1—Segovia, the generation evacuation transformer 400/132 kV tripped in the 132 kV level. This caused the loss of 94 MW of generation and the loss of 26.5 Mvar in reactive power absorption.

All facilities downstream from this installation were commissioned prior to the entry into force of the RfG.

No oscillographic records from the protection system that operated were provided. However, an oscillographic record was delivered to the Expert Panel from the 400/132 kV transformer at 400 kV TS 1—Segovia, in which the voltage levels at both 400 kV (connection point) and 132 kV can be observed at the moment of the trip. The phase-to-ground voltage measured at the 400 kV level at the moment of the trip was 256.0 kV, which corresponds to 443.4 kV phase-to-phase. The phase-to-ground voltage measured at the 132 kV level was 84.7 kV, which corresponds to 146.7 kV phase-to-phase. The settings received for the overvoltage function include a voltage threshold of 1.1 pu with a time delay of 0.5 seconds.

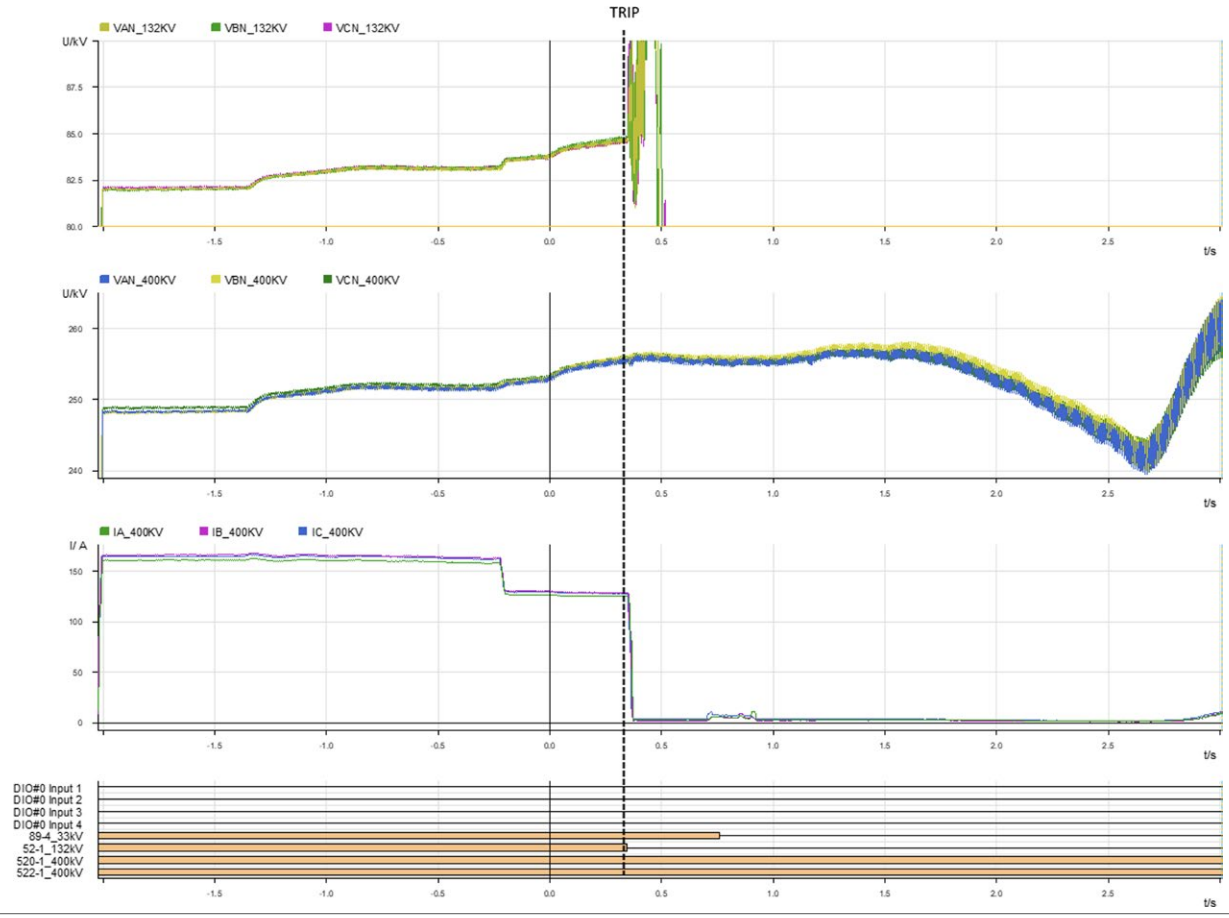


Figure 3-35: Voltage at the 400 kV and 132 kV voltage levels of the TS 1—Segovia generation transformer and transformer current at the 400 kV level

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 230 MW
- V2: 0 MW
- V3: 0 MW
- V4: 0 MW
- V5: 0 MW

This trip occurred when the voltage measurement was above the defined threshold as applicable to these installations (435 kV). Therefore, it appears that this disconnection is aligned with the applicable requirements.

3.2.1.3.9 Event ID 5b3: 220 kV TS 1—Cáceres 37.5 MW Disconnection

At 12:33:17.975, one PV plant connected to 220 kV TS 1 – Cáceres was disconnected. This caused the loss of 37.5 MW of generation and 7 Mvar in reactive power absorption.

It is the understanding of the Expert Panel that this PV plant must comply with Order TED 749/2020.

To determine the timing of the generation loss, events provided by the responsible control centre were used. It was verified that approximately at that moment, a voltage step is observable in the PMU measurements from 220 kV TS 4—Cáceres. This is the closest PMU available, although it is located slightly less than 100 km away.

No oscillographic records from the protection system that operated were delivered to the Expert Panel. However, an event record has been received showing that the trip was triggered by the operation of the transformer overvoltage protection.

According to the information received, the trip was caused by an overvoltage condition. The setting of this protection function at the time of the incident was 110 % of 220 kV, with a time delay of 1.5 seconds. Likewise, the owner has reported that the voltage reached at the moment of the trip was 250.39 kV.

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 106 MW
- V4: 0 MW
- V5: 0 MW

In this event, the voltage measurement at the point of connection was not available at the instant of the trip. Moreover, both the voltage threshold configured in the protection relay (1.1 pu with a 1.5-second delay) and the voltage value at the moment of the trip reported by the facility owner (250.39 kV) do not seem to be in line with the applicable requirements.

The facility owner has reported that the overvoltage protection settings were modified after the incident, resulting in a voltage threshold of 1.16 pu and a time delay of 1 second.

3.2.1.3.10 Event ID 5b4 220 kV TS 3—Badajoz 71.9 MW Disconnection

At 12:33:18.020, three PV plants connected to 220 kV TS 3—Badajoz were disconnected due to the tripping of an upstream element. This caused the loss of 71.9 MW of generation and 16 Mvar in reactive power absorption. It is the understanding of the Expert Panel that all PV plants connected to this point of connection must comply with Order TED 749/2020.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel.

However, oscillographic records were received from the PV plants coinciding with the trip of the upstream facility. These records were likely synchronised using the nearest PMU frequency to time-align data. They show that the phase-to-ground voltage at the plant's location reached 147.2 kV, which corresponds to 254.9 kV phase-to-phase. High-resolution records are not available at the connection point (220 kV TS 3—Badajoz).

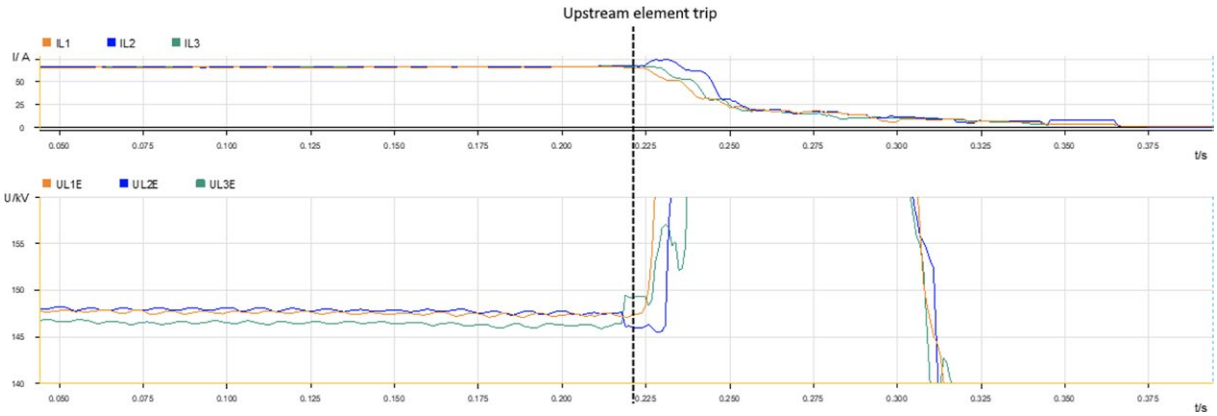


Figure 3-36: OScillographic record from one PV plant with a point of connection in 220 kV TS 3—Badajoz

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 124 MW
- V4: 0 MW
- V5: 0 MW

There is no available voltage measurement at the point of connection at the instant of the trip, nor information on whether the voltage at that point exceeded 1.15 pu and for how long it remained above that value. The voltage at one of the plants that was disconnected due to the trip of an upstream facility was indeed above the applicable required threshold; however, the duration for which it remained above that threshold is unknown.

3.2.1.3.11 Event ID 6a1 220 kV TS 2—Seville 3 MW Disconnection

At 12:33:18.102, the generation-transmission interface facility tripped at the end corresponding to the generation evacuation network and additionally sent a remote trip signal to the transmission network substation end (220 kV TS 2—Seville). This caused the disconnection of 3 MW of PV plants. It is the understanding of the Expert Panel that this plant must comply with Regulation (EU) 2016/631 (RfG).

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel. However, an oscillographic record is available from the end connected to the transmission network, where it can be observed that the voltage at the moment of the trip was 142.3 kV, which would orrespond to a phase-to-phase voltage of 246.5 kV.

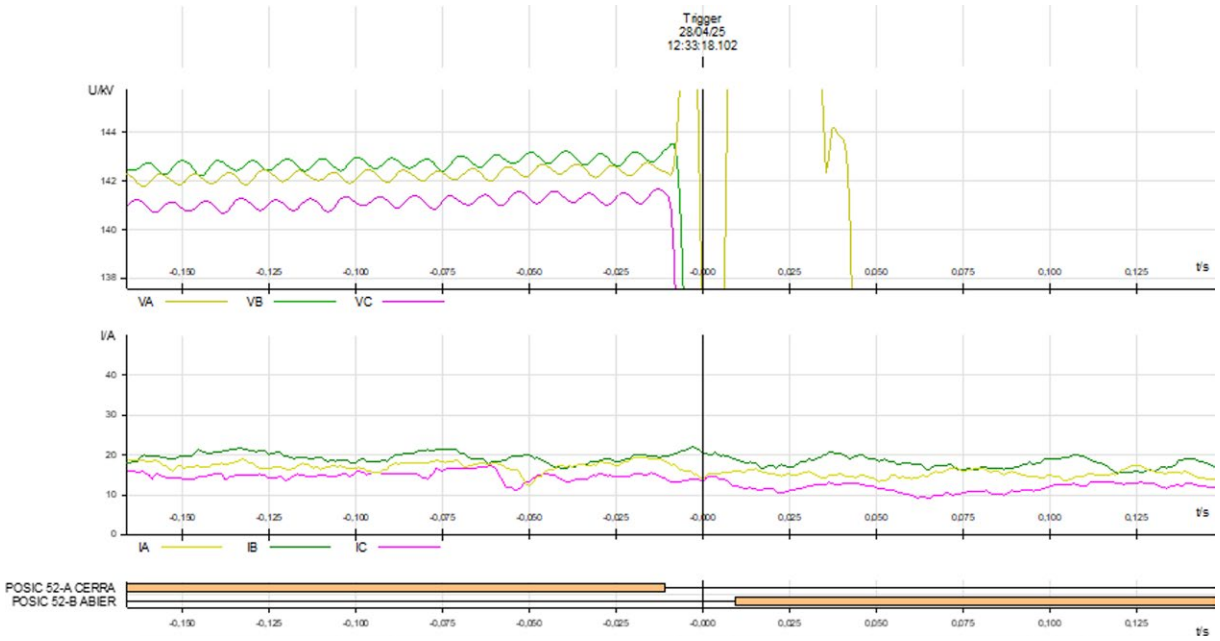


Figure 3-37: Oscillographic record from "220 kV TS 2—Seville" collector bay

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 100 MW
- V3: 0 MW
- V4: 0 MW
- V5: 0 MW

The disconnection of this generation collector installation occurs when the voltage at the point of connection was below the applicable voltage threshold (253 kV). Therefore, it appears that this trip diverges from the applicable requirements.

3.2.1.3.12 Event ID 6a2 220 kV TS 3—Badajoz 20 MW Disconnection

At 12:33:18.220, 20 MW of progressive reduction of power injection was observed due to a link facility at a 220 kV substation located in the province of Badajoz.

It is the understanding of the Expert Panel that all PV plants connected to this point of connection must comply with Order TED 749/2020.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel.

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 292 MW
- V4: 0 MW
- V5: 0 MW

There is not enough information available to assess this power reduction.

3.2.1.3.13 Event ID 6a3 400 kV TS 3—Badajoz 16 MW Disconnection

At 12:33:18.360, a 16 MW reduction was observed due to a link facility at substation 400 kV TS 3—Badajoz and a 4 Mvar absorption reduction. This was measured by a PMU that is connected at the transmission end of the line, but there is no information on which exact plant was disconnected.

It is the understanding of the Expert Panel that all PV plants connected to this point of connection must comply with Order TED 749/2020.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel. During the preparation of this final report, information was received from some plants that evacuate through this interconnection facility; however, none of them correspond to the generation loss identified at this moment.

Figure 3-38 shows the power and voltage measured by the PMU installed at that line position. Subsequent reductions in power injection down to zero can also be observed. The phase-to-phase voltage at the moment of the first observed disconnection was 447.7 kV.

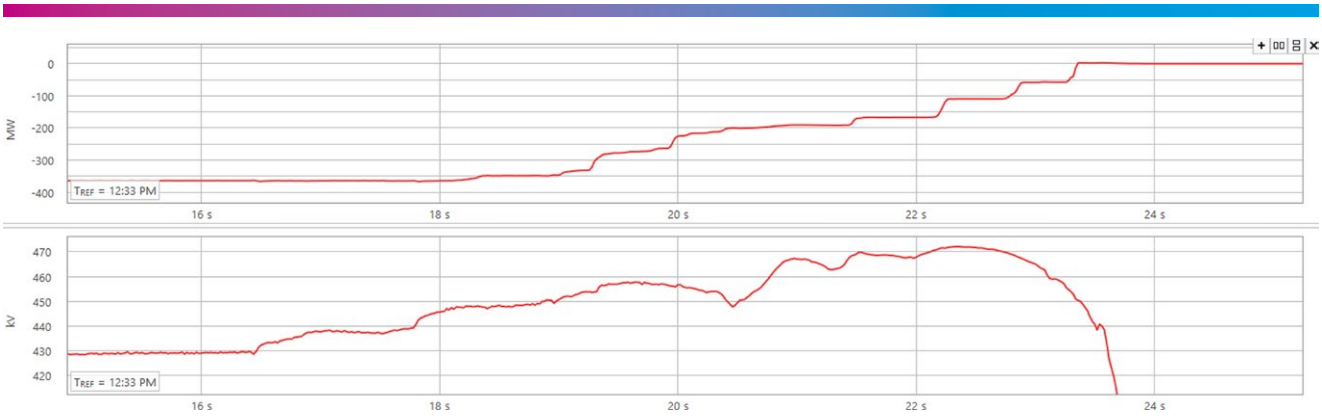


Figure 3-38: PMU measurements at the 400 kV TS 3—Badajoz collector bay

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 369 MW
- V4: 0 MW
- V5: 0 MW

At the moment of the trip, the voltage was above 440 kV. This is a level that a V3 category generation module should maintain for at least 60 minutes without disconnection in accordance with the applicable requiremnets. The system frequency at the instant of the trip was above 49.80 Hz.

However, the TED 749/2020 Order also establishes a transient overvoltage withstand requirement defined by the curve shown in Figure 3-21, which depends on the reached voltage level.

According to PMU measurements, the voltage at 400 kV TS 3–Badajoz exceeds the 440 kV threshold at 12:33:17.580, which is 780 ms before the trip. The voltage measured at the point of connection (400 kV TS 3—Badajoz) when it tripped was 447.7 kV. Considering this measured voltage, for the trip to meet the established threshold, at least 763 ms should have elapsed between the instant the 440 kV threshold is exceeded and the moment of the trip and 780 ms elapsed.

Therefore, it can be considered that this trip is aligned with the applicable requirements.

3.2.1.3.14 Event ID 6a4 400 kV TS 2—Cáceres 41 MW Disconnection

At 12:33:18.410, a PV plant connected to 400 kV TS 2—Cáceres tripped. This PV plant was generating 41 MW. It is the understanding of the Expert Panel that this plant must comply with Order TED 749/2020.

No oscillographic records from the protection system that operated were delivered to the Expert Panel. However, an oscillographic record from a transformer protection has been received, in which the disconnection instant of the plant can be observed, nevertheless, it does not include voltage measurements (Figure 3-39). The settings of the overvoltage function that operated have been received, and the settings are as follows: 440 kV (1.1 pu) with a time delay of 1.5 seconds as first stage and 460 kV (1.15 pu) with a time delay of 0.6 seconds as second stage.

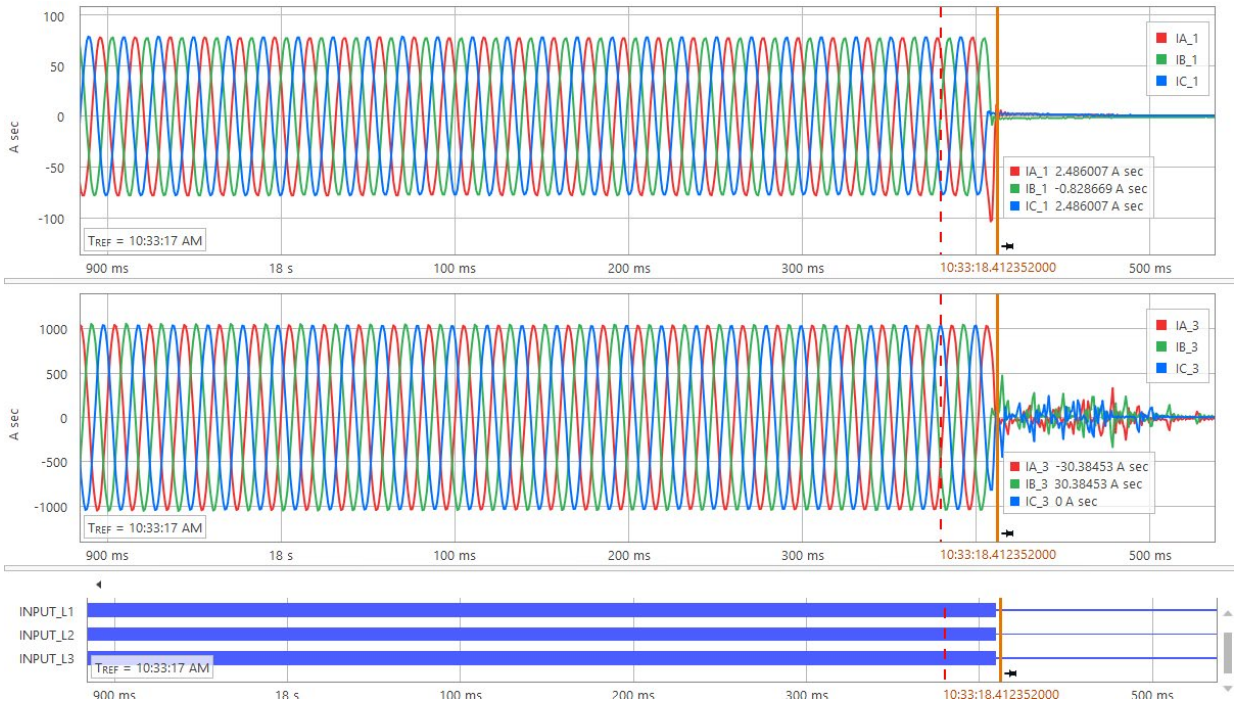


Figure 3-39: Transformer protection oscillography record from the PV plant

The phase-to-phase voltage at the moment of the first observed disconnection was 443.7 kV at 400 kV TS 2—Cáceres.

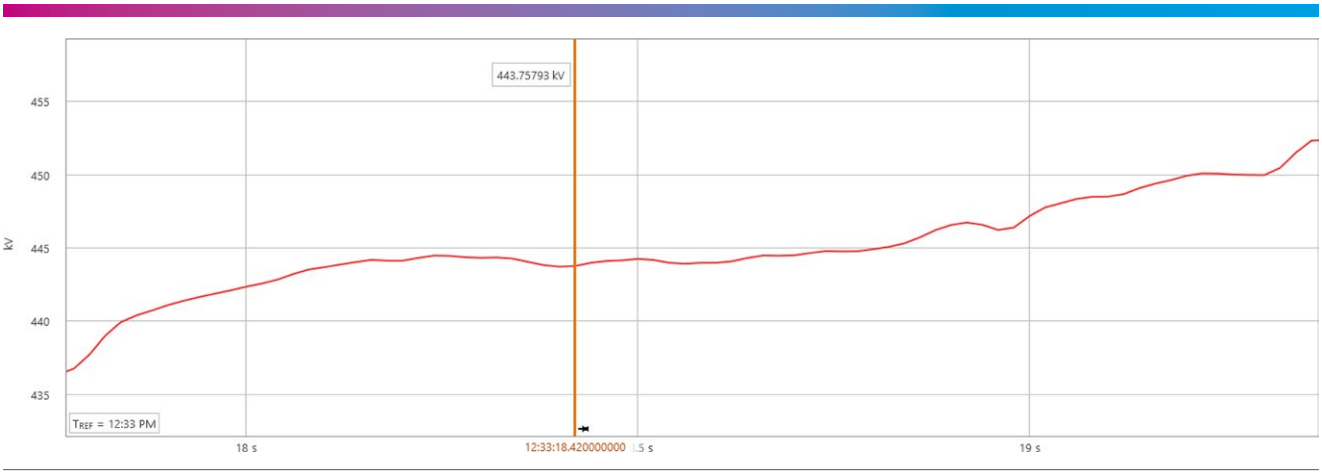


Figure 3-40: PMU voltage measurement at 400 kV TS 2—Cáceres

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 42 MW
- V4: 0 MW
- V5: 0 MW

At the moment of the trip, the voltage was above 440 kV. The requirements applicable to the category defined in this document as V3 specify that this category must withstand 440 kV for at least 60 minutes. The system frequency at the instant of the trip was above 49.80 Hz. However, the TED Order also establishes a transient overvoltage withstand requirement, defined by the curve shown in Figure 3-21 and dependent on the reached voltage level.

According to PMU measurements, the voltage at 400 kV TS 2—Cáceres exceeds the 440-kV threshold at 12:33:17.860, which is 550 ms before the trip. The voltage measured at the point of connection (400 kV TS 3—Badajoz) when it tripped was 444.3 kV.

Considering this measured voltage, for the trip to meet the established threshold, at least 898 ms should have elapsed between the instant the 440 kV threshold is exceeded and the moment of the trip. However, only 550 ms elapsed.

Therefore, it is considered that this trip diverges from the applicable requirements.

According to the applicable requirements, with a voltage of 440 kV, the plant must remain connected for at least 60 minutes, whereas the settings at the time of the incident have caused the plant to disconnect after 1.5 seconds when the voltage reached 440 kV it appears that the settings are not aligned with the applicable requirements. It should also be taken into account that the possible hysteresis present in the relay, as explained in event ID 5a2, will cause that with an overvoltage threshold set at 1.1 pu it is not possible to meet the applicable requirements. Furthermore, it must be taken into account that the validation of these voltage levels must be assessed at the connection point in the transmission network, not locally in the evacuation substation. Therefore, given that the voltage at the transmission substation and at the evacuation substation is not exactly the same, a sufficient safety margin must be ensured so that, as long as 440 kV is not exceeded in the transmission network, the plant must not disconnect.

3.2.1.3.15 Event ID 6a5 400 kV TS 1—Murcia 63 MW Disconnection

At 12:33:18.540, a PV plant connected to a 132 kV distribution network close to 400 kV TS 1—Murcia tripped when it was generating 63 MW. It is the understanding of the Expert Panel that this plant must comply with the Spanish Operational Procedure 1.4.

The settings of that relay have not been received.

The cause of the trip was overvoltage. The phase-to-phase voltage at the 132 kV substation was 147.3 kV.

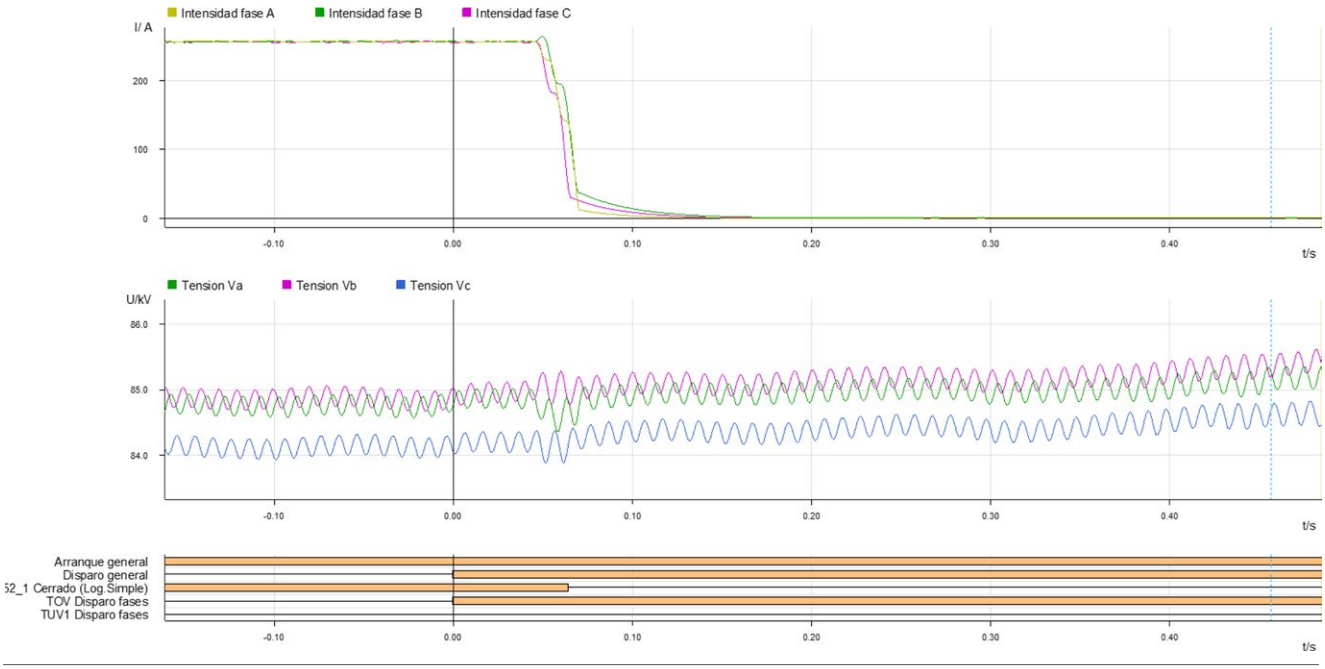


Figure 3-41: Oscillographic record from the PV plant

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 0 MW
- V4: 0 MW
- V5: 74 MW

Since this plant belongs to category V5, there is no regulation defining the voltage levels that the plant must withstand without disconnection.

3.2.1.3.16 Event ID 6a6 220 kV TS 1—Cádiz 26 MW and ID 6a7 220 kV TS 1—Cádiz 128 MW Disconnection

At 12:33:18.630, a transformer connected to 220 kV GS 1—Cádiz tripped, and 25.7 MW were lost. After 50 ms, the evacuation line to 220 kV TS 1—Cádiz tripped at the generation end, and an additional 127.5 MW were disconnected. There are wind farms and PV plants connected to 220 kV TS 1—Cádiz through this evacuation line. Some of these plants were commissioned before the RfG came into force, others afterwards, and some after the entry into force of the TED 749 Order.

No oscillographic records from the protection system that operated, nor the settings of that relay, were delivered to the Expert Panel. However, an oscillographic record was received from the bay protection system at the circuit breaker corresponding to the evacuation line towards the transmission network.

There are no high-resolution records from the transmission substation that is the point of connection. However, the phase-to-ground voltage recorded at the generation substation was 143.3 kV, which corresponds to a phase-to-phase voltage of 248.2 kV.

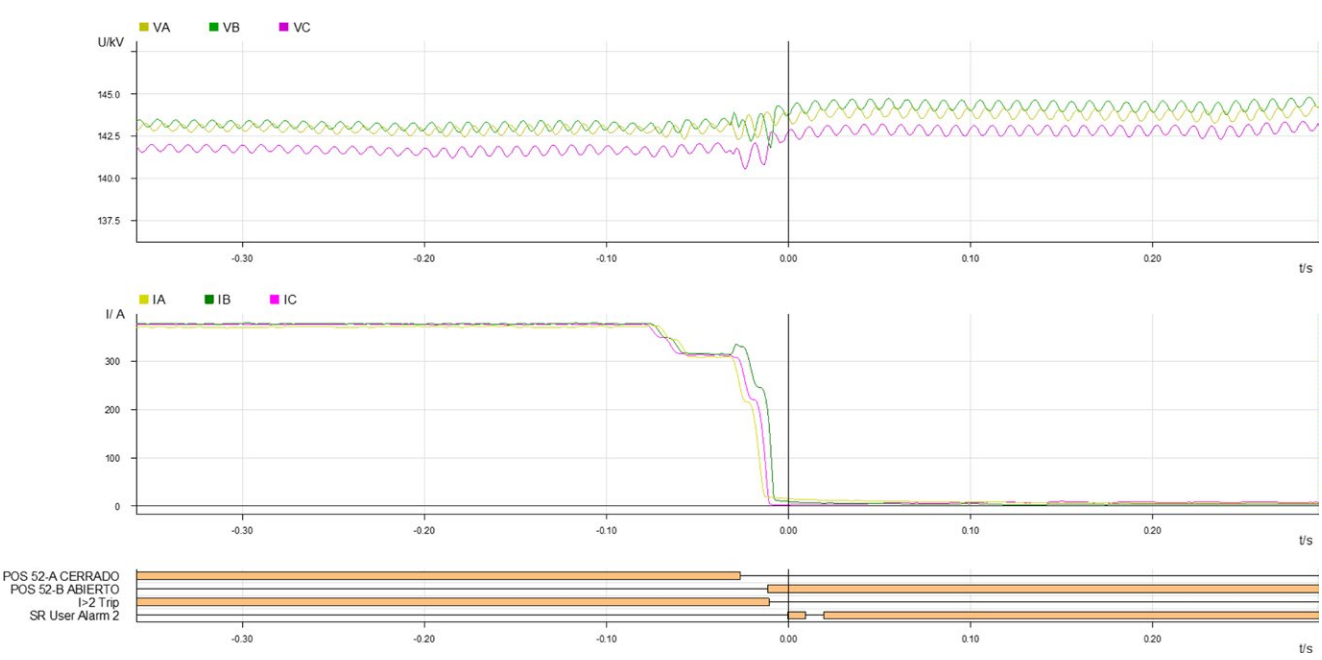


Figure 3-42: Oscillographic record from the 220 kV GS 1—Cádiz bay to TS 1—Cádiz

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 100 MW
- V2: 8 MW
- V3: 88 MW
- V4: 0 MW
- V5: 0 MW

There is no voltage measurement available at the point of connection coinciding with the trips; however, voltage measurement is available at the end of the generating plant's collector substation, which is only separated by 175 meters. In addition, current and voltage measurements are available, and therefore, together with the parameters of that line, they allow calculating the voltage at the transmission substation end.

Prior to the first disconnection, the voltage measured at the end of the generating plant's collector substation was 142.9 kV phase to ground (247.5 kV phase to phase).

If, based on this voltage and the active and reactive power flow through the line, the voltage at the transmission substation end is calculated, the resulting value is the same, 142.9 kV phase to ground (1.125 pu and 247.5 kV phase to phase), as the difference would be only 0.0005 kV phase to ground. This is consistent with the fact that the two substations are separated by only 175 meters of line. This trip corresponds to the tripping of a transformer. Downstream of it there is a facility classified as category V2, which therefore cannot be disconnected until the voltage exceeds 253 kV (1.15 pu), given that its connection point is at 220 kV. Therefore, this trip is not in accordance with the applicable legislation.

Prior to the second disconnection, which corresponds to the trip of the coupling facility at the end of the generating plant's collector substation, the voltage measured at the collector substation end was 143.3 kV phase to ground (248.2 kV phase to phase). If, based on this voltage and the active and reactive power flow through the line, the voltage at the transmission substation end is calculated, the resulting value is again the same, 143.3 kV phase to ground (1.128 pu and 248.2 kV phase to phase). Downstream of this coupling facility there are generation plants classified as category V3, which must remain connected for at least 60 minutes provided that the voltage does not exceed the threshold of 253 kV (1.15 pu). Therefore, it appears that the trip of this coupling facility diverges from the applicable requirements.

3.2.1.3.17 Event ID 6a8 400 kV TS 1—Málaga 154 MW Disconnection

At 12:33:18.846, the generation–transmission interface line tripped at the end corresponding to the generation evacuation network and additionally sent a remote trip signal to the transmission network substation end (400 kV TS 1—Málaga). This caused the disconnection of several PV plants with a total production of 154 MW.

It is the understanding of the Expert Panel that all PV plants connected to this point of connection must comply with Order TED 749/2020.

The received settings indicate that the overvoltage function is configured at 439.56 kV phase-to-phase (1.0989 pu) with a time delay of one second. Additionally, the oscillographic record from the tripping relay was delivered to the Expert Panel, showing that at the moment of the trip, the voltage of all three phases was above 1.1 pu. Figure 3-43 shows the oscillographic record that tripped the line at the generation end.

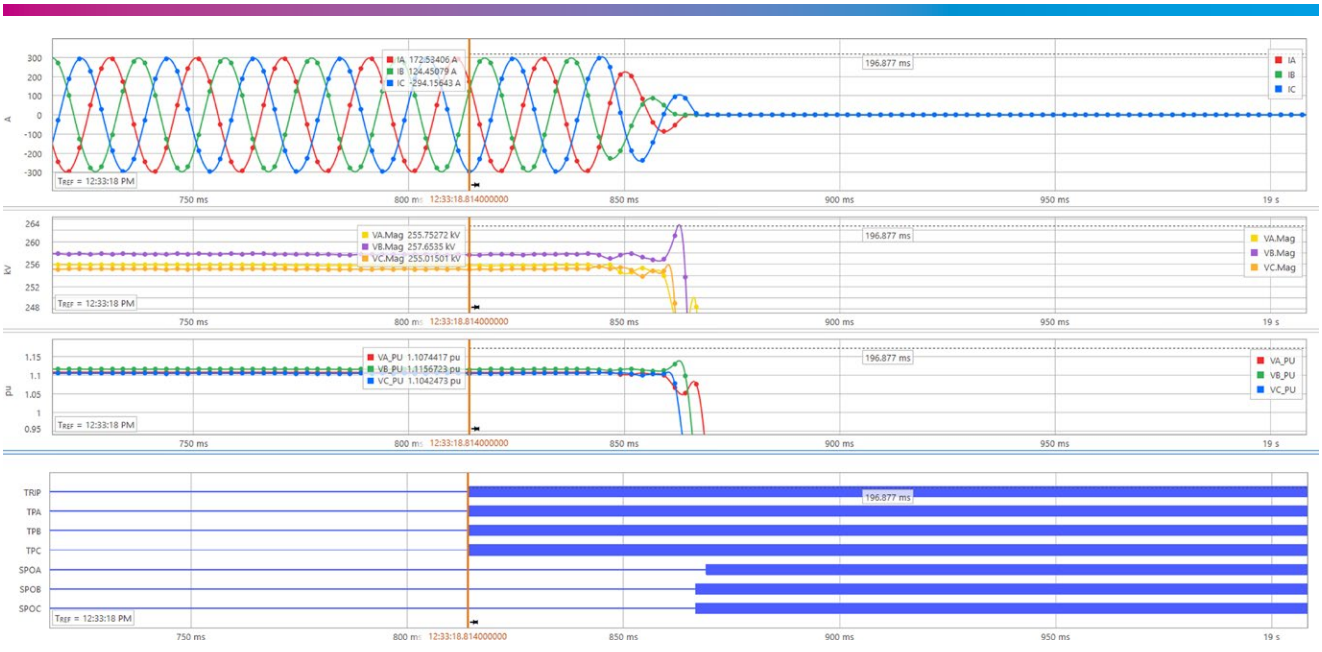


Figure 3-43: Line protection oscillographic record from the generation end

Figure 3-44 shows the oscillographic record obtained at the transmission end. The phase-to-phase voltage at the time of the trip was 448.6 kV.

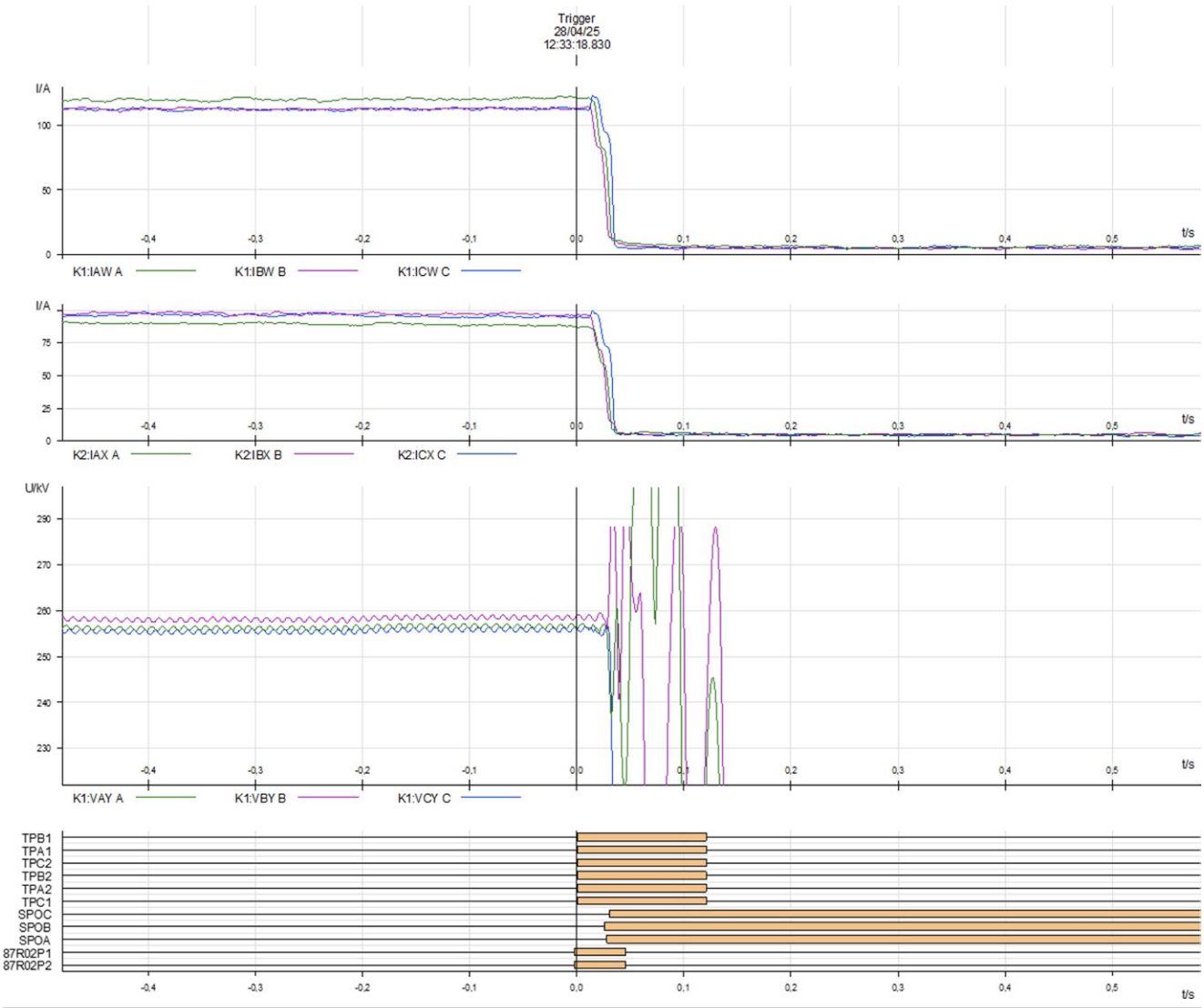


Figure 3-44: Oscillographic record from 400 kV TS 1 – Málaga, collector bay (TPx1: breaker 1 trip of each phase, TPx2: breaker 2 trip of each phase, SPOx: Single pole open of each phase, 87R02P1: transfer trip received through communication channel 1, 87R02P2: transfer trip received through communication channel 2)

The distribution overvoltage withstand requirements for generation connected downstream of the tripped element is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 174 MW
- V4: 0 MW
- V5: 0 MW

The settings implemented in the protection that tripped appear to diverge from the requirements applicable to this facility.

The applicable standard requires withstanding 440 kV for at least 60 minutes; however, the protection is configured to trip when the voltage reaches 439.56 kV for one second. In addition to being below the applicable required threshold, it must also be taken into account that the setting should include a sufficient margin for measurement errors and, furthermore, since the relay does not measure the voltage at the point of connection, the overvoltage function threshold that has to consider this in order to not pick up until the voltage at the point of connection actually exceeds the specified limit.

The threshold implemented in the protection, expressed in secondary values, is 1.11 pu of the nominal voltage on the secondary side of the voltage transformer.

However, since the nominal voltage of the voltage transformer is 396 kV rather than 400 kV, the voltage threshold defined for the primary side corresponds to the previously indicated value (1.0989 pu).

The voltage at the instant of the trip was above the value defined in line with the applicable requirements, which specifies that the installation must withstand up to 440 kV for at least 60 minutes, with the system frequency above 49.8 Hz. Therefore, it appears that this disconnection is most likely aligned with the applicable requirements. However, it is not possible to assess alignment with the transient overvoltage curve, as no recording is available that allows identifying the moment at which 440 kV was exceeded, and the pick up threshold is set below 440 kV. Based on the voltage measured at the point of connection at the trip instant, 448.6 kV, the protection should not have operated until at least 796 ms had elapsed after the 440 kV threshold was surpassed.

3.2.1.3.18 Event ID 6b 400 kV TS 1—Cuenca 530 MW Disconnection

At 12:33:18.951, the generation–transmission interface line tripped at the end corresponding to the generation evacuation network and additionally sent a remote trip signal to the transmission network substation end (400 kV TS 1—Cuenca). This caused the disconnection of several PV plants with a total production of 530 MW.

It is the understanding of the Expert Panel that all PV plants connected to this point of connection must comply with Order TED 749/2020.

The settings delivered to the Expert Panel for these relays show two stages for the overvoltage protection function: a first stage with a voltage threshold of 1.088 pu and a time delay of 60 minutes, and a second stage with a voltage threshold of 1.11 pu and a time delay of 1.2 seconds. The COMTRADE record of the relay that tripped at the generation substation end shows that the maximum phase-to-ground voltage measured at the time of the trip was 257.8 kV (1.116 pu), which corresponds to a phase-to-phase voltage of 446.6 kV.

Figure 3-45 shows the record of the generation side.

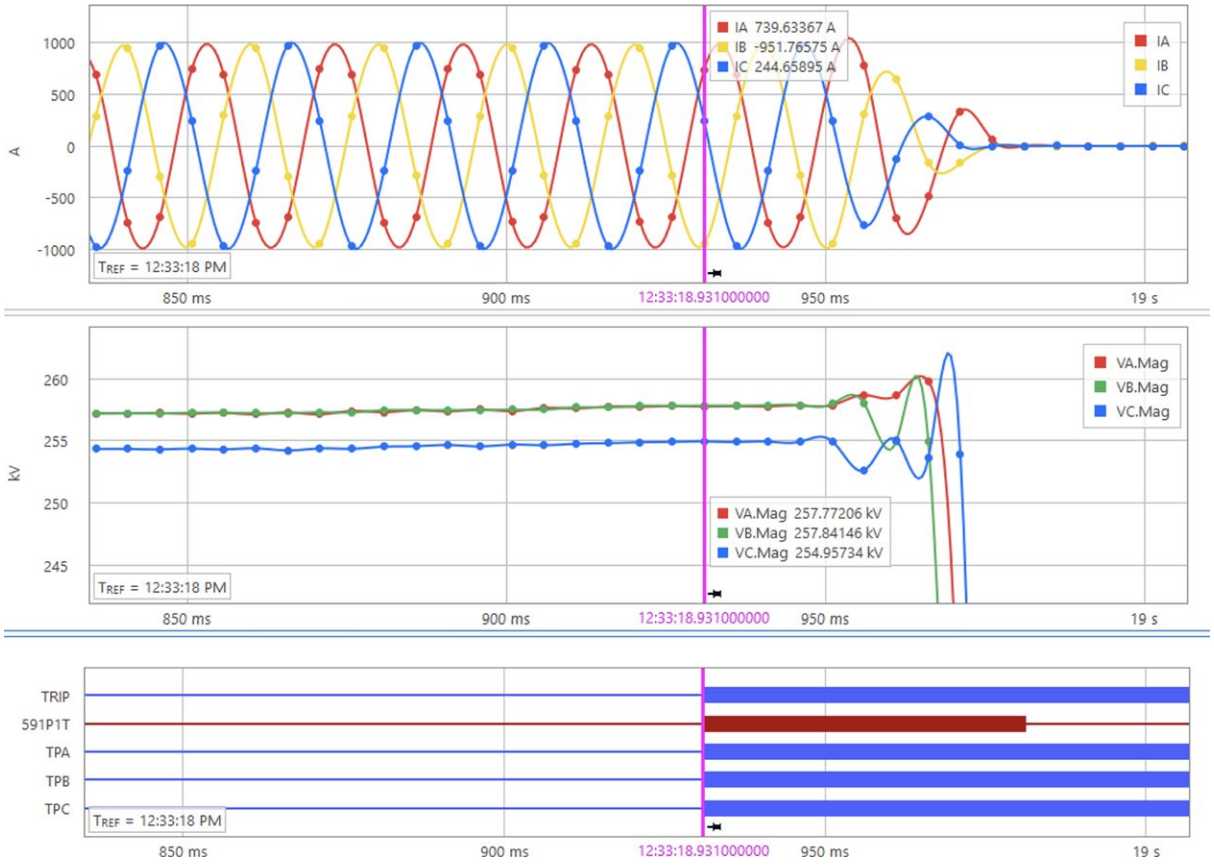


Figure 3-45: Oscillographic record from 400 kV GS 1—Cuenca, TS 1—Cuenca bay (591P1T: overvoltage trip)

Figure 3-46 shows the oscillographic record corresponding to the transmission substation end. It shows that the phase-to-ground voltage measured at the moment of the trip was 259.0 kV, which corresponds to a phase-to-phase voltage of 448.6 kV.

It should be noted that the ripple observed in the image in the RMS voltage values is caused by the fact that the system frequency at that moment was not 50 Hz, but approximately 49.8 Hz. This necessitated adjusting the time window used for filtering and calculating this value to align with the actual system frequency. However, the visualisation software is not capable of performing this adjustment. The same behaviour has been observed in previous records.

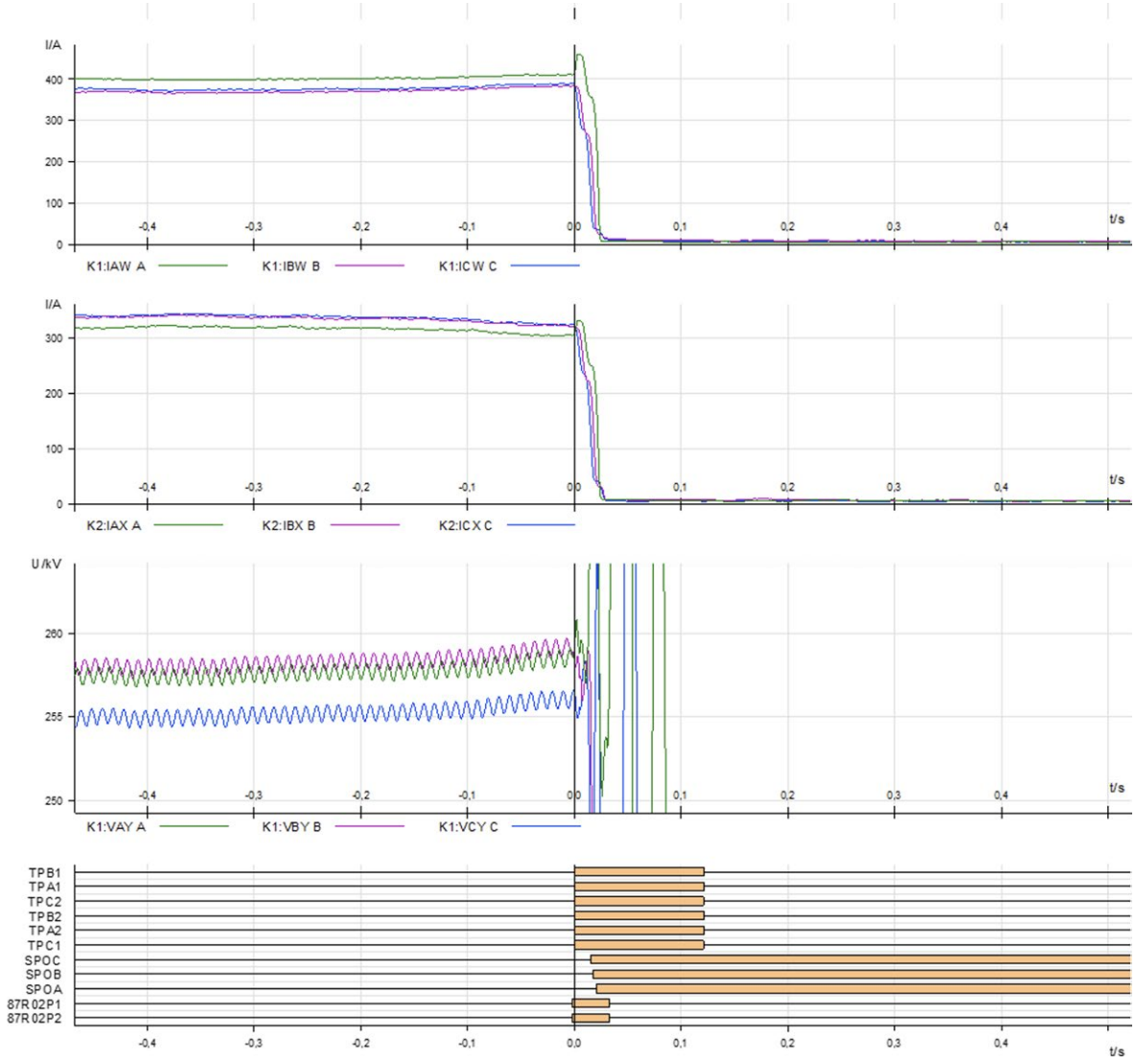


Figure 3-46: Oscillographic record from 400 kV TS 1–Cuenca transmission substation, GS 1–Cuenca bay (TPx1: breaker 1 trip of each phase, TPX2: breaker 1 trip of each phase, SPOx: Single pole open of each phase, 87R02P1: transfer trip received through communication channel 1, 87R02P2: transfer trip received through communication channel 2)

The distribution overvoltage withstand requirements for generation connected to the 400 kV TS 1–Cuenca substation is as follows:

- V1: 0 MW
- V2: 0 MW
- V3: 664.85 MW
- V4: 0 MW
- V5: 0 MW

According to the applicable requirements, the installation must remain connected for at least 60 minutes up to the threshold of 440 kV. This trip occurred when the voltage was above the threshold. No recordings with sufficient pre-fault duration are available to determine the exact moment at which the voltage exceeded the 440 kV threshold so it is not possible to assess alignment with the transient overvoltage curve. It seems likely that this trip is aligned with the applicable requirements.

In accordance with the applicable requirements, the voltage setting threshold should be above 1.1 pu. However, the first stage configured for the overvoltage protection function is 1.088 pu with a 60 minute delay it appears that these settings diverge from the applicable requirements. This stage also appears to be configured as if the plant were required to trip after 60 minutes when the voltage threshold defined in accordance with the applicable requirements is exceeded.

It is important to clarify at this point that the applicable requirements do not mandate tripping once the thresholds and associated delay times are exceeded; rather, they require the installation to remain connected at least until those thresholds and times are reached. The plant may remain connected for a longer period or withstand higher voltages if it has the capability to do so.

3.2.1.3.19 Conclusions

Of the 19 trips detected between 12:32:57 and 12:33:19, 18 of which occurred after 12:33:16, it is considered that 9 of them diverged from the applicable requirements and caused the loss of 1866.7 MW. For three of the trips (125.6 MW) there is insufficient information to evaluate them, and another 3 trips (802 MW) seem aligned with the applicable requirements, but it is not possible to assess whether they meet the requirements regarding transient overvoltages.

Only 4 of the trips (195.87 MW) seem aligned with the applicable requirements.

Settings for the overvoltage function were received for 8 protection devices out of the 19 trips analysed. All of the settings received diverge from the applicable requirements.

3.2.1.4 Generation Disconnections in Portugal

In Portugal, the disconnection of generating units connected to the transmission grid that occurred between 12:33:23 and 12:33:37 was caused by a frequency drop below 47.5 Hz.

This response seems aligned with the applicable requirement outlined in the previous chapter.

Concerning generating units connected to the distribution grid, some of them disconnected before 12:33:23, and the total amount provided by the DSO were:

F-setting (Hz)	PV (MW)	Wind (MW)	Other (MW)	Time delay time (ms)
49.5 & 50.5	167	71	337	≈ 150 (with CB)
49.8 & 50.2	2	109	210	≈ 150 (with CB)

Table 3-4: Disconnection of generation units on the DSO grid before 12:33:23

Explanation: Even though connections points protection is set to F ≤ 47.5 Hz and F ≥ 51.5 Hz, some power plants have their internal protection settings configured at 49 Hz and 51 Hz, or other more restrictive values, which are unknown to the DSO.

These internal protection relays were configured by the manufacturer during the old plant's commissioning and have remained unchanged since then. E-REDES PT, in coordination with the national authorities and power plant owners, is working to harmonise these settings.

3.2.1.5 Generation Disconnections in France

Below is a detailed list of impacts on generation units and network elements.

Tripping time	Generation unit	Type	P lost (MW)	Q lost (Mvar)	Voltage level (kV)	Cause of the trip
12:33:19.850	Hydro 1 South-West of France	Hydro	3.5	0	63	Overcurrent protection tripped
12:33:19.850	Hydro 2 South-West of France	Hydro	3.5	0	63	Overcurrent protection tripped
12:33:19.850	Hydro 3 South-West of France	Hydro	3.5	0	63	Overcurrent protection tripped
12:33:20.900	Hydro 4 South-West of France	Hydro	-35	0	225	Pump decoupling due to underfrequency protection trip
12:33:23.307	Hydro 5 South-West of France	Hydro	28	-17	225	Loss of electrical excitation of stator
12:33:23.460	Hydro 6 South-West of France	Hydro	-210	27	400	Pump decoupling due to a low power (147 MW)
12:33:23.863	Hydro 7 South-West of France	Hydro	6	?	150	Loss of electrical energisation
12:33:23.857	Hydro 8 South-West of France	Hydro	1.8	0	63	Loss of the 63 kV line due to an overcurrent protection (1,145 A low current)
12:33:23.857	Hydro 9 South-West of France	Hydro	1.8	0	63	Loss of the 63 kV line due to an overcurrent protection trip
12:33:30	Hydro 10 South-West of France	Hydro	5	-2	63	Internal protection trip
12:33:35.759	Nuclear 1 South-West of France	Nuclear power plant	1,290	-430	400	Internal Protection trip
12:35:53.189	Hydro 12 South-West of France	Hydro	17	-2	63	Overcurrent protection
12:36:22.098	Hydro 13 Sout West of France	Hydro	17	0		Overcurrent protection
12:36:22.098	Hydro 14 South West pf France	Hydro	17	0		Overcurrent protection
12:39:55.121	Hydro 15 South West of France	Hydro	6	0	67	Internal protection trip
14:42:08.320	Hyrdo 16 South West of France	Hydro	6	0	67	Internal protection trip
12:42:36:637	Hydro 17 south West of France	Hydro	3	0	70	Manual opening of CB (instruction from RTE)
12:45:08.903	Hydro 18 South West of France	Hydro	10	0	259	Manual opening of CB (instruction from RTE)
12:45:06.035	Hydro 19 South West of France	Hydro	10	0	259	Manual opening of CB (instruction from RTE)
12:45:01.552	Hydro 20 Sputh West of France	Hydro	10	0	259	Manual opening of CB (instruction from RTE)

Table 3-5: List of production tripping in France

3.2.1.5.1 Nuclear 1 South-West of France

The nuclear reactor Nuclear 1 South-West (1,290 MW) disconnected at 12:33:35.759.

Response of French nuclear power plants to the Iberian blackout of April 28, 2025

As seen in chapter 3.1.2.1.1 the transient during the loss of synchronism triggered significant voltage and frequency oscillations. On the France-Spain interconnection, power oscillations of several thousand MW occurred for a few seconds.

The frequency wave received in the southwest was dampened with distance. The power plants Nuclear 1 South-West and Nuclear 2 South-West received the strongest wave. It ultimately caused Nuclear 1 South-West to scram.

The Nuclear 1 South-West site comprises two 1,300 MW units, only one of which was online at the time.

At Nuclear 2 South-West, the frequency wave only caused a temporary drop in power in two reactors. At the Nuclear 2 South-West site, three of the four 900 MW units were in service.

The attenuation of the wave was such that no automatic actions were triggered at other French nuclear plants.

The scram of Nuclear 1 South-West and the power drop at Nuclear 2 South-West

Nuclear 1 South-West experienced significant and rapid fluctuations in secondary parameters (generator frequency, active power, voltage), which generated a transient variation in the pressure of the first turbine wheel and activated the alternator overspeed protection. The almost complete closure of the steam valves occurred twice within the space of one second and thus led to a significant drop in secondary power. The insertion of control rods followed to balance the power of the nuclear core with that of the secondary system, i.e. the turbine and alternator. This caused a significant reduction in the neutron flux.

The duration and depth of this drop in neutron flux within the core exceeded the threshold for triggering the neutron flux reduction protection system, leading to the automatic shutdown of the reactor.

This protection system was originally designed to be triggered in the event of an unexpected control rod drop.

The frequency and voltage remained within design limits, so islanding was not initiated.

At Nuclear South West 2, the steam valves on the 900 MW units close more slowly. Thus, the drop in secondary and subsequently nuclear core power was less sudden and the protection against excessive negative variations in neutron flux was not triggered. Secondary power always remained above 50 % of its nominal value.

Apart from these events, no additional protection systems were activated at either of the two plants most exposed to the frequency oscillation.

3.2.1.5.2 Solar 1 South-West of France

Production decreased from 16 to 7 MW. As the voltage increased, production progressively recovered.

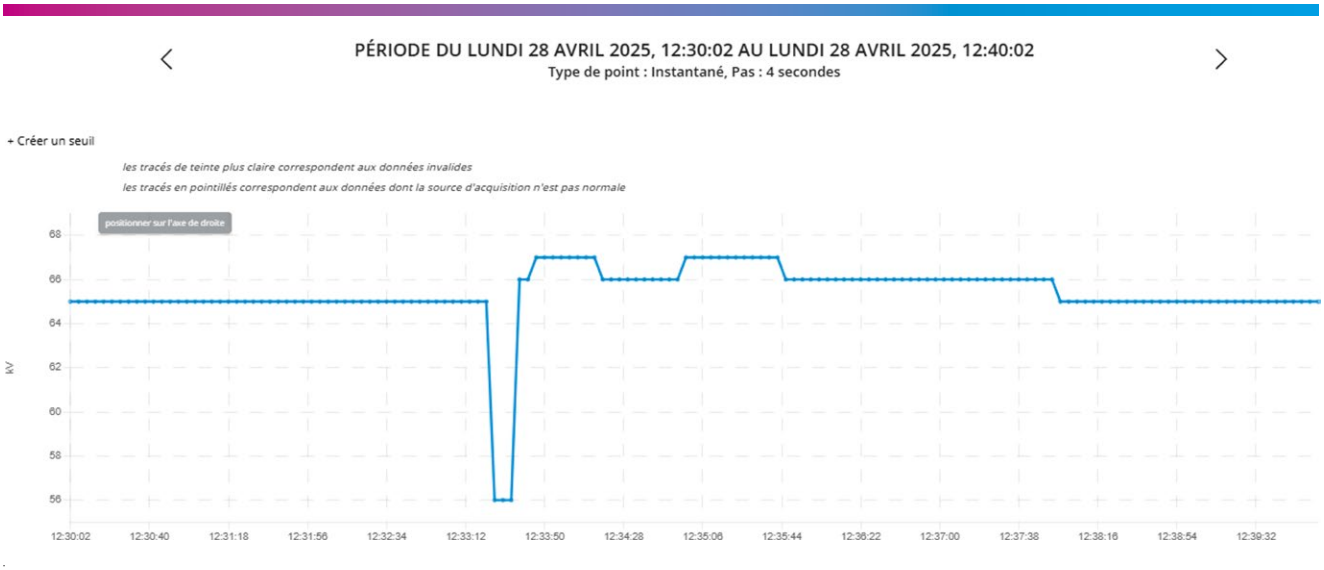


Figure 3-47: Solar 1

3.2.1.5.3 Solar 2 South-West of France

There was a decrease in production from 23 to 4 MW, along with an increase in the voltage (see Figure 3-48).

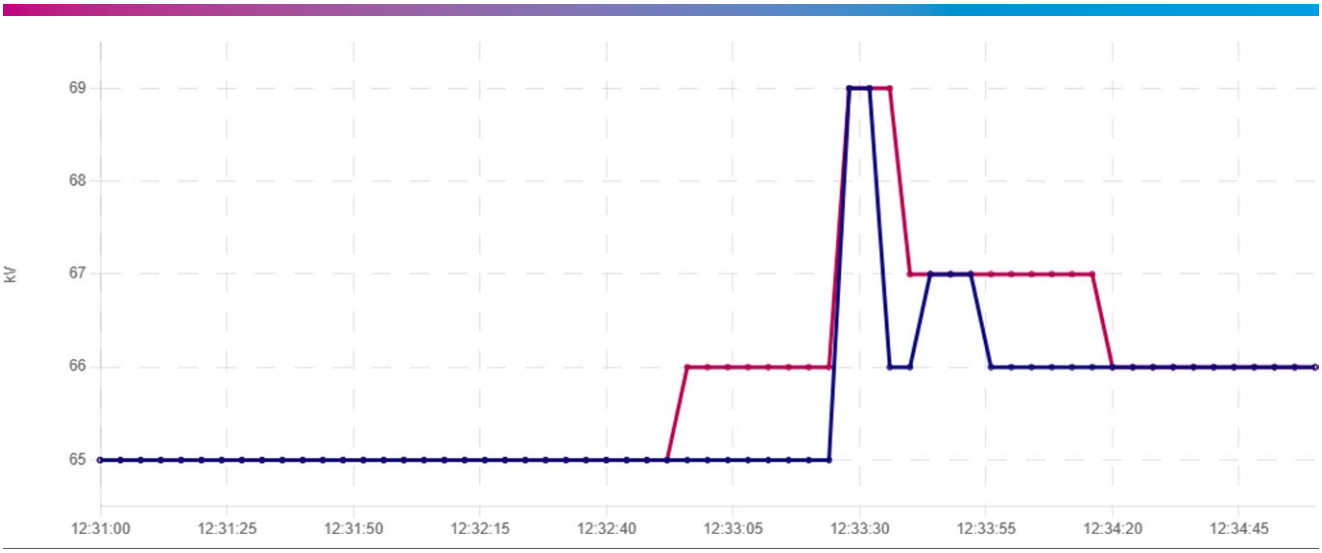


Figure 3-48: Solar 2a (blue) and Solar 2b (red)

3.2.1.5.4 Battery 1 South-West of France

A rise in voltage was seen at 12:33:26, as shown in Figure 3-49.

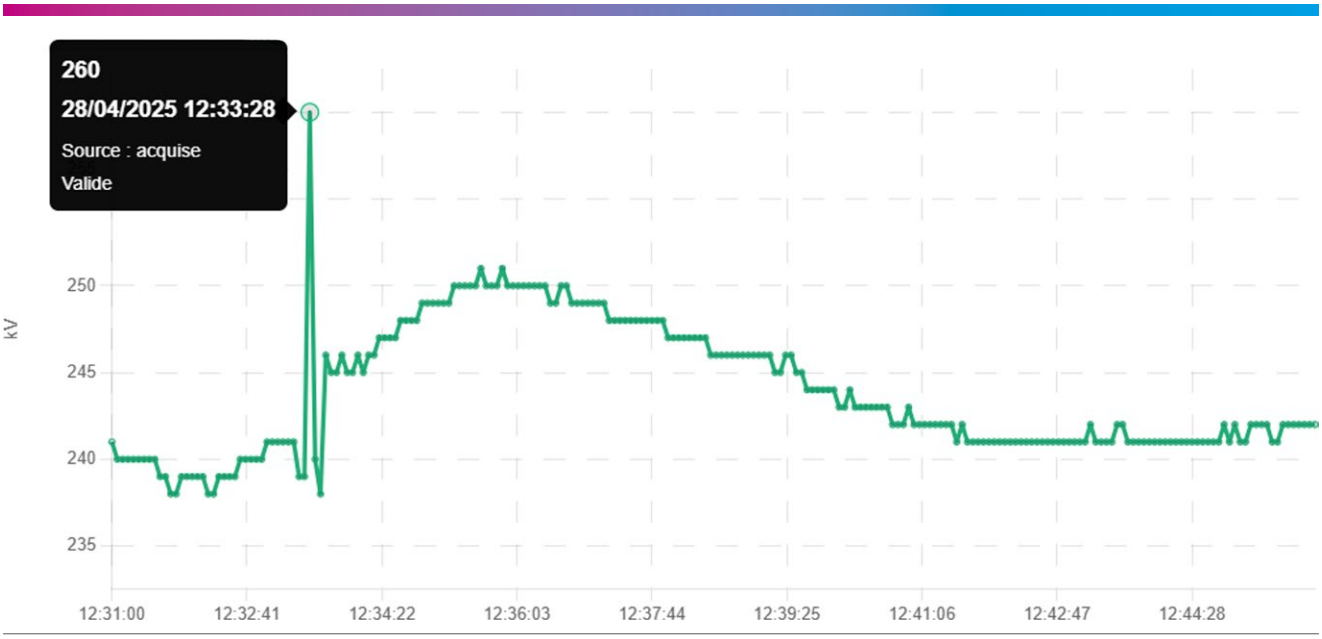


Figure 3-49: Battery 1

3.2.2 Transmission System Protection Behaviour During the Incident

The events that will be analysed are listed in Table 3-6. The identification of the events is in line with the sequence of the events presented in Table 3-1.

Event	Time (CEST)	Substation A (TSO)	Substation B (TSO)	Voltage Level (kV)	Asset Type	Relay Trigger	Cause of Tripping	Tripped Phases	Relay Parameter Settings
9	12:33:19.971	Arganda (RE)	Loeches (RE)	220	OHL	OV	Incorrect measurement from phase B voltage transformer in Loeches substation	3 phase	V > 120 % Vnominal time delay = 1 s
16a	12:33:20.229	Puerto de la Cruz (RE)	Beni Harchane (ONEE)	400	OCHL-TIE	UF	Low frequency function at Morocco side and transfer trip to Spain	3 phase	f < 49.5 Hz
16b	12:33:20.473	Puerto de la Cruz (RE)	Melloussa (ONEE)	400	OCHL-TIE	UF	Low frequency function at Morocco side and transfer trip to Spain	3 phase	f < 49.5 Hz
31a	12:33:21:407	Baixas (RTE)	Vic (RE)	400	OHL-TIE	OST	Loss of synchronism	3 phase	2 beats
31b1	12:33:21.437	Argia (RTE)	Arkale (RE)	220	OHL-TIE	OST	Loss of synchronism	3 phase	2 beats
31b2	12:33:21.535	Argia (RTE)	Hernani (RE)	400	OHL-TIE	OST	Loss of synchronism	3 phase	2 beats
42	12:33:23.076	Valdecaballe-ros (RE)	Maguilla (RE)	400	OHL	OV	Overvoltage	3 phase	V > 120 % Vnominal time delay = 1 s
52	12:33:23.954	Mogadouro (REN)	Central da Valeira (REN	220	OHL	DIST	Distance Z1 trip due to unstable voltage conditions protection	3 phase	Z < 80 % line
55b	12:33:25.144	Cartelle (RE)	Lindoso (REN) (ckt 2)	400	OHL-TIE	DIST	Z1 trip due to voltage and frequency collapse	3 phase	Z < 80 % line
55c	12:33:25.183	Cartelle (RE)	Lindoso (REN) (ckt 1)	400	OHL-TIE	DIST	Z1 trip due to voltage and frequency collapse	3 phase	Z < 80 % line
55d	12:33:25.325	Cedillo (RE)	Falagueira (REN)	400	OHL-TIE	DIST	Z1 trip due to voltage and frequency collapse	3 phase	Z < 80 % line
55e	12:33:25.410	Saucelle (RE)	Pocinho (REN)	220	OHL-TIE	DIST	Z1 trip due to voltage and frequency collapse	3 phase	Z < 80 % line
56	12:33:25.434	Lagoaça (REN)	Armamar (REN)	400	OHL	DIST	Distance Z1 trip due to unstable voltage conditions	3 phase	Z < 80 % line
55f	12:33:26.165	Aldeadávila (RE)	Pocinho (REN) (ckt 1)	220	OHL-TIE	DIST	Z1 trip due to voltage and frequency collapse	3 phase	Z < 80 % line

Table 3-6: Performance of the protection system of lines in the sequence of events, abbreviations: OHL: overhead line, OCHL: overhead-cable hybrid line, OHL-TIE: overhead line tie line, OCHL-TIE: overhead-cable hybrid line tie line, DIST: distance protection, UF: under frequency protection, OST: Loss of synchronism protection, OV: overvoltage protection

3.2.2.1 Event ID 9: 220 kV Arganda (RE)—Loeches (RE)

At 12:33:19.971, the Arganda (RE)—Loeches (RE) 220 kV overhead line (OHL) tripped. The overvoltage function caused the trip; during the high voltage situation, an incorrect measurement from the phase B voltage transformer in the Loeches substation was the cause of the overvoltage function trip. During the tests performed on the capacitive voltage transformer after its disassembly, short circuits were detected in some capacitive elements. This resulted in the transformation ratio not responding as expected.

The relay protection system of the OHL operated according to its settings (1.2 pu, one-second delay).

In Figure 3-50, the upper chart shows the voltage measured at the Loeches substation, where the overvoltage was detected, and the lower chart shows the voltage measured at the Arganda substation. Arganda relays did not trip, but the breaker opened because they received a transfer trip signal.

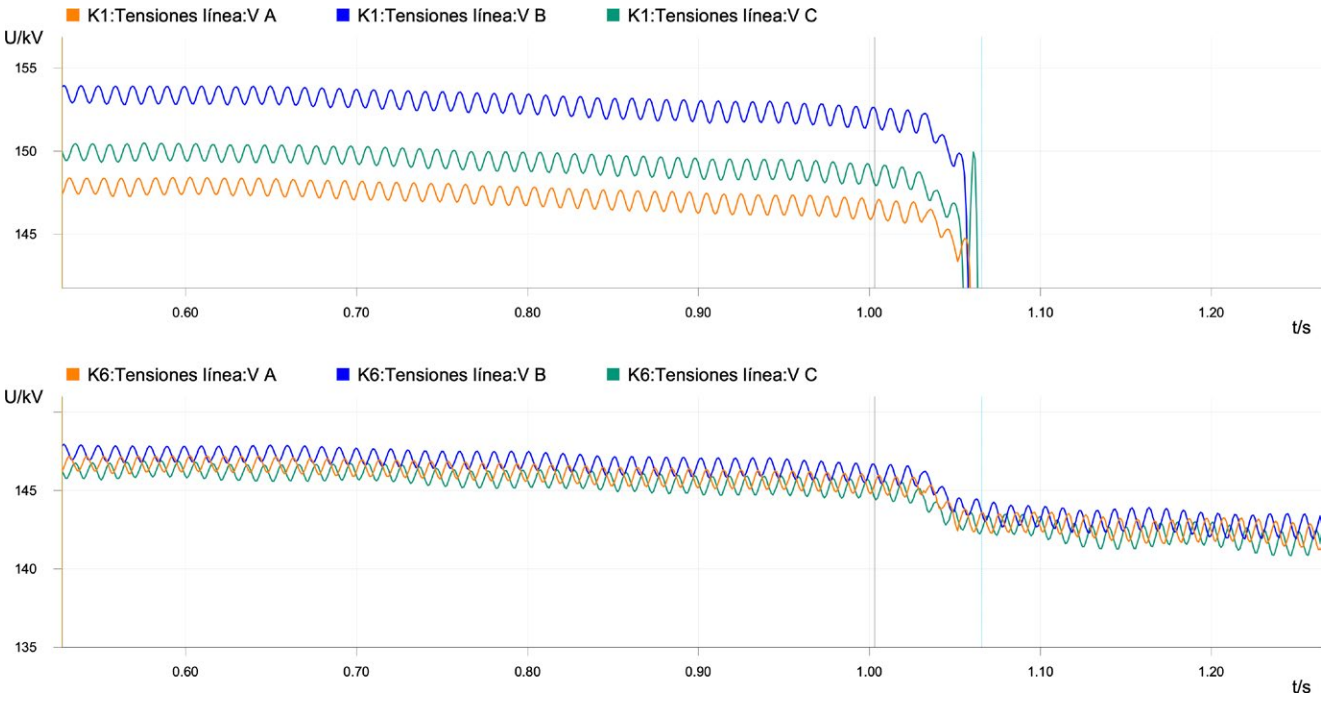


Figure 3-50: RMS Voltages in Loeches substation (upper chart) and in Arganda substation (lower chart)

Figure 3-50 shows that the voltage measurements in the different phases at the Loeches substation vary significantly from one phase to another. This anomaly suggests that the voltage transformer was exhibiting abnormal behaviour. Additionally, the fact that no other lines in the substation tripped, when all have the same overvoltage settings, leads to the conclusion that the voltage measurement at the Loeches substation was not correct.

This trip was undesired, as the primary voltage was below Red Eléctrica's overvoltage trip threshold criteria. However, at the moment of the trip, the voltage had already risen above the normal operating limits of the system and even exceeded the values that generators commissioned after 8 January 2021 are required to withstand before tripping. Furthermore, the disconnection of this 220 kV line had no impact on the development of the incident.

3.2.2.2 Event ID 16a: 400 kV Puerto de la Cruz (RE)–Beni Harchane (ONEE)

At 12:33:20.229, the Puerto de la Cruz (RE)–Beni Harchane (ONEE) 400 kV tie-line opened in Beni Harchane due to an underfrequency protection and also sent a direct transfer trip to the Spanish side (Puerto de la Cruz) that opened 161 ms later.

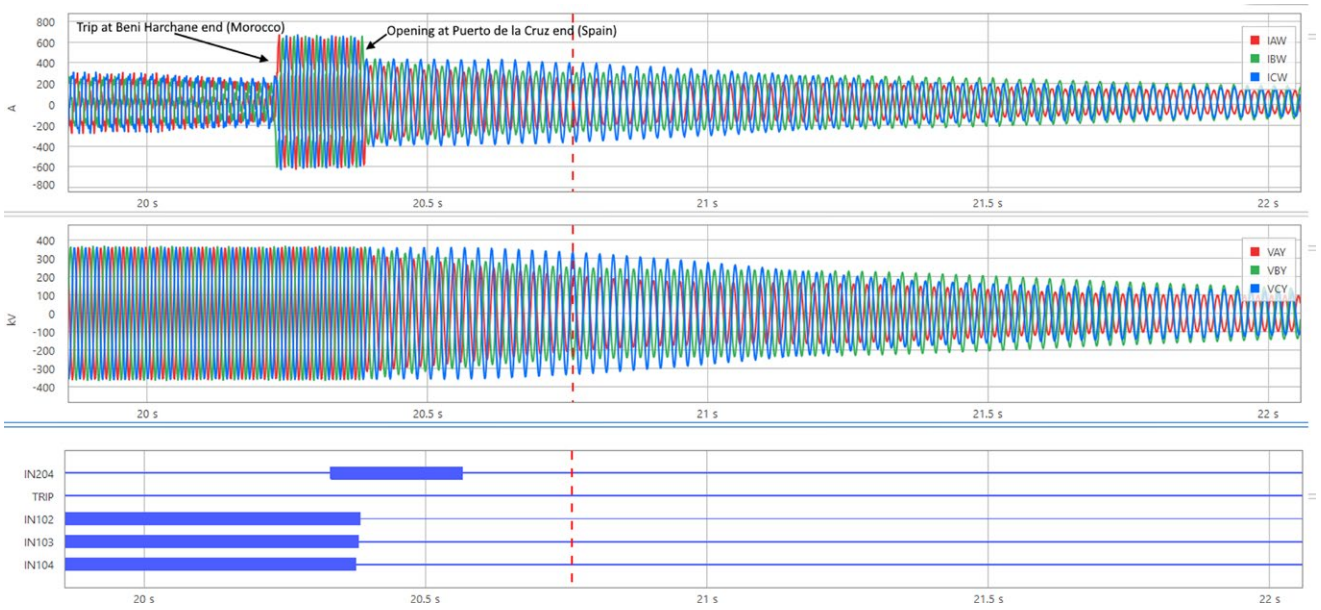


Figure 3-51: Beni Harchane bay at Puerto de la Cruz 400 kV substation relay recording (IN102, IN103, and IN104: breaker position, IN204: direct transfer trip reception)

This trip was consistent with the settings implemented on the Moroccan side. On the Spanish end, there is no under-frequency tripping function; however, a loss-of-synchronism protection is in place. Nevertheless, during this incident, no loss of synchronism occurred between the Iberian and Moroccan systems.

3.2.2.3 Outage ID 16b: 400 kV Puerto de la Cruz (RE)–Melloussa (ONEE)

At 12:33:20.473, the Puerto de la Cruz–Melloussa 400 kV line opened in Melloussa due to an underfrequency protection and also sent a direct transfer trip to the Spanish side (Puerto de la Cruz) that opened 91 ms later.

Figure 3-52 shows the recording of the protection installed in the Melloussa bay at the Puerto de la Cruz 400 kV substation.

The recording shows how the Moroccan end of the line tripped and indicates that it sent a direct transfer trip to the Spanish end. In addition, Figure 3-52 shows a trip signal, but this operation occurred when the breaker was already opened. This relay operation was caused by the discharge of the submarine cable on the reactor, as there was current with a low voltage.

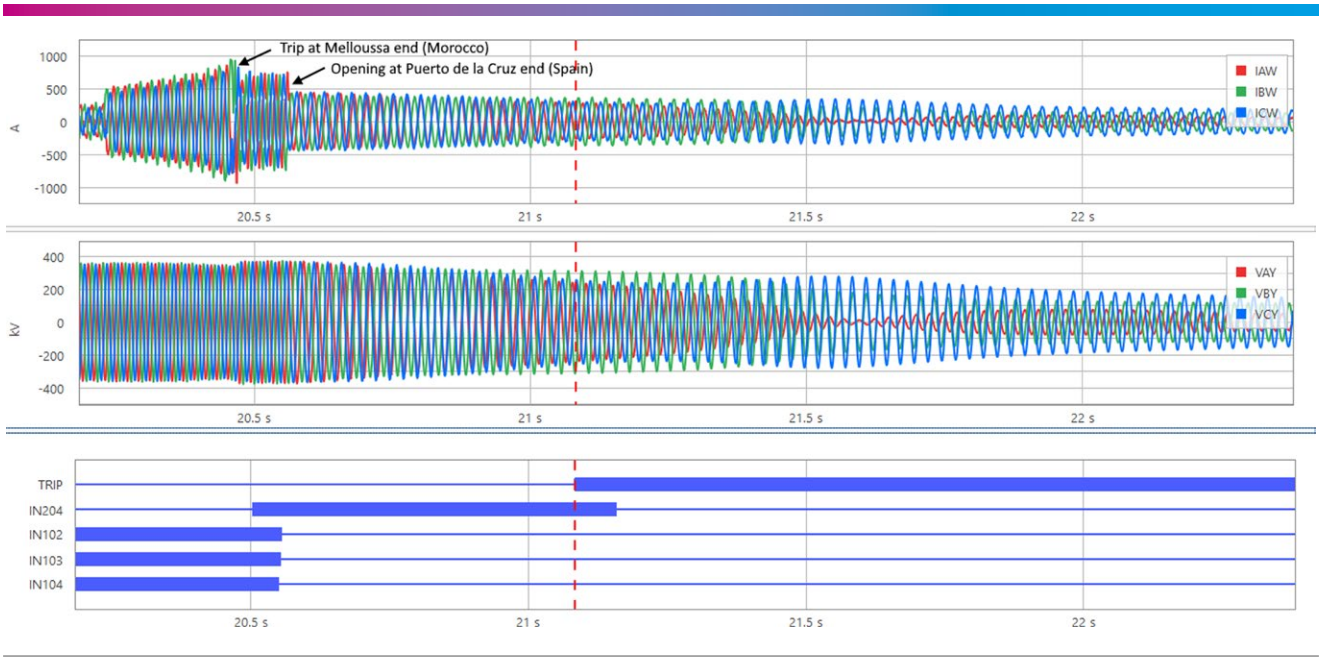


Figure 3-52: Melloussa bay at Puerto de la Cruz 400 kV substation relay recording (IN102, IN103, and IN104: breaker position, IN204: direct transfer trip reception)

This trip was consistent with the settings implemented on the Moroccan side. On the Spanish end, there is no under-frequency tripping function; however, a loss-of-synchronism protection is in place. Nevertheless, during this incident, no loss of synchronism occurred between the Iberian and Moroccan systems.

3.2.2.4 Event ID 31a: 400 kV BAIXAS (RTE)—Vic (RE)

At 12:33:21.407, the Baixas (RTE)—Vic (RE) 400 kV line opened in Baixas (RTE) due to the tripping of the loss of synchronism protection. The Vic (RE) end tripped 35 ms later. The loss of synchronism protections on both sides are set for two beats (swing), which means that they trip after two swings (see Figure 3-53 and the explanation accompanying Figure 3-55).

Figure 3-53 shows the oscillography recorded by the distance protection of Baixas bay at Vic 400 kV substation because loss of synchronism protections are not able to record oscillography.

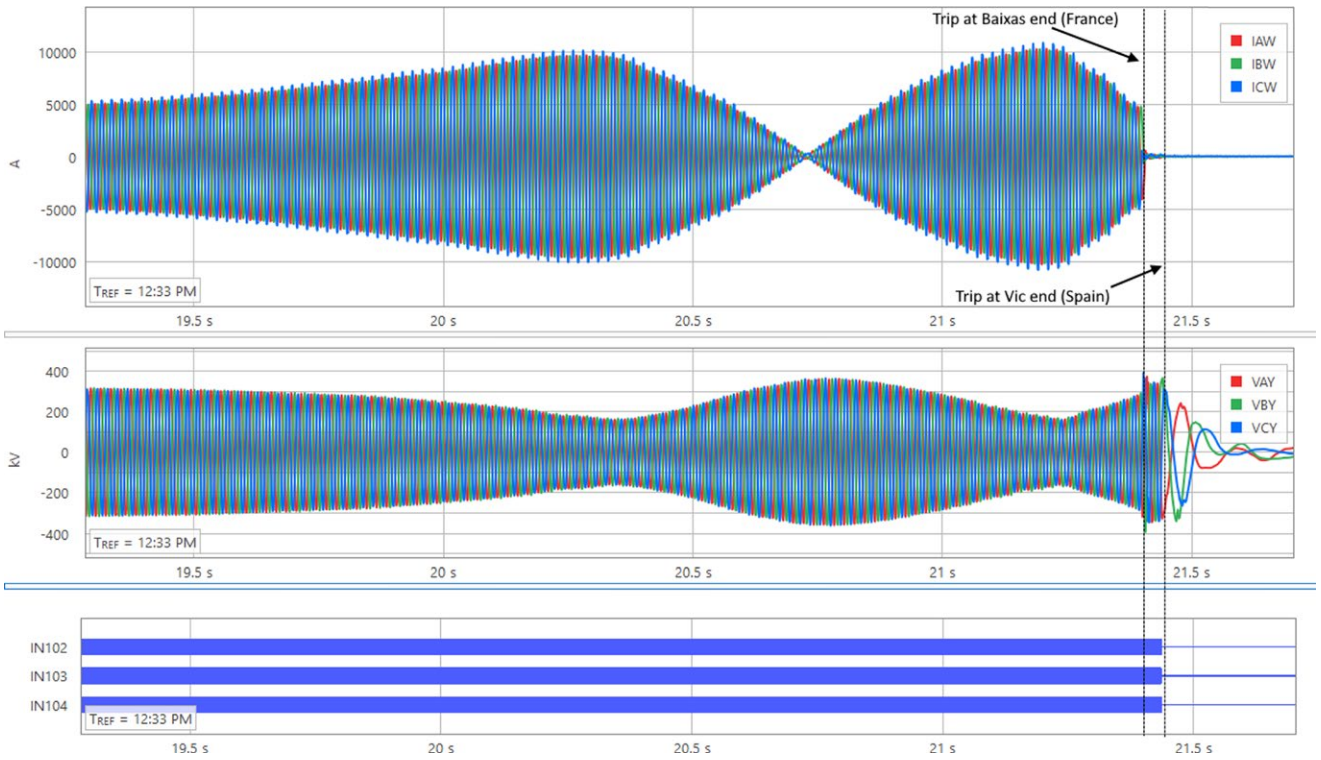


Figure 3-53: Baixas bay at Vic 400 kV substation distance relay recording (IN102, IN103 and IN104: breaker position).

Figure 3-54 illustrates the impedance evolution observed at Baixas bay at the Vic 400 kV substation using PMU data. It depicts the initial conditions, the changes following the first and second generation loss (events 3 and 4a, respectively), and the subsequent evolution at one-second intervals after the second generation loss.

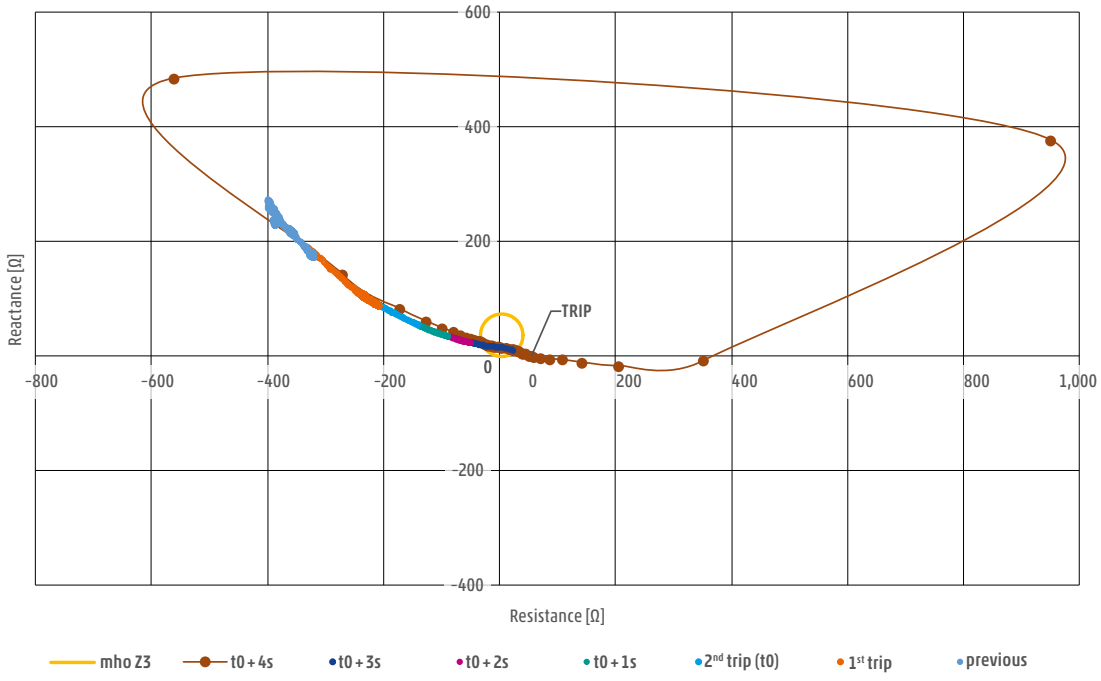


Figure 3-54: Baixas bay at the Vic 400 kV substation: Impedance evolution using PMU data

Explanation of the Operation of DRS Loss of Synchronism Protection

DRS loss of synchronism protection permanently measures the voltage peak values. When the relay measures at least 10 consecutive decreasing peaks followed by four consecutive increasing peaks, it detects one beat if the minimum voltage reached is below a configured voltage threshold. After detecting the configured number of beats, the relay is ready to trip, and it will trip after a time delay that starts when the average RMS voltage during the oscillation is exceeded, as seen in Figure 3-52.

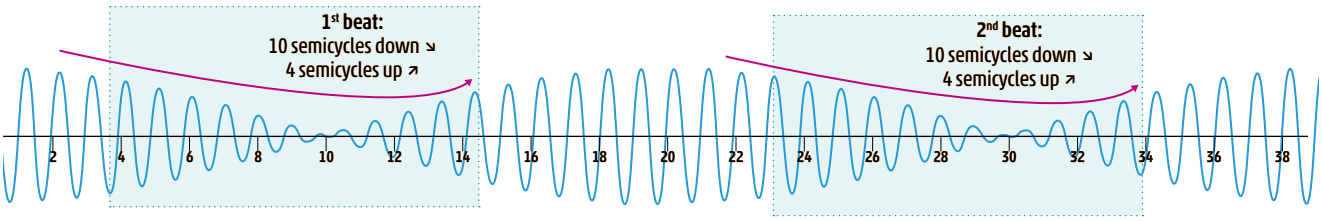


Figure 3-55: DRS loss of synchronism protection operating principle

Both the trip on the French side and the one on the Spanish side are considered appropriate and aligned with the protection system design. The disconnection of this line, followed by the tripping of the remaining France-Spain AC interconnections, enabled the isolation of the disturbance, preventing it from affecting the French system. This, in turn, allowed for a faster re-energisation of part of the Spanish system from France.

3.2.2.5 Event ID 31b1: 220 kV ARGIA (RTE)—Arkale (RE)

At 12:33:21.437, the Argia (RTE)—Arkale (RE) 220 kV line opened in Arkale (RE) due to the tripping of the loss of synchronism protection. The Argia (RTE) end did not trip because of the voltage fall (trip on the Spanish side); therefore, the loss of synchronism protection lost the count of the pulse (see Figure 3-56).

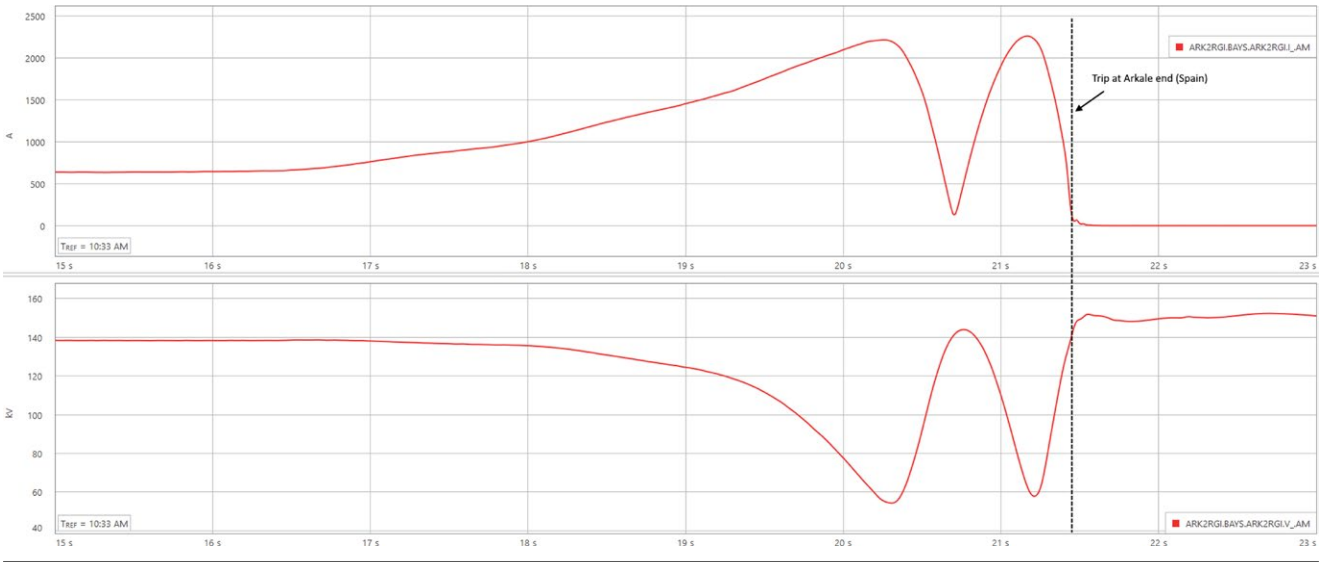


Figure 3-56: Argia bay at Arkale 220 kV PMU data

Figure 3-57 illustrates the impedance evolution observed at Argia bay at the Arkale 220 kV substation using PMU data. It depicts the initial conditions, the changes following the first and second generation loss (events 3 and 4a, respectively), and the subsequent evolution at one-second intervals after the second generation loss.

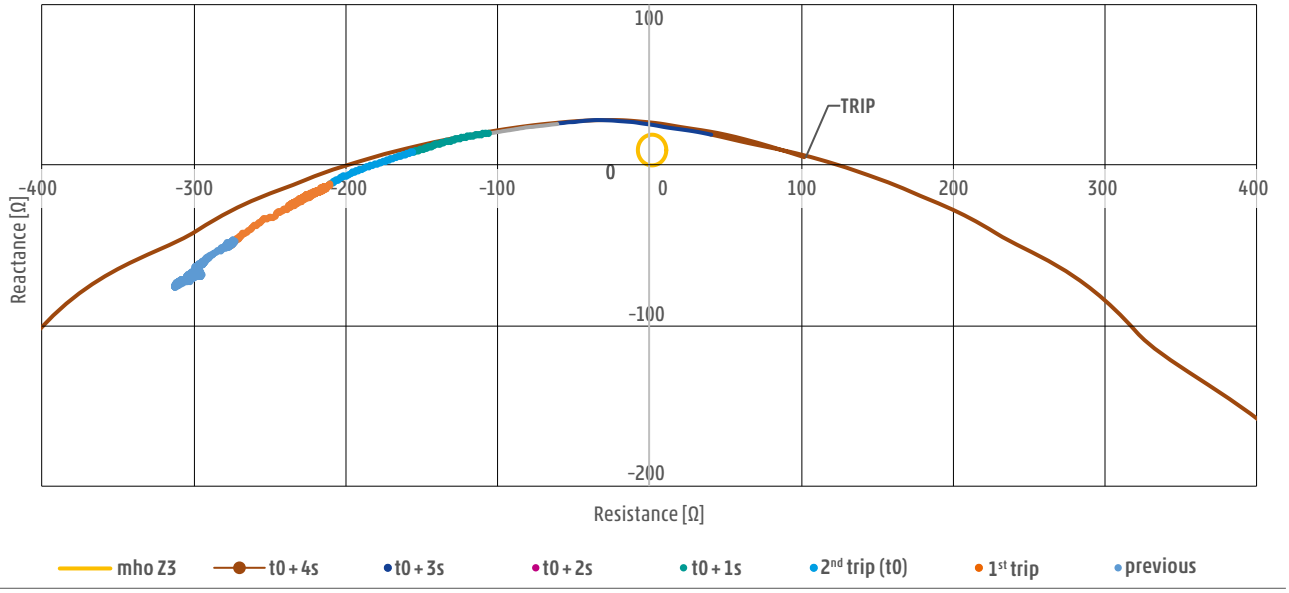


Figure 3-57: Argia bay at Arkale 220 kV substation: Impedance evolution using PMU data

The trip on the Spanish side is considered to be consistent with the protection system design. The absence of trigger from the RTE protection (DRS Loss of synchronism) is due to a fall of voltage. Indeed the DRS protection works by couting 10 semicycles and 4 semicycle up to count a beat.

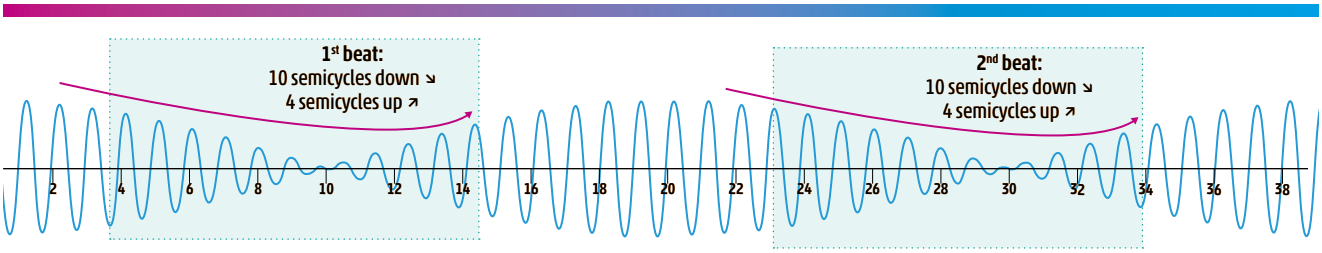


Figure 3-58 : principle of the DRS.

In the french case, the DRS need to count two beats before trig. However, on the Argia—Arkale line, there was a fall in voltage as we can see in Figure 3-59:

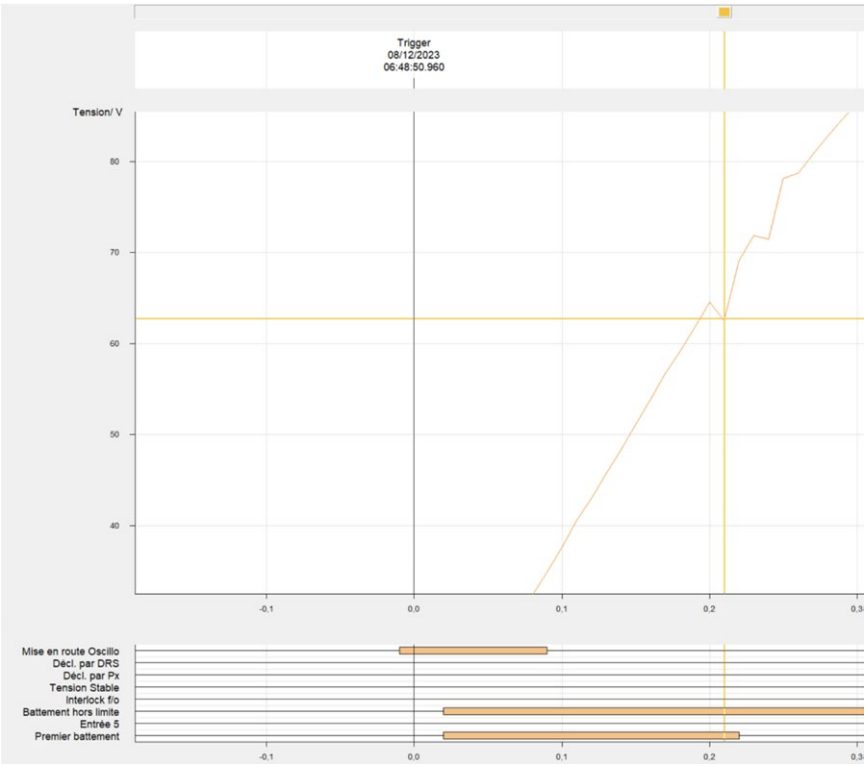


Figure 3-59 : Voltage fall as seen in Argia

As previously indicated, the tripping of the France–Spain AC interconnection lines enabled the isolation of the French system from the disturbance, preventing it from being significantly affected. Moreover, it facilitated a faster re-energisation of part of the Spanish system from France.

3.2.2.6 Event ID 31b2: 400 kV ARGIA (RTE)—Hernani (RE)

At 12:33:21.535, the Argia (RTE)—Hernani (RE) 400 kV line opened in Hernani (RE) due to the tripping of the out-of-step function of the distance protection, whose operating principle is based on monitoring the trajectory of the impedance seen by the relay. The Argia (RTE) end did not trip because of the voltage fall (trip on the Spanish side); therefore, the loss of synchronism protection lost the count of the pulse.

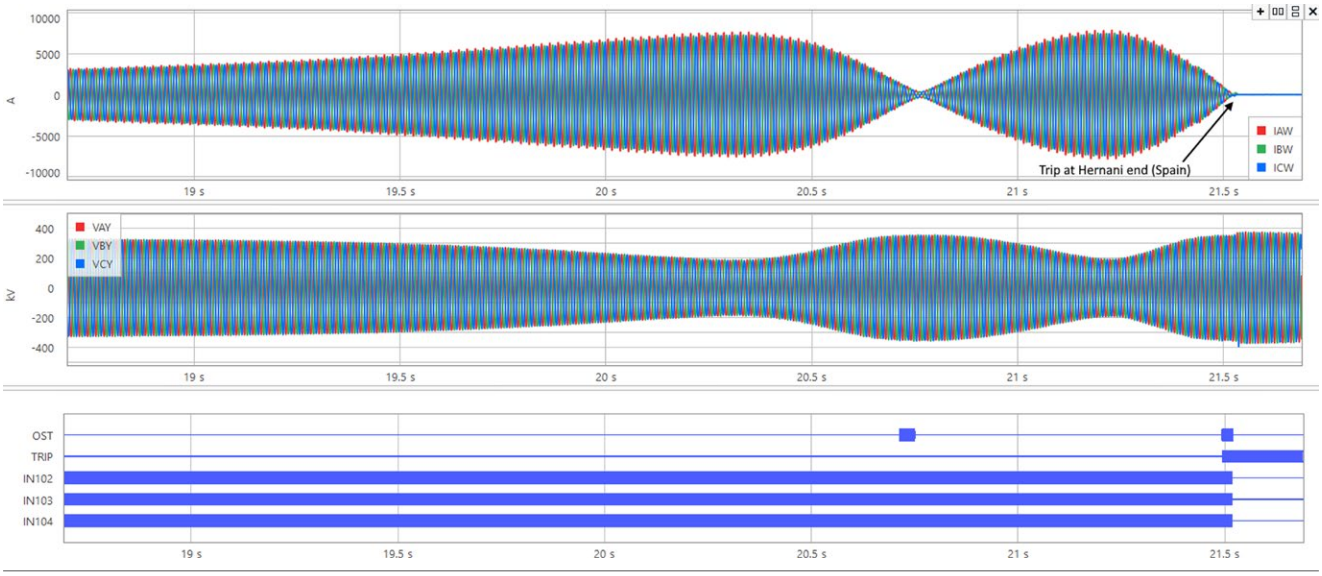


Figure 3-60: Argia bay at Hernani 400 kV substation recording (IN102, IN103, and IN104: breaker position)

Figure 3-61 illustrates the impedance evolution observed at Argia bay at the Hernani 400 kV substation using PMU data. It depicts the initial conditions, the changes following the first and second generation loss, and the subsequent evolution at one-second intervals after the second generation loss.

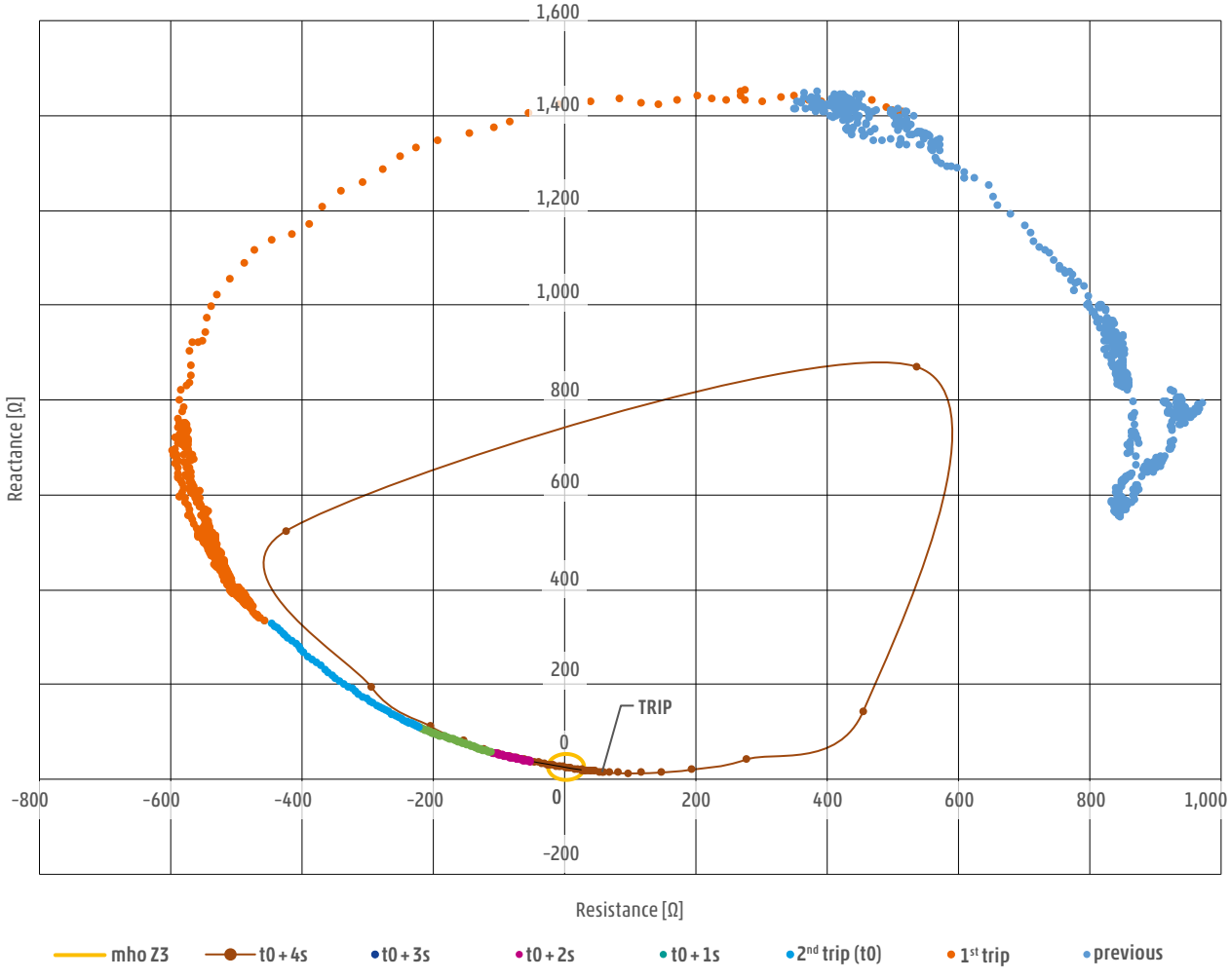


Figure 3-61: Argia bay at Hernani 400 kV substation: Impedance evolution using PMU data

The trip on the Spanish side is considered to be appropriate and consistent with the protection system design. The absence of trigger from the RTE protection (DRS Loss of synchronism) is due to a fall of voltage. Same situation as explained in the section 3.2.2.5

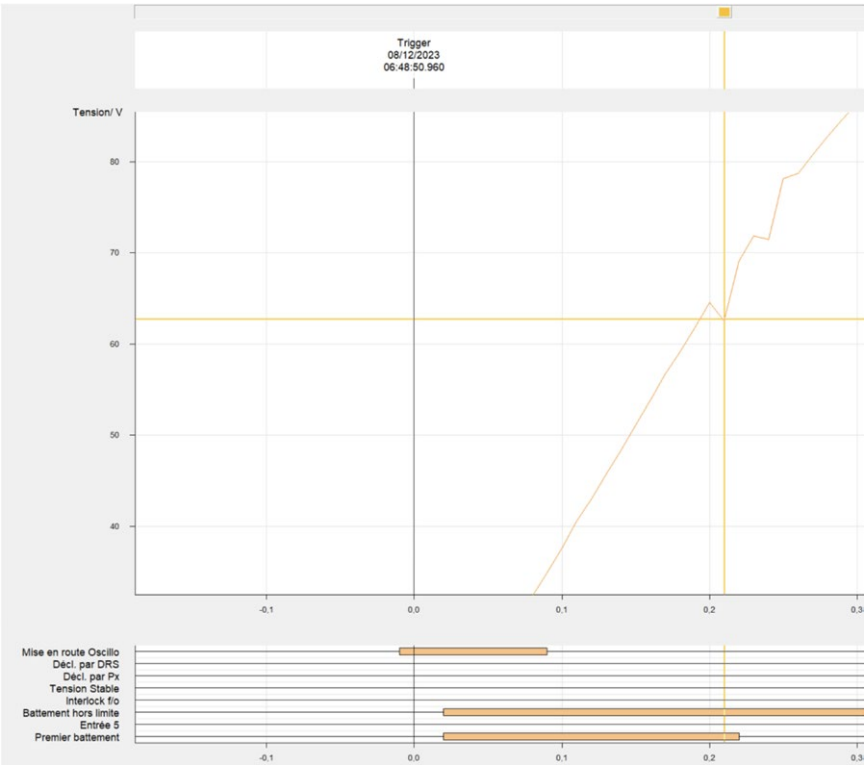


Figure 3-62: Hernani bay at Argia 400 kV substation recording

As previously indicated, the tripping of the France–Spain AC interconnection lines enabled the isolation of the French system from the disturbance, preventing it from being significantly affected. Moreover, it facilitated a faster re-energisation of part of the Spanish system from France.

3.2.2.7 Other Transmission Trips in Spain

After 12:33:23.750, with the frequency already below 44 Hz, several transmission network lines tripped due to the operation of zone 1 of the distance protections. Two main factors caused the operation of these protections: very low system frequency, sometimes even outside the limits established for their correct operation, and low voltages, in some cases almost zero while there was still current through those lines. When these protection systems tripped, the system conditions were so degraded that, although there was no short circuit in these elements, the operation of the protections cannot be considered incorrect. In addition, these trips did not contribute to aggravating the incident because, at that moment, the Iberian electrical system was not recoverable.

As an example of this, there are the trips on the Spanish side of six tie-lines with Portugal, as shown in Table 3-6. Figure 3-63 shows the trip of the Lindoso 2 bay in the Cartelle 400 kV substation.

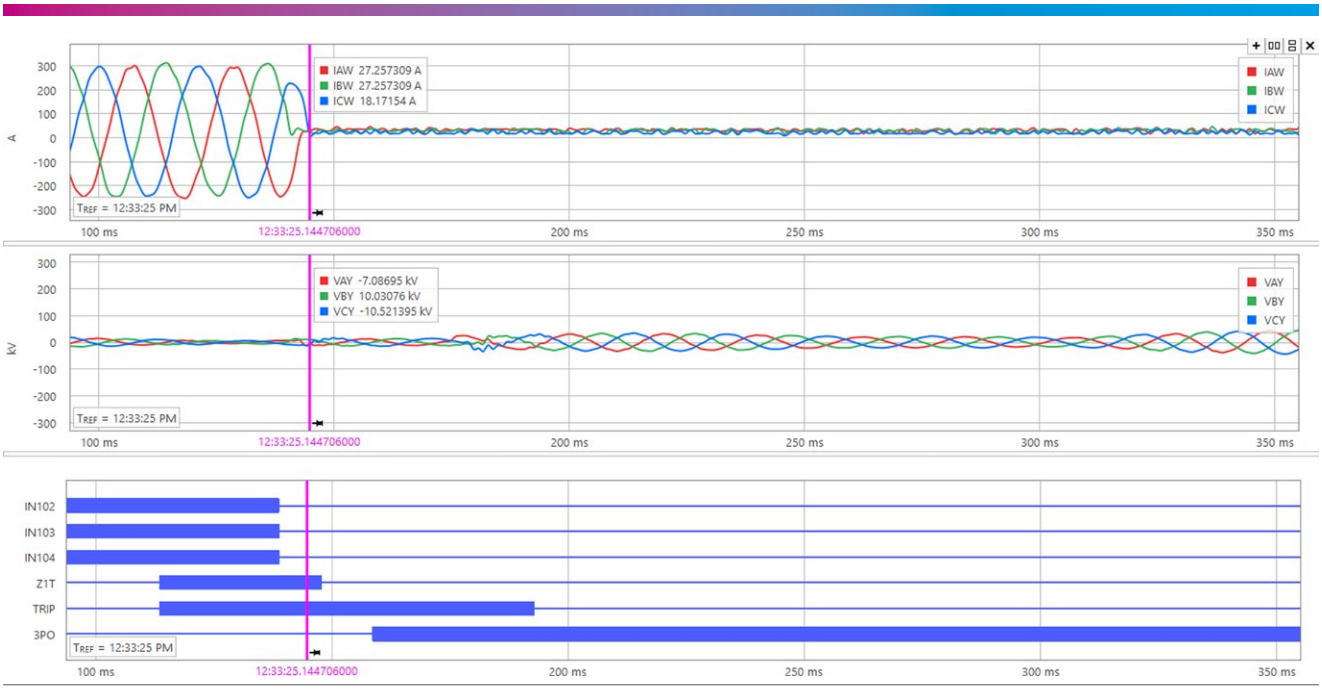


Figure 3-63: Lindoso 2 bay at Cartelle 400 kV substation relay recording (IN102, IN103, and IN104: breaker position, Z1T: distance zone 1 trip, 3PO: 3 pole open)

All these trips, triggered by the operation of zone 1 of the distance protection function, occurred under highly degraded system conditions. Although such operations diverge from the intended purpose of this protection–namely, to clear internal faults within the line–they are considered unavoidable due to the prevailing conditions: frequency values outside or near the relays design frequency limits, voltage levels close to zero, and the presence of current flowing through the lines. These trips had no impact on the development of the incident, as by the time they occurred, the frequency had already dropped below 44 Hz and the voltage had collapsed, rendering the system unrecoverable.

After 12:33:27, all circuit breakers belonging to the transmission network in Spain tripped due to the undervoltage function.

This function trips all the circuit breakers with a time delay of four seconds when the voltage drops below 65 % of the nominal voltage. The purpose of this function is to open all circuit breakers in an area when a blackout occurs in that area to facilitate a faster and more orderly service restoration. In some circuit breakers, this function is disabled because of the design of the service restoration plans; in those cases, these circuit breakers remained closed.

All these trips triggered by the undervoltage protection function are considered to be appropriate, consistent with the protection system design, and contributed to the subsequent restoration of service.

3.2.2.8 Other Transmission Trips in Portugal

Within the Portuguese transmission network, the protection systems do not have the following protection functions enabled at the transmission level: out-of-step, over/underfrequency (except at 60 kV for UFLS), and over/undervoltage (except for undervoltage protection applied to shunt reactors and overvoltage protection applied to shunt capacitors). As a result, during the blackout event, no trips were initiated by these functions.

Two transmission lines tripped due to the operation of zone 1 of the distance protection function. This trip was caused by unstable voltage conditions. Under these circumstances, the distance protection algorithm was unable to maintain a reliable voltage reference, which compromised its ability to operate correctly.

Although no fault was detected in the system, the behaviour observed does not indicate a malfunction of the protection system. Instead, it reflects a limitation in the algorithm's performance under low-voltage dynamic conditions.

As illustrated in Figure 3-64, voltage and current oscillations were recorded prior to the trip. These oscillations differ from a typical power swing. In this case, the voltage drop is accompanied by a simultaneous current drop, and the same pattern is observed during current increases. The affected lines were Lagoaça–Armamar 400 kV (trip at 12:33:25.434) and Mogadouro–Central da Valeira 220 kV (trip at 12:33:23.95). These trips did not contribute to the deterioration of system conditions, as they occurred after the voltage collapse.

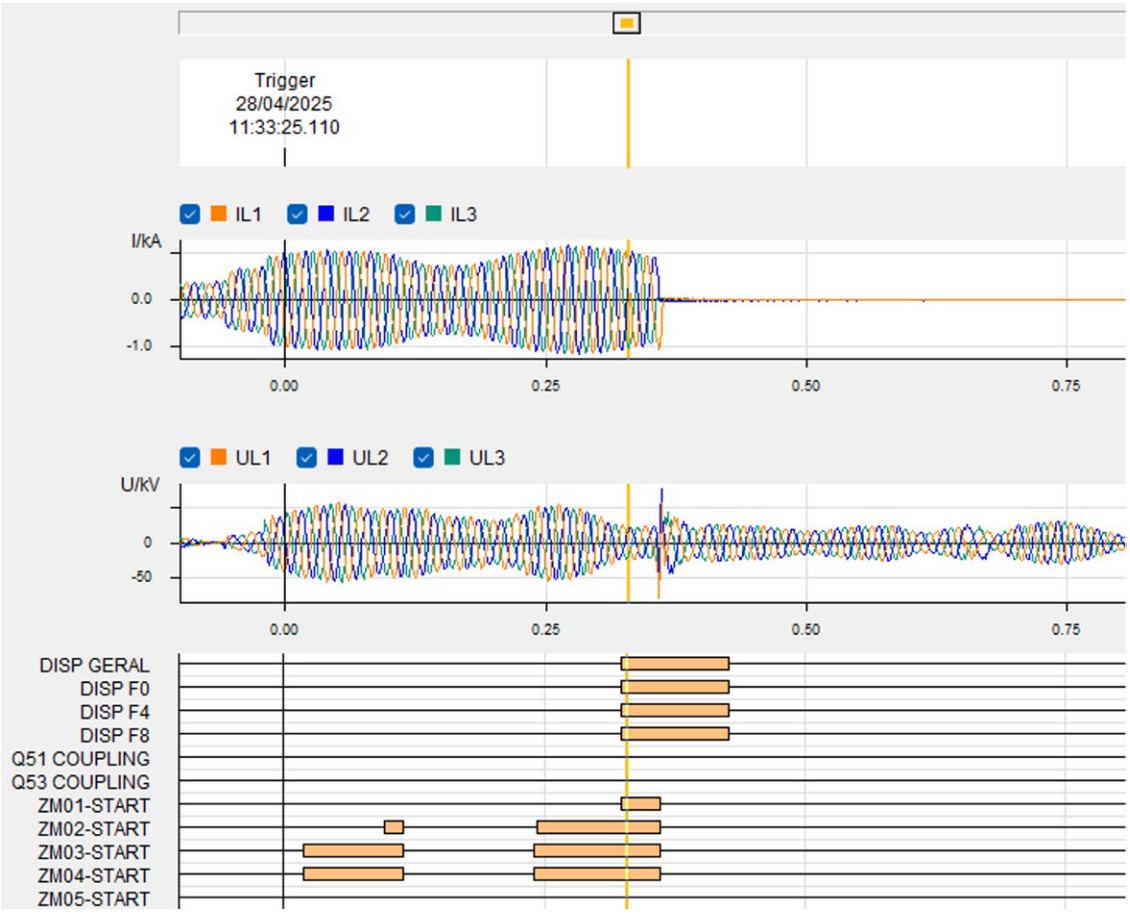


Figure 3-64: Z1 Trip of Lagoaça–Armamar line in the Lagoaça 400 kV substation due to unstable conditions

The remaining transmission network circuit breakers that were energised prior to the blackout were opened by a control logic known as "zero voltage cut-off, top = 6 s" This function issues a trip command to the circuit breaker if the measured voltage remains below 30 % of the nominal value for a continuous duration of six seconds.

3.2.2.9 Other Transmission Trips in France

Time (CEST)	Substation A (TSO)	Substation B (TSO)	Voltage Level (kV)	Asset Type	Relay Trigger	Cause of Tripping	Tripped Phases	Relay Parameter Settings
12:33:20:298	Mousserolles (RTE)	Borderes et Lamesant (RTE)	63	OHL	DIST	Distance protection tripped the line incorrectly.	3 phase	overreach zone Z < 120 % line
12:33:20:518	Mouguerre (RTE)	Lussagnet (RTE)	63	OHL	OST	Loss of synchronism	3 phase	1 beat
12:33:20:551	Dax (RTE)	Transformer 1 400/63 kV	63	OCHL	OST	Loss of synchronism	3 phase	1 beat
12:33:20:572	Aire-Sur-Adour (RTE)	Transformer 3 400/63 kV	63	OHL	OST	Loss of synchronism	3 phase	1 beat
12:33:20:583	Midour (RTE)	Usson (RTE)	63	OCHL	OST	Loss of synchronism	3 phase	1 beat
12:33:21:280	Issel (RTE)	Marsillon (RTE)	400	TR	OST	Loss of synchronism	3 phase	2 beats
12:33:21:280	Issel (RTE)	Orlu (RTE)	400	TR	OST	Loss of synchronism	3 phase	2 beats
12:33:21:382	Lavelanet (RTE)	Bus coupler	63	OHL	OST	Loss of synchronism	3 phase	2 beats
12:33:21:404	Berge (RTE)	Porta (RTE)	225	OHL	OST	Loss of synchronism	3 phase	2 beats
12:33:21:410	Nentilla (RTE)	Cantegrit (RTE)	150	OHL	OST	Loss of synchronism	3 phase	2 beats
12:33:21:418	Argia (RTE)	Berge (RTE)	225	COUPLER	OST	Loss of synchronism	3 phase	2 beats
12:33:21:423	Latour-de-Carol (RTE)	PORTA (RTE)	63	OHL	OST	Loss of synchronism	3 phase	2 beats
12:33:21:427	Marsillon (RTE)	CANTEGRIT (RTE)	225	OHL	OST	Loss of synchronism	3 phase	2 beats
12:33:21:451	Marsillon (RTE)	BERGE (RTE)	225	OHL	OST	Loss of synchronism	3 phase	2 beats

Table 3-7: List of tripped lines in France

3.2.2.9.1 63 kV MOUGUERE (RTE)-MOUSSEROLLES (RTE)—LE BOUCAU (RTE)

At 12:33:20:298, the Mouguerre (RTE)—Mousserolles (RTE)—Le Boucau (RTE) 63 kV line opened in the Mousserolles substation. The distance protection tripped the line because the measured impedance crossed the characteristic. In this case, the protection should have been in blocking state (power swing blocking) and not tripped.

RTE investigated and the bandwidth of the power swing blocking band of this protection was too short in comparison of the pumping's frequency. This explains why only this protection and not the others distances protections trigged in this case.

3.2.2.9.2 63 kV MOUGUERRE (RTE)—GUICHE (RTE)

At 13:33:20.518, Mouguerre (RTE)—Guiche (RTE) 63 kV OHL opened in the Mouguerre substation due to the loss of synchronism protections function trip. The loss of synchronism protection function is set to trip on the first beat (swing) of voltage if the voltage dip reaches the preset value. The protection system of the OHL tripped according to its settings.

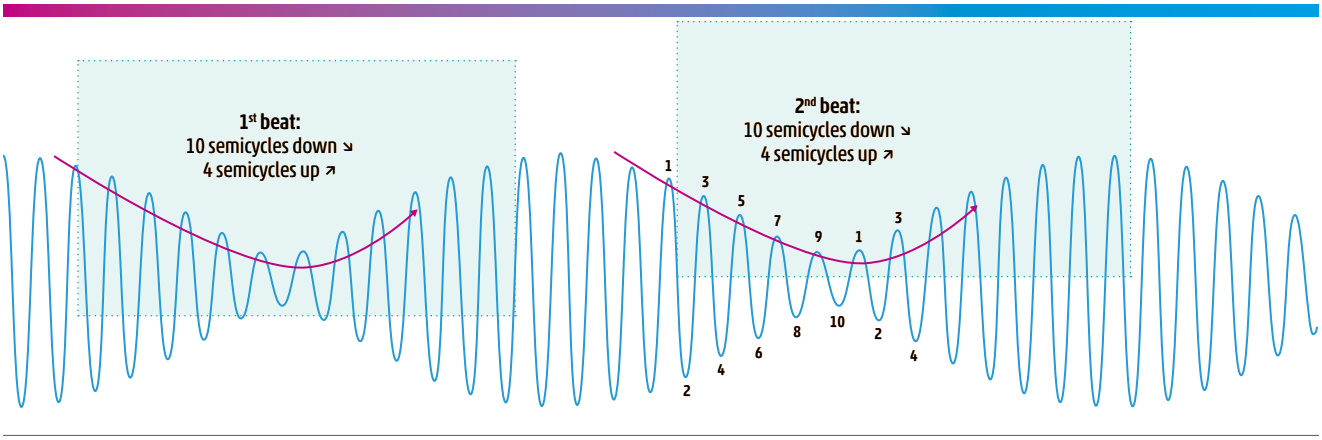


Figure 3-65: DRS loss of synchronism protections operating principle

Explanation of the operation of the loss of synchronism protection: the loss of synchronism protection permanently measures the voltage peak values. When the relay measures at least 10 consecutive decreasing peaks followed by four consecutive increasing peaks, it detects one beat if the minimum voltage reached is below a configured voltage threshold. After detecting the configured number of beats, the relay is ready to trip, and it will trip after a time delay that starts when the average RMS voltage during the oscillation is exceeded, as seen in Figure 3-65.

3.2.2.9.3 63 kV DAX (RTE)—ARRIOSSES (RTE)

At 13:33:20.551, Dax (RTE)—Arriosses (RTE) 63 kV OHL opened in the Dax substation due to the loss of synchronism protection function trip. The loss of synchronism protection function is set to trip on the first beat (swing) of voltage if the voltage dip reaches the preset value. The protection system of the OHL tripped according to its settings.

3.2.2.9.4 63 kV AIRE SUR ADOUR (RTE)—BORDERES ET LAMESANT (RTE)

At 13:33:20.572, Aire sur Adour (RTE)—Borderes et Lamesant (RTE) 63 kV OHL opened in the Aire sur Adour substation due to the loss of synchronism protection function trip.

The loss of synchronism protection function is set to trip on the first beat (swing) of voltage if the voltage dip reaches the preset value. The protection system of the OHL tripped according to its settings.

3.2.2.9.5 63 kV MIDOUR (RTE)—LUSSAGNET (RTE)

At 13:33:20.583, Midour (RTE)—Lussagnet (RTE) 63 kV OHL opened in the Midour substation due to the loss of synchronism protection function trip.

The loss of synchronism protection function is set to trip on the first beat (swing) of voltage if the voltage dip reaches the preset value. The protection system of the OHL tripped according to its settings.

3.2.2.9.6 400 kV/63 kV Transformer 1 at ISSEL (RTE) substation

At 12:33:21:290, the 400 kV/63 kV Transformer 1 at Issel (RTE) substation tripped on 400 kV due to the loss of synchronism protection function operation. The protection tripped on the second beat (swing) of voltage on 400 kV because at the first beat, there was not enough voltage dip according to the settings, as shown in Figure 3-66 the protection system of the transformer operated according to its settings.

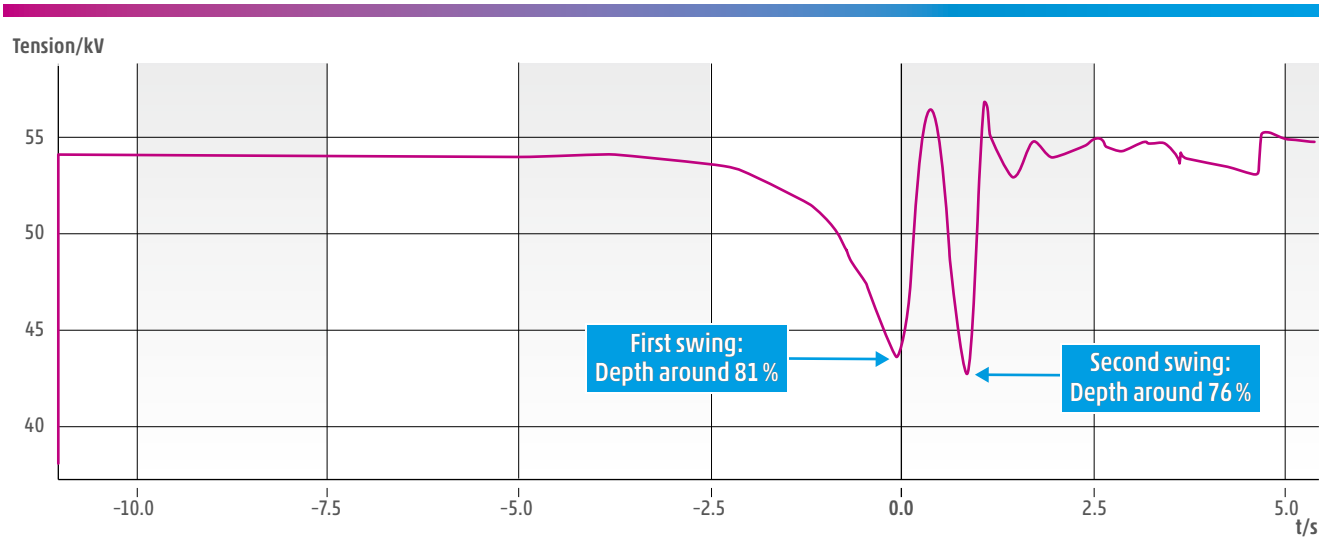


Figure 3-66: The voltage dip in the first and second swing on the Transformer 1 at Issel (RTE) substation.

3.2.2.9.7 400 kV/63 kV Transformer 3 at Issel (RTE) substation

At 12:33:21:310, the 400/63 kV Transformer 3 at Issel (RTE) substation tripped on 400 kV due to the loss of synchronism protection function operation. The protection tripped on the second beat (swing) of voltage on 400 kV because at the first beat, there was not enough voltage dip according to the settings. The protection system of the transformer operated according to its settings.

3.2.2.9.8 63 kV LAVELANET (RTE)—USSON (RTE)

At 13:33:21.382, Lavelanet (RTE)—Usson (RTE) 63 kV OHL opened in the Lavelanet substation due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage because at the first beat, there was not enough voltage dip according to the settings, as shown in Figure 3-67. The protection system of the OHL tripped according to its settings.

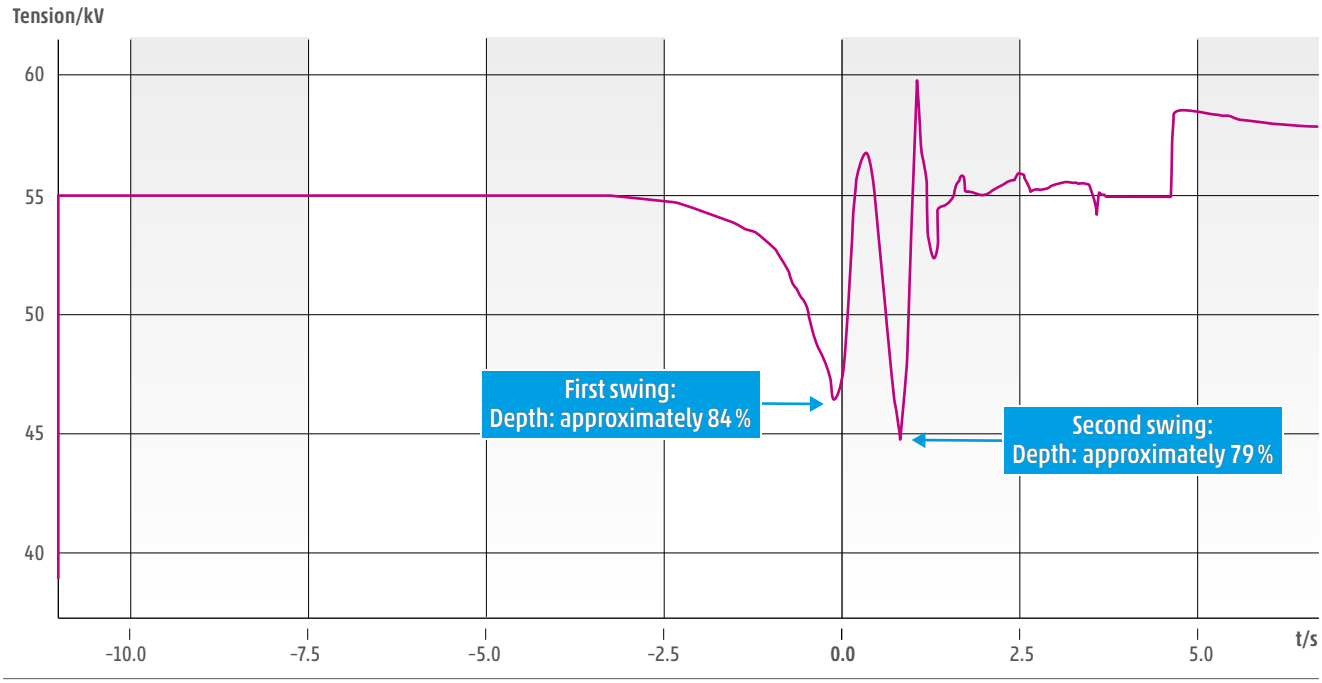


Figure 3-67: The voltage dip in the first and second swing at Lavelanet (RTE) substation.

3.2.2.9.9 225 kV BERGE (RTE)—MARSILLON (RTE)

At 13:33:21.404, Berge (RTE)—Marsillon (RTE) 225 kV OHL opened in the Berge substation due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage at 65 % voltage dip. The beats (swings) are much deeper in the Berge substation than on the other end of the line. Figure 3-68 shows that the first beat was not deep enough to trigger the protection. The protection triggered on the second beat, which is much deeper, in line with its settings.

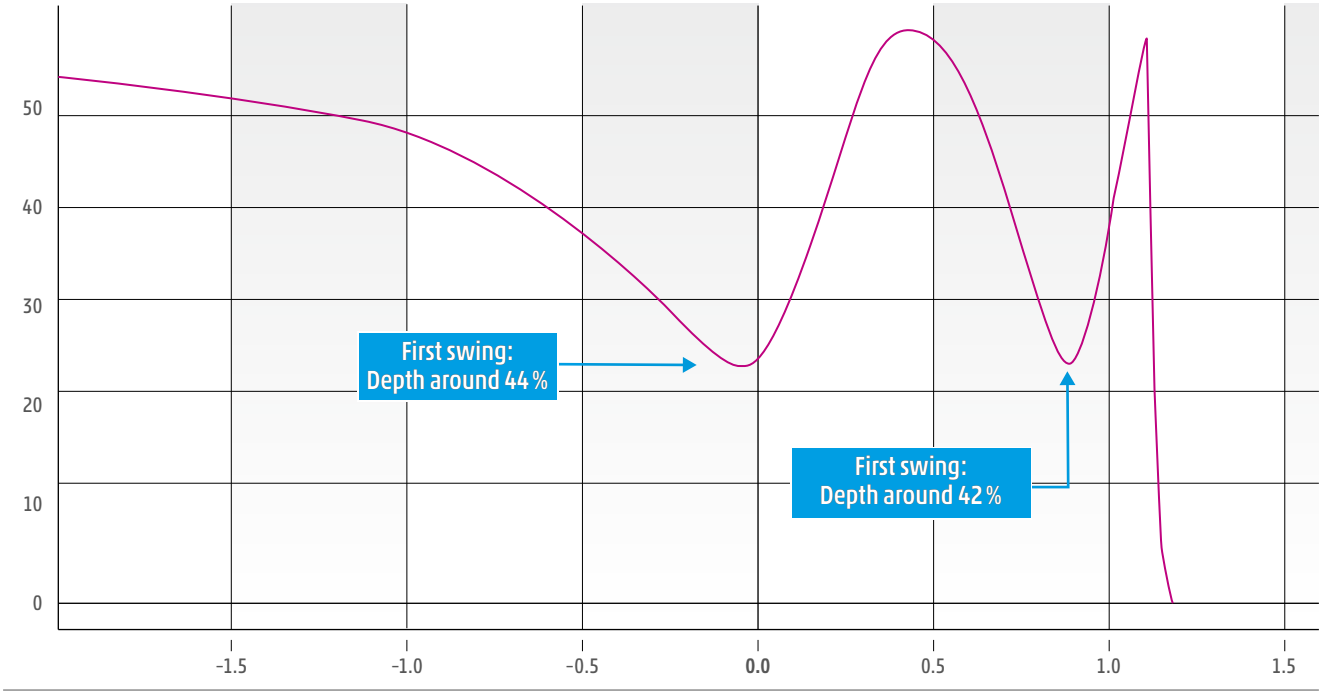


Figure 3-68: Voltage pulse on the Berge—Marsillon lines

3.2.2.9.10 150 kV NENTILLA (RTE)—ORLU (RTE)

At 13:33:21.410, Nentilla (RTE)—Orlu (RTE) 150 kV OHL opened in Nenti substation due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage because at the first beat, there was not enough voltage dip according to the settings. The protection system of the OHL tripped according to its settings.

3.2.2.9.11 225 kV ARGIA (RTE)—Bus Coupler

At 13:33:21.418, Argia (RTE) 225 kV bus coupler circuit breaker opened due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage. The protection system of the OHL tripped according to its settings.

3.2.2.9.12 63 kV LATOUR-DE-CAROL (RTE)—PORTA (RTE)

At 13:33:21.423, Latour-de-Carol (RTE)—Porta (RTE) 63 kV OHL opened in the Latour-de-Carol substation due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage because at the first beat, there was not enough voltage dip according to the settings. The protection system of the OHL tripped according to its settings.

3.2.2.9.13 225 kV MARSILLON (RTE)—CANTEGRIT (RTE)

At 13:33:21.427, Marsillon (RTE)—Cantegrit (RTE) 225 kV OHL opened in the Marsillon substation due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage. The protection system of the OHL tripped according to its settings. The loss of synchronism protection of Cantegrit had tripped previously, but subsequently opened the breaker at 12:33:21:437.

3.2.2.9.14 225 kV MARSILLON (RTE)—BERGE (RTE)

At 13:33:21.451, Marsillon (RTE)—Berge (RTE) 225 kV OHL opened in the Marsillon substation due to the loss of synchronism protection function trip. The protection tripped on the second beat (swing) of voltage. The protection system of the OHL tripped according to its settings.

3.2.2.10 Conclusions

The France-Spain AC interconnection lines trips are aligned with the protection system design. The disconnection of these lines enabled the isolation of the disturbance, preventing it from affecting the French system. This, in turn, allowed for a faster re-energisation of part of the Spanish system from France.

The Morocco–Spain interconnection lines trips are aligned with the protection system design in the Moroccan side. The disconnection of these lines enabled the isolation of the disturbance, preventing it from affecting the Moroccan system. Given the size of the Moroccan power system and the weak Spain–Morocco interconnection, with only two transmission lines, maintaining both systems interconnected would not have prevented the blackout and would likely have caused a voltage collapse in the Moroccan system. This isolation, in turn, allowed for a faster re-energisation of part of the Spanish system from Morocco.

One of the internal trips in the Spanish transmission network occurred before reaching, in the primary measurement, the voltage level at which the overvoltage protection function is expected to operate according to its settings (264 kV), due to an error in the voltage measurement. However, given the security margin applied with respect to the voltage level at which generator disconnections may begin even for units with the highest voltage requirements (253 kV), the trip took place once the voltage threshold required for generation had already been exceeded. Therefore, this trip did not worsen the incident, both because of the voltage level at which it occurred and because of the limited operational significance of the line that tripped.

Once the system was no longer recoverable, with very low voltage and low frequency levels, trips of transmission network elements occurred that are considered unavoidable, as system conditions were already highly degraded and the protection equipment cannot deal with these conditions. These trips are also deemed to have had no impact on the development of the incident, since at the moment they occurred the system was already unrecoverable.

No trips occurred on the Portugal side of the Portugal–Spain interconnections. The protection functions on the Portuguese side remained stable under low-voltage and low-frequency dynamic conditions and correctly refrained from operating, as no faults were detected on the interconnection circuits. This behaviour is consistent with the design of the Portuguese protection system.

In Portugal, two internal transmission lines tripped due to the operation of zone 1 of the distance protection function. These trips were triggered by highly unstable voltage conditions after the system had already reached the point of no return, preventing the distance protection algorithm from maintaining a valid voltage reference.

The France-Spain interconnection AC lines trips are aligned with the protection system except for the Argia–Hernani, the Argia–Arkalé lines and the Argia busbar coupler in 225 kV. This problem is currently identified by RTE and has been addressed for the other protection.

The disconnection of the AC interconnection enabled the isolation of the disturbance, preventing it from affecting the French system. This, in turn, allowed for a faster re-energisation of part of the Spanish system from France.

Concerning the French 63 kV lines. Many out of step protection tripped before the interconnection lines. This is considered appropriate by RTE.

Concerning the only distance protection that tripped in the South-west, the situation is known by RTE and have been addressed.

3.3 System Defence Plan

Commission Regulation (EU) 2017/2196 of 24 November 2017 established the Network Code on Electricity Emergency and Restoration (NC ER). This regulation outlines the rules for (i) safeguarding operational security, (ii) preventing the propagation and deterioration of an incident, (iii) avoiding widespread disturbance and the blackout state, and (iv) allowing the efficient and rapid restoration of the electricity system from emergency and/or blackout states.

Section 2 of the regulation establishes the requirements for the design and implementation of the System Defence Plan. In particular, Article 15 sets out the requirements for the automatic underfrequency control scheme.

3.3.1 Spanish System Defence Plan

The Spanish defence plan for a situation involving a sudden drop in system frequency due to an imbalance between production and consumption includes the following two strategies, in accordance with Article 15 of the NC ER:

» 1. Automatic low frequency pump-storage disconnection plan

» 2. Automatic low frequency demand disconnection strategy.

The first strategy consists of two steps, with activation thresholds of 49.5 Hz and 49.3 Hz. The second strategy defines six frequency thresholds, beginning at 49 Hz and decreasing in 0.2 Hz intervals down to 48 Hz. At each threshold, portions of demand are progressively disconnected, as shown in Table 3-8 and Figure 3-69.

Step	Threshold (Hz)	Demand disconnected (% of total load)
1st	49.0	6
2nd	48.8	9
3rd	48.6	8
4th	48.4	8
5th	48.2	7
6th	48.0	7

Table 3-8: Low Frequency demand disconnection plan

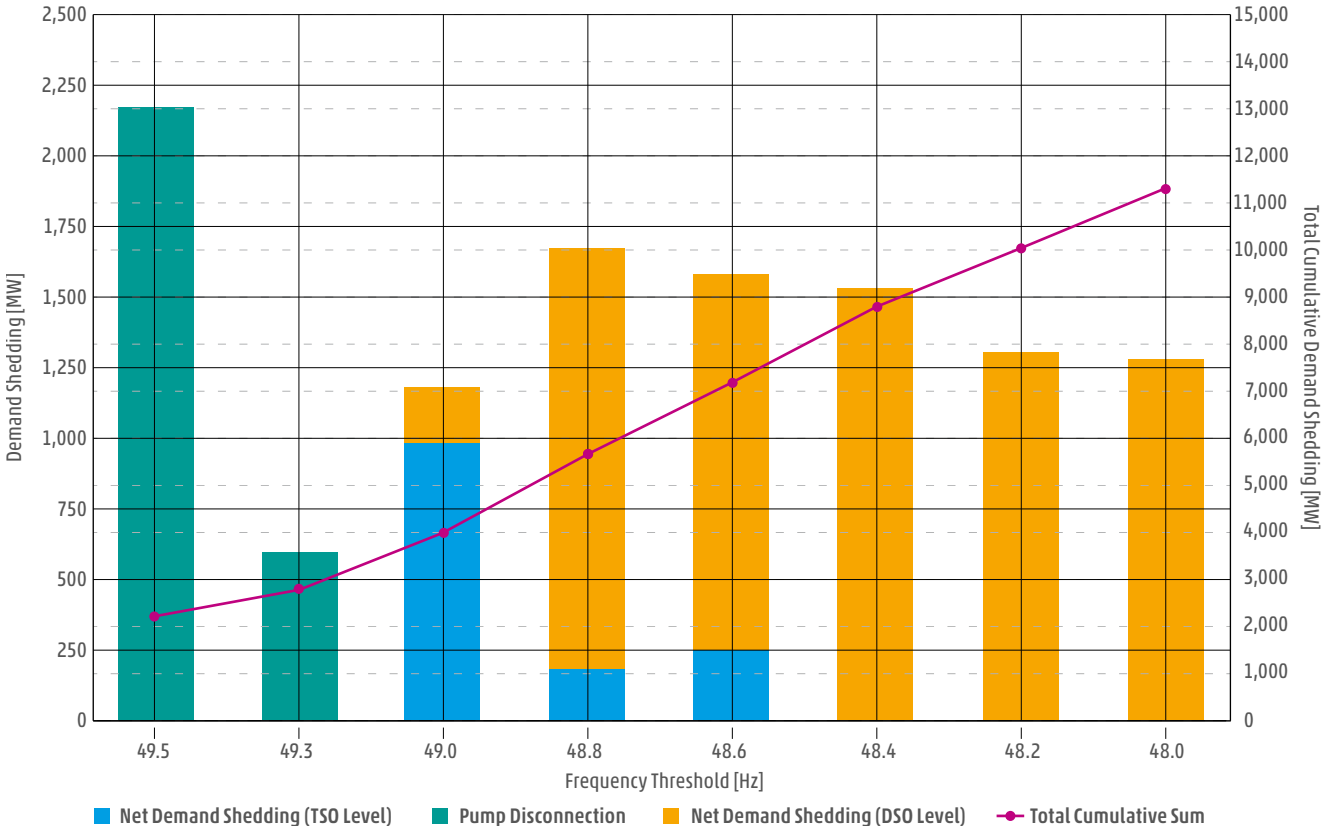


Figure 3-69: Activation of the Low Frequency Demand Disconnection Plan for RE.

The Spanish defence plan has incorporated demand across different frequency steps, taking into account its geographical location to ensure a balanced impact on both congestion and voltage levels. The plan does not have specific measures related to voltage; it is primarily designed to maintain the balance between generation and demand, which is reflected in the change of frequency.

Per the Spanish defence plan, the LFDD demand disconnected at each frequency level is, in some cases, physically disconnected at the connection point of specific load units, while in others, it is physically disconnected at the distribution connection points. This means that a specific section of the distribution system is disconnected in its entirety, regardless of whether active/reactive power is being injected or absorbed by that section.

3.3.1.1 Automatic Low Frequency Disconnection of Pump Storage

Before the incident, the total power from pumping units was 3,031 MW, of which 2,365 MW came from pumping associated with the first frequency threshold (49.5 Hz) and 666 MW with the second frequency threshold (49.3 Hz).

At 12:33:20.180, the system frequency fell below 49.5 Hz, triggering the first step of the automatic low frequency disconnection of pump-storage units. The disconnection associated with this initial threshold was effective between 12:33:20.133 and 12:33:20.800. At that time, 24 pumping units were available, 20 of which were tripped. However, two of these were disconnected due to unrelated causes, before the operation of the underfrequency relay. Of the remaining four units, one was disconnected at 12:33:23.888 due to loss of synchronism. No information is available for the disconnection of the remaining three. The total pump-storage power disconnected at this frequency threshold amounts to **2,168 MW** out of 2,365 MW (91 %).

At 12:33:20.500, the frequency fell below 49.3 Hz, triggering the activation of the second step. At this stage, six pump units were available. However, one pump-storage unit disconnected at 12:33:24.000 due to undervoltage issues. At this frequency threshold, the total power disconnected amounts to **588 MW**.

Considering both frequency thresholds, the pump storage disconnection amounts to **2,756 MW**.

Table 3-9 shows the pump storage disconnection for both steps.

Pump-storage	Step [Hz]	P Tripped
Pump 1	49.5	Y
Pump 2	49.5	Y
Pump 3	49.5	Y
Pump 4	49.5	No information available
Pump 5	49.5	No information available
Pump 6	49.5	Y
Pump 7	49.5	Y
Pump 8	49.5	Y
Pump 9	49.5	Y
Pump 10	49.5	Y
Pump 11	49.5	Y
Pump 12	49.5	Y
Pump 13	49.5	No information available
Pump 14	49.5	Y
Pump 15	49.5	Y
Pump 16	49.5	Y
Pump 17	49.5	Y
Pump 18	49.5	Y
Pump 19	49.5	Y
Pump 20	49.5	Y
Pump 21	49.5	Y
Pump 22	49.5	Y
Pump 23	49.5	Y
Pump 24	49.5	Y
TOTAL at 49,5 Hz (MW)		2,168
Pump 25	49.3	Y
Pump 26	49.3	Y
Pump 27	49.3	Y
Pump 28	49.3	failed
Pump 29	49.3	Y
Pump 30	49.3	Y
TOTAL at 49,3 Hz (MW)		588
TOTAL (both steps) (MW)		2,756

Table 3-9: Pump storage disconnection for the Spanish system

3.3.1.2 Low Frequency Automatic Demand Disconnection

On 28 April 2025 at 12:30:00, the total consumption in Spain reached **29,631 MW**, representing the total load estimate closest to the instant of LFDD activation. From 12:33:20.600 onwards, the six frequency steps of the low frequency automatic demand disconnection plan were activated as follows:

- At 12:33:20.600, the system frequency dropped below 49.0 Hz, triggering the activation of the first step of load shedding. During this step, a total of 1,175.9 MW of load was disconnected.
- At 12:33:20.760, the system frequency dropped below 48.8 Hz, triggering the activation of the second step of load shedding. The total load shed in this step was 1,669.1 MW.
- At 12:33:21.000, the system frequency dropped below 48.6 Hz, triggering the activation of the third stage of load shedding. The total amount of load disconnected in this stage was 1,574.6 MW.
- At 12:33:21.380, the system frequency dropped below 48.4 Hz, triggering the activation of the fourth step of load shedding. The total load shed in this step was 1,523.9 MW.
- At 12:33:21.820, the system frequency dropped below 48.2 Hz, triggering the activation of the fifth step of load shedding. The total load shed in this step was 1,294.3 MW.
- At 12:33:22.040, the system frequency dropped below 48.0 Hz, triggering the activation of the sixth step of load shedding. The total load shed in this step was 1,267.1 MW.

Notably, two DSOs reported that a few underfrequency relays had not yet been adapted to the agreed low frequency demand disconnection scheme. As a result, 71.03 MW were disconnected at 48.7 Hz, although they are currently programmed to activate at 48.6 Hz. Therefore, the 71.03 MW at 48.7 Hz were added to the 48.6 Hz frequency threshold.

The total disconnected demand was **8,504.9 MW**, representing approximately 29 % of the prior total consumption. This value falls short of the theoretical target set by the defence plan, which establishes a range with a minimum of 38 %. This includes the load from consumers connected at the transmission grid and at the distribution grid. Table 3-10 presents the net demand disconnected during the six steps.

Step	Threshold (Hz)	Load disconected (MW)	Actual (% of total load)	Plan (% of total load)
1st	49.0	1,176	4.0	6
2nd	48.8	1,669	5.6	9
3rd	48.6	1,575	5.3	8
4th	48.4	1,524	5.1	8
5th	48.2	1,294	4.4	7
6th	48.0	1,267	4.3	7
	TOTAL	8,505	28.7	45.0

Table 3-10: Demand disconnection for the Spanish System

3.3.2 Portuguese System Defence Plan

The frequency drop activated the underfrequency defence plan of the Portuguese electrical system, in accordance with Article 15 of Regulation (EU) 2017/2196 issued by the Commission on Emergency and Restoration. The plan foresees progressive load shedding based on frequency thresholds, described as follows:

- From 49.8 Hz to 49.3 Hz, the defence plan automatically shed the hydroelectric units in pumping mode.
- At 49.2 Hz, the defence plan shed the electro-intensive industrial load at the TSO and DSO level.
- From 49.0 Hz to 48.0 Hz, the defence plan shed distribution-connected consumers.

According to REN and E-REDES data, the total shed active power was:

- Hydroelectric units in pumping mode: 2,098 MW
- Electro-intensive industrial consumers: 218 MW
- Distribution network consumers: 1,955 MW.

On 28 April 2025 at 12:33:00, the total consumption in Portugal amounted to 5,865 MW.

The preliminary estimation of the total demand disconnected as part of LFDD, between 49.0 Hz and 48.0 Hz, amounted to 1,955 MW, representing 33.3 % of the prior total consumption. Considering the 218 MW of electro-intensive industrial consumers disconnected at 49.2 Hz, the total disconnected demand represents 37.1 % of prior total consumption. This value falls short of the theoretical target set by the defence plan, which establishes a range with a minimum of 38 %. Details can be found in Table 3-11, Table 3-12, Table 3-13, and Figure 3-70.

The Portuguese defence plan is designed to shed consumption in different frequency steps. Geographical dispersion of the consumption to be shed in each frequency step is assured, mitigating the impacts of congestions and voltages in the grids.

Per the Portuguese defence plan, the LFDD demand disconnected at each frequency level is, in some cases, physically disconnected at the connection point of specific load units, while in others, it is physically disconnected at the distribution grid and at transmission connection points with the distribution grid. This means that a specific section of the distribution system is disconnected in its entirety, regardless of whether active/reactive power is being injected or absorbed by that section.

Pump-storage	Step (Hz)	P previous (MW)	P tripped (MW)
Pump 1	49.8	18	18
Pump 2	49.8	18	18
Pump 3	49.8	124	124
Pump 4	49.8	0	0
Pump 5	49.8	0	0
Pump 6	49.8	221	221
Pump 7	49.8	0	0
Pump 8	49.8	0	0
TOTAL at 49.8 Hz		381	381
Pump 9	49.7	0	0
Pump 10	49.7	113	113
Pump 11	49.7	337	337
TOTAL at 49.7 Hz		450	450
Pump 12	49.6	0	0
Pump 13	49.6	0	0
Pump 14	49.6	219	219
Pump 15	49.6	219	219
TOTAL at 49.6 Hz		438	438
Pump 16	49.5	0	0
Pump 17	49.5	114	114
Pump 18	49.5	74	74
Pump 19	49.5	75	75
Pump 20	49.5	207	207
TOTAL at 49.5 Hz		470	470
Pump 21	49.4	0	0
Pump 22	49.4	0	0
Pump 23	49.4	0	0
Pump 24	49.4	0	0
TOTAL at 49.4 Hz		0	0
Pump 25	49.3	30	30
Pump 26	49.3	0	0
Pump 27	49.3	329	329
TOTAL at 49.3 Hz		359	359
TOTAL (all steps)		2,098	2,098

Table 3-11: Pump storage disconnection for the Portuguese system

Threshold (Hz)	Electro-intensive load disconnected (MW)
49.2	218

Table 3-12: Disconnection of electro-intensive industrial consumers for the Portuguese System

Threshold (Hz)	Load disconnected (MW)	Real (% of total load)	Planned (% of total load)
49.0	315	5.3	6.7
48.8	293	5.0	6.6
48.6	315	5.3	6.9
48.4	323	5.5	6.6
48.2	282	4.8	6.4
48.0	427	7.3	9.7
TOTAL	1,955	33.3	42.9

Table 3-13: Demand disconnection as part of the LFDD plan for the Portuguese system (preliminary estimation)

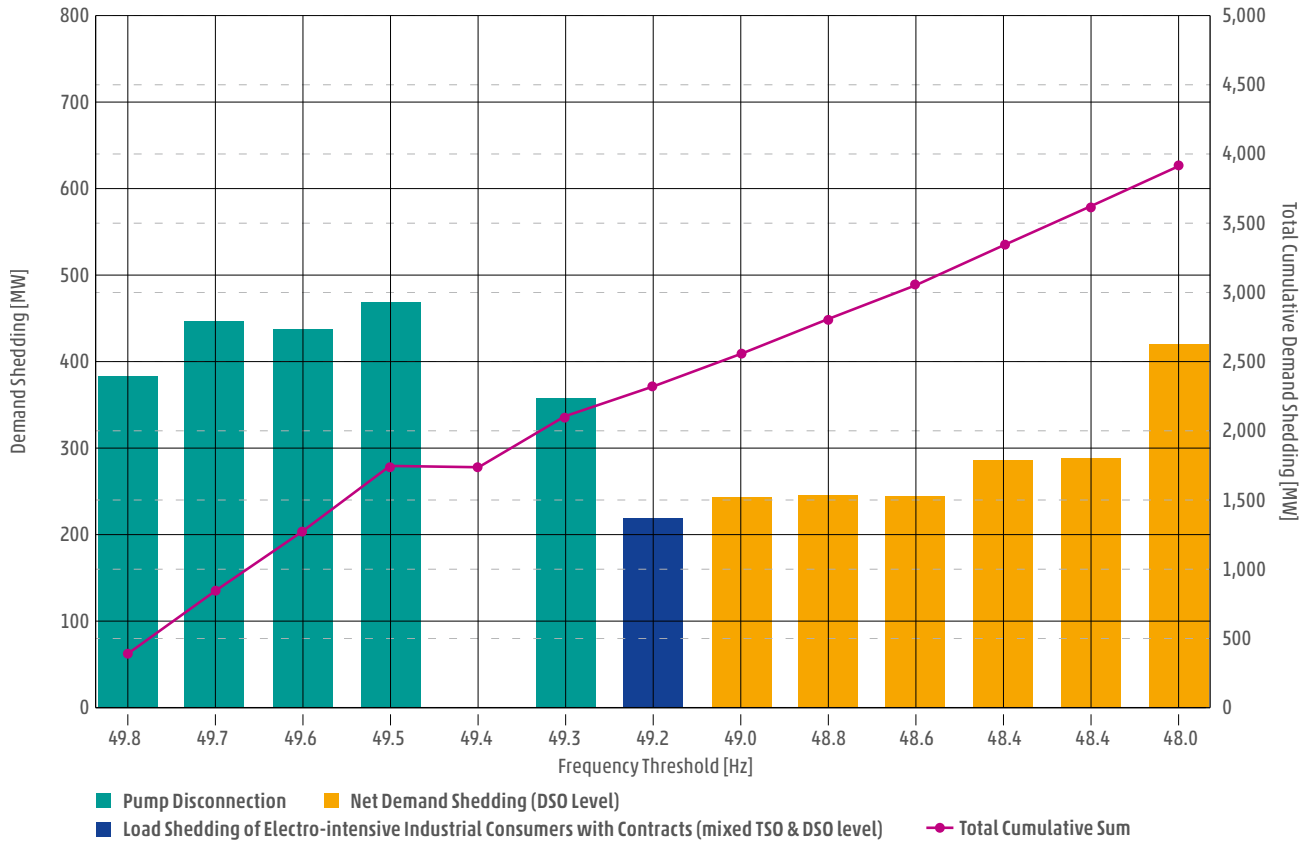


Figure 3-70: Activation of the low frequency demand disconnection plan for REN

3.3.3 French System Defence Plan

The French defence plan of RTE includes the following mechanisms:

» A load shedding plan on a frequency criterion aimed at fighting frequency collapses, which sheds 45 % of the total load in six steps between 49 and 48 Hz.

Step	Threshold (Hz)	Demand disconnected (% of total load)
1st	49.0	5
2nd	48.8	8
3rd	48.6	8
4th	48.4	8
5th	48.2	8
6th	48.0	8

Table 3-14: Low Frequency demand disconnection plan

» Local relays called DRS (Débouclage suite à Rupture de Synchronisme – see Section 3.2.2.4), a French acronym for area islanding protections in case of loss of synchronism, should detect any loss of synchronism and immediately isolate the ill network portion from the healthy one. To accomplish this, the defence plan defines a synchronism zone, with loss of synchronism protection (DRS) installed at the border of this zone.

As presented in Section 3.2, the French defence plan worked as expected. Many losses of synchronism protection (DRS) were triggered at the boundary of the synchronism zone defined in the defence plan, as shown in Figure 3-71.

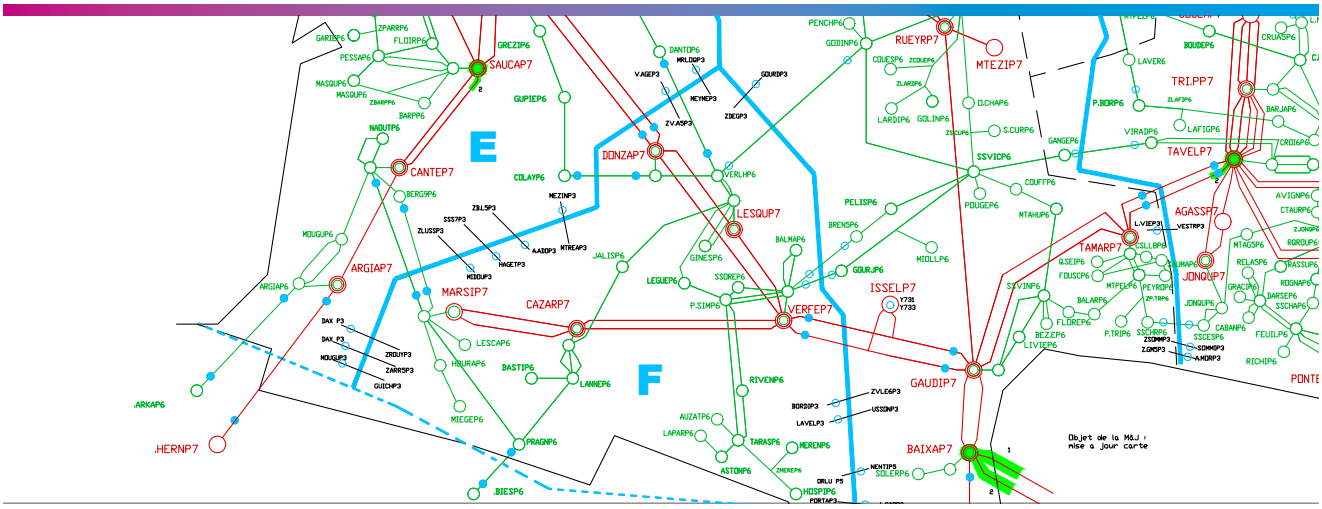


Figure 3-71: Defence plan synchronism zone on the southwest of France

On the other hand, some load shedding based on the frequency criterion occurred in the Basque Country, in accordance with the plan.

3.3.4 Overview of Pump-Storage Shedding

Before the incident, 5,129 MW of pump-storage units were connected: 3,031 MW in the Spanish network and 2,098 MW in the Portuguese network. As a result of the underfrequency condition, 2,756 MW of pump storage disconnected in Spain and 2,098 MW of pump storage disconnected in Portugal, with a total power of 4,854 MW, to support the restoration of the generation–demand balance.

Table 3-15 shows the frequency thresholds and the pump storage disconnected in each step. Notably, the load shedding relays of some units experienced delays in their proper operation.

System	Frequency threshold (Hz)	Disconnected load (MW)
PT	49.8	381
PT	49.7	450
PT	49.6	438
ES	49.5	2,168
PT	49.5	470
PT	49.4	0
ES	49.3	588
PT	49.3	359
	TOTAL	4,854

Table 3-15: Pumped storage tripped per frequency threshold

3.3.5 Overview of Load Shedding

Due to the underfrequency condition in the Iberian Peninsula, all the frequency thresholds defined in the system defence plan in Portugal and Spain were reached, and the pump-storage and load shedding were activated to try to restore generation–demand balance. In particular:

- In Spain, 8,505 MW of consumption were disconnected, including 1,402.5 MW from industrial consumers connected to the transmission grid and 7,102.4 MW of load in the distribution network.
- In Portugal, 2,173 MW of consumption were disconnected, including 218 MW from industrial consumers and 1,955 MW of load in the distribution network.

The effective load shedding in Portugal and Spain reached 10,678 MW, as shown in Table 3-16 and Table 3-17.

The values for the transmission load shedding in the Spanish system were obtained mainly from COMTRADE files and verified against previous SCADA values. There are no COMTRADE files available from load shedding in the distribution network in the Portuguese and Spanish systems. Distribution load shedding values were provided by the DSOs.

	Electro-intensive Industrial Consumers (MW)		Other Load (MW)		
Threshold (Hz)	PT	ES	PT	ES	TOTAL
49.2	218				218
49.0			315	1,176	1,491
48.8			293	1,669	1,962
48.6			315	1,575	1,890
48.4			323	1,524	1,847
48.2			282	1,294	1,576
48.0			427	1,267	1,694
TOTAL	218		1,955	8,505	10,678

Table 3-16: Load shedding for ES and PT

Threshold (Hz)	DSO1 (ES) (MW)	DSO2 (ES) (MW)	DSO3 (ES) (MW)	DSO4 (ES) (MW)	DSO5 (ES) (MW)	E-REDES (PT) (MW)	Total DSO (MW)
49.0	85.1	23.7			97.0	315.0	520.8
48.8	529.9	190.0			767.5	293.0	1,780.4
48.7	49.6				21.4		71.0
48.6	423.9	195.7		5.2	628.1	315.0	1,567.9
48.4	633.8	216.8		21.6	651.7	323.0	1,846.9
48.2	412.3	220.3	60.4	12.2	589.1	282.0	1,576.3
48.0	544.1	218.2	0.7	11.6	492.5	427.0	1,694.1
TOTAL	2,678.7	1,064.7	61.1	50.6	3,247.3	1,955.0	9,057.4

Table 3-17: Distribution load shedding per frequency threshold

3.4 Impact of the Blackout on System Users on the French Network

The incident of 28 April 2025 resulted in a full blackout in Spain and Portugal, interrupting the power supply of all consumers in both countries. The impact of the incident was much more limited in France, as described in this section.

This impact was mainly concentrated in the southwest area of France.

Regarding the load, two substations at the distribution level in the Basque Country were impacted by the dynamic phenomena during the collapse phase.

The first substation was shed due to the frequency load shedding defence plan.

The second substation lost supply due to the activation of the loss of synchronism protection.

The total unsupplied power amounted to 7 MW. System users were restored within 15 – 20 minutes.

Regarding the impact on industrial system users, approximately 15 experienced total or partial process interruptions due to the activation of their own protections (related to voltage or oscillations).

Some suffered equipment damage, while others experienced a slow restart following necessary cleaning procedures. Additional users experienced voltage dips.

On the generation side, one nuclear power plant experienced an automatic interruption of the reactor, and around 20 other generation facilities connected to the transmission grid, as well as additional power plants on the distribution grid, tripped.

This was related either to:

- Low transient frequency
- High current level on the stator side
- Loss of synchronism

Other generation facilities tripped due to high voltage levels. This was the case for two hydraulic power units, as well as some PV and wind turbines. The two hydropower pumps were consuming around 200 MW.

Generation facilities connected to networks below 45 kV were disconnected due to the activation of protections, primarily related to overvoltages.

Finally, a few batteries that were connected before the event were disconnected very briefly. The total loss of supplied power was minor (around 5 MW), and the restart took between one and three minutes.

4 ANALYSIS OF RELEVANT FACTORS

This chapter presents detailed technical analyses of the various phenomena and actions that preceded the blackout, with the purpose of understanding precisely the causes of the incident. All these analyses are based on the factual data collected and presented essentially in Chapters 2 and 3.

Based on the results of these technical analyses, the Expert Panel established the root cause tree describing the key factors that eventually led to blackout. The root cause tree is presented in the last section of this chapter.

It should be noted that these technical analyses focus on the Spanish system, as the facts described in chapters 2 and 3 indicate that the origin of the blackout lies in Spain, not in Portugal.

The chapter is structured according to the following sections :

4.1 Voltage control

4.2 Oscillations

4.3 Analysis of TSO-DSO flows

4.4 System Defence Plans

4.5 Alarm handling

4.6 Root Cause Tree

4.1 Voltage control

4.1.1 Structure of the section

This section discusses voltage control.

- The first subsection (4.1.2) discusses the reactive power resources that collectively were unable to respond to the voltage increase. It addresses technical and regulatory aspects and suggests future improvements.
- The second subsection (4.1.3) describes voltage management aspects that contributed to the loss of voltage control.
- The third subsection (4.1.4) describes voltage trends on 28 April, explains the voltage increase in more detail, and considers the conditions in the morning and on a previous day.
- The fourth subsection (4.1.5) presents the static-model voltage sensitivity analyses that were performed.
- The recommendations regarding voltage control referenced throughout this chapter are, like other recommendations, located in Chapter 9.
- The additional areas for attention/improvement identified during the investigation are listed in the conclusions of Chapters 4.1.2 and 4.1.3.

This section complements the data already included in Chapter 2.

4.1.2 Reactive power contribution by various assets

This section describes the capabilities of various assets to provide reactive power to the system on 28 April.

4.1.2.1 Integrated network components (shunt reactors, STATCOMs)

On the day of the incident, only one STATCOM was in operation. Another was undergoing final testing before commissioning, while others were planned for the future and were not yet operational at the time of the incident. Only the available STATCOM was considered in the security analysis.

There are around 90 shunt reactors installed in the Spanish grid, corresponding to a reactive power capacity of 13,300 MVAr. Of these, 43 shunt reactors (with a total of 6,000 MVAr) were manoeuvred for voltage control in the period 09:00 –12:30 on 28 April (84 switching operations in total, 40 in the south).[30] In Spain, shunt reactors are operated by RE manually.[31]

Some shunt reactors were disconnected and reconnected during or around oscillation episodes due to the low and high voltages experienced during voltage swings, especially during the low voltages during the oscillations.

Following the oscillation episodes, between 12:20 and 12:25, three shunt reactors were disconnected and four transmission lines were connected, increasing the grid's reactive power production. To address the resulting voltage rise, five shunt reactors were connected between 12:25 and 12:28. As a result of these switching manoeuvres, the voltage returned to within allowed limits by 12:30.

No additional shunt reactors were connected from 12:28 until the incident. Throughout the voltage rise from 12:32:00 to 12:33:16 and the blackout, 7,700 MVAr of shunt reactors remained connected and 5,600 MVAr remained disconnected. The share of connected reactors was particularly low in southern Spain, where voltages had been particularly low after the oscillations, as shown in Table 4-1.

The following chart summarises the status of shunt reactors.

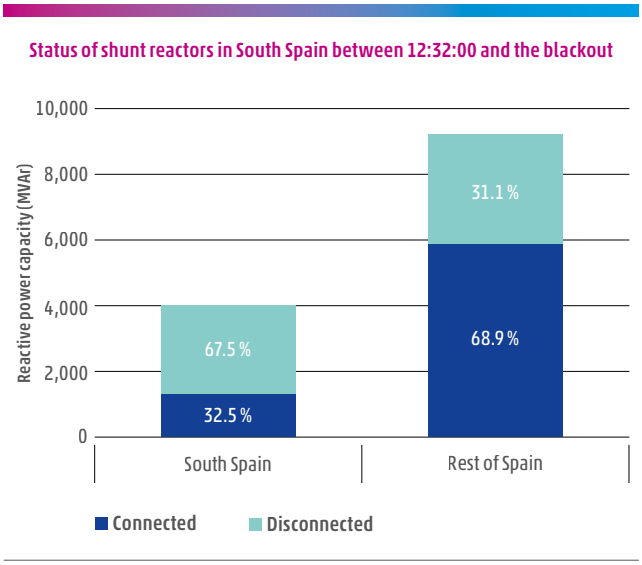


Figure 4-1: Status of shunt reactors between 12:32:00 and the blackout.

As the actions to connect shunt reactors were manual, given the short lead time, the shunt reactors were not manually connected during the minute-long voltage rise (see Figure 4-8).[32] RE considered, both on 28 April and previously, that the assets intended to absorb fast voltage variations were conventional generators, STATCOM, and the HVDC link, rather than shunt reactors. The Expert Panel issued related recommendations in Chapter 9.

Area	Reactive power capacity of connected shunt reactors	Reactive power capacity of disconnected shunt reactors
Southern Spain	1,300 MVAr (32.5 %)	2,700 MVAr (67.5 %)
Other regions of Spain	6,400 MVAr (68.9 %)	2,900 MVAr (31.1 %)
Total	7,700 MVAr (58 %)	5,600 MVAr (42 %)

Table 4-1: Connection status of shunt reactors in Spain between 12:32:00 and the blackout

30 The chronology of voltage control actions is provided in Table 2-2 in Section 2.1.2, and the status of shunt reactors is summarised in Section 2.6.4.1.

31 The Expert Panel has not comprehensively assessed the approach across all TSOs. Nevertheless, several TSOs participating in the Panel indicated that they only have manual control of shunt reactors, while RTE indicated they have a hybrid approach combining manual and automatic control.

32 Furthermore, as described in Section 4.1.3.2 (Alarm handling), voltage remedial actions were not included in the operator's handbook.

4.1.2.2 Generators in Spain

This section focuses on the reactive power provisions by generators connected to the transmission network in Spain.[33]

4.1.2.2.1 Introduction

In terms of how power system users provide reactive power, this report distinguishes three reactive power control modes (generally, not limited to Spain):

- Reactive power control mode (where the unit assumes a reactive power setpoint and provides a set value (in MVAr) of reactive power injection or absorption);
- Voltage control mode (where the unit assumes a voltage setpoint that can be defined either at the connection point or at the generator's low-voltage side and injects or absorbs reactive power as needed to move the voltage towards that setpoint);
- Fixed power factor (where the unit assumes a power factor setpoint and injects or absorbs reactive power proportional (i.e. capacitive power factor) or inversely proportional (i.e. inductive power factor) to its instantaneous active power output).

With regard to power park modules of Type C and above, Article 21(3)(d) of the Commission Regulation 2016/631 (NC RfG) provides that the power park module shall be capable of providing reactive power automatically by either of the three modes and that the relevant system operator may specify which of the three reactive power control mode options and setpoints to apply.[34] Nevertheless, this requirement does not apply to the power park modules connected before the application of the NC RfG.

4.1.2.2.2 Implementation in Spain

On the day of the incident, the requirements concerning reactive power provision by generators in Spain varied by generator type:

- A. RES generators (either connected to the transmission or the distribution grid) followed a **fixed power factor range**. The national regulation[35] defined the power factor range, while for some RES generators above 5 MW, the TSO[36] set a specific range for each power plant to reflect system needs, resulting in different values across plants. The power factor for facilities connected to the distribution network was agreed upon by the TSO in coordination with the relevant DSO.
- B. Conventional/Synchronous generators above 30 MW
  - i. By default, they were mandated to provide reactive power according to a **function Q(U-actual, P-installed, U-reference)**, where the non-real-time voltage reference (U-reference) was defined for the electrical point not shared with other generation units (usually corresponding to the connection point) – as detailed in Operational Procedure PO 7.4 (2000)[37]
  - ii. Some generators were also able to volunteer to receive **direct voltage setpoints** from the system operator.

33 CNMC indicated that in Spain, the regulatory framework is complex, as regulatory technical operational details are proposed by the TSO and must be approved by the relevant regulatory authorities before entering into force. CNMC is responsible for markets and system services, while the Ministry is responsible for security criteria and grid connection requirements.

34 The utilisation of the requirements in Spain is discussed in Section 4.1.2.3.

35 Royal Decree 413/2014 of 6 June, which regulates the activity of producing electrical energy from renewable energy sources, cogeneration, and waste.

36 This means that the fixed power setpoint was explicitly sent from the system operator to each unit, in this case via email. While the communication is direct, there is nevertheless no direct data link – i.e. there is no direct setpoint in the sense of the system operator's instruction automatically applying to the generator's control system. In Portugal, voltage control is implemented via setpoints sent through the SCADA systems of the TSO and the generator. Generators should be capable of operating in three modes (reactive power control setpoint, voltage control setpoint, power factor control). After receiving a voltage setpoint, it is up to the power plant control (PPC) to manage the plant's resources to deliver the desired output at the grid interconnection point. Older wind power plants function on a fixed tan fi mode (between –0.2 and 0.2), according to a fixed setpoint determined by the TSO. These setpoints were communicated via email and are not changed regularly.

37 Resolution of 10 March 2000, of the Secretary of State for Industry and Energy, which approves the operating procedure of the system (P.O. 7.4) "Complementary service for tension control of the transmission network".

A process to amend Operational Procedure PO 7.4 (2000) was initiated by RE in 2021 and approved by the national regulatory authority (CNMC) in June 2025, after the incident.[38] By 2023, the process had led to the development of a pilot market-based voltage control project in two regions of Spain. After the pilot projects and regulatory sandboxes concluded, participating conventional generators were allowed, on a voluntary basis, to follow a direct voltage setpoint sent by the system operator to the participating unit in real time.

This meant that some generators were able to transition, in real time, between the optional direct voltage setpoint control mode (pilot project, direct voltage setpoint) and the default non-real-time voltage reference mode (Operational Procedure PO 7.4).

The following subsections provide a more detailed description of reactive power provision in Spain by generators.

4.1.2.3 Reactive power provision by RES generators in Spain

4.1.2.3.1 Technical aspects

As indicated in point A in Section 4.1.2.2.2, RES generators were set to follow a fixed-power-control mode. This was a key factor in the incident for two reasons:

- First, it meant that resources operating in fixed-power-factor mode **injected a proportional (fast) reactive power ramp,** and thus a voltage ramp, **into the system** for every (fast) active power ramp (e.g. a schedule change).
- Second, **fixed-power-factor-mode resources** provided **no reactive power support** that would compensate for voltage fluctuations, since their reactive power output did not respond to conditions in the grid.

The Expert Panel has issued a related recommendation, which can be found in Chapter 9. Further takeaways are detailed in the summary of conclusions in Section 4.1.2.7.

4.1.2.3.2 Applicable requirements

The operational requirements for certain aspects of RES power plants are based on the national Spanish Royal Decree in force since 2014 (RD 413/2014 on renewable energy sources). These requirements specify the power factor range for RES installations, set between 0.98 inductive and 0.98 capacitive (see Article 7e of the Royal Decree). Based on provisions in Article 7e, RE also instructed some power plants to operate within a narrower range (e.g. 0.98 – 0.99 inductive) deemed more in line with system needs.[39] In effect, **RES power plants followed a fixed power factor** within the required range. Article 7e of the Royal Decree did not explicitly mandate any other approach to voltage control for RES.

The decree also appears to provide for the option of a **RES-specific voltage control service** (see paragraph ii of Article 7e) **or for RES participation in the regular voltage control service** (see paragraph iii of Article 7e), but these regulatory developments had not been finalised.[40, 41] The Royal Decree, as in force on the relevant date, does not appear to assign the TSO or the national regulatory authority[42] any direct competence to modify its provisions, **particularly regarding the voluntary nature of participation in these services.**

For the Expert Panel's observations on the applicable requirements, please refer to Chapter 9.

38 The amendment proposed in 2021 had to be approved by the national regulatory authority (CNMC), which in 2019 was granted the power to regulate non-frequency services, among others. The NRA's assessment with stakeholders identified several difficulties, including a conflict with other national rules outside the competence of the NRA – specifically regarding the obligation for RES to provide dynamic (i.e. non fixed-power-factor-based) voltage control – and a lack of evidence to discard a market-based approach due to economic inefficiency, as required by Article 40 of Directive (EU) 2019/944. To continue developing the voltage control framework, two pilot projects were launched in 2022 and 2023: the first tested a market-based voltage control service in two regions of Spain, and the second explored the voltage control capabilities of consumers. Based on the findings of the pilot projects, the TSO proposed an amendment to the initial proposal in March 2024, which was subsequently consulted on, revised, and approved in June 2025.

39 If the plant was connected to the distribution grid, the instruction was agreed upon with the DSO.

40 The process related to paragraph iii had begun but had not reached agreement. A separate process related to paragraph ii never took place but there was a proposal of voltage control service provided by RES in 2011, under the scope of a different Royal Decree (RD661/2007) and which appears to have not been finalised.

41 Pilot projects did take place.

42 I.e. CNMC, the national regulatory authority designated according to Article 57 of Directive (EU) 944/2019.

4.1.2.4 Reactive power provision by conventional generators in Spain

As indicated in point B.i of Section 4.1.2.2.2, the default requirements for reactive power provision by conventional generators at the time of the incident were defined by Operational Procedure PO 7.4 (2000).[43]

At the same time, as indicated in point B.ii, some generators had the option to receive direct voltage setpoints from the TSO.

4.1.2.4.1 Technical aspects

The basic service described in Operational Procedure PO 7.4 requires each generator to set its reactive power output based on a function described in the Operational Procedure. The generator sets its reactive power output based on:

- The system voltage reference set by the TSO (set at 405 – 410 kV on 28 April, as it had been for several years beforehand). While the voltage reference for Operational Procedure PO 7.4 was generally set for the whole system, it was temporarily set independently for some generators.
- The actual voltage at the generator connection point
- The installed active power of the generator

Thus, under the applicable requirements outlined in Operational Procedure PO 7.4, a generator was to determine its reactive power output itself, according to the formula set out in Operational Procedure PO 7.4, and considering the global voltage reference published by the TSO.

In some cases, a specific voltage reference for an individual generator could be sent via telephone communication,[44] but this was not the default approach. In contrast, if the generator participated in the pilot project, a direct voltage setpoint was sent to each participating generator each minute.

Under the requirements of Operational Procedure PO 7.4, the system operator was responsible for calculating the expected reactive power provision by generators. Unlike in the pilot project, this reference was not sent to generators in real time; instead, it was used ex post to determine remuneration, penalties, and monthly fulfilment reports, which were shared with the respective generators. Generators were also expected to calculate the reference themselves, but the TSO did not conduct close-to-real-time monitoring of differences between its calculated reference (using the Operational Procedure PO 7.4), and the generator's actual reactive power output.

The Expert Panel issued related recommendations in Chapter 9.

4.1.2.4.2 Applicable requirements

The Operational Procedure PO 7.4 was published in 2000. On 28 April 2025, the process to amend the procedure was ongoing, but the original version still applied. In June 2025, the new version was approved.

CNMC's interpretation of the Operational Procedure PO 7.4 is included in Section 2.6.5.1. The Operational Procedure PO 7.4 (2000), as applicable on 28 April 2025, established the rights and responsibilities of generators for providing reactive power. It required generators to provide a minimum reactive power contribution (absorbing or injecting, depending on the voltage) whenever the voltage was outside a margin around the voltage reference set by the TSO.

According to the Procedure, in order to determine penalties or payments related to the voltage control service, the service is considered delivered if reactive power meeting these requirements is provided in at least 75 % of the samples (i.e. nine out of 12 measurement samples taken in an hour, according to the Procedure).

The Procedure PO 7.4 prescribed penalties for generators not complying with the procedure. However, it appears that the penalties were not applicable in practice, since the economic scheme required by the Operating Procedure PO 7.4 had not been developed.

43 As noted above, the generators that participated in the pilot project were able to alternate between a) providing reactive power considering the direct voltage setpoint in the pilot project and b) providing voltage control considering the Operational Procedure 7.4 (2000).

44 It was issued on 28 April for one generator outside the southern region.

While these aspects were not raised by third parties during this investigation, the Expert Panel identified several potential unclarities in the Operating Procedure PO 7.4 (2000), namely:

» a) It appears that Annex 6 of the Operational Procedure includes a chart referenced in the main text that aims to explain the expected reactive power setpoints. The chart refers to voltage values between 380 kV and 420 kV but does not refer to voltages between 420 kV and 435 kV. Furthermore, while the capability of the AVR[45] can be inferred by the TSO from the structural information provided by the generator, it appears that the Procedure also does not explicitly describe the required speed of reactive power response, which instead must be inferred from the provisions describing validation.

» b) The Operational Procedure does not appear to define the measurement and sampling approach for assessing compliance with the obligation (Article 10). Specifically, it does not define which values (instantaneous or average over the 5-minute interval) of voltage or reactive power output should be used by either the TSO or the generator.

The Operational Procedure allowed all providers to apply for derogation from the technical restrictions for reactive power provision in certain operating points. This application was to be presented to the Ministry. In cases approved by the Ministry following an NRA report, the restrictions would be communicated to the system operator for its consideration. It is the understanding of the Expert Panel that some generators have submitted applications for technical restrictions to the competent authority (the Ministry); however, to the Expert Panel's knowledge, the Ministry has not responded.

4.1.2.4.3 Assessment of reactive power provision

The scope of this investigation and the mandate of the Expert Panel do not grant it the authority, nor does it have any intention, of interpreting the applicable legislation or determining liabilities for non-compliance with legal requirements. Within the limits of its mandate, and for the purposes of identifying the causes of the incident, the Expert Panel sought to understand the actual technical behaviour of relevant generators with respect to reactive power on 28 April.

As depicted in the relevant figures,[46] on 28 April, the actual reactive power output of some conventional synchronous generators that were providing reactive power diverged from the TSO's reference, which it established in accordance with the Operational Procedure PO 7.4. It was also observed that for several generators, during the morning hours of 28 April (i.e. between 9 am until the incident), actual reactive power output (based on RE SCADA data) aligned with reference values (Q-reference, also calculated from RE SCADA data according to the formula in Article 10 and the associated graph in the Operational Procedure PO 7.4[47]) in less than 75 % of the samples.

It is noted that the number of conventional generators providing reactive power is set by the TSO according to its daily security assessments of system needs after the results of the market.[48]

The Expert Panel's in-depth examination focused on three thermal conventional generators connected in the south – the region where voltage conditions first deteriorated and the first overvoltage disconnections occurred.

RE provided records showing that the three thermal conventional generators in the south had, at least once in the year preceding the incident and under different grid conditions, delivered reactive power exceeding the TSO's reference values.

In seeking to understand the discrepancies between the days, the Expert Panel reached out to the owners of two of the power plants. Due to time constraints, by the time the investigation was finalised, the Expert Panel was only able to request generator information for these two thermal power plants in the south, and not for any other plants in Spain.

45 Automatic voltage regulator.

46 See Section 2.6.6.

47 This is without prejudice to any potential derogations from the requirements, which the Expert Panel is not competent to assess.

48 This is further described in Section 2.6.8.

According to the generator owner of one, the generator in question performs differently depending on the manual configuration, which includes changing the voltage setting at its terminal (i.e. the generator follows a different capability curve reflected in a different function Q(U)). This change can be implemented manually by the generator on its own initiative or after a suggestion by the TSO.

Based on information provided by owners for the assessed past day, RE sent the generator an advance instruction to increase reactive power absorption. However, it appears this instruction was not sent for this generator for 28 April. According to the owner, this explains why the provision of nominal reactive power was different (higher) in the past day compared to 28 April.

The Expert Panel also observed that the rate of change of reactive power output of another conventional generator[49] operating on the day followed a linear trajectory that was slower than the faster changes in the TSO's reactive power reference.[50] After steep changes in the reference (driven by steep changes in voltage), the generator's reactive power output began ramping towards the reference. However, given the ramp slope, reaching the new reactive power reference would have required a certain amount of time (several minutes).

RE provided a record implying[51] that the same thermal conventional generator did, on at least one other occasion less than one year before the incident and under different grid conditions, provide a faster reactive power response than on 28 April. Conversely, the generator indicated that its response time on 28 April was similar to the past day, and that it was set according to the vendor's specifications.

The Expert Panel did not systematically assess whether the output of these power plants had already occasionally or regularly diverged from the TSO's reference in the period preceding 28 April 2025.

The Expert Panel did not investigate the issue further due to time constraints.

The reactive power contribution of conventional generators was initially presented in the factual report in an aggregated way, along with a reference. After further investigation, the Expert Panel has determined that the original charts should be amended with regard to the "Q-reference" to ensure greater consistency when representing aggregated regional charts.[52] One change involved the original charts not considering the period during which a power plant was following a direct voltage setpoint pilot project. The second change addressed the representation of Q-reference: the original charts set Q-reference to 0 MVAr when the voltage was within the specified range, instead of indicating that in practice, any Q output – not only 0 MVAr – was expected.

The updated charts (Figures 2-84 to 2-86) thus:

- Set Q-reference = Q-actual, where the actual voltage relevant for the generator is within the voltage reference
- Set Q-reference = Q-actual, where a generator is following a direct voltage setpoint pilot project

4.1.2.4.4 Findings

The Expert Panel found several potential unclarities and areas potentially requiring further specification in the applicable requirements for voltage control by conventional generators. These include the absence of an explicit requirement on response speed, the lack of reliable immediate reaction due to the 75 % hourly fulfilment target, and the absence of economic consequences in case of underperformance.

Furthermore, the actual reactive power output of several conventional power plants on the day reached the TSO's expected reference according to the applicable requirements in the Operational Procedure PO 7.4 less than 75 % of the samples (hourly), and there was no close-to-real-time monitoring of this divergence.

The combination of these factors resulted in the **lack of effective voltage control support by the conventional generators.**

For the Expert Panel's observations on the applicable requirements, please refer to Chapter 9.

49 For confidentiality reasons, this generator is not identified here, but its reactive power contribution is included in the aggregated reactive power charts in 2.6.6.

50 The system operator pointed out that the primary response should be faster, per the AVR characteristics provided during commissioning, and that this generator had been programmed in the morning to provide voltage control.

51 The time granularity of the figure did not enable an accurate assessment.

52 See Section 2.6.6.

4.1.2.5 Generators <1 MW connected to mid- and low-voltage networks

The Expert Panel did not reach substantive conclusions regarding reactive power provision in its investigation of the behaviour of generators connected at voltage levels lower than 220 kV, or the voltage trends in these networks.[53] This is due, among other factors, to low visibility and limited data availability. More details on the behaviour of small generators (and associated recommendations) are provided in Chapter 4.3.

4.1.2.6 Loss of reactive power provision by disconnected elements

4.1.2.6.1 Downstream networks

The Expert Panel found that the design of some downstream networks[54] in Spain was not aligned with system needs, resulting, among other issues, in more challenging voltage control. In some cases, voltage control in these networks was insufficient due to a combination of factors, including generators operating in fixed power factor control mode, fast generation ramps, the physically slower changes of tap-changing transformers, and downstream element disconnection settings.

In particular, the Panel investigated a case in a substation in the south (400 kV TS 1 – Granada). The downstream network connected to the TSO grid at the 400 kV level. It is important to note that this network is neither transmission nor distribution: it is owned by the generators. A tap-changing transformer connected the 400 kV and 220 kV levels. Following a voltage rise across both sides of a tap-changing transformer (which remained at a given tap setting throughout the fast voltage rise), downstream system components disconnected due to overvoltage (the overvoltage disconnection setting at the disconnected element was 1.1 p.u., with no delay). At the time of disconnection, the voltage was not exceeded on the 400 kV side. This occurred because the tap position, remaining unchanged during the voltage rise, resulted in a transmission ratio that produced a higher p.u. voltage value at the 220 kV side than at the 400 kV side. In this case, the tap position (despite the presence of an on-load automatic tap changer) worsened the situation, as it caused the voltage on the 220 kV side to be higher.

The overvoltage trip led to the disconnection of all downstream generation, resulting in the loss of reactive power absorption from the disconnected units.[55]

The rise in voltage that led to the disconnection (partly due to fixed-power-factor generators changing setpoints) occurred faster than the physical or design constraints of tap-changing transformers, meaning that changing a single tap position could take longer. As a result, the combination of transformer tap changes and fixed-power-factor generators was insufficient to withstand fast voltage changes.

This points to the need for solutions regarding the design and coordination of the downstream networks. It appears that, due to the network design, the downstream network owner's responsibility to ensure that generators remain connected while the voltage at the connection point is within limits was not fulfilled.

Regarding protection coordination, it is worth noting that the protection setting (zero time delay at 1.1 p.u.) had not been reviewed by RE, since the coordination with transmission protection was not deemed necessary. (In contrast, such coordination is needed for protections that operate under short-circuit conditions, as these must be cleared by tripping the minimum number of elements.) In the original communication of structural information to RE, a 1-second delay for the activation of voltage protection was indicated. During the investigation of the incident, the Expert Panel received information that the delay was actually zero seconds. However, RE indicated that this change had not been communicated in advance.[56]

53 Note: The Spanish Royal Decree referred to in Footnote 6 is also applicable to mid- and low-voltage networks.

54 For the purposes of this report, "downstream network" refers to a third-party-owned grid in Spain (up to 220 kV) that extends from the connection point with the TSO grid, across a third-party-owned tap-changing transformer, to various generators.

55 Note: Similar phenomena of downstream generation in Spain disconnecting due to the tripping of upstream third-party-owned grid components also occurred in the 2021 **system separation** (p. 34)

56 See Chapter 3 (Disconnections) for more details.

Regarding the operation of tap-changing transformers, the transformers between the transmission and distribution grids in Spain have been governed by an agreement between RE and DSOs since 2022. The agreement specifies that taps are operated by distribution operators, and that the TSO may issue an order to block tap-changing transformers if voltages at certain reference points exceed set thresholds for 15 minutes. Since the voltage fluctuations on 28 April were faster than 15 minutes, and since the reference points only contained a subset of the nodes in the system, no order to block tap-changing transformers was issued.

Regarding protection settings in general, the investigation further revealed that under Spain's national network codes, the upper limit of the allowed voltage operating range (435 kV) was reaching the threshold at which some pre-NC RfG generators were able to immediately disconnect (435 kV). This means that no margin existed. The margin could be restored by either reducing the allowed operating range or by raising the disconnection requirements for pre-RfG generators. For post-RfG generators in Spain, the threshold was 440 kV and the margin was 5 kV. The margin could be increased by reducing the allowed operating range.

The Expert Panel issued a related recommendation in Chapter 9.

4.1.2.6.2 The HVDC

The behaviour of the 2 × 1000 MW VSC-type HVDC between Spain and France is discussed in Chapters 2 and 3 of the report.

The HVDC's reactive power contribution is described in Chapter 2.7. Following the blackout in Iberia, the HVDC system between Spain and France disconnected and active power transfer was lost. At the same time, the VSC HVDC had also been providing voltage control at both converter stations independently.

However, together with the loss of the DC connection, reactive power support (voltage control) was lost at both ends (including in the station at the French side, where the grid otherwise remained in operation).

The Expert Panel issued a related recommendation in Chapter 9.

4.1.2.7 Summary of conclusions

The following table summarises the capabilities and implementation aspects regarding reactive power provisions of various assets. It assesses the ability of the reactive power assets to respond to an approximately 1-minute-long voltage rise (aiming to reflect the period 12:32:15 – 12:33:15; see Figure 4-8), both in terms of technical capability, and the actual response on 28 April, given the implementation and context.

Reactive power asset	Technical capability to respond to a ~1-min voltage rise	Implementation in Spain on the day	Expectation to respond to a ~1-min voltage rise	Explanation (Remarks)	Related recommendations
RES generators	Technically possible	Fixed power factor	Not possible	In fixed power factor mode, reactive power output is based on active power output, not on voltage	Reco. 1 Reco. 2
Conventional generators, default framework (PO 7.4)	Technically possible	Reactive power output set by generator based on formula in PO 7.4 (2000)	Explicit response speed not defined PO 7.4 (and must be inferred from the combination of the validation formula in PO 7.4 and the structural data provided by generators at commissioning or updated after a significant change	» Non-fulfilment was allowed in 25 % of each hour » No close-to-real-time monitoring or systemic real-time feedback between TSO and generator	
Conventional generators, optional framework (voltage setpoint project)	Technically possible	Real-time voltage setpoint received from TSO	Yes (if technically feasible)	*(Participation not mandatory; most conventional generators subject to PO 7.4)*	
Shunt reactors	Technically possible	Manually operated	Limited	Manual operation requires lead time for decision-making	Reco. 2 Reco. 3
STATCOM	Technically possible	One in operation	Possible	Depending on configuration	Reco. 2

Table 4-2: Summary of reactive power contribution by various assets.

The following two figures compare the actual (realised) absorption of reactive power against the potential of each asset[57] before the voltage rise (12:32) and at approximately 1 minute into the voltage rise (12:33).

Figure Note: Actual reactive power absorption is the actual SCADA measurement in case of generators and the reactive power capacity of connected shunt reactors. "Remaining capacity" is the difference between the actual absorption and the "reference", where:

- For PO 7.4 generators, the reference is calculated by the System Operator based on provisions in PO 7.4 (2000). The data points (not shown in the chart) are 74 MVAr for actual and 146 MVAr for remaining capacity.
- For RES generators, the reference takes the total instantaneous active power of >5 MW RES and assumes a power factor of 20 % inductive was set for all these generators.
- For shunt reactors, the reference is the total installed capacity of shunt reactors (both connected and disconnected).

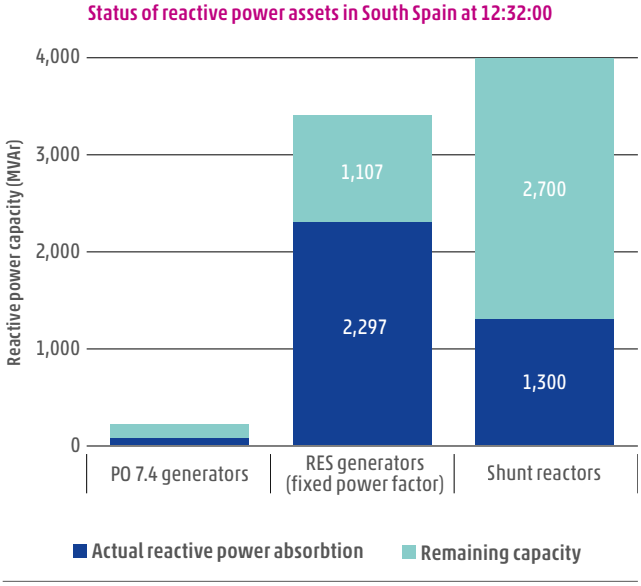


Figure 4-2: Status of voltage control by selected reactive power assets at 12:32:00.

Figure Note: Actual reactive power absorption is the actual SCADA measurement in case of generators and the reactive power capacity of connected shunt reactors. "Remaining capacity" is the difference between the actual absorption and the "reference", where:

- For PO 7.4 generators, the reference is calculated by the System Operator based on provisions in PO 7.4 (2000).
- For RES generators, the reference takes the total instantaneous active power of >5 MW RES and assumes a power factor of 20 % inductive was set for all these generators.
- For shunt reactors, the reference is the total installed capacity of shunt reactors (for both connected and disconnected).

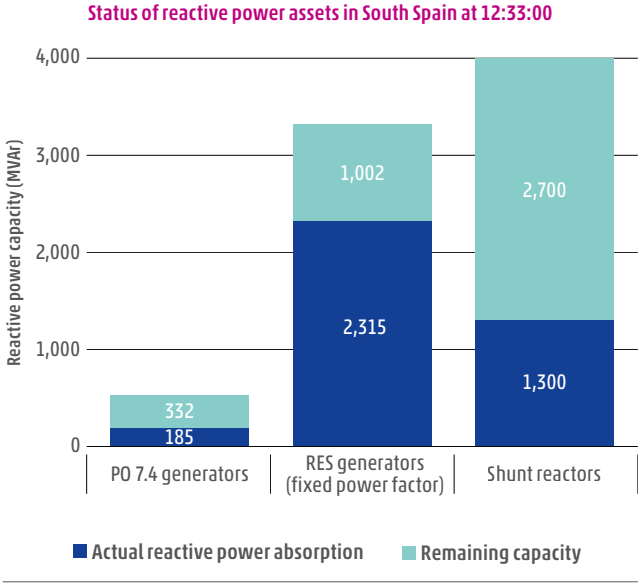


Figure 4-3: Status of voltage control by selected reactive power assets at 12:33:00.

The figures (considering the underlying assumptions), show that of the three reactive power assets shown for Spain, the largest remaining capacity for voltage control, in absolute capacity terms, is available in shunt reactors. At 12:33:00, the not realised reactive power potential of Operational Procedure PO 7.4 generators is numerically equal to connecting three 150 MVAr shunt reactors, while the remaining capacity of RES corresponds to seven such shunt reactors.

57 See assumptions regarding potential in the note under the figure.

As discussed in this section, the operation of shunt reactors in Spain was manual. Given the large number of installed shunt reactors (around 90 across Spain), a fast utilisation of these assets using manual decisions is prohibitively difficult. Achieving a more effective utilisation of available shunt reactors to address system voltage needs (e.g. quickly connecting the most relevant[58] assets when necessary) would thus require either automatic or semi-automatic systems.

During the investigation, the French TSO RTE indicated that in France, automatic utilisation of shunt reactors (and capacitors) is implemented for a large share of their reactors (around 50 % of the assets). These shunt reactors and capacitors can be operated either:

- by manual remote control, or
- automatically, whereby the assets connect/disconnect automatically at preset voltage thresholds (at or close to the normal voltage limits), based on a local voltage measurement and with a response time of 5 – 10 seconds.

While not implying that this is a universal solution, RTE further explained that in its implementation, there are two operating modes on the automated assets: a regulation mode and a safety mode. The regulation mode is activated from the control centre, while the safety mode is always active and cannot be deactivated. More specifically:

- In regulation mode, the shunt reactor's activation thresholds are within the normal voltage range, which for RTE is 200 – 245 kV (e.g., a shunt reactor would connect at 244 kV).
- In safety mode, a shunt reactor would connect when the voltage is outside the normal range (e.g., a shunt reactor would connect at 248 kV).

Furthermore, the settings are adapted to each case after dynamic studies. When there are several assets in the same electrical substation or nearby, the settings must prevent simultaneous switching on/off.

The following figure shows a schematic of the functioning of the automatic asset activation system.

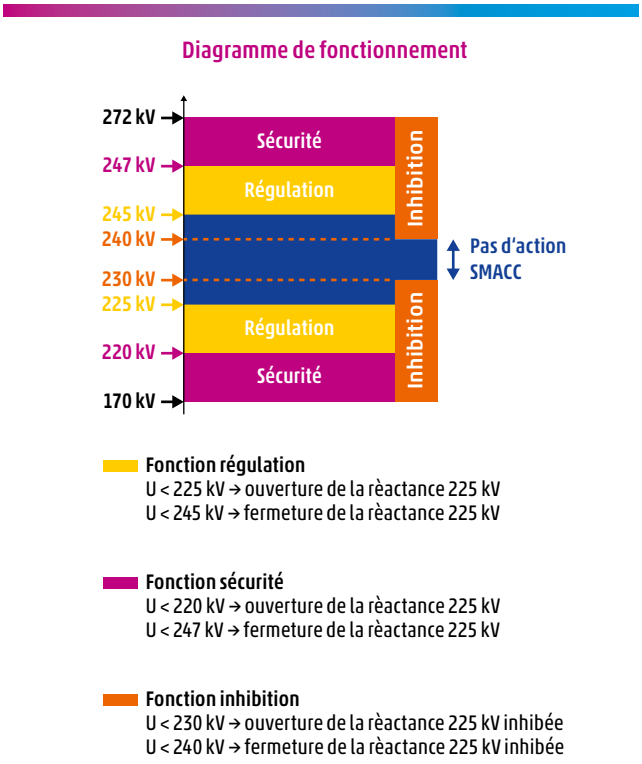


Figure 4-4: Schematic of the functioning of the automatic asset activation system. (Source: RTE)

The connection status of shunt reactors in Spain was also a feature in the system separation incident in 2021.[59] Before the split occurred (due to cascaded trips of transmission network elements that led to an active power imbalance of 2.5 GW in the Iberian Peninsula), some shunt reactors in Spain had been in a disconnected state due to low voltages, and they remained in a disconnected state for some time after the system separation, when overvoltages occurred.[60] No recommendation regarding shunt reactors was issued. After the voltage collapse in South-East Europe in 2024, a recommendation[61] was issued to analyse an automatic emergency control concept for shunt capacitors and inductors during high or low voltage situations.

The relevant new recommendation can be found in Chapter 9 of this report.

58 In the scope of this investigation, Coreso performed a static sensitivity analysis to assess voltage sensitivity resulting from shunt reactor connection at node 400 kV TS1—Badajoz, where the second overvoltage disconnection occurred on 28 April 2025 at 12:33:16. At that time, the last SCADA voltage measurement before disconnection was 434 kV. The analysis (subject to limits and caveats) indicated a voltage decrease of around 3 – 4 kV for each shunt reactor connected. (This analysis is referenced in Section 4.1.5).

59 **Final report** on the Continental Europe Synchronous Area separation on 24 July 2021.

60 Section 4.2 (p. 26 – 27) of the **report** summarising the connection/disconnection actions, and Section 4.3 (p. 32) of the **report**.

61 **Final report** on the grid incident in South-East Europe on 21 June 2024. See Section 9.2.7 (p. 137) of the **report**.

With regard to reactive power contribution by synchronous conventional generators and RES generators, the Expert Panel observed that different approaches were implemented in other control areas. In Portugal,[62] synchronous generators by default receive a voltage or reactive power setpoint by phone directly from the TSO, and are required to deliver reactive power in accordance with the setpoint, within restrictions defined by a P-Q area and a power factor range. Hydro power plants provide the service not only in generation mode but also in pumping/synchronous power mode. One hydro power plant is specifically contracted to provide voltage control even when not producing active power – it functions as a synchronous compensator, and the service uses a reactive power setpoint. RES generators in Portugal (solar power plants with installed capacity above 1 MW, as well as newer wind power plants) receive real-time setpoints from the TSO's SCADA, most commonly a voltage setpoint. Notably, solar PV generators provide voltage control service both during the day and at night (i.e. even in the absence of the primary energy source).

The relevance of providing comprehensive European cross-border guidance on voltage control practices is discussed in a recommendation in Chapter 9.

Regarding the reactive power potential of inverter-based RES, the figures above assume operation within the constraints of a fixed-power-factor control mode. If a different voltage control mode is implemented, a higher potential might be achieved, as the reactive power output would not be directly linked to the instantaneous active power output but rather to the capabilities of the inverter. Such technical decisions also enable further implementation options, such as e.g. the night use of solar PV for voltage control and/or the possibility of inverter-based assets operating as STATCOMs, as described in a recommendation in Chapter 9.

Area of attention: The Expert Panel suggests assessing reactive power flows between the distribution and transmission grids, while also considering the control and design of the relevant tap-changing transformers.

Area of attention: The Expert Panel suggests that distribution operators consider the most appropriate voltage control mode from distribution-connected generators. The relevance of this suggestion depends on local specifics and is relevant insofar as enabled by the applicable requirements.

4.1.3 Voltage management

This section addresses concepts related to voltage management from the operational perspective. The section discusses:

- The link between operational limits and disconnection limits
- The handling of alarms
- Managing reactive power capabilities to cope with expected reactive power fluctuations

62 See Section 2.6.5.2.

4.1.3.1 Operational limits and disconnection thresholds

The range of voltages expected in a power grid (the operating voltage range) generally considers the voltage withstand capabilities of system users and includes appropriate safety margins.

As described in Section 3.2.1.1, the requirements applicable in Spain enabled the generators connected to the 400 kV network in Spain to disconnect when the voltage exceeded 435 kV (older connection dates) or 440 kV (newer connection dates).

On the day of the incident, the allowed upper limit of the operating voltage range in Spain was 435 kV.[63] This meant that the **safety margin between the allowed voltage operating range and the voltages at which generators could disconnect was low or non-existent.**

The Spanish Operational Procedure P.O. 1.4 (1998)[64] and the exceptions introduced in European regulations specifically for Spain enabled the possibility for the Spanish TSO to operate the transmission grid at 435 kV (this is described in Section 2.6.1). In other EU Member States, the allowed operating range is up to 420 kV.

The Expert Panel identified the limited safety margin as a contributing factor to the incident. A related recommendation is included in Chapter 9.

4.1.3.2 Alarm handling

Detailed information on alarm handling at all levels of the system is included in Section 4.5.

The overvoltage alarm thresholds at 400 kV nodes in Spain were, depending on the node, at thresholds of 420 kV, 430 kV, or 435 kV. For RE, voltage-related (curative or close-to-real-time preventive) remedial actions were not defined in an operator handbook.[65] RE indicated that the actions are nevertheless known by all operators due to continuous training programs.

There are alarm thresholds defined for 220 kV nodes (at values of 240, 235, and 246 kV). However, for global monitoring of the system, only 400 kV pilot nodes are used (i.e. these actually generate specific voltage alarms).

Between 12:00 and 12:30, 1,026 alarms below the low-voltage alarm thresholds (375 – 395 kV), 78 of which were not in RE's transmission network, and 137 alarms above the high-voltage alarm thresholds (420 – 435 kV), eight of which were not in RE's transmission network, were triggered in RE SCADA, in part due to voltage oscillations.

During the voltage rise between 12:25 and 12:29, 24 430 kV alarms were triggered and five shunt reactors were connected, after which the voltage was mostly below 420 kV by 12:30.

After the 12:30 mark, the first 430 kV alarm was triggered at 12:33:01. This was only 15 seconds before the second disconnection (Event 4 in Table 3.1) and the overvoltage cascade, meaning that manual remedial actions such as disconnecting lines or connecting shunt reactors were not possible.

There was one 435 kV alarm at 12:27:46. No other 435 kV alarm was triggered before the blackout.

The SCADA alarm system usually does not distinguish voltage values in the three phases and therefore provided no input to operators regarding the voltage values of individual phases.[66]

63 While the allowed range was 435 kV, RE indicated that the preferred operation was below 420 kV.

64 Resolution of 30 July 1998, of the Secretary of State for Energy and Mineral Resources, which approves a set of technical and instrumental procedures necessary to carry out the adequate technical management of the electrical system.

65 For comparison, in REN, such actions were defined in a handbook.

66 For example, at 12:33:11, one of the zero-to-phase voltages was around 5 kV higher than the positive sequence voltage (435 kV in Phase B-to-ground, vs 430 kV in positive sequence). At 12:33:11, the voltage measurement in PMU ARN4ALD (close to the node where disconnections occurred 5 seconds later) reached 430 kV in positive sequence voltage. However, at this time, one of the zero-to-phase voltages (Phase B) was already at 435 kV, as voltages in Phase B were notably higher than in the other two phases. The asymmetry was present throughout 12:00 – 12:33 (differences between two phases up to 9 kV).

4.1.3.3 Managing reactive power capabilities to cope with expected reactive power fluctuations

A combination of factors affecting the voltage control capabilities of the power system occurred simultaneously on 28 April.

As discussed in Section 4.1.2.1, shunt reactors in Spain were operated manually, meaning that there was a decision-making lead time before they could be (re-) connected.

As discussed in Section 4.1.2.3, the RES generators followed a fixed-power-factor control mode and thus were unable to respond to voltage fluctuations.

As discussed in Section 4.1.2.4, the synchronous conventional power plants were following the default Operational Procedure PO 7.4 framework, while one plant voluntarily (temporarily) took part in a direct voltage control set-point pilot project.

At 12:30, the following conventional power plants were connected in the southwestern (Andalusia, Castilla-La Mancha, Extremadura) part of Spain:

- One nuclear power unit with a total installed capacity of around 1,000 MW
- Two combined-cycle power units with a total installed capacity of around 800 MW
- One hydro unit with a total installed capacity of around 70 MW

The rest of the generation mix in this region was inverter-based.

After the incident, RE noted that some conventional power plants falling under the Operational Procedure PO 7.4 framework did not perform as expected. As per the design of the Operational Procedure PO 7.4 regime (generators follow Operational Procedure PO 7.4 voltage references), no real-time setpoint was sent directly to the generators. Hence, RE was unable to instruct conventional generators to change their reactive power behaviour in real time. RE can instruct a change to the voltage reference in real time via a telephone call. One such call did take place on 28 April (for a unit outside the southwest-centre region).

On the other hand, for one generator investigated in more detail by the Expert Panel, the generator indicated that on the assessed past day, when its reactive power absorption was higher, it had received advanced instructions from RE to increase reactive power absorption. However, no such instruction was sent to this generator on 28 April.

Due to the oscillatory incidents in the power system (oscillations), the system operator dispatched an additional conventional unit in Andalucia in real time to increase the combined cycle generation equipped with power system stabilisers, aiming to increase damping of additional oscillations that could have appeared after these two oscillatory incidents. Although the connection order was issued at 12:26, the start-up time of the unit (and for combined-cycled units[67] in general, ranging from 2 to 8 hours depending on the shutdown time) did not allow sufficient time to connect to the system, as the incident occurred before the start-up.

As discussed in Section 4.1.2.1, only one STATCOM was in operation in Spain that day, while another was performing its test before commissioning.

An additional factor regarding the voltage control management, as mentioned in Section 4.1.2.6.1, was that some renewable generation disconnections in Spain occurred faster than expected (while the voltage at the connection point was inside the limit). As these plants operated with a fixed power factor absorbing reactive power, these disconnections added to the increase in the power system voltage. Another factor involving voltage control management was that a certain amount of domestic PV disconnected after each oscillation and reconnected a few minutes later, causing some impact on voltages. This is discussed in more details in chapter 4.3.

The combination of these factors meant that the system's reactive power capabilities on the day were unable to absorb the voltage rise after 12:32.

The Expert Panel issued a relevant recommendation in Chapter 9.

67 Power system stabilisers in Spain have been declared and adjusted in cooperation with Red Eléctrica in most combined cycle units, in one nuclear power unit, and in a hydro-power plant in the Galician region.

4.1.3.4 Summary of conclusions

Topic	Related recommendation
Operational limits and disconnection thresholds	Recommendation 4
Managing reactive power capabilities	Recommendation 2

Table 4-3: Summary of conclusions

Area of attention/improvement: The Expert Panel suggests that TSOs make meaningful use of information on the reactive power control capabilities of power-generating modules to inform their operational planning process.

4.1.4 Sequence of events in voltage control

This subsection provides a detailed explanation of the voltage rise on 28 April, linking it to several reactive-power-related factors discussed above and considering the conditions from a previous day. The subsection builds on the related section in Chapter 2 and adds data where relevant.

4.1.4.1 Voltage control on 28 April

This subsection examines voltage conditions on 28 April from 09:00 onwards.

4.1.4.1.1 Morning (09:00–12:00)

Voltages were fluctuating on the morning of 28 April. At 10:49:44, voltage in the Olmedilla node reached 433 kV. Between 10:40 and 10:50, Spain's total commercial net position decreased by 0.8 GW.

At 11:47:18, voltage in Olmedilla reached 435.7 kV, for a few seconds, exceeding the allowed operational range[68] in 0.7 kV. Between 11:40 and 11:50, the total commercial net position of Spain decreased by 0.8 GW.

68 This information complements section 1.2 of the factual report, which stated that voltages remained fully within limits throughout the morning.

4.1.4.1.2 Half-hour after 12:00

In the period 12:00:00 – 12:25:00, voltages oscillated with amplitudes of up to 30 kV peak-to-peak (at the 400 kV level), during two notable periods of oscillation. No voltage values were outside the allowed range. From 12:00 to 12:20, six lines were connected to damp oscillations.

In the period 12:20:00 – 12:25:00, during one of the oscillation episodes, three shunt reactors (a total of 400 MVAr) were disconnected due to low voltage swings.

In addition, four 400 kV overhead lines in Spain, which were open for voltage control, were connected to increase damping and reduce the oscillations, according to the bilateral "oscillation and low damping" mitigation protocol.

In the period 12:25 – 12:29, after the second oscillation, a rise in voltage occurred, and five shunt reactors (total 700 MVAr) were connected, two of which were in Andalucia. No voltage out of range was reached.

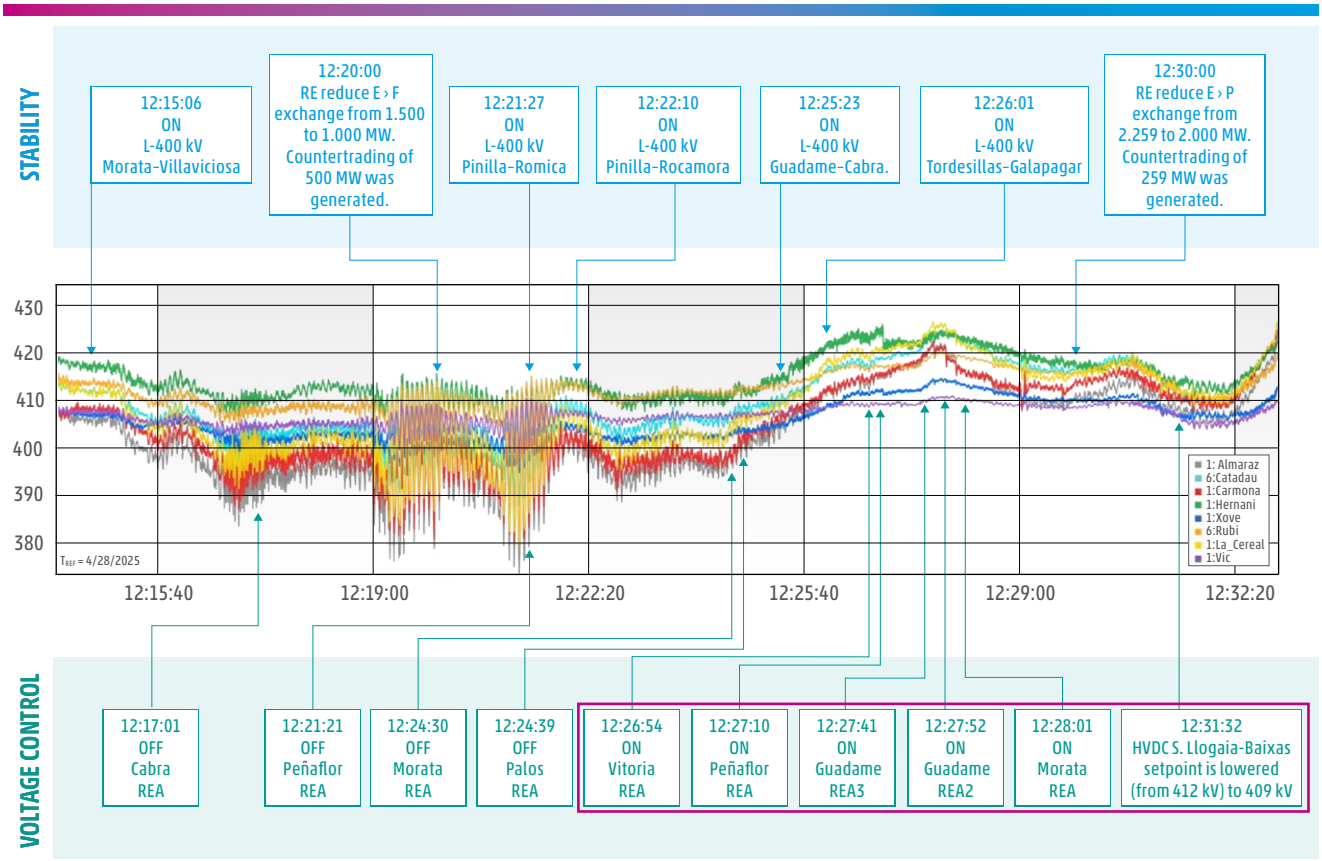


Figure 4-5: Summary of voltage control and stability events from 12:15 onwards.

4.1.4.1.3 From around 12:30 to the blackout

4.1.4.1.3.1 Active power setpoint changes

At 12:30, there was a 0.4 – 0.5 GW change in generation schedules expected, reflected in a scheduled reduction of export from Spain to Portugal of around 0.4 – 0.5 GW.[69] Furthermore, from 12:29 onwards, there was downwards aFRR demand in Spain of up to 0.4 GW.[70]

As described in Section 4.1.2.3, RES units in Spain followed a fixed power factor. Therefore, each change of active power setpoint meant a change in reactive power injection, either proportionally (capacitive power factor) or negatively proportionally (inductive power factor).[71]

As most RES units followed an inductive power factor, downwards changes in active power injection meant an decrease in their reactive power absorption.

In addition, changed active power setpoints of certain generators also changed patterns of flows within the bidding zone, and hence additionally affected reactive power conditions through a changed loading of various transmission lines.

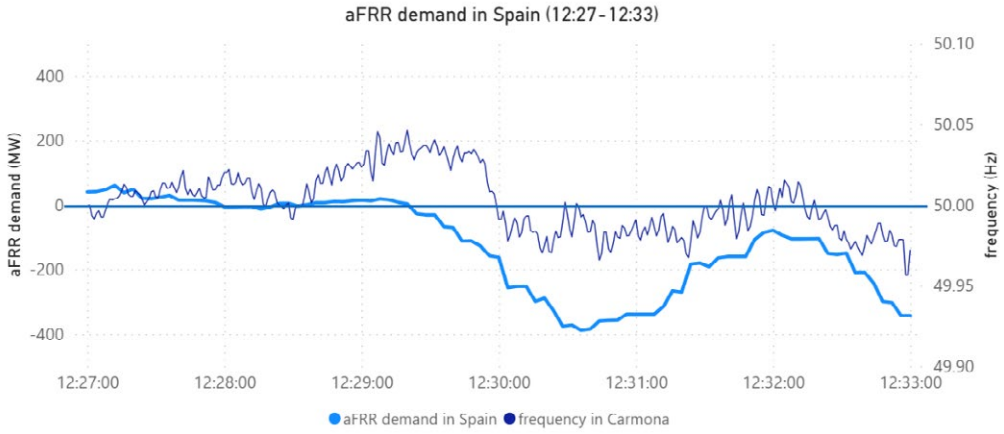


Figure 4-6: aFRR demand in Spain.

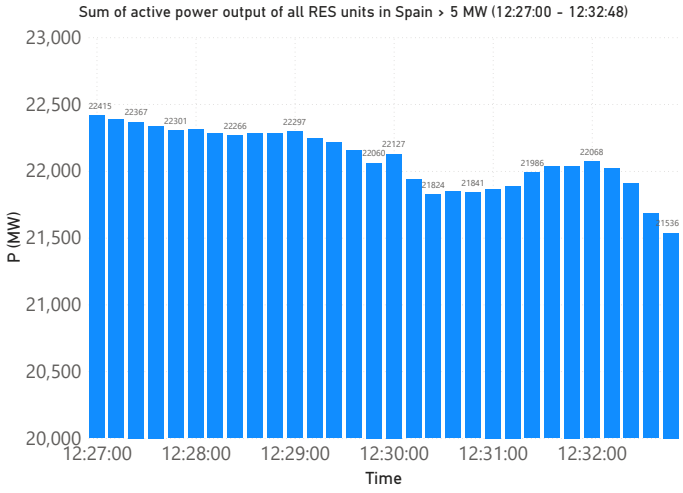


Figure 4-7: Sum of active power output of >5 MW RES generators in Spain.

69 See Figure 2-24 in Section 2.3.4.

70 See Figure 4-6 above and the full figure in Section 2.3.4.

71 In Figure 4-11, the change of active and reactive setpoints is depicted for three selected power plants in 12:32:00 – 12:32:48.

At 12:32:00, voltage was within ranges.[72] As seen in Figure 4-7 above, the output of RES generators decreased by around 0.9 GW from 12:27 to 12:32:48. In particular, around 0.5 GW of that decrease occurred from 12:32:00 to 12:32:48, concurrent with an increase in downwards aFRR demand, as seen in Figure 4-6.[73] Furthermore, in this period, the 0.2 GW events described in Chapter 3 as 2a and 2b also were subject to decreasing setpoints, and their corresponding SCADA data showed a rapid decrease to zero infeed.

Per the portfolio-based bidding in aFRR, it was not possible to know in real time which exact units would change setpoints in response to aFRR demand.

4.1.4.1.3.2 Voltage rise

From 12:32:00 onwards, concurrent with the 0.5 GW change in RES generator output, voltages in numerous nodes increased.

The following chart shows voltages across 400 kV PMUs, with a particularly steep rise registered in PMU TAB4LIT (Andalusia, southern Spain), coloured black in the figure.

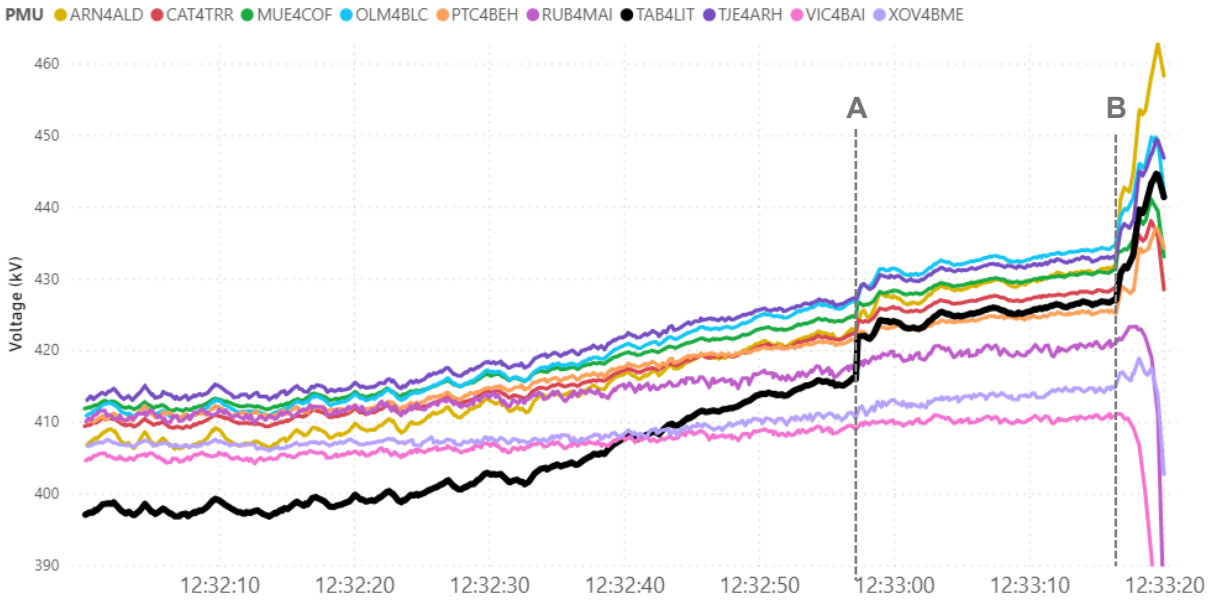


Figure 4-8: Voltages in a set of 400 kV PMUs.

72 409.1 kV at Carmona, 397.1 kV at Tabernas, and 405.4 kV at Santa Llogaia. In the period 12:29:00–12:32:00, the voltage was largely decreasing.

73 It is noted that in some periods, downwards aFRR demand in Spain is present while frequency is below 50 Hz.

4.1.4.1.3.3 Evolution of event "A"

A substation close to PMU TAB4LIT is 400 kV TS1–Granada located in Andalusia. The following figure focuses on the voltage rise in this substation. It was in this substation that the first identified overvoltage disconnection occurred at 12:32:57 (marked "A" in the chart above; corresponds to Event 3 in Table 3.1).

Figure 4-9 shows voltage, active power, and reactive power measured in the substation from 12:00:00 to 12:32:56. Figure 4-10 shows that in the minute before the disconnection, the voltage measured in the substation increased from 400.6 kV at 12:32:12 to 417.2 kV at 12:32:56, i.e. at 22.6 kV/min.

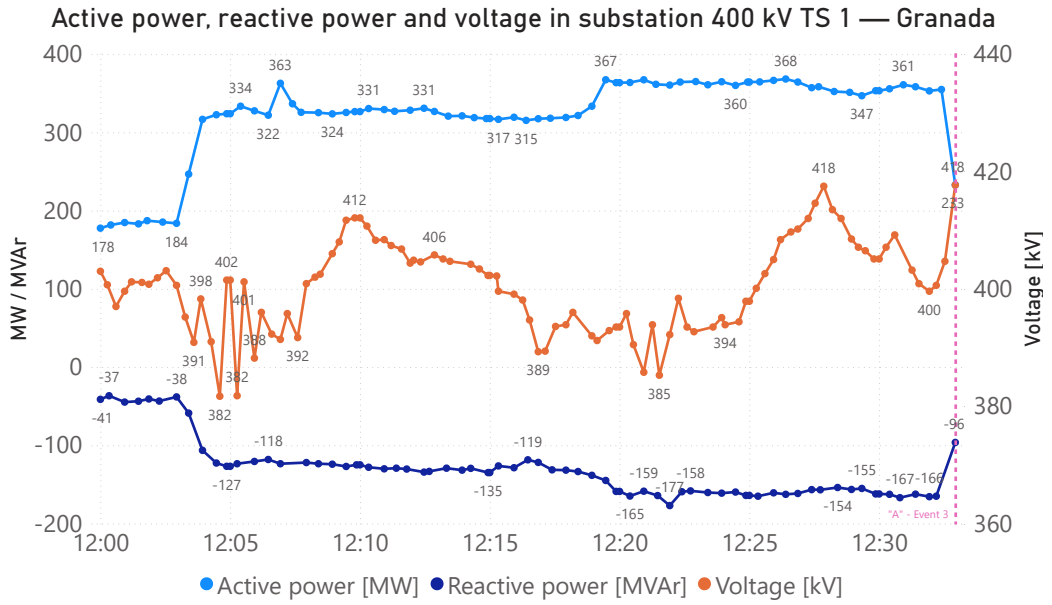


Figure 4-9: P, Q, and V in 400 kV TS 1–Granada, 12:00:00 -12:32:56, SCADA data.

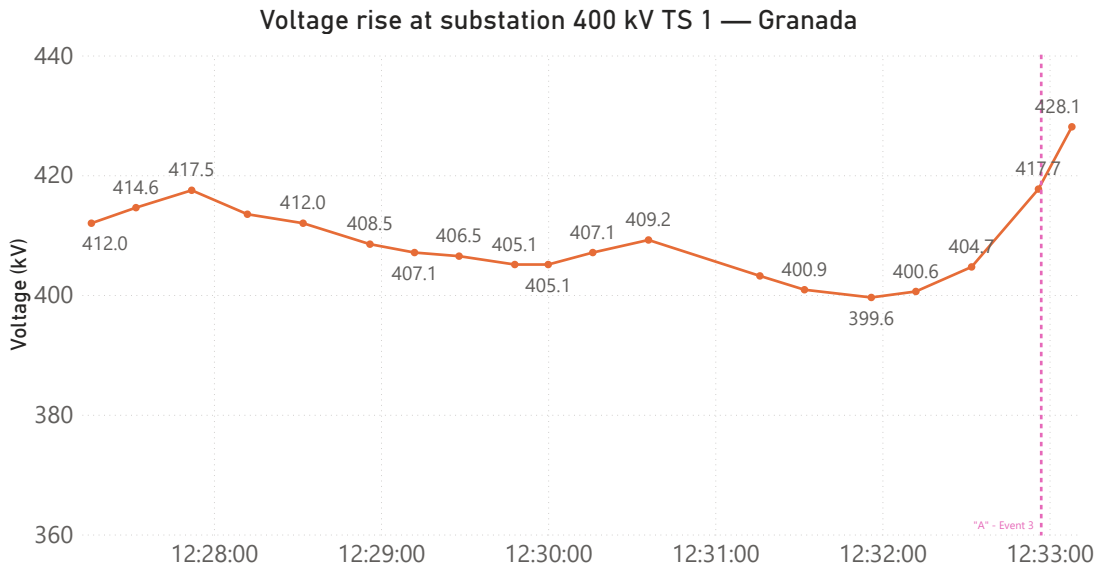


Figure 4-10: Voltage in 400 kV TS 1–Granada, zoomed in to 12:27:16 -12:32:56, SCADA data.

During the period of this voltage rise, the reactive power in the substation rose from –165 MVAr to –96 MVAr. The slope of the reactive power rise was +94 MVAr/min. The ratio of voltage change vs reactive power change was a 0.24 kV increase per 1 MVAr, showing the sensitivity of the voltage change to a change in injection (or to other factors not captured here). This stability aspect could be subject to further academic and practical research on phenomena occurring in systems with a high share of inverter-based resources.

The Expert Panel has identified three example generators in 400 kV TS 1—Granada (total capacity 150 MW) which, between 12:32:24 and 12:32:48, ramp down, changing their active power output [74] by 78.8 MW (from 123.4 to 41.6 MW). At the same time, due to the fixed-inductive-power-factor control mode, their reactive power absorption decreased by 33.3 MVAr (from –35.5 MVAr to –2.2 MVAr).

Figure 4-11 shows both the active and reactive power output of the three units and the voltage in 400 kV TS 1—Granada.

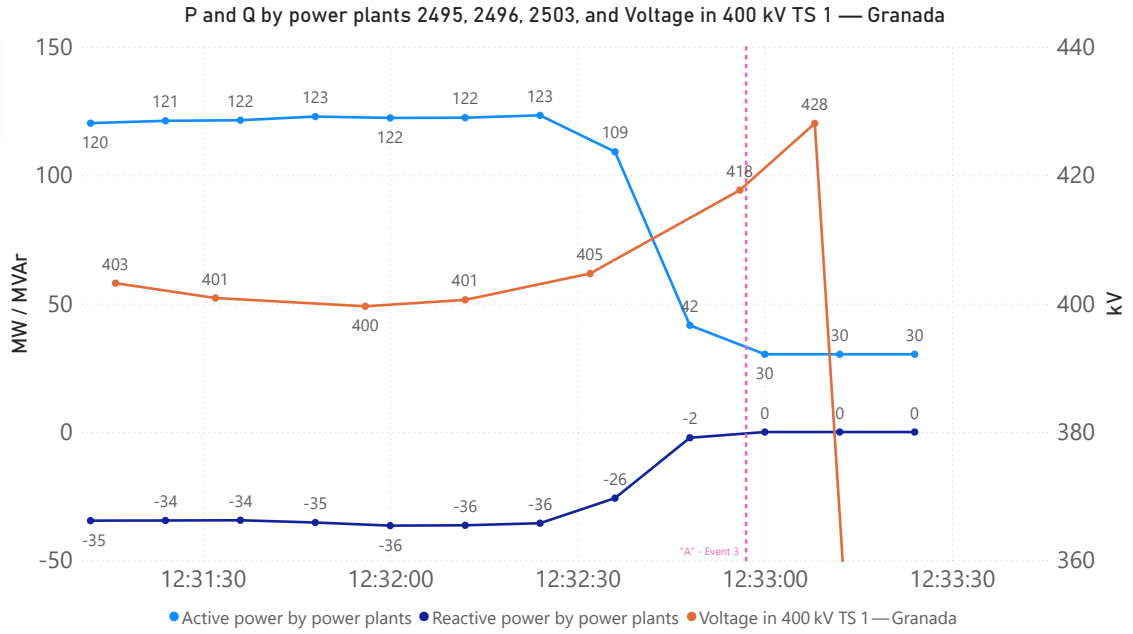


Figure 4-11: P, Q, of three RES units, and V in 400 kV TS 1—Granada, SCADA data.

At 12:32:57, the 220 kV side of the 400/220 kV transformer in 400 kV TS 1—Granada tripped.[75] The voltage in the substation registered before the trip was 417.9 kV (within the operational margins), whereas the p.u. voltage on the other side of the transformer was higher, for reasons described in Section 4.1.2.6.

The trip led to the loss of all downstream generation in 400 kV TS 1—Granada, and the loss of the remaining reactive power absorption by the connected generators (the total reactive absorption in 400 kV TS 1—Granada was –96 MVAr at 12:32:56 and 0 MVar in the next time stamp at 12:33:00).[76]

74 According to RE SCADA data.

75 See Section 3.2.1.3.1.

76 This is also seen as the vertical surge in voltage in Figure 4-8 above.

4.1.4.1.3.4 Evolution of event "B"

The rise in voltage continued for around 19 seconds, before another set of generators disconnected at substation 400 kV TS 1—Badajoz (corresponds to Event 4a in Table 3.1). The voltage rise in the substation is shown in Figure 4-12.

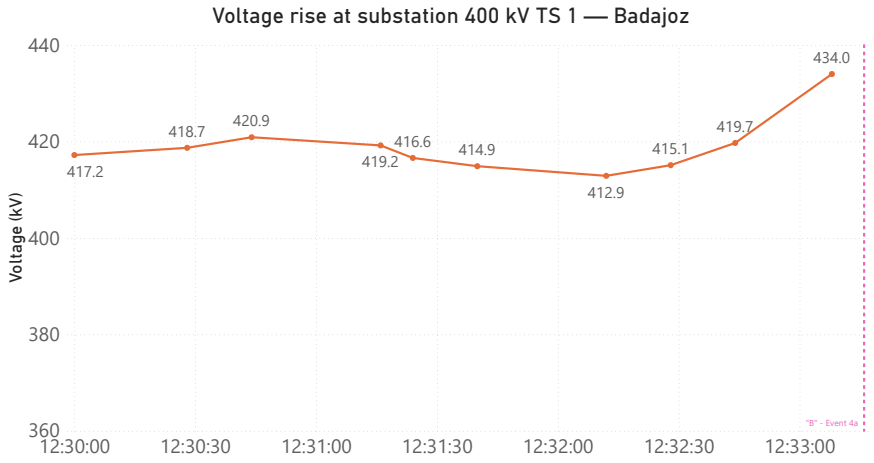


Figure 4-12: Voltage in 400 kV TS 1—Badajoz, SCADA data.

With this disconnection, 165 MVAr of reactive power absorption was lost.

In the closest PMU (ARN4ALD), the voltage at 12:33:16.420 was 431.7 kV when expressed as the positive symmetrical component of all three phases. However, the phase-to-ground voltage B was 436.9 kV.

The reading of voltage in Phase B was around 9 kV higher than Phase A, reflecting the asymmetry of the voltage system or measurement errors.[77] This is relevant if overvoltage protection triggers based on any of the three voltages. Figure 4-13 shows the phase voltages measured in PMU ARN4ALD.

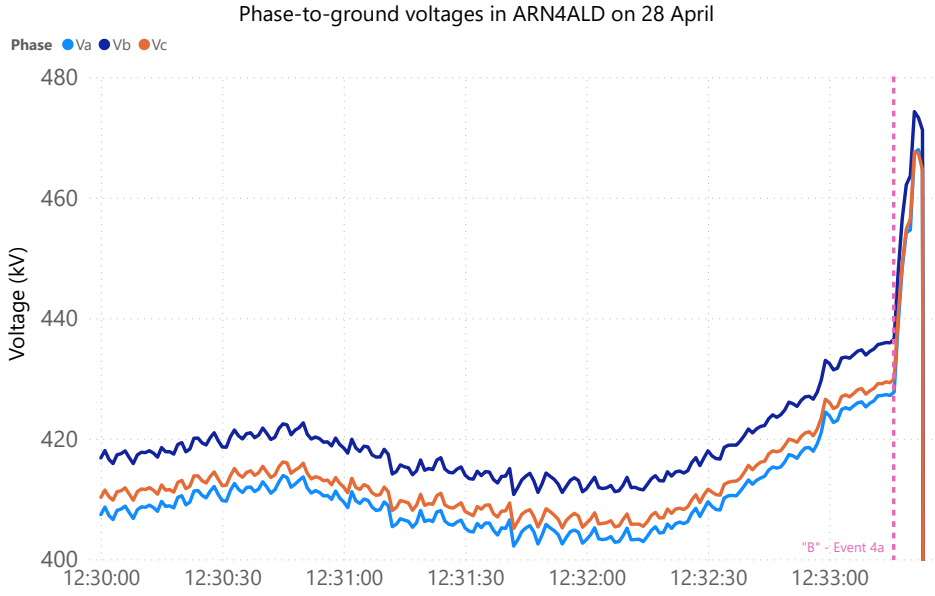


Figure 4-13: Phase-to-ground voltages in ARN4ALD, PMU data.

77 See also Section 3.1.1, which assigns 3 kV of the difference to a measurement error in Phase B.

4.1.4.1.3.5 After event "B"

The voltage control section did not investigate the voltage conditions after Event B (in the 2.5 seconds from Event B until 12:32:19, 12 further overvoltage disconnections occurred). The details of the disconnections are described in the Disconnections section 3.2.1.3.

A cascade of overvoltage disconnections led to frequency phenomena described in other chapters and, eventually, a blackout occurred.

4.1.4.2 Voltage control in the days preceding 28 April

To better understand the circumstances of the incident, particularly the presence and impact of some of the factors related to the event on 28 April, the Expert Panel also assessed voltage fluctuations that occurred on certain previous days. In particular, the Expert Panel assessed 22 April, when an overvoltage event occurred, alongside changes in generation schedules, cross-border flows, and the connection of conventional power plants and shunt reactors. However, different circumstances meant that an incident of the same scale did not occur and no ICS investigation took place.

On 22 April, overvoltage disconnections of some generation and demand facilities occurred in Spain. At 19:00, the commercial net position of Spain decreased by 1.6 GW, reflecting changes in generation schedules and also manifesting as changes in cross-border flows. There was a co-occurrence of a decrease in solar power (due to the sun going down) and conventional power plants connecting.

At around 18:58, concurrent with the change in schedules, voltages started rising in some nodes of the grid. The voltages rose for around 3 minutes, until 440 kV was reached in Phase B at PMU ARN4ALD at 19:01:30. After that, in some seconds, voltages rose above 450 kV and continued climbing for around half a minute. At the peak (19:02:27), a value of 452 kV (up to 455 kV in Phase B) was registered in PMU ARN4ALD. Around 3 minutes after 19:00, the conventional power plants reached technical minimum power and became able to control the voltage.[78] By 19:05, the situation was remedied, as RE connected 22 shunt reactors.

It is worth noting that on 22 April, no significant oscillation and no low-voltage situation was observed before the voltage rise (as it was on 28 April).

Figure 4-14 shows the positive symmetrical component voltages in a set of PMUs.

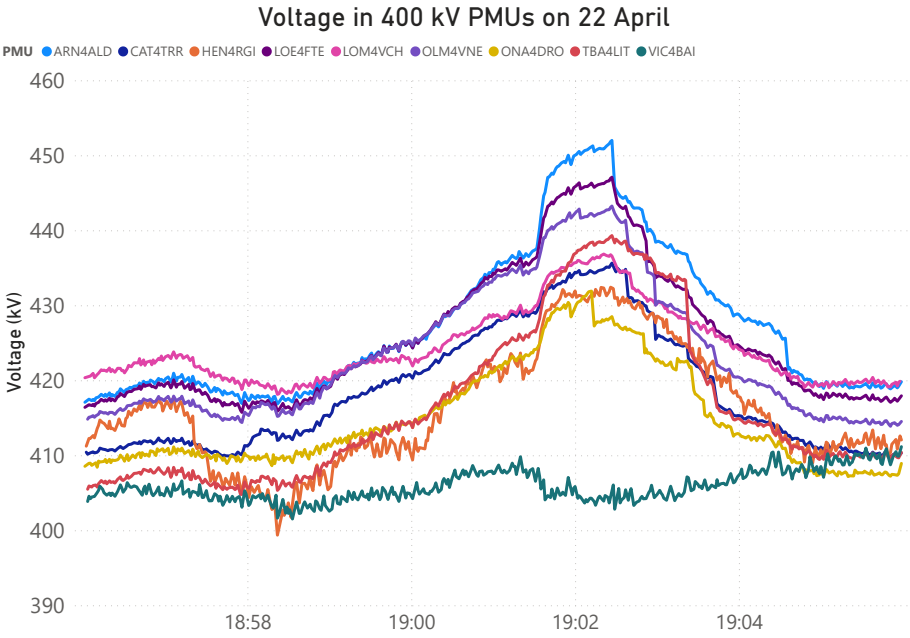


Figure 4-14: Voltage in a set of PMUs on 22 April.

78 A lesson from the event was that a longer overlap is needed, meaning that the ramping time of conventional units should be taken into account, so that connecting these resources is considered earlier in operational planning.

Notably, phase asymmetry was present in the measurements[79] in PMU ARN4ALD (as on 28 April) – the phase-to-ground voltage reading of Phase B was around 5 kV above the other two phases throughout the event, as seen in Figure 4-15.

While phase asymmetry depends on several factors and is not extraordinary in itself, it is considered here in the context of the possibility that overvoltage protection was triggered based on a single phase-to-ground or phase-to-phase voltage.

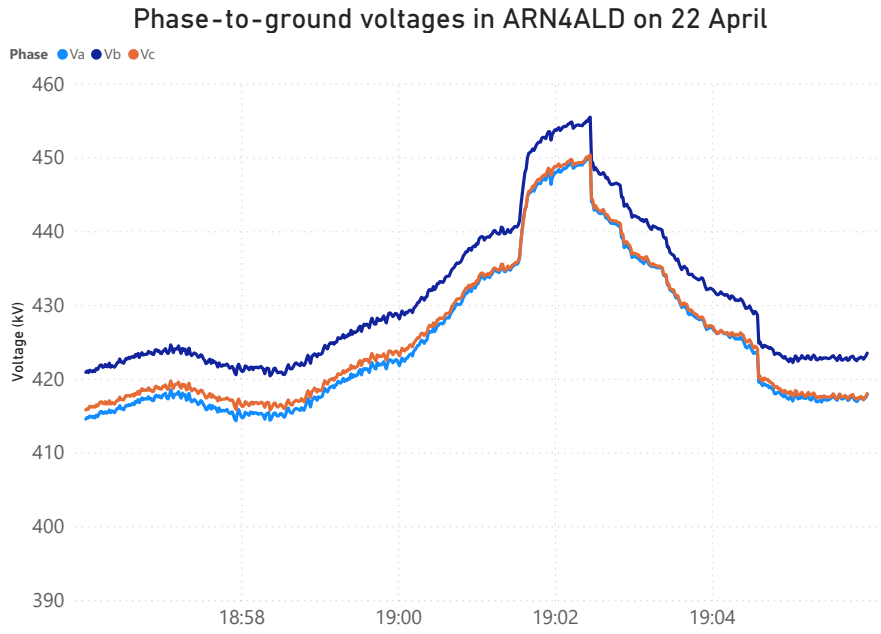


Figure 4-15: Phase-to-ground voltages in ARN4ALD on 22 April.

While some overvoltage disconnections of generators occurred on 22 April, the conditions on the day did not lead to an overvoltage cascade leading to a blackout.

79 See also Section 3.1.1, which assigns 3 kV of the difference to a measurement error in Phase B.

4.1.5 Sensitivity analyses

In the scope of the voltage control workstream, certain sensitivity analyses were performed by Coreso on real-time snapshot grid models (RTSN) shared by TSOs. Where possible, sensitivity analyses were performed separately in both UCTE and CGMES formats.

Summary:

» **1.** The analyses were performed with the following limitations:

- **a.** RTSNs are steady-state representations of the grid, meaning that dynamic analyses are not possible.
- **b.** UCTE models are generally less relevant for voltage analyses due to the format's limitations
- **c.** Unavailability of fully comprehensive RTSNs in a commonly agreed-upon modelling format in the Continental Europe Synchronous Area
- **d.** Some elements are occasionally modelled differently in RTSNs compared to DACF and IDCF grid models; therefore, Coreso experienced some issues involving the modelling of some "complex" grid elements, which had to be addressed with the concerned TSOs.
- **e.** The three provided RTSN IGMs from the SWE TSOs were merged for time stamp 12:30 with the following considerations:
  - i. For UCTE, the RE IGMs are not synchronised, meaning that the 12:23 snapshot was used as the closest to the 12:30 time stamp;[80]
  - ii. For CGMES, the REN IGM was leading to voltage divergence. Instead, the REN IDCF file was used.

» **2.** Outcome of the sensitivity analyses:

- **a.** In the base case, some differences in voltage values between RTSN and SCADA are observed.
- **b.** Analysis 1: Voltage setpoint 407.5 kV applied to RES (aiming to emulate behaviour in voltage control mode):
  - i. Dispersion of voltages reduces (interquartile range is 6.6 kV in the base case vs 3.6 kV in the sensitivity in UCTE and 8.5 kV vs 7.9 kV in CGMES) in both UCTE and CGMES.
- **c.** Analysis 2: Voltage setpoint 407.5 kV applied to conventional generators:
  - i. Dispersion of voltages reduces (interquartile range 6.6 kV base case, 4.1 kV in the sensitivity)
  - ii. Only performed in UCTE, since conventional generators at CGMES are modelled at the voltage level where they are actually connected (i.e. 30 kV), necessitating remote voltage control, which could not be performed.
- **d.** Analysis 3: Modifying tap position in node 400 kV TS 1—Granada and assessing voltage sensitivity:
  - i. Linear regression results in voltage sensitivities to tap changes of +1.19 kV/ increased tap (for the 220 kV voltage side) and –0.78 kV/ increased tap (400 kV voltage side) (performed in CGMES).
- **e.** Analysis 4: Voltage sensitivity to connecting a shunt reactor in node 400 kV TS 1—Badajoz 400 kV:
  - i. Calculation yielded a 3 – 4 kV change for each 150 MVAr absorption added (0.023 kV/MVAr) in both UCTE and CGMES. In UCTE, the shunt reactors are modelled as an equivalent load.
- **f.** Analysis 5: Modifying P and Q injection in node 400 kV TS 1—Granada and calculating voltage change:
  - i. Max voltage variation calculated at 3 – 4 kV in UCTE and CGMES, substantially less than the real SCADA measurement (16 – 17 kV).
  - ii. In UCTE, voltage rise was predominantly driven by changes in Q, whereas in CGMES, voltage changes occurred similarly after changing either P or Q injection.

80 In addition, in UCTE, some fixes, such as changes in nodes representation, were applied to ensure convergence. In CGMES, RTSN IGM by REN was not included in the RTSN CGM due to voltage divergence, so the IDCF IGM was used.

4.2 Oscillations

4.2.1 Scope of the Chapter

This chapter aims to analyse in detail, using various methodologies and data sources, the oscillatory phenomena that occurred from around 12:00 until the cascading events that led to the blackout. Summarising the main physical concepts allows us to understand, to the greatest extent possible, oscillatory phenomena. Oscillatory phenomena were also observed in the system on 28 April before 12:00, as well as on other days in April 2025, but with a very low amplitude that did not pose a significant risk to the system. This section, however, focuses on 12:00 – 12:30, when two very prominent oscillation episodes occurred.

Setting aside the rigorous scientific definition already reported in Chapter 2, we will focus on the physics of the oscillations observed, which involved both phenomena particularly active in a specific area of the Iberian grid, yet involving a vast geographical area, as well as inter-area oscillations involving the entire Continental Europe (CE) grid.

There is a common denominator in all oscillatory phenomena, which allows us to classify the scenario as more prone to triggering an oscillation; however, it does not guarantee their occurrence, as oscillations can arise from mechanisms associated with chaotic system behaviour.

The following factors can contribute to potential oscillations:

- Generator reactance relative to the electrical system to which they are connected: This can be compared to the "electrical distance" from the system. If particularly high, it causes generators to operate at suboptimal points for controlling oscillations. Reactance depends on the short-circuit power at the connection point, which is influenced by network topology (e.g. it increases when more transmission lines are out of service, or when the ratio of conventional rotating generators delivering short-circuit power to static generators is low).
- The transmission angle between the generators and the grid, which is influenced by system topology
- Operating points (when a large area is under export conditions, the angle spreads) and voltage profiles (which induce under-excitation operation of generators)
- The presence of static and dynamic compensation elements (e.g. reactors, STATCOMs, condensers)
- Dynamic stabilisation features and their tuning (power system stabilisers on rotating machines, HVDC POD, STATCOM POD, static generators POD)
- Eventual IBR converter-driven instability

We will analyse the dynamics of the oscillations in detail, assessing their extent and identifying the most active areas, and confirming the hypotheses and findings through root-mean-square (RMS) simulations, short-circuit power evaluation, and small-signal analysis. These results will inform actions and recommendations aimed at mitigating the occurrence of such phenomena in the future.

4.2.2 Construction of Analytical Framework for Oscillatory Analysis

To emulate the inter-area and 0.63 Hz phenomena, the full Continental Europe Synchronous Area (CE SA) transmission system was considered. A detailed model of the Spanish peninsular electric system (SPES) was used, based on a state estimator static case and supplemented with dynamic models of system generation, HVDC, and FACTS. For the inter-area oscillation study, data from 12:21 were used to represent SPES operating and topology conditions during the event, while the scenario at 12:08[30] was employed to reproduce SPES operating conditions during the 0.63 Hz oscillation. The rest of the CE system was modelled with the "Dynamic model of continental Europe v2". As this model represents a different time horizon and operating point than 28 April 2025, it was adjusted to mimic the operating conditions (generation, load, system topology, etc.) at the time of interest. This adjustment was performed [7] using the data available at 12:00 in the ENTSO-E Transparency Platform, which provides 1-hour resolution data. This hour was selected to reflect operating conditions immediately prior to the oscillations.

This model can represent the system's oscillatory behaviour in terms of electromechanical modes, but the base damping must be properly adjusted. This was achieved by modifying variables outside the Spain and Portugal portions of the system, with the French and remaining CE SA regions calibrated to reproduce the system's damping values at the two studied time stamps on 28 April 2025.

The analyses in this chapter use two complementary methodologies: eigenvalue analysis, which operates in the frequency domain, and RMS simulation, which operates in the phasor domain. Eigenvalue analysis provides information on the existing electromechanical modes in the system and enables the extraction of quantitative information regarding the effect of different actions undertaken by Red Eléctrica to mitigate the oscillations. RMS simulation provides qualitative and illustrative information and helps analyse the 0.63 Hz oscillation, which does not appear in the model, supporting the hypothesis that this oscillation was not a natural mode of the system but rather the result of forced or induced generator behaviour.

30 The procedure for constructing the analytical framework for oscillatory analysis begins with a consistent snapshot, appropriately adapted to represent the grid configuration at the time of the oscillation.

4.2.3 Validation of the Analytical Model

Once the electromechanical model of the network was created, a validation was performed to verify its adequacy in representing oscillatory phenomena.

4.2.3.1 0.2 Hz Inter-Area Oscillation

Regarding the East–Centre–West inter-area oscillation, the validation process considered two different tests:

- Stimulating the system with a perturbation (a "step" of imbalance between load and production)
- Injecting stochastic noise on the loads to reproduce the natural ambient oscillation

The contingency involves simulating the loss of a large load in the CE power system, which triggers the inter-area oscillation. It is relevant to mention that such event did not occur during the 28 April 2025 incident. However this report assumes a significant power deviation, such as a large load disconnection, to generate an RMS simulation with appreciable inter-area oscillations.

With reference to the ambient oscillation mentioned before, a stochastic process associated with several loads – specifically, an Ornstein–Uhlenbeck (OU) process – was used to begin fluctuations. In the literature, the OU process is used to emulate typical frequency variations under normal operating conditions [8]. It is characterised by a mean value, a standard deviation, and an autocorrelation value. Typical standard deviations are around 0.5 – 2 %, whereas the autocorrelation typically ranges between 0.01 Hz and 1 Hz, depending on the size of the network and the penetration of non-programmable generation [9]. In this case study, for implementation reasons, the process was applied to a small number of loads; thus, to trigger the inter-area oscillation, the standard deviation of the OU process associated with these loads was increased to 30 %. In future studies, a process with a more realistic standard deviation value (0.5 – 2 %) might be applied to all pertinent system loads to verify whether, under small damp conditions, the inter-area oscillations can be triggered without any contingency. The autocorrelation value used in this work is 0.1 Hz, which aligns with other studies.

The OU process is associated with some relevant loads (in terms of active power consumption) in several countries. Some are at the edges of the CE SA transmission system, such as Spain and Turkey, and others are in the central part of the system. In this manner, the inter-area oscillation is triggered quite naturally.

All the loads in the SPES and the CE SA transmission system are assumed to follow an IZ model, with active power exhibiting the dynamic behaviour of a constant current model and reactive power following a constant impedance model. This is a common choice for transient stability analysis, as it captures the main characteristics of the average load dynamics while keeping computations simple due to the constant Z component [10].

The modal analysis of the CE model at 12.21 naturally provides information about the electromechanical modes of the system, including the East–Centre–West mode. The damping ratio of this mode, after adjusting the study model, is 0.28 %. This value is consistent with the initiation of an oscillation at that time. This case, which will be referred to as "A_0", represents the scenario at 12:21 pm, when the 0.2 Hz inter-area oscillation was observed. This case will be used as a reference throughout this document.

Mode	Mode characteristics	
	f_{mode} (Hz)	ζ (%)
East-Centre-West mode	0.20	0.28

Table 4-4: requency and damping ratio of the inter-area East–Centre–West oscillation mode. Reference case A_0.

Figure 4-16 shows the mode shape associated with generator speeds (also known as observability) obtained using DIgSILENT PowerFactory's "Eigenvalue Analysis" tool. These mode shapes represent the positions of the right eigenvector for the East–Centre–West inter-area mode and indicate which machines in the power system oscillate in phase or in the opposite phase.

It can be seen that the expected mode is approximated in a realistic manner. In fact, the SPES electrical frequencies are clustered together, led by approximately 180º by the frequencies of the central area. Turkey's electrical frequency leads the SPES by approximately 50 – 60º. Thus, the calibrated system realistically resembles the expected behaviour during an East–Centre–West inter-area oscillation.

For the sake of completeness, the following shows the electrical frequency oscillations in several buses of the system, as captured by an RMS simulation using the methodology described in the previous subsections.

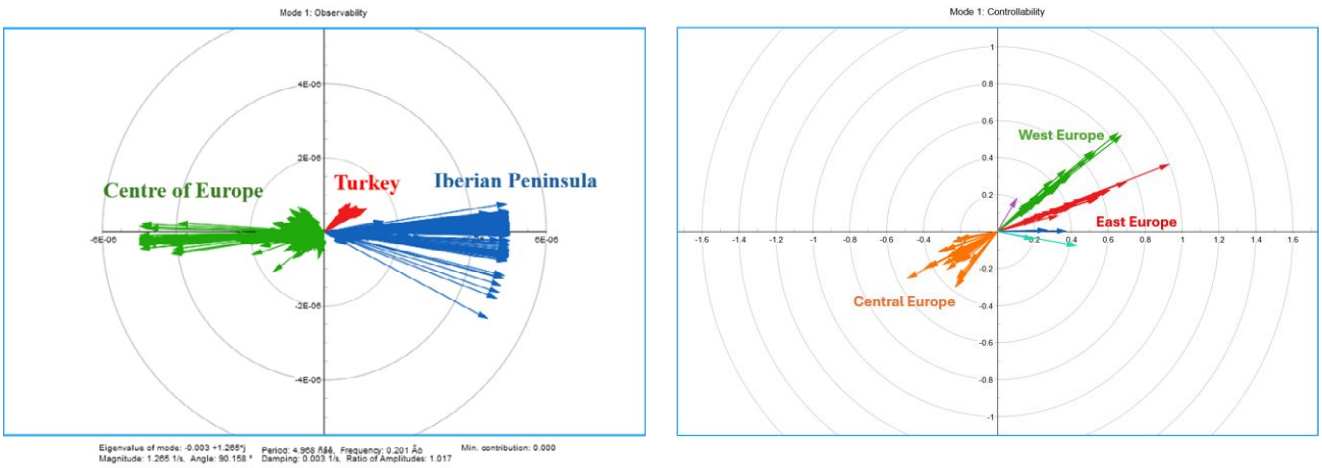


Figure 4-16: Mode shape of East-Centre-West 0.2 Hz mode (observability and controllability)

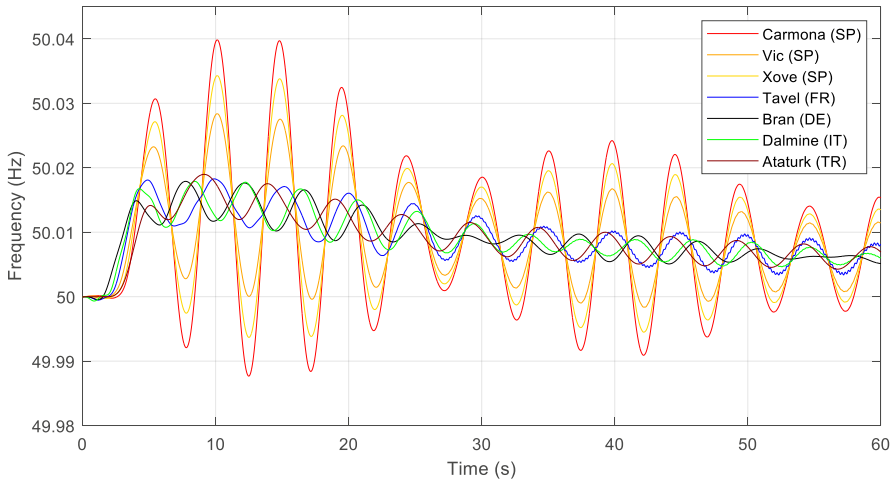


Figure 4-17: Inter-area oscillation triggered by the load contingency at simulation time = 1 s.

Figure 4-17 shows the inter-area oscillations obtained with the simulation model. Similar to the results reported in Figure 2-63 of [11], the frequencies in the Spanish system (Carmona, Vic, and Xove) show the most oscillations. The frequencies in the central part of the CE system, such as those at the Bran (Germany) and Dalmine (Italy) buses, present a phase difference of 180º. Consistently, the frequency at the Tavel (France) bus exhibits an intermediate phase position. The frequency at the Ataturk (Turkey) bus leads the Spanish frequency by approximately 60º.

As observed by the PMU frequency measurements during this incident, the damping ratio is not constant. This is expected, as oscillations cause substantial variations in the network's operating point, which in turn lead to corresponding changes in the damping ratio.

Thus, these results seem to reproduce the East–Centre–West inter-area oscillations registered by PMUs between 12:19 and 12:22.

Figure 4-18 shows that the OU stochastic process can trigger inter-area oscillation. This process also excites other inter-area modes with higher frequency, as it occurs in real power systems. However, the general East–Centre–West mode is maintained, with the SPES frequency lagging central Europe by approximately 180º and Turkey by approximately 40 – 60º.

Thus, introducing the OU process enables the realistic reproduction of inter-area oscillations, supporting the idea that inter-area oscillations can occur even without a major incident.

Overall, the RMS simulations – both simulating a contingency and incorporating an OU stochastic process – yield consistent results and can successfully capture the system's oscillatory behaviour when the 0.2 Hz oscillation occurred.

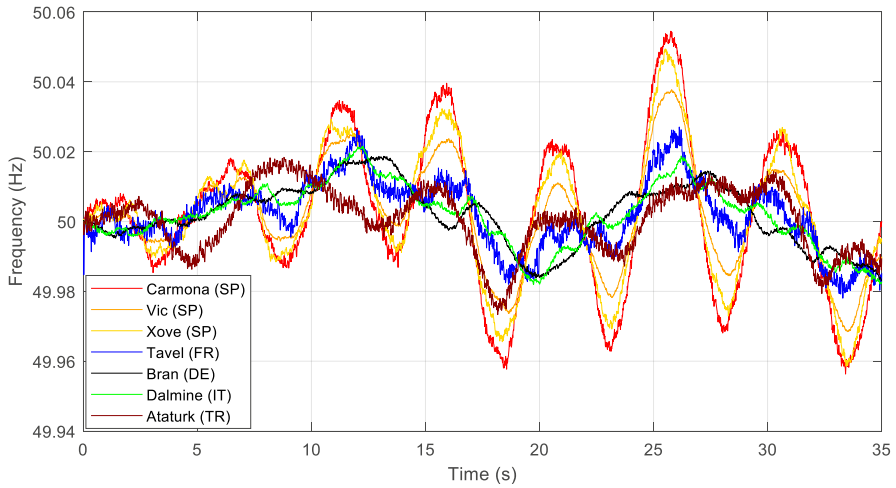


Figure 4-18: Inter-area oscillation triggered by the OU stochastic process.

4.2.3.2 0.63 Hz Oscillation

This section outlines the assumptions used to reproduce the 0.63 Hz oscillations. Since this oscillation differs from an inter-area oscillation, validation focused on the HV active node during the oscillation, where an artificial signal was injected to observe system behaviour. The explanation and classification of the oscillation will be discussed in more detail, along with the techniques used to find the "most active" node, where the so-called "plant A" is located. The simulation is designed to verify that by injecting a forcing signal in the bay of plant A, it is possible to reproduce general system behaviour comparable to the actual recordings.

As high-resolution data for the active and reactive power injected into the node under observation was unavailable, the adopted strategy was to reconstruct the injected signal using a PMU located in a nearby substation to capture the oscillatory trend, combined with SCADA data to obtain the average values of the active and reactive power injected by the plant.

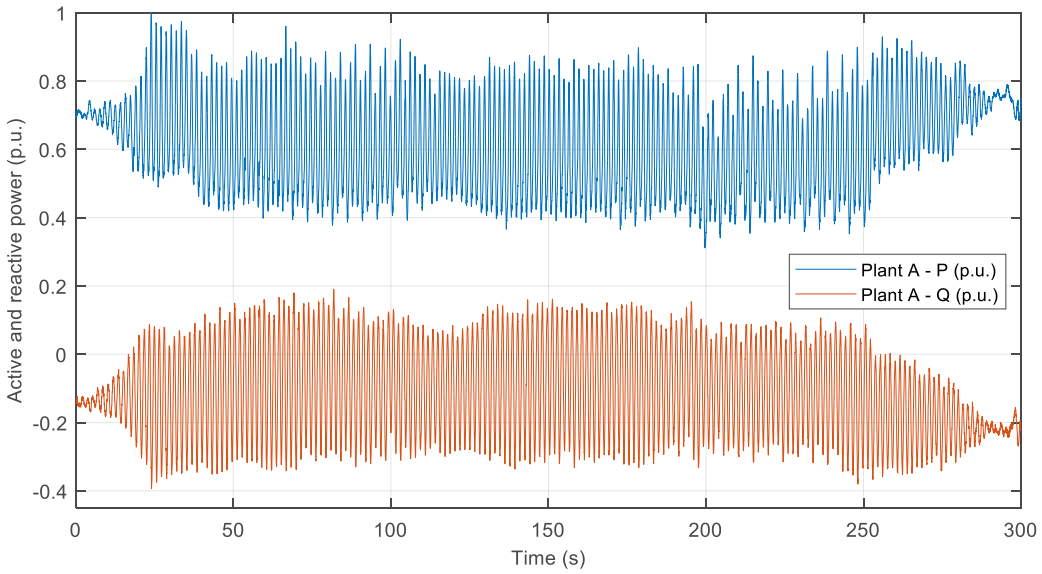


Figure 4-19: Synthetic signal used to trigger the 0.63 Hz oscillations (from 12:03.00 to 12:08:20).

The network model was adjusted to resemble the operating conditions during the 0.63 Hz oscillations, focusing on the connected lines, the control mode of the VSC–HVDC INELFE-1 when the oscillations occurred (angle difference control, ADC), and the active power interchange between Spain and France (≈1,350 MW).

Regarding the CE SA system, since the event occurred between 12:03 and 12:08, the closest available data from the Transparency Platform is at 12:00, which was therefore used for analysis.

Figure 4-20 shows the frequency of several buses across the CE SA transmission system with the injection of the previously mentioned synthetic signal. For comparison, the same buses used in the 0.2 Hz inter-area oscillation are displayed.

A comparison of Figure 4-20 and Figure 4-21 clearly shows that the injected synthetic signal causes the system to oscillate.

The primary frequency component excited is 0.63 Hz, but, as explained in the next paragraphs, the inter-area frequency of 0.2 Hz is also partially excited.

The simulation was conducted for 300 s. As detailed below, this allows monitoring of the system's frequency response when the synthetic signal is injected, showing that perturbations injected at the Badajoz node reproduce frequency responses in SPES buses similar to those recorded by real PMU measurements.

Figure 4-21 provides a closer look at the first 20 s of the simulation. It can be observed that during the first 10 s, when the amplitude of the synthetic signal is relatively small, the buses in the Iberian Peninsula begin oscillating according to their typical inter-area mode frequency (0.2 Hz), with small peak-to-peak oscillations of ≈5 – 7 mHz. After 10 s, the amplitude of the synthetic signal increases, and the frequencies of the peninsular system begin to exhibit the non-typical 0.63 Hz mode. Consistent with PMU measurements, this phenomenon occurs predominantly in the Iberian system.

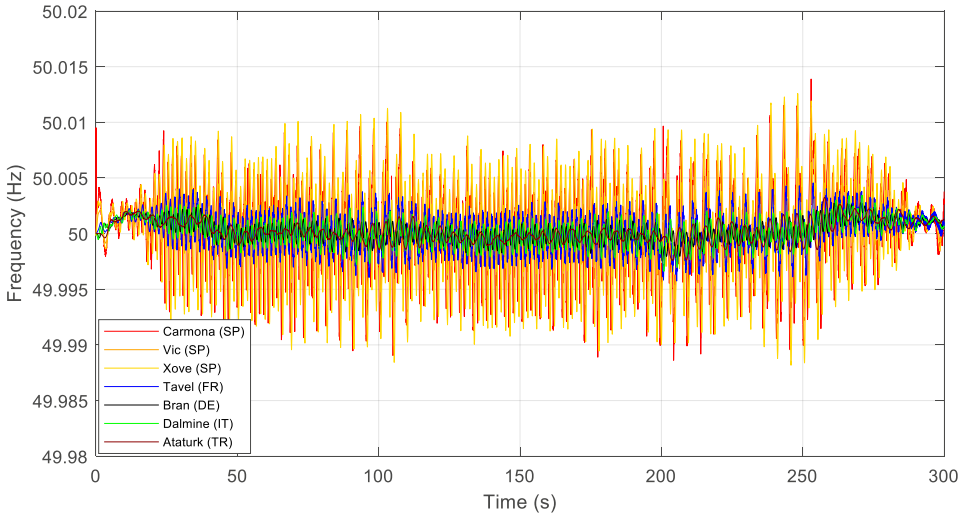


Figure 4-20: 0.63 Hz oscillations with the synthetic signal injected through plant A.

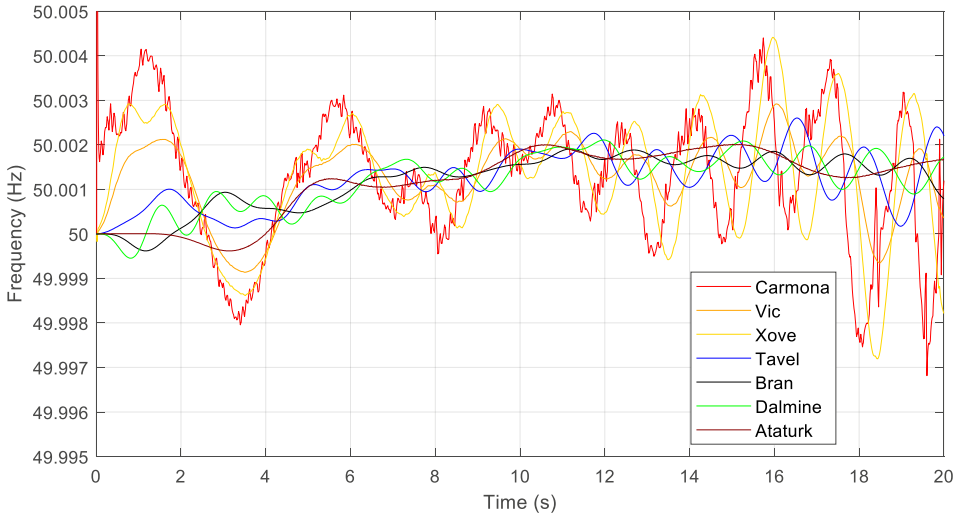


Figure 4-21: 0.63 Hz oscillations triggered by the synthetic signal injected through the plant A (simulation time ranging from 0 s to 20 s).

Figure 4-22 presents an inspection of the subsequent 30 s, showing how the presence of the 0.63 Hz component increases its weight in system frequencies as the amplitude of the synthetic signal increases. The oscillations now reach peak-to-peak values of ≈18 mHz.

Figure 4-23 shows a magnified portion of system frequencies during the simulation time from 230 s to 250 s. At this stage, the amplitude of the oscillations reaches peak-to-peak values of ≈24 mHz.

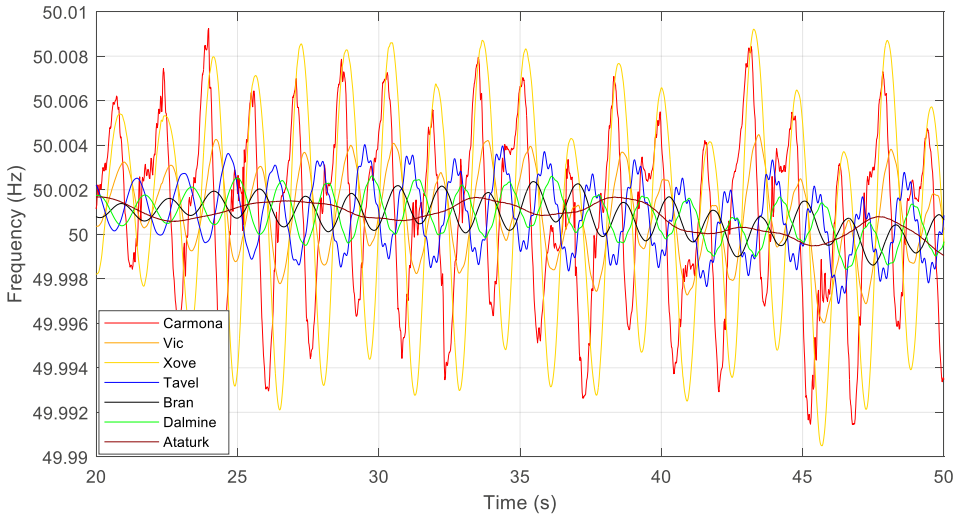


Figure 4-22: 0.63 Hz oscillations with the synthetic signal injected (simulation time ranging from 20 s to 50 s).

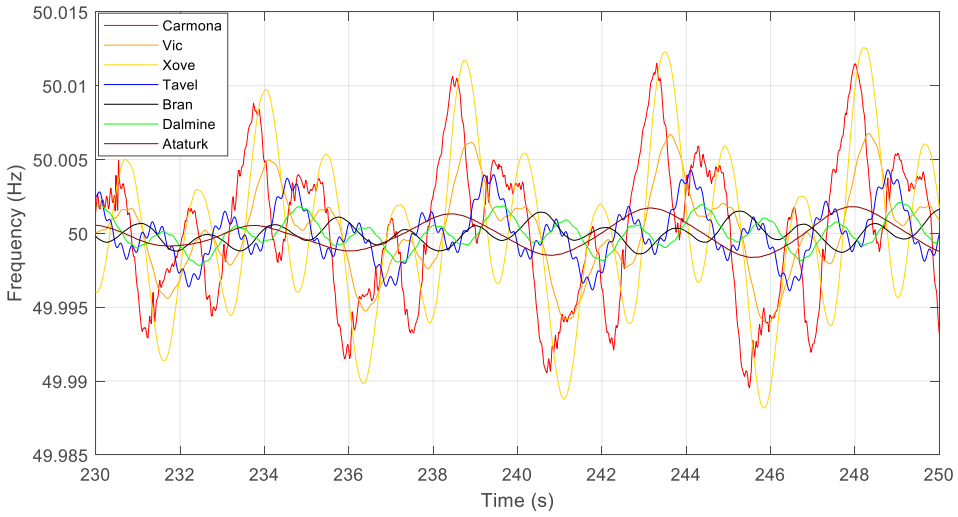


Figure 4-23: 0.63 Hz oscillations with the synthetic signal injected (simulation time ranging from 230 s to 250 s).

The electrical frequency at the Carmona bus is shown in Figure 4-24 (a), along with its frequency spectrum obtained using the fast Fourier transform (FFT) applied to the full 300 s simulation (b). As expected, the frequency component around 0.63 Hz, corresponding to the injected signal, shows the highest magnitude across the simulation, followed by a 0.2 Hz component associated with the inter-area mode that also appears.

To analyse how these two frequency components vary over time, a 40 s sliding window FFT was applied, using a time offset of 10 s. The results are reported in Figure 9 (c). It can be observed that the 0.63 Hz component has a higher magnitude during the first part of the simulation, while the 0.2 Hz component increases towards the end. Consistent with Figure 4-23, where both the 0.2 Hz and 0.63 Hz components are clearly visible, the two components have comparable magnitudes between 200 s and 250 s of simulation time.

The simulation results for the CE SA system indicate that the 0.63 Hz oscillation could potentially excite the inter-area mode. In the last 20 s of the simulation, as the disturbance decreases in magnitude, the 0.63 Hz component correspondingly decreases rapidly. In contrast, the 0.2 Hz component decreases more slowly, as it is an intrinsic mode of the system, and can therefore persist for a longer time. Figure 4-24 also shows a 1.2 Hz oscillation component (also excited by the 0.63 Hz oscillation), which is consistent with the analyses in Subsection 4.2.4.

The conclusive findings of this simulation stage, confirming the PMUs recording analyses, are as follows:

- The 0.63 Hz is also able to excite the 0.2 Hz inter-area oscillation.
- Interrupting the 0.63 Hz oscillation injection halts the oscillation in the system, confirming the "artificial/forced" origin of the phenomenon (as shown in Subsection 4.2.5.1).

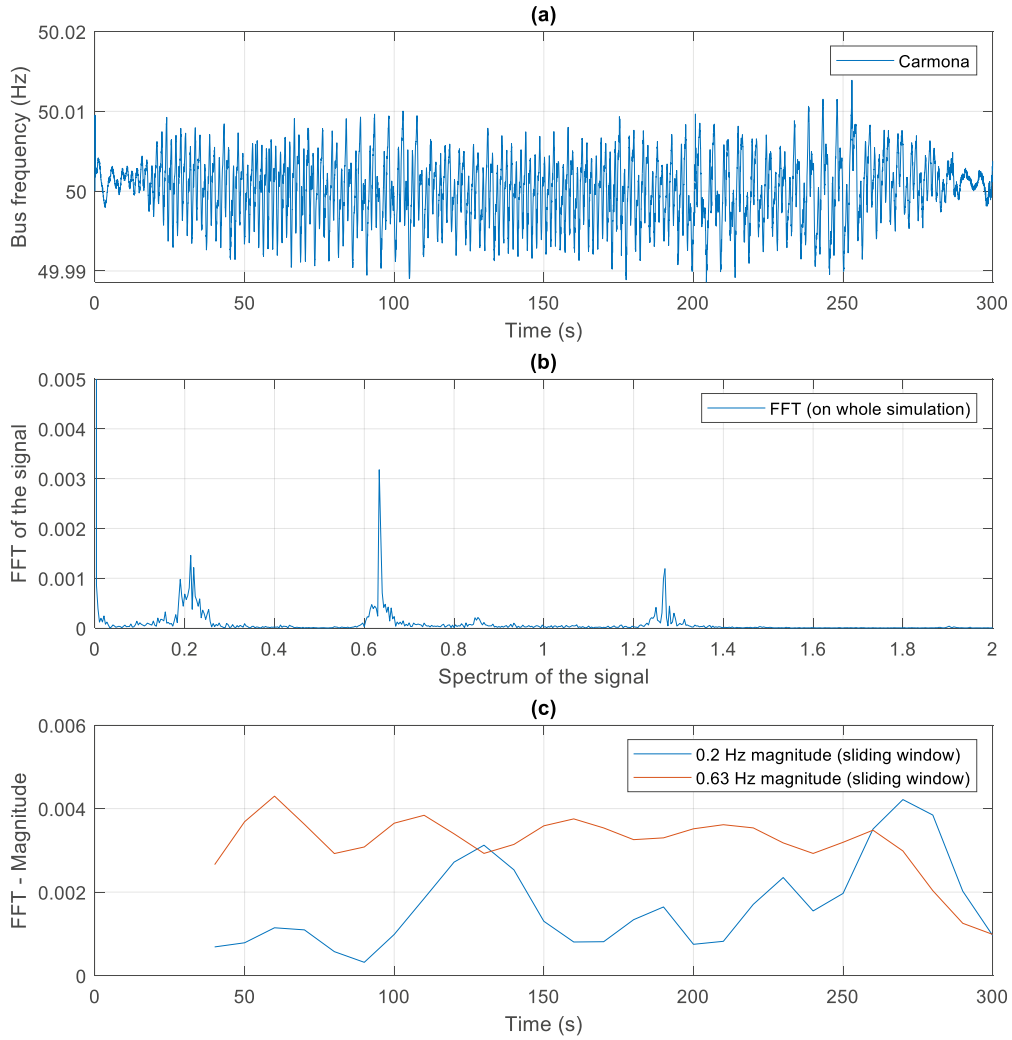


Figure 4-24: Frequency spectrum of the whole signal (Carmona bus electrical frequency), and magnitude over time for the two main frequency components applying a sliding window FFT.

4.2.4 Analysis of Plausible Causes

4.2.4.1 First Oscillation

The first oscillation at 0.63 Hz requires correct classification. An analysis strategy was adopted to reach a logical and plausible conclusion. No high-resolution (with a sampling rate higher than 1 s) recordings were provided from the distribution grid level or main production sites. However, the extensive monitoring network of Red Eléctrica based on PMUs, along with the contribution of several locations across CE, enabled a detailed analysis of the oscillation.

The following steps were investigated in detail to confirm the hypothesis of a "non-natural" 0.63 Hz oscillatory frequency.

» 1. Analyse all PMUs available measurands during the event and in the past (selected PMUs and selected time frames)

» 2. Process historical data provided by Red Eléctrica's modal analysis tool

» 3. Examine any specific past event

» 4. Check with the electromechanical model

» 5. Run an additional check with a different methodology using SCADA measurements

» 6. Ask the more active power plants for any significant information

Following these steps, it is expected that the most plausible explanation for the first oscillation will be identified. The topology of the grid before the first oscillation is depicted in the figure below.

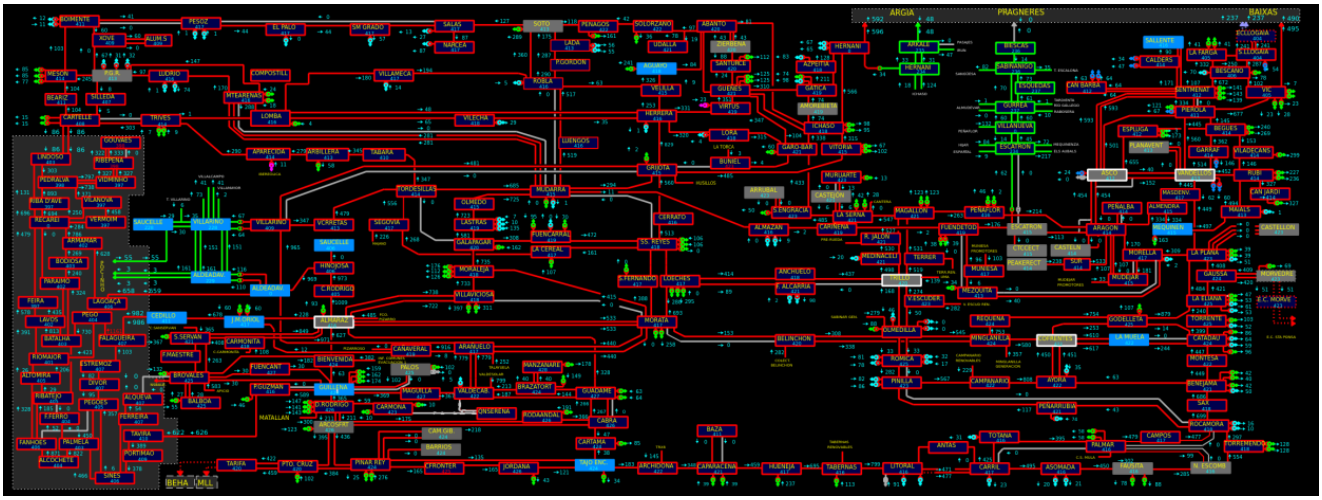


Figure 4-25: Grid topology before the first oscillation (in white the lines in open status)

4.2.4.1.1 Analysis of Available Measurands

A complete set of CE PMU data was collected, comprising 128 voltage and frequency measurands with a 100 ms sampling rate.

The modal analysis was conducted using the Unscented Kalman Filter (UKF) algorithm, which is particularly effective for analysing nonlinear systems.

The algorithm calculates, for each time sample, the dominant modes, amplitudes, and damping ratios for the analysed measurand set. It can also display the mode shapes, showing the relative activity of each state variable for a selected mode. The amplitude (normalised) provides the contribution of each single state variable to the mode, while the angle indicates the phase displacement of each state variable active in the mode. Plotting these phase displacements on a geographical chart makes it possible to determine whether a location oscillates in phase or in opposition to others.

Figure 4-26 shows the raw measurand analysed (frequencies), the modal dominant modes (Mode 1: 0.63 Hz and Mode 2: 0.2 Hz), the mode amplitude, and the damping ratios. The figure clearly illustrates the dominance of Mode 1. Figure 4-27 shows the mode shapes of the two identified modes. The 0.2 Hz mode exhibits the well-known displacement where the Iberian grid and Turkey oscillate against central Europe, while the 0.63 Hz mode is primarily confined to the Iberian Peninsula, with only a very small amplitude in the CE system.

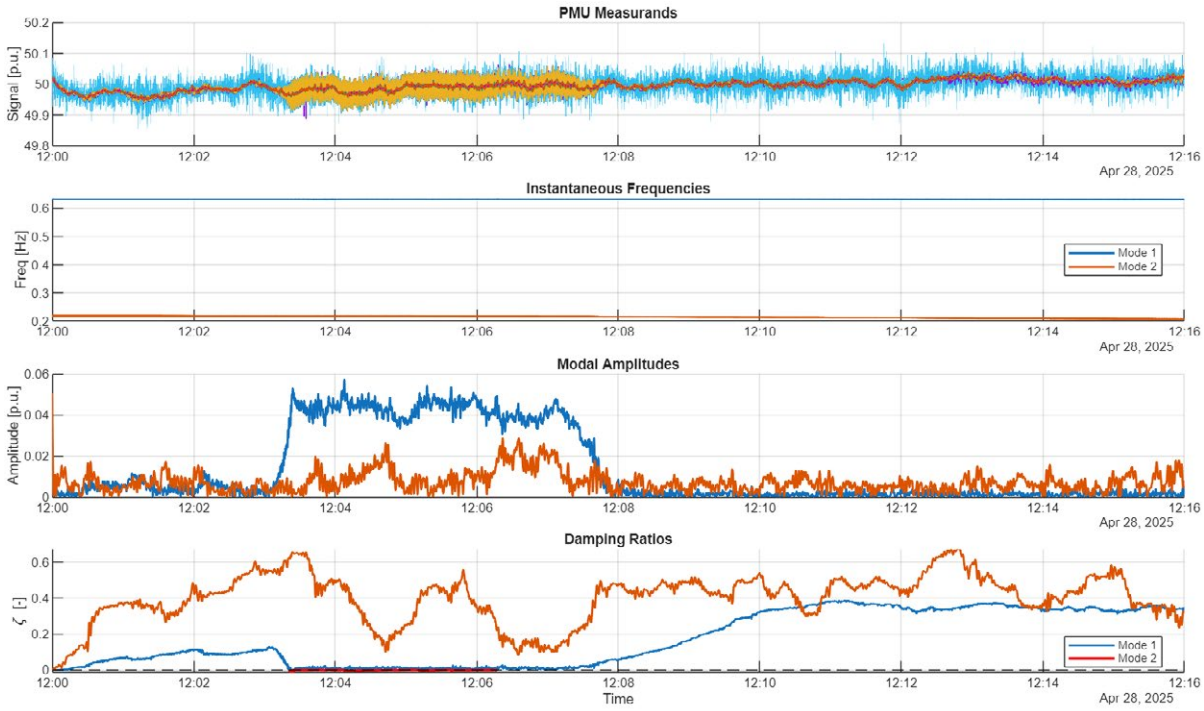


Figure 4-26: UKF algorithm applied to the first oscillation (sampling rate 100 ms).

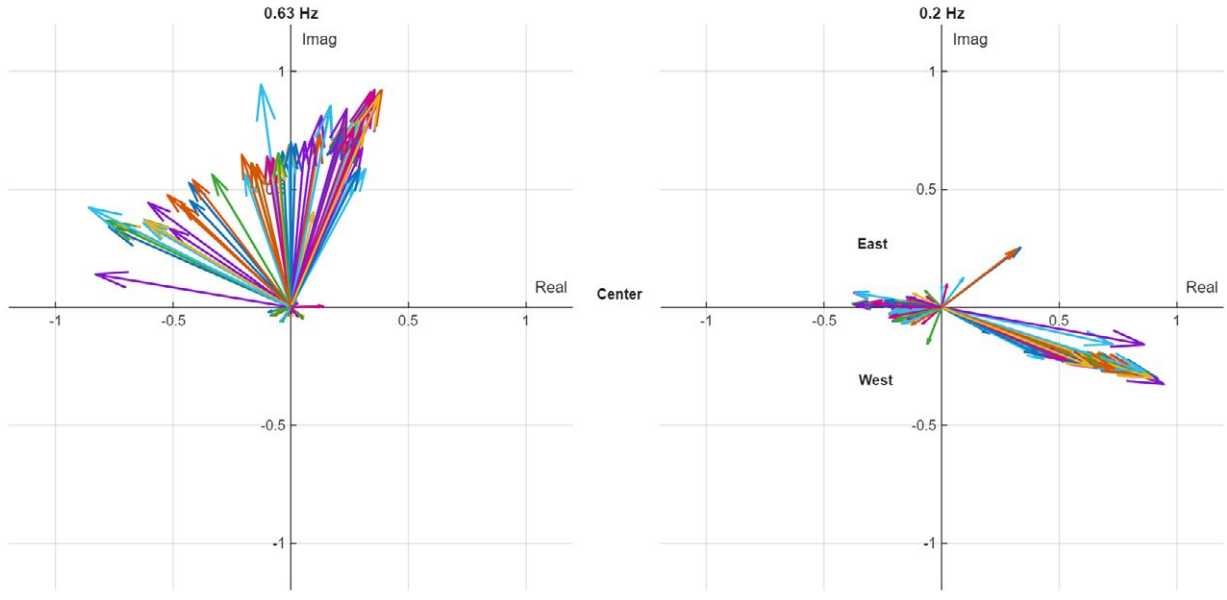


Figure 4-27: Mode shape of 0.2 Hz and 0.63 Hz.

The geographical displacement shown in Figure 4-28 clearly shows that the 0.63 Hz mode is mainly located in the Iberian Peninsula. Repeating the mode shape analysis on voltages (400 kV and 220 kV expressed in pu of rated value) again reveals the clear presence of the mode in the Iberian Peninsula.

The 0.2 Hz mode shape clearly shows the classical inter-area angle displacement, with large portions of the system (east and west) oscillating in phase opposition (180°) against central Europe. The 0.63 Hz mode is characterised by vectors that are in phase (voltages) or, in the case of frequency, with an angle spread of around 90 – 100°.

Based on the results, the following observations can be made:

- A wide geographical analysis confirms that the 0.63 Hz mode is located primarily in the Iberian Peninsula.
- The mode shape does not follow the typical inter-area pattern. The voltage amplitude shows pronounced activity in a specific area of Spain, between Carmona and Almaraz.

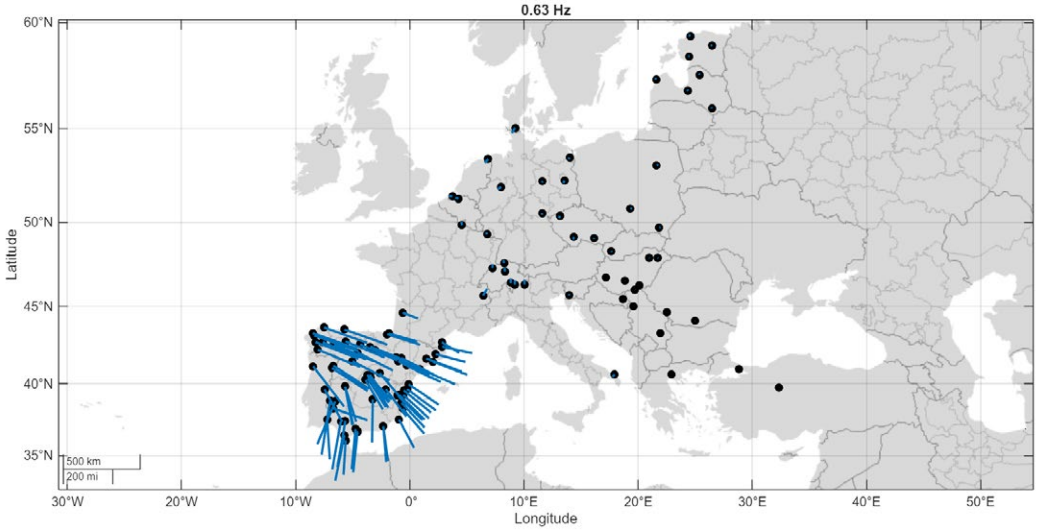


Figure 4-28: Geographical mode shape of 0.63 Hz.

4.2.4.1.1.1 Analysis of Historical Data

Once the specific local characteristics of the 0.63 Hz have been clarified, the next step is to examine a longer period to identify any evidence of a dominant mode that may normally be present.

To verify whether the 0.63 Hz mode is naturally present in the system, the results of the modal analysis of frequency between January and April 2025 were examined at different substations in the Spanish peninsular power system: Xove (northwest), La Cereal (centre), Cofrentes (east), and Carmona (south).

The algorithm used for the modal calculation processes 20-minute windows, and the analysis window is shifted every 20 s. Therefore, every 20 s it provides results based on the ambient modal analysis of the preceding 20 minutes.

The algorithm used for the modal calculation processes 20-minute windows, and the analysis window is shifted every 20 s. Therefore, every 20 s it provides results based on the ambient modal analysis of the preceding 20 minutes. The algorithm is capable of detecting oscillation modes with frequencies between 0.04 and 4 Hz. The number of occurrences represents the number of those 20-minute windows in which the mode is present (considering for each 20 minutes time frame all the modes that are present over a predefinied threshold 2 mHz). The following figure shows the number of detected oscillations for each frequency range at those substations. All detected occurrences are shown, regardless of the damping with which they were identified. Since the algorithm relies on ambient measurements, the oscillation amplitude does not need to be large, nor does the damping need to be low, for the mode to be detected and characterised, making it well-suited for characterising the modes present in the system.

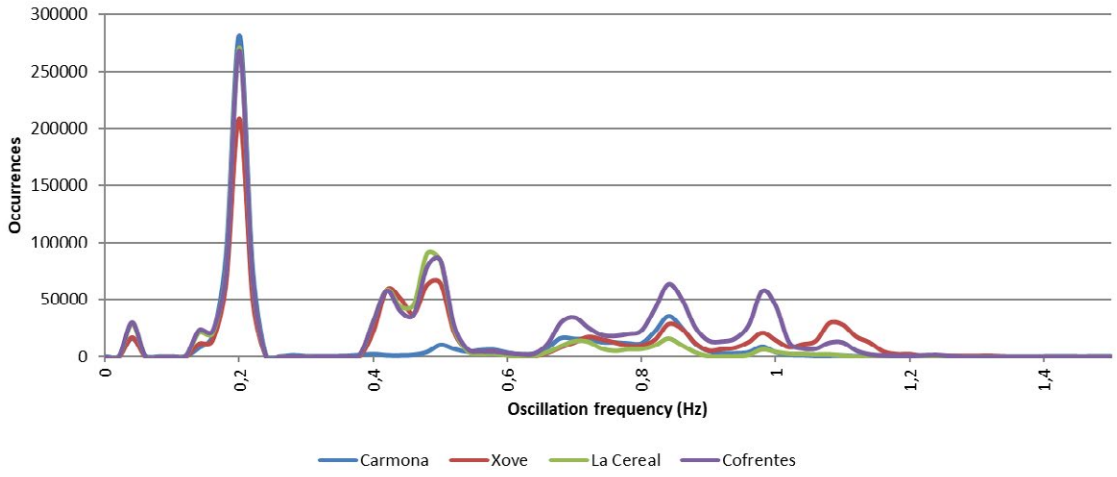


Figure 4-29: Number of occurrences of modes between 0.01 and 1.5 Hz detected between January and April 2025. (Source: Commercial tool for control room modal analysis). Analysis done by third part tool.

Likewise, for the same frequency range, the occurrences with damping below 5 % were examined to determine whether the mode appeared only occasionally in the system but exhibited low damping.

The following figure shows all detected occurrences with calculated damping below 5 %. The vertical scale has been reduced to one-quarter of that in the previous figure to better highlight the number of occurrences.

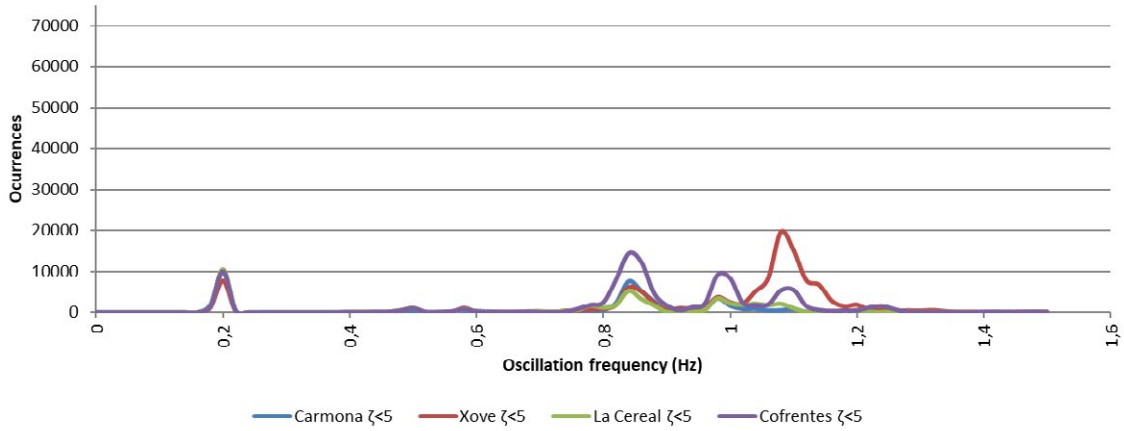


Figure 4-30: Number of occurrences of modes between 0.01 and 1.5 Hz with damping lower than 5 % detected between January and April 2025.

Analysis of the two previous figures indicates that the 0.63 Hz oscillation mode is not naturally present (or, in other words, its low occurrence confirms the absence of a clear "dominance") in the system. The inter-area East–Centre–West oscillation mode, with a frequency around 0.2 Hz, can be clearly observed, as well as another inter-area mode around 0.5 Hz, which corresponds to an oscillation mode between Morocco and the Iberian Peninsula. Additional local modes can then be observed between 0.8 Hz and 1.1 Hz.

In the 0.60 – 0.65 Hz interval, this modal analysis, performed using the frequency measurement from Carmona, found 48 occurrences between 10:00 and 11:00 on 28 April 2025 (out of 180 possible), 140 occurrences between 11:00 and 12:00 (out of 180 possible), and 101 occurrences between 12:00 and 13:00 (out of 101 possible), since from 12:33:33 on 28 April there are no data available to perform modal analysis.

An additional analysis was conducted using a different algorithm, Fast Stochastic Subspace Identification (FSSI), on data from four PMUs recorded in April 2025 before the incident.

The selected locations are reported in the figure below.

Figure 4-31: Number of occurrences of modes between 0.01 and 1.5 Hz with a damping lower than 5 detected between January and April 2025.

Each point in the graphs represents an estimate of the algorithm. The size of the point is proportional to the normalised energy (with 1 representing the total energy of the estimates for a specific time), and the colour of the point indicates the damping. Each analysed time frame covers 1 hour (sampling rate 100 ms), with the x-axis ticks representing 10-minute intervals. The graph on the left spans 00:00 to 01:00, and the graph on the right spans 12:00 to 13:00, allowing comparison of night and day scenarios, including a festive day (Sunday 6 April).

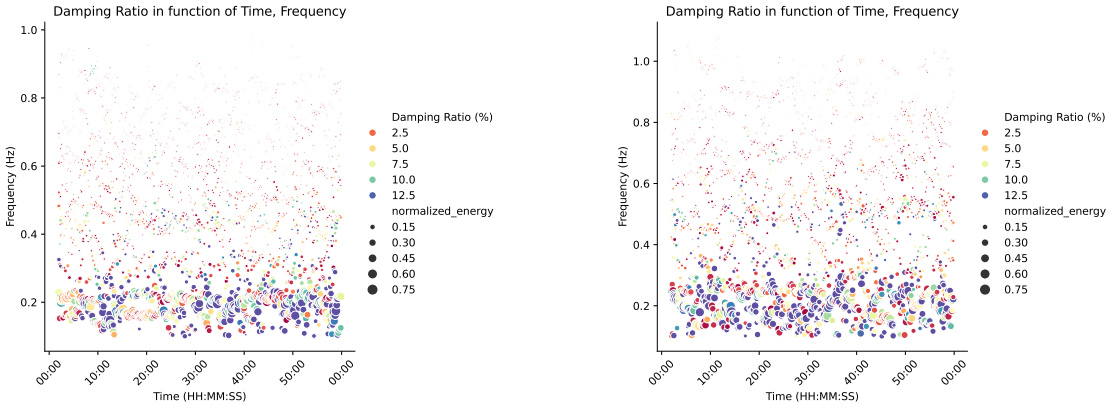


Figure 4-32: FSSI analysis on 1 April 2025.

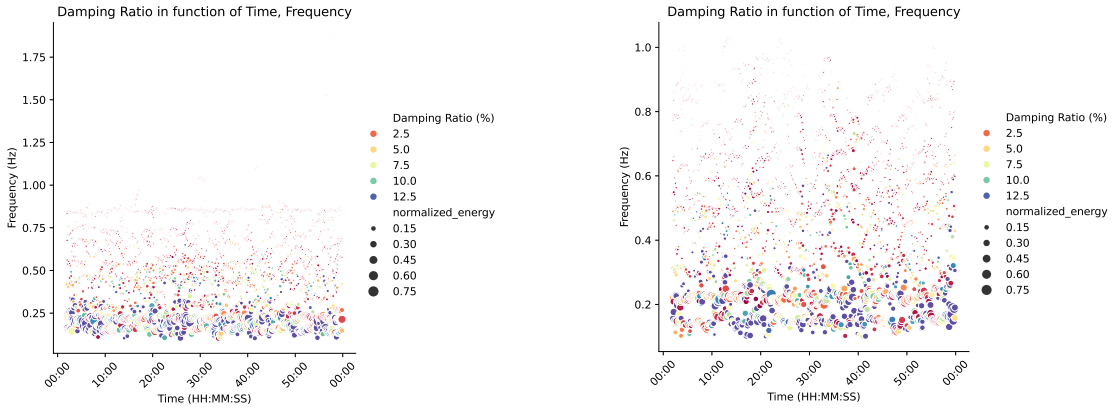


Figure 4-33: FSSI analysis on 2 April 2025.

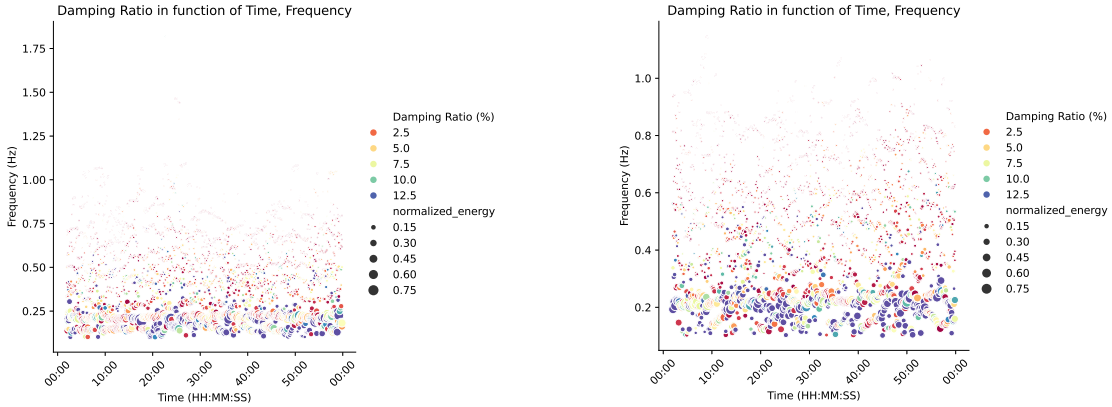


Figure 4-34: FSSI analysis on 4 April 2025.

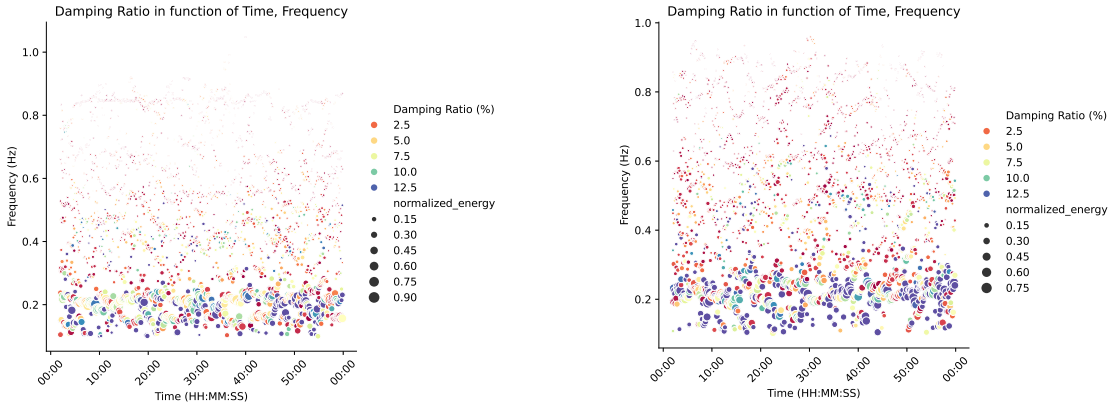


Figure 4-35: FSSI analysis on 6 April 2025.

Overall, although some estimates appear around 0.6 Hz (also on 8.4.2025, a short 0.63 Hz oscillation with peak-to-peak amplitude 15 mHz is noted in PMU data at 14:06, after a 2.6 GW upward ramp in generation schedules at 14:00), they do not show strong consistency, and no clear 0.6 Hz mode is visible. In conclusion, the FSSI analysis confirms occasional low damping at 0.85 Hz (2 and 4 April) but shows no evidence of a 0.6 Hz mode.

4.2.4.1.1.2 Analysis of Event Data

There are several methods for locating the source of an oscillation based on measurement data. With the available data, no method can provide by itself definitive proof of the source. Therefore, multiple methods were applied in this analysis to check for coherent results. For the sake of brevity, these methods are not described here; the reader is referred to the bibliography for further details.

Dissipative Energy Flow Method

The dissipative energy flow (DEF) method [12] was applied to the Red Eléctrica PMU data between 12:00 and 12:15 for the 0.63 Hz frequency.

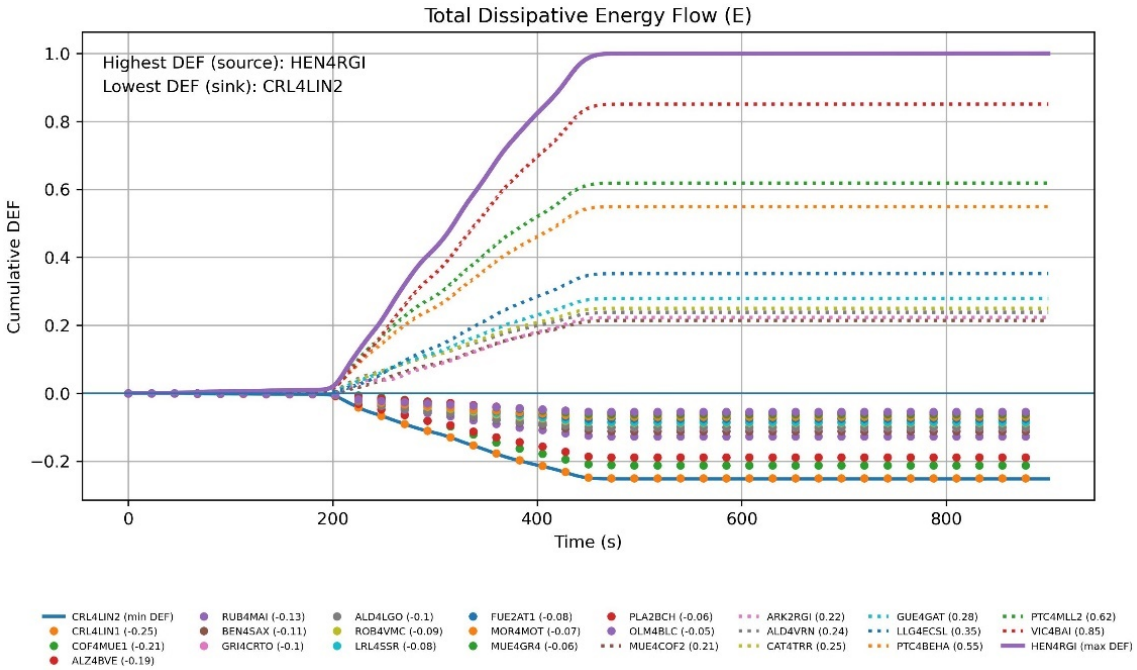


Figure 4-36: DEF against time.

DEF values are indicated in brackets in the legend. T = 0 corresponds to time = 12:00.

DEF values are plotted on the map in Figure 4-37.

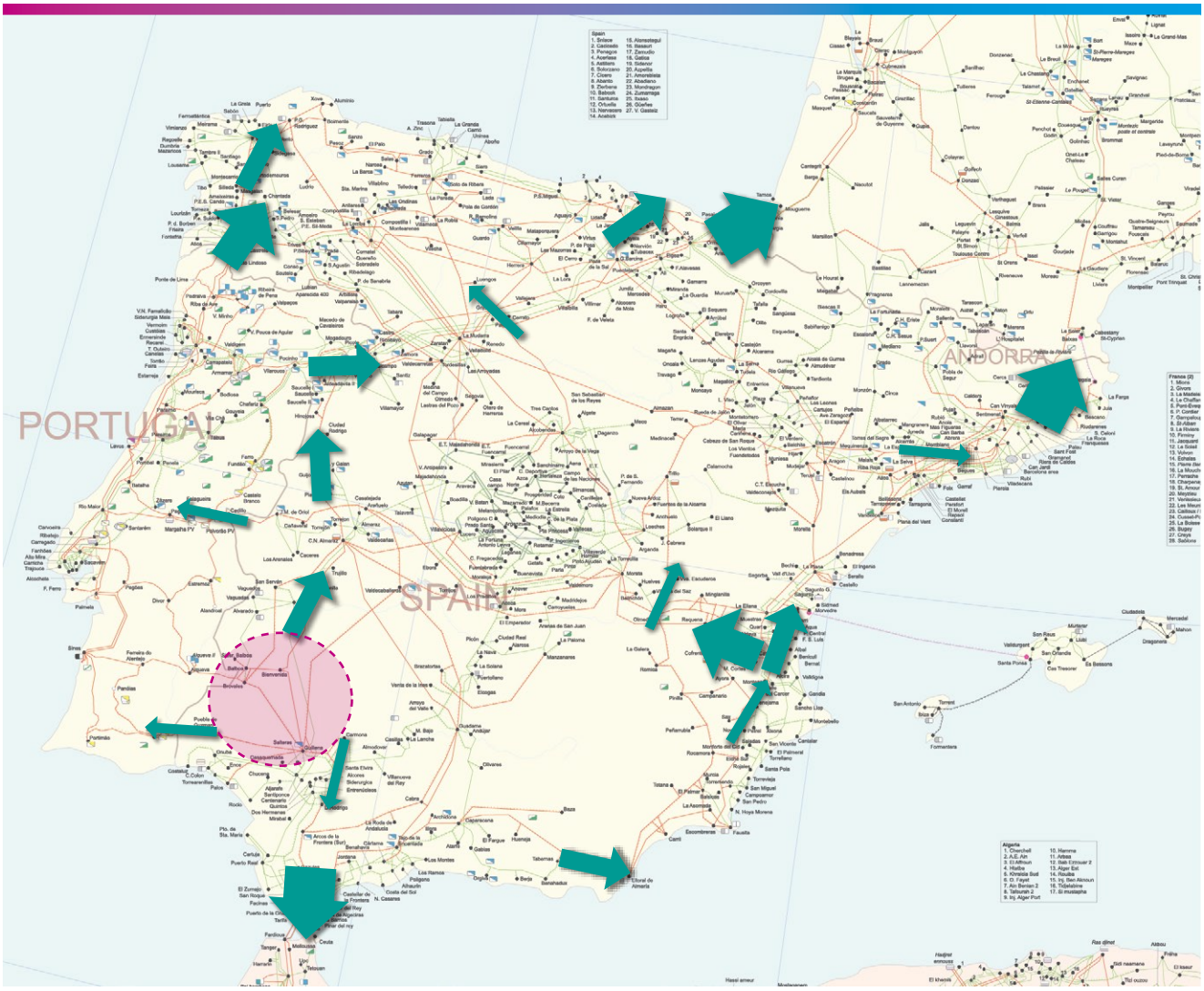


Figure 4-37: DEF calculated on Red Eléctrica PMUs.

The width of the black arrows is proportional to the computed DEF. Only |DEF| > 0.1 are shown. When PMUs are too close, the DEF are added: For example, the DEF on the east of the border between France and Spain is the sum of the DEFs from the 400 kV AC Baixax–Vic PMU and the PMUs from the HVDC between Spain and France. We can observe that the Iberian Peninsula as a whole is exporting oscillation energy to France (DEF: 2.39) and Morocco (DEF: 1.16).

Focusing more closely, another area (DEF: 0.5) – circled in red on the map – is also exporting oscillation energy and could be the location of the oscillation source. Since this area is not fully covered by PMUs, it is difficult to compare the DEF of the likely source with the DEF exported from the Iberian Peninsula. However, it remains possible that the source in this area interacted with other sources in the Iberian Peninsula summing up with interaction with all generators a final total exported energy.

Presence of Harmonics

The presence of harmonics of the fundamental oscillation frequency (0.63 Hz) can be used to locate the source of forced oscillations [13]. As these harmonics propagate less far from the source, the PMU where they are highest is likely the closest to the source.

The FSSI [14] method was applied to the Red Eléctrica PMU data on active and reactive power between 12:00 and 12:15 to analyse the modes and search harmonics.

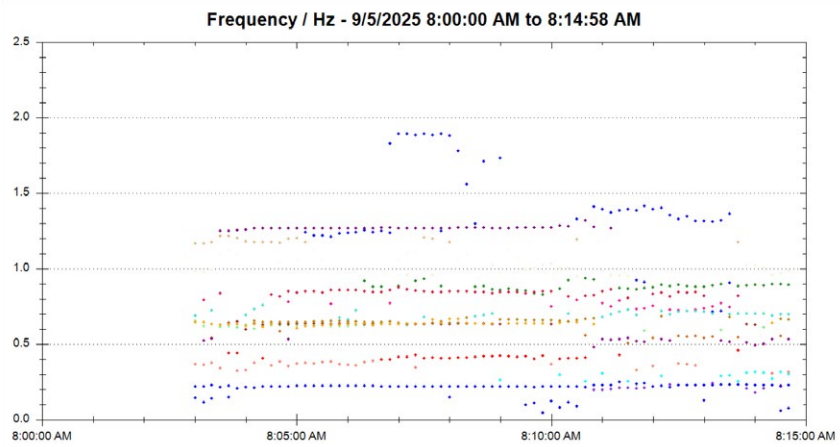


Figure 4-38: Frequency of modes on P signal (Hz).

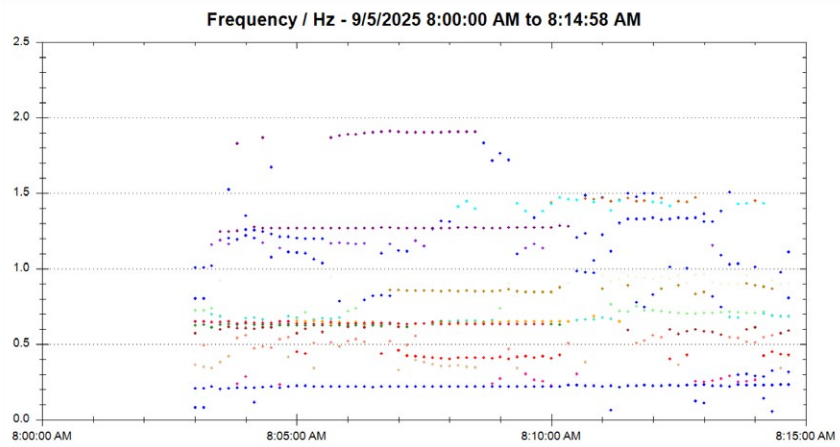


Figure 4-39: Frequency of modes on Q signal (Hz).

The 0.63 Hz component is present and has the highest energy during the period, but estimates are also observed at 1.26 Hz (Harmonic 2) and 1.89 Hz (Harmonic 3), in both P and Q measurements. However, the energy of these harmonics is quite low. Table 4-5 provides the ratio of the maximum energy of each harmonic to the maximum energy of the fundamental.

	P signal (%)	Q signal (%)
Harmonic 2	3.2	1.2
Harmonic 3	0.2	0.2

Table 4-5: Ratio of the maximum energy of each harmonic to the maximum energy of the fundamental.

Harmonic 3 is too weak, with few estimates, to be considered. Figure 4-40 illustrates the mode shape of the Harmonic 2 P signal:

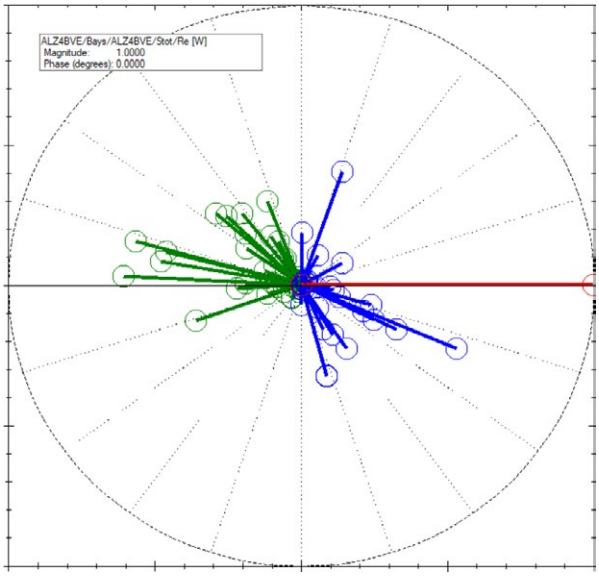


Figure 4-40: Mode shape of P signal.

The PMU with the highest observability for this 1.2 Hz mode is Almaraz–Bienvenida, which is connected close to the generator oscillating (deducted by other nodes PMUs measurands and SCADA measurands) in the Badajoz region.

Figure 4-41 shows the mode shape of the Harmonic 2 Q signal:

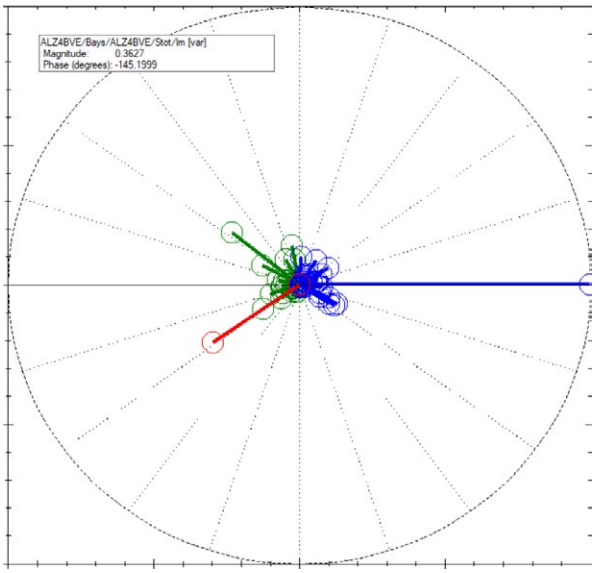


Figure 4-41: Mode shape of Q signal.

The Almaraz–Bienvenida PMU is in red and has the second-highest amplitude. The highest amplitude is observed at the Vic–Baixas PMU (horizontal blue line on the right), located on one of the 400 kV AC lines between France and Spain.

The presence of Harmonic 2 on the Q signal at this location could be linked to the proper reactive power modulation of the POD-Q controller on the HVDC link, mainly in France, as the POD-Q was deactivated in Spain 51 s after the onset of the 0.63 Hz oscillations.

SCADA Data Generation Analysis

Another way to locate the source of forced oscillations is to use SCADA data. Several methods exist, but for this work, "SCADA-Based Detection and Analysis of Oscillations with Inferential Statistics" [15] was employed, for which open-source code is available.[31]

The method was developed by the IEEE Forced Oscillation Task Force and is commonly used to identify a" more active generator. While this generator is not necessarily the main source of the oscillation, it is the most active plant identified during the oscillation.

The analysis was performed on all P and Q SCADA data from generators above 50 MW (144 in total). The plots below illustrate the measurands considered as probable sources of oscillations. The plots on the left show the raw data from measurements, while the plots on the right represent the data after a detrending stage. The detrending stage aims to remove long-term trends, such as the ramping up of a generator, which can obscure the detection of faster oscillations. The detrending method used here is the differencing method: the detrended value for sample *n* is calculated by subtracting the raw value of sample *n-1* from the raw value of sample *n*.

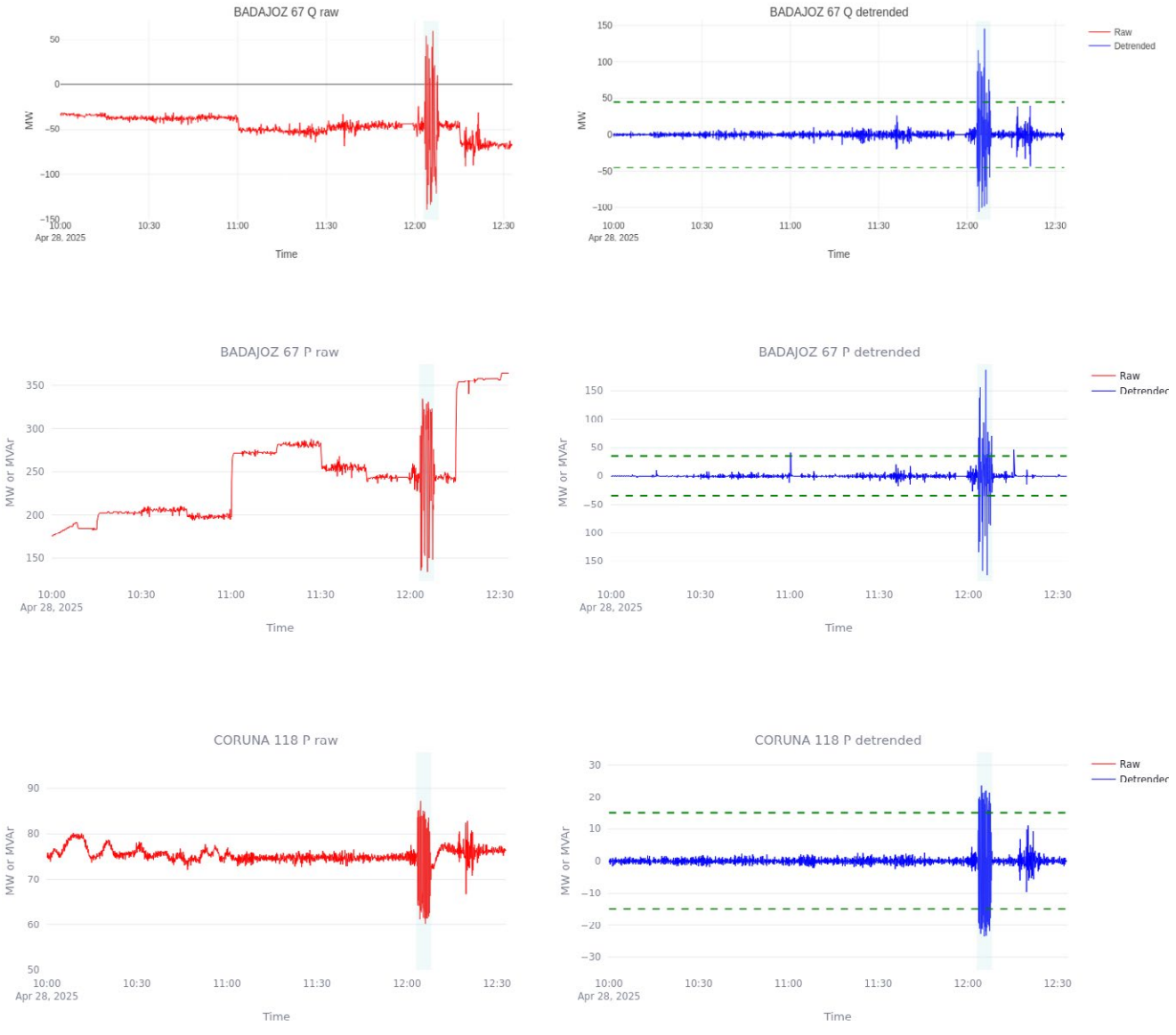


Figure 4-42: Source location software output.

31 GitHub – rte-france/oscillation-analysis-with-scada-data: **Python implementation of a method to locate the source of forced oscillations in a power system.**

In the first iteration, the tool identifies the generator located in the Coruna province – a combined cycled gas turbine (CCGT) in northern Spain – as the most active. However, this disappears in the next iteration. It is worth noting that, by design, the algorithm converges in several iterations. In this case, it detects a high energy value in the first iteration, which is subsequently surpassed by another node. This is confirmed by analysing the group recordings, where some energy is indeed detected during the oscillation, but with a lower amplitude than the node identified at the end of the procedure.

The amplitude of the active power is negligible but indicates that the local oscillation in the Badajoz province triggers a "response" in the opposite side of the system.

The observed generator in Badajoz is the same as the one identified in Subsection 2.5.6.3 (Figure 2.59). An examination of the correlation between oscillations and the reactive power exchange reveals a clear correlation of reactive power at 0.63 Hz.

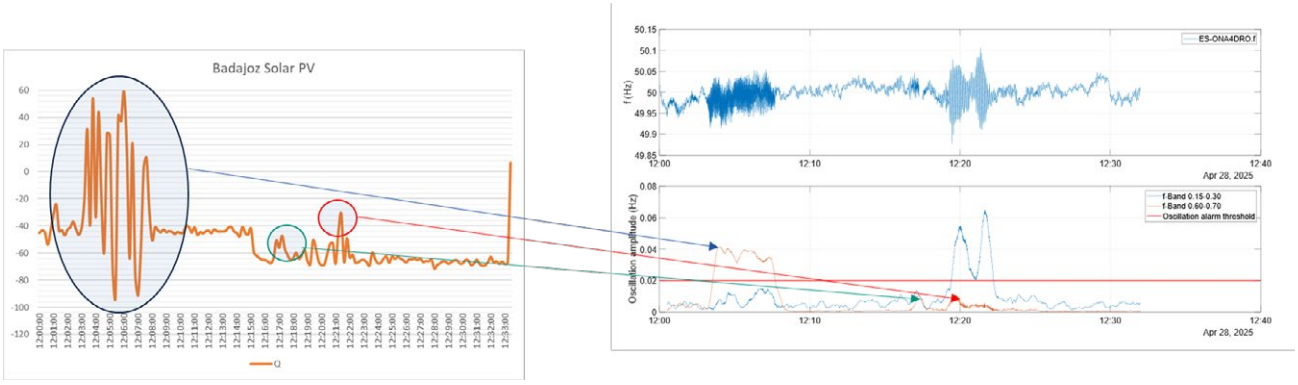


Figure 4-43: Correlation between oscillations (FFT on sliding window) and reactive power (SCADA measurement 4 s).

Conclusion from Measurement Analysis

The DEF method, harmonic analysis (at least for the P signal), and SCADA data all point to consistent localisations.

The output of the specific analysis algorithms aimed at identifying a possible source of the oscillation converged on the same substation mentioned in Chapter 2.

4.2.4.1.2 Summary of Causes and Possible Improvements

All the analyses carried out in this chapter confirm the plausible classification of the 0.63 Hz forced oscillation within the family of *converter-driven* stability phenomena.

The oscillation has the maximum observability near a generator in the Badajoz province, as noted in Chapter 2. Based on the analysed historical data and simulations, no natural dominant mode is observed at this frequency. The analysis of the measurands (PMU and SCADA) using the oscillation source location method points to the same area.

In addition, the simulations were able to replicate the oscillatory effect.

On the one hand, according to the information provided to EP, no alarms or anomalies related to the oscillatory behaviour were found in the plant controller event list. In addition, the plant was classified as an "existing generator" under the RfG code. In contrast, the other generator connected to the same node, which exhibited a lower response to the oscillation, was commissioned in 2021, meaning that the RfG provisions apply.

Instability phenomena in inverters due to the system reactance seen by the generator are well known [16] [17] [18] [19], and real events have also confirmed that old inverters can experience instability issues, especially when running at intermediate power levels. It was also observed that during the period 12:03 – 12:15:12, the inverter was not operating at maximum power, as the available RES energy was higher and the inverter was therefore limiting its electrical output.

When the inverter increased its P setpoint at 12:16, its oscillatory behaviour became less pronounced. Moreover, given the large size of the generator, it can have a relevant impact on grid dynamics.

After collecting and evaluating all these elements, the most plausible conclusion is to classify this oscillation as forced/converter driven: the plant interacted with the other generators in the area, dominating the oscillation.

Finally, the fact that a single generator was the most active during the oscillation at 0.63 Hz suggests that a converter-driven phenomenon plausibly induced a forced oscillation.

4.2.4.2 Second Oscillation

In this section, the inter-area oscillation at 0.2 Hz will be analysed using PMU measurements. Subsection 4.2.5 will investigate the effectiveness of the single damping countermeasure and explore possible improvements to oscillatory stability. Below the system topology before the second oscillation.

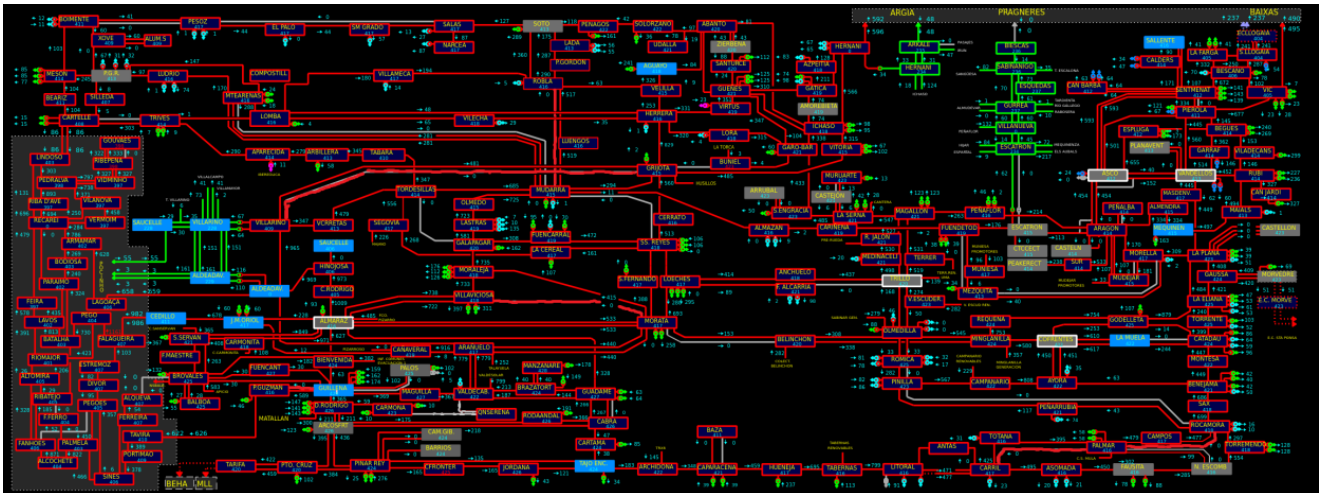


Figure 4-44: Grid topology before the second oscillation (in white the lines in open status).

4.2.4.2.1 Analysis of Available Measurands

The modal analysis of the frequency measurands shows a significant participation of the Iberian Peninsula oscillation relative to central Europe and the eastern region. The northeast side of the system is observed to be in phase opposition. This behaviour is consistent with the characteristics of the ECW mode, especially when considering the recent interconnections with UC–MO and Baltic states.

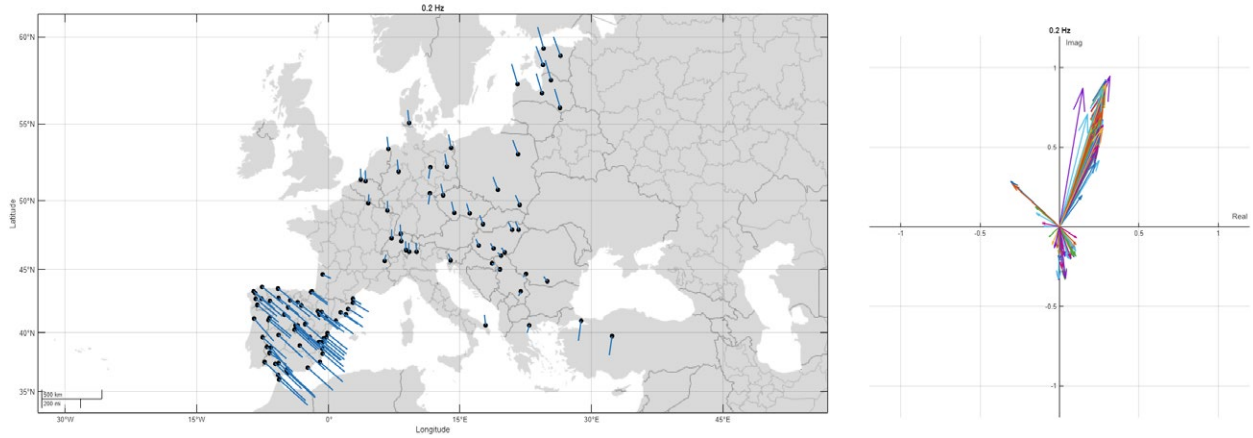


Figure 4-45: Mode shape of 0.2 Hz.

4.2.4.2.2 Summary of Causes

The East-Centre-West mode is the inter-area mode in CE SA with the lowest damping on average. It was triggered on several occasions, and strong oscillations occurred. The PMU and oscillation monitoring tools in place are sufficient to detect abnormal mode activity and initiate remedial actions.

This second oscillation arose from a combination of factors that together enabled the phenomenon, as described in the introduction to this chapter. The high transmission angle (around 80° between Carmona and S. Llogaia),[32] some lines being out of service, and the insufficient action of a few power system stabilisers created the preconditions for the triggering of the oscillations. It must be emphasised that predicting the onset of an oscillation is very difficult, as it also depends on complex, sometimes chaotic interactions between generators, loads, and the grid.

However, managing this phenomenon solely in real time places significant pressure on the control room. Alleviating this burden requires the capability to anticipate the damping of oscillations. Moreover, to avoid the existence of a low damping mode in the CE SA, it is necessary to study the evolution of inter-area modes over the coming years and make the right investment decisions to improve their damping. Lastly, the difficulty of reproducing the event, along with the need to integrate a detailed model with a simplified one, with all the uncertainties it creates, must be addressed to improve the efficiency of post-mortem analysis. All of this can be achieved by maintaining an up-to-date CE SA dynamic model, allowing TSOs to conduct effective post-mortem analysis and planning studies.

4.2.5 Effectiveness of Measures Against Oscillations

The following section aims to assess the effectiveness of various measures against oscillations. First, Subsection 4.2.5.1 assesses the measures applied to the 0.63 Hz forced oscillation. Subsection 4.2.5.2 details the impact of the applied measures on the 0.2 Hz inter-area oscillation.

Subsection 4.2.5.3 then analyses how additional measures, which were not available during the incident, could impact the 0.2 Hz inter-area oscillation. Subsection 4.2.5.4 summarises the findings and provides recommendations to ensure proper damping of oscillations.

4.2.5.1 Assessment of Measures Against Forced Oscillations

This section considers the oscillations characterised by a 0.63 Hz frequency that occurred between 12:03 and 12:08. The calibration of the base case scenario used to reproduce this phenomenon, hereafter referred to as F_0 for simplicity, is detailed in Subsection 4.2.3.2. This scenario reflects the actual network conditions at 12:03 and is compared with other operating scenarios in the following subsections.

Since the eigenvalue analysis did not detect any poorly damped modes in the 0.4 – 0.8 Hz frequency range, the 0.63 Hz mode does not seem to be inherent to the system. In fact, these oscillations could not be reproduced by simulating a large active power deviation or introducing the aforementioned OU stochastic process.

Thus, this section does not provide the eigenvalue results. Instead, RMS simulations are used during the sensitivity analysis to check the effectiveness of the actions taken by Red Eléctrica. A forced oscillation in the active and reactive power injections at 0.63 Hz was simulated at the most active node detected in Subsection 4.2.4.1. The 0.63 Hz oscillation in P and Q injections were modelled based on PMU data from nearby lines. The average values of the P/Q injections were obtained from SCADA data. This study focuses on analysing the impact of forced oscillation introduced by a power park module on the overall oscillatory stability of the power system.

32 It is worth recalling that the transmission angle is directly related to the active power flow, so any internal or external exchange ramp or deviation causes an angle deviation.

4.2.5.1.1 Effectiveness of Actions Taken by Red Eléctrica

This subsection assesses the effectiveness of the actions taken by Red Eléctrica to mitigate 0.63 Hz oscillations – reducing the active power interchange between Spain and France, switching the VSC–HVDC INELFE-1 from an ADC to a CDC control strategy, increasing the active power flow from 180 MW to 500 MW per link (to reduce the flow through AC interconnection lines), and connecting several transmission lines [11]. The following cases are compared:

- F_0: Base case: no corrective actions
- F_{TOT}: All corrective actions applied

Their joint effect is assessed by conducting an RMS simulation, which is reported in Figure 4-46. As expected, although the protocol for addressing low damping is not designed for forced oscillations, the mitigation measures have a small but positive effect on the 0.63 Hz oscillations, as F_{TOT} exhibits smaller oscillation amplitudes compared to F_0. However, compared to the inter-area oscillation study presented in Subsection 4.2.5.2, the effectiveness of these actions is relatively limited. This result is expected: if the source of the 0.63 Hz oscillation is not removed, these actions have limited effectiveness, significantly influencing only the 0.2 Hz oscillation component of the oscillations, which is intrinsically present in the CE SA system.

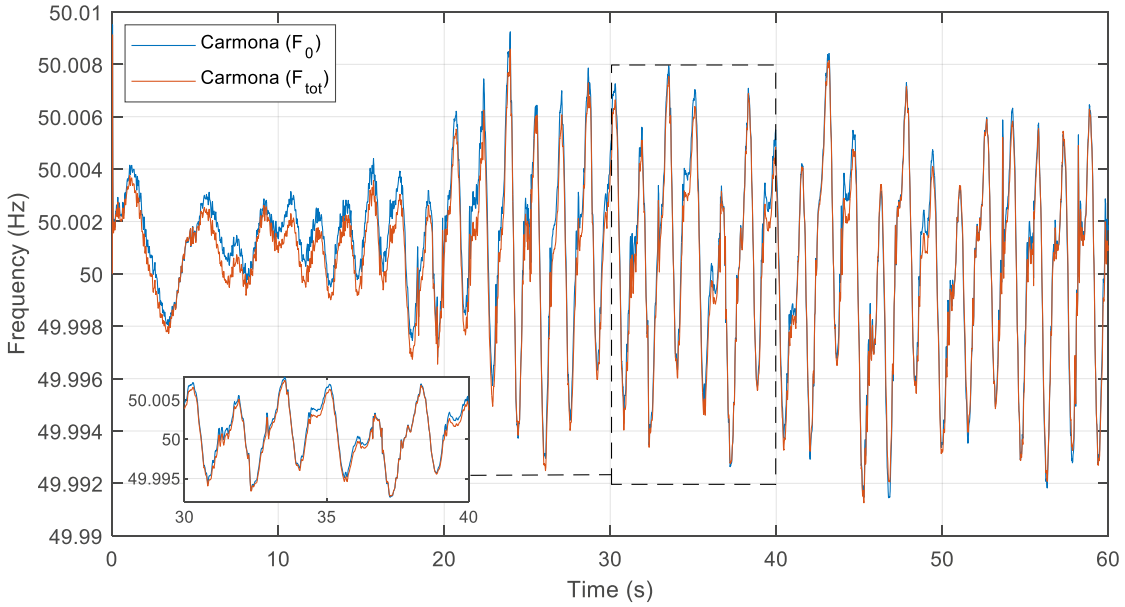


Figure 4-46: Impact of actions taken by Red Eléctrica on the 0.63 Hz oscillations – F_0 vs F_{TOT}.

Figure 4-47 shows the active and reactive power variations of VSC–HVDC INELFE-1 due to POD-P and POD-Q controls, respectively. The F_{TOT} scenario exhibits higher ΔP and ΔQ because the frequencies at the AC terminals of the VSC–HVDC link are oscillating.

Therefore, the POD-P and POD-Q controllers provoke P/Q modulation precisely to damp oscillations.

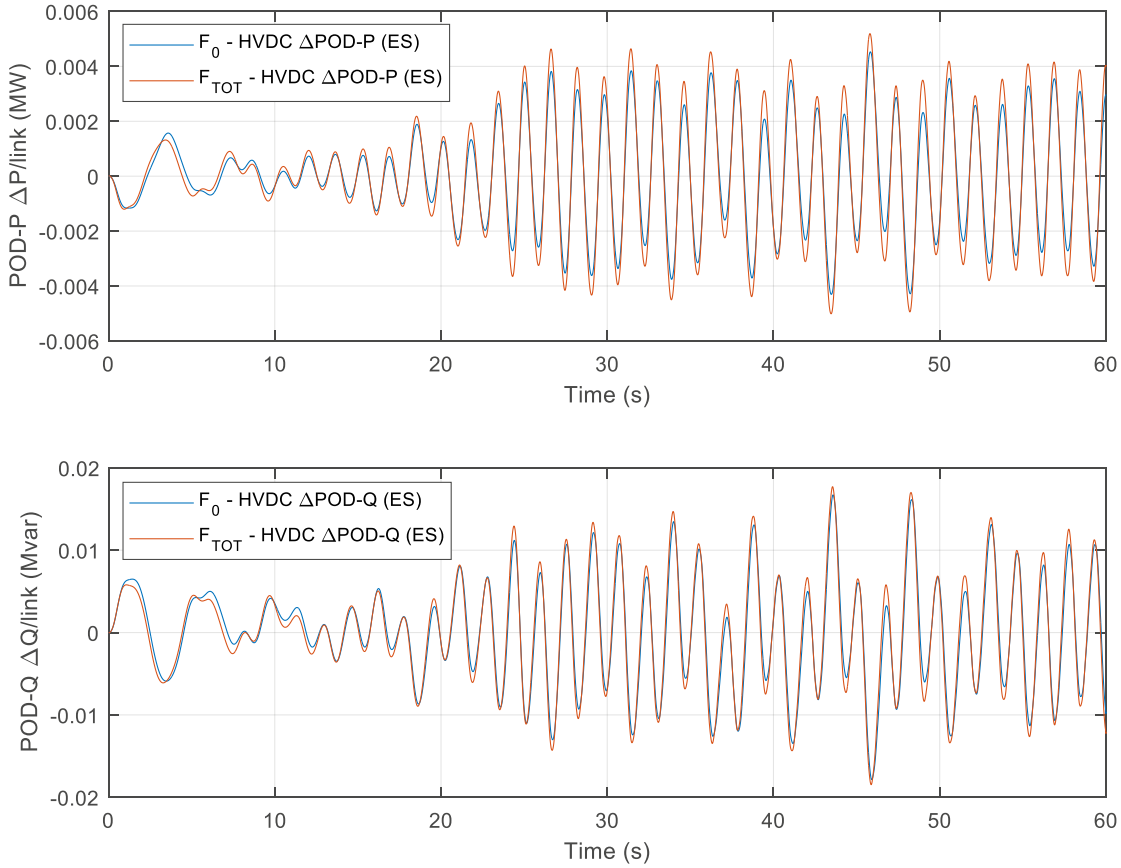


Figure 4-47: ΔP and ΔQ (per link) due to VSC-HVDC INELFE-1 POD controls – F_0 vs F_{TOT}.

The active power of the VSC-HVDC link (exported from Spain to France) is shown in Figure 4-48. Active power through the VSC-HVDC is higher in the F_{TOT} scenario than in the base case (F_0) because the active power setpoint of the VSC-HVDC link was increased as one of the corrective actions. In both cases, the active power through the link varies due to P modulation by the POD-P controller, as discussed previously. The behaviour of the INELFE-1 VSC-HVDC link in the presence of the 0.63 Hz oscillation is considered satisfactory.

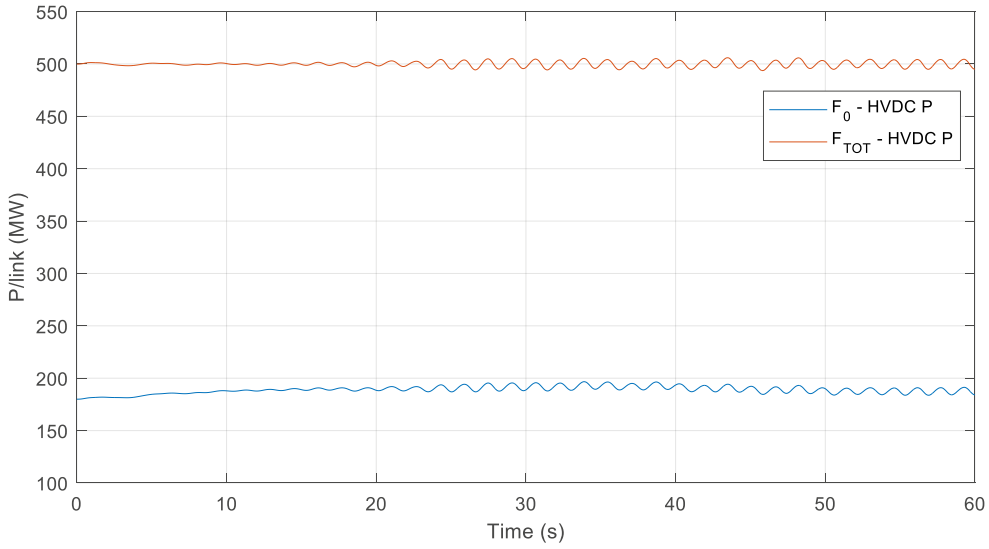


Figure 4-48: P and Q (per link) of VSC-HVDC INELFE-1 – F_0 vs F_{TOT}.

Figure 4-49 shows the Q of Vitoria's STATCOM 150 Mvar during the F_0 scenario (with POD control activated), an alternative scenario similar to F_0 but with POD control deactivated, and the aforementioned F_{TOT} scenario (also with POD control activated).

The reactive power injection of Vitoria's STATCOM changes due to its voltage controller and its POD-Q controller, to damp power oscillations. The behaviour of Vitoria's STATCOM 150 Mvar in the presence of the 0.63 Hz oscillation is considered satisfactory.

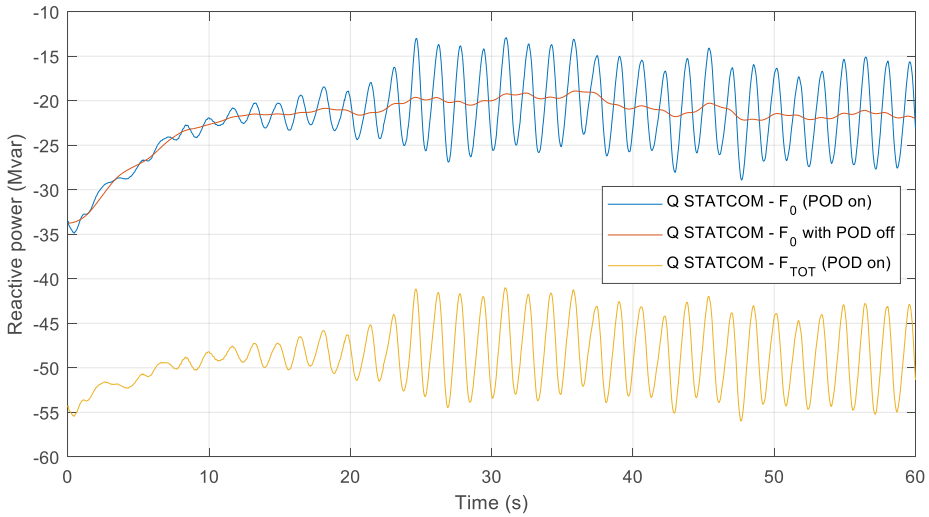


Figure 4-49: Q provided by Vitoria's STATCOM during the 0.63 Hz oscillations – F_0 vs F_0 with POD deactivated vs F_{TOT}.

4.2.5.1.2 0.63 Hz Oscillation Decay in 10 Seconds

In this case study, the artificially injected oscillation is supposed to stop at simulation time 30 s (approximately 10 s after its amplitude became significant). The aim of this simulation is to determine whether the system continues oscillating at the same frequency – in other words, whether this oscillation mode is intrinsic to the electric network or purely related to the forced injection.

The results shown in Figure 4-50 suggest that this type of propagation cannot persist across the CE SA system once the source is shut down. In fact, immediately after 30 s in the simulation, the oscillations are almost immediately damped. This result aligns with the eigenvalue analysis, which did not identify any poorly damped modes near 0.63 Hz.

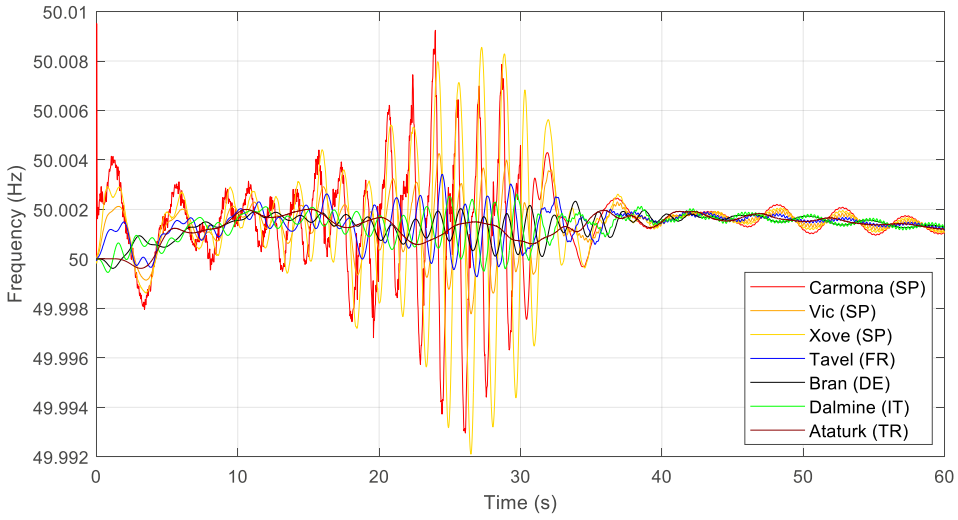


Figure 4-50: Trend of 0.63 Hz oscillations across the CE SA system if the injected signal is shut down.

4.2.5.1.3 Inter-Area Resonance from the 0.63 Hz Oscillation

This case study analyses the potential resonant interactions between forced oscillations and electromechanical inter-area modes. To achieve this, the damping ratio of the CE SA system is reduced (with respect to the one in scenario F_0) by acting on the calibration measures described in Subsection 4.2.2.

The simulation results reported in Figure 4-51 suggest that the inter-area oscillation can be triggered by the 0.63 Hz oscillation if the damping ratio of the inter-area mode is sufficiently small.

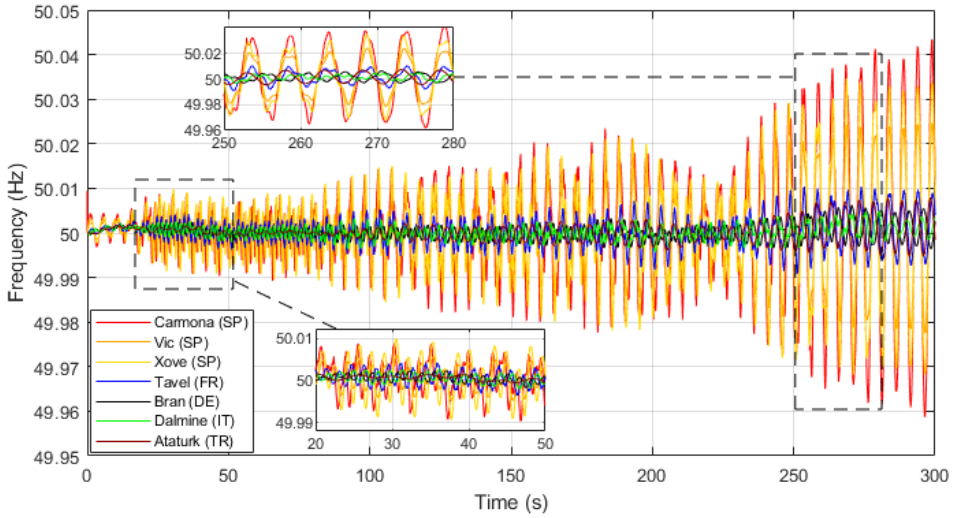


Figure 4-51: F_0 scenario with a small damping ratio of the inter-area mode.

In fact, by examining the leftmost magnified portion of the figure, both the 0.63 Hz and the inter-area oscillations can be observed, whereas the same section in Figure 4-51 shows a more pronounced 0.63 Hz component. In the rightmost magnified portion, representing the simulation time from 250 s to 280 s, the inter-area oscillation is clearly triggered, while the forced oscillation, still injecting the signal (see Figure 4-19), is less easily detectable.

This resonance between different oscillatory modes has been previously studied [20] [21], and the present work corroborates this concept in a very large model representative of the CE SA system.

4.2.5.1.4 Discussion

The results of the previous subsections allowed the following general conclusion to be drawn:

Forced oscillations can also excite natural electromechanical modes in the power system if their damping ratio is low, with frequencies in a similar range. Therefore, oscillations in different system variables might have two components: forced oscillations and electromechanical oscillations. Measures designed to damp electromechanical oscillations (e.g. POD controllers in devices, power-exchange reductions, manoeuvres, etc.) can influence forced oscillations.

As discussed previously, a forced oscillation cannot be completely mitigated unless the source of the oscillation is disconnected. However, since forced oscillations excite electromechanical modes in the power system, the corrective measures used to damp electromechanical oscillations can also help mitigate oscillatory behaviour in the presence of forced oscillations, as discussed in previous subsections.

4.2.5.2 Assessment of Measures Against Inter-Area Oscillations

This section examines the East–Centre–West 0.2 Hz inter-area oscillation that occurred between 12:19 and 12:22. In the following section, the applied measures during this phase of the incident are analysed:

- Performance of INELFE-1 VSC–HVDC link (2 x 1,000 MW)
- Effect of manoeuvres taken by Red Eléctrica to dampen the East-Centre-West mode inter-area oscillation of 0.2 Hz
- Performance of Vitoria's STATCOM 150 Mvar

4.2.5.2.1 INELFE-1 VSC–HVDC Link

This subsection analyses the behaviour of the France–Spain VSC–HVDC interconnection link INELFE-1 (2 x 1,000 MW) during the incident on 28 April 2025 [22]. The analysis presented in this subsection focuses on the impact of INELFE-1 VSC–HVDC on damping inter-area oscillation.

The operating mode of the VSC–HVDC link at 12:19 was as follows:

- The power order was distributed among the links according to their respective link capabilities (MULTI-LINK control mode).
- The Baixas converter station in France was set as the Master Station for Active Power Control (PCONMASTER).
- There was active power flow from Spain to France, with power control proportional to the angle difference between both ends of the link (ADC).
- The converter stations at Sta. Llogaia were operating in voltage control mode (V-mode), in which the HVDC link contributes to voltage control by providing reactive power exchange with the system through an AC voltage target setpoint.
- Power oscillation controllers were activated by active power modulation (POD-P) and reactive power modulation (POD-Q).[33]

Figure 4-52 shows a general diagram of the INELFE-1 VSC-HVDC link with the names of the variables to aid in interpreting the analysis in the following sections. Converter station A is located at the Spanish end (Sta Llogaia 400 kV substation) and converter station B is at the French end (Baixas 400 kV substation).

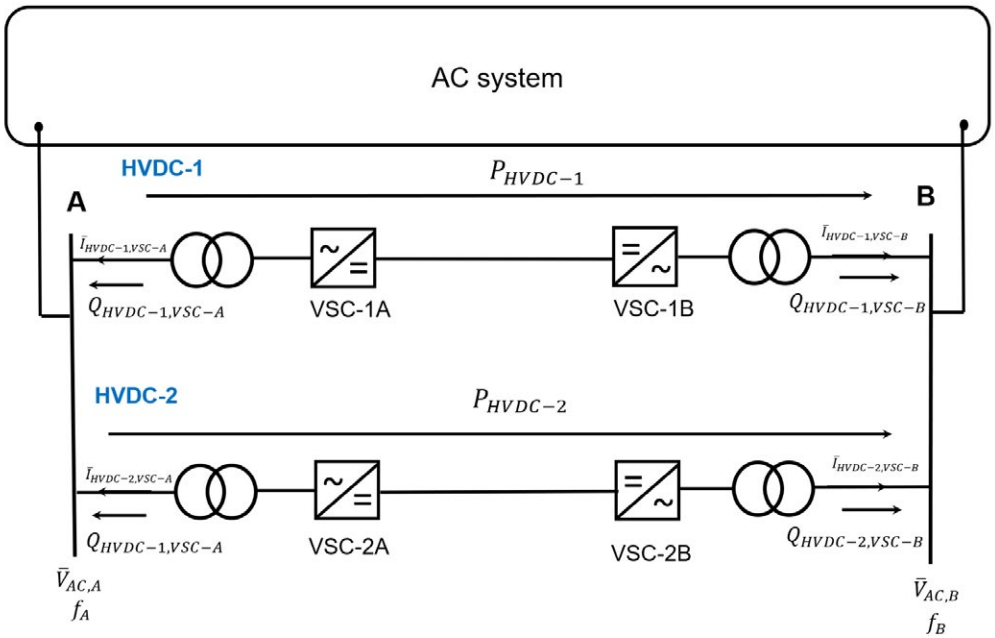


Figure 4-52: INELFE-1 VSC–HVDC link.

33 At 12:19, the HVDC was operating at constant power, with POD-Q disabled by the link protection (see Chapter 2).

The active power setpoint of link HVDC-*i* (with positive values corresponding to power exported from Spain to France) is given by:

$$P^{*}_{HVDC-i} = P_0 + \Delta P^{ref,ADC} + \Delta P^{ref,POD} \quad (1)$$

where:

- P_0 is a constant-power term.
- $\Delta P^{ref,ADC}$ is the supplementary active power setpoint of ADC control. This term is given by equation (2).
- $\Delta P^{ref,POD}$ is the supplementary active power setpoint of the POD-P controller.

$$\Delta P^{ref,ADC} = \frac{K}{1+s\tau}(\delta_A - \delta_B) \quad (2)$$

The setpoint of the reactive power injection of converter VSC-j of link HVDC-i is given by:

$$Q^{*}_{HVDC-i,VSC-j} = Q^{ref} + \Delta Q^{ref,POD} \quad (3)$$

where:

- If the converter is in Q-mode, $Q^{ref} = Q_0$ is a constant reactive power setpoint.
- If the converter is in V-mode, $Q^{ref} = Q^{ref,V}$ is the reactive power setpoint provided by the voltage controller.
- $\Delta Q^{ref,POD}$ is the supplementary reactive power setpoint provided by the POD-Q controller.

Figure 4-53 shows the block diagrams of the POD-P and POD-Q controllers for the INELFE-1 VSC–HVDC link. The input signal of the POD-P controller is the frequency difference between the AC terminals of the VSC–HVDC link. The input signal of the POD-Q controller is the frequency deviation at the AC terminal of the converter station relative to the nominal frequency.

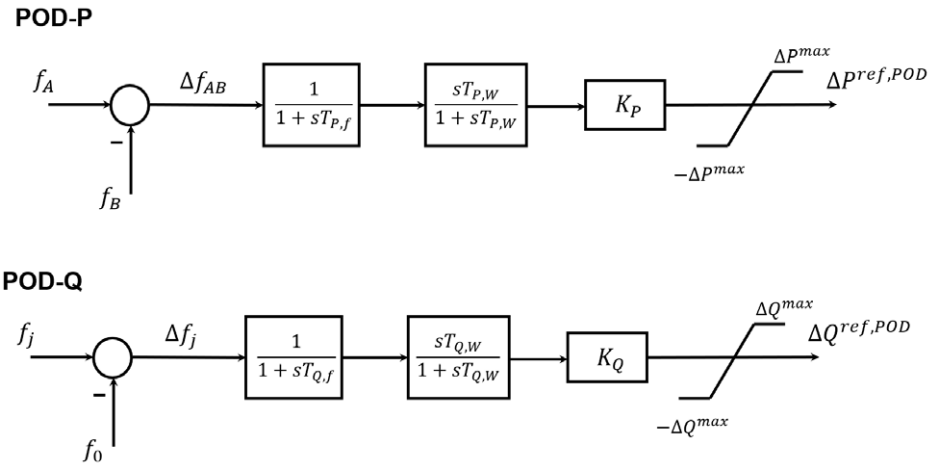


Figure 4-53: INELFE-1 VSC–HVDC link. Block diagrams of POD-P and POD-Q controllers.

After the first 0.63 Hz oscillation, the total active power exchange between Spain and France was reduced, the active power flow of the VSC–HVDC link was increased to reduce the power flow through the parallel AC lines, and ADC control in the VSC–HVDC link was switched to constant power control (CPC). In addition, the POD-Q controller was automatically disabled during the first oscillation (0.63 Hz) to prevent the controller from malfunctioning [22]. The converter stations of the VSC–HVDC link continued in V-mode during the incident. These are the VSC–HVDC link operating conditions for the scenario employed for the analysis of the East–Centre–West inter-area oscillation.

4.2.5.2.1.1 Operation of the VSC–HVDC link (ADC to CPC and Psetpoint)

As described in [11] and [23], the VSC–HVDC INELFE-1 was operated with a transmitted active power of 1,000 MW (500 MW per link) and with the CPC control mode activated; the rationale behind this selection is as follows:

- By increasing the transmitted active power through the VSC–HVDC link from approximately 360 MW to 1,000 MW, the P flowing through the AC interconnection lines is decreased, thus increasing the damping ratio of the inter-area mode.
- By switching from the ADC, which emulates an AC line and thus exacerbates the inter-area oscillations, to CPC control, the damp ratio is increased.

To confirm these statements, several simulations were conducted, using both the eigenvalue analysis and RMS simulations.

The following cases were compared:

- B_0: Base case + HVDC in ADC mode + 180 MW/link + ES–FR 1,355 MW
- B_1: Base case + HVDC in ADC mode + 180 MW/link + ES–FR 875 MW
- B_2: Base case + HVDC in CPC + 180 MW/link + ES–FR 875 MW
- B_3: Base case + HVDC in ADC mode + ES–FR 875 MW
- A_0/B_4: Base case (HVDC in CPC mode + 180 MW/link + ES–FR 875 MW). Notice that case B_4 corresponds to case A_0: the base case considered for the scenario for the analysis of the 0.2 Hz inter-area oscillation.

The related damping ratios of the inter-area mode are reported in Table 4-6, providing the case in which the CPC and ADC are used and the P setpoint of the VSC–HVDC INFELFE-1 is changed. Additionally, a fourth case (B_0) is considered, where the operation point prior to the inter-area oscillation is analysed. This includes the ADC mode of the VSC–HVDC system, the interchanged power through the VSC–HVDC links of 180 MW (per link), and the overall interchanged P between Spain and France (approximately 1,350 MW). In this manner, the operation scenario of the VSC–HVDC system at 12:06 is replicated. This study enables the analysis of the combined effect of the actions taken by Red Eléctrica on the controls of the VSC–HVDC system and the active power interchanged between Spain and France.

Case	Control Mode	P set point, per link (MW)	Total ES–FR interchange (MW)	Inter-area mode		Variation
				ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
B_0	ADC	180	≈1,355	-1.03	0.20	-1.30
B_1	ADC	180	≈875	-0.15	0.20	-0.42
B_2	CPC	180	≈875	0.05	0.20	-0.22
B_3	ADC	500	≈875	0.15	0.20	-0.12
B_4	CPC	500	≈875	0.28	0.20	-

Table 4-6: Sensitivity analysis – damping effect of VSC–HVDC INELFE-1 CPC and ADC control modes, active power setpoint, and ES–FR interchange on inter-area oscillations.

The results show that the operating measures applied to the INELFE-1 VSC–HVDC link and the total interconnection between Spain and France positively contributed to damping the inter-area oscillation:

- Reducing the total active power exchange between Spain and France increases the damping ratio of the inter-area oscillation.
- Increasing the active power flow through the VSC–HVDC link also increases the damping ratio of the inter-area oscillation by reducing the power flow through the parallel AC lines of the interconnection.
- The change from ADC mode to CPC mode in the VSC–HVDC link increases the damping ratio of the inter-area oscillation.

Figure 4-54 shows the frequency at the 400 kV Carmona substation for some relevant cases, obtained from the RMS time-domain simulation. The results confirm the conclusions of the eigenvalue analysis: the control actions the operational measures applied to the INELFE-1 VSC–HVDC link, as well as the total interconnection between Spain and France, contributed to damping the inter-area oscillation. This is evidenced by the reduction in oscillations observed in the frequency measurements when these measures were implemented.

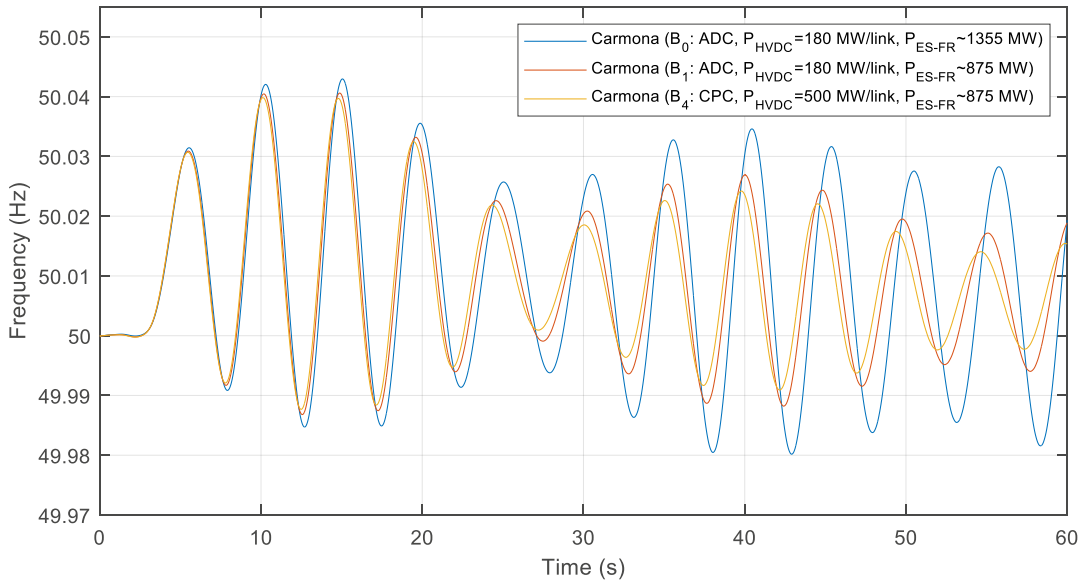


Figure 4-54: Impact of Pmode control and P setpoint of the VSC-HVDC INELFE-1 link on the inter-area oscillations - B_0 vs B_1 vs B_4 (load contingency).

4.2.5.2.1.2 POD Controllers

This subsection provides the results of a sensitivity analysis on the effectiveness of the POD-P and POD-Q controllers of the VSC–HVDC INELFE-1. The objective is to analyse the effectiveness of the POD-P and POD-Q controllers in damping the 0.2 Hz inter-area oscillation observed during the incident. For the analysis, the VSC–HVDC link is in CPC mode and the converter stations in V-mode, as in the reference case A_0.

The following cases were compared:

- C_0: Base case A_0 + HVDC POD-P off and POD-Q off
- A_0/C_1: Base case (with POD-P on and POD-Q off), corresponding to case A_0
- C_2: Base case + POD-P off and POD-Q on
- C_3: Base case + POD-P on and POD-Q on

The results in Table 4-7 demonstrate that both the POD-P and POD-Q controllers of the INELFE-1 VSC–HVDC link are effective in damping the inter-area oscillations. In this scenario, the POD-Q controller is more effective than the POD-P controller. This is due to the 400 kV Vic–Baixas line (which runs parallel to the VSC–HVDC link) and the limited effectiveness of the POD-P controller, which uses as its input the difference between the two terminals of the link. These results were confirmed by the RMS time-domain simulation (Figure 4-55 and Figure 4-56).

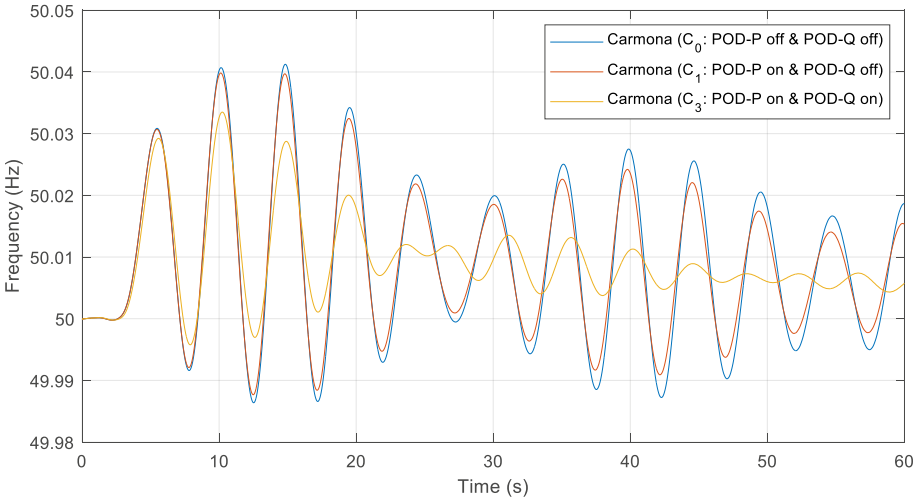


Figure 4-55: Impact of the POD controller of VSC–HVDC INELFE-1 on the inter-area oscillations – C_0 vs C_1 vs C_3 (load contingency).

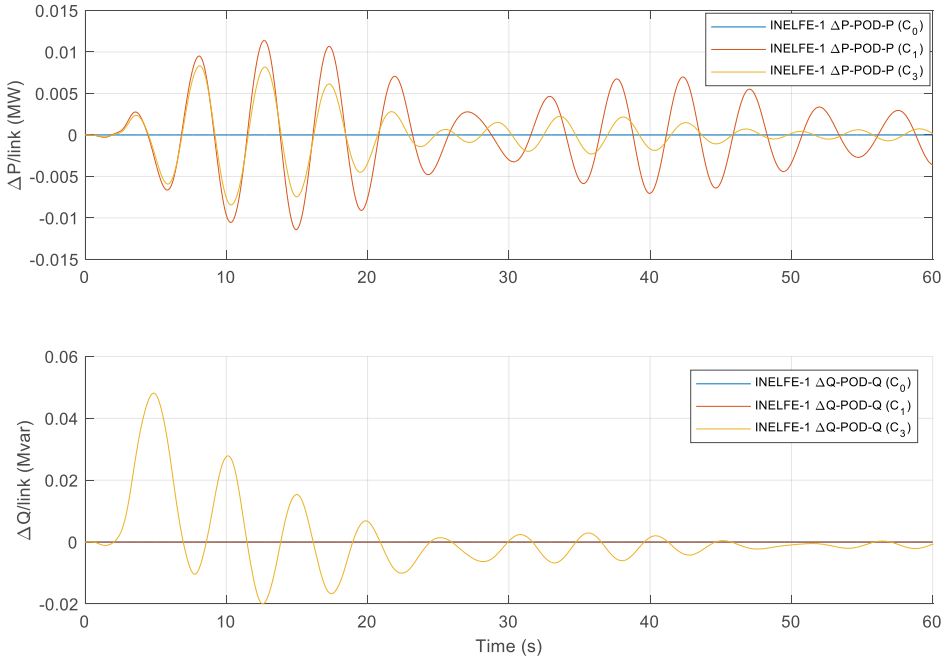


Figure 4-56: References of POD-P (ΔP) and POD-Q (ΔQ) of VSC-HVDC INELFE-1- C_0 vs C_1 vs C_3 (load contingency).

Case	Inter-area mode		Variation
	ς (%)	f_{mode} (Hz)	Δς (%)
C_0: Base case A_0 + HVDC POD-P off & POD-Q off	– 0.18	0.20	-
C_1: Base case (with POD-P on & POD-Q off) (A_0)	0.28	0.20	0.46
C_2: Base case + POD-P off & POD-Q on	1.91	0.24	2.09
C_3: Base case + POD-P on & POD-Q on	2.28	0.24	2.46

Table 4-7: Sensitivity analysis – damping effect of VSC-HVDC INELFE-1 POD-P and POD-Q controls (activated only on the Spanish side) on inter-area oscillations.

In 2022, a task force between RTE and Red Eléctrica (the French and Spanish TSOs) was formed to improve the performance of the POD-P controller for the INELFE-1 VSC-HVDC link [24]:

- The POD-P gain was increased from Kp = 500 MW/Hz/link to Kp = 2,500 MW/Hz/link to increase its effectiveness.
- Modifications were made to certain logic conditions to enable the use of the POD-P controller in conjunction with the ADC of the INELFE-1 VSC-HVDC interconnector.

The impact of increasing the POD-P gain is now analysed. For this purpose, the following cases were compared:

- C_0: Base case A_0 + HVDC POD-P off and POD-Q off
- C_5: Base case, with POD-P on and Kp = 500 MW/Hz (per link) and POD-Q off; old POD-P controller settings
- A_0/C_1: Base case, with POD-P on and Kp = 2,500 MW/Hz (per link) and POD-Q off; current POD-P controller settings; corresponding to case A_0

Table 4-8 shows the damping ratio of the East–Centre–West inter-area oscillation. As expected, the results demonstrate that a higher POD-P controller gain (Kp = 2,500 MW/Hz/link) results in a higher damping ratio of the critical inter-area oscillation.

Case	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
C_0: POD-P off	– 0.18	0.20	-
C_5: POD-P on, Kp = 500/Hz/link (old settings)	0.12	0.20	0.30
C_1: POD-P on, Kp = 2,500/Hz/link (current settings)	0.28	0.20	0.46

Table 4-8: Sensitivity analysis – VSC-HVDC INELFE-1 – POD-P controller, impact of the gain.

These results show that the modifications made in 2022 to the POD-P controller of the INELFE-1 VSC–HVDC interconnector improved its performance and its capability to damp inter-area oscillations in the CE power system.

4.2.5.2.2 Effect of Topology Changes Taken by Red Eléctrica to Dampen the East-Centre-West Mode 0.2 Hz Inter-area Oscillation

As detailed in Table 2-2, Red Eléctrica took several topology actions between 12:02 and 12:25 to dampen the 0.6 Hz oscillation and the 0.2 Hz inter-area oscillation (connection of lines) and control voltage (disconnection of shunt reactors). In this section, a sensitivity analysis is conducted to evaluate the effectiveness of these manoeuvres in mitigating the inter-area oscillation at 0.2 Hz (connection of lines). The analysis considers each measure separately as well as their combined effect, and the total effect is also presented. As previously mentioned, to construct the model of the Spanish transmission system, this study relies on data from 12:21 obtained via the state estimator. Consequently, all line switching operations occurring between 12:02 and 12:21 are reverted. Lines were connected to damp oscillations in the power system. In this study, each line is then individually reconnected to evaluate its contribution to damping the 0.2 Hz inter-area oscillation.

A reference case, L0, excluding these topology changes, serves as a benchmark case. It should be noted that L_0 does not match the calibrated base case A_0, as seven lines differ in their connection status. In addition, because the active power control of the INFELFE-1 VSC–HVDC link employs the AC line emulation logic at the onset of the switching operations, this control mode is assumed for this sensitivity analysis. As this represents the normal operation of the HVDC link, this operation scenario seems more relevant for evaluating the impact of topology changes.

Table 4-9 provides the frequency of the inter-area mode f_{mode} (Hz), its damping ζ (%), and the corresponding variation with respect to the L0 base case. As mentioned before, these results were obtained using the Modal/Eigenvalue Analysis tool in a commercial software environment.

Case	Zone of Spanish electric system	Inter-area mode		Variation
		ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
L_0: Base case (all lines disconnected)	-	0.16	0.20	-
L_{TOT}: all aforementioned lines are considered connected	-	0.36	0.20	0.20

Table 4-9: Sensitivity analysis – damping effect of connecting several HV lines on inter-area oscillations.

The results in Table 4-9 show that the topology actions implemented by Red Eléctrica had a beneficial impact. When assessed individually, their effect on damping varies, ranging from negligible changes in the damping ratio $\Delta\zeta$ (10^{-3} %) to more appreciable values of 10^{-1} %. The L_{TOT} simulation case, which considers all actions together, resulted in a non-negligible 0.20 % increase in damping ratio.

To illustrate this, an RMS simulation comparing the L_0 and L_{TOT} cases is presented below. The analysed frequency corresponds to the Carmona bus in southern Spain. The simulation covers a 40 s time window, considering both the load disconnection contingency and the OU stochastic process discussed in Subsection 3.1.

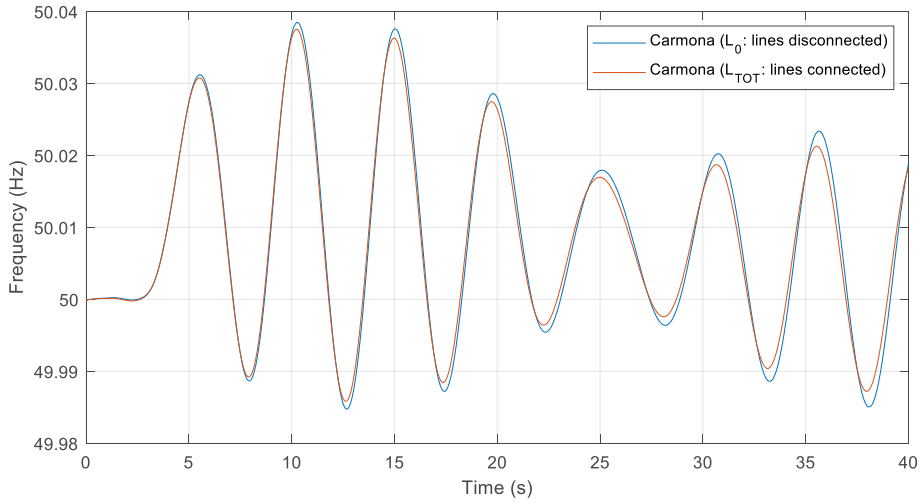


Figure 4-57: Impact of manoeuvres implemented by Red Eléctrica on the inter-area oscillations – L_0 vs L_{TOT} (load contingency).

Figure 4-57 reports the inter-area oscillations triggered by a large load disconnection event. In this case, no OU stochastic process is considered. It is possible to observe how an event of this kind can induce inter-area oscillation, with similar characteristics in terms of mode frequency (around 0.2 Hz) and a varying damping that oscillates between positive and negative values, with values consistent with those provided by the eigenvalue analysis. Figure 4-58 shows the RMS simulation for the inter-area oscillation triggered by the OU process associated with the 20 loads described previously.

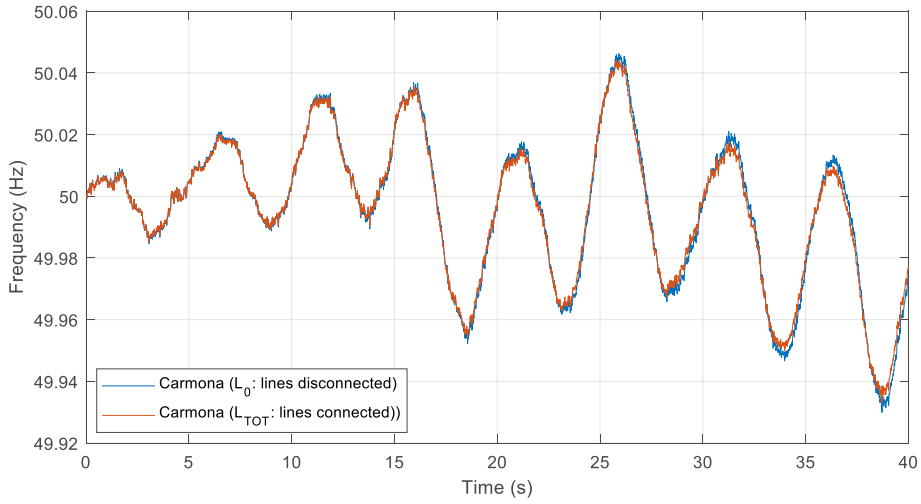


Figure 4-58: Impact of manoeuvres implemented by Red Eléctrica on inter-area oscillations – L_0 vs L_{TOT} (OU stochastic process).

The RMS results support the qualitative insights derived from the eigenvalue analysis: the implemented topology manoeuvres contribute positively to the mitigation of inter-area oscillations. Consistent with the eigenvalue analysis, the manoeuvres have a beneficial, if limited, effect, which appears reasonable given the scale of the CE system. Regarding the varying value of the damping ratio observed in both Figure 4-57 and Figure 4-58, similar considerations can be derived with respect to Subsection 3.1.

4.2.5.2.3 Performance of Vitoria's STATCOM 150 Mvar and POD Controller

This subsection assesses the effectiveness of the POD-Q control of Vitoria's STATCOM 150 Mvar. Vitoria's STATCOM can be controlled in voltage control mode (V-mode) or with constant reactive power control mode (Q-mode). Figure 4-59 shows the block diagram of the POD-Q controller of Vitoria's STATCOM 150 Mvar. The input to the POD-Q controller is the frequency deviation measured at the AC point of connection relative to the nominal value, defined as $\Delta\omega = \omega - \omega_0$ (pu) (with $\omega_0 = 1$ pu). The output of the POD-Q controller is a supplementary reactive current setpoint.

The reactive current injection reference for the STATCOM is given by:

$$i_Q^* = i_Q^{ref} + \Delta i_Q^{ref,POD} \qquad (4)$$

where:

- $\Delta i_Q^{ref,POD}$ is the supplementary reactive current setpoint provided by the POD-Q controller.
- If the STATCOM is operating in Q-mode, the reactive power control provides a reactive current setpoint of i_Q^{ref}, calculated as a function of the constant reactive power setpoint $Q^{ref} = Q_0$.
- If the STATCOM is operating in V-mode, the voltage control generates a reactive power reference, which is then translated into a reactive current reference i_Q^{ref}.

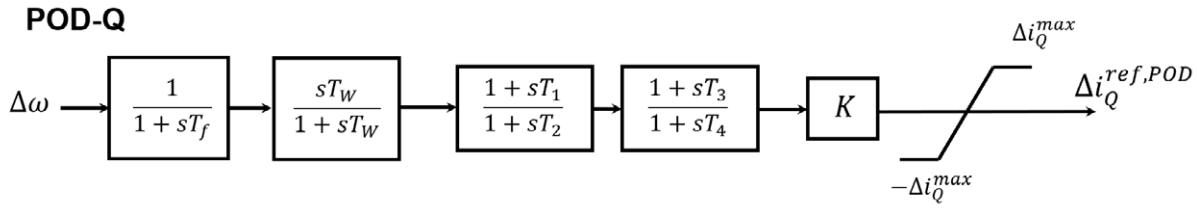


Figure 4-59: Vitoria's STATCOM 150 Mvar – POD-Q controller. Block diagram.

The following cases were compared for the V-mode/ Q-mode of Vitoria's STATCOM 150 Mvar:

- D_0: Base case: no STATCOM connected
- D_1: STATCOM with POD-Q off
- A_0/D_2: STATCOM with POD-Q on, corresponding to A_0

Table 4-10 shows the results when the STATCOM is in V-mode. The base case D_2 (A_0) is used as a first reference point, comparing the damping ratio to a case where the STATCOM is not connected (D_0) and a case where it is connected and the POD-Q control is deactivated (D_1). Note that case D_2 (A_0) is the representative scenario of the real situation during the event, when it was properly activated. It can be observed that if the STATCOM is not connected to the system (D_0), the damping ratio does not differ significantly from that in case D_2, with the STATCOM connected and POD on.

This result is unsurprising, as it is well established in the literature that introducing a new element without POD control into the system typically reduces the damping ratio due to increased grid stiffness and changes in load flow on nearby branches.

In fact, as shown in Table 4-10, if the POD-Q is not activated, as in case D_1, the damping ratio decreases by 0.08 % relative to the base case, proving the effectiveness of the control. It is worth highlighting that the activation of POD-Q control makes the presence of the STATCOM non-deteriorating relative to the base case. It must be noted that this STATCOM was planned with the main purpose of controlling voltage in the area. The POD function was added later.

Case	V-mode		
	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
D_0: No STATCOM connected	0.28	0.20	-
D_1: STATCOM, POD-Q off	0.19	0.20	– 0.08
D_2: STATCOM, POD-Q on (A_0)	0.28	0.20	0

Table 4-10: Sensitivity analysis – damping effect of the Vitoria STATCOM POD-Q control with V-mode control on inter-area oscillations.

The RMS simulation shown in Figure 4-60, in which the oscillation is triggered by the outage of the aforementioned load in the eastern part of the CE SA system, confirms the results obtained from the eigenvalue analysis. Specifically, when the POD of the STATCOM is out of service, the frequency at the Carmona bus exhibits larger fluctuations.

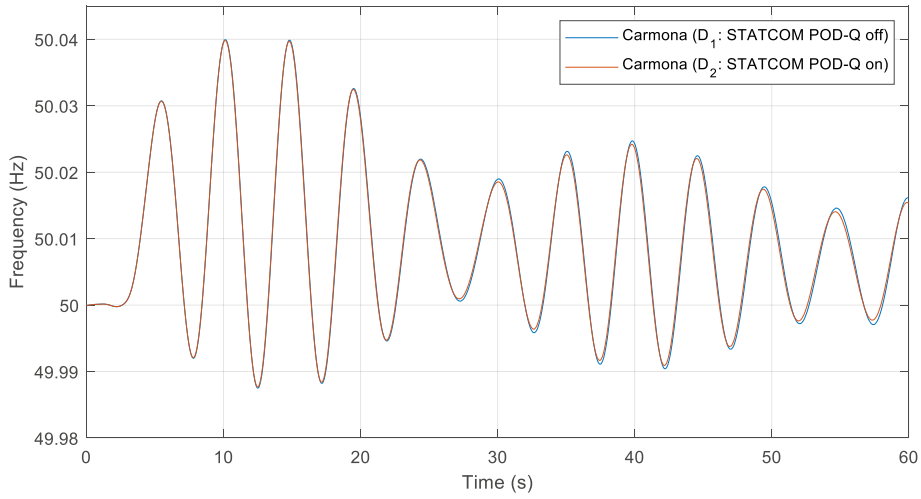


Figure 4-60: Impact of POD-Q control of Vitoria STATCOM on inter-area oscillations – D_1 vs D_2 (load contingency).

Figure 4-61 reports the reactive power provided by the Vitoria STATCOM with the POD activated and deactivated. It can be observed that the reactive power injection of the STATCOM presents greater variations when its POD controller is activated, specifically to damp inter-area oscillations. Notice that the reactive power injection of the STATCOM also varies when the POD is off, because it is in V-mode.

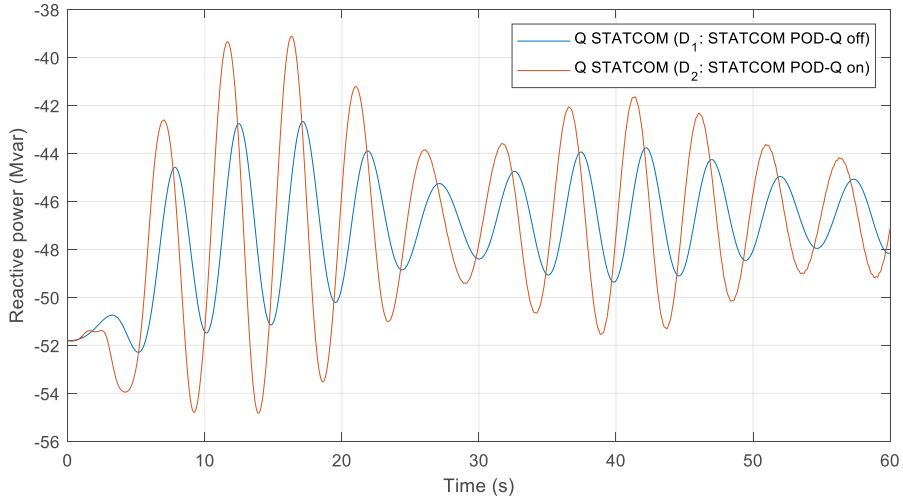


Figure 4-61: Q variation considering the Vitoria STATCOM in V-mode, POD on vs POD off – D_1 vs D_2 (load contingency).

To thoroughly assess the effectiveness of the STATCOM POD control, an eigenvalue analysis is conducted considering the Q-mode control of the STATCOM. The results presented in Table 4-11 show that, in this case as well, the POD control is effective and improves the damping capability of the inter-area oscillation system. The $\Delta\zeta$ of 0.06 % is comparable to that observed in the V-mode control case, confirming the consistency of the results.

Case	Q-mode		
	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
D_0: No STATCOM connected	0.28	0.20	-
D_1: STATCOM, POD-Q off	0.28	0.20	0.00
D_2: STATCOM, POD-Q on	0.34	0.20	0.06

Table 4-11: Sensitivity analysis – damping effect of Vitoria STATCOM POD-Q control with Q-mode control on inter-area oscillations.

A comparison of Table 4-10 and Table 4-11 reveals that, in terms of dampening inter-area oscillations, the Q-mode is more effective when the POD-Q controller is turned on. It must be emphasised that one of the primary functions of the STATCOM is to maintain voltage within a desired range, which can be achieved only by activating the V-mode control in conjunction with the POD-Q controller. The introduction of new STATCOMs into the Iberian system will therefore require additional adjustment following the described approach.

4.2.5.3 Assessment of Additional Potential Measures Against Inter-Area Oscillations

In this subsection, several additional sensitivity analyses are reported. These studies cover different operating scenarios based on what occurred on the day of the incident. As before, particular attention is devoted to the influence of several factors on the damping ratio of the inter-area mode, including the effect of power system stabiliser (PSS) control systems in both CCGT and nuclear power plants (NPPs), the increased presence of synchronous generators (SGs) in the Spanish transmission system, the impact of the system's equivalent inertia constant, the power flow through the AC Spain–France interconnection lines, and the potential role of synchronous condensers (SCs) and STATCOMs in Spanish transmission system current plans in mitigating future inter-area oscillations.

4.2.5.3.1 Effect of PSSs in CCGT Power Plants

In the case of CCGT power plants in service during the incident, the effect of their PSS is evaluated by deactivating the PSS and evaluating the damping ratio ζ relative to base case A_0. Additionally, for the sake of completeness, the combined effect of the PSS on all online CCGTs is computed. The results are reported in Table 4-12.

Case	Zone of Spanish electric system	Inter-area mode		Variation
		ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
A_0: Base case	–	0.28	0.20	–
$CCGT_{TOT}$: All PSSs put out of service	–	–0.84	0.20	–1.12

Table 4-12: Sensitivity analysis – effect of CCGT power plant PSS.

As expected, putting the PSS out of service reduces the damping ratio of the inter-area mode. Considering all PSSs out of service simultaneously provides a noticeable damping ratio decrease of –1.12 %, highlighting the importance of this stabilising control system and the necessity of proper tuning.

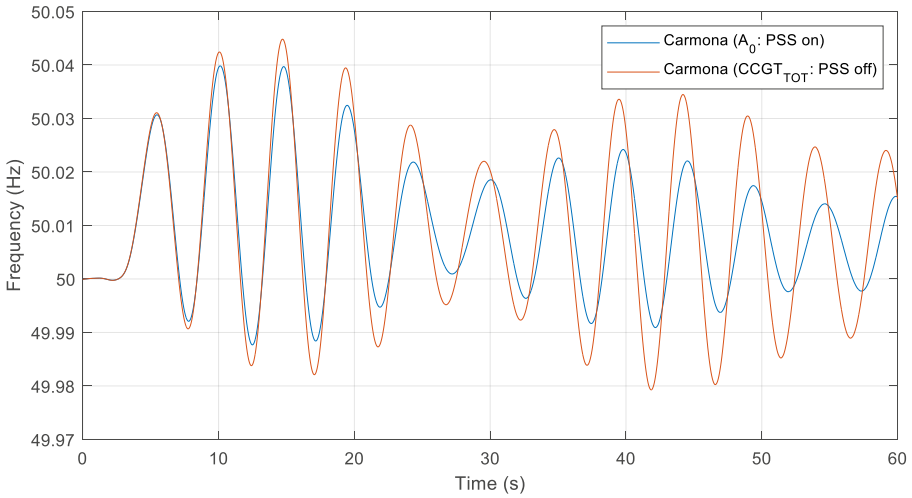


Figure 4-62: Impact of CCGT power plants PSS on inter-area oscillations – A_0 vs $CCGT_{TOT}$ (load contingency).

4.2.5.3.2 Effect of PSSs in NPPs

In this case study, the impact of a hypothetical PSS in NPPs on the damping ratio of the inter-area oscillation is assessed. As these power plants are not equipped with PSSs, it is important to mention that their tuning process is preliminary, yet consistent. In fact, the tuning process is usually handled by a dedicated working group and may take several months to complete. Therefore, if these stabilising devices were to be installed in NPPs, simulations using the final PSS parameters could yield slightly different results. Nevertheless, the outcomes obtained from the eigenvalue analysis tool are presented in Table 4-13.

Case	Zone of Spanish electric system	Inter-area mode		Variation
		ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
A_0: Base case	-	0.28	0.20	-
NPP_{TOT}: All PSSs in service	-	1.55	0.19	1.27

Table 4-13: Sensitivity analysis – effect of PSSs on connected NPPs.

As before, the combined effect of the PSSs in all NPPs is considered, resulting in a damping ratio increase of 1.27 %. Although based on preliminary PSS tuning, these results highlight the potential importance of these devices in stabilising the system during electromechanical oscillations.

As for other sensitivity analyses, the RMS simulation confirms the results obtained by the eigenvalue analysis. Figure 4-63 clearly shows that introducing PSSs into NPPs would help mitigate inter-area oscillations.

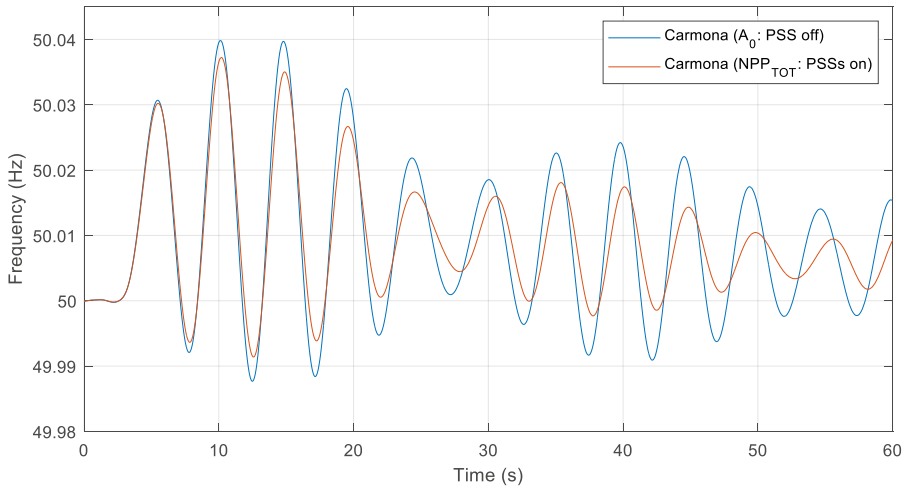


Figure 4-63: Impact of NPP PSSs on inter-area oscillations – A_0 vs NPP_{TOT} (load contingency).

4.2.5.3.3 Additional Synchronous Generators in the SPES System

This subsection assesses the effect of adding additional SGs to the Spanish transmission system on the damping ratio. For this purpose, 5 SGs – equipped with primary frequency control, AVR, and tuned PSS – are included in the calibrated model. An equivalent active power value is reduced in nearby CIG power plants. This allows the power flow through the interconnection AC lines to be maintained, avoiding any significant bias in this analysis, as power flow has a notable impact on the inter-area damping ratio.

The damping ratio provided by DIgSILENT PowerFactory's Eigenvalue Analysis tool was used for quantitative analysis. Table 4-14 reports the simulation results for their combined effect. The effect of connecting these additional generators with PSS is similar to that of activating and tuning the PSS in NPPs.

As expected, adding additional SG with correctly tuned PSSs to the Spanish transmission system helps mitigate the inter-area oscillation.

Case	Zone of Spanish electric system	Inter-area mode		Variation
		ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
A_0: Case base	-	0.28	0.20	-
SG_{TOT}: The 5 SGs connected	-	1.39	0.19	1.12

Table 4-14: Sensitivity analysis – effect of including additional synchronous generators on damping ratio

4.2.5.3.4 Effect of Inertia

In this sensitivity study, the influence of the system's equivalent inertia on the inter-area oscillation is assessed. For this purpose, the inertia of all SGs connected to the SPES is doubled and tripled. Obviously, increasing the inertia constant of each SG without changing any other parameter is not realistic.

It would be more realistic to act on the number of SGs connected to the system. However, this would introduce other effects besides the inertia constant, such as the presence of additional control systems (PSS, AVR, and primary frequency control), whereas this sensitivity analysis is focused on isolating the effect of inertia alone on the damping ratio.

Case	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
A_0: Base case	0.28	0.20	-
Doubled H	1.21	0.18	0.93
Tripled H	3.03	0.17	2.76

Table 4-15: Sensitivity analysis – effect of inertia on damping ratio.

The results presented in Table 4-15 suggest that doubling the value of the equivalent inertia constant of the Spanish system increases the damping ratio of the inter-area mode by 0.93 %, while tripling the inertia value results in an increase of 2.76 %. These results are unsurprising, as it is well established in the literature that low inertia levels can promote inter-area oscillations. However, even with a tripled equivalent inertia of the Spanish system, the damping ratio remains relatively small, at around 3 %. It can be observed that this magnitude is comparable to that obtained by tuning and activating PSSs on synchronous units or implementing other measures.

Therefore, increasing inertia solely to damp inter-area oscillations is not a realistic option. These results indicate that while the role of inertia is non-negligible, it is not sufficient on its own to avoid the occurrence of inter-area oscillations.

Additionally, it can be observed that the frequency of the inter-area mode decreases as the system inertia increases. This can be explained by considering the equivalent mass of the Spanish system: as this mass increases, oscillations become slower.

4.2.5.3.5 Effect of Power Flow Through AC Interconnection Lines Between Spain and France

This subsection assesses the effects of the power flow through the AC interconnection lines between Spain and France by changing the active power production of several CIGs. The results, considering four different interchange values, are reported in Table 4-16. These include decreases of approximately 200 MW and 400 MW, as well as increases of approximately 200 MW and 450 MW.

It should be noted that the interchange is considered positive when Spain is exporting towards France, and negative when importing power from France. The same convention is applied to active power variations ΔP.

As expected, reducing the power flow through the AC interconnections helps mitigate inter-area oscillations. Conversely, increasing the interchange power decreases the damping ratio. This result is expected and consistent with theoretical results reported in the literature [10].

Case	ΔP interchange through AC lines (MW)	Inter-area mode		Variation
		ζ (%)	f_{mode} (Hz)	Δζ (%)
A_0: Base case	-	0.28	0.20	-
CIG ΔP: –250 MW	–192	0.75	0.20	0.48
CIG ΔP: –500 MW	–394	1.18	0.20	0.90
CIG ΔP: +250 MW	210	–0.14	0.20	–0.41
CIG ΔP: +500 MW	446	–0.52	0.20	–0.79

Table 4-16: Sensitivity analysis - effect of power flow through AC interconnection lines (Spain-France)

4.2.5.3.6 Effect of Synchronous Condensers

This sensitivity analysis assesses the effect on the damping ratio of including the eight SCs included in Spanish transmission system 2020 – 2026 MAP2 plan (July 2025), which will be installed in the SPES over the next few years.

As expected, including the SCs has a favourable effect on oscillation mitigation. It is worth noting that in this study, it is assumed that the SCs are equipped with PSSs, which is a common procedure.

Nevertheless, the PSSs must still be calibrated, as several SC characteristics are still unknown (SCs are currently under commission). In this analysis, the SCs are represented using a generic model; therefore, the results should be considered preliminary. However, similar outcomes are expected when employing more detailed models. As shown in Figure 4-64, the RMS simulation yields consistent results, with SCs showing a favourable influence on damping inter-area oscillations.

Case	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	Δζ (%)
A_0: Case base	0.28	0.20	-
A_0+ 8 SCs (planification 2020-2026 MAP2, July 2025)	0.97	0.20	0.69

Table 4-17: Sensitivity analysis - effect of SCs on damping ratio

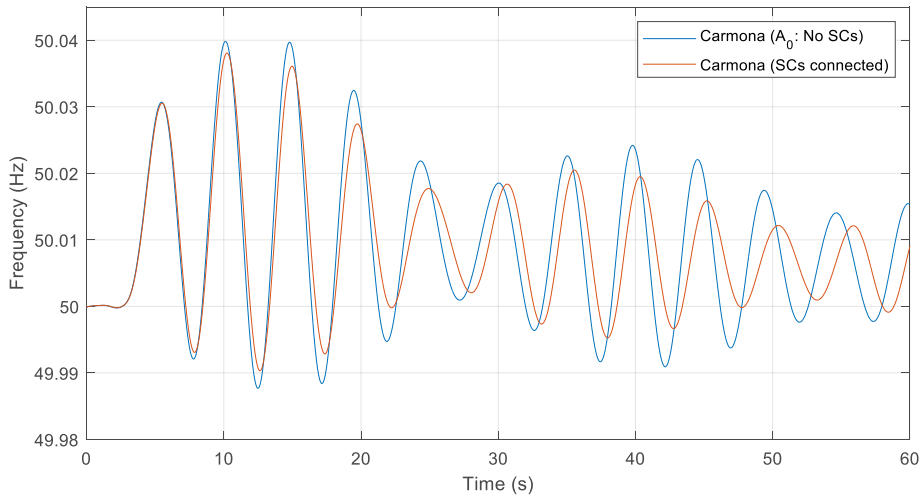


Figure 4-64: Impact of SCs on inter-area oscillations (load contingency).

It is worth mentioning that the inertia constant of each SC is 6 s (related to the nominal apparent power of 250 MVA), which corresponds to an additional 1.5 GW of kinetic energy available in the system for each SC. This increase in inertia, along with the presence of PSS and AVR controls, contributes positively to the improvement of the damping ratio.

4.2.5.3.7 Impact of STATCOMs

This subsection analyses the impact of the STATCOMs in the Iberian Peninsula power system on inter-area oscillation damping:

- STATCOM Vitoria 150 Mvar (North) (May 2024)
- STATCOM Tabernas 150 Mvar (South) (July 2025)
- STATCOM Lousame 150 Mvar (North) (Jan 2026)
- STATCOM Moraleja 150 Mvar (Centre) (Feb 2026)

Vitoria's STATCOM was the only one in service during the incident of 28 April 2025, with the remainder commissioned over the following months. The four STATCOMs are equipped with POD controllers that operate through reactive power modulation. Under normal operating conditions, the STATCOMs are operated in V-mode, which is the control mode assumed in the simulations.

The following cases were compared:

- E_0: No STATCOMs connected
- E_1: All STATCOMs connected with POD-Q off
- E_2: All STATCOMs connected with POD-Q on

In the simulations, the STATCOMs are in V-mode control.

Table 4-18 shows the damping ratio of the East–Centre–West inter-area oscillation in the CE power system. As expected, STATCOMs with POD controllers help damp the critical inter-area oscillation.

Case	V-mode		
	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
E_0: No STATCOMs connected	0.28	0.20	-
E_1: All STATCOMs connected with POD-Q off	0.19	0.20	–0.09
E_2: All STATCOMs connected with POD-Q on	0.81	0.20	0.53

Table 4-18: Sensitivity analysis – effect of the STATCOMs and their POD controllers.

4.2.5.3.8 Impact of VSC Converters with POD Controllers (BESS, UCAP, E-STATCOMs, etc.)

This subsection analyses the impact of generic voltage source converters (VSC) equipped with POD-P/POD-Q controllers on inter-area oscillation damping. These generic VSC converters could represent different technology enablers, such as:

- An energy storage system (ESS) with a power converter, such as a battery energy storage system (BESS) or ultracapacitor (UCAP), equipped with POD-P/POD-Q controllers
- E-STATCOMs: POD-P/POD-Q controllers

Two VSC converters with grid-following (GFL) control, located arbitrarily within the Iberian Peninsula power system, were analysed:

- VSC-A (100 MVA): Located at Carmona 400 kV in the southern area of the Ibernia Peninsula
- VSC-B (100 MVA): Located at La Cereal 400 kV in the central area of the Ibernia Peninsula power system

These two locations were selected for illustration purposes in this study. Generic dynamic models of the VSC converters and their controllers were used. Saturation parameter of POD-P/POD-Q controllers is assumed to be ±10 % (% relative to the rating power base).

Table 4-19 shows the damping ratio of the East–Centre–West inter-area oscillation in the CE power system. The results show that the POD-P and POD-Q controllers of both VSC converters are effective in damping the critical inter-area oscillation in the CE power system. In general, POD-P controllers are more effective than POD-Q controllers. In addition, VSC converters can have both POD-P/POD-Q controllers activated simultaneously, producing significant improvements.

Therefore, technology enablers based on VSC converters, such as BESSs, UCAPs, or E-STATCOMs, could be an attractive solution to help damp inter-area oscillations in the CE power system through POD-P/POD-Q controllers.

Case	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
Base case: PODs off in VSC-A and VSC-B	0.28	0.20	-
VSC-A: POD-P	1.35	0.20	1.07
VSC-A: POD-Q	0.38	0.20	0.10
VSC-A: POD-PQ	1.45	0.20	1.17
VSC-B: POD-P	0.99	0.20	0.71
VSC-B: POD-Q	0.40	0.20	0.12
VSC-B: POD-PQ	1.11	0.20	0.83
VSC-A, VSC-B: POD-PQ	2.12	0.20	1.84

Table 4-19: Sensitivity analysis – effect of VSC converters with POD-P/POD-Q controllers.

4.2.5.3.9 Impact of Several Devices to Be Included in the Future

This subsection assesses the joint effect of the devices scheduled for inclusion in the SPES over the next few years, i.e. the eight SCs (see Subsection 4.2.5.3.6), four STATCOMs (see Subsection 4.2.5.3.7), and two generic VSC systems equipped with POD controls (see Subsection 4.2.5.3.8). Furthermore, for this analysis, the POD-Q of VSC–HVDC INELFE-1 is assumed to be in service. While the impact of these devices/controls on the inter-area modes is assessed individually in the previous subsections, this analysis aims to compare the scenario of the incident of 28 April 2025 at 12:21 against a hypothetical future scenario where these elements are in service.

The eigenvalue analysis results in Table 4-20 show a significant increase in damping ratio, around 2.6 %, indicating that these devices would have a non-negligible positive effect in damping inter-area oscillations.

The time-domain simulation in Figure 4-65 confirms the previous observations, as the oscillations are damped more rapidly and therefore reach smaller amplitudes.

Of course, this analysis does not aim to provide a fully comprehensive representation of a future SPES scenario, as the network model is based on the state estimator of 28 April 2025 at 12:21 (as detailed in Subsection 4.2.2), rather than on network planning assumptions. Nevertheless, these results provide a useful indication of the combined impact of the aforementioned devices.

Case	V-mode		
	Inter-area mode		Variation
	ζ (%)	f_{mode} (Hz)	$\Delta\zeta$ (%)
A_0: Base case (as of 28 April 2025 at 12:21)	0.28	0.20	-
Hypothetic future scenario	2.90	0.24	2.62

Table 4-20: Sensitivity analysis – Effect of the SCs, STACOMs and other VSC equipped with POD controls

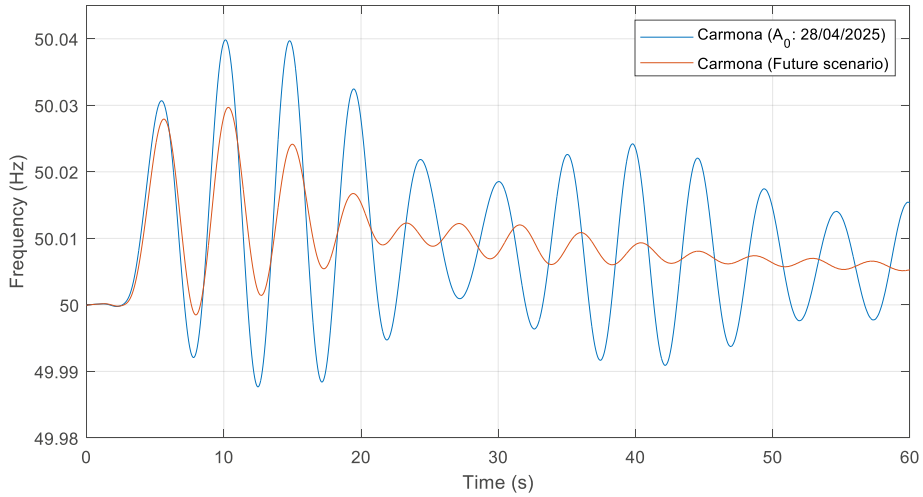


Figure 4-65: Impact of SCs, STACOMS, and other VSCs equipped with POD controls (load contingency).

4.2.5.4 Conclusion

This chapter aimed to analyse the oscillations, the effects of applied countermeasures on both forced and inter-area oscillations during the event. In addition, potential measures for future implementation were assessed. Based on the simulations conducted and results obtained, it can be concluded that:

Forced oscillations

- The applied measures had a minor but positive impact on the observed **forced oscillation** of 0.63 Hz during the incident, even though these measures were not designed to dampen oscillations in this frequency range. This includes changing the control mode of INELFE-1, decreasing exports in the AC system, and topological changes (see Subsection 4.2.5.1.1).
- The most effective way to prevent forced oscillations is to localise their source and apply targeted countermeasures. This might include activating or deactivating controls, changing unit setpoints, or disconnecting the sources of forced oscillations. For this purpose, operational oscillation monitoring and detection must be improved,and effective countermeasures must be ready.Neither was available in the system during operation on 28 April.

Inter-area oscillations

- The operational measures applied during the incident to mitigate **inter-area oscillations** can be confirmed as effective (see Subsection 4.2.5.2). This includes the control mode and POD control of INELFE-1, reduction of exports in the AC system, topological changes, and the performance of the Vitoria STATCOM.
- The POD-P/POD-Q controllers of the INELFE-1 HVDC link have a positive impact on damping inter-area oscillations. The POD-Q controller was automatically disabled during the first 0.6 Hz oscillatory event due to a logic condition in the controller (repeated saturations in upper/lower POD-Q limits) to prevent it from malfunctioning. It would have been desirable to have the POD-Q controller enabled during the second 0.2 Hz inter-area oscillation. Thus, it should be ensured that the POD-Q is active during similar events, either through the use of alarms for POD-Q deactivation, automated reactivation of the controller, or by studying ways to avoid deactivation when limits are reached.
- Several additional measures to improve damping of inter-area oscillations have been identified, which have a significant impact. This includes installing and tuning of PSS on nuclear power plants, installation of SCs, installation of additional STATCOMs with POD-Q and VSC with POD-P/-Q. Activating additional synchronous machines or increasing the level of inertia also has a positive impact on the damping of inter-area oscillation, but cannot be considered as an efficient option taking into account the more efficient effect of planned synchous condensers and STATCOMs.

The analysis finally highlighted the following necessary actions at the system level:

» 1. introduce TSO coordinated inspections and tests at the field level to verify and certify (eventually with the involvement of third party certification bodies) the stability and mandatory services required by the Grid Code of generators BESS and in the future of new types of loads (electrolyzers, data centers, etc.);

» 2. intensify high-frequency and power quality monitoring tools at the level of individual relevant plants (generators, loads) and significant sub-transmission/distribution areas;

» 3. deepen the behavior of converter driven instability with system studies and laboratory tests, including laboratory and site acceptance tests to new generators and significant loads;

» 4. evaluate, with ad hoc studies, the introduction of defense systems to manage automatic containment actions and improve oscillation damping;

» 5. perform periodic tuning of power system stabilisers (PODs);

» 6. with reference to existing generators, evaluate any needed improvement of power quality monitoring system and needed intervention on plant control systems to improve stability;7. extend the mandatory requirement of Power System Stabiliser on new or existing power plants having a significant sensitivity on the system damping of oscillations;

» 8. to support system stability, it is needed to improve requirements for demand/generation/HVDC and FACTS to avoid converter driven instability phenomena.

4.3 Analysis of Flows Between TSO and DSO Grids in Spain

4.3.1 Scope of the Chapter and Data Basis

The scope of this chapter is to analyse the relevant factors that influenced the TSO–DSO flows in Spain, particularly after the oscillatory periods and during the 12:32 –12:33 timeframe on 28 April 2025.

To perform the analysis, the following data from Spain has been used:

- TSO–DSO exchanges per region (P, Q) // 4 sec resolution
- Load per region (P) // 4 sec resolution
- aFRR activations (P) // 4 sec resolution
- Datasets from two PV inverter manufacturers for each region of Spain
  - Dataset manufacturer M#1, aggregated with 5 min resolution
  - Dataset manufacturer M#2, aggregated with 10 sec resolution
- DSO data from selected feeders // 12 sec resolution
- TSO data from selected feeders // 8 sec resolution

Additionally, to analyse the effect on voltage dependency, typical load parameters from the literature have been used.

4.3.2 Relevant Factors Influencing TSO–DSO Flows

The changes in TSO–DSO flows that occurred on 28 April 2025 (particularly after the two oscillatory periods and between 12:32 and 12:33) are the result of complex interactions among various factors within the distribution and transmission grids.

Since no single factor can fully explain the deviating TSO–DSO flow changes on 28 April 2025 (see Figure 2.13), it is important to provide an overview that summarises the relationships among the relevant factors influencing TSO–DSO flows (Figure 4-66). This chapter focuses on the description of the relevant factors in the distribution grids (bullet points in black-coloured box). However, it should be noted that these factors also apply to transmission grids with directly connected grid users (bullet points in grey).

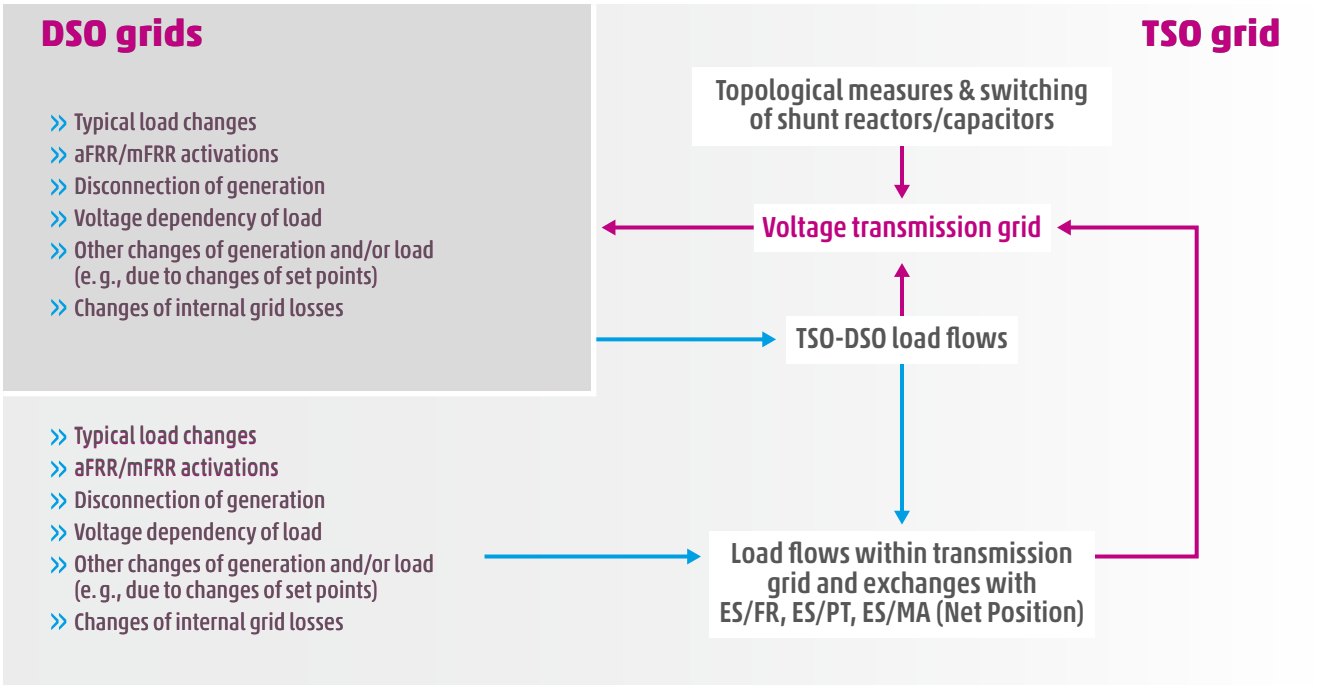


Figure 4-66: Overview of relationships among relevant factors influencing the TSO–DSO flows.

Changes in the distribution grids, such as the activation of reserves (aFRR, mFRR), the disconnection of generation units (including embedded generation units <1 MW), or other effects (e.g. changes of set points of generation units within BRPs or P/U control of embedded generation units), influence the TSO–DSO flows, which in turn have an impact on the voltage and the regional flows across the transmission system.

The impact of global TSO–DSO flow changes on regional flows (and, hence, transmission line loadings and reactive power demand) across the Spanish transmission system on 28 April 2025 is illustrated in Figure 4-67. This figure shows that the active power flows through two 400 kV transmission lines (Eliana-Plana 400 kV and D. Rodrigo-Guillena 400 kV, whose locations are indicated in Figure 4-68[30]) are negatively correlated with changes in total TSO–DSO flows. In other words, the active power flows through the transmission lines and the total TSO–DSO flows (and load) behave almost in the opposite direction.

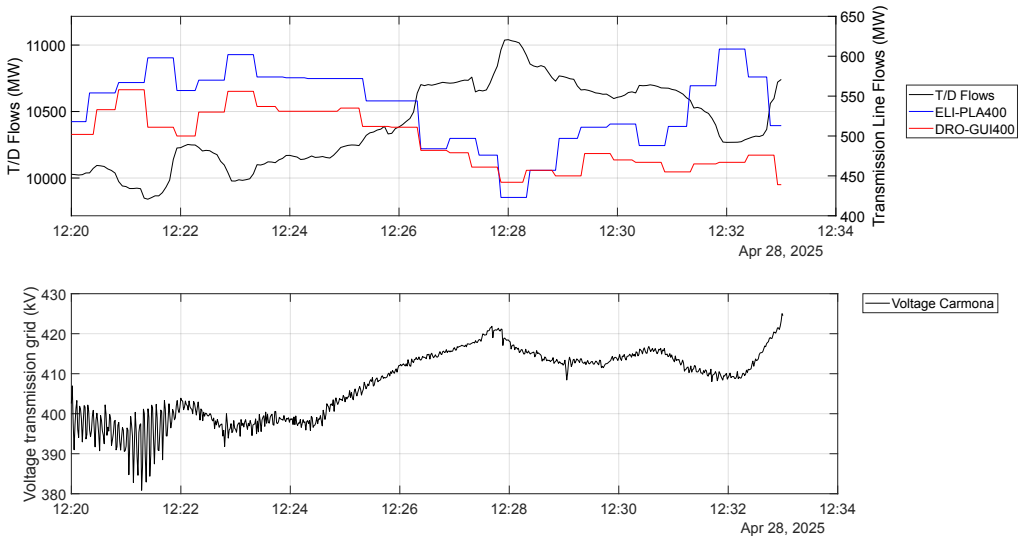


Figure 4-67: Total TSO–DSO flows, active power flows through two transmission 400 kV lines (Eliana-Plana 400 kV and D. Rodrigo-Guillena 400 kV) and voltage at 400 kV Carmona substation (Source: Red Eléctrica)

Figure 4-68: Location of the two selected transmission 400 kV lines (Eliana-Plana 400 kV and D. Rodrigo-Guillena 400 kV), in blue (Source: Red Eléctrica)

30 These two transmission lines were selected, because the increase in TSO-DSO flow was more prominent in the South of Spain. Transmission line D. Rodrigo-Guillena 400 kV is located in the South of Spain, while transmission line Eliana-Plana 400 kV is located on the axis that connects the South with the Northeast of Spain.

Since there is also a certain amount of feedback of the voltage from the transmission grid to the distribution grids – e.g., voltage dependency of loads or disconnections of grid users due to voltage variations at medium- or low-voltage DSO feeders – a fully exhaustive (retrospective) determination of all factors influencing the TSO–DSO flows is challenging.

In the next sections, detailed analyses of the following (possible) factors influencing the TSO–DSO flows and load have been conducted:

- aFRR/mFRR activations;
- disconnection of generation units <1 MW within the DSO grids;
- voltage dependency of load.

4.3.3 Analysis of Relevant Factors Influencing the TSO–DSO Flows

4.3.3.1 Reserve Activations (aFRR, mFRR)

As shown in Figure 4-69, there is no clear correlation between aFRR activations and the sum of TSO–DSO flows in Spain during the 12:00 – 12:33 time period. The lack of correlation can be explained by the fact that most aFRR providers are on the transmission grid, with fewer than 20 % in distribution grids.

Although no mFRR activation data (with the same time resolution as the available aFRR activation dataset) was available for analysis, similar conclusions can be drawn regarding the correlation between mFRR activations and the sum of TSO–DSO flows within Spain for the 12:00 – 12:33 time period.

An assessment of the mFRR involved between 12:00 and 12:30 shows that the share of mFRR in the distribution grids is around 10 – 20 %, while between 12:30 – 12:45 the share is <10 %. Therefore, it is reasonable to apply the conclusions drawn from aFRR to mFRR (Table 4-21).

Timestamp	Requirement*	Flow PT>ES	Offers (Spain)*	Share in DSO grids (%)**
[12:00, 12:15)	630 MW	150 MW	480 MW	19 %
[12:15, 12:30)	990 MW	150 MW	840 MW	16 %
[12:30, 12:45)	450 MW	150 MW	300 MW	9 %

Table 4-21: Information regarding mFRR (MARI) in Spain

(*) A value > 0 means a positive reserve (upwards direction).
(**) Estimation is necessary because certain programming units (UPs) have physical units connected to both the transmission and distribution grids. The UP's owner determines how the requirement is allocated among its units.

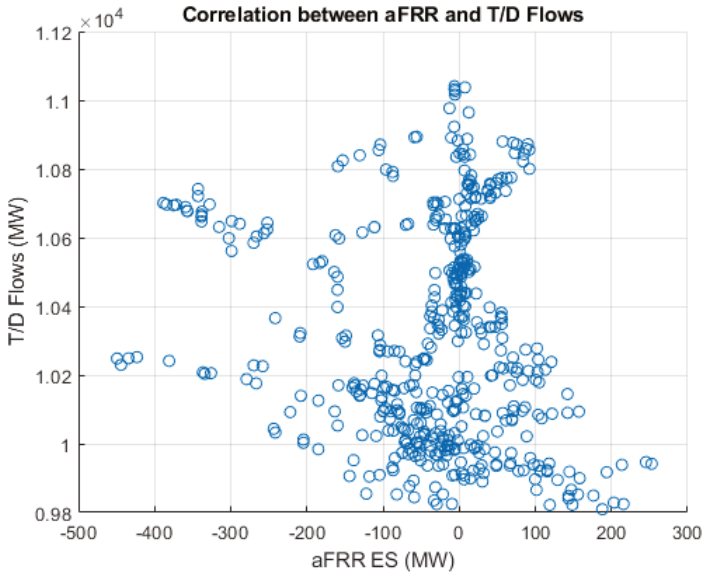


Figure 4-69: Correlation between aFRR activations (x-axis, (+) sign = positive/upwards aFRR activation, (-) sign = negative/downwards aFRR activation) and sum of TSO-DSO flows within Spain (y-axis) (Source: Red Eléctrica)

4.3.3.2 Disconnection of Generation <1 MW ("Embedded Generation")

4.3.3.2.1 Description of data used and assumptions

In the course of preparing this Final Report, the Expert Panel received data from two PV inverter manufacturers, enabling the analysis of possible disconnections of embedded generation units <1 MW on 28 April 2025. Both datasets contain only information about inverters connected to low-voltage (LV) grids.

The dataset provided from PV inverter manufacturer #1 (M#1) includes the following information, which is aggregated per region in 5-minute intervals:

- Number of reporting devices
- Aggregated nominal active power capacity
- Aggregated active and reactive power
- Number of overvoltage warnings
- Number of overvoltage trips
- Number of undervoltage trips
- Number of overfrequency trips
- Number of underfrequency trips

The dataset provided from PV inverter manufacturer #2 (M#2) includes the following information, which is aggregated per region in 10-second intervals:

- Number of reporting devices
- Number of overvoltage trips
- Number of undervoltage trips
- Number of overfrequency trips
- Number of underfrequency trips

The standard protection settings for overvoltage trips of inverters are the same for both manufacturers:

- U>: 253 V/1.5 s
- U>>: 264 V/0.2 s

The current Spanish grid code sets a minimum grid reconnection time of 180 seconds (3 minutes) after tripping. Furthermore, the PV inverter manufacturer M#2 explained that an additional delay of approximately 60 seconds is required before reconnection takes place, during which the inverter performs several technical tests. Therefore, in the context of this analysis (see in particular Section 4.3.3.2.2.3), it is considered that the total time between disconnection and reconnection of an inverter is 240 seconds (4 minutes).

The installed PV inverter capacity in Spain reported by the two manufacturers amounts to approximately 690 MW. Red Eléctrica estimates[31] that the total installed PV capacity in Spain in April 2025 for facilities <1 MW was approximately 6,478 MW,of which approximately 75 % (4,860 MW) is connected to LV grids. Hence, the combined dataset provided by the two manufacturers represents approximately 14 %[32] of the total installed PV inverter capacity connected to LV grids in Spain.

Red Eléctrica estimates that, on 28 April 2025, from 10:00 onwards, the actual aggregated production from PV facilities <1 MW was approximately 4,447 MW (69 % of the installed capacity). Applying the same ratio only to PV facilities connected to LV grids, it follows that the PV facilities connected to LV grids had an aggregated production of approximately 3,340 MW.

4.3.3.2.2 Analysis

This section presents an analysis of the data provided by the two inverter manufacturers. The analysis focuses on inverter trips from 10:00 until the blackout, aiming to assess whether there is a link between these disconnections, the transmission grid voltage and changes in the TSO–DSO flow. In addition, this section includes an estimate of the total disconnected PV production resulting from these trips.

4.3.3.2.2.1 Analysis of disconnections aggregated in 5-minute intervals

As explained in Section 4.3.3.2.1, inverter manufacturer M#1 provided their data in 5-minute intervals, while inverter manufacturer M#2 provided their data in 10-second intervals. To compare the two datasets, the data received from manufacturer M#2 were therefore aggregated into 5-minute periods (by summing the data from 30 consecutive 10-second intervals).

Global view

Figures 4-70 and 4-71 show, for each 5-minute period, the proportion of inverter trips due to overvoltage, separately for the two manufacturers. This proportion is calculated across all regions, i.e., for Spain as a whole. The figures also show the voltage evolution on the 400 kV grid (top) and the changes in TSO–DSO flows (bottom).

For both datasets, similar correlations (trends) can be observed between the number of overvoltage-related trips, grid voltage and changes in TSO–DSO flows. For example, it appears that a higher proportion of overvoltage trips is registered during the 11:05 – 11:10 period, during which the TSO–DSO flows and the grid voltage increased significantly and a first oscillatory episode took place around 11:06. A similar trend/correlation is observed during the 11:25 – 11:50 period, as well as during and/or after the two major oscillatory periods (12:05 – 12:10 and 12:25 – 12:30).

Hence, it cannot be excluded that a certain share of PV inverter overvoltage trips was associated to these three oscillation episodes. However, the 5-minute aggregation of overvoltage-related trips is not sufficiently detailed to accurately assess the possible link between these phenomena.

Furthermore, because the blackout occurred in the middle of a 5-minute period (12:30 – 12:35), an analysis based on a 5-minute aggregation does not allow for deriving any correlations between the PV inverter trips and the phenomena that occurred during the incident.

31 This estimation is based on the following: 72 % corresponds to installations reported by distribution system operators to Red Eléctrica and which have their self-consumption status incorporated into their electricity supply contracts. The remaining 28 % corresponds to estimations made by Red Eléctrica.

32 The PV capacity reported by Red Eléctrica is the total peak capacity of PV panels, while the manufacturers report the aggregated peak capacity of their PV inverters. The capacities of PV panels and PV inverter for a given facility are usually different, but this difference can vary significantly from one facility to another. Therefore, in absence of information about the typical difference between the PV inverter and PV panel capacities, the Expert Panel considers them as being equivalent in the context of this analysis.

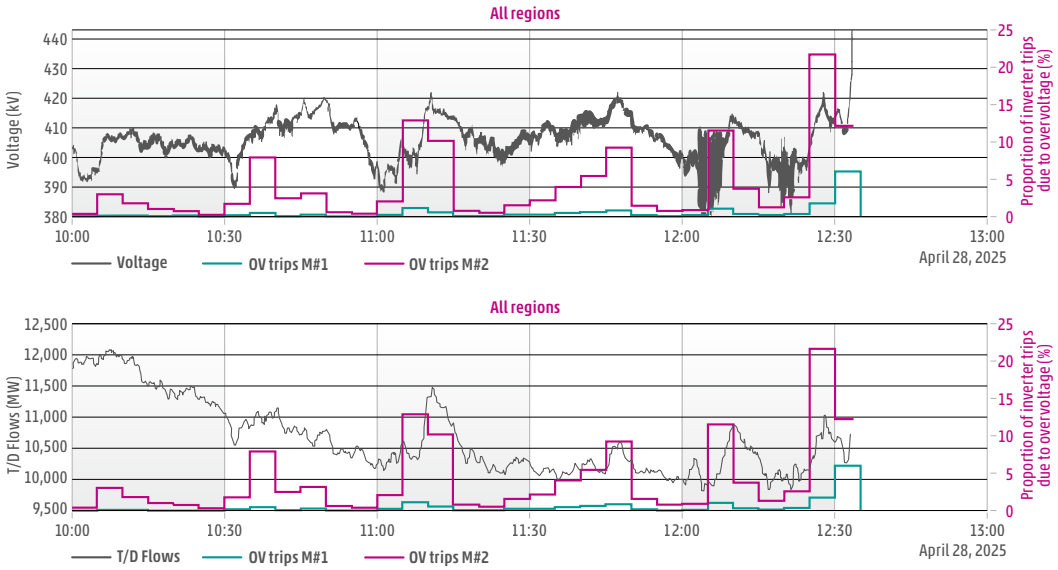


Figure 4-70: 10:00 –12:35 time period/Correlation between voltage at 400 kV Carmona substation, sum of TSO–DSO flows and proportion of inverter trips in Spain due to overvoltage (Source: Red Eléctrica, PV inverter manufacturers M#1 and M#2)

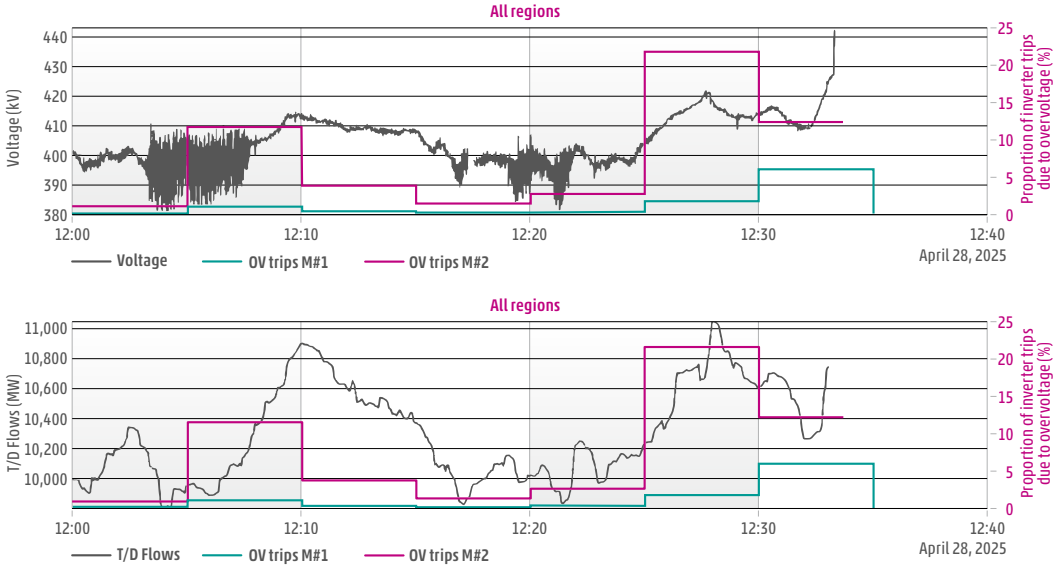


Figure 4-71: 12:00 –12:35 time period/Correlation between voltage at 400 kV Carmona substation, sum of TSO–DSO flows and proportion of inverter trips in Spain due to overvoltage (Source: Red Eléctrica, PV inverter manufacturers M#1 and M#2)

To gain a complete overview of the voltage-related disconnection behaviour of the inverter fleets of both inverter manufacturers, Figure 4-72 presents the proportion of inverter trips due to undervoltage during the 10:00 –12:35 period.

It appears that the number of undervoltage trips is close to zero throughout the period, except in the last time interval (12:30 –12:35), when the voltage collapsed just before the blackout. Therefore, it can be concluded that undervoltage trips did not play a significant role in changes in TSO–DSO flows.

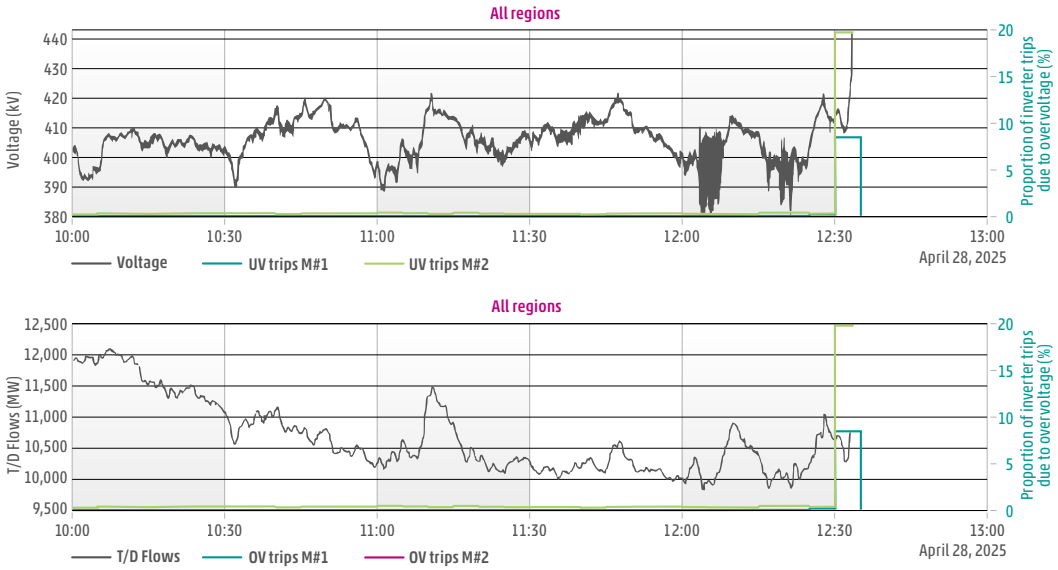


Figure 4-72: 10:00 –12:35 time period/Correlation between voltage at 400 kV Carmona substation, sum of TSO–DSO flows and proportion of inverter trips in Spain due to undervoltage (Source: Red Eléctrica, PV inverter manufacturers M#1 and M#2)

To conclude the analysis of the inverter data aggregated by 5-minute intervals, it is important to note that, while the trends observed above are similar for both datasets, the disconnection rates due to overvoltage reported by the two inverter manufacturers are significantly different, at all timeframes (manufacturer M#2 reports many more trips than manufacturer M#1). The Expert Panel does not have any information that would explain this difference, and researching the reasons exceeds the scope of the Expert Panel's investigation.

Regional view

This section analyses differences between regions in Spain in terms of installed PV capacity, inverter trips and TSO–DSO flow changes.

Figure 4-73 illustrates the regional distribution of the installed capacity of the inverter fleet of both manufacturers, compared with Red Eléctrica's estimated regional distribution of the total PV capacity connected to LV grids. Although some differences can be seen among the three maps, the distribution of the inverter fleets of both manufacturers appears similar to that of the entire population of LV-connected PV units across the country.

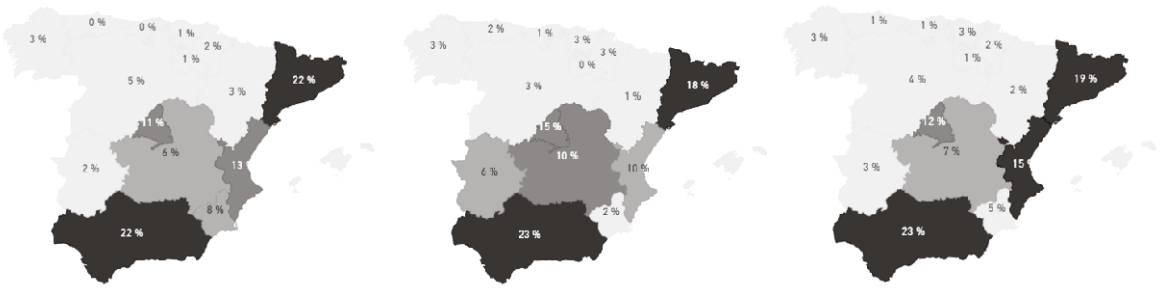


Figure 4-73: Regional distribution (as a percentage) of the fleet of PV inverters of both manufacturers (left: manufacturer M#1; middle: manufacturer M#2; right: Red Eléctrica's estimation of the total PV capacity connected at low-voltage)

Figures 4-74 and 4-75 compare the regional distribution of the overvoltage trips reported by the two inverter manufacturers with the regional distribution of the change in TSO–DSO flows measured by Red Eléctrica during the 5-minute periods of 12:05 – 12:10 and 12:25 – 12:30, respectively.

From Figure 4-74, it appears that the proportion of overvoltage trips is by far the highest in Andalusia, which is also the region with the largest change in TSO–DSO flow during the concerned time period. However, the share of trips in Andalusia relative to the total number of trips in Spain is much higher in both inverter datasets than the share of flow change in Andalusia relative to the total TSO–DSO flow change across Spain. In other regions, greater differences between datasets are visible, which can be partially explained by differences in the proportions of installed inverters across regions, as seen in Figure 4-74.

Furthermore, comparing the maps on the right side of Figures 4-74 and 4-75 prompts the conclusion that, although the percentages are different, there is relatively good correlation between the regions with the largest share of small PV units and the regions with high TSO–DSO flow changes during the concerned time period, namely Andalusia, Cataluña, Valencia, Madrid and Castilla La Mancha.

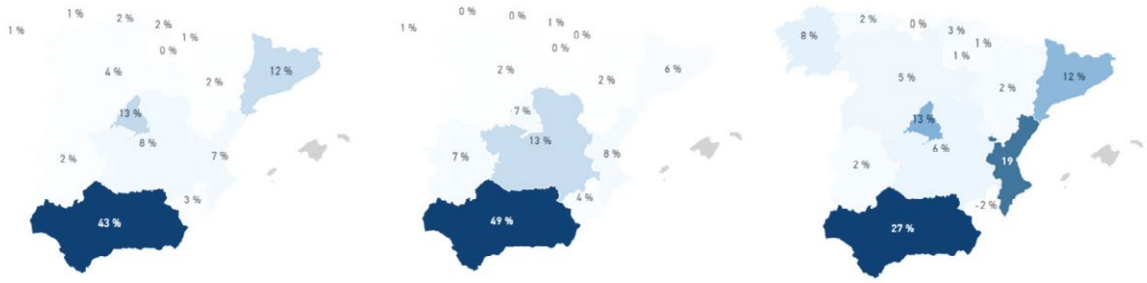


Figure 4-74: 12:05 – 12:10 time period/left and centre: Regional distribution of the PV inverter overvoltage trips (as a percentage of all inverters) reported by M#1 and M#2, respectively; right: regional distribution of the TSO–DSO flow changes as a percentage of the total TSO–DSO flow change throughout Spain

Similar observations can be made for the 12:25 – 12:30 time period, as illustrated in Figure 4-75. There is generally better alignment between the percentages of trips reported by the two manufacturers per region and the regional distribution of flow changes.

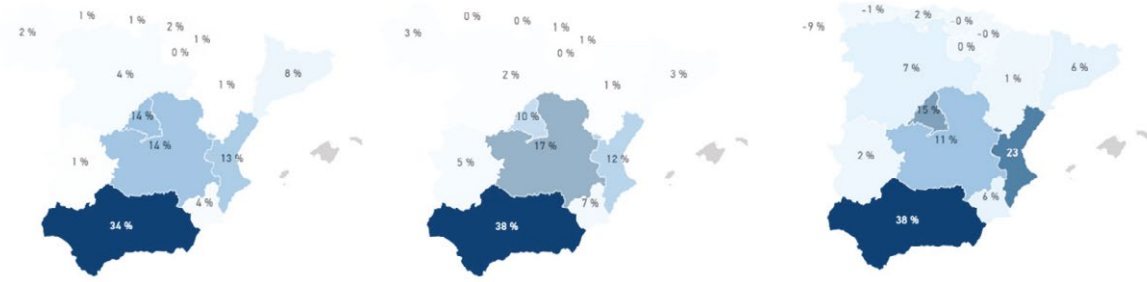


Figure 4-75: 12:25 – 12:30 time period/left and centre: Regional distribution of the PV inverter overvoltage trips (as a percentage of all inverters) reported by M#1 and M#2, respectively; right: regional distribution of the TSO–DSO flow changes as a percentage of the total TSO–DSO flow change throughout Spain

4.3.3.2.2.2 Analysis of disconnections with a 10-second resolution

The data from manufacturer M#2 were provided at 10-second granularity, allowing a much finer analysis of the possible link between inverter trips, grid voltage and TSO–DSO flow changes.

Figures 4-76, 4-77 and 4-78 show the proportion of inverter trips from manufacturer M#2 due to overvoltage, for each 10-second period for three periods of different durations (10:00 –12:35, 12:00 –12:35 and 12:30 –12:35).

This proportion is once again calculated for all regions together, i.e., for Spain as a whole. The figures also show the voltage evolution on the 400 kV grid (top) and the changes in TSO–DSO flows (bottom).

Thanks to the higher temporal resolution, it is possible to confirm that the majority of overvoltage trips are correlated with episodes of voltage increases, particularly during the 11:08 –11:12 period and after the two major oscillatory periods (12:07 –12:10 and 12:25 –12:28).

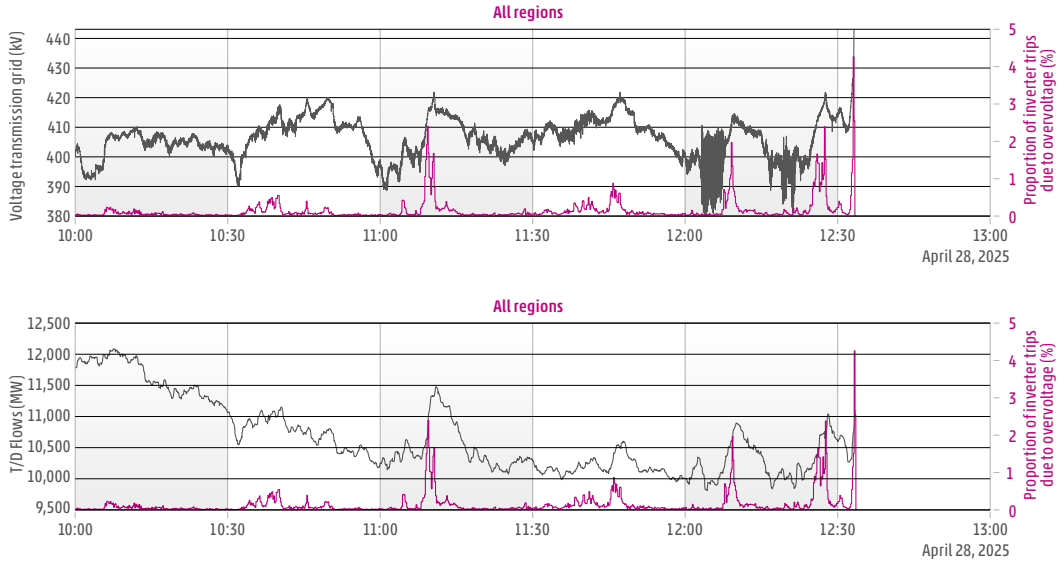


Figure 4-76: 10:00 –12:35 time period/Correlation between voltage at 400 kV Carmona substation, sum of TSO–DSO flows and proportion of overvoltage inverter trips (10 sec interval) from manufacturer M#2 (Source: Red Eléctrica, inverter manufacturer M#2)

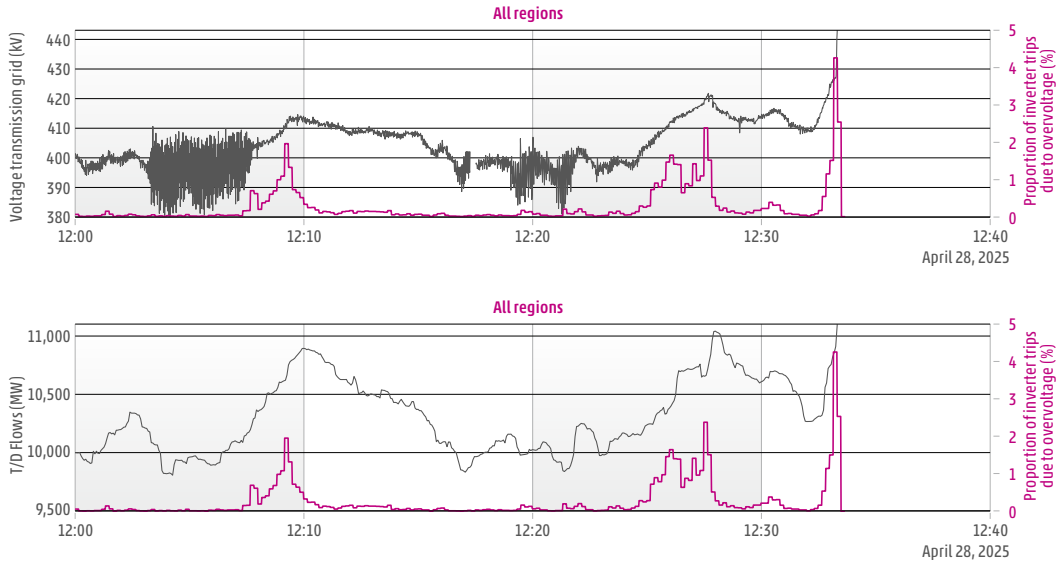


Figure 4-77: 12:00 –12:35 time period/Correlation between voltage at 400 kV Carmona substation, sum of TSO–DSO flows and proportion of overvoltage inverter trips (10 sec interval) from manufacturer M#2 (Source: Red Eléctrica, inverter manufacturer M#2)

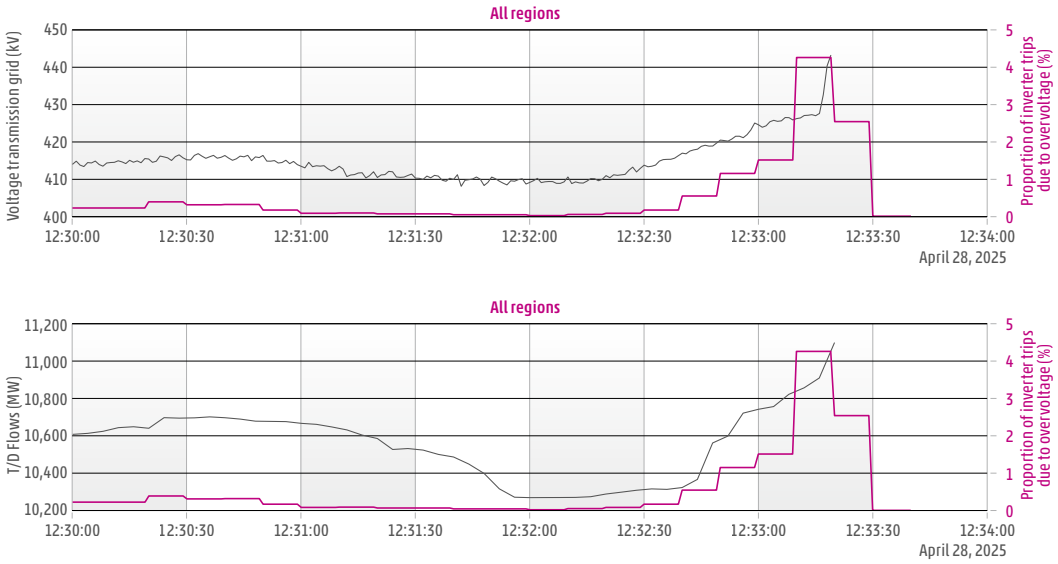


Figure 4-78: 12:30 –12:35 time period/Correlation between voltage at 400 kV Carmona substation, sum of TSO–DSO flows and proportion of overvoltage inverter trips (10 sec interval) from manufacturer M#2 (Source: Red Eléctrica, inverter manufacturer M#2)

4.3.3.2.2.3 Estimation of the total amount of disconnected generation (units <1 MW) based on the 10-second resolution dataset from inverter manufacturer M#2

The purpose of this subsection is to estimate the extent to which it is possible to correlate the TSO–DSO flow changes with the total amount of disconnected generation units <1 MW.

The methodology used, described below, is based on several assumptions and simplifications in the absence of further information. These factors should therefore be taken into account when interpreting the results.

The methodology used is based on the following assumptions and simplifications:

» **Dataset:** To estimate the total amount of disconnected generation <1 MW at a given moment, a relatively accurate timestamp for each inverter trip is required. Therefore, the dataset of manufacturer M#1 cannot be used in this context, as a 5-minute temporal resolution is insufficient. On the contrary, the 10-second resolution available in the dataset from manufacturer M#2 is appropriate. Hence, only the dataset from manufacturer M#2 can be used to perform this estimation. This does not imply that the data from M#2, particularly regarding the number of trips, are necessarily more accurate than those from M#1. In particular, the Expert Panel does not know whether the tripping behaviour of the M#2 inverter fleet overestimates, underestimates, or is in line with the actual total disconnected generation of the entire population of PV units <1 MW.

» **Reconnection time of tripped inverters:** As described in Section 4.3.3.2.1, the minimum time between the disconnection after a trip and the automatic reconnection of an inverter is equal to approximately 240 seconds (4 minutes). For the sake of simplicity, in the context of this methodology, it is assumed that every tripped inverter always[33] reconnects after a fixed time of 4 minutes. While trip/disconnections of inverters at e.g., timestamp 12:00:00 lead to an increase of the total amount of disconnected generation (i.e., a decrease of production infeed to the DSO grid), automatic reconnections of the same inverters at timestamp 12:04:00 lead to the decrease of this same value (i.e., an increase of production infeed to the DSO grid).

33 This simplification assumes that the grid parameters (voltage and frequency measured at inverter terminals) immediately return into the required normal operating ranges after a disconnection has happened. Under real conditions, this might often not be the case, which means that the total time between disconnection and reconnection of a tripped inverter can last significantly longer.

» **Reference value regarding actual aggregated production from PV facilities <1 MW:** Since the applied dataset from M#2 solely contains information about inverters connected to LV grids, the Expert Panel decided – in line with Section 4.3.3.2.1. – to consider 3,340 MW (69 % of the actual aggregated production from PV facilities <1 MW) as the reference value of the actual aggregated PV production from 10:00 onwards on 28 April. With this approach, it can be ensured that the disconnection behaviour of the LV-connected inverters in M#2 is linked only to the estimated actual generation from PV facilities <1 MW with a connection point at the same voltage level. However, this also means that the applied methodology can lead to an underestimation (by approximately 30 %) of the total amount of disconnected generation <1 MW, since the potential effect of voltage variations on medium-voltage (MV) connected inverters is not considered in the calculation.

In order to estimate the total amount of disconnected generation <1 MW the following calculation steps are applied:

» 1. Calculate sum of trips (= overvoltage + [undervoltage + overfrequency + underfrequency][34]) for each 10-second window.

» 2. To take into account the disconnection and reconnection behaviour of tripped inverters, apply a sliding window with a span of 4 minutes (= 24 periods of 10 seconds) and create the sum of all trips that have been calculated in step 1) within this sliding window.

» 3. Calculate the proportion of disconnected inverters from Step 2 relative to the total number of inverters in the fleet from M#2/

» 4. Extrapolate the percentage value from Step 3 to the estimated total PV production at the low-voltage level (3,340 MW; see Section 4.3.3.2.1).

The result of this methodology is shown in Figure 4-79 (bottom graph), which correlates the estimated amount of disconnected generation based on the proposed methodology with the TSO–DSO flows. It appears that the significant changes in the TSO–DSO flows during the 11:05 – 11:10, 12:05 – 12:10, and 12:25 – 12:30 periods occur nearly in parallel with the changes in the estimated disconnected generation.

Since both y-axes in the bottom graph of Figure 4-79 have the same scale, it is also possible to estimate the ratio of disconnected generation compared to the overall TSO–DSO flow changes for the relevant time periods:

Time period	Change in TSO-DSO flows	Estimated actual disconnected generation (peak)	Ratio
11:05 - 11:10	~ 1.150 MW	~ 550 MW	~ 48 %
12:05 -12:10	~ 1.000 MW	~ 430 MW	~ 43 %
12:25 -12:30	~ 900 MW	~ 620 MW	~ 69 %

Table 4-22: Comparison between changes in TSO-DSO flows and estimated actual disconnected generation (peak)

34 Even though these trips do not play a significant role, they are included for the sake of completeness.

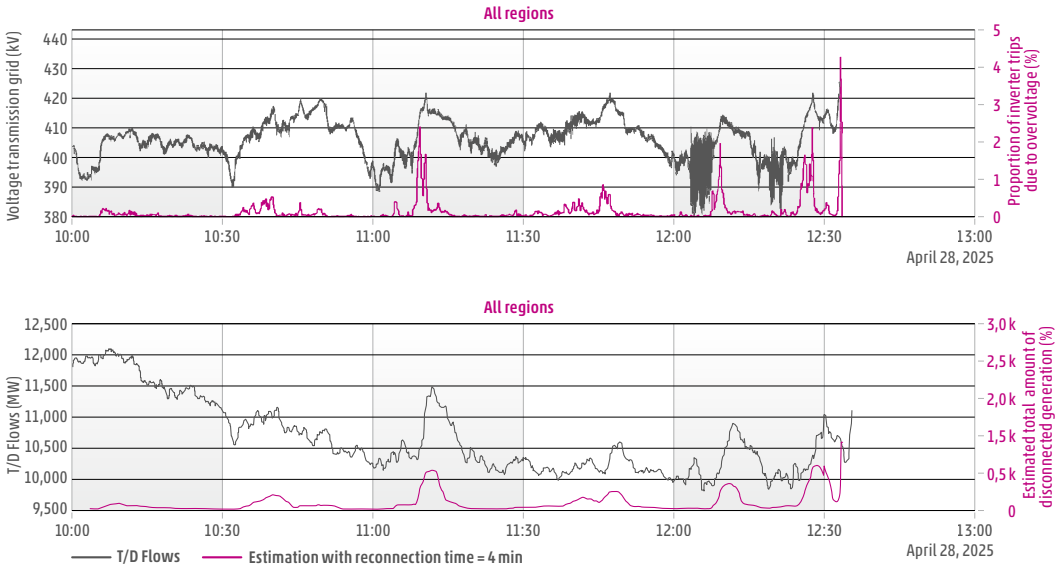


Figure 4-79: Estimation of the total amount of disconnected generation (units <1 MW connected to LV grids) based on the applied methodology and its underlying parameters

Figure 4-80 is a zoom of Figure 4-79 focusing on the period from 12:31:30 to 12:33:15. From this figure, it can be estimated that the disconnected generation of LV-connected PV units increased by approximately 77 MW (= ~231 MW – ~154 MW) in the timeframe from 12:32:00 to 12:32:57, corresponding to Event 1 (see Table 3-1), which comprises a portfolio of 317.30 MW.

Therefore, while respecting all addressed assumptions, simplifications and uncertainties described above, it can reasonably be concluded that the net load increase corresponding to Event 1 is partially due to the disconnection of PV units connected to the LV grid.

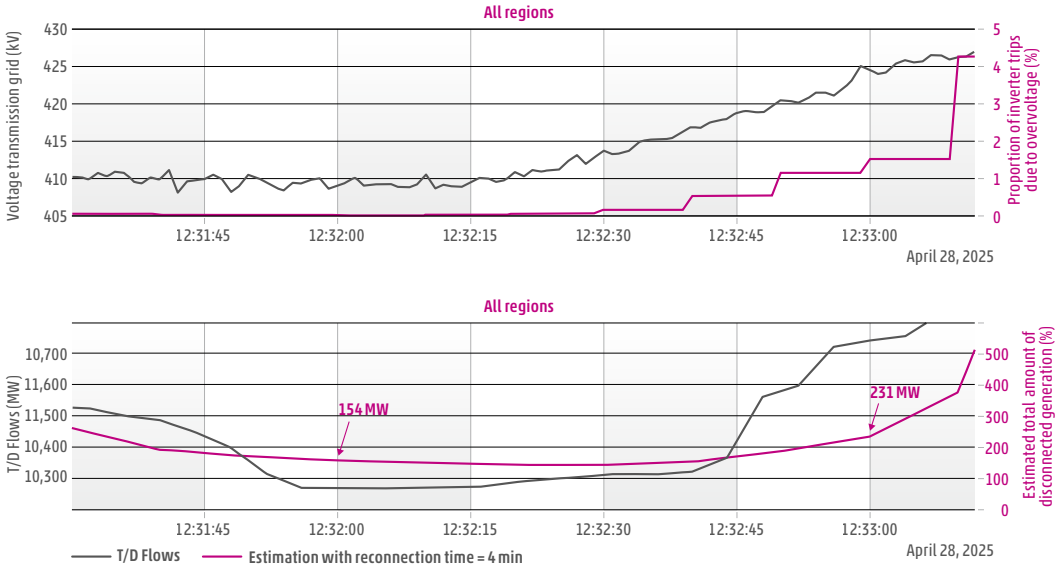


Figure 4-80: Identification of the additionally disconnected generation (units <1 MW) in the 12:32:00 -12:32:57 timeframe of Event 1 based on the applied methodology and its underlying parameters

4.3.3.2.2.4 Sensitivity of the amount of disconnected generation to the inverters' reconnection time

As described in Section 4.3.3.2.2.3, a fixed reconnection time of 4 minutes has been applied to estimate the total amount of disconnected generation <1 MW.

The purpose of this subsection is to estimate the influence of the reconnection time $T_{Reconnect}$ on the estimated total amount of disconnected generation <1 MW. For this purpose, four different reconnection times were selected:

- 10 seconds (equivalent to an immediate reconnection since the M#2 dataset has a temporal granularity of 10 seconds)
- 1 minute
- 2 minutes
- 3 minutes
- 4 minutes

Figure 4-81 shows the effect of the selected reconnection time on the total amount of disconnected generation <1 MW.

A more detailed zoom on the 11:00 – 11:30 timeframe in Figure 4-82 reveals that $T_{Reconnect}$ significantly influences the peak in total disconnected generation for $T_{Reconnect}$ values between 10 seconds and 2 minutes. When $T_{Reconnect}$ increases further, the peak continues to increase but at a smaller rate of change, while the amount of disconnected generation is maintained for a longer time

However, the estimation results with a selected $T_{Reconnect}$ of just 10 sec (= almost immediate reconnection) and 1 min clearly indicate that the peak values of disconnected generation are significantly lower compared to the estimation variants mentioned before. Essentially, the same applies to the persistence of the disconnected generation.

Taking these findings into account, the fundamental question arises as to what extent the choice of $T_{Reconnect}$ and the design of the automatic reconnection procedure of small PV inverters in general may affect the interoperability of the TSO and DSO networks.

It therefore seems appropriate to perform further analysis of the automatic reconnection behaviour of small PV inverters (particularly for events with a high rate of disconnection events).

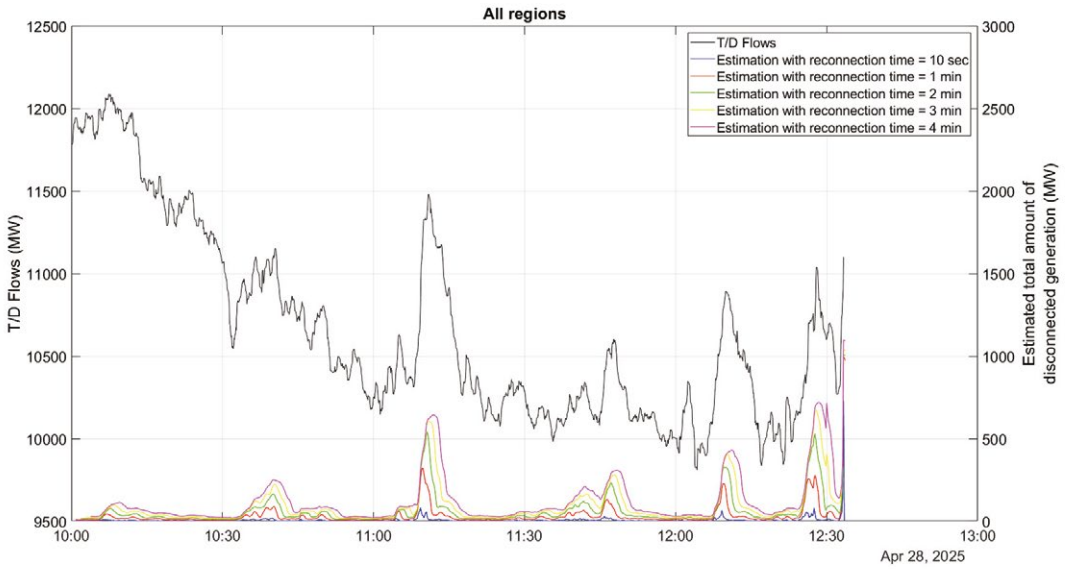


Figure 4-81: Sensitivity analysis regarding the effect of the selected reconnection time on the total amount of disconnected generation <1 MW for the 10:00 – 12:35 timeframe

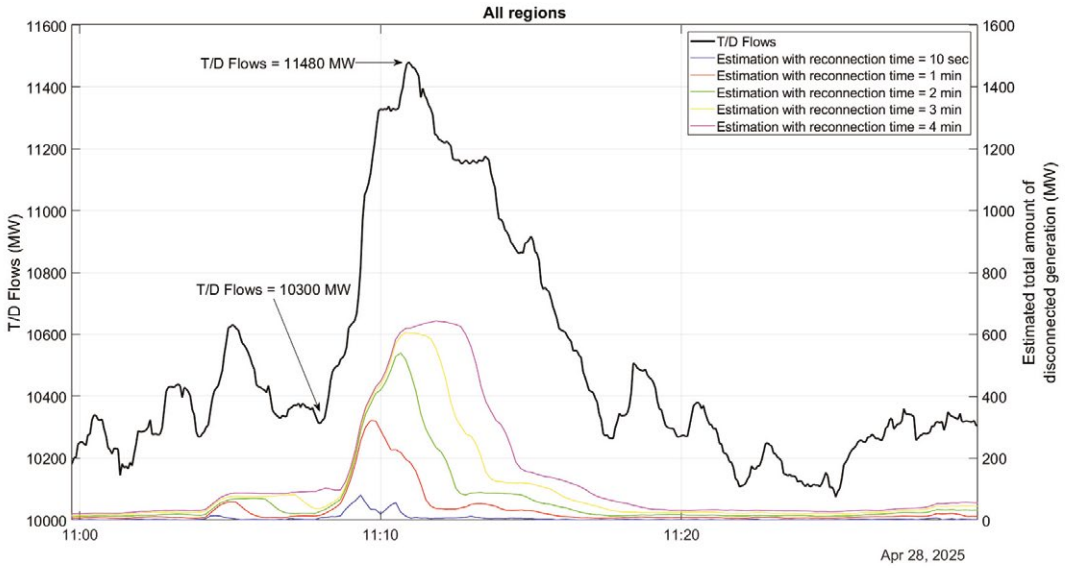


Figure 4-82: Sensitivity analysis regarding the effect of the selected reconnection time on the total amount of disconnected generation <1 MW, enlarged for the 11:00 – 11:30 timeframe

4.3.3.3 Voltage Dependency of Loads

4.3.3.3.1 Introduction

The voltage dependency of loads is an **inherent property** of electrical power systems. It is rooted in the fundamental design of electrical devices and describes how a device's power consumption varies with changes in the supply voltage.

In large power systems, the total load is usually a time-varying mix of thousands of different electrical devices. While some electrical devices always draw the same amount of power, others consume more or less depending on whether the voltage is higher or lower than the nominal voltage.

According to the well-known ZIP model, loads can be categorised into the following three categories (Figure 4-83):

- Constant impedance (Z): Current changes directly with voltage (e.g., heating, lighting).
- Constant current (I): Current remains constant, so power changes linearly with voltage.
- Constant power (P): The device draws constant power by adjusting current as voltage changes, often seen in regulated electronics, which can cause instability.

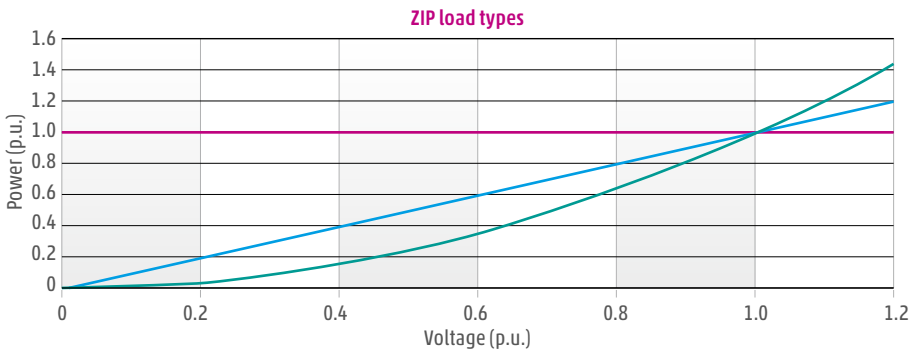


Figure 4-83: ZIP load types

Instead of splitting a load into three separate parts (like the ZIP model), the exponential load model uses only a single exponent to define the load's behaviour:

$$P = P_0 \left(\frac{V}{V_0}\right)^{\alpha} \text{ and } Q = Q_0 \left(\frac{V}{V_0}\right)^{\beta}$$

Where P_0, Q_0 and V_0 and represent the nominal (rated) values for voltage and active/reactive power, V represents the actual voltage and α and β are coefficients depending on the load mix and time.

Typical exponent values for the exponential load model representing different load types are shown in Table 4-23.

Load class	coefficient	coefficient
Residential		
Summer	1.2	2.9
Winter	1.5	3.2
Commercial		
Summer	0.99	3.5
Winter	1.3	3.1
Industrial	0.18	6.0

Table 4-23: Typical exponent values for the exponential load model

4.3.3.3.2 Analysis of measurements

As shown in Figure 4-84, some correlation is observed between the (averaged) voltage at the 400 kV level and the sum of TSO–DSO active and reactive power flows within Spain for the 12:00 – 12:33 time period.

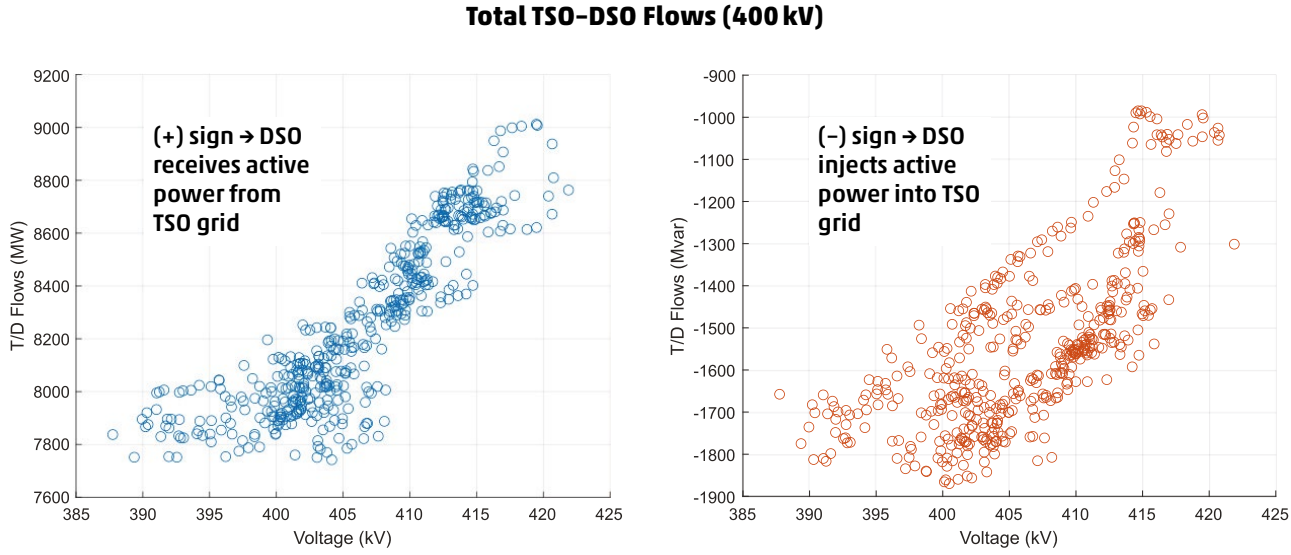


Figure 4-84: Correlation between (average) voltage at 400 kV level (x-axis) and sum of TSO–DSO active and reactive power flows within Spain (y-axis; Source: Red Eléctrica)

4.3.3.3.3 Correlation of measurements and simplified exponential load models

To evaluate the influence of voltage dependency using available measurements from DSOs and RE, two versions of a simplified exponential load model for different load classes have been set up.

Load class	Share (%)	coefficient	coefficient
Residential			
Summer	100	1.2	2.9
Winter	-	1.5	3.2
Commercial			
Summer	0	0.99	3.5
Winter	-	1.3	3.1
Industrial	0	0.18	6.0

Table 4-24: Load model #1 - Pure residential Z load, calibrated with "summer" coefficients

Load class	Share (%)	coefficient	coefficient
Residential			
Summer	100/3	1.2	2.9
Winter	-	1.5	3.2
Commercial			
Summer	100/3	0.99	3.5
Winter	-	1.3	3.1
Industrial	100/3	0.18	6.0

Table 4-25: Load model #2 - Mixture of residential, commercial and industrial load

As can be seen in Figures 4-85, 4-86, 4-87 and 4-88, some correlation is found between the simplified load models and the measurements provided by DSOs and Red Eléctrica for the 09:00 –12:33 time period:

- Substation – City of Zaragoza/Aragón region (Figure 4-85)
- Substation – City of Barcelona/Cataluña region (Figure 4-86)
- Substation – City of Sevilla/Andalusia region (Figure 4-87)
- Substation – City of Madrid/Madrid region (Figure 4-88)

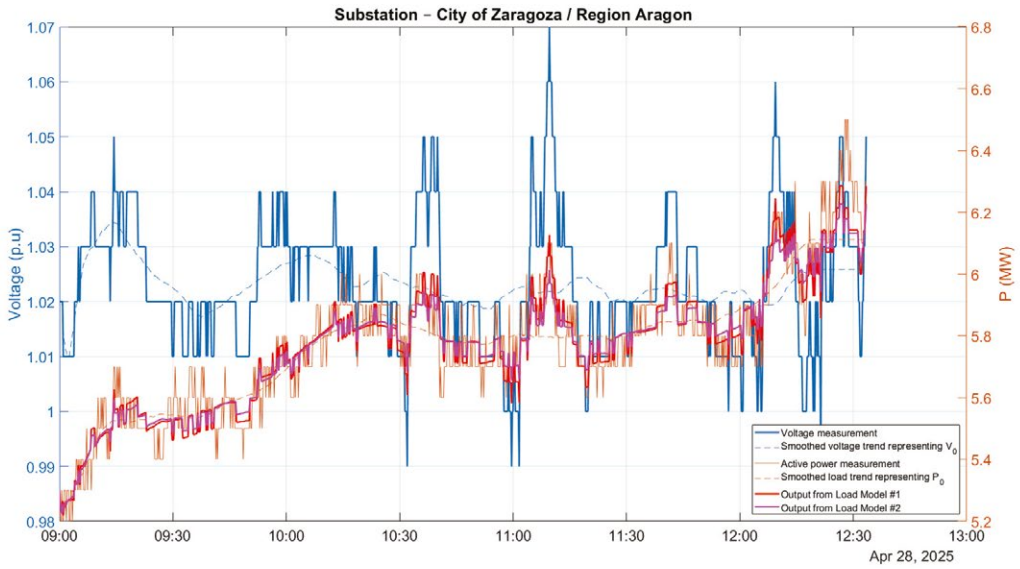


Figure 4-85: Substation – City of Zaragoza/Aragón region – interfacing point with low embedded RES share (e.g., rooftop PV)/Correlation between voltage (left y-axis), TSO-DSO flows and the outputs of the load models (right y-axis) Source: DSO Entity

As shown in Figure 4-85, at the peak of voltage around 12:28 (peak of demand on TSO–DSO flows) the load models estimate a lower value than the active power measurements, which may be explained by other factors (as explained in Section 4.3.2) in parallel to voltage dependency.

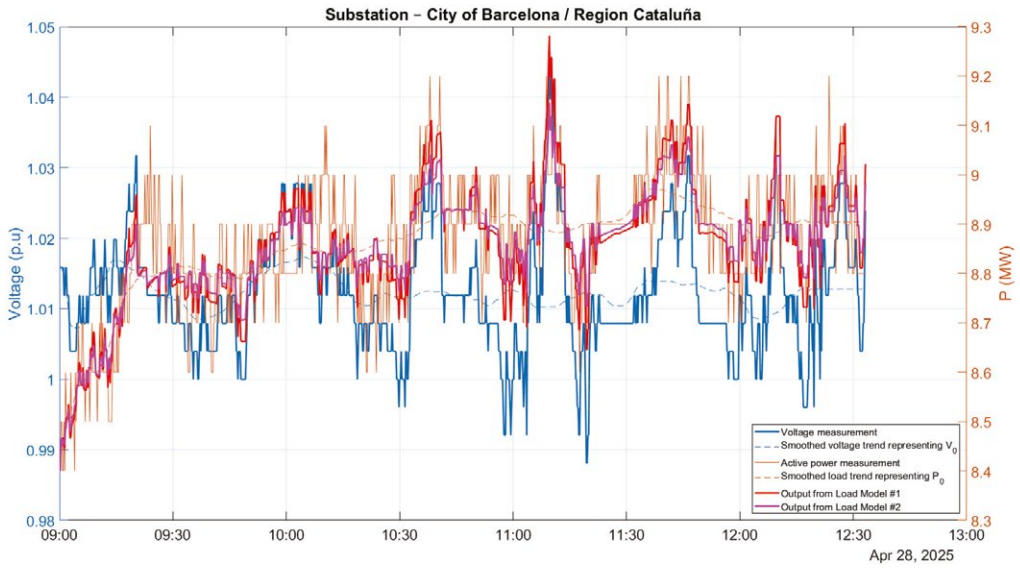


Figure 4-86: Substation – City of Barcelona/Cataluña region – interfacing point with low embedded RES share (e.g., rooftop PV)/Correlation between voltage (left y-axis), TSO-DSO flows and the outputs of the load models (right y-axis) Source: DSO Entity

As shown in Figure 4-86, the active power measurements tend to be higher than the load model estimations (e.g., 9:15 –10:30 or 11:30 –12:30).

However, at the peak of voltage around 11:10 (peak of demand on TSO–DSO flows), the load models estimate higher values than the active power measurements.

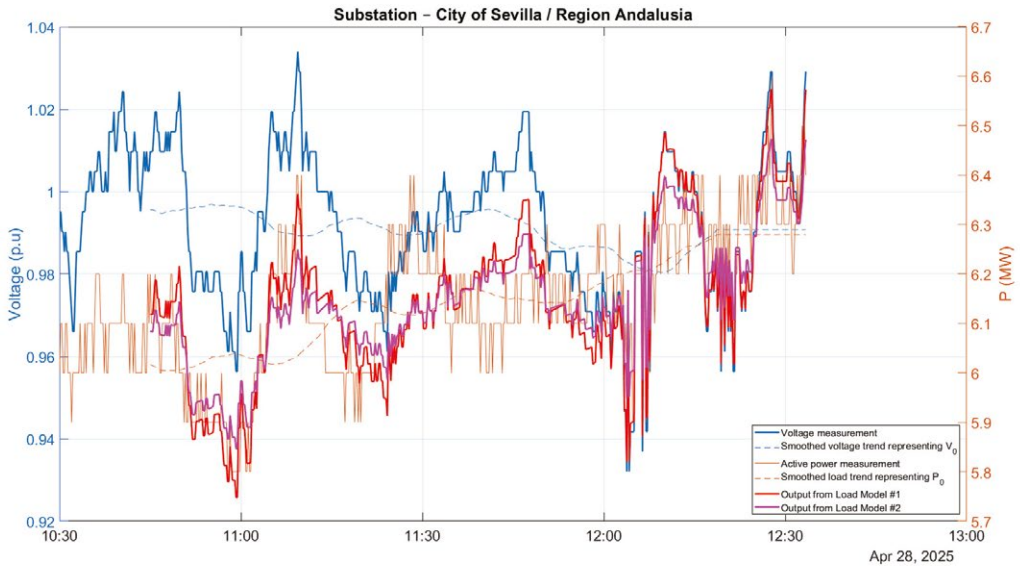


Figure 4-87: Substation – City of Sevilla/Andalusia region – interfacing point with low embedded RES share (e.g., rooftop PV)/Correlation between voltage (left y-axis), TSO-DSO flows and the outputs of the load models (right y-axis) Source: DSO Entity

As shown in Figure 4-87, during the 11:25 –11:35 and 11:50 –12:05 intervals, the load models underestimate the load. In contrast, during 11:35 –11:50, 12:10 –12:15 and 12:25 –12:30, the load models overestimate the load.

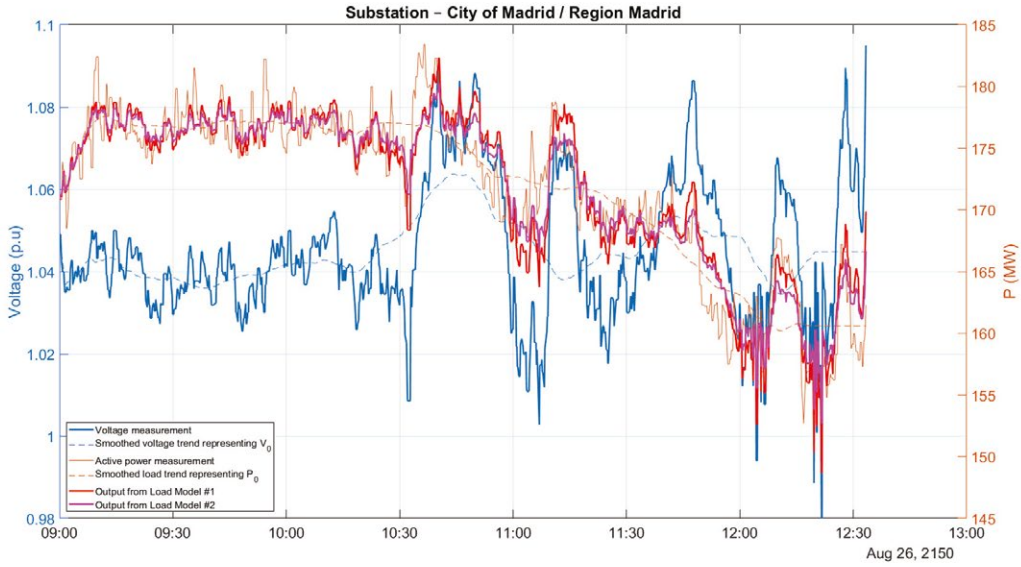


Figure 4-88: Substation – City of Madrid/Madrid region – interfacing point with high-medium embedded RES share (e.g., rooftop PV)/Correlation between voltage (left y-axis), TSO-DSO flows and the outputs of the load models (right y-axis) Source: Red Eléctrica

As shown in Figure 4-88, between 11:45 and 12:00, the load models estimate higher values than the active power measurements. The same applies after 12:30.

4.3.4 Conclusions

The analyses developed in Chapter 4.3 allow drawing the following conclusions related to the observed changes in TSO–DSO flows on 28 April 2025:

- Changes in TSO–DSO flows can only slightly be linked to aFRR/mFRR activations.
- Changes in TSO–DSO flows are, to a certain extent, linked to overvoltage tripping of generation units <1 MW connected to low-voltage grids:
  - the proportion of overvoltage trips shows a high correlation with the increase in voltage on the transmission grid;
  - the estimated total amount of disconnected generation is strongly correlated with the changes in TSO–DSO flows;
  - the total amount of disconnected generation is highly dependent on the actual number of trips and on the total time between the disconnection and the reconnection of the disconnected inverters;
  - the methodology used to estimate the total amount of disconnected generation is based on several assumptions and simplifications.

These factors should therefore be taken into account when interpreting the results.

- The Expert Panel recommends performing further analysis of the voltage-related disconnection and automatic reconnection of small PV inverters (see Recommendation 11 in Chapter 9).
- Because voltage dependency of loads is an inherent property of power systems, it can also be linked to a certain extent to the load increase at the DSO level (and hence changes in TSO–DSO flows):
  - A correlation is established between a simplified load model and DSO measurements.
  - However, detailed models, information on load composition, and additional representative DSO measurements would be needed to accurately model the detailed behaviour of all TSO/DSO interfacing points (including the location and load/generation composition, especially the share of PV supply at the LV level).
- Furthermore, based on the calculations performed to estimate the total amount of disconnected generation, it appears that the net load increase of 317 MW observed between 12:32:00 and 12:32:57 (Event 1 in Table 3-1) is partially due to the disconnection of PV units connected to the LV grid. However, due to the assumptions made and the uncertainties in the available data, it is not possible to precisely determine what proportion of this net load increase can be attributed to LV-connected PV units, nor whether estimates for LV-connected units can be applied to MV-connected inverters.

4.4 System Stability and Defence Plans

The purpose of the analyses outlined in this section is to elucidate the phenomena that occurred in the final stages preceding the blackout of the Iberian Peninsula and to present various sensitivity studies concerning the principal external variables that may have been pivotal to the event, from both frequency and voltage stability standpoints.

4.4.1 Methodological Framework for Analysis

The simulation model was developed starting from the latest available state-estimator snapshot prior to the blackout event, time-stamped at 12:25. This snapshot ensures full alignment with the actual system conditions preceding the incident, in particular with respect to system topology, power flows, and voltage profile, thereby establishing a robust and realistic initial operating point for the dynamic simulations.

The static snapshot includes a complete and detailed representation of both topology and operating point of the Spanish and Portuguese power systems, along with the portions of the French and Moroccan grids close to the Spanish borders. The remainder of the European grid is represented through equivalent generators at the model boundaries.

Based on this initial model, several adjustments were needed to make it suitable for RMS dynamic simulation purposes. Synchronous generator-based power plants were represented in the greatest possible detail, including step-up transformers and auxiliary loads, ensuring an accurate depiction of their electrical interface with the transmission network. Renewable generation plants were modelled as individual generators, including generator busbars, transformers, auxiliary loads, etc. In addition, very small generation units connected to medium and low voltage levels of the distribution network were estimated and modelled as aggregated equivalent generators at the distribution level. Throughout this adaptation process, the voltage magnitudes, voltage angles, and line power flows of the initial snapshot were preserved.

Once the study model was adapted to the level of detail required for dynamic studies, comprehensive dynamic models were incorporated to reproduce the behaviour of all relevant system components. Each generator was associated with a specific dynamic model based on owner and manufacturer information, technology type, and applicable grid code requirements, including the controls necessary to provide the prescribed response to disturbances, as well as protection models aligned with the robustness requirements defined by the regulations.

In addition, dynamic models of load behaviour were implemented, along with area-based load-shedding protection schemes, where the percentage of disconnection was updated based on the most recent estimated real values. Line protection models for loss of synchronism at the interconnections with France were included, as well as dynamic representations of HVDC links, STATCOM devices, and additional system operator models developed for reporting and result analysis.

With respect to HVDC representation, an equivalent model at the points of connection was modelled, providing a consistent and robust reproduction of the HVDC's dynamic behaviour during the event.

Overall, this modelling framework enables reliable RMS simulations, ensuring consistency with the validated steady-state operating point while accurately capturing the dynamic interactions relevant for voltage stability assessment.

For the sake of full transparency, it is important to mention that the construction of such a detailed representative model of the Iberian Peninsula and neighbouring power system required the Expert Panel to strictly collaborate with the affected TSOs, particularly Red Eléctrica, which were deeply and primarily involved in this process. These results could not have been achieved without strong and transparent collaboration, while simultaneously ensuring the confidentiality of third-party information relevant to system security. The Expert Panel monitored all developments and guided the investigation by defining modelling requirements, validating assumptions and intermediate results, and ensuring methodological consistency and technical rigour throughout the entire analysis process.

4.4.2 Frequency Stability

This section outlines the analyses and conclusions on frequency stability, based on RMS dynamic simulations conducted using the detailed model described in section 4.4.1.

The simulations allow for the reproduction of frequency evolution during the incident and assess the role of inertia and the behaviour of the defence plan during the event.

4.4.2.1 Reproduction of the Frequency Trajectory During the Incident

4.4.2.1.1 Adjustments to the Snapshot

The model described in section 4.4.1.1 was used to analyse the dynamic behaviour of the system in the final phases of the event, starting from 12:33:16. To ensure the highest level of accuracy, the steady-state equilibrium point was further adjusted to reflect the system conditions at that specific time. These changes were made as a series of modifications and load flow convergences of the initial snapshot:

» 1. The production of generation technology was adjusted to match the generation mix at 12:33:16.

» 2. The international exchanges with France, Portugal, and Morocco were adjusted to the values provided by PMU measurements for 12:33:16.

» 3. Power plants based on renewable energy sources, cogeneration, and waste-to-energy were converted to the fixed power factor control mode, according to Royal Decree 413/2014 ($\cos\delta = 0{,}985$ inductive).

» 4. A set of topological changes was modelled to reflect the state of the transmission network at 12:33:16, including tap changes of the Arkale phase-shifting transformer, line closures and openings, and reactor connections/disconnections carried out by the TSO between 12:25:00 and 12:33:16, as indicated in the incident chronology in Chapter 3.

» 5. The disconnection of the transformer in the province of Granada that occurred at 12:32:57 was modelled, as well as the loss of the active and reactive generation connected to it (Event 3 in Chapter 3).

» 6. The active and reactive power injections of the generation subject to the requirements applicable under P.O. 7.4 (synchronous generators of >30 MW) were modelled according to the measurements available for each unit as 12:33:16.

» 7. Demand by zones was adjusted to reflect the power variation in demand transformers between 12:25:00 and 12:33:16.

» 8. PMU measurements (63 buses at 400 kV and 220 kV) were used as a reference to adjust the voltage profile for the transmission network of the adapted model to better represent the situation during the event.

4.4.2.1.2 Validation of the Model and Simulation Results

To validate the model and build confidence in its ability to effectively replicate system behaviour, a simulation was run to reproduce the evolution of electrical quantities in the moments leading up to the incident. Specifically, a dynamic simulation beginning with the disconnection of generation at Badajoz (Event 4 in the list presented in Chapter 3) was carried out, and the evolution of the main system variables was compared with the available PMU measurements.

It is important to note that in the simulation, all generators disconnected automatically through the triggering of their individual protection models. Only the tripping of a combined cycle power plant (12:33:20.100), the L/Loeches–Arganda 220 kV line (12:33:19.971), and of the Spain–Morocco interconnections (Puerto de la Cruz–Beni Harchane 400 kV at 12:33:20.229 and Puerto de la Cruz–Mellousa 400 kV at 12:33:20.473) were time-based disconnections due to the lack of proper protection models and/or the lack of information about the cause of the disconnection.

Figure 4-89 and Figure 4-90 show the frequency and voltage at the La Cereal 400 kV substation, located north of Madrid near the centre of the Iberian Peninsula, compared to the real-world PMU measurement at the same location. As shown, the simulation model captures the general trend of the frequency, showing an accurate match between the simulation and the real-world measurement data.

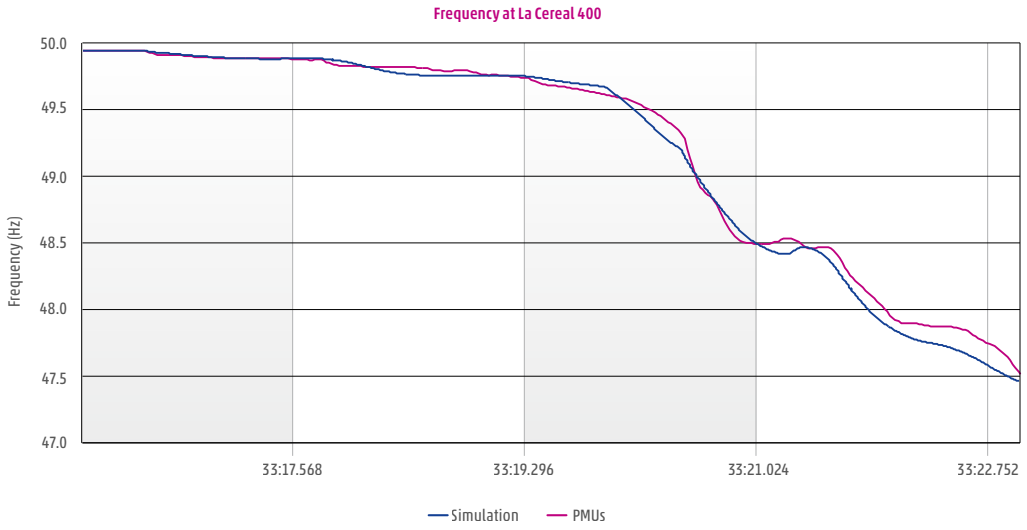


Figure 4-89: Transient response of the simulation model and the PMU measurements of frequency at the La Cereal 400 kV substation.

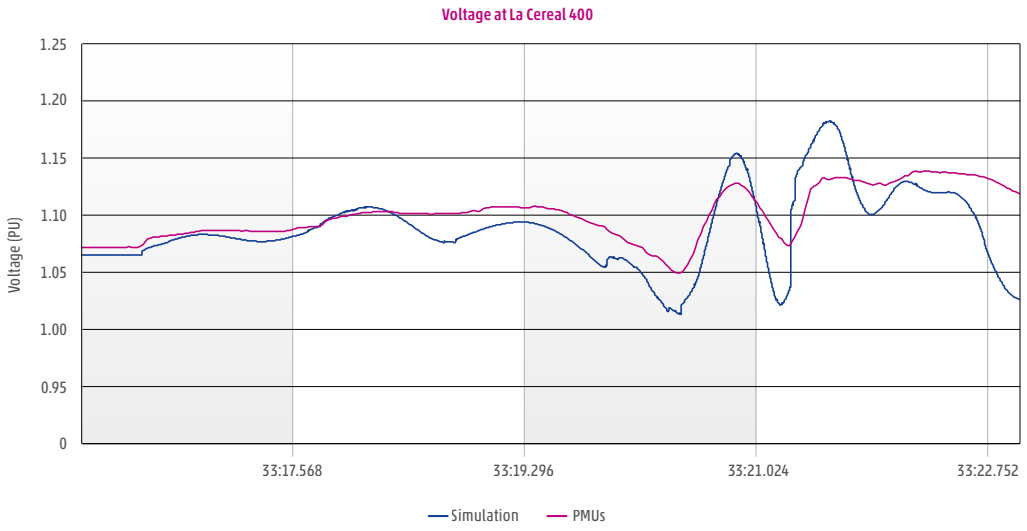


Figure 4-90: Transient response of the simulation model and the PMU measurements of voltage at the La Cereal 400 kV substation.

Figure 4-91 shows the Spain–France active power exchange. As can be seen, both curves reproduce similar dynamics, showing the progressive increase in imports from France caused by the generation loss, followed by the subsequent loss of synchronism of the AC interconnection lines.

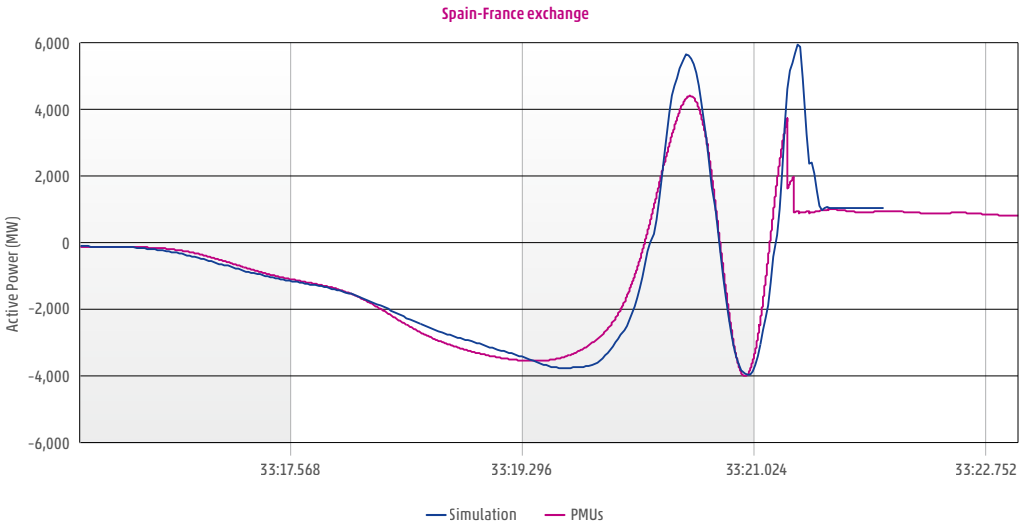


Figure 4-91: Transient response of the simulation model and the PMU measurements of active power exchange between Spain and France.

Figure 4-92 shows the accumulated generators and pumps disconnected throughout the simulation, disaggregated into very small wind and PV units embedded in the distribution networks (orange); hydro, solar thermal, and large wind and PV power plants (green); and pumps (blue).

At t = 0.470 s in the simulation (corresponding to 12:33:16.470), the initial trip of the Badajoz power plant occurs (in green). Subsequently, embedded generation continues to disconnect due to overvoltage protection tripping (in green). After a couple of seconds, the rate of disconnection increases, with non-embedded generation also starting to disconnect.

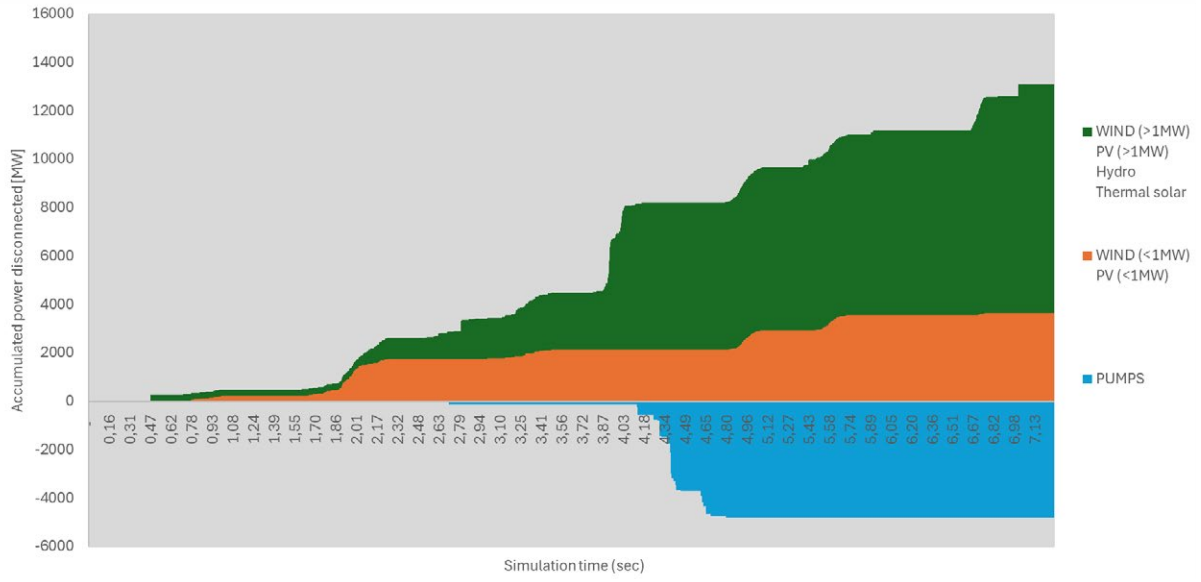


Figure 4-92: Disaggregation of generation and pumps disconnected throughout the simulation: (green) hydro, thermal solar, PV (>1 MW) and wind (>1 MW); (orange) wind and FV (<1 MW); (blue) pumps.

4.4.2.2 Role of the Defence Plan Performances

This simulation aims to investigate the impact of a "perfect" load-shedding scenario on the incident. In this case, "perfect" load shedding refers to the exact implementation of shedding percentages at each activation threshold. To perform this variant, load-shedding protection models were modified to include the theoretical percentages of demand that had to be disconnected at each step (slightly increased). Figure 4-93 and Figure 4-94 show the Spain–France active power exchange and the frequency at the La Cereal 400 kV substation for the base case (blue), the case with the theoretical load-shedding percentages (green), and the PMU measurements (red).

Initially, both simulations are identical, and the blocking of load-shedding protections due to reverse power flow does not affect the initial disconnection of generation as imports increase and synchronism is lost. However, at around t = 5 s (12:33:21.000), the simulations begin to diverge, with the simulation using theoretical load-shedding percentages exhibiting a slightly slower frequency decrease caused by the increased load shedding. However, the loss of synchronism remains minimally affected, and at the end of the simulation, the frequency at La Cereal is still decreasing rapidly. Hence, even "perfect" load shedding would not have avoided the blackout.

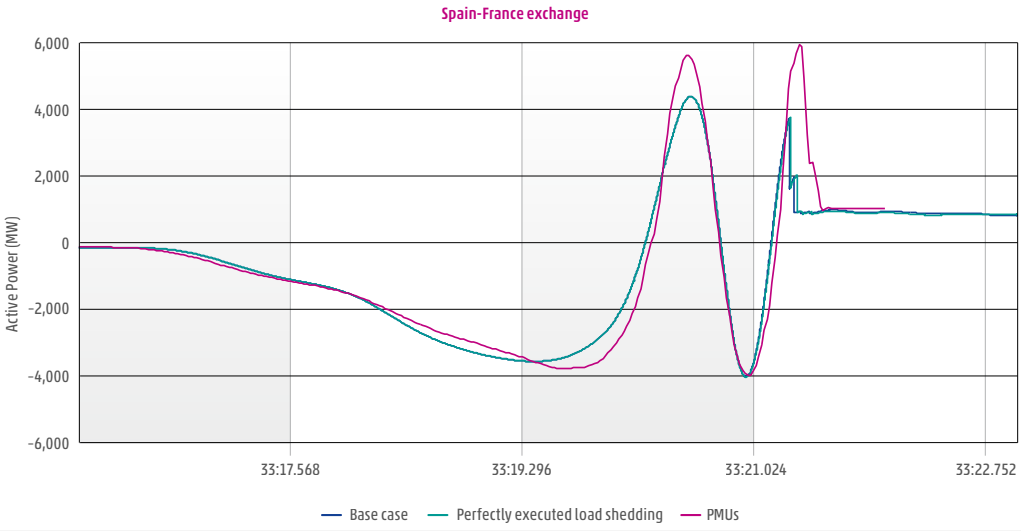


Figure 4-93: Transient response of the base case, the variant with perfectly executed load shedding, and the PMU measurements of active power exchange between Spain and France.

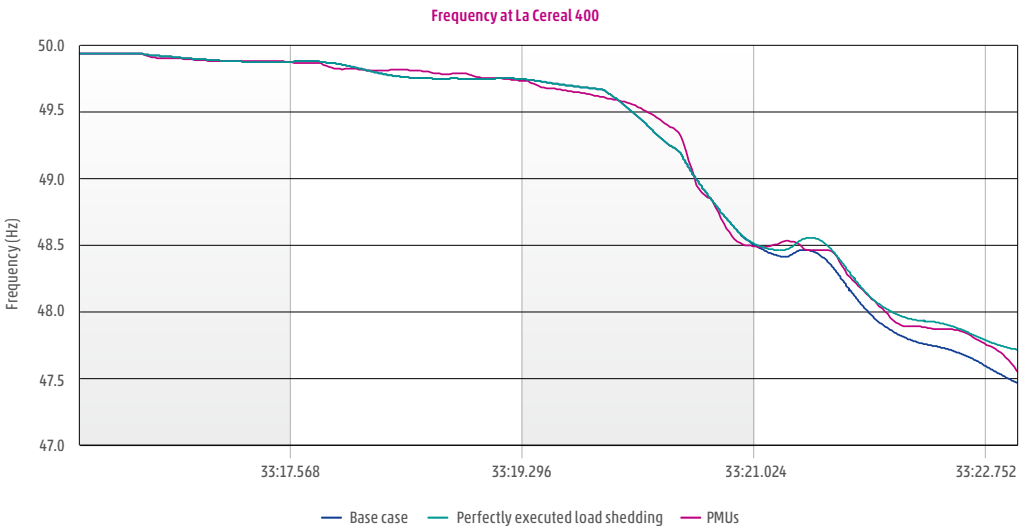


Figure 4-94: Transient response of the base case, the variant with perfectly executed load shedding, and the PMU measurements of the frequency at the La Cereal 400 kV substation.

4.4.2.3 Role of Inertia

Role of increasing the total system inertia

This simulation aims to investigate how additional system inertia could have affected the transient evolution during the incident. For this purpose, the dynamic models of the conventional generators, turbines, and other rotating masses were modified to increase the inertia constant. After these modifications, the total system inertia increased +20 % compared to the base case. Figure 4-95 and Figure 4-96 show the Spain–France active power exchange and the frequency at the La Cereal 400 kV substation for the base case (blue), the case with a +20 % increase in total system inertia (green), and the PMU measurements (red). The two simulations are very similar, and only a slight delay in the loss of synchronism can be observed. These results indicate that additional system inertia had a minimal impact on the degree of system evolution in the final moments of the incident. To better understand Figure 4-96, it is worth to note that a higher inertia makes the peninsular system inject and absorb energy with the rest of Europe in a slower (and slightly stronger) way during the loss of synchronism than in the base case. This makes the transient unloading of the system higher and, therefore, the overvoltage disconnections happens before than in the base case, making the frequency evolve in a steeper way towards blackout.

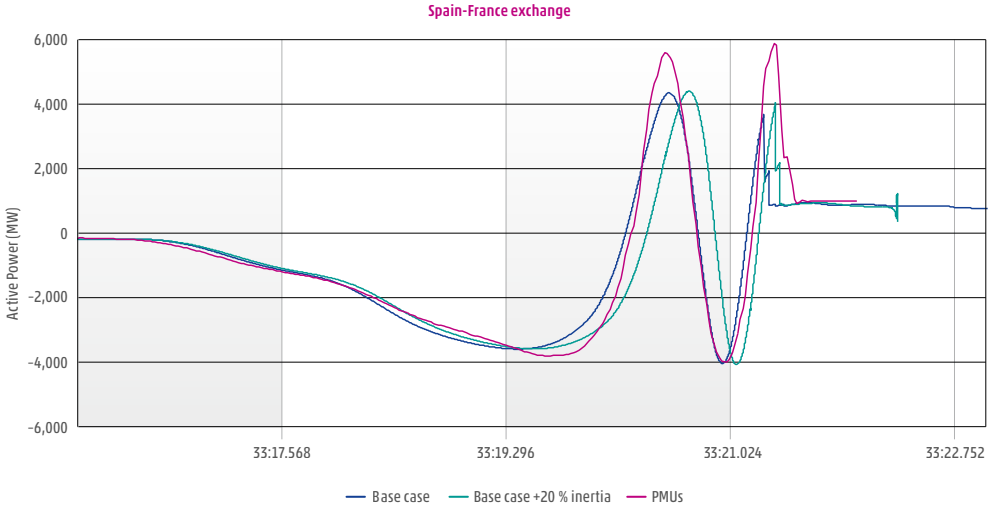


Figure 4-95: Transient response of the base case, the variant with additional system inertia, and the PMU measurements of active power exchange between Spain and France.

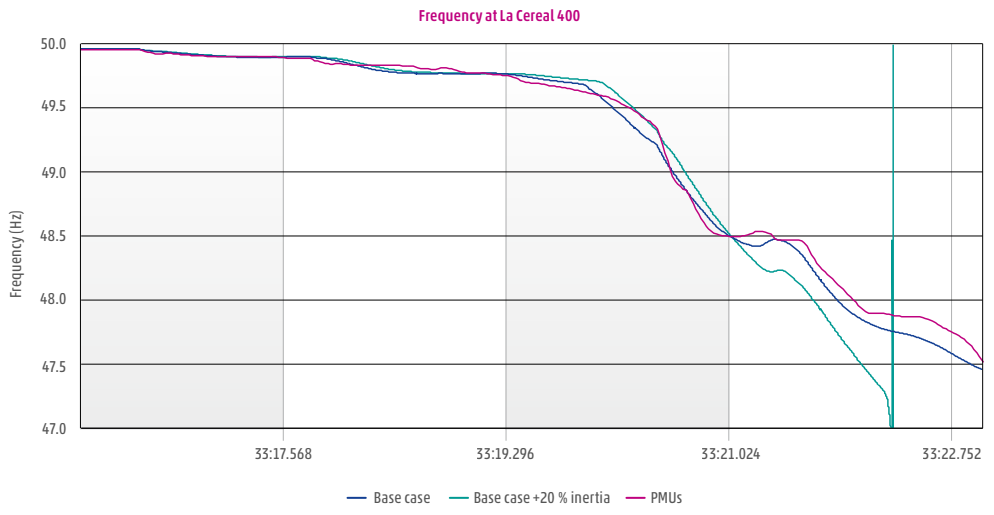


Figure 4-96: Transient response of the base case, the variant with additional system inertia, and the PMU measurements of the frequency at the La Cereal 400 kV substation.

4.4.2.4 Role of HVDC Power Reversal Mode

This simulation aims to investigate the impact of the HVDC power reversal mode on the Spain–France exchange, the loss of synchronism, and the evolution of frequency. This functionality allows for rapid inversion of the power flow on the HVDC link to counteract the overall system imbalance. To simulate this functionality, it is assumed that after the exchange reaches its maximum import capacity from France to Spain, the HVDC power reversal mode activates and immediately changes from exporting approximately 1 GW to importing 1 GW. Figure 4-97 and Figure 4-98 show the Spain–France active power exchange and the frequency at the La Cereal 400 kV substation for the base case (blue), the case with the HVDC power reversal mode (green), and the PMU measurements (red).

Before the activation of the HVDC power reversal mode, both simulations are identical. At t = 3.5 s (12:33:19.5000), the power reversal mode is activated and the Spain–France active power exchange exhibits a sharp step, increasing the import to 4.7 GW. However, the simulation results show that the loss of synchronism is not affected by the sudden increase in imports, and the frequency drops significantly faster due to generator disconnections caused by fast transients in the electrical quantities. These results indicate the activation of the HVDC power reversal mode when the Spain–France exchange reaches its maximum in the import from France to Spain does not have a positive impact on the incident.

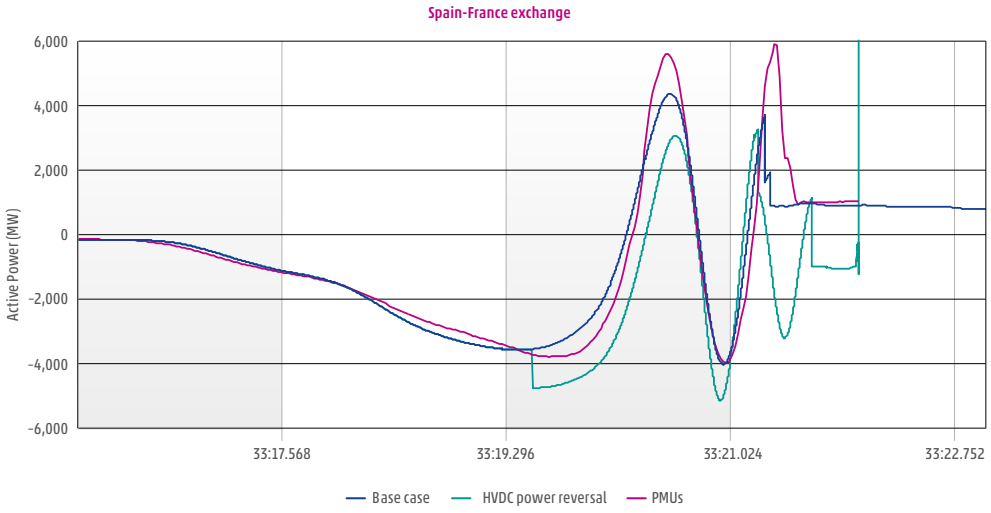


Figure 4-97: Transient response of the base case, the variant with the HVDC power reversal mode, and the PMU measurements of active power exchange between Spain and France.

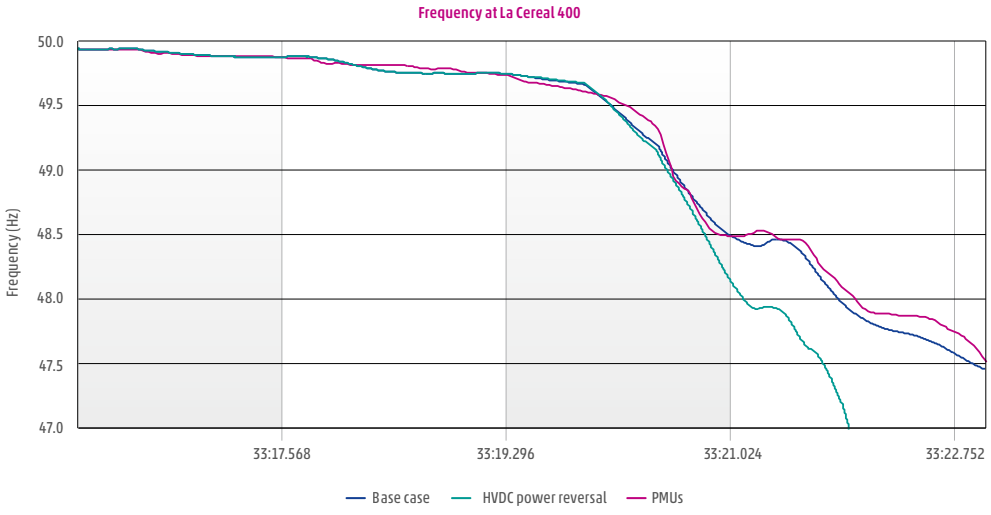


Figure 4-98: Transient response of the base case, the variant with the HVDC power reversal mode, and the PMU measurements of the frequency at the La Cereal 400 kV substation.

4.4.2.5 Conclusion Regarding Frequency Stability

The above analyses indicate that, even under the assumption of perfect execution of defence plan actions, neither these measures nor additional inertia would have been sufficient to prevent system collapse. Furthermore, a rapid intervention on the HVDC interconnection between France and Spain aimed at restoring the Iberian Peninsula's system balance could have actually worsened system behaviour due to fast transient phenomena.

4.4.3 Voltage Stability

This section outlines the analyses and conclusions regarding the voltage stability perspective. This study employed two different models to efficiently assess phenomena related to voltage dynamic behaviour across different characteristic time scales. In particular, the model described in section 4.4.1 was used to reproduce the final moments of the incident (section 4.4.3.2), providing a detailed representation of the dynamic behaviour of all components. Conversely, a quasi-dynamic model was specifically developed to reconstruct the voltage evolution in the minutes leading up to the incident, allowing for a better understanding of elements that contributed to voltage volatility. Additional details about this model are provided in the sections below.

4.4.3.1 Analysis of Voltage Transient Between 12:25:00 and 12:33:16

The aim of this section is to simulate the entire time window between 12:25:00 and 12:33:16 to gain insight into the voltage variations before the high voltage cascade that occurred at 12:33:16.

4.4.3.1.1 Description of the Model

As previously noted, the simulation of these long-term phenomena is based on a quasi-dynamic model of the system, built using the open-source tool Dynawo.[30] An anonymised version of the static snapshot described in section 4.4.1 was imported to prevent the disclosure of sensitive third-party information while allowing the Expert Panel to conduct independent analyses. The simulations were performed using the DynaFlow[31] approach implemented in Dynawo.

Specifically, the approach used in this study allows for long-term time-domain simulations even when detailed dynamic models of components (such as generators or loads) are not available. It relies on simplified time-domain representations, allowing an approximation of the system's dynamic response without requiring in-depth knowledge of the internal dynamic behaviour of each individual component. This makes the approach suitable for analysing the overall system evolution over longer time scales; however, it is not intended to accurately reproduce the fast dynamics that characterise the final seconds of the incident, where more detailed models are required to accurately capture system behaviour. Therefore, the divergence of this quasi-static model by real measurands in the last part of simulated transients is not considered.

4.4.3.1.2 Base Case

The snapshot at 12:25:00 is the initial point of the simulation. The main events affecting the voltage have been forced into the simulation:

- Variations in active power loads in Portugal and Morocco to match the measured flow exchange with Spain.
- Variations in distribution loads (active and reactive power) in Spain to match the measured exchanges from TSO to DSO. These simulated variations include both load and generation variations at the DSO level.

30 Linux Foundation for Energy: **https://github.com/dynawo/dynawo**.

31 The France–Spain border: A case study advocating for the use of simplified time-domain simulations for steady state calculation: **https://doi.org/10.1016/j.epsr.2022.108410**.

- Variations of active and reactive power of photovoltaic and wind generation under 50 MW connected to the transmission system, to match measured active and reactive power. Other generation categories connected to transmission were not modified, as their variations were minimal during this period.
- 12:25:23: Closing of the Guadame–Cabra 400 kV line
- 12:26:01: Closing of the Tordesillas–Galapagar 400 kV line
- 12:26:54: Closing of the Vitoria shunt reactor
- 12:27:10: Closing of the Penaflor shunt reactor
- 12:27:41: Closing of the Guadame 3 shunt reactor
- 12:27:52: Closing of the Guadame 2 shunt reactor
- 12:28:01: Closing of the Morata shunt reactor
- 12:32:57: Event ID 3 400 kV TS1–Granada, 355 MW disconnection
- 12:33:16.4: Event ID 4a 400 kV TS1–Badajoz, 582 MW disconnection
- 12:33:16.9: Event ID 4b 400 kV TS2–Badajoz, 145 MW disconnection
- 12:33:17: Event ID 5a1 400 kV TS1–Segovia, 23 MW disconnection
- 12:33:17.5: Event ID 5b1 400 kV TS1–Badajoz and 5b2 220 kVTS1–Huelva, 152 MW disconnection
- 12:33:18: Event ID 5a2 400 kV 400 kVTS1–Sevilla, 550 MW disconnection

All active power variations are compensated for in France. For example, if the exchange from Spain to Portugal decreases, the exchange from Spain to France is increased.

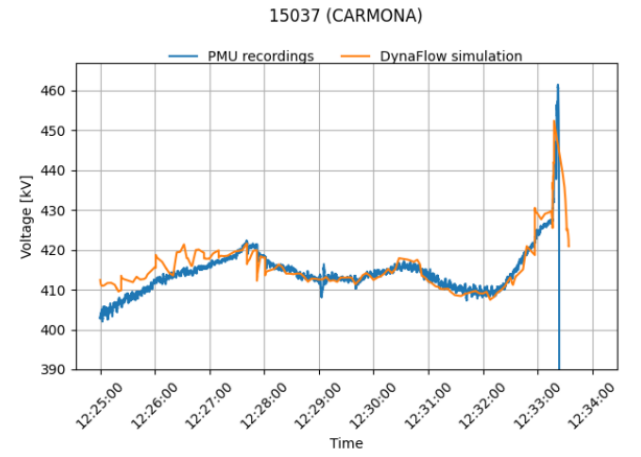


Figure 4-99: Voltage at the 400 kV substation in southern Spain: simulation vs real-world measurement

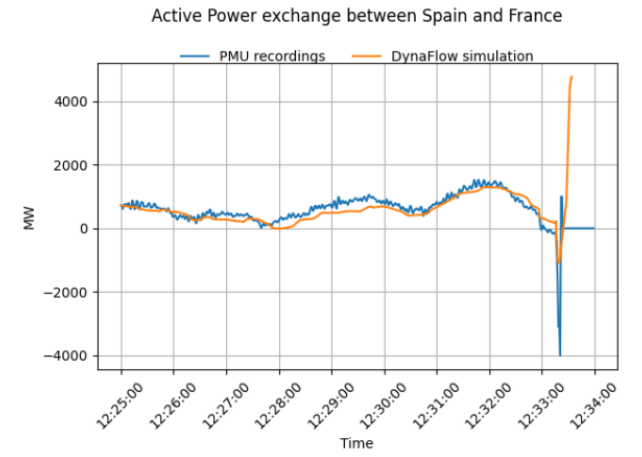


Figure 4-100: Active power flow on the Spain-France border: simulation vs real-world measurement.

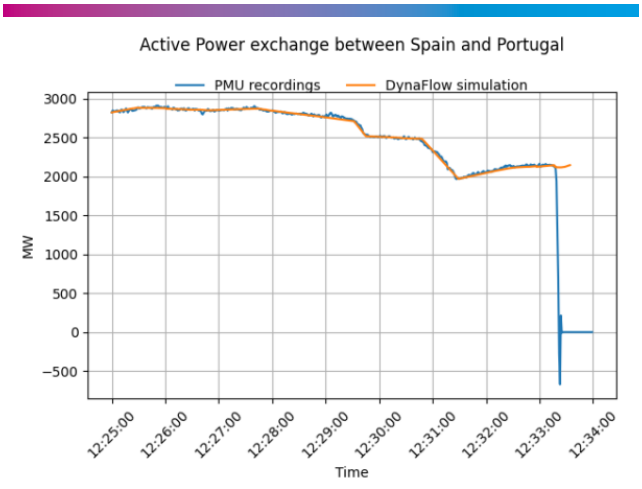


Figure 4-101: Active power flow on the Spain-Portugal border: simulation vs real-world measurement.real-world measurement

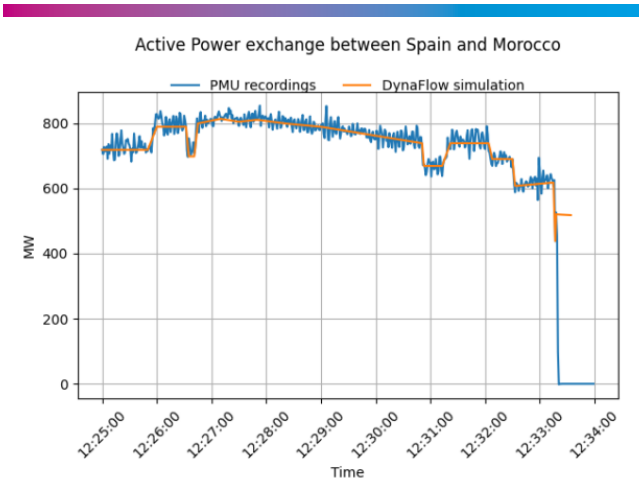


Figure 4-102: Active power flow on the Spain-Morocco border: simulation vs real-world measurement.real-world measurement

To verify the quality of the simulation, Figure 4-99 depicts the voltage trajectory at one of the main 400 kV substations in southern Spain. Similarly, Figure 4-100, Figure 4-101, and Figure 4-102 show the active power flow on the Spain–France and Portugal–Morocco borders, respectively.

The simulation closely matches the measurements, capturing the main characteristics of the event: a voltage increase between 12:25 and 12:28, followed by a reduction in voltage until 12:31:40, and then a sharp voltage increase.

At 12:32:57, the 400 kV TS 1–Granada tripped (Event ID 3), along with other disconnections (Event IDs 4a, 4b, 5a1, 5b1, 5b2, 5a2) close to 12:33:16.

The main observation is that the exchange with Portugal decreases throughout the simulation, with a sharp drop of 500 MW between 12:30:48 and 12:31:29, increasing the exchange from Spain to France while simultaneously reducing voltage. These results provide an initial indication of the sensitivity of this exchange to voltage volatility in southern Spain, where resources for voltage regulation were limited.

4.4.3.1.3 Parameters Affecting Voltage Variations

Starting from the base case, several simulation variants were developed to better understand the influence of the different exogenous factors affecting system behaviour. In each variant, a single exogenous variation was applied individually, while all other conditions remained the same as in the base case.

This approach allows the effect of each factor to be isolated and its individual contribution to the observed voltage evolution to be evaluated. By comparing the results of each variant with the base case, the relative importance of each exogenous driver in shaping the voltage profile during the minutes leading up to the incident can be assessed.

Analysis 1: No variation in the exchange between Spain and Morocco

In this analysis, the power exchange between Spain and Morocco was assumed to remain constant at the value observed from the snapshot at 12:25.

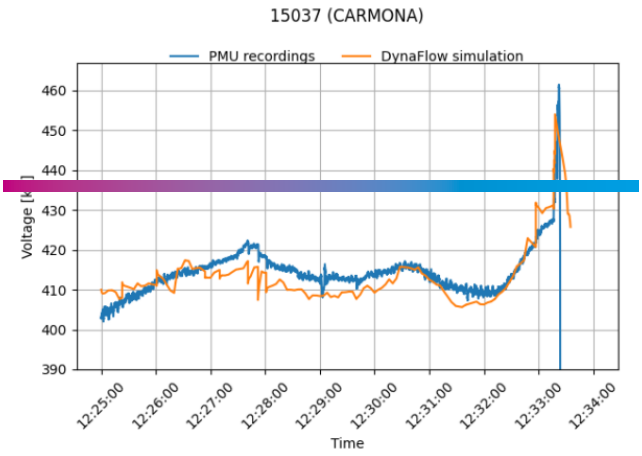


Figure 4-103: Fixed exchange on the Spain–Morocco border: voltage at the 400 kV substation in southern Spain.

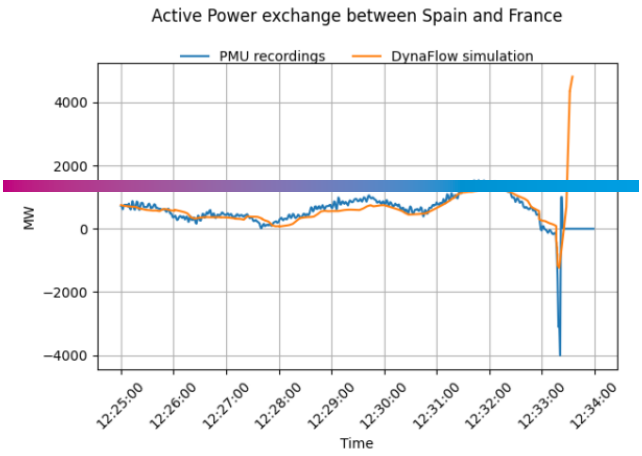


Figure 4-104: Fixed exchange on the Spain–Morocco border: active power flow at the France–Spain border.

Although the variations in power exchange between Spain and Morocco are relatively small (around 180 MW, see Figure 4-103), they lead to non-negligible voltage changes, highlighting a slight sensitivity of the voltage to this flow.

In particular, the power flow on the Spain–Morocco border is directly correlated with voltage. Maintaining a higher value has little impact on the power flow on the opposite Spain-to-France border (as power variations there are minimal), but increases local flows, which in turn lowers voltages in southern Spain.

Analysis 2: No variation in the exchange between Spain and Portugal

In this analysis, the power exchange between Spain and Portugal was assumed to remain constant at the value observed from the snapshot at 12:25.

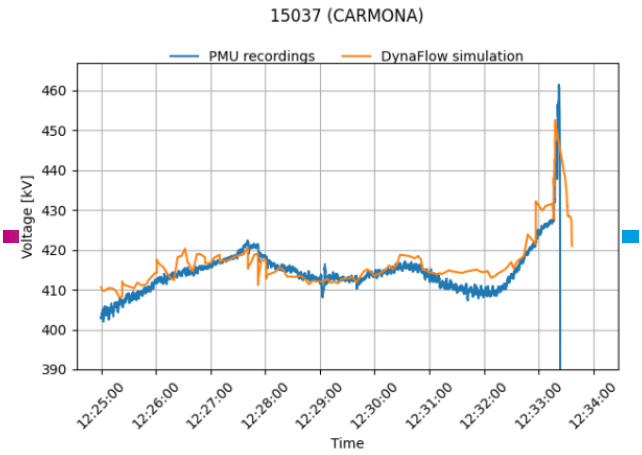


Figure 4-105: Fixed exchange on the Spain-Portugal border: voltage at the 400 kV substation in southern Spain.

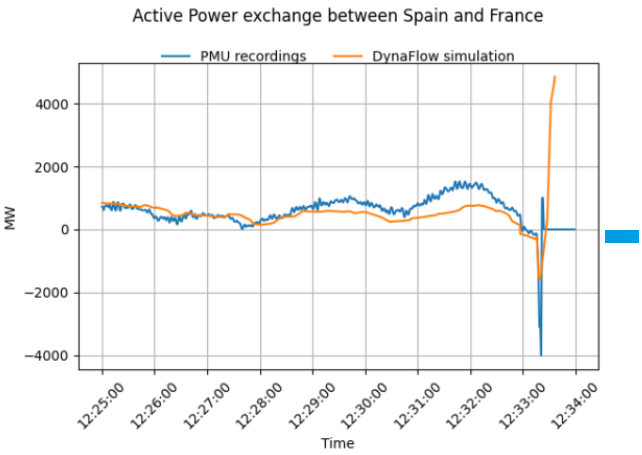


Figure 4-106: Fixed exchange on the Spain-Portugal border: active power flow at the France-Spain border.

Figure 4-101 shows that variations in power flows on the Spain–Portugal border exceed 900 MW over the time window of this analysis. In particular, a rapid change of about 500 MW started at 12:30:48. These fluctuations cause voltage deviations in southern Spain (specifically, an increase in voltage levels), demonstrating the significant impact of cross-border flows on voltage in the area.

The simulations results depicted in Figure 4-105 and Figure 4-106 allow to conclude that the Spain to Portugal power flow was negatively correlated to the voltage variation, since keeping a higher power flow on this border lowers the power flow on the opposite Spain to France border, with the effect of higher voltages in southern Spain.

Analysis 3: No variation in exchange from TSO to DSO

In this analysis, the power exchange between TSO and DSO was assumed to remain constant at the value observed from the snapshot at 12:25.

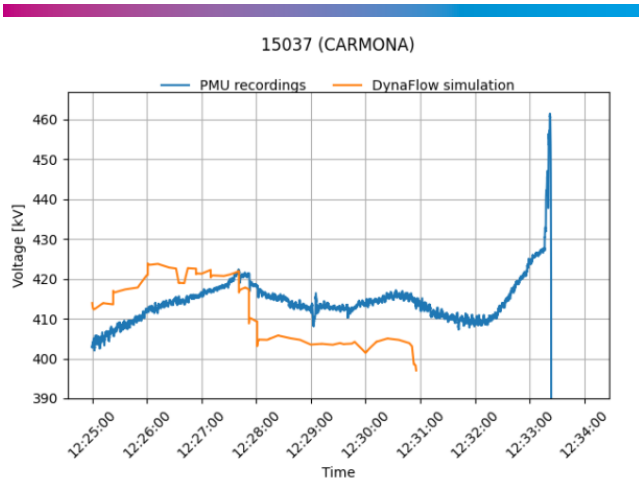


Figure 4-107: Fixed exchange TSO-DSO: voltage at the 400 kV substation in southern Spain.

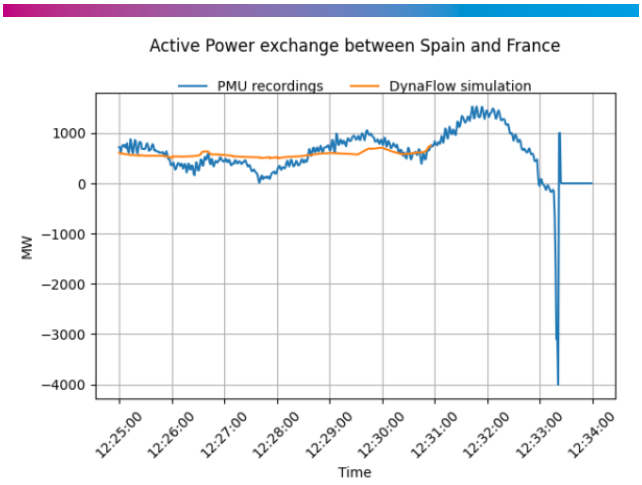


Figure 4-108: Fixed exchange TSO-DSO: active power flow at the France-Spain border.

The variation in TSO-to-DSO power flow is around 800 MW, comparable to the Spain–Portugal exchange variations. However, its impact on voltage is considerably stronger. Figure 4-108 clearly shows that the TSO-to-DSO exchange is a key driver of the voltage rise.

This strong sensitivity makes the simulation more challenging if adjustments to the other electrical quantities driving the model are not considered. However, it is worth noting that the aim of this study is not to perfectly reproduce the exact system trajectory, but rather to highlight the main drivers influencing the observed voltage deviations.

Analysis 4: No variation in generation below 50 MW

In this analysis, the production of PV and wind generators with a capacity below 50 MW in Spain was assumed to remain constant at the value observed from the snapshot at 12:25.

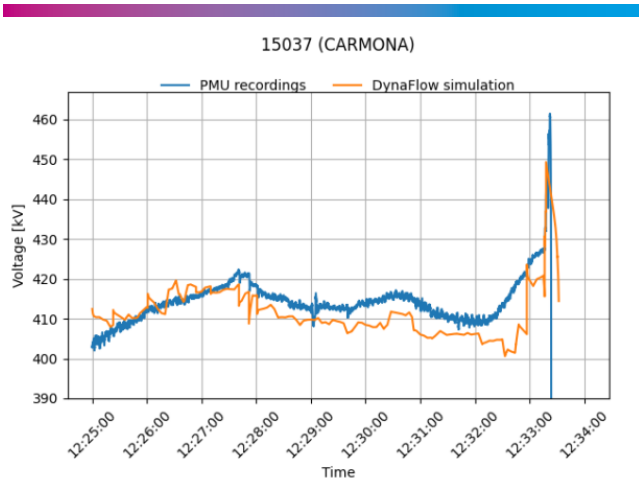


Figure 4-109: Fixed generation from DSO-connected PV: voltage at the 400 kV substation in southern Spain.

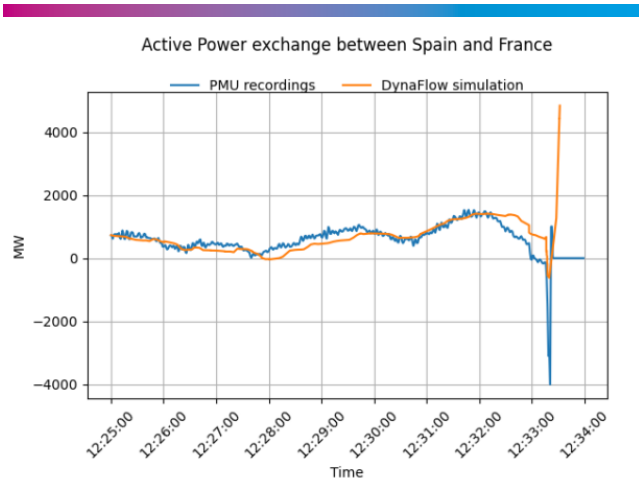


Figure 4-110: Fixed generation from DSO-connected PV: active power flow at the France-Spain border.

The variation in PV generation connected to DSOs consists of a reduction in injected power of up to 400 MW and a reduction in absorbed reactive power of up to 70 MVAr.

The simulation results depicted in Figure 4-109 and Figure 4-110 show a negative correlation between voltage levels and the power injection from distributed generation, consistent with the behaviour observed for the TSO–DSO exchange, as expected.

Analysis 5: Voltage variations without closing 400 kV lines for oscillation damping improvement

In this analysis, a variant of the base case was developed in which the closing of 400 kV lines intended to improve oscillation damping (10 lines from 12:07 to 12:26) was not applied. The switching actions performed during the event were therefore removed from the simulation to evaluate their influence on voltage behaviour.

As expected, Figure 4-111 shows that the topological changes implemented to counteract the oscillations have a non-negligible impact on voltage levels in southern Spain. The closing of the 400 kV lines reduces the network impedance and redistribute the active and reactive power flows of lines, which in turn increase the voltage profile in the area.

This observation confirms that the corrective actions taken by the operators to successfully dampen the oscillations, in accordance with the procedures in place and discussed in Section 4.2, had an impact on the voltage level in the South of Spain.

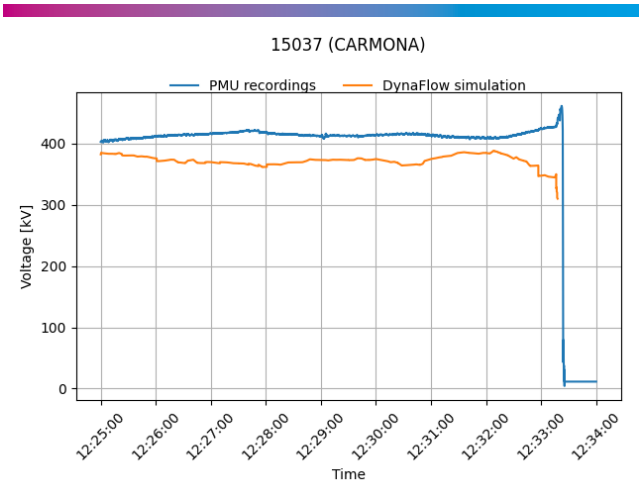


Figure 4-111: No line disconnection to counteract oscillations: voltage at the 400 kV substation in southern Spain.

Analysis 6: Additional reactive power margins from conventional generation

The aim of the simulation is to evaluate the voltage transient assuming additional reactive power margins to the generators spinning at 12:32, showing as additional reactive power absorption could contribute to stabilise the voltage behaviour of the system.

The simulation results reported in Figure 4-112 are obtained with the long term analysis approach that take into account in steps by 1 second the dynamic evolution of the system. It is important to underline that, although the dynamics of the AVR are represented, the model does not capture the existing effects of AVR limiters but considers the physics of the generators adopting as reference capability of the generators a square curve, available by the snapshots. These assumptions are not limiting the simulation but necessary to give the needed general evaluation about system needs. The simulation shows how additional margin could improve the reactive power management increasing the total physically available reserve.

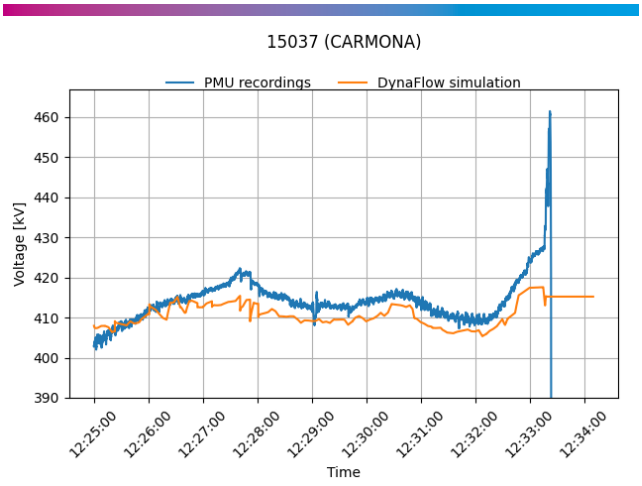


Figure 4-112: Increase of margin on generators.

Analysis 7: Connection/disconnection of shunt reactors

In this simulation, the connection of a shunt reactor (the Caparacena 400 kV shunt reactor, rated at 150 MVAr) is considered. The reactor is automatically connected shortly before Event ID 3, confirming the beneficial impact of reactor reconnection. To implement a control strategy effectively in real operations, dedicated studies are needed to evaluate whether the reactors can be manoeuvred immediately without compromising asset integrity, substation personnel safety, and any other EMT implications. It should be noted that the simulation still exhibits divergence at the end of the time horizon (after 12:33:16). This occurs because all other contributions affecting the voltage, active power, and reactive power evolutions are kept identical to those in the base case.

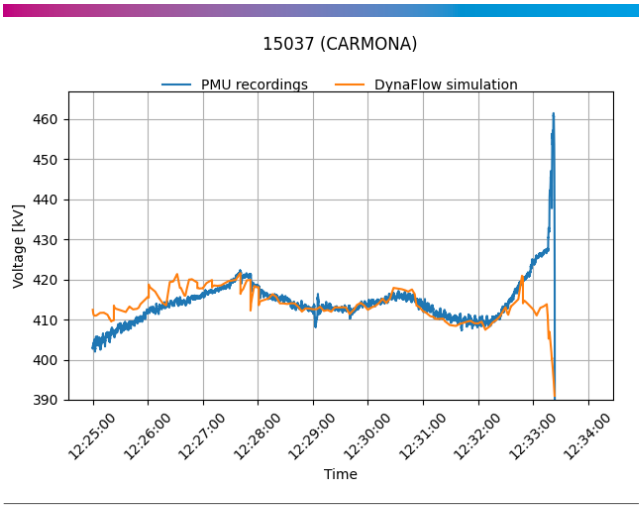


Figure 4-113: Increase of margin on generators.

4.4.3.2 Analysis of Voltage Transient from 12:33:16

The model described in section 4.4.2.1 was also used to derive conclusions about voltage stability. In particular, several simulations were performed to assess the effects on system behaviour of having additional reactive power margins for voltage control and how this could mitigate the negative effects of defence system actions on voltage profiles. This last set of simulations was done with an RMS algorithmic approach in order to study with a different approach the evolution of voltage transient.

4.4.3.2.1 Role of Voltage Regulation from Additional Synchronous Condensers

To further explore the concept of reactive power margins for voltage regulation, an additional variant was developed to investigate the impact of additional dynamic voltage support provided by new synchronous condensers distributed across the transmission network. A total of eight 250 MVA synchronous condensers were modelled in the base case, including the step-up transformer, according to the Spanish Network planning modification (Modificación de aspectos puntuales (MAP) 2021–2026[32]) approved on 14 July 2025. The dynamic model of the synchronous condensers and step-up transformers was based on standard models from the library with typical parameters for the technology.

After these changes, a dynamic simulation was conducted for the same conditions as in the base case simulation. Figure 4-114 and Figure 4-115. show the Spain–France active power exchange and the frequency at the La Cereal 400 kV substation for the base case (blue), the case with eight new synchronous condensers (green), and the PMU measurements (red).

Similar to the previous analyses, the increased voltage control capabilities provided by the synchronous condensers create a safer situation to address the initial incidents, avoiding cascading generation trips and allowing the system to reach a stable operating point (as seen in the frequency and the Spain–France exchange) compared to the base case simulation and PMU measurements.

32 https://www.miteco.gob.es/es/energia/novedades/novedad-000280.html

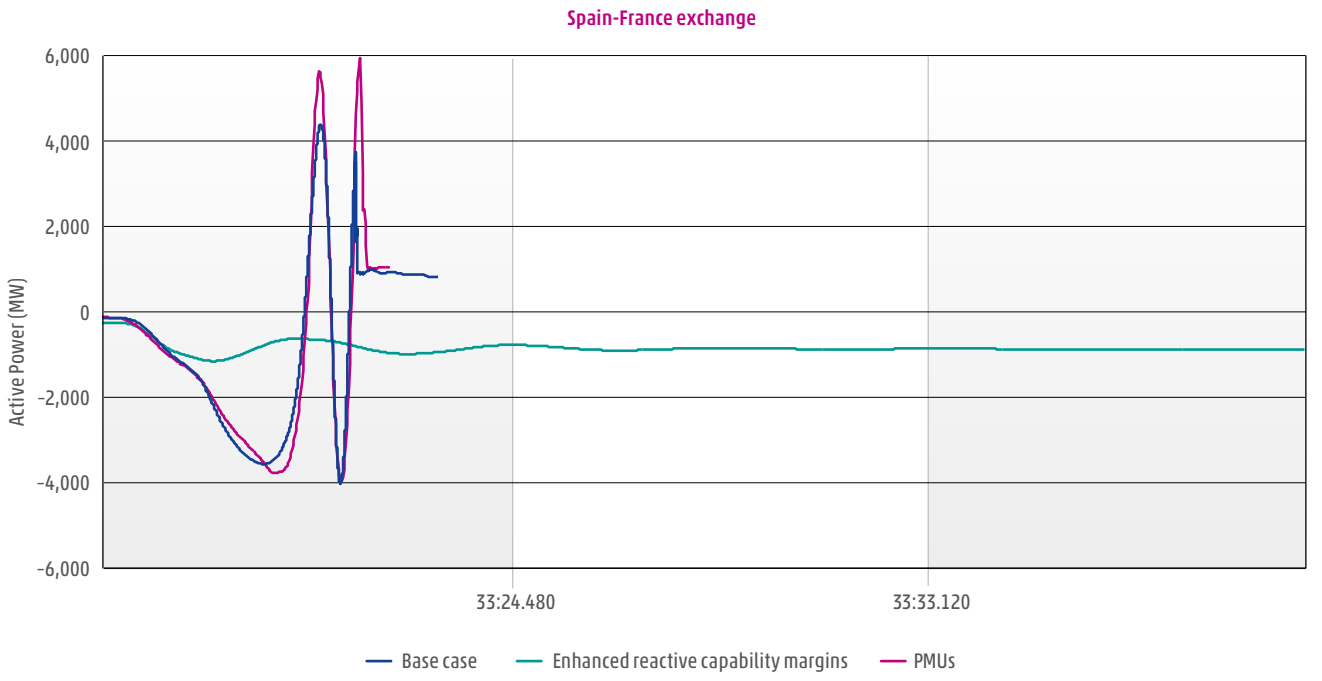


Figure 4-114: Transient response of the base case, the variant with eight new synchronous condensers, and the PMU measurements of active power exchange between Spain and France.

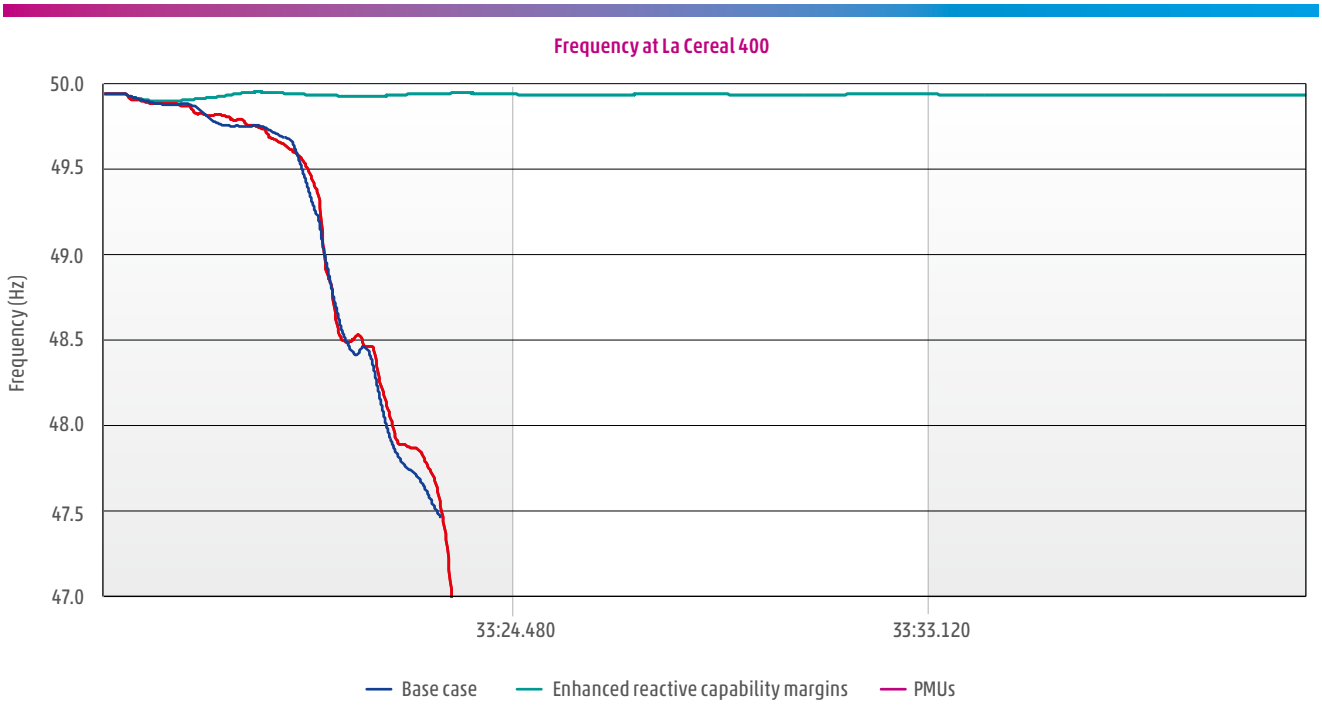


Figure 4-115: Transient response of the base case, the variant with eight new synchronous condensers, and the PMU measurements of active power exchange between Spain and France.

4.4.3.2.2 Role of Defence System Effects on Voltage Transient

This simulation aims to investigate the effect of blocking underfrequency load-shedding protections when voltages rise above a critical threshold. During the event, load-shedding protections were activated to mitigate the active power imbalance and slow the frequency drop caused by the cascade generation tripping. However, an undesired effect was the reduction of litne loading and, consequently, the voltage increases at the nearby buses as lines became increasingly capacitive. To perform this variant, the load-shedding dynamic models were deactivated. Figure 4-116 and Figure 4-117 show the Spain–France active power exchange and the frequency at the La Cereal 400 kV substation for the base case (blue), the case with the load-shedding protection blocked (green), and the PMU measurements (red).

Initially, the simulations are identical, and the blocking of load-shedding protections does not affect the initial disconnection of generation with the import increase and the subsequent loss of synchronism. However, around t = 5 s (12:33:21.000), the simulations begin to diverge. The simulation with load-shedding protections blocked shows a significantly faster decrease of frequency, reaching 47 Hz several seconds earlier than observed in the PMU measurements. These results indicate that, despite having a potentially negative effect on increasing voltage due to the reduction of line loading, the activation of load-shedding protections has a significantly positive effect on system frequency, reducing the active power imbalance and mitigating the frequency drop.

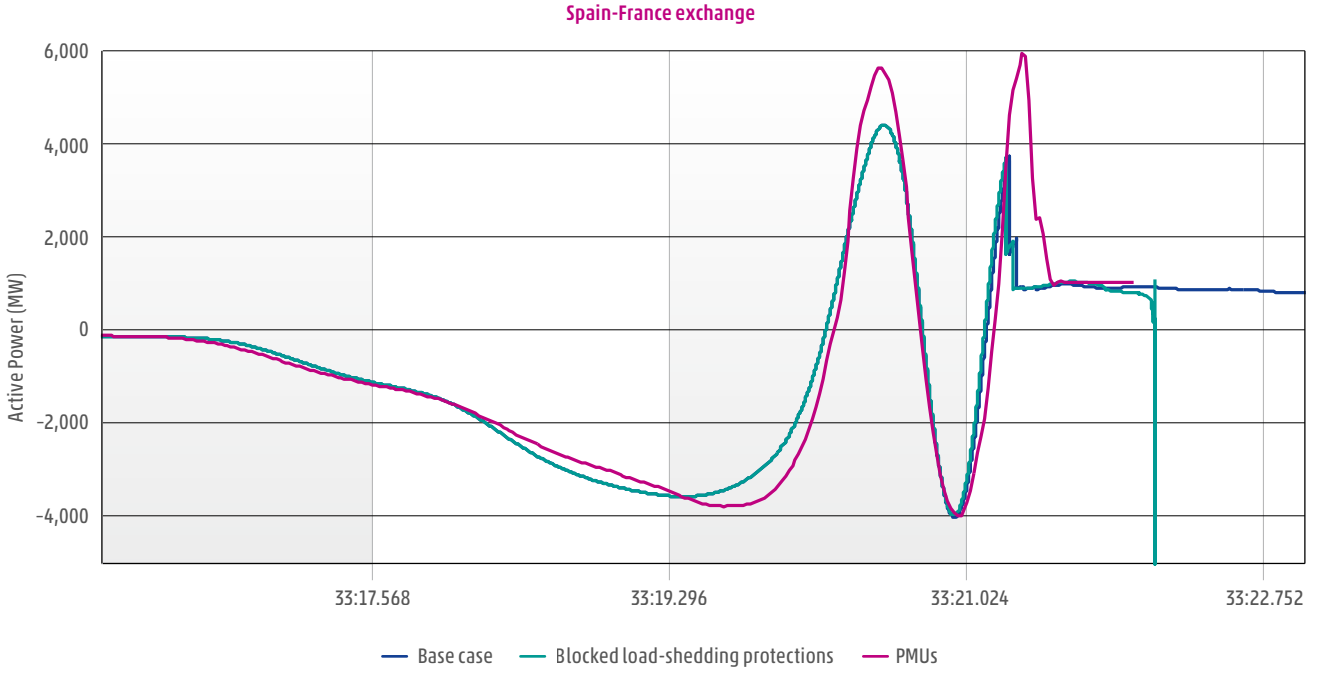


Figure 4-116: Transient response of the base case, the variant with load-shedding protection blocked, and the PMU measurements of active power exchange between Spain and France.

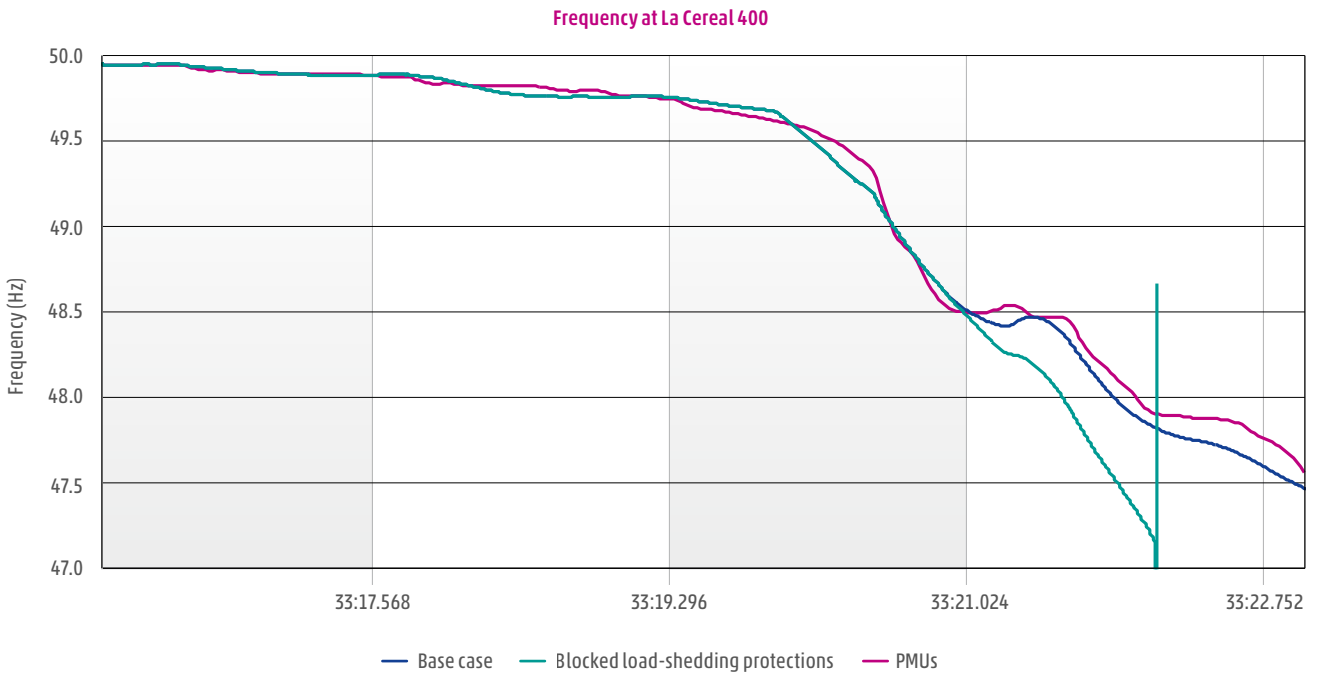


Figure 4-117: Transient response of the base case, the variant with load-shedding protection blocked, and the PMU measurements of the frequency at the La Cereal 400 kV substation.

4.4.3.3 Short-Circuit Power

In this chapter, the values of the three-phase short-circuit power are provided for relevant substations of the Spanish peninsular system at 11:53, and compared with the values during the previous year (2024).

In the table below you can find the percentage of the time (in 2024) where the value of Short circuit power was lower than the value of Short circuit power on 28 April at 11:53

Node	Short-Circuit Power (11:53) [MVA]	SC Power (11:53) Percentile 2024 [%]
SS REYES 400 kV	10,555.3	28.5
ELIANA 400 kV	11,167.3	47.2
HERNANI 400 kV	10,158.4	85.2
VIC 400 kV	8,830.1	0.1
RUBI 400 kV	11,104.3	39.4
MESON V. 400 kV	9,664.5	86.1
GUILLENA 400 kV	12,271.2	56.4
BIENVENIDA 400 kV	8,385.7	13.4
SS REYES 220 kV	9,657.4	27.5
ELIANA 220 kV	8,001.9	75.3
HERNANI 220 kV	6,372.2	81.5
VIC 220 kV	4,963.6	30.0
RUBI 220 kV	10,107.1	9.6
MESON V. 220 kV	7,330.5	74.0

Table 4-26: Short-circuit power in 400 kV substations in the Spanish peninsular system and comparison with 2024 values.

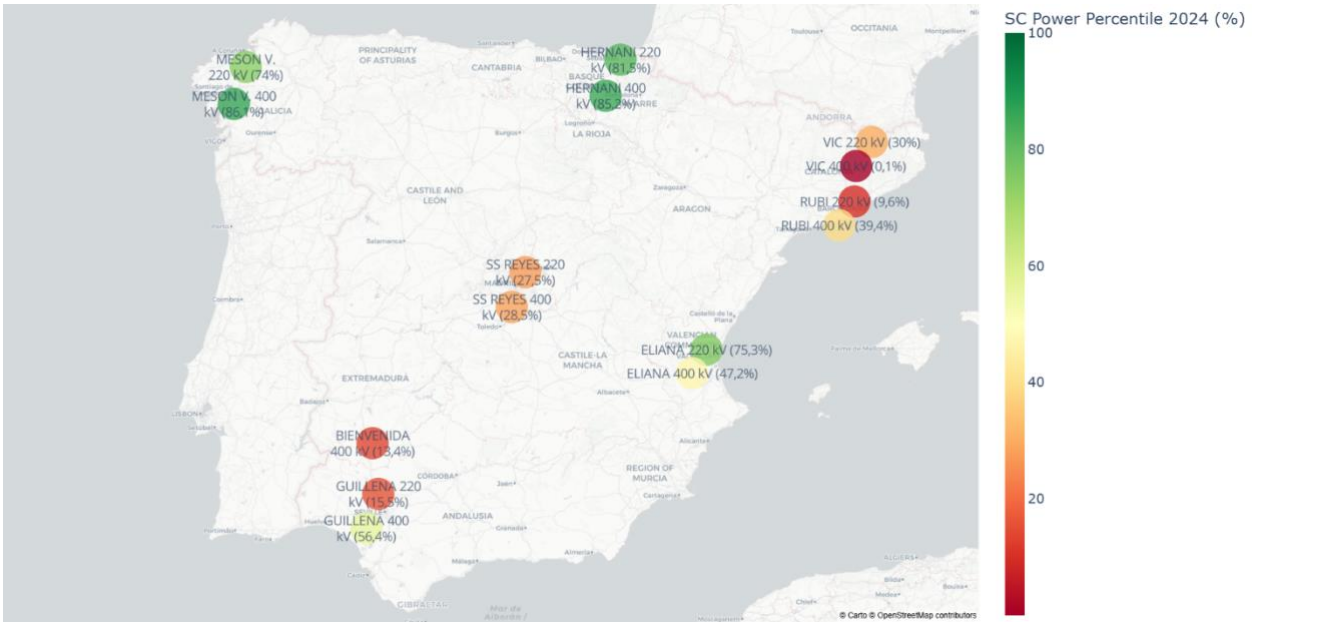


Figure 4-118: Short-circuit power in the 400 kV and 220 kV substations in the Spanish peninsular system and comparison with 2024 values.

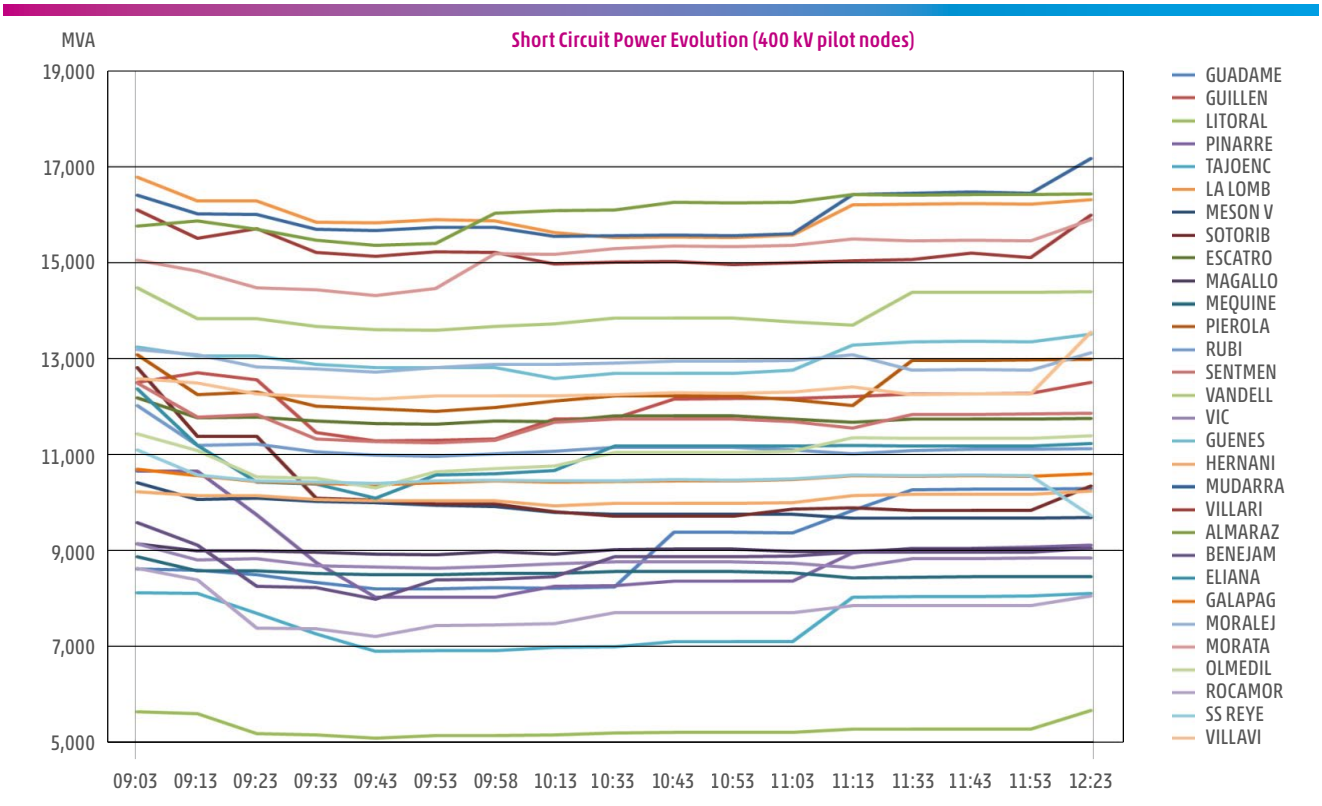


Figure 4-119: Short-circuit power evolution in the Spanish 400 kV network (pilot nodes).

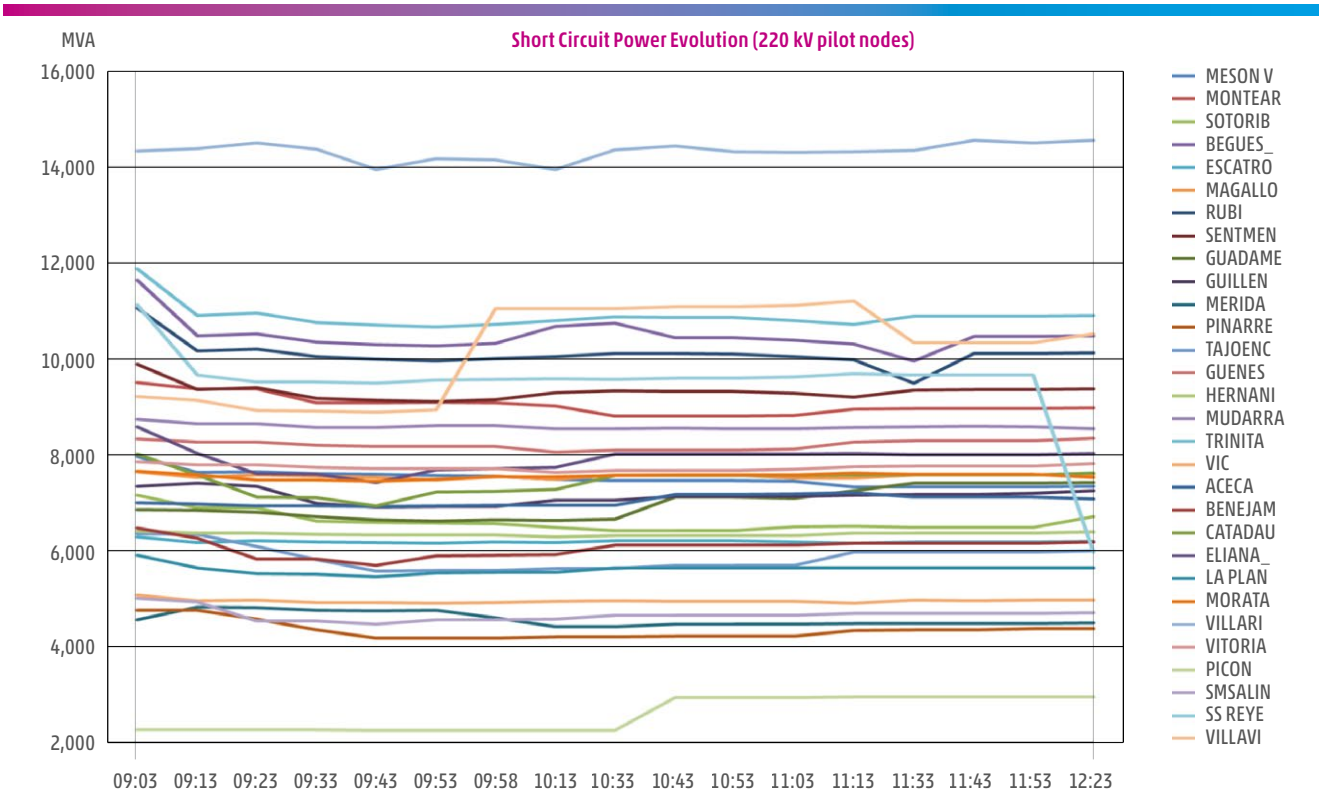


Figure 4-120: Short-circuit power evolution in the Spanish 220 kV network (pilot nodes).

Additionally, to gain a better understanding of the electrical situation of the Bienvenida 400 kV substation in Badajoz, the calculation of the operational short-circuit ratio (OSCR: ratio between the short-circuit power of a substation and the power infeed by inverter-based generators connected to it) is represented in Figure 4-121.

Inverter manufacturers for both generation and demand have typically considered values such as 10 and 6 in the past, and as low as 3 with the latest inverters, as the lowest limit for stable plant operation. The OSCR values on the morning of 28 April were higher than these technical references.

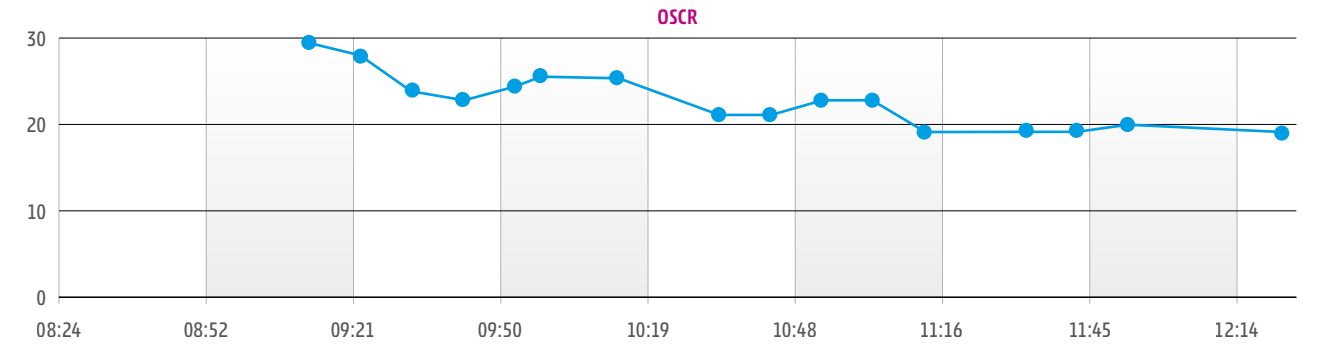


Figure 4-121: Evolution of OSCR at the Bienvenida 400 kV substation.

4.4.3.4 Conclusions Regarding Voltage Stability simulations

The voltage stability was investigated with a set of simulations aimed at obtaining the system sensitivity to different voltage control strategies.

The simulations demonstrated that, at time of the incident, the voltage of the Spanish power system was sensitive to any variations – whether related to exchanges with Portugal, France or Morocco, TSO-to-DSO flows, or generation below 50 MW – resulting in significant voltage fluctuations.

According to the results, the real capacity of the system to manage a certain excess of capacitive reactive power is the key point that explains the loss of voltage control.

In detail, the results of simulations can be summarised as follows:

- exchanges between Spain, Marocco and France confirm that voltage is strongly sensitive to the power flows
- exchange with DSOs shows a more evident sensitivity on voltage than the previous one
- reclose some transmission lines, which is a correct and effective maneuver to reduce oscillations, had the side effect to generate additional reactive power, worsening the voltage profiles
- automatic connection of shunt reactors during the voltage rise would have helped maintaining voltage within acceptable limits. By limiting the voltage increase, this action prevents further disconnections and therefore avoids cascading effects that could ultimately lead to a blackout.
- Increase the quantity of reactive power absorbed by synchronous machines (enlarging capabilities into the simulations to obtain a wider support by generators or adding to the rotating park synchronous condensers) confirms a good control of voltage, avoiding the rise out of control. That could be obtained adding more synchronous generators or in future, more synchronous condensers/ STATCOMs, enabling voltage regulation by static generators

4.4.3.5 System Defence Performance

This section evaluates the performance of low-frequency demand disconnection (LFDD) in the Iberian Peninsula power system during the incident.

4.4.3.5.1 TSO Level

Red Eléctrica provided the oscillographic records for all load-shedding relays installed at transmission network substations.

A detailed analysis of this information enables accurate verification of the load shed at transmission positions for each step, as well as identification of the exact instant of the trip, the amount of load shed, and the frequency value at the moment of pickup and tripping of the load-shedding relays.

Likewise, an exhaustive assessment of the behaviour of the load-shedding relays was carried out, confirming that in all cases, the relays operated correctly according to their intended design, in line with the pickup thresholds and time delays established for each step of the load-shedding scheme.

The blocking function in case of reverse active power flow was activated as designed in those LFDD relays installed on the transmission network that require it. Therefore, it can be concluded that the performance of the LFDD at transmission level positions was correct and in accordance with the design.

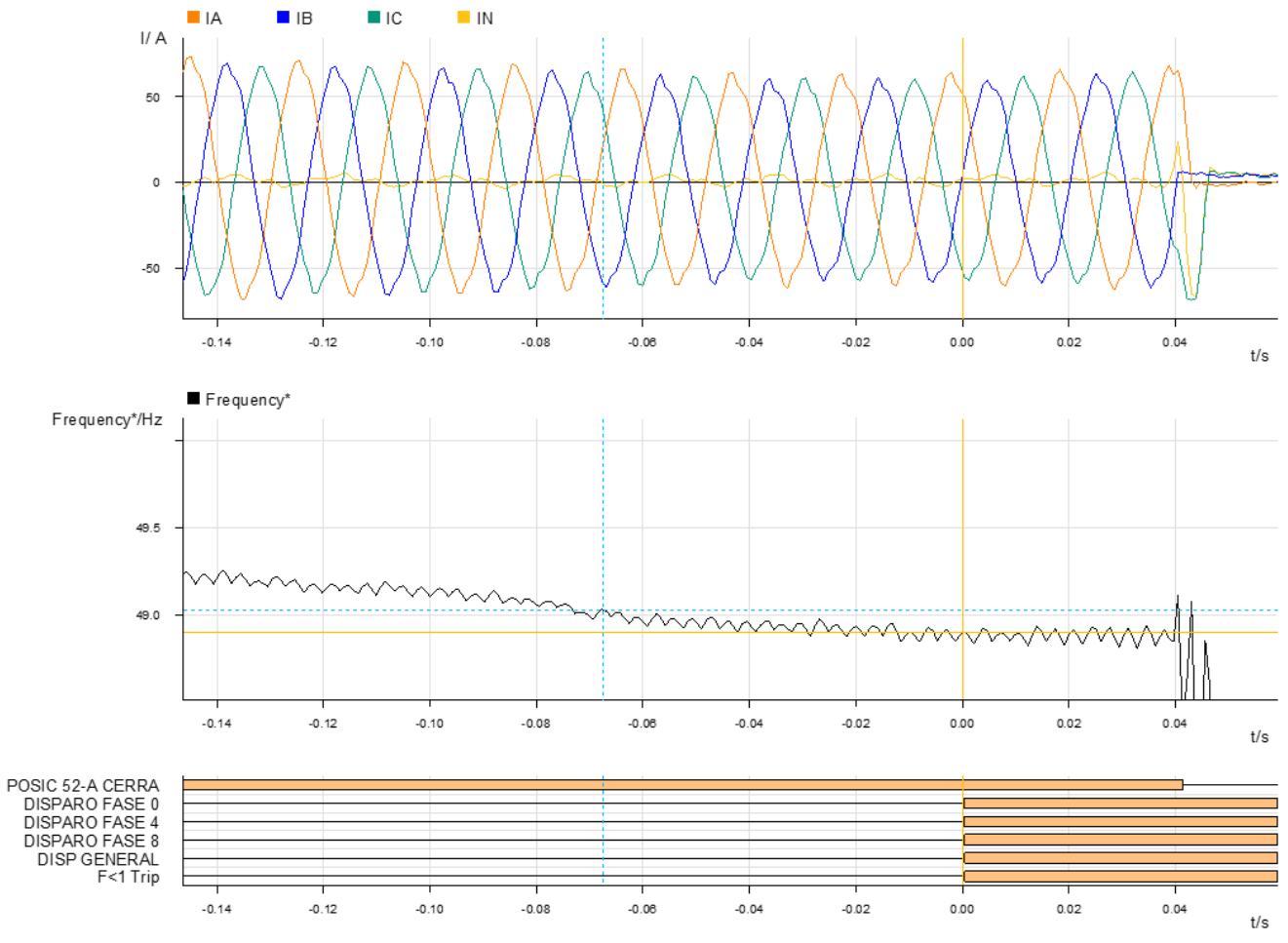


Figure 4-122: Example of oscillographic record.

4.4.3.5.2 DSO Level

The DSOs provided the load-shedding demand data associated with each step at each distribution network position. Some also provided the instant of disconnection.

The overall assessment of the LFDD was satisfactory. In general, no deviations from the applicable requirements were observed in the LFDD operation by DSOs. Nevertheless, an analysis of the information received by DSOs revealed certain limitations that may serve as a starting point for potential improvements in future evaluations of the system defence plan:

- Oscillograph records from every individual load-shedding relay that operated were not provided to the Expert Panel, making it impossible to thoroughly verify their individual performance.
- The number of LFDD relays with blocking functionality for reverse active power flow in the distribution network is very limited, as this requirement has only applied since the Demand Connection Network Code came into force in 2016.
- Most DSOs reported having very limited means to detect the volume of unwanted disconnection of DER less than 1 MW along with loads, due to their volatile influence over time. This is especially important for selecting feeders with lower DER penetration to avoid disconnecting generation.

4.4.3.6 Conclusion Regarding LFDD Performance

LFDD is a system protection scheme designed to automatically shed predefined loads when the system frequency falls below set thresholds. Its main objective is to prevent a widespread system collapse by rapidly reducing demand to restore frequency and stop its decline.

In this incident, although the LFDD scheme operated according to design, its contribution was not sufficient to preserve system stability under the specific conditions of this event. The incident was not just caused by a generation deficit and the resulting frequency drop, but was accompanied by a widespread overvoltage scenario, both prior to the activation of the LFDD and during the load disconnection following the activation of all steps, as explained in previous sections.

During the period in which the underfrequency load-shedding steps were activated, it was observed that, despite fully shedding the demand associated with each step, the system frequency continued to drop, because the generators were simultaneously tripping due to overvoltage. In addition, the voltages continued to rise for two reasons.

On one hand, the load shedding reduced currents, causing voltage drops in line and cable sections, resulting in a higher residual voltage and higher reactive power generated by the transmission lines. On the other hand, the simultaneous disconnection of generators reduced active voltage control and reactive power absorption, resulting in higher residual voltage.

LFDD was designed and activated to protect the system against frequency drops in line with the applicable requirements. However, it was not intended to stabilise the system under conditions of widespread overvoltage across the network.

Simulations indicate that if the LFDD relays had been blocked due to overvoltages – which means no load shedding for frequencies below 49.0 Hz – the frequency would have collapsed even faster than in reality. The actual LFDD scheme – without blocking in case of overvoltages – did not worsen the outcome. On the contrary, it managed to delay the system collapse, which could unfortunately not be avoided due to other effects for which the LFDD scheme was not designed.

4.5 Management of alarms related to voltage

The provisions applicable to voltage control and reactive power management under the Commission Regulation (EU) 2017/1485 of 2 August 2017 establishing a guideline on electricity transmission system operation (SO GL) provide as follows:

- Each TSO is responsible for maintaining voltage at transmission system connection points within the defined steady-state ranges during normal operating conditions, as provided in Article 27 *Obligations of all TSOs regarding voltage limits*.
- If the voltage at any connection point deviates from these ranges, the TSO must implement appropriate voltage control and reactive power management actions to restore compliance, in accordance with Article 29 *Obligations of all TSOs concerning voltage control and reactive power management in system operation*.

4.5.1 Alarms at the TSO level

This section addresses alarm management within the TSO control centres of RTE, REN and Red Eléctrica.

TSO control centres receive real-time alarms in their SCADA systems whenever voltage levels at transmission grid nodes approach or exceed the voltage limits defined by the applicable requirements on national[30] and European levels respectively, based on measured data collected at intervals ranging from one minute to seconds. These thresholds are customised to meet each TSO's voltage control requirements, such as applying stricter limits due to contracts with transmission-connected consumers or producers or HV equipment, or based on node categorisation for voltage monitoring or alignment with the TSO–DSO voltage control agreement.[31]

In real-time operation, alarms are configured to signal when operational thresholds for loading or voltage are reached, as defined by established procedures and practices. When the measured voltage at a substation exceeds the specified threshold, an alarm is displayed on the dispatcher's interface and may trigger additional alerts in the control room (e.g., audible signals such as a bell).

Subsequently, alarm management practices were analysed within the control centres of TSOs, DSOs and generation facilities. In addition, an analysis was conducted of the voltage alarms received from 12:00:00 to 12:33:17 at the control centres, and the voltage manual control actions taken by operators.

Tables 4-27 to 4-29 show the defined threshold values for voltage alarm activation for each TSO:

Nominal voltage	Min. forbidden threshold [kV]	Close to max. [kV]	Max. forbidden threshold [kV]
400 kV	380	419.5	420
225 kV	200	244.7	245
150 kV	125	164.7	165
90 kV	82.8	96.9	97.2
63 kV	58	67.7	68

Table 4-27: Thresholds defined for RTE's voltage alarms

Nominal voltage	Undervoltage limit [kV]	Overvoltage limit [kV]
400 kV	380	420
220 kV	209	245
150 kV	142	165

Table 4-28: Thresholds defined for REN's voltage alarms

30 The applicable national regulation in Spain is the PO 1.4, in Portugal is the "Manual de Procedimentos da Gestão Global do Sistema do setor elétrico" (MPGGS) and in France is the Code de conduite du réseau de transport C1-1 - French Regulation: Décret No. 2003-588 Article 2 - EU Regulation: No. 2017/1485 Article 27 and Annex II Table 2.

31 In accordance with Article 29(5) of Regulation (EU) 2017/1485.

		Low-voltage alarm limits		High-voltage alarm limits	
Nominal Voltage	**Type of node**	**First level [kV]**	**Second level [kV]**	**First level [kV]**	**Second level [kV]**
400 kV	Transmission network nodes	390		430	
	Pilot nodes for voltage control in the transmission network	395	375	420	435
	Pilot nodes for voltage agreement (TSO-DSO)	380	375	420	435
	Border and near interconnections nodes	395	360	420	435
220 kV	Transmission network nodes	200		240	
	Pilot nodes for voltage agreement (TSO-DSO)	205	198	235	246
	Border and near interconnections nodes	198		246	

Table 4-29: Thresholds defined for Red Eléctrica's voltage alarms[32]

In general, these voltage alarm thresholds remain constant across all system states, both in real-time operation and in operational planning. However, it is foreseen that temporary extended limits may be applied under exceptional circumstances, such as during top-down restoration strategies for very short periods of time, or, in the case of RTE, under N 1 conditions as specified in the applicable regulations.

The TSOs have also implemented additional tools for voltage control monitoring:

- RTE provides a voltage heat map that issues audible and visual alarms when under- or over-voltage limits are exceeded, with a geographical representation.
- REN uses a dedicated SCADA display showing all busbar voltage values in a tabular format. When limits are exceeded, the corresponding values blink in red.
- Red Eléctrica operates complementary screens that display measurements from generation units with dynamic voltage control capability, the status of discrete elements, and voltage trends at pilot nodes in each zone. Operators can also access zonal maps displaying voltage values received from each substation. When the first level limits are exceeded, the corresponding values turn red.

The TSOs have simulation tools capable of performing N and N-1 analyses in real time. RTE and Red Eléctrica analyse short-term scenarios to anticipate random reactive power losses caused by any element tripping.

For alarm management, the TSOs primarily rely on SCADA data rather than PMU data or other sources. In the absence of direct measurements or if necessary for specific tools, estimated or calculated data may be used.

Regarding alarm prioritisation, each TSO has implemented its own approach using different classification methods, such as alarm levels, colour codes or audible or visual alerts, to help operators quickly identify and prioritise actions. Alarms function independently of the generator type connected to each node and are triggered immediately once the threshold is exceeded. As shown in the defined thresholds, the alarm thresholds defined by RTE, REN and Red Eléctrica are stricter than the limits for remaining connected without damage or disconnection – under both steady-state and temporary conditions – applicable to generation connected to the transmission network, as defined by the applicable requirements (Sections 3.2.1.1 and 3.2.1.2). This ensures an operational margin in line with the applicable requirements. In RTE's case, the starting time above the threshold is displayed.

Operators continuously monitor voltage levels and alarms to implement preventive measures before threshold limits are reached. If limits are exceeded, corrective actions are carried out in accordance with the applicable requirements, established based on, among others, national (TEN-101 and TEN-102) or regional instructions for RTE; operational handbooks and training documentation for REN, and national (PO 7.4) and European regulations (SO GL) and training documentation for Red Eléctrica. Voltage control actions are executed through SCADA systems, market platforms or by phone, depending on the measure to be applied. If any action affects the DSO network or a specific generator,

32 Pilot nodes for voltage control are nodes distributed throughout the transmission network and represent the voltage profile of each area. Pilot nodes for voltage agreement represent the voltage at the connection point between the TSO and the DSO. Border and near interconnections nodes represent the voltage close to interconnections.

Red Eléctrica informs the affected DSO or Generation Control Centre (GCC) in advance. When established by the applicable requirements or mutual agreement, certain actions, such as load transfers performed by the DSO or voltage and reactive power adjustments carried out by the GCC, are executed at the TSO's request.

Each operator acts within its scope of responsibilities: the TSO maintains transmission network voltage within the limits defined by the applicable requirements, the DSO ensures supply quality in the distribution network, and GCCs operate their generators to align with the requirements applicable to the voltage control, following TSO voltage instructions in the case of REN or RTE, or as established by the applicable requirements (PO 7.4 and RD 423/2014) in the case of Red Eléctrica.

When a TSO, DSO or GCC detects a voltage issue, they notify the need to coordinate the necessary actions or apply national procedures and agreements (such as TSO–DSO voltage control agreements) under the TSO's guidance. Coordination in these cases can be undertaken by phone or via tools that link the different control centres.

4.5.2 Alarms at the DSO level

This section contains the information that relevant distributors[33] provided regarding the management of voltage alarms in their control centres. The DSO control centres receive alarms related to voltage levels in their respective distribution networks, with varying alarm thresholds among distributors.

The reported values and criteria are summarised below, whereby one DSO corresponds to the Portuguese electricity system, while the remaining ones belong to the Spanish electricity system:

- **DSO 1:** Has two alarm levels, normally set at ±1.05 p.u. and ±1.07 p.u. with respect to the operating voltage, indicating that these ranges may vary at certain points in the network.
- **DSO 2:** Has three alarm levels for undervoltage and overvoltage, defined within the limits defined by the applicable requirements and adjustable according to the connected load, always below ±1.10 p.u. with respect to the nominal voltage.
- **DSO 3:** Confirms the existence of voltage and reactive power alarms without specifying thresholds. It indicates that these vary depending on the network area and are defined with sufficient tolerance to allow the necessary operations to keep voltage within the limits defined by the applicable requirements.
- **DSO 4:** Has one alarm level for overvoltage (1.07 p.u. with respect to nominal voltage) and two levels for undervoltage (0.93 p.u. and 0.60 p.u. with respect to nominal voltage), applied uniformly.
- **DSO 5:** Sets alarms at 1.04 p.u. from normal operating values for each transformer and busbar. In some cases, these thresholds are readjusted to meet the operational requirements of certain facilities.

DSOs do not specify whether these alarms are coordinated with the trip thresholds of the generation connected to their network. Some also state that the generation connected to their network must align with the thresholds defined by the applicable requirements.

If alarm thresholds are exceeded, DSOs indicate that the primary solution to resolve voltage deviations is the on-load tap changer of their transformers (automatic or manual), while they also have capacitor banks. If the high-voltage side is significantly deviated, they contact the TSO to take appropriate actions or to apply the TSO–DSO voltage control agreement at the interconnection point. In particular, the Portuguese DSO indicates that it may take additional actions, such as changing the reactive power control mode or the setpoint of RfG-compliant type B, C and D generators, as well as transferring or switching off loads. Additionally, DSO control centres have alarms for transformer tap status (extreme tap position). In one specific case, supervision has also been implemented for changes in short-term international exchange programmes between Spain and neighbouring systems, due to their impact on regional voltage.

33 For the purposes of this assessment, a targeted sample of relevant distributors – those connected to transmission network with a large volume of customers (> 100.000, in the case of DSOs connected to the Spanish electricity system) – was contacted. It should be noted that the Spanish electricity system comprises more than 300 distributors.

In general, DSO control centres have established alarm prioritisation based on purpose, generating both visual and audible alerts. These alarms are triggered based on information received through each control centre's SCADA system.

Regarding the use of other tools, two DSOs describe the use of heat maps, real-time evolution graphs of electrical magnitudes or graphical representations of these with applied colour codes. Additionally, one DSO reports that its control centre performs day-ahead contingency analyses.

4.5.3 Alarms at the GCC level

This section presents the analysis of GCC responses regarding voltage control alarm management. The Expert Panel requested information from 12 control centres, selected as being the most representative, from which 10 responses were received: four from centres managing generation in Portugal and six in Spain.

Some GCCs have real-time voltage measurements through their SCADA systems, obtained from signals transmitted by local control units or connection nodes. In most cases, GCCs receive voltage alarms for conventional generation and hydro, but not for renewables.

Voltage-related alarms in GCCs are primarily triggered by the following causes:

- Activation of protections associated with generating units (under/overvoltage trips), reported in four of the GCCs analysed.
- In some cases, deviation of the reactive power delivery or voltage levels from the limits defined by the applicable requirements, as observed in five of the GCCs analysed.
- Exceeding alert thresholds defined according to the protection settings of generating units, reported in one of the GCCs analysed.

In most control centres, the type of generation unit determines the criteria for voltage alarms and actions by control centre operators. For example, in conventional units, voltage is typically managed on site rather than via generator control centres. In the case of hydro plants, voltages are managed in the control centres. Renewable generation does not have voltage violations but rather alarms are raised in the case of over/under voltage trips.

Some control centres report no alarms for certain generators, as voltage control in these cases is performed locally. However, other GCCs implement multiple types of alarms. Typically, these criteria are applied consistently across all plants within the same technology.

Based on the type of alarm received, certain GCCs have indicated that operators may implement one of the following actions:

- Restore the lost generator.
- Request information or instructions from the relevant system operator.
- Request information from the generator.

Each GCC has established its own alarm prioritisation based on mechanisms such as audible alarms, colour codes and flashing indicators, depending on criticality. In general, alarms are received through each control centre's SCADA system.

Additionally, some GCCs indicate that if the operator identifies anomalous voltage values or any disturbance, they contact the TSO's control centre to report the situation and, if necessary, adapt operations in accordance with the TSO's instructions.

4.5.4 Analysis of voltage alarms recorded in Red Eléctrica control centres and their operational handling

In this section, the voltage control alarms received at the Red Eléctrica control centres between 12:00 and 12:33:17 are analysed, as are the voltage-control and stability-related switching operations carried out on the transmission network.

For the period between 12:00 and 12:23, the analysis covers only the alarms recorded at the 400 kV level. From 12:25 onwards, the assessment includes all alarms received at both 400 kV and 220 kV to provide a more in-depth evaluation of the moments leading up to the blackout.

Each period has been divided into time intervals during which voltage alarms were triggered. Between these intervals, no additional alarms were triggered, except for isolated activations that cleared quickly.

Figure 4-123 illustrates the distribution of each of the control areas referenced in this analysis.

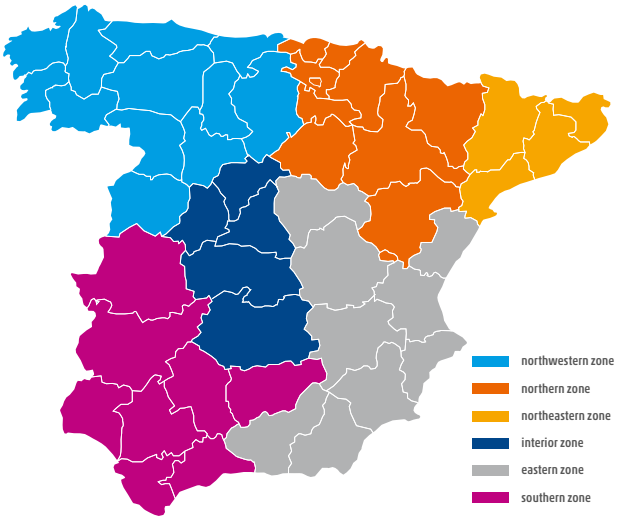


Figure 4-123: Map of Red Eléctrica´s control areas

From 12:00 to 12:07:43 [7 min 43 sec]

A total of 678 low-voltage alarm threshold events were recorded during this period: 258 in the southern zone, 247 in the eastern zone and 174 across the remaining zones. Of these, 53 alarms corresponded to nodes outside the transmission network, where the TSO is not responsible for voltage control. Six nodes activated a combined total of 122 alarms, while the remaining 78 nodes activated a total of 122 alarms.

In the eastern zone, three alarms registered voltage values below 375 kV. All cleared within 30 seconds, remaining outside the limits of PO 1.4 but within the thresholds established in the SO GL. Alarm activations and clearances occurred continuously alongside voltage control actions carried out across the transmission system.

Voltage control actions taken in each zone during this period included:

- Shunt reactors switched off: two in the south, one in the interior and two in the north.
- Lines switched on: one in the interior.
- HVDC voltage setpoint increased: raised to 412 kV in the northwest.

Stability control actions taken in each zone during this period included:

- Lines switched on: one in the south, two in the east, and two in the northwest

From 12:09:17 to 12:10:29 [1 min 12 sec]

A total of 27 logs related to high-voltage alarm thresholds were triggered: 12 in the northern zone, 10 distributed between the southern and eastern zones, and the remainder in the interior and northeastern zones. One corresponded to a node located outside the scope of TSO voltage management.

All alarms remained below 430 kV, with the maximum recorded voltage value reaching 422.4 kV at a node in the southern zone. No additional control actions were required, and most alarms returned to below their threshold values within a few seconds to 5 minutes.

From 12:14:48 to 12:23:22 [8 min 34 sec]

A total of 348 low-voltage alarm threshold events were recorded across the transmission network: 125 in the eastern zone, 139 in the southern zone, 65 in the interior zone, and the remainder in the northern and northeastern zones. Among these, three alarms registered voltage values below 375 kV, all of them occurring in the southern zone. Two of these corresponded to the same node located outside the transmission network, where the TSO is not responsible for voltage management. During this period, five nodes activated 64 alarms (4 of them located in the southern zone), while the remaining 67 nodes activated 67 alarms.

Overall, 25 alarms originated from nodes outside the transmission network, of which 22 occurred in the southern zone. As in previous periods, alarms were activated and cleared as voltage control actions were executed across the transmission network.

The following voltage control actions were applied during this period:

- Shunt reactors switched off: one in the southern zone and one in the northern zone.

Stability control actions taken in each zone during this period included:

- Lines switched on: one in the southern zone, two in the interior zone and two in the eastern zone.

From 12:25:24 to 12:28:05 [2 min 41 sec]

A total of 398 high-voltage alarm threshold events were recorded across 231 nodes, with 70 in the interior zone, 54 in the northern zone, 49 in the southern zone, 36 in the northwestern zone, 20 in the eastern and northeastern zones and two in the RTE network. Thirty-seven of these alarms corresponded to 27 nodes outside the transmission network, which fall outside the TSO's management scope. Consequently, a total of 204 transmission network nodes exhibited alarm events, on the order of 30 % of the nodes comprising that network. Dispatchers implemented remedial voltage control actions to restore voltage to alarming thresholds.

Among the affected nodes, 47 recorded voltage values of 420 – 424 kV, 15 of 430 – 433 kV and one node in the southern zone exceeded 435 kV. This 435 kV alarm event was cleared within 16 seconds after voltage control actions were applied. At 220 kV, 49 nodes registered values of 235 – 237 kV, 124 nodes registered values of 240 – 243 kV, and two nodes in the RTE network exceeded 245 kV.

It should be noted that some alarms are configured to activate when voltage levels approach the limits defined by the applicable requirements. Aside from the alarm activation above 435 kV, none of the alarms within the transmission network under Red Eléctrica's supervision exceeded the limits defined by the applicable requirements (Section 2.6.1).

Voltage values returned below the alarming threshold between a few seconds and 6 minutes after the following voltage control actions:

- Shunt reactors switched on: two in the southern zone, two in the northern zone and one in the interior zone.
- HVDC voltage setpoint adjustment: reduced to 409 kV in the northwestern zone.

The intervals between the first alarm and the first action executed by the operator in the northern and southern zones were 1 min 33 sec and 2 min 11 sec, respectively.

From 12:32:39 to 12:33:17 [38 sec]

During this interval, 445 logs related to high-voltage alarm thresholds were triggered across 259 nodes: 83 in the interior zone, 75 in the southern zone, 44 in the northern zone, 28 in the eastern zone, 25 in the northwestern zone, and four in the northeastern zone. Twenty-five alarms originated from 19 nodes outside the transmission network, where the TSO is not responsible for managing voltage control. Therefore, a total of 240 nodes in the transmission network triggered an alarm event, representing approximately 35 % of all nodes in that network.

Within the transmission network, 42 nodes registered voltage values of 420 – 427 kV, 22 of 430 – 433 kV and one node in the northwestern zone reached 435.2 kV (with no generation connected at that location). At 220 kV, 23 nodes recorded voltages of 235 – 237 kV, 170 nodes recorded voltages of 240 – 245 kV, and one node in the southern zone exceeded 245 kV (generation is connected to this node; the recorded voltage was 247.1 kV).

Due to the brief duration of this 38-second event, operators were unable to execute any voltage control actions.

It should be noted that some alarms are configured to activate when voltage levels approach Spanish regulatory thresholds. Aside from alarm activation at voltages above 435 kV and 245 kV, none of the alarms within the transmission network under Red Eléctrica's supervision exceeded the limits defined in the applicable regulation (Section 2.6.1).

4.5.5 Analysis of voltage alarms recorded in REN control centres and their operational handling

The following section presents the analysis of the alarms received at REN's control centres at the 400 kV level (no alarms were recorded at the 220 kV level during this period).

From 12:04:00 to 12:05:46 [1 min 46 sec]

During this period, 10 undervoltage alarms were triggered across two nodes at the 400 kV level, reaching a minimum value of 378.7 kV. Alarm activations and clearances occurred continuously throughout this interval.

Manual actions from operators were not needed because of the nature of oscillations resulting in undervoltage spikes.

From 12:19:17 to 12:21:18 [2 min 1 sec]

During this period, 14 undervoltage alarms were triggered across five nodes at the 400 kV level, reaching a minimum value of 377.9 kV.

As in the previous interval, manual actions from operators were not needed because of the nature of oscillations resulting in undervoltage spikes. Alarm activations and clearances continued occurring throughout this period.

From 12:32:45 to 12:33:17 [32 sec]

During this interval, 56 overvoltage alarms were triggered across 16 nodes at the 400 kV level, all exceeding 420 kV.

Alarms are managed by operators assessing the validity of an alarm, preparing remedial actions considering their consequences and then taking action. The operator did not have sufficient time to take voltage control actions and follow alarm management process.

4.5.6 Analysis of voltage alarms recorded in RTE control centres and their operational handling

According to the TEN/101 handbook, RTE only considers alarms from the voltage heatmap, which is refreshed every 5 minutes and generated from calculated or estimated voltages.

Alarms registered on the voltage heat map

At 12:10, one node exceeded the threshold, registering a voltage of 245.9 kV, and another was close to the threshold, with a voltage of 244.7 kV.

At 12:15, one node registered a voltage close to the threshold, with a value of 244.7 kV.

Alarms registered on SCADA

No voltage-related alarms were registered in the control centres between 12:00:00 and 12:33:17.

4.5.7 Analysis of voltage alarms recorded in distribution control centres

In this section, the alarms reported by one distribution control centre in Spain are analysed, since the Expert Panel had not received information from the other relevant DSOs.

The following periods are noteworthy:

From 12:05:39 to 12:10:29 [4 min 50 sec]

A total of 1,024 alarms were recorded across 144 nodes due to voltage levels exceeding the distributor's threshold, mainly at 45 kV (583 alarms) and 15 kV (374 alarms). The remaining alarms corresponded to different voltage levels (220, 132, 50 and 20 kV). The maximum voltage value recorded during this interval was 1.09 p.u. at the 45 and 220 kV levels. They also include alarms associated with transmission grid nodes located at the boundary with the distribution grid. Some alarms cleared within the interval itself, albeit only briefly.

After this period, additional high-voltage-related activations and clearances occurred, most of which were resolved within seconds.

From 12:21:25 to 12:28:16 [7 min 51 sec]

A total of 2,179 alarms were triggered due to voltage levels exceeding the distributor's limit. The majority occurred at 45 kV (933 alarms), 220 kV (574 alarms) and 15 kV (626 alarms), while the remaining ones corresponded to other voltage levels (132, 66, 50 and 20 kV). During this period, the alarms were triggered at 290 nodes, with recorded maximum voltage values of 1.09 p.u. at 15 kV and 45 kV, and 1.10 p.u. at 220 kV.

The dataset also includes alarms at transmission network nodes bordering the distribution network. During this interval, some alarms briefly returned to normal.

Once this period ended, additional activations and clearances associated with high voltage were recorded, with most clearing within a few seconds.

From 12:32:42 to 12:33:17 [35 sec]

A total of 947 alarms were recorded for exceeding the voltage levels defined by the distributor, mainly at 220 kV (410 alarms), 45 kV (409 alarms), and 15 kV (101 alarms), while the remainder were at 132, 66, and 20 kV. The affected area comprised 198 nodes, with maximum recorded voltages values of 1.10 p.u. at 45 kV and 1.11 p.u. at 220 kV. The alarms also include those associated with transmission network nodes at the boundary with the distribution network.

No analysis has been conducted regarding the voltage control actions applied by the distributor during any of the three periods.

4.5.8 Analysis of voltage alarms recorded in generation control centres

As noted in the previous analysis, generation control centres have only specific alarms related to voltage control in very specific cases.

In this context, the alarms recorded in the systems of one of the control centres for hydro plants located in Spain were analysed, which revealed the following:

From 12:00 to 12:07:30 [7 min 30 sec]

The control centre received 76 alarms for voltage below the nominal value, affecting six nodes (five in the southern zone and one in the northwestern zone) at different voltage levels: 400, 132, 66 and 11 kV. The minimum voltage values recorded during this interval were 0.88 p.u. at the 11 kV and 0.93 p.u. at 132 kV levels.

From 12:08:46 to 12:15:02 [6 min 16 sec]

A total of 26 alarms for voltage above the nominal value were recorded at 220, 132, 45 and 25 kV, affecting four nodes (two in the southern zone, one in the interior zone and one in the northwestern zone). The maximum voltage values recorded during this interval were 1.10 p.u. at the 45 kV level and 1.08 p.u. at the 25 kV level.

From 12:17:06 to 12:21:42 [4 min 36sec]

Another 26 alarms for voltage below nominal were received, affecting six nodes (five in the southern zone and one in the northwestern zone), matching those from the first period. The voltage levels involved were 400, 132, 66 and 11 kV. The minimum voltage values observed during this period were 0.89 p.u. at the 11 kV level and 0.94 p.u. at the 132 kV level.

Across these first three intervals, alarms were repeatedly activated and cleared at the same nodes, indicating voltage variations occurring close to the configured thresholds.

From 12:25:38 to 12:30:43 [6 min 5 sec]

During this interval, 26 alarms for voltage above the nominal value were recorded at 220, 132, 45 and 33 kV, corresponding to five nodes (three in the northwestern zone, one in the southern zone and one in the interior zone), some of which were also affected during the second period. The maximum voltage values observed during this interval were 1.1 p.u. at the 45 kV level and 1.09 p.u. at the 220 kV level.

From 12:32:46 to 12:33:17 [31 sec]

In the final interval, 11 alarms for voltage above the nominal value were recorded at 220, 45, 33 and 15 kV, corresponding to four nodes (three in the northwestern zone and one in the interior zone). These nodes were affected during previous periods. The maximum voltage values observed during this interval were 1.10 p.u. at the 45 kV level and 1.09 p.u. at the 220 kV level.

It is worth noting that across all intervals with voltages above the nominal value, the 45 kV node accounted for the majority of alarms, around 80 %.

4.5.9. Conclusions from alarms recorded in Red Eléctrica's control centre and their operational handling

- The number of voltage alarms was approximately four times the number of affected nodes. This is consistent with the configuration of alarms, which are issued per busbar measurement and, for 400 kV pilot nodes, for two alarm thresholds. At the end, more than one alarm was activated on the same node, indicating that the alarm log did not necessarily reflect a widespread impact across all nodes, but rather multiple indications of a single voltage condition.
- The deactivation of alarms is finalised only when the voltage leaves the ±2,5 kV deadband. This means that voltages dropped below the high-voltage alarm threshold or exceeded the low-voltage alarm threshold faster than the durations extracted from the alarm log.
- Two of the analysed time intervals showed a significant number of alarm events triggered below the low-voltage alarm thresholds, with 678 and 348 activations, respectively. Only a small subset of alarms (0.6 %) was triggered at voltages below 375 kV, and all were cleared rapidly within 30 seconds.
- The activation of most alarm events triggered above the high-voltage alarm thresholds occurred at voltages below 430 kV (76.1 %), with a limited number at 430 – 435 kV (22.6 %) and only sporadic activations above 435 kV (1.1 %). Regarding alarms for voltages above 435 kV:
  - Between 12:25 and 12:28, one alarm event was triggered due to a voltage exceeding 435 kV in the southern zone. It was cleared within 16 seconds after implementing voltage control remedial actions.
  - In the final analysed period, between 12:32 and 12:33:17, two alarms were activated, with the voltage reaching 435.2 kV at a node in the northwestern zone, both associated with the two busbars of the same substation, which have no generation units connected.
  - It should be noted that voltage alarm thresholds are configured to activate when voltage levels approach the limits defined by the applicable requirements in Spain.
- In the last two periods analyzed, the transmission network nodes that triggered at least one alarm event represented approximately 30 % and 35 % of the total nodes, respectively, whereas in the preceding periods the number of affected nodes was significantly lower, on the order of 10 %. During the same time interval on the previous Monday, 21 April 2025, between 12:00 and 12:33:17, a total of 37 alarm events were recorded across 18 nodes, including one node located outside the transmission network, this corresponds to approximately 2.5 % of all transmission network nodes.
- Voltage control remedial actions taken by the operator (coupling and decoupling of shunt reactors, opening and closing of lines, setting HVDC voltage setpoints) were effective in bringing voltage back within the alarming threshold within minutes in most cases (typically less than 5 minutes).
- The manual corrective actions were carried out within humanly realistic time frames. The measured response times between the first recorded alarm exceeding the high-voltage alarm threshold and the execution of the first voltage control remedial action 1 min 33 s in the northern zone and 2 min 11 s in the southern zone take into account the time required for the operator to detect the alarm, analyse the issue, assess the most appropriate action while taking its impact into account, and implement it. In the final analysed period, Red Eléctrica's dispatchers were unable to implement manual voltage control actions due to the very short duration of the event (38 seconds). In such timeframes, only dynamic voltage control – i.e., generator voltage regulation together with STATCOM and HVDC voltage control – can effectively mitigate voltage deviations.

» Alarm log distribution exhibited clear zonal clustering: the southern and eastern zones repeatedly stood out for low-voltage alarm events, while the interior, southern and northern zones showed the greatest concentration of high-voltage alarm episodes.

» A non-trivial portion of alarms were triggered outside the transmission network, for which the TSO is not responsible for voltage control management. They were mainly registered in the south. Furthermore, it should be noted that not all non-transmission network nodes generate voltage alarms at Red Eléctrica's control centres, so the actual number of alarm activations in the non-transmission network may have been higher than Red Eléctrica could observe.

4.5.10 Conclusions from alarms recorded in REN's control centre and their operational handling

» The number of 400 kV voltage alarms exceeded the number of affected nodes because alarms are triggered per busbar and at two identical thresholds, leading to multiple alarms often referring to the same issue. No alarms at 220 kV were triggered during this period.

» Within this interval, two consecutive periods were identified in which 10 and 14 undervoltage alarms were recorded, reaching a minimum voltage value of 377.9 kV. This was followed by a third period with 56 overvoltage alarms, all above 420 kV and affecting a larger number of nodes compared with the previous undervoltage periods. Throughout the entire interval between 12:00 and 12:33:17, no voltage control actions were carried out by the control centre, as the alarms cleared within a few seconds and dispatchers are not expected to perform manual actions in such a short timeframe.

4.5.11 Conclusions from alarms recorded in RTE's control centre and their operational handling

Before 12:33:17, the voltage levels at the 225 and 400 kV RTE south-west substations were within normal limits, and no specific action was taken.

As a result of the tripping, the voltage values exceeding the thresholds were processed and recorded as expected in the tools (voltage heat map and SCADA).

4.5.12. Conclusions from alarms recorded in the analysed distribution control centre

» During this period, a total of 4,150 alarms were recorded due to voltage levels exceeding the distributor's voltage threshold, affecting 308 nodes. The maximum recorded voltage was 1.10 p.u. in the distribution network and 1.11 p.u. at the 220 kV level, for which the DSO is not responsible for voltage control.

» In the first interval, most alarms occurred at 45 kV and 15 kV. In the second interval, while 45 kV and 15 kV remained the most affected levels, there was a significant increase in alarms at 220 kV. In the third interval, the majority of alarms were concentrated at 220 kV and 45 kV.

» Although these intervals coincide with periods during which voltage alarms exceeding high-voltage thresholds were recorded in the transmission network, in the first interval, both the duration and the number of affected nodes were considerably higher than those recorded in the transmission network.

» No analysis has been conducted regarding the voltage control actions applied by the distributor during any of the three intervals.

4.5.13 Conclusions from alarms recorded in the analysed generation control centre

- The generation control centre recorded a total of 165 alarms related to voltage deviations from the nominal value, including 102 above nominal and 63 below nominal. These alarms were triggered in a small but recurring group of nodes, mainly in the southern and north-western areas.
- The voltages below nominal occurred across two intervals, affecting the same six nodes, with minimum voltage values of 0.88 p.u. at the 11 kV level and 0.93 p.u. at the 132 kV level.
- Voltages above the nominal value were recorded across three periods, affecting four to five nodes per interval, with maximum voltage values of 1.10 p.u. at the 45 kV level and 1.09 p.u. at the 220 kV level. Several of these nodes appeared repeatedly across different periods, indicating a notable level of recurrence.
- These intervals correspond to the time frames in which voltage alarms were triggered in the transmission network.

4.5.14 Summary from recorded voltage alarms and their operational handling

The high number of alarm logs recorded in Red Eléctrica's control centres is explained by the alarm system's configuration: multiple alarms were triggered at the same node, and several were recorded at nodes outside the transmission network, beyond the TSO's voltage control responsibility.

Most voltage alarm events were triggered at values within the voltage limits defined by the applicable requirements, and most low-voltage alarms occurred during the oscillatory period. Before 12:33:17, only sporadic activations occurred at voltages above 435 kV (1.1 % of the total events above high voltage alarm thresholds) or below 375 kV (0.6 % all lowvoltage alarm threshold events) but cleared in a few seconds.

In the last two periods analyzed, about 30 % and 35 % of the transmission network nodes triggered at least one alarm event, whereas in the prior periods the proportion was substantially lower, around 10 %. During the same time interval on the previous Monday, the share of transmission network nodes that activated at least one voltage alarm was only 2.5 %.

The dispatcher's corrective actions restored voltage within thresholds within a few minutes. During the assessed period, between 12:25 and 12:28, the response times between the first alarm event above the high voltage threshold and the execution of the first remedial voltage control action are considered operationally reasonable, as they include the time needed for the operator to detect the alarm, analyse the issue and determine and implement the most appropriate action.

In the final analysed period, after 12:32, Red Eléctrica's dispatchers were unable to implement manual voltage control actions due to the very short duration of the event (38 seconds). In such time frames, only dynamic voltage control can effectively mitigate voltage deviations.

With respect to the alarm events recorded in REN's control centre, the number of 400 kV voltage alarms exceeded the number of affected nodes because alarms are triggered per busbar and at two identical thresholds. Throughout the entire timeframe, no manual voltage control actions were taken by the dispatchers, as the undervoltage alarms cleared within seconds and the short duration of the overvoltage final event prevented dispatchers from performing control actions.

The distribution control centre received only alarms associated with voltage levels exceeding the distributor's established limits, coinciding with the time intervals in which high voltage alarm threshold events were recorded in the transmission network. The duration of the events in the distribution network was longer than that observed in the transmission network.

The GCCs also recorded alarm events related to under- and over-nominal voltage values at a recurring group of nodes, corresponding to the same time periods during which voltage alarms were triggered in the transmission network.

Area of attention/improvement: The Expert Panel suggests that all TSOs undertake any developments that they deem necessary in the design of their alarm system and associated operators' responses to further enhance their ability to detect and address any relevant phenomena.

4.6 Root Cause Tree

The description of the events that occurred and led to the blackout on 28 April 2025, as detailed in chapters 2 and 3 of this report, combined with the technical analyses described in the sub-chapters 4.1 to 4.5, allow the Expert Panel to identify the main elements that caused the incident and the relationships between them. The result is the Root Cause Tree that is presented in Figure 4-124. The methodology applied to find the proper relation between the elements of the incident is the so-called Root Cause Analysis (RCA), which is a method of problem-solving used for identifying the root causes of faults or problems. This methodology systematically asks the question "why?" an event happened until all causes are identified, starting in the present case from the question "Why was there a system blackout on 28 April 2025?"

The methodology followed here is loosely based on the Five Whys exercise which is a questioning technique for going beyond symptoms of problems to identify the underlying or root causes of a problem. It follows several logical steps:

- State the problem as accurately and briefly as possible;
- List the possible causes of the problem by asking the question "Why?" or "Why did that happen?";
- For each of the causes, again ask the question "Why?", and list answers again;
- Continue this process at least 5 times or until having reached the source of the problem, the lowest level;
- This level is where something can be done about avoiding the event in the future.

The identified causes can then be analysed further to see whether this is only a circumstance which eventually contributed to the chain of events or if the concerned cause shall be subject to improvement and recommendations. It must be stressed that not all causes will identify an issue; some causes might be unavoidable, like for instance the fact that load varies with voltage, they will not be catalogued as a cause requiring further improvements.

In Figure 4-124, the blue boxes refer to the causes for which the panel considers it useful to make one or more recommendations aimed at improving the resilience of the European electricity system. These recommendations are explained in Chapter 9 of this report. The number of the recommendation related to a specific cause is mentioned next to the box describing the cause. The structure of the tree is divided into two parts and the chain of the events can be explained accordingly. The bottom part of the tree shows the identified issues (several of these are structural, some of them coincidental or event-driven) which contributed to the fast voltage increase that occurred from around 12:32:00 onwards. From these identified elements the tree highlights with blue colour those ones which can be influenced by proper measures and subject to recommendations. The upper part of the tree represents the last sequence of the event which relates to system defence and final steps before the blackout. Some of these elements are also subject of recommendations which are also highlighted with blue colour.

Root cause tree

The blue boxes correspond to factors that contributed to the incident and on which the Expert Panel is making recommendations.

The white boxes correspond to additional elements or explanations that allow to understand how the incident developed.

(x) Link to recommendation number

Abbreviations

aFRR	automatic Frequency Restoration Reserve
BRP	Balancing Responsible Party
CESA	Continental Europe Synchronous Area
DRS	loss of synchronism protection (Débouclage suite à Rupture de Synchronisme)
PSS	Power System Stabiliser
Q-reference	reference reactive power (as defined in § 2.6.6)
RES	Renewable Energy Sources
RfG	Requirements for Generators Network Code
SPS	Special Protection Scheme

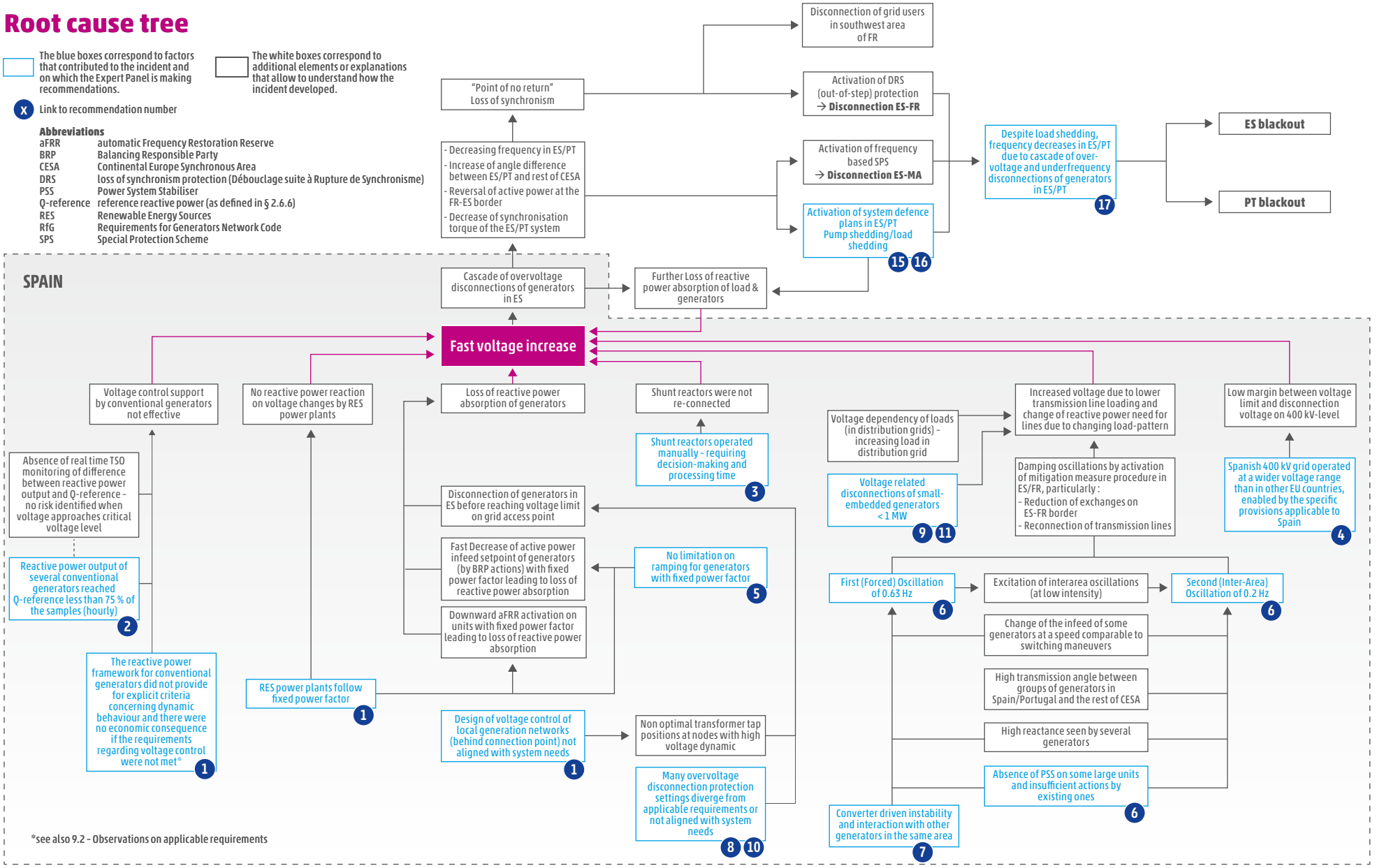


Figure 4-124: Root cause tree linked with recommendations

Although the role of the different elements of the tree is explained as part of a detailed technical analysis in the chapters 4.1 – 4.5, some further explanation on what is the relation of those elements to each other and how they contributed to the blackout is necessary due to the complexity of the incident. The order of the list below as well as the arrangement of the boxes and branches in the root cause tree do not reflect the importance of these key factors.

1. The reactive power output of several conventional generators reached Q-reference in less than 75 % of the samples (hourly).

The Operational Procedure 7.4 (2000) applicable in Spain on the day of the incident determined the minimum reactive power contribution (absorbing or injecting depending on the voltage) by large conventional generators when the voltage is outside of a margin around the voltage reference set by the TSO. Based on the result of performed evaluations of SCADA-data of RE several conventional units provided less reactive power support than the calculated values based on the procedure without prejudice to any potential derogations. More reactive power absorption would have led to a decrease in voltage.

This factor contributed to a non-effective voltage control by conventional generators, especially in some regions.

2. The reactive power framework for conventional generators did not provide for explicit criteria concerning dynamic behaviour and there were no economic consequence if the requirements regarding voltage control were not met.

The framework in place the day of the incident did not include explicit specifications for the dynamic voltage control. Moreover, the Procedure prescribed penalties in the event that generators would not comply with the procedure; however, the penalties were not applicable in practice, since the economic scheme required by the Operating Procedure had not been developed by the designated entity. The existence of a financial scheme could have incentivised the generators to implement improvements in voltage control performance.

This factor contributed to a non-effective voltage control by conventional generators, especially in some regions.

3. RES power plants follow fixed power factor.

A fixed power factor control mode means that the reactive power contribution is inherently proportional to active power injection. RES plants in Spain followed this mode.

First, this meant that for every active power change (e.g. schedule change) the generators injected into the system a proportional reactive power change (and hence a voltage change);

Secondly, fixed-power-factor-mode resources provided no reactive power support that would compensate voltage fluctuations, since their reactive power output did not respond to conditions in the grid.

4. Design of voltage control of local generation networks (behind connection point) not aligned with system needs.

The design of some non-regulated local generation networks in Spain (incl. voltage control/disconnection settings) was not aligned with system needs, meaning, among others, a more challenging voltage control and higher risk of disconnection. Voltage control in such networks was insufficient in some cases, due to a combination of fixed power factor control mode of generators, fast generation ramps, physically slower changes of tap-changing transformers.

5. No limitation on ramping for generators with fixed power factor.

Fast decrease of active power infeed of generators with fixed power factor lead naturally to a reduction of reactive power absorption. Thus, fast ramping down of such units resulted in fast increase of voltage as well.

6. Shunt reactors are operated manually – requiring decision-making and processing time.

Shunt reactors in Spain were operated manually. The time needed to manually connect shunt reactors is longer than what is needed when voltage changes very fast. Due to the speed at which voltage rose after 12:32:00, disconnected shunt reactors could not be connected in time due to the needed processing time for manual switching.

7. Many overvoltage disconnection protection settings diverged from applicable requirements or were not aligned with system needs.

Some generation units (or connecting transformers or lines) disconnected at their voltage level before the voltage disconnection conditions at the transmission connection point and time delay conditions were met. This led to a loss of absorption of reactive power, resulting in a steep increase in voltage.

8. Converter-driven instability and interaction with other generators in the same area

A converter-driven instability combined with interaction with other generators in the same area led to forced oscillations which endangered the system as frequency and voltage can severely fluctuate then within seconds.

9. Absence of PSS on some large units and insufficient actions by existing ones.

Power System Stabilisers (PSS) installed on generators side are able to damp oscillations. The more PSS installed the better oscillations can be mitigated. Not all large generation units were equipped with PSSs and the action of the existing PSSs was insufficient to damp the oscillations.

10. First (forced) Oscillation of 0.63 Hz and second (inter-area) Oscillation of 0.2 Hz

Oscillations are a major threat to system security. Thus, initiating mitigation measures in order to damp oscillations has highest priority from operational perspective. On the day of the incident, the dispatchers implemented mitigation measures during both oscillatory episodes, which successfully damped the oscillations but contributed to increase the voltage.

11. Voltage-related disconnections of small embedded generators < 1 MW

Based on data of two inverter companies, and subject to the assumptions and simplifications explained in chapter 4.3, it can be considered that there was a significant number of overvoltage trips of small PV units connected to the low-voltage DSO grids. These units tripped during periods of high voltage ramps in the transmission network but also during and around periods of high voltages and voltage fluctuations (slow/fast changes, oscillations).

This led to changes in load flow pattern of the transmission grid resulting in voltage changes.

12. Spanish 400 kV grid operated at a wider voltage range than in other EU countries, enabled by the specific provisions applicable to Spain.

Operating the system with a voltage range 380 – 435 kV, while generators were to remain connected up to 435 kV or 440 kV (depending on commissioning date) meant that the voltage margin was low or non-existent.

13. Activation of system defence plan in ES/PT–pump shedding/load shedding.

Load and pump shedding as part of the system defence plan aim to reduce the consumption of active power in order to stop and restore frequency drops. When disconnecting grid users, that absorb reactive power at the same time, it naturally increases the voltage. Thus, the activated load shedding helped the system to stabilise frequency, but on the other hand some automatic disconnections, e.g. pump power plants absorbing reactive power at time of disconnection, have intensified the overvoltage problem.

14. Despite load shedding, frequency decreases in ES/PT due to cascade of overvoltage and underfrequency disconnections of generators in ES/PT.

As the effect of further loss of generators due to overvoltage was higher than loss of load due to activated load shedding, frequency continued to decrease until system totally collapsed and blackout occurred.

5 RESTORATION PROCESS

The objective of this chapter is to explain the restoration process in the systems affected by the incident, namely the Portuguese and Spanish systems. In the case of the French system, the incident only impacted a local area, as described in Section 5.4.

This chapter is divided into ten sections:

5.1 Preconditions and strategies for the restoration process

5.2 The restoration sequences in the Spanish system

5.3 The restoration sequences in the Portuguese system

5.4 French local area restoration by RTE

5.5 Generation and load recovery

5.6 Steps after system restoration

5.7 Market restoration

5.8 Behaviour of distributed energy resources and load during restoration

5.9 Voice communication and remote control problems during restoration

5.10 Conclusion

5.1 Preconditions and strategies for the restoration process

Restoring electrical grids in continental Europe after the occurrence of a significant disruption is a complex and coordinated process. TSOs work collaboratively in accordance with Regulation (EU) 2017/2196, which sets out the Network Code on Electricity Emergency and Restoration (hereinafter NC ER), to re-establish electricity supplies through a mixture of "top-down" and "bottom-up" approaches. The top-down approach utilises voltage from neighbouring, healthy parts of the network, while the bottom-up approach relies on the black-start capability of power plants. Black-start capability enables certain power plants to start up independently without any external power supply, which is vital for gradual restoration when no external voltage is available.

5.1.1 Red Eléctrica preconditions and strategies for the restoration process

The restoration process in Spain used a combination of top-down and bottom-up restoration strategies. For the top-down strategy, support was activated through the neighbouring countries, creating three zones with stable frequency and voltage that gradually increased and subsequently synchronised with the remaining areas. Specifically, the following supports were activated:

- Support from France via the interconnections through the País Vasco
- Support from France via one of the interconnections through Cataluña
- Support from Morocco via one of the interconnections through Andalucía

In the bottom-up strategy, implemented in parallel, re-energisation was initiated using hydropower plants (HPP) with black-start capability.

The restoration plan for the Spanish peninsular (i.e. mainland) electrical system divides the network into seven specific areas, as shown in Figure 5-1. Each area contains at least two HPPs with black-start capability, with one serving as backup to the other, in line with the applicable requirements.

Restoration process timeline

- At 12:33, restoration service plans were activated in all seven areas.
- At 12:34, REN was contacted to confirm that the entire Portuguese electrical grid was without voltage, as indicated by the Red Eléctrica control system. Additionally, it was confirmed that REN had initiated the process of restoring the Portuguese electrical system.

» At 12:36, Red Eléctrica communicated the situation in its control area to all the other European TSOs, changing the system state from normal to blackout via the ENTSO-E Awareness System (EAS). Almost immediately, the state was changed again to the restoration state.

» At 12:36, RTE was contacted to confirm that their transmission network had not experienced any incidents, as indicated by the Red Eléctrica control system. Additionally, RTE was requested to send support to Red Eléctrica through the 400 kV Argia—Hernani and Baixas—Vic lines.

» At 12:44, market activities were suspended, in line with the applicable requirements. In coordination with the NEMO (OMIE-Operador del Mercado Ibérico de Energía), intraday markets and adjustment services markets were suspended.

» At 12:47, L'Office National de l'Électricité et de l'Eau potable (ONEE) in Morocco was contacted to confirm that their transmission network had not experienced any incidents, as indicated by the Red Eléctrica control system. It was also agreed that ONEE would provide support of up to 100 MW through one of the submarine cables that connects Morocco with Spain.

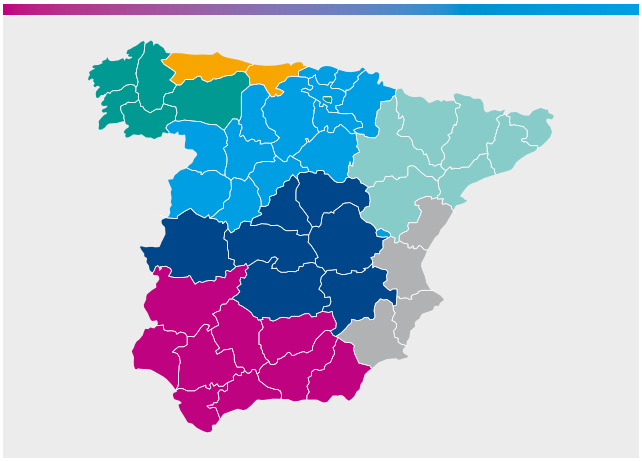


Figure 5-1: Specific restoration plans for the Spanish system – Red Eléctrica

5.1.2 REN: preconditions and strategies for the restoration process

In Portugal, the "Plano Nacional de Reposição de Serviço" ("National Service Restoration Plan") was activated. This procedure sets out the restoration strategies to be adopted following a generalised blackout of the Portuguese system. The strategies used differ depending on whether the generalised shutdown is national (top-down restoration, with support from the Spanish system) or Iberian (bottom-up restoration, without support from the Spanish system).

In line with the applicable requirements, the Portuguese system has two power plants with black-start capability, one serving as backup to the other. The National Service Restoration Plan provides for the restoration of the Portuguese system from one of these power plants, or from the start-up of both, depending on operational conditions.

The restoration was started by the two power plants with black-start system capability in Portugal (the bottom-up strategy): the Combined Cycle Gas Turbine Powerplant (CCGT) 1-Norte, located near the Oporto area, and HPP 1-Centro, near Lisbon.

To restore the system as quickly as possible, the dispatch centre operators (located in Sacavém) and the network operation centre operators (located in Vermoim) were divided into two teams: the first team was in charge of establishing an electrical island with HPP 1-Centro, and the second team was in charge of establishing an electrical island with CCGT 1-Norte.

In a phone call at 12:34, it was verified with Red Eléctrica that the blackout had affected the entire Iberian Peninsula.

At 12:38, the Portuguese system state was changed from normal to blackout in the EAS. At 17:04, it was changed to the restoration state.

As a result of this incident, REN requested that the NEMO (OMIE):

- Cancel "in advance" the first intraday auction for 29 April
- Halt negotiations on the continuous intraday market
- Cancel "in advance" the second intraday auction for 29 April
- Cancel "in advance" the third intraday auction for 29 April

Additionally, REN suspended the transactions on:

- TERRE (Trans European Replacement Reserves Exchange): The hour between 13:00 and 14:00 on 28 April was the last period during which transactions were established on TERRE.
- MARI (Manually Activated Reserves Initiative): The quarter-hour between 12:45 and 13:00 on 28 April was the last period during which transactions were established on MARI.
- The National aFRR capacity market and national redispatch markets, for the remaining periods on 28 April and all periods on 29 April.

5.1.3 RTE: preconditions and strategies for the restoration process

There was only one 63 kV substation (SS) without voltage on the RTE grid (Arriosse), as shown in Figure 5-2. This had a limited impact on customers, as by 12:49 they were already being supplied via the DSO network. Consequently, the restoration plan was not activated, and this SS was re-energised at 13:57.

The nuclear power plant that tripped also had a limited impact. The restoration plan was therefore not activated, and the plant was re-energised on the next day, 29 April, at 16:15.

Figure 5-2: Location of Arriosse SS (63 kV) on the RTE grid

5.2 The restoration sequences in the Spanish system

This section describes the restoration sequences in Spain in detail, from the initial actions through to the total restoration of the Spanish system.

Due to confidentiality constraints, the names of the generation units are anonymised for external publication.

5.2.1 Overview of the Spanish system restoration

Restoration process timeline

- At 12:43, the L-400 kV Argia–Hernani interconnection was energised, initiating the top-down re-energisation strategy, as established in the procedure for mutual support between the Spanish and French systems after incidents. This marked the beginning of the restoration of the Northern zone, using voltage from France.
- At 12:55, HPP Extremadura 1 was started up using its black-start capability, to supply auxiliary services to a nuclear power plant and stabilise the load at the 220 kV Talavera SS. At 13:56, the group tripped, losing the created electrical island. The events of the bottom-up restoration in the Extremadura 1 island are described in detail in section 5.2.2.
- At 13:00, HPP Duero[34] 1 was started up using its black-start capability, initiating the Duero 1 electrical island, which was designed to supply critical loads, other HPPs, and synchronisation with the Northern zone. At 13:12, the group tripped. It was started up again at 14:20. The events of the bottom-up restoration in the Duero 1 island are described in detail in section 5.2.3.
- At 13:04, voltage was received from Morocco, initiating the restoration of the Southern zone. It was agreed that ONEE would provide support up to 100 MW through one of the submarine cables connecting the two systems. Following a disconnection, caused by an underfrequency relay trip at 14:27, voltage was restored from Morocco at 14:34. The network energisation from the 400 kV Tarifa SS was restarted to enable voltage supply to the thermal groups in the area. The events of the top-down restoration scenario in the Southern zone are described in detail in section 5.2.4. After analysis, it became clear that the cause of the trip was an issue in the Moroccan system.
- At 13:06, HPP East Asturias was started up using its black-start capability. At 13:23, the group tripped, preventing the initiation of the East Asturias electrical island from the start-up of this HPP. The events of the bottom-up restoration in the East Asturias island are described in detail in section 5.2.5.
- At 13:26, HPP Galicia was started up using its black-start capability, initiating the Galicia electrical island, which was designed to supply load in Galicia, critical loads, and to energise the auxiliary services of several thermal power plants (TPPs). The events of the bottom-up restoration in the Galicia island are described in detail in section 5.2.6.
- At 13:35, the L-400 kV Baixas–Vic interconnection was switched on in Cataluña, starting the top-down re-energisation strategy in the Cataluña area, as established in the procedure for mutual support between the Spanish and French systems after incidents. This marked the beginning of the restoration of the Eastern zone, using voltage from France. The main objectives were to supply auxiliary services for the TPPs, supply critical loads, and synchronise with the other areas.
- At 13:52, HPP Levante was started up using its black-start capability, initiating the Levante electrical island, which was designed to energise the auxiliary services of the TPPs in the northern Levante area. The events of the bottom-up restoration in the Levante island are described in detail in section 5.2.7.
- At 14:11, HPP Duero 2 was started up using its black-start capability, initiating the Duero 2 electrical island, which was designed to supply critical loads, other HPPs, and synchronise with the other islands. The events of the bottom-up restoration in the Duero 2 island are described in detail in section 5.2.8.
- At 14:11, HPP Cantabria was started up using its black-start capability, initiating the Cantabria electrical island, to supply load in the Cantabria area and auxiliary services for several TPPs, and synchronise with the islands created in the Asturias area. The events of the bottom-up restoration in the Cantabria island are described in detail in section 5.2.9.

34 HPP Duero 1 and Duero 2 are in the same area, with one serving as backup of the other.

» At 14:22, HPP Extremadura 2, with black-start capability, was started up using a soft-start process[35], to supply the auxiliary services of a nuclear power plant and to create the Extremadura 2 electrical island, thereby enabling load stabilisation and the connection of the remaining HPPs and TPPs. The events of the bottom-up restoration in the Extremadura 2 island are described in detail in section 5.2.10.

» At 14:28, HPP West Asturias was started up using its black-start capability, initiating the West Asturias electrical island in the 132 kV distribution network. This was designed to supply load in the Asturias area, connect other HPPs, and synchronise with the East Asturias and Cantabria electrical islands. The events of the bottom-up restoration in the West Asturias island are described in detail in section 5.2.11.

» At 15:01, HPP Aragón was started up using its black-start capability, initiating the Aragón electrical island, which was designed to connect other nearby HPPs and supply the auxiliary services of several TTPs. The events of the bottom-up restoration in the Aragón island are described in detail in section 5.2.12.

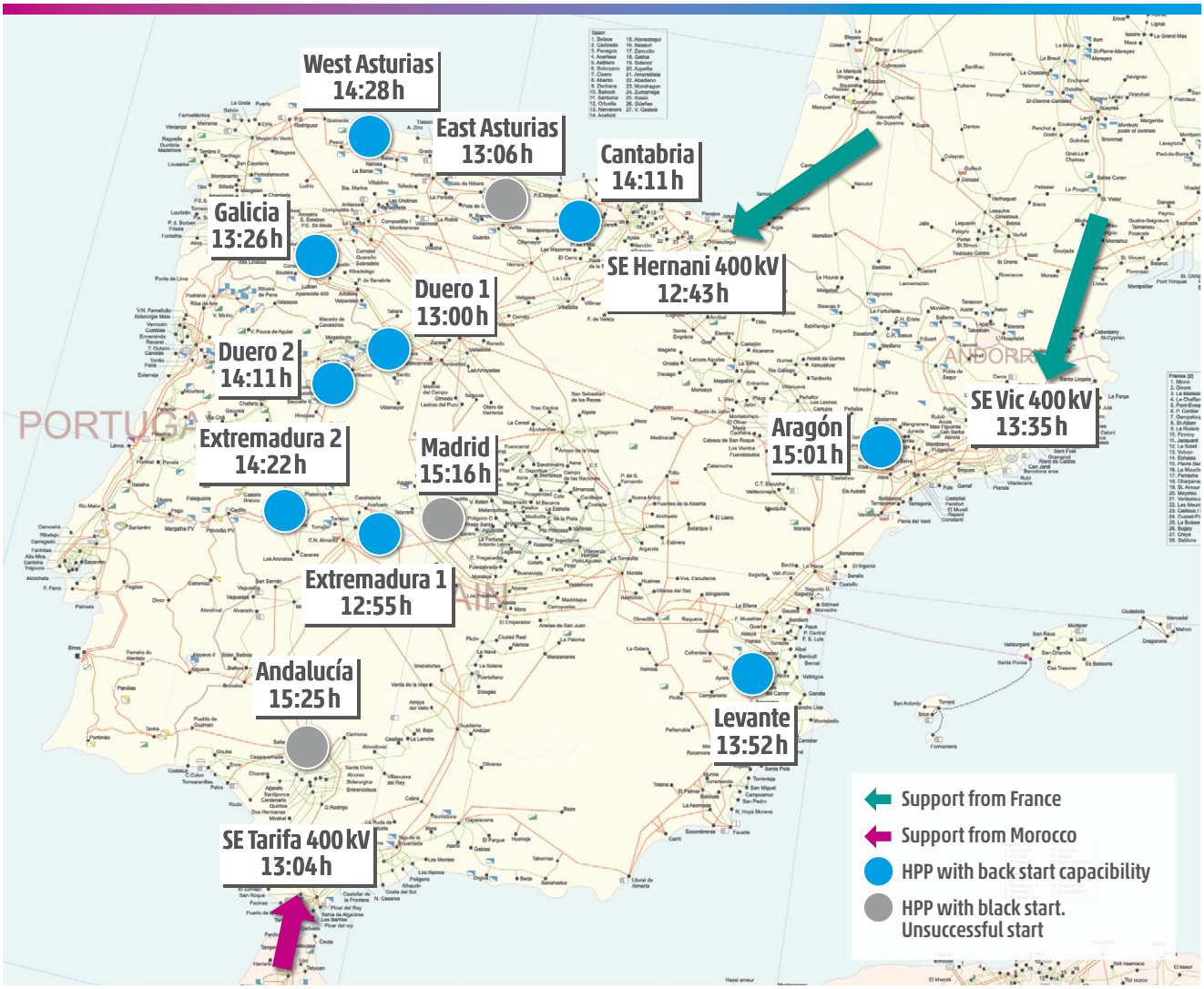


Figure 5-3: Starting times of HPPs with black-start capability and the support from interconnections with France and Morocco - Red Eléctrica

35 The process of energising the grid with gradually increasing voltage.

» At 15:07, all nuclear power plants confirmed having external supply for their auxiliary services.

» At 15:25, an attempt was made to start the available HPP with black-start capability in Andalucía, without success. Taking into account the proximity and progress of the Southern zone, the re-energisation of the Andalucía area from the zones with stable voltage and frequency was prioritised in order to speed up the restoration of the zone. The events of the bottom-up restoration in the Andalucia island are described in detail in section 5.2.13.

» At 15:16, HPP Madrid was started in the 132 kV distribution network, using its black-start capability. Despite several attempts, it was not possible to create the island, which was intended to supply the auxiliary services of a TPP and some stabilisation loads in the Toledo area. The events of the bottom-up restoration in the Toledo island are described in detail in section 5.2.14.

To speed up the restoration process, re-energisation from those zones with stable voltage and frequency to the other zones was prioritised as much as possible.

At 15:14, the first TPP was connected in the Southern zone.

Figure 5-4 shows the evolution of the restoration in the transmission network and black-start islands through to 15:30.

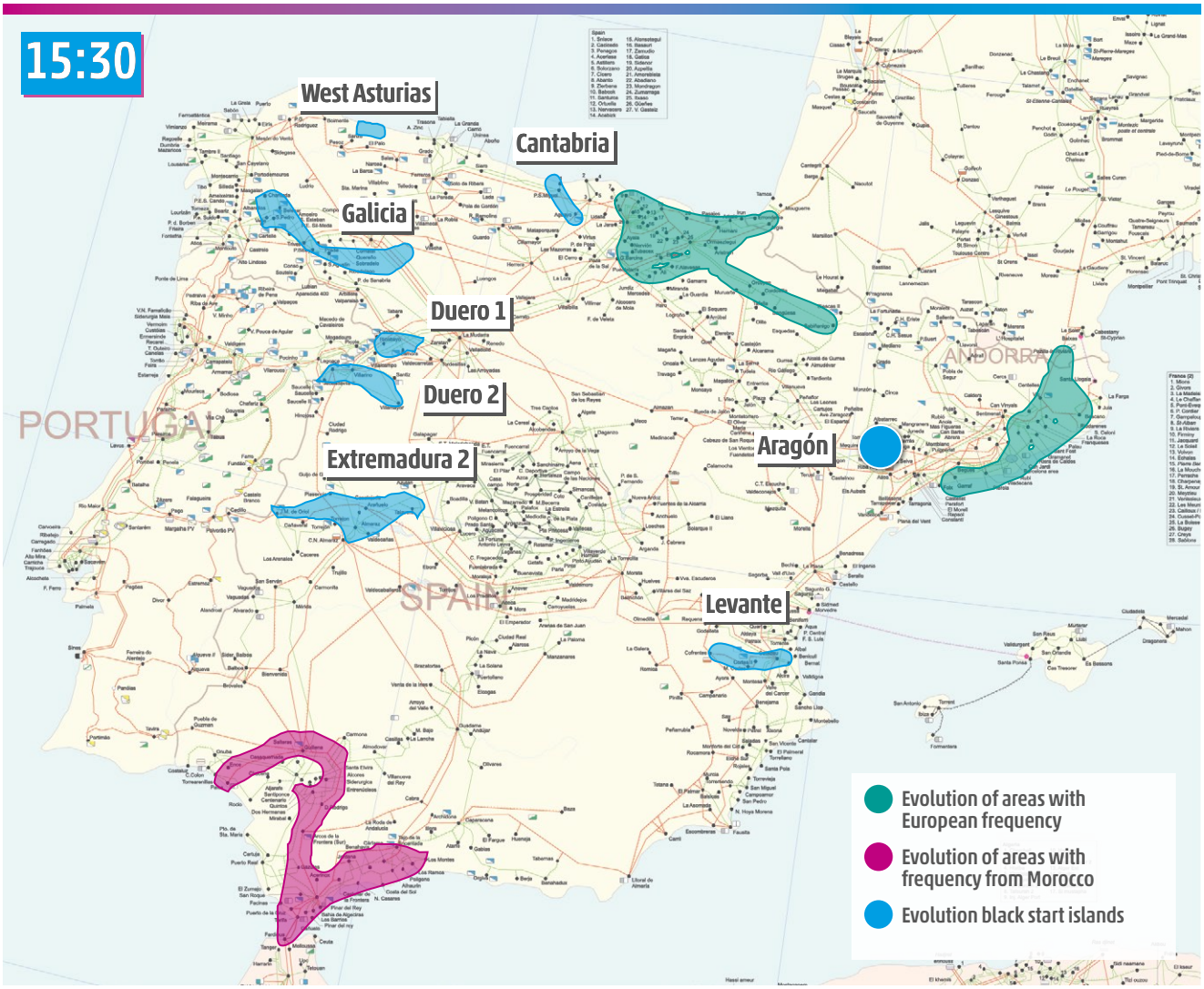


Figure 5-4: Evolution of the restoration in the transmission network and black-start islands through to 15:30 (1,952 MW of load supplied) – Red Eléctrica

Restoration process timeline

- At 15:59, one link in the HVDC Baixas—Sta. Llogaia interconnection was switched on.
- At 16:03, HPP Levante tripped, losing the Levante electrical island.
- At 16:06, the Cantabria HPPs tripped, losing the Cantabria electrical island. Taking into account the proximity and progress of the Northern zone, the re-energisation of the Cantabria area from the zones with stable voltage and frequency was prioritised in order to speed up the restoration of this zone.
- At 16:20, the first TPP was connected in the Northern zone.
- At 16:21, the L-400 kV Aguayo—Abanto was energised from the Northern zone. Subsequently, the Cantabria HPPs were re-connected. From this point, following synchronisation with the frequency in continental Europe, the restoration of the Cantabria area and the supply of auxiliary services to TPPs in the area were resumed.
- At 16:34, the L-400 kV La Plana—Masdenvergue was switched on to initiate the restoration process in the Levante zone, with stable voltage and frequency from the Eastern zone. Subsequently, the auxiliary services of the TPPs in the northern Levante area were supplied.
- At 16:43, a new attempt was made to create the electrical island from HPP Levante, without success. At 17:23, the 400 kV SS La Muela was energised from the Eastern zone, and the HPPs were connected to it.
- At 16:54, the Northern zone was synchronised with the Aragón electrical island through the L-220 kV Escatrón—Villanueva. The situation is represented in Figure 5-5.

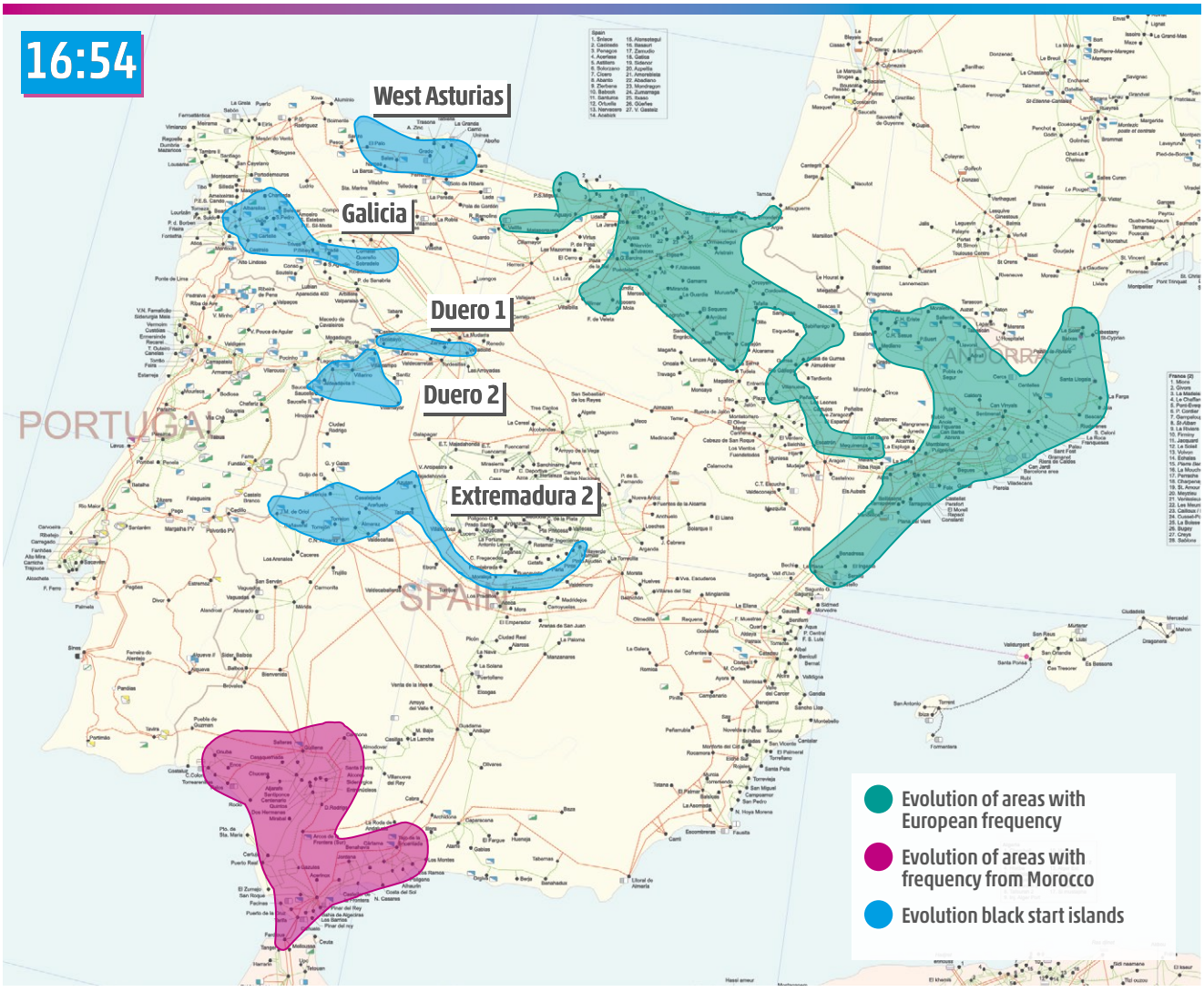


Figure 5-5: Evolution of the restoration in the transmission network and black-start islands through to 16:54 (3,303 MW of load supplied) - Red Eléctrica

» At 17:01, after several failed attempts to synchronise the Duero 1 electrical island and the Duero 2 electrical island, it was decided to disconnect HPP Duero 1. Synchronisation was not possible due to the frequency difference between the islands and the inability of the groups to regulate it.

» At 17:49, the Duero 2 electrical island was synchronised with the Northern zone through the L-220 kV Arroyadas–Tordesillas, connecting the Duero area to continental Europe's frequency. The situation is represented in Figure 5-6.

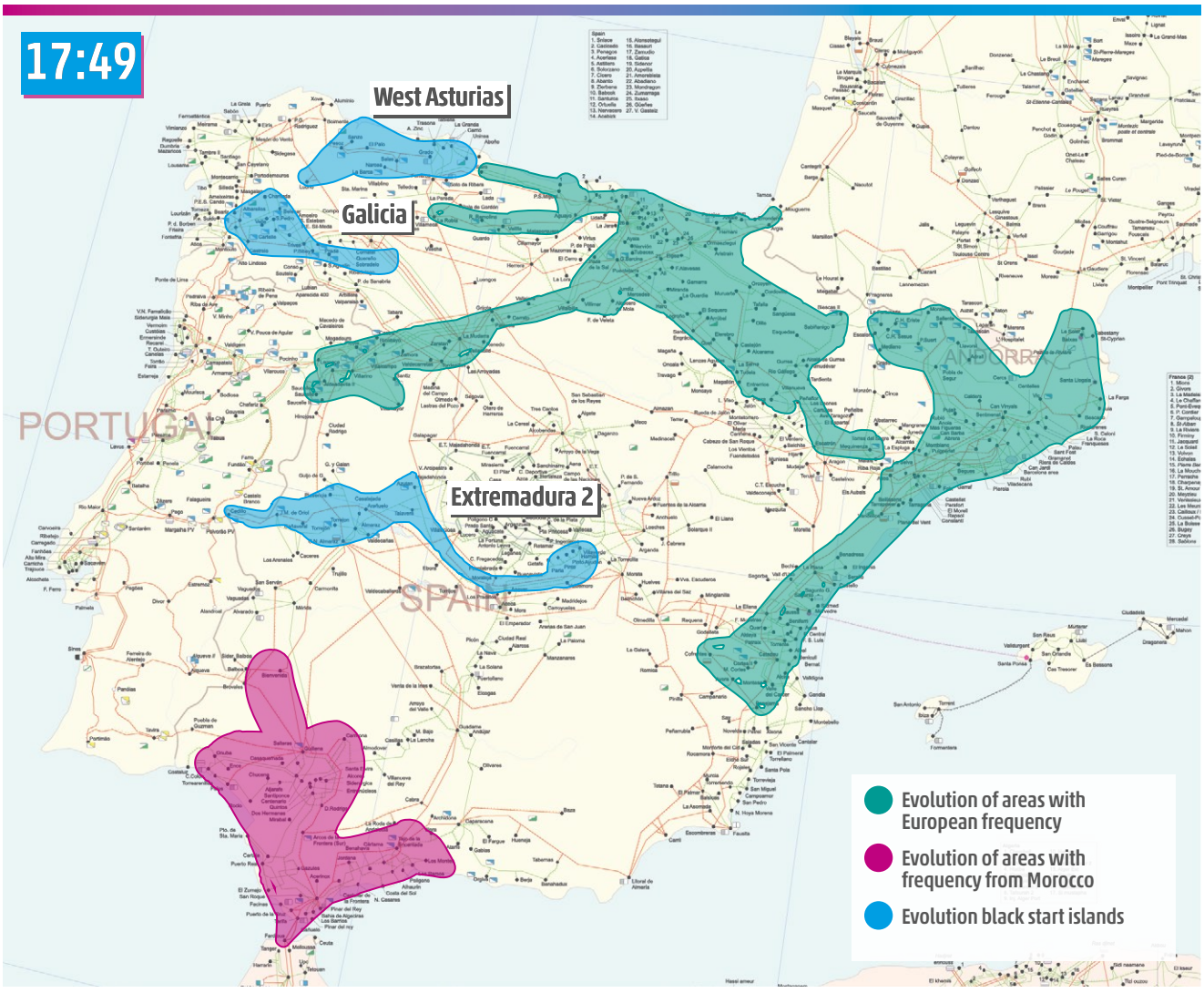


Figure 5-6: Evolution of the restoration in the transmission network and black-start islands through to 17:49 (5,565 MW of load supplied) – Red Eléctrica

» At 18:16, the L-400 kV Bienvenida – Almaraz was switched on, synchronising the Southern zone with the Extremadura 2 electrical island. By this time, several HPPs were connected within the Extremadura 2 electrical island and had extended their supply to the Madrid area. The situation is represented in Figure 5-7.

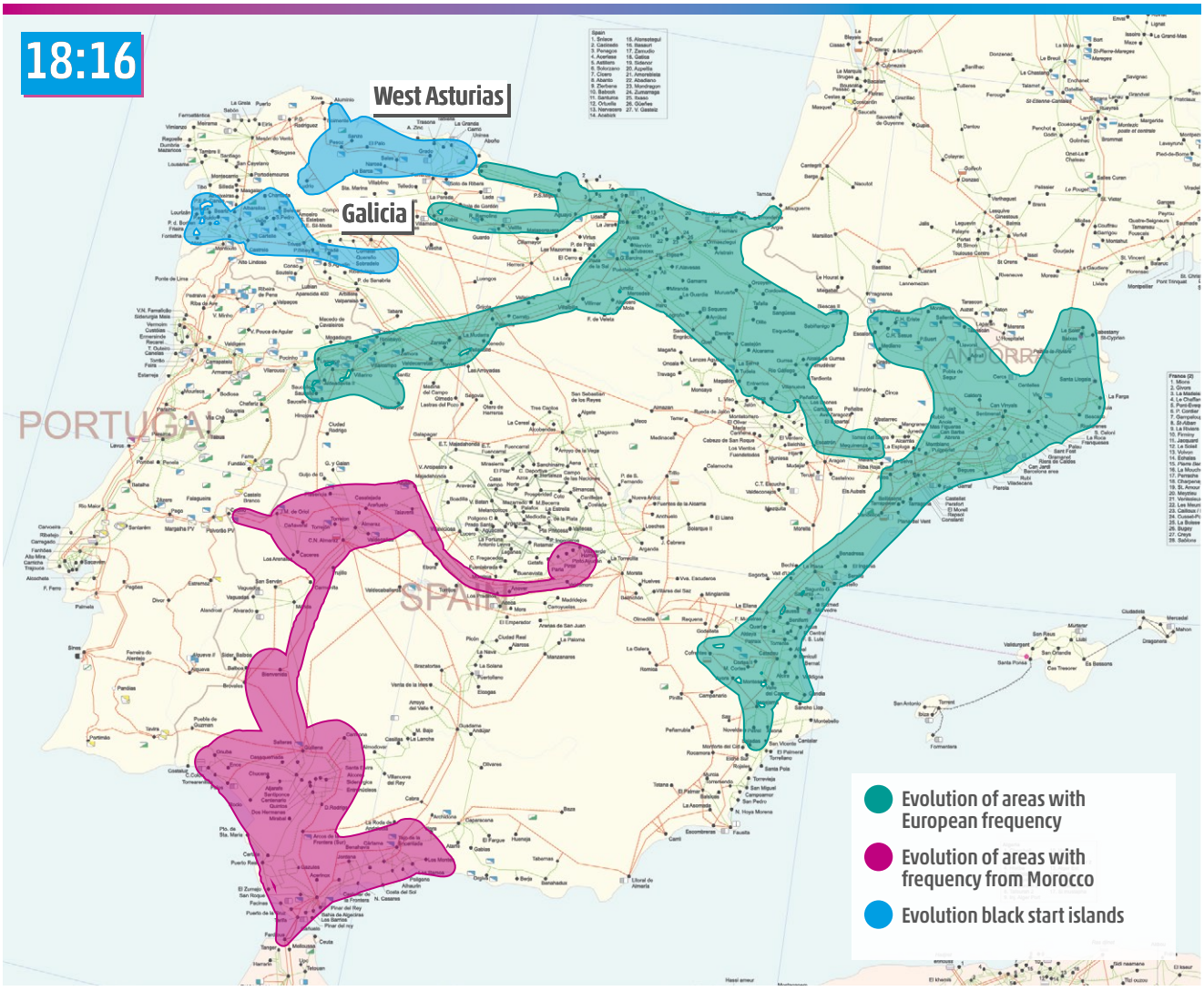


Figure 5-7: Evolution of the restoration in the transmission network and black-start islands through to 18:16 (6,809 MW of load supplied) – Red Eléctrica

» At 18:36, the L-220 kV Aldeadávila – Pocinho 1 interconnection was switched on, sending voltage from the Northern zone to the Portuguese electrical system for the first time in the restoration process. From this point onwards, REN received voltage with continental European frequency.

» At 18:43, the West Asturias electrical island, initiated in the Asturias distribution network, was synchronised with the Northern zone through the ATP1 220/132 kV Siero. The situation is represented in Figure 5-8.

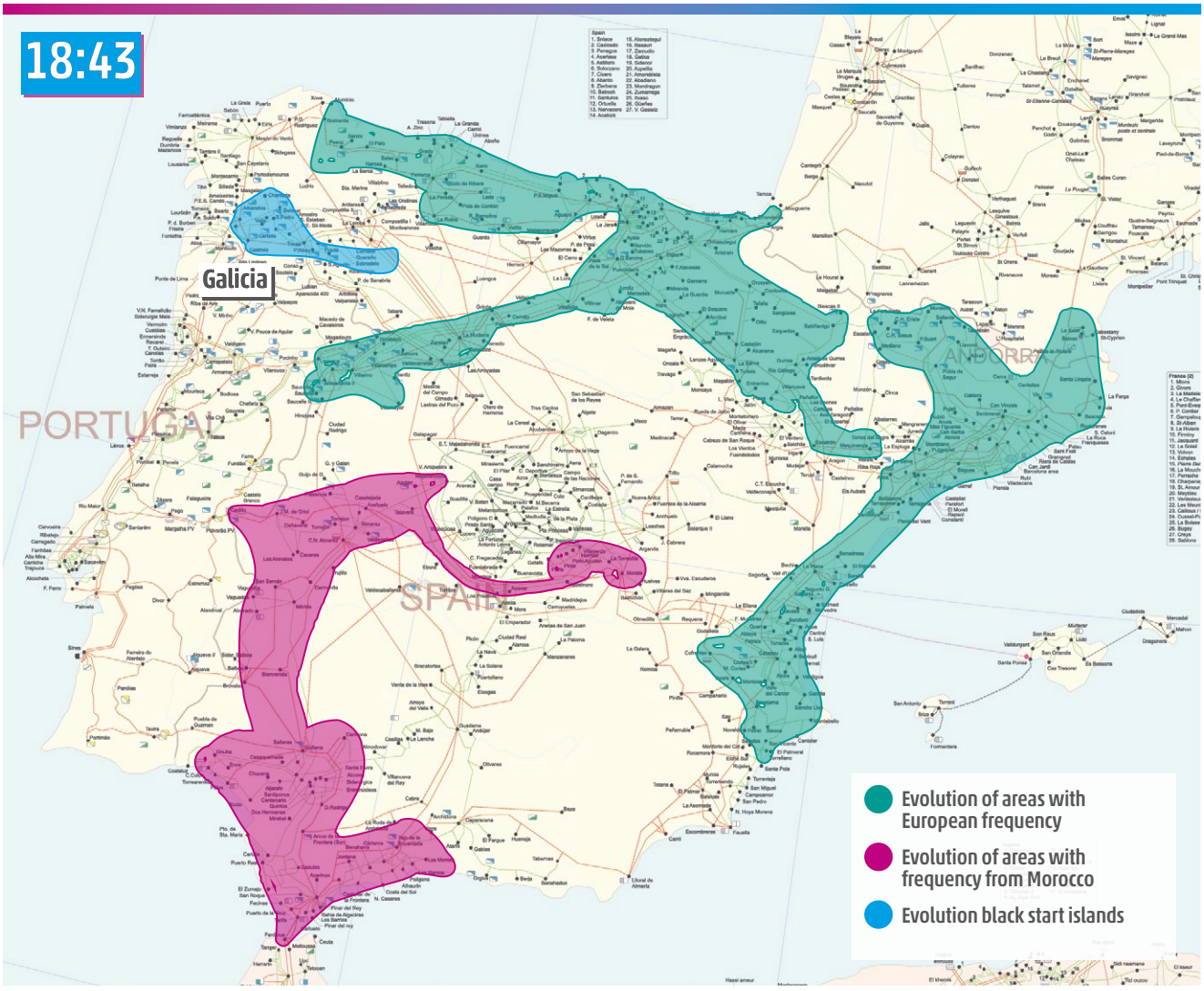


Figure 5-8: Evolution of the restoration in the transmission network and black-start islands through to 18:43 (7,470 MW of load supplied) - Red Eléctrica

» At 19:15, the Galicia electrical island tripped. Due to the proximity of voltage from the Northern zone, it was decided not to restart the island.

» At 19:18, the 220 kV Tordesillas-Valparaíso–Arbillera link was switched on to initiate the re-energisation of the Galicia area from the Northern zone, using stable voltage and frequency to speed up the restoration of the zone. The situation is represented in Figure 5-9.

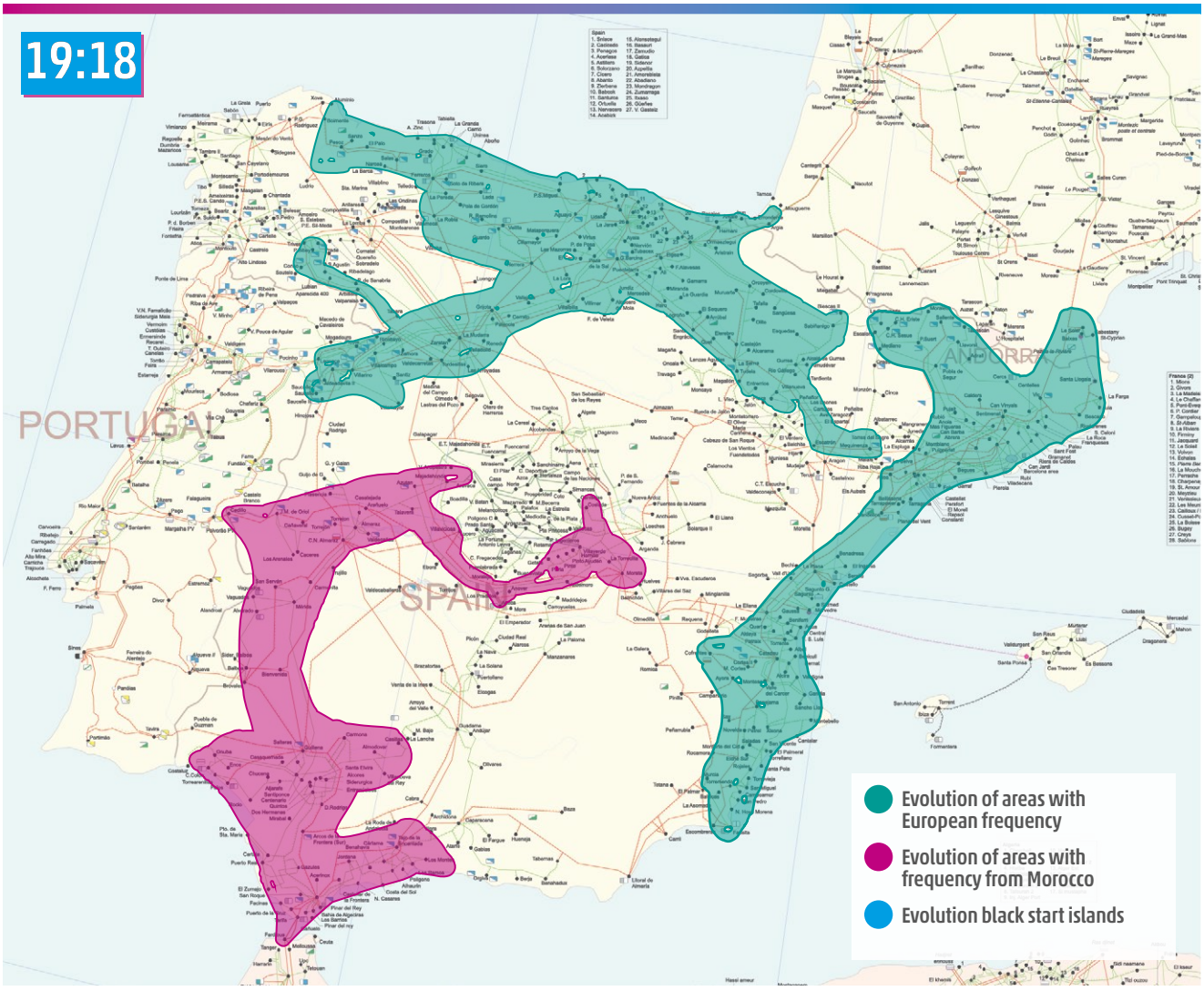


Figure 5-9: Evolution of the restoration in the transmission network and black-start islands through to 19:18 (8,612 MW of load supplied) - Red Eléctrica

» At 19:32, the L-400 kV Almaraz–C. Rodrigo was switched on, synchronising the Northern zone, which at the time primarily comprised País Vasco, Navarra, La Rioja, Castilla León, Cantabria, Asturias, and Galicia, with the Southern zone, composed of Andalucia, Extremadura, Castilla la Mancha, and Madrid. The synchronisation was accomplished without problems. The situation is represented in Figure 5-10.

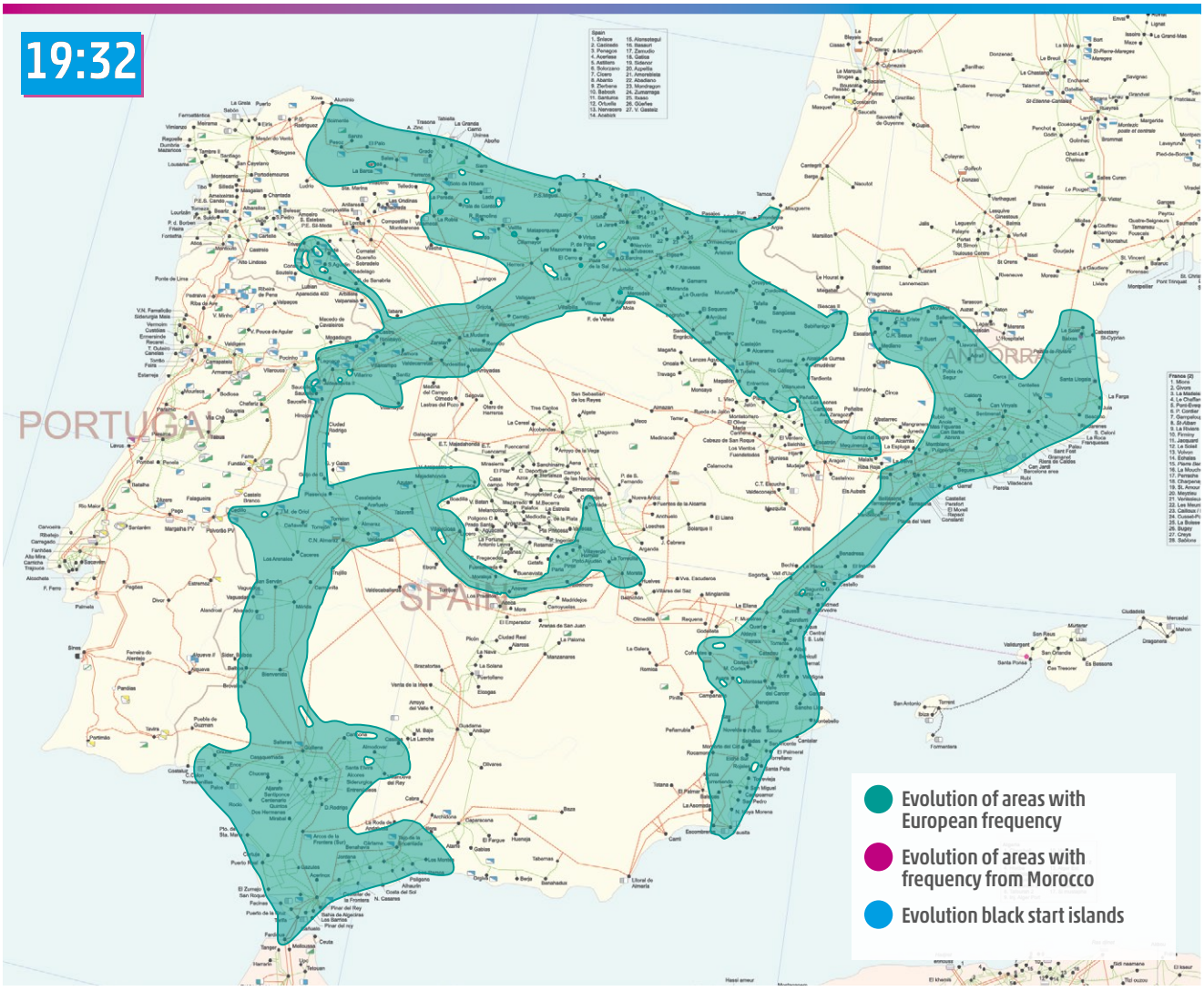


Figure 5-10: Evolution of the restoration in the transmission network through to 19:32 (8,827 MW of load supplied) – Red Eléctrica

» At 19:53, the L-400 kV Peñaflor—Aragón was switched on, connecting the Northern and Eastern zones and creating a single restoration area in the Spanish peninsular electrical system. The situation is represented in Figure 5-11.

» At 19:57, the L-220 kV Saucelle—Pocinho was switched on, reinforcing the interconnection between the Northern zone of Portugal with Spain. At 19:58, the L-220 kV Aldeadávila—Pocinho 2 was also switched on.

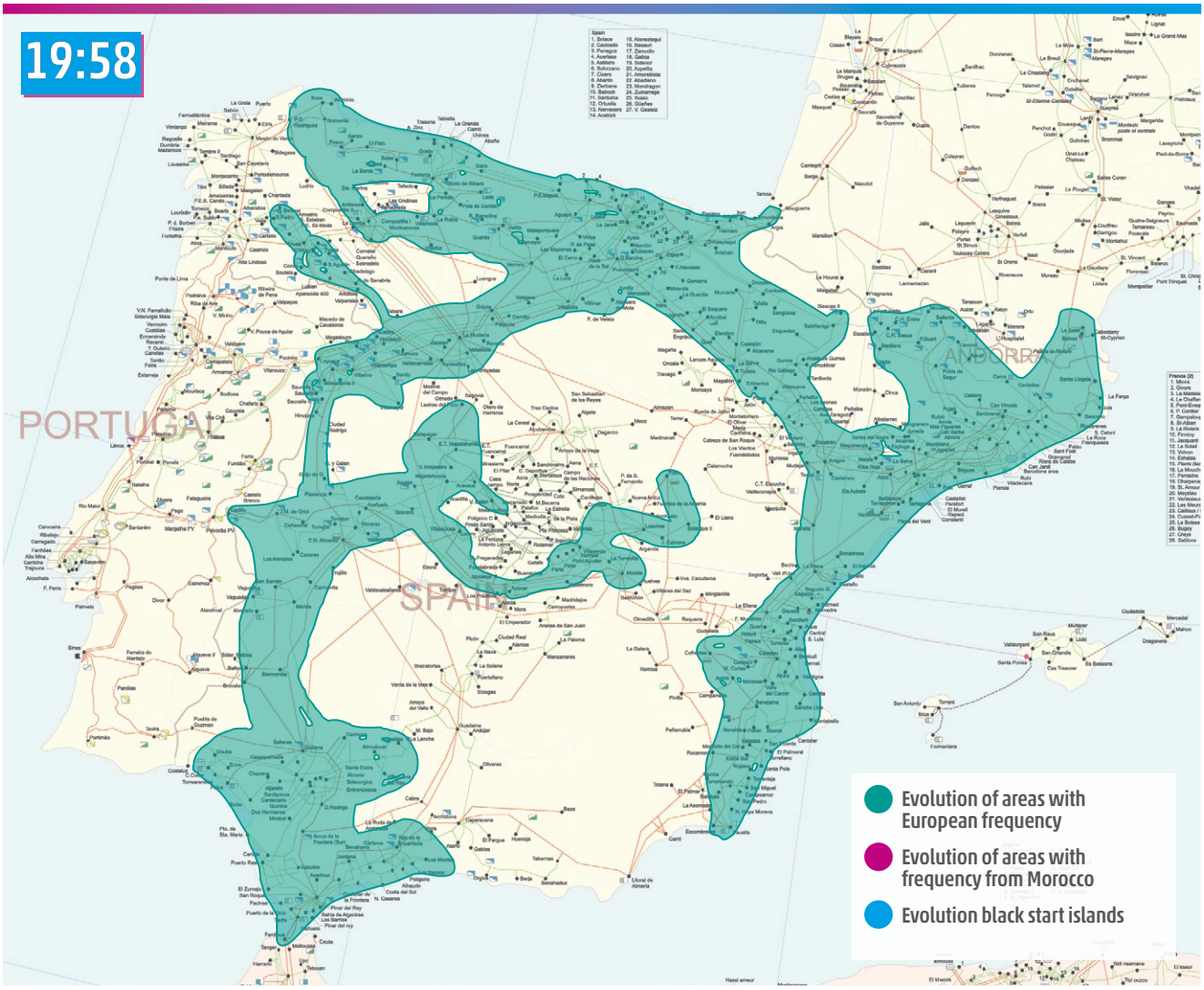


Figure 5-11: Evolution of the restoration in the transmission network through to 19:58 (9,566 MW of load supplied)

With all the areas of the Spanish peninsular system connected to the European synchronous system, the restoration process for the transmission network accelerated substantially. The autonomy of the transmission network equipment ensured proper functioning of the voice communication system, telemetry, and remote control until the completion of the restoration process.

Cooperation between Red Eléctrica, REN, and RTE, as well as the distribution and generation control centres, was continuous throughout the entire restoration process.

It should be noted that the restoration of the load in some areas of the Spanish system progressed more slowly due to problems with the voice communication system at some of the distribution and generation control centres. This made it difficult to contact these centres from the Red Eléctrica control centres. Additionally, some of these distribution companies experienced issues in receiving telemetry and remote-control signals at their facilities.

From this moment onwards, the rest of the available TPPs and HPPs were progressively connected to the system.

Below are some of the most notable milestones:

- At 21:20, 13 TPPs were connected to the Spanish peninsular electrical system.
- At 21:58, the second link of the HVDC Baixas–Sta. Llogaia interconnection began to exchange power in AC emulation mode.
- At 21:34, the L-400 kV Tavira–Puebla de Guzmán was switched on to reinforce the interconnection between Spain and Portugal in the Southern zone.

Figure 5-12 shows the evolution of the restoration in the transmission network through to 22:33.

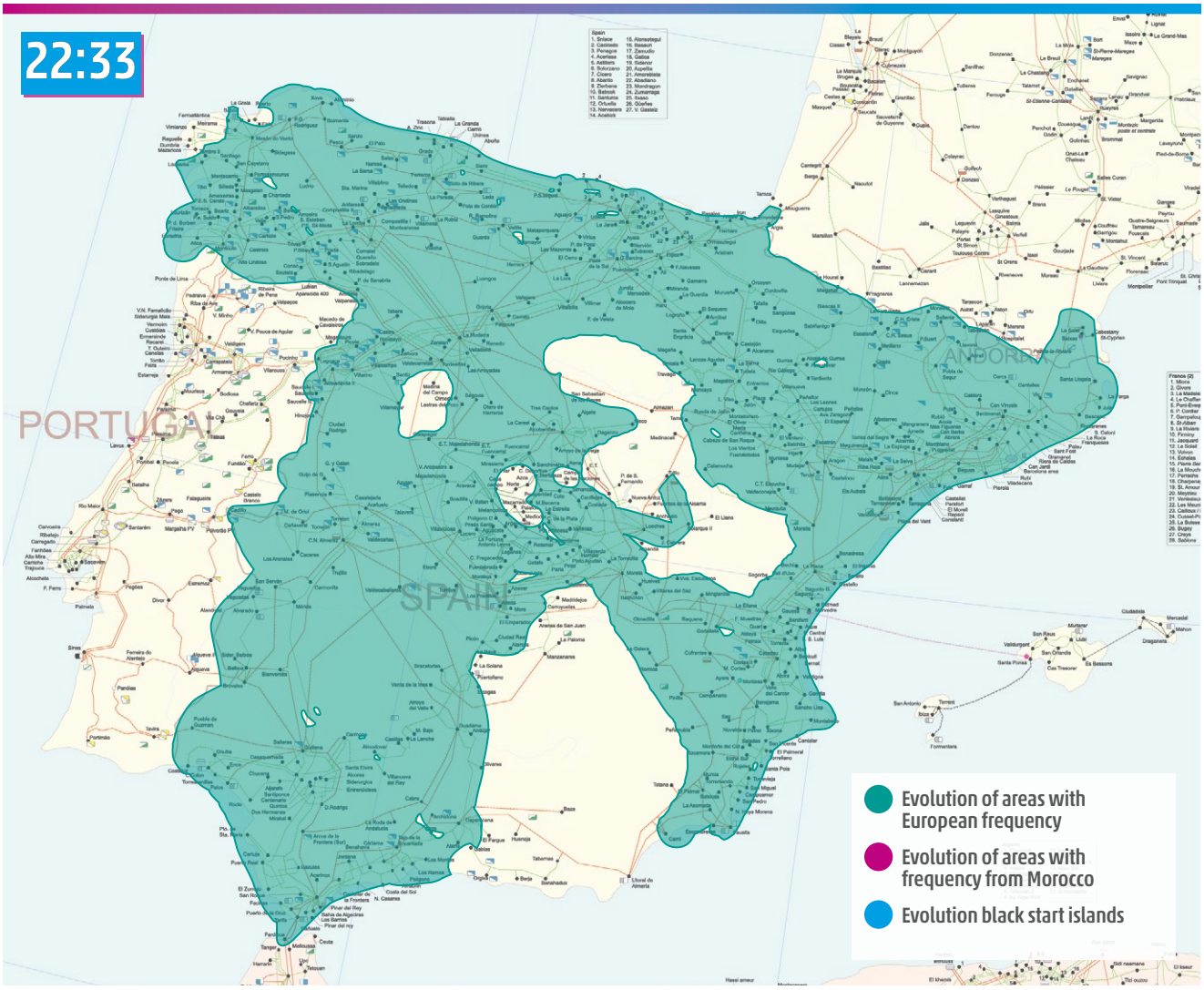


Figure 5-12: Evolution of the restoration in the transmission network through to 22:33 (14,760 MW of load supplied)

- At 23:32, 21 TPPs were connected to the Spanish peninsular electrical system.
- At 00:06 on 29 April, the master aFRR controller was started, initiating load-frequency control.
- From 01:38 onwards, the Renewable Energy Control Centre (CECRE) progressively sent instructions to allow the production of renewable, cogeneration, and waste (RCW) generation units. At 07:05, instructions were released to all RCW generation units.
- At 02:18, it was agreed with ONEE to end the emergency energy transfer from Morocco.
- At 02:59, in coordination with REN, the EAS state was changed from restoration to emergency.
- At 03:09, 31 TPPs were connected to the Spanish peninsular electrical system.
- At around 04:00, 100 % of the transmission network was restored.
- At 10:56, voltage was sent to Andorra through the double circuit 132 kV Margineda–Adrall.

» At 11:39, market participants were informed through the markets information system (ESIOS) that, in coordination with the NEMO, market activities were to be restored.

» At 14:13, the Peninsula–Balearic interconnection was energised.

» At 14:34, the result of the daily market for 30 April was published.

» At 14:36, in coordination with REN, the EAS state was changed from emergency to alert.

5.2.2 Bottom-up restoration of the Extremadura 1 island in Spain

» At 12:55 on 28/4/2025, HPP Extremadura 1 was started up using its black-start capability, to supply auxiliary services to a nuclear power plant and stabilise the load at the 220 kV Talavera SS. According to the restoration plan, HPP Extremadura 2, with black-start capability, is designated as the priority unit for creating this island, which extends to Madrid; however, it was initially unavailable due to a remote-control issue (see section 5.2.9). With the main strategy (starting up Extremadura 2), more load would have been available for supply.

» At 13:51, once the voltage had become available, another hydropower plant in the same area (HPP Castilla-La Mancha G2) synchronised with this island. HPP Castilla-La Mancha G2 had a minimum stable active power higher than the load in the island at the moment of synchronisation. The DSO in this area was not able to increase the load in time (probably due to autonomous reconnection of a small RCW < 1 MW).

» At 13:56, the HPP Extremadura 1 group tripped due to power reversal protection, losing the created electrical island.

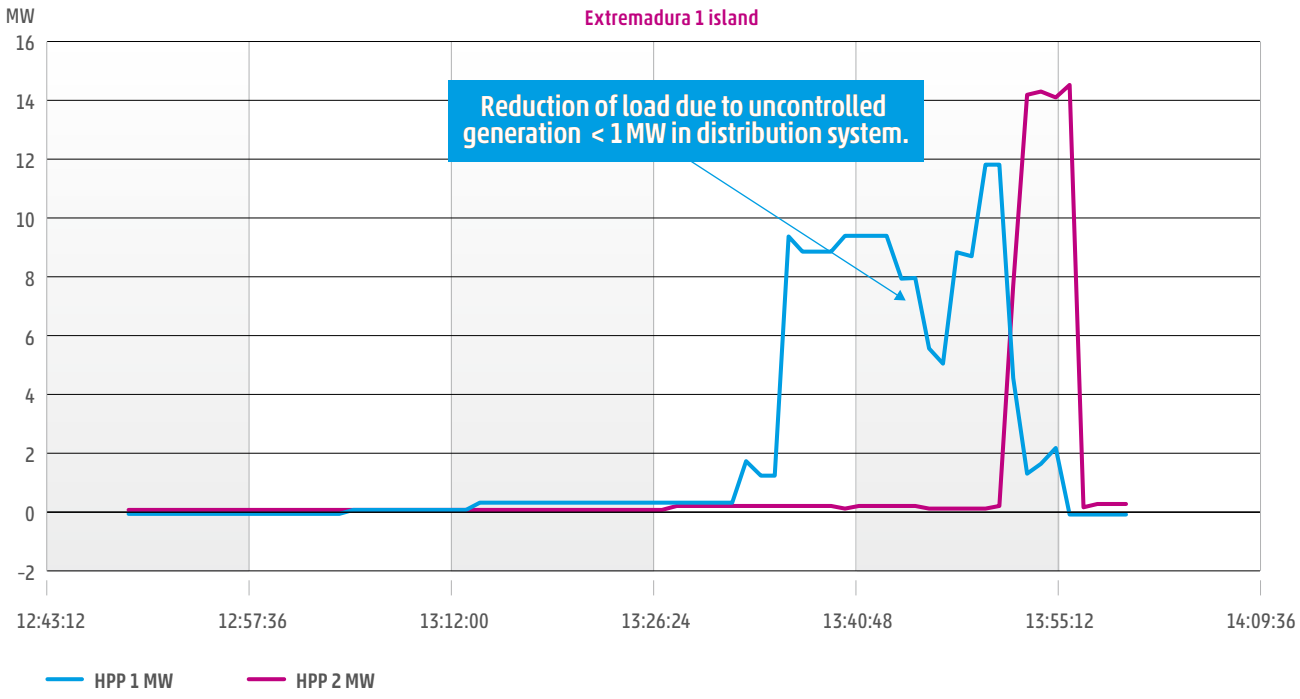


Figure 5-13: Evolution of active power in island Extremadura 1 between 12:43 and 14:09

5.2.3 Bottom-up restoration of the Duero 1 island in Spain

» At 13:00 on 28/4/2025, HPP Duero 1 was started up using its black-start capability, initiating the Duero 1 electrical island.

» At 13:12, the HPP Duero 1 group tripped due to severe vibrations when operating at low load. It was started up again at 14:21.

» At 17:01, after several failed attempts to synchronise the Duero 1 electrical island and the Duero 2 electrical island, it was decided to disconnect HPP Duero 1. Synchronisation was not possible due to the frequency difference between the two islands (see Figure 5-14) and the inability of the groups to regulate this. Several attempts were made to synchronise the islands by coupling large load blocks in the Duero 2 island to reduce the frequency, but these efforts were unsuccessful.

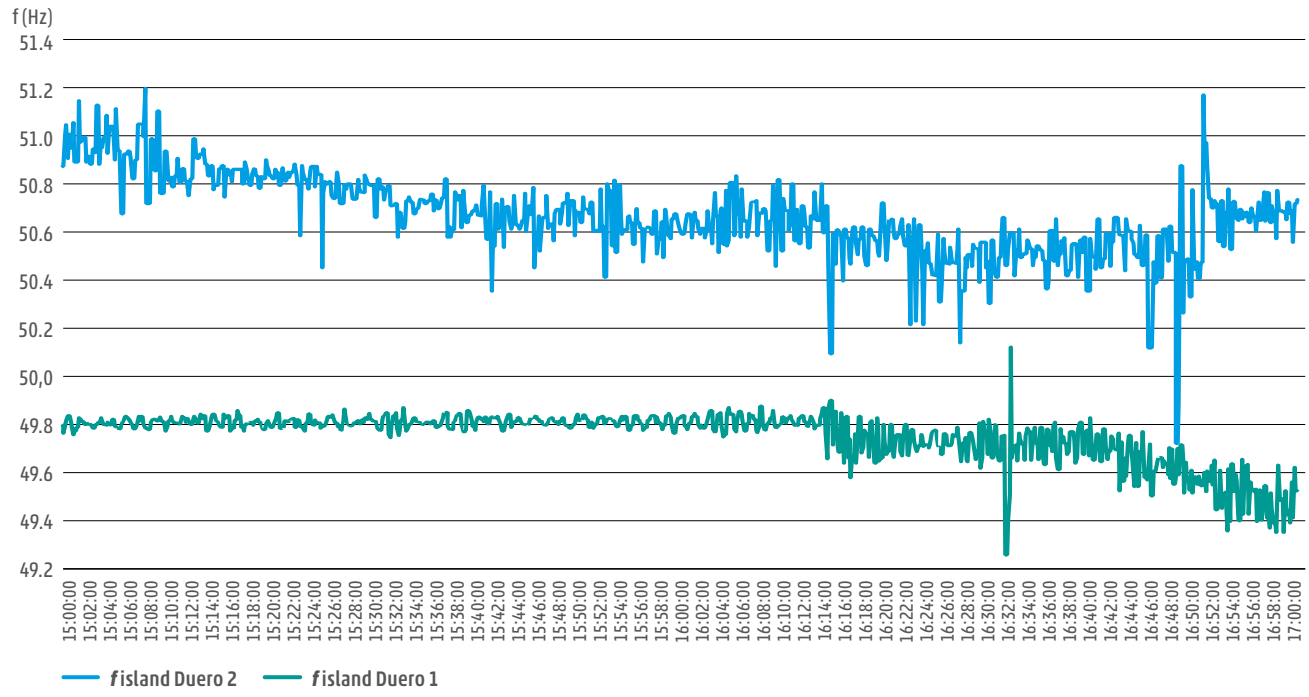


Figure 5-14: Evolution of the frequency in the Duero 1 and 2 islands between 15:00 and 17:01

5.2.4 Top-down restoration of the Southern zone in Spain

- At 13:04 on 28/4/2025, voltage was received from Morocco, initiating the restoration of the Southern zone.
- At 14:27, the load in the Southern zone supplied from Morocco had reached approximately 55 MW when an underfrequency relay tripped the cable at 49.50 Hz, as shown in Figure 5-15.
- The under-frequency situation at 14:27 was due to an imbalance in the power system of Morocco, since there was no additional load increase in the Spanish system at that moment.
- At 14:34, the network energisation from the 400 kV Tarifa SS was restarted to enable voltage supply to the thermal groups in the Southern area of Spain.
- The frequency in the Moroccan system, including the Southern island of Spain, was very volatile, because the link between Morocco and the large continental European system was not available at that time.
- At 19:32, the L-400 kV Almaraz–C. Rodrigo was switched on, synchronising the Southern zone with the Northern zone.
- The Expert Panel recommends to the TSOs of Spain and Morocco that they investigate the possibilities for temporarily changing the under-frequency threshold on the interconnector during the initial steps of the top-down restoration scenario to a less conservative value, equal to (or a few millihertz above) the first load-shedding frequency threshold in the Moroccan system. It also recommends that they provide for the possibility of switching back to the original frequency threshold when the Spanish system is reconnected to the Continental European system.

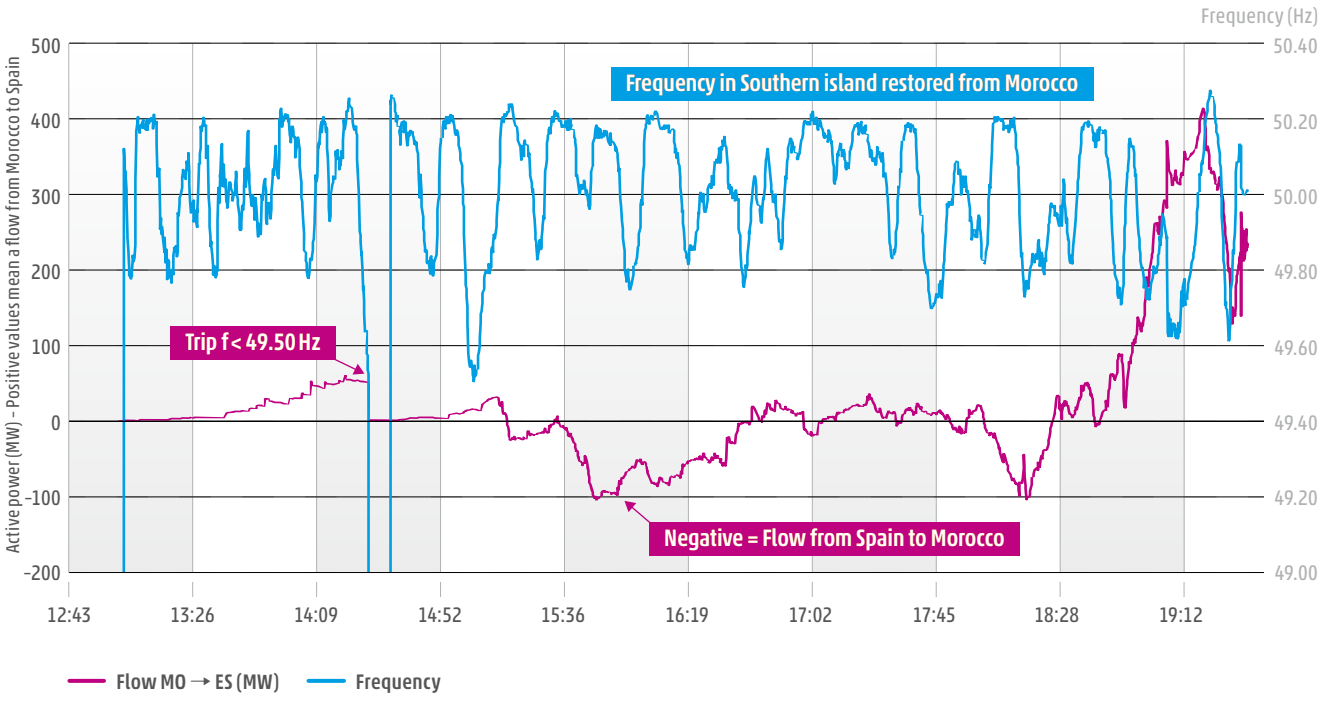


Figure 5-15: Active power exchange between Morocco and Spain (left axis) and frequency of the Southern island in Spain, measured at the Spanish substation Puerto de la Cruz (right axis)

5.2.5 Bottom-up restoration of the East Asturias island in Spain

» At 13:06 on 28/4/2025, unit 2 of HPP East Asturias was started up using its black-start capability, to energise a 132 kV distribution system in the East Asturias area. Figure 5-16 shows the active and reactive power exchange between unit 2 and the island grid.

» At 13:23, unit 2 of HPP East Asturias was tripped by an under-frequency protection at 49.50 Hz, causing the collapse of the East Asturias electrical island.

» This frequency drop corresponded to an increase in active power production by unit 2 of HPP East Asturias, most probably due to a load increase at that moment in the distribution system. The exact size of the load step could not be identified from the recordings.

» The setting of the under-frequency protection to disconnect the production unit at a threshold of 49.50 Hz deviate from the applicable requirements under the Spanish operational procedure, which prescribes that the underfrequency protection of generation facilities must be coordinated with the frequency-based load-shedding system. This means that production facilities may only disconnect from the grid if the frequency falls below 48 Hz, with a minimum delay of 3 seconds.

» The Expert Panel regrets that neither the operator of the production unit nor the DSO that was in charge of operating the distribution system in East Asturias were able to provide recorded measurements of the frequency in the island during the restoration period. The Expert Panel therefore recommends improving observability for TSOs and DSOs of any production and storage facilities involved in scenarios for the bottom-up restoration of distribution systems.

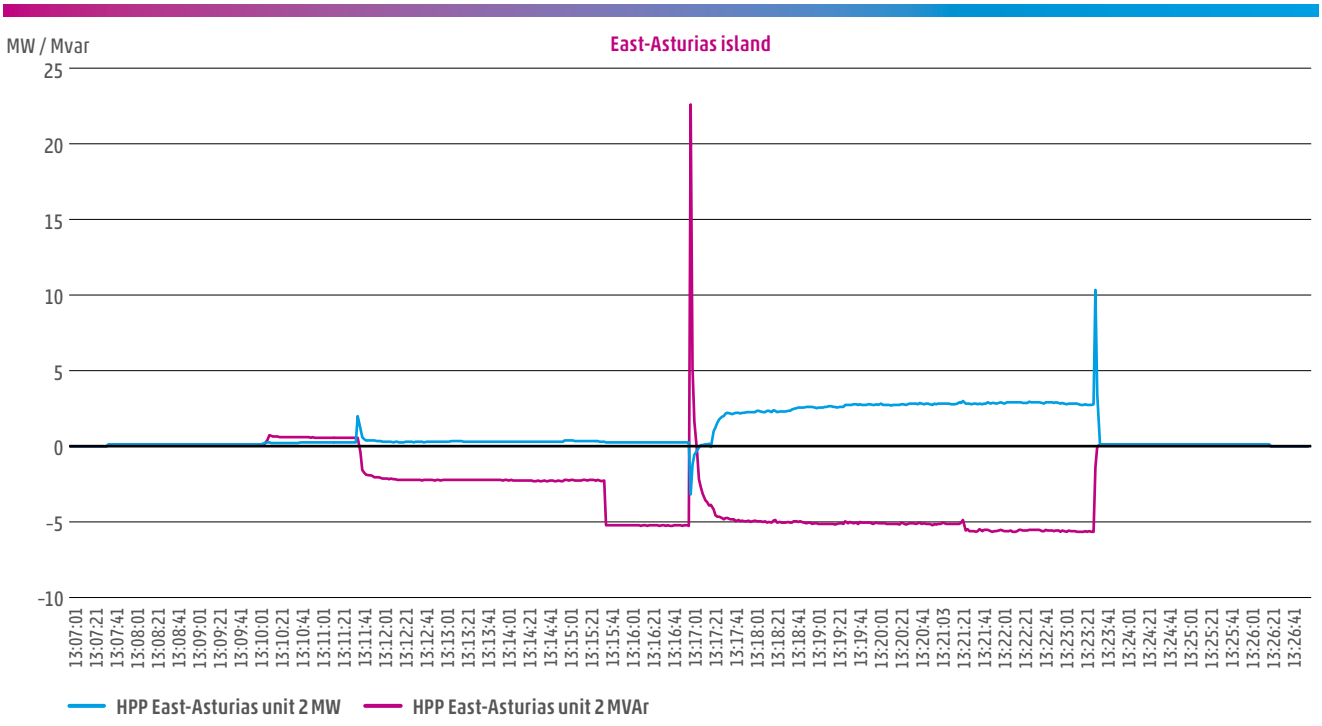


Figure 5-16: Evolution of active and reactive power in the East Asturias island between 13:06 and 13:26

5.2.6 Bottom-up restoration of the Galicia island in Spain

- At 13:26 on 28/4/2025, unit 1 of HPP Galicia 1 was started up using its black-start capability, initiating the Galicia electrical island, which was designed to supply load in Galicia and critical loads, and to energise the auxiliary services of several thermal power plants (TPPs).
- With the progressive creation of the black-start island and the connection of multiple loads, unit 3 of another hydropower plant, HPP Galicia 2, was synchronised at approximately 16:02; this unit was also intended to absorb reactive power.
- Due to insufficient frequency control by unit 1 of HPP Galicia 1 (see Figure 5-17), which could not be resolved upon request by the TSO, frequency control was switched to unit 3 of HPP Galicia 2 at 16:11. Unit 1 of HPP Galicia 1 was assigned a fixed active power setpoint.
- At 16:18, unit 3 of HPP Galicia 2 was tripped by the overexcitation protection (ANSI 24), also known as "Volt-over-Hertz (V/Hz)" protection. Prior to this, a brief increase in grid voltage was recorded, indicating a deficiency in voltage regulation, possibly caused by the activation of the under-excitation limiter on unit 3 of HPP Galicia 2, or by a manual voltage regulation malfunction. After the trip, the plant operator of HPP Galicia 2 informed the TSO that a single unit at the plant could not control voltage and frequency at the same time. Unit 1 of HPP Galicia 1 took over additional active power injection, thereby initially stabilising the island.
- At 16:27, unit 2 of HPP Galicia 2 was synchronised first, with the aim that it would take over frequency control from unit 1 of HPP Galicia 1.
- At 17:46, unit 3 of HPP Galicia 2 was synchronised, with a defined active power setpoint and voltage regulation.

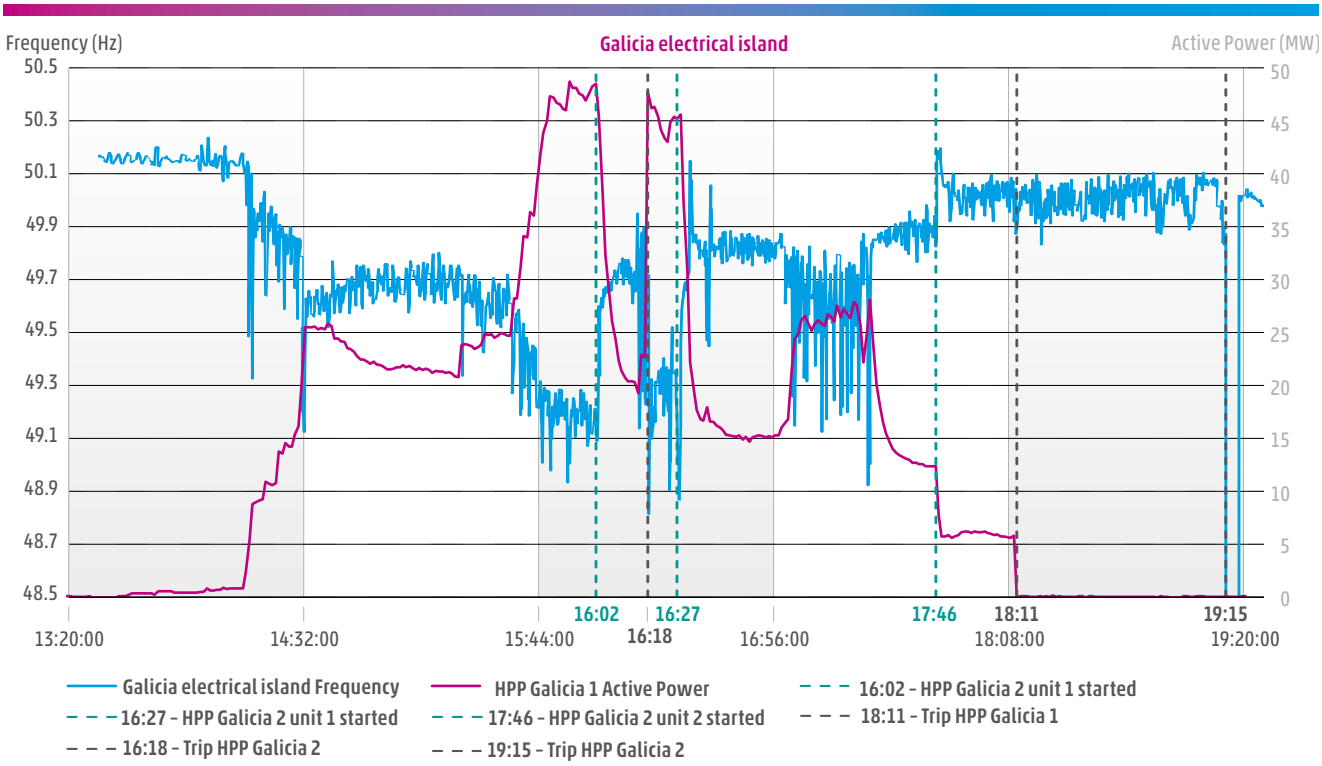


Figure 5-17: Evolution of active power and frequency in the Galicia electrical island between 13:20 and 19:30

» At 18:11, unit 1 of HPP Galicia 1 was tripped due to power reversal protection. According to Figure 5-17, the synchronisation of unit 3 of HPP Galicia 2 led to a reduction in the active power output of unit 1 of HPP Galicia 1. This indicates issues with the active power-frequency control in unit 1 of HPP Galicia 1, as the assigned fixed active power setpoint could not be maintained.

» At around 18:17, the DSO informed the TSO that, due to a remote-control problem, no further loads could be connected in the Galicia island. The only remaining option for expanding the grid island, in the absence of any additional loads, was synchronisation with a neighbouring part of the system that was already energised.

» Prior to the synchronisation attempt, around 18:40, several transmission lines were disconnected to reduce the reactive power absorption of the HPPs. After the disconnection, voltage variations appeared, probably due to an unstable behaviour in the AVR of the HPPs. During the oscillation, at around 19:15, a transmission line was connected, and units 2 and 3 of HPP Galicia 2 tripped due to an overvoltage event.

» With this, the Galicia electrical island had completely collapsed. Due to the proximity of voltage from the Northern zone, it was decided not to restart the island.

5.2.7 Bottom-up restoration of the Levante island in Spain

» The restoration plans (PRS-0-003 "Restoration of Levante Zone" and PRS-0-010 "Supply of nuclear power plants auxiliary services") provide for the creation of two separate islands:

- **Island 1:** use unit 1 of HPP Levante 1 with black-start capability to energise one busbar ("busbar 1") in the nearby 400 kV substation and further energise the rest of the Levante zone. Additionally, this group would be used to supply the auxiliary services of power plant HPP Levante 2 in order to connect more groups to this Levante island.
- **Island 2:** use unit 2 of HPP Levante 1 with black-start capability to energise another busbar ("busbar 2") in the same nearby 400 kV substation, to energise the auxiliary services of the NPP in the northern Levante area.

» At 13:06 on 28/4/2025, unit 1 of HPP Levante 1 was started up using its black-start capability. According to the producer, this group supplied the auxiliary services of the nearby HPP Levante 2, through a 20 kV link. This went on until 29 April, with no trips or incidents.

» At 13:52, unit 2 of HPP Levante 1 was started up using its black-start capability, initiating the Levante electrical island, which was designed to energise the auxiliary services of the NPP in the northern Levante area.

» At 14:02, the 400 kV/66 kV transformer at the nuclear power plant was energised at nominal voltage. Unit 2 of HPP Levante 1 was able to withstand the transformer inrush current. However, although a soft start re-energisation procedure was foreseen, the operators of the TSO and the producer finally agreed with the nuclear power plant to energise the transformer at nominal value.

» At 14:08, the producer was requested by the TSO to connect unit 1 of HPP Levante 1 to "busbar 1" of the nearby 400 kV substation to create island 1. For unknown reasons, the high voltage breaker of this group could not be closed, meaning that "busbar 1" remained without voltage.

» At 14:36 and 14:50, the producer tried to energise "busbar 1" by closing the high voltage breaker of unit 2 of the nearby HPP Levante 2, which was started using auxiliary power from unit 1 of HPP Levante 1. Both attempts were unsuccessful, leaving "busbar 1" at zero voltage.

» At 14:54, the producer tried unsuccessfully to energise "busbar 1" by closing the high voltage breaker of unit 1 of HPP Levante 2, leaving "busbar 1" still at zero voltage.

» At 15:19, due to the unsuccessful attempts to energise "busbar 1" from the 3 above-mentioned units, the TSO decided to close the busbar coupler between "busbar 1" and "busbar 2", with the aim of continuing the restoration process to the rest of the Levante zone. Busbar 2 was already energised from unit 2 of HPP Levante 1. From this point onwards, the TSO progressed the restoration with one island instead of two separate islands.

» At 15:29, the TSO energised a 40 km 400 kV overhead line, causing unit 2 of HPP Levante 1 to increase reactive power absorption from 11 Mvar to 45 Mvar, as shown in Figure 5-18. The black-start unit initially remained stable after this line connection.

» The energisation of the 400 kV line at 15:29 caused the voltage at the nearby 400 kV substation to increase to 452 kV, as shown in Figure 5-19.

» At 16:00, another network component in the distribution grid was connected, leading to a further increase of reactive power absorption by unit 2 of HPP Levante 1 (from 45 to 50 Mvar) and a further voltage increase (up to 455 kV).

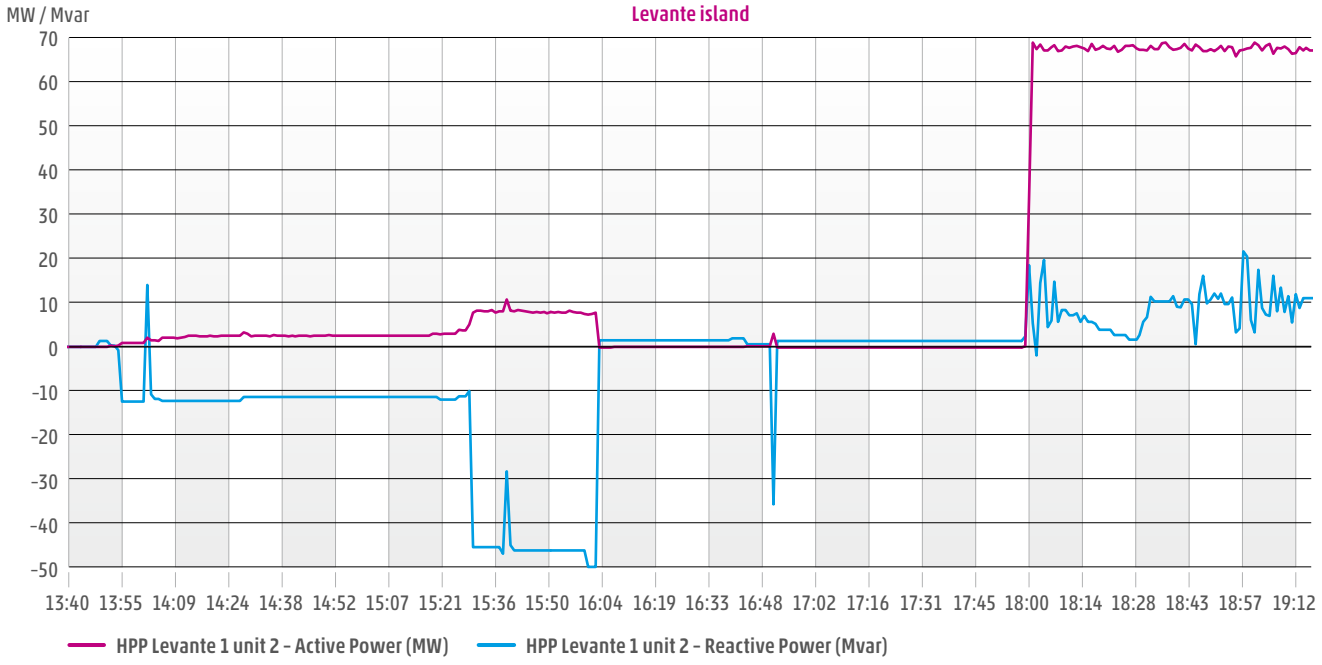


Figure 5-18: Active and reactive power of HPP Levante 1 - unit 2

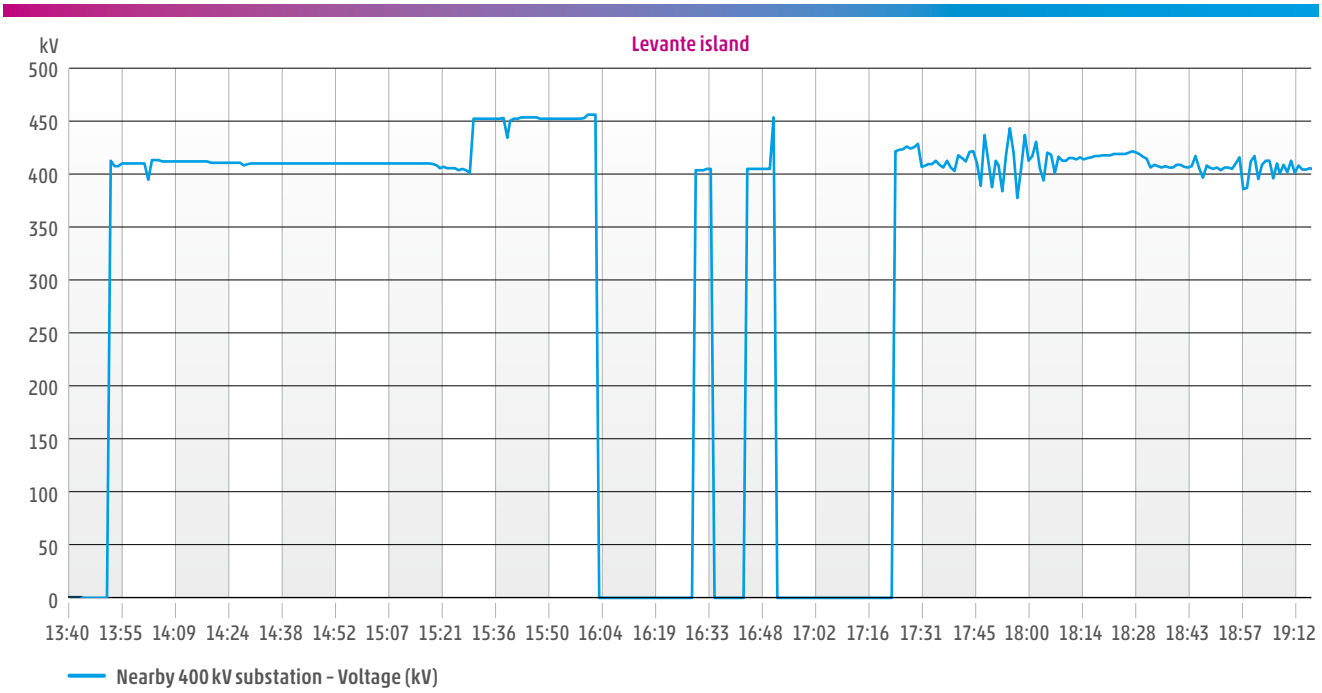


Figure 5-19: Voltage at the nearby 400 kV substation

» At 16:03, unit 2 of HPP Levante 1 tripped, due to the action of the "Volt-over-Hertz" (V/Hz) protection set at 1.1 p.u., losing the Levante electrical island. The generator was no longer able to absorb more reactive power and the UEL (Under Excitation Limit) was activated. The reactive power absorption limitation raised the voltage level and the generator tripped on the V/Hz function.

» At 16:30 and at 16:43, two new attempts were made to create the Levante electrical island from HPP Levante 1 unit 2, but the unit tripped each time due to Volt-over-Hertz protection.

» At 17:23, the nearby 400 kV substation was energised from the Eastern zone, and at 17:59, HPP Levante 1 unit 2 was connected to it. After synchronisation, damped voltage oscillations were observed at the nearby 400 kV substation.

» The TSO and the producer are currently reviewing the restoration plan for the Levante area and clarifying the technical possibilities of power plants with black-start capability.

» In general, the Expert Panel recommends that HPPs are started at voltages under the nominal value, as stated in the restoration plans, to allow the operation of the 400 kV lines at reduced voltage. This would limit reactive power production by the grid elements, and avoid activation of the V/Hz machine protection during the first restoration steps.

5.2.8 Bottom-up restoration of the Duero 2 island in Spain

» At 14:11 on 28/4/2025, unit 1 of HPP Duero 2 was started up using its black-start capability, initiating the Duero 2 electrical island, which was designed to supply critical loads, other HPPs, and synchronise with the other islands.

» At 17:49, the Duero 2 island synchronised with the Northern zone via the 220 kV line Tordesillas-Arroyadas.

» At 18:46, the Duero 2 island was decoupled from the Northern zone due to the tripping of the 220 kV Arroyadas–Renedo line (see Figure 5-20), which was caused by stability issues arising from the very long 220 kV corridor and high power flow between the Northern zone and the Duero 2 island. The Duero 2 electrical island did not collapse, but several 220 kV lines and generating units within the Duero 1 island were disconnected.

» At 18:54, synchronisation was quickly restored through the 220 kV Arroyadas–Renedo line. Subsequently, reinforcement of the interconnection between the zones was carried out via the 400 kV network.

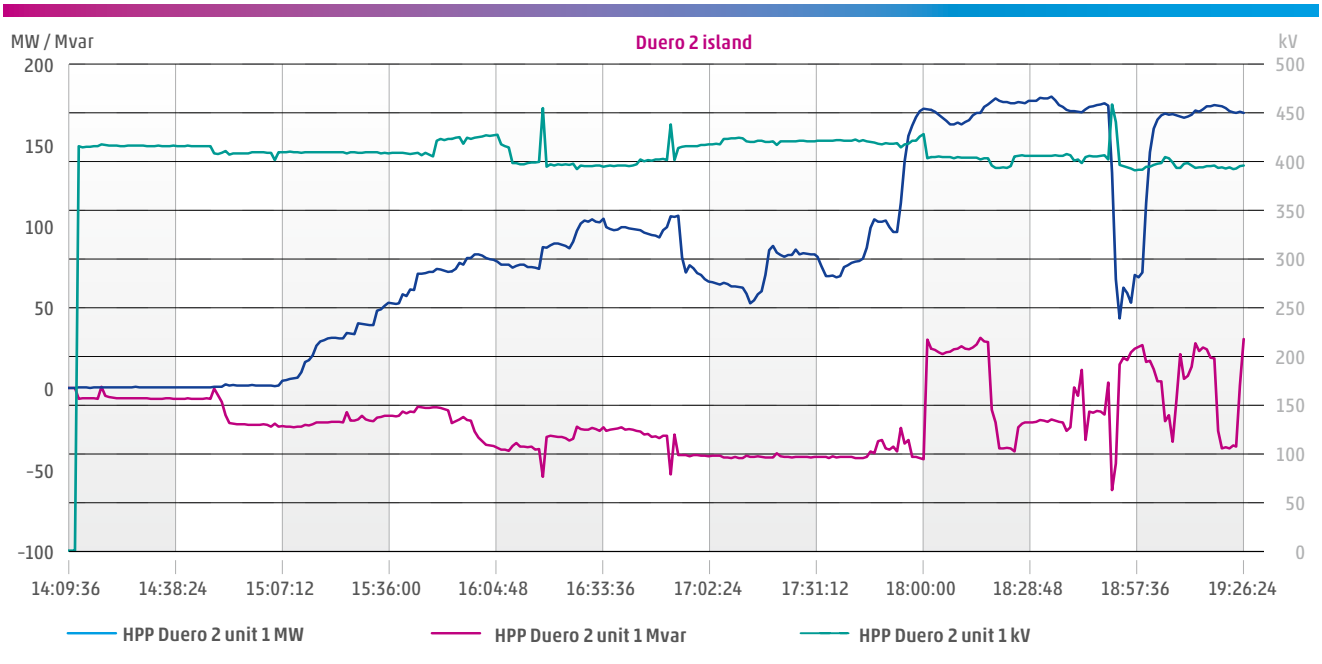


Figure 5-20: Evolution of voltage, active and reactive power in the Duero 2 island

5.2.9 Bottom-up restoration of the Cantabria island in Spain

- At 14:05 on 28/4/2025, unit 2 of HPP Cantabria was started up using its black-start capability, initiating the Cantabria electrical island, which was designed to supply load in the Cantabria area, supply auxiliary services for several TPPs, and synchronise with the islands created in the Asturias area. It is established in the Restoration Plan to start this island with the largest possible number of groups operating in parallel.
- At 15:23, HPP Cantabria unit 1 was connected to stabilise the island, as shown in Figure 5-21.
- At 15:53 and 15:57, respectively, the step-up transformers of units 3 and 4 of HPP Cantabria were magnetised from the 220 kV side, causing an inrush current, which was supplied from units 1 and 2. Units 3 and 4 were not synchronised to the island formed by units 1 and 2.
- As from 15:53, the active power of unit 1 started to increase, reaching 20 MW at 16:02, while at the same time the active power of unit 2 was pushed back to zero, and even to negative values.
- Figure 5-22 shows that the frequency remained close to 51 Hz during this period. This behaviour suggests that an additional load of approximately 10 MW was connected to the island and that there was a problem with the load sharing between units 1 and 2.
- At 16:02, an unknown event triggered a frequency decrease leading to slow oscillations of increasing amplitude in the active power between units 1 and 2. This led, at 16:06, to the disconnection of unit 2 due to reverse power flow and of unit 1 due to over frequency, with the loss of the entire Cantabria electrical island.
- Considering the proximity and progress of the Northern zone, the re-energisation of the Cantabria area from the zones with stable voltage and frequency was prioritised in order to speed up the restoration of this zone.

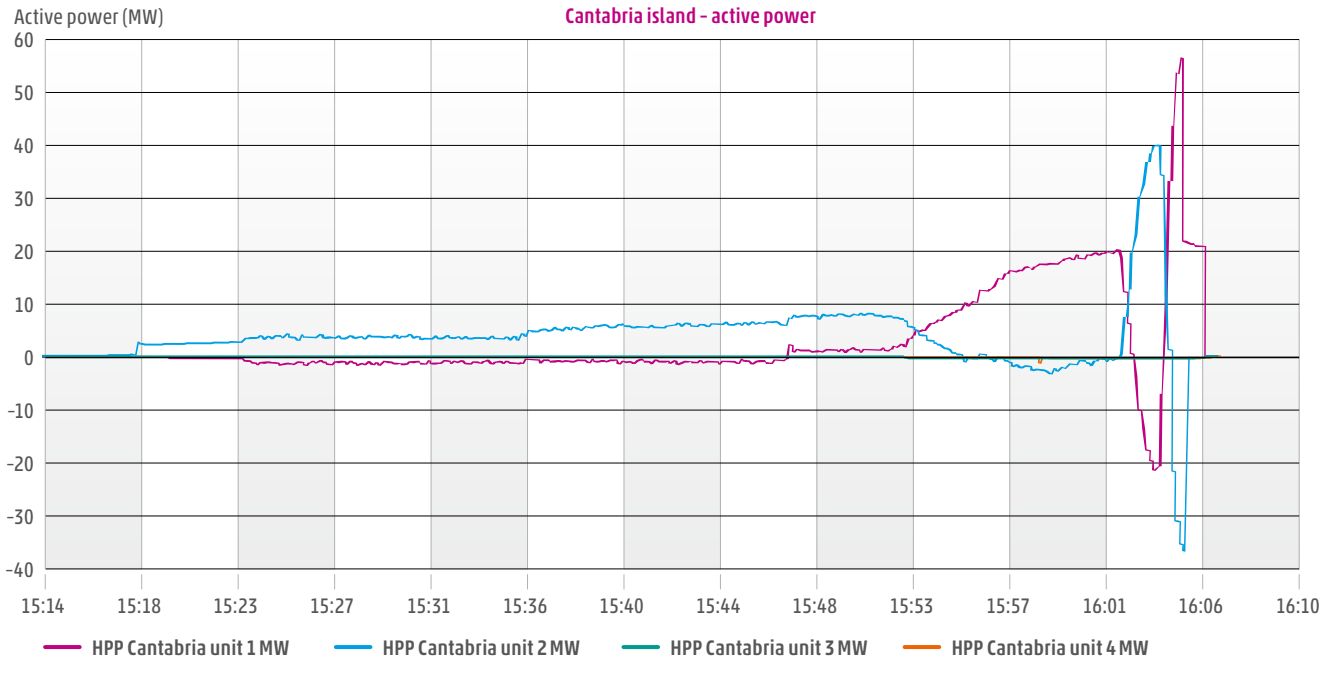


Figure 5-21: Active power in the Cantabria island

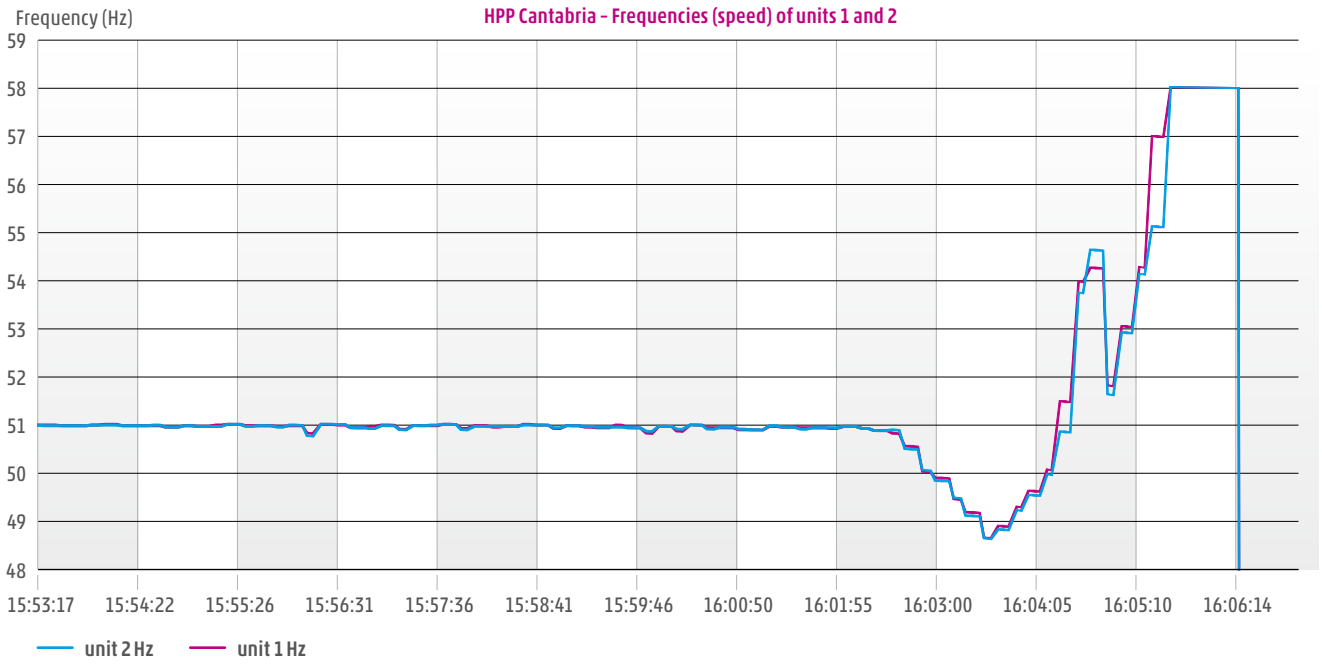


Figure 5-22: Frequencies of units 1 and 2 of HPP Cantabria

5.2.10 Bottom-up restoration of the Extremadura 2 island in Spain

» At 14:21 on 28/4/2025, unit 3 of HPP Extremadura 2, with black-start capability, was started up using a soft-start process, to supply the auxiliary services of a nuclear power plant and to create the Extremadura 2 electrical island, thereby enabling load stabilisation and the connection of the remaining HPPs and TPPs.

» Figure 5-23 shows the active and reactive power of the units involved in the island (unit 3 of HPP Extremadura 2 and units 1 and 2 of HPP Castilla-La Mancha) from its start-up at 14:21 until the synchronisation of the island with the Southern zone at 18:16.

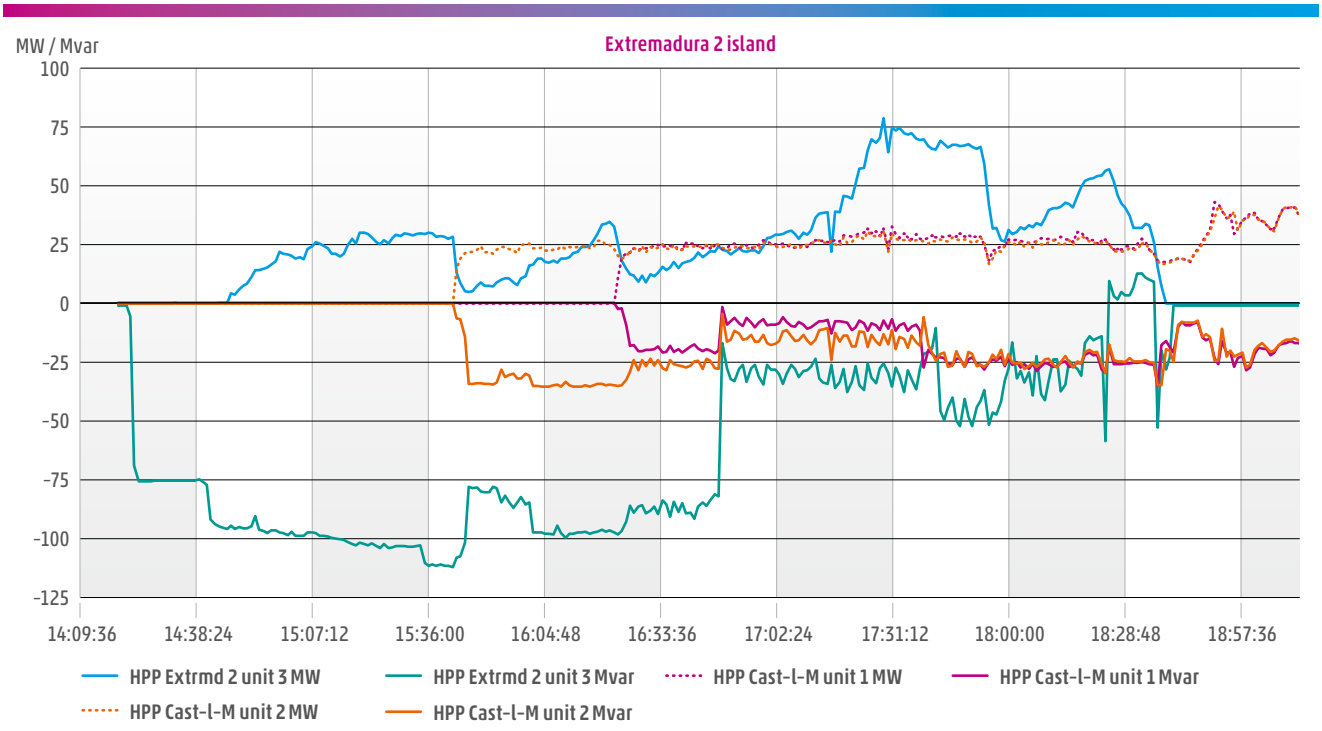


Figure 5-23: Active and reactive power in the Extremadura 2 island

» At 18:16, the L-400 kV Bienvenida–Almaraz was switched on, synchronising the Extremadura 2 electrical island with the Southern zone. Meanwhile, the supply had extended to the Madrid area.

» At 18:35, after synchronisation with the Southern zone, unit 3 of HPP Extremadura 2 tripped on reverse power, with the island frequency at 50.12 Hz at the moment of the trip. This did not cause any issues for the progress of the restoration. According to the producer, the generators with black-start capability regulate frequency and cannot change their operating mode while running. The producer recommends shutting down the black-start units once the island has reached a sufficient size to be stable on its own, through the primary regulation of the generators that compose it. However, the Expert Panel recommends that the producer investigate the technical possibilities of having the capability to change the operating mode of the black-start generators while they are running.

5.2.11 Bottom-up restoration of the West Asturias island in Spain

» At 14:32 on 28/4/2025, unit 2 of HPP West Asturias I was started up using its black-start capability, initiating the West Asturias electrical island in the 132 kV distribution network. This was designed to supply load in the Asturias area, connect other HPPs, and synchronise with the East Asturias and Cantabria electrical islands.

» At 14:46, unit 2 of HPP West Asturias I tripped due to low pressure in the oil system.

» At 14:57, unit 2 of HPP West Asturias I was started again in island mode.

» From 15:29 to 18:18, several groups were synchronised to the island. The sequence was as follows:

- At 15:29: unit 1 of HPP West Asturias II
- At 15:48: unit 2 of HPP West Asturias II
- At 17:44: unit 3 of HPP West Asturias II
- At 18:18: unit 1 of HPP West Asturias I

» The power output of the groups until synchronisation with the Northern zone is represented in Figure 5-24. It can be seen in this figure that after synchronisation, the power output of the groups decreases. This reduction of power was due to the significant difference in frequencies between the West Asturias island and the Northern zone at the moment of synchronisation. It is recommended that the parameters of the parallel switching devices located in the distribution system be reviewed.

» At 18:43, the West Asturias electrical island, initiated in the Asturias distribution network, was synchronised with the Northern zone through ATP1 220/132 kV Siero.

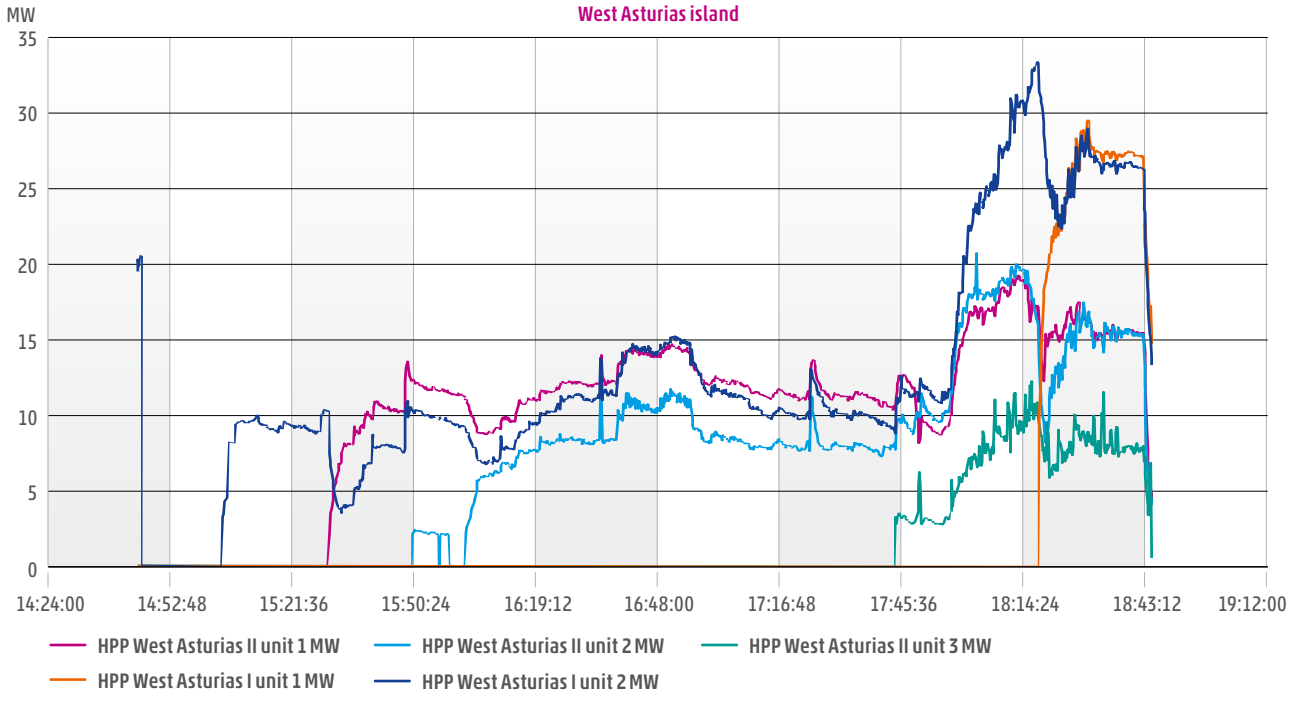


Figure 5-24: Active power in the West Asturias island

5.2.12 Bottom-up restoration of the Aragón island in Spain

- At 15:01 on 28/4/2025, unit 3 of HPP Aragón was started up using its black-start capability, initiating the Aragón electrical island, which was designed to connect other nearby HPP and supply the auxiliary services of several TPPs.
- Figure 5-25 shows the active and reactive power exchanged between unit 3 of HPP Aragón and the Aragón island, as well as the voltage amplitude measured at the connection point between the power plant and the high voltage network.
- From 16:44, oscillations emerged in the voltage, and were also visible in the HPP's speed before and after the synchronisation of the Aragón island with the Northern zone. The HPP's active and reactive power remained almost constant.
- No reasons for the oscillations, observed by the TSO and DSO, could be identified.
- At 16:54, the Aragón electrical island was synchronised with the Northern zone through the L-220 kV Escatrón–Villanueva. The synchronisation was accomplished using a synchro-check device with a difference of frequencies setting lower than 150 mHz.

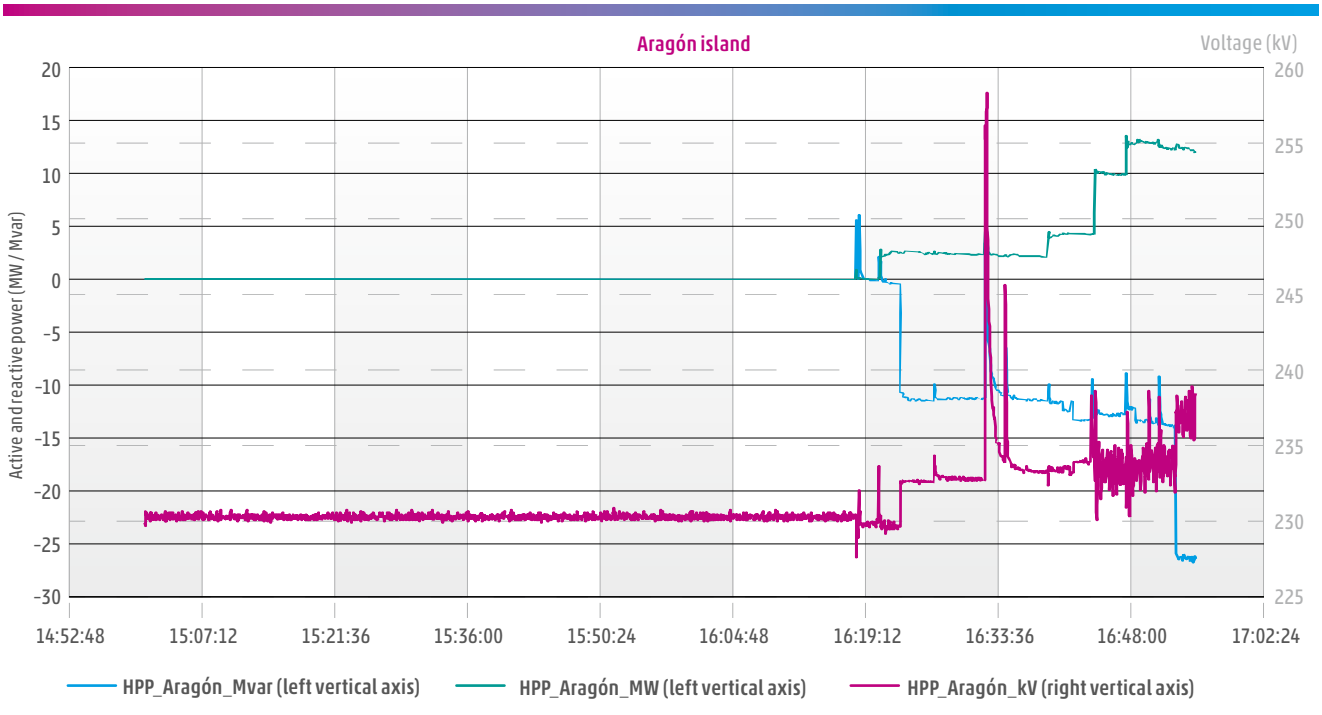


Figure 5-25: Active and reactive power and voltage in the Aragón island

» Figure 5-26 shows the active power and frequency in the Aragón island.

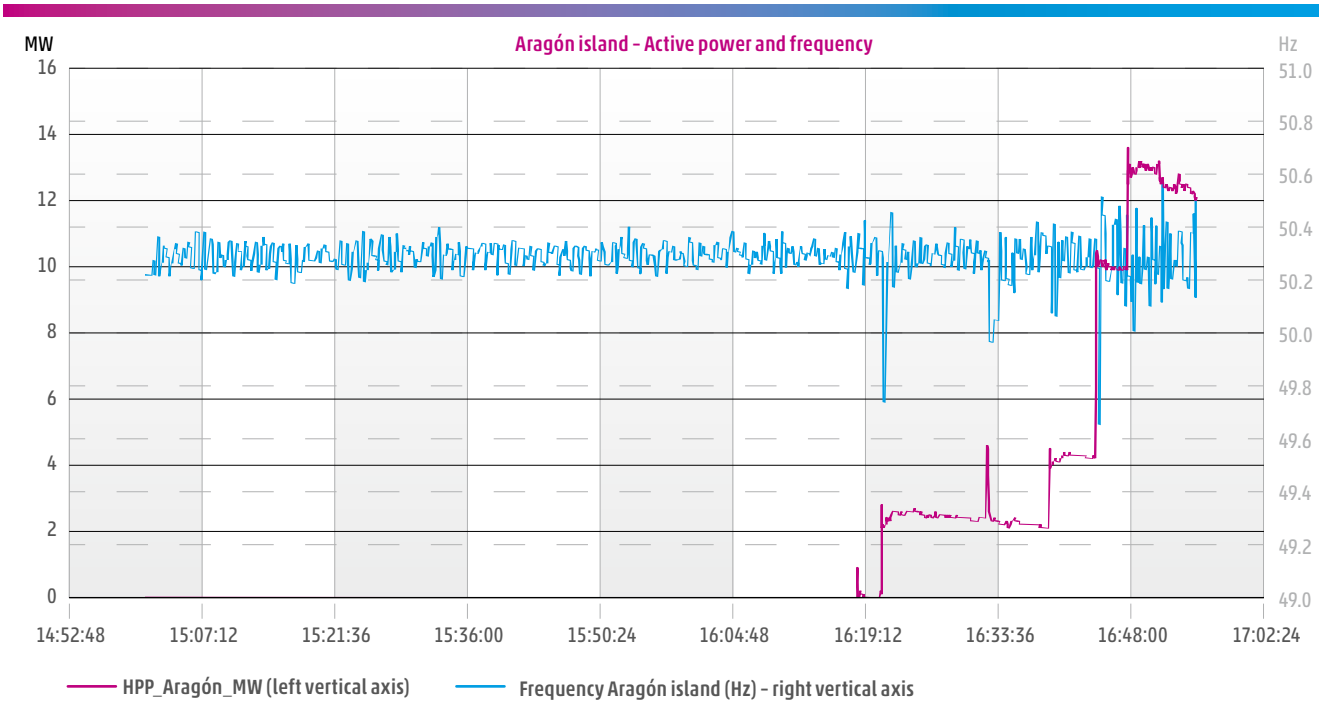


Figure 5-26: Active power and frequency in the Aragón island

5.2.13 Bottom-up restoration of the Andalucía island in Spain

» At 13:35 on 28/4/2025, an attempt was made to start the hydropower plant HPP Andalucía 1, with black-start capability, without success. The other planned black-start power plant, HPP Andalucía 2, was unavailable.

» According to the owner of HPP Andalucía 1, the black-start mode start-up sequence stopped because the instrumentation did not detect any lubrication oil flow to the mechanical bearings of the group. Future black start tests should seek to solve this island mode problem.

» At 14:05, the second black start attempt of HPP Andalucía 1 also failed. The island mode start-up sequence stopped for the same reasons as during the first attempt at 13:35.

» Considering the proximity and progress of the Southern zone, the re-energisation of the Andalucía area from the Southern zone was prioritised in order to speed up the restoration.

» At 16:59, HPP Andalucía 1 successfully connected to the Southern zone.

5.2.14 Bottom-up restoration of the Toledo island in Spain

- At 15:16 on 28/4/2025, HPP Madrid was started in 132 kV distribution network, using its black-start capability. The objective was to supply the auxiliary services of a TPP and some stabilisation loads in the Toledo area.
- At 15:19, a technical problem during the switching on of the auxiliary services led to the tripping of the unit.
- At 15:36, after the restart of the black-start unit, an initial connection of loads – approximately 30 % of the machine's nominal power – caused the unit's speed to drop and subsequently led to the unit being tripped by the turbine controller.
- Further attempts to create a stable black-start island were unsuccessful for various reasons:
  - 16:01 – Trip due to overvoltage, accompanied by variations in the grid voltage
  - 16:22 – Trip due to speed drop, coinciding with load connection
  - 17:57 – Trip by the protection system due to an external cause
- Despite several attempts, it was not possible to create the island. Due to the advanced stage of grid restoration in the other zones, priorities were reassessed.

5.3 The restoration sequences in the Portuguese system

This section describes the restoration sequences in Portugal in detail, from the initial actions through to the total restoration of the Portuguese system.

Due to confidentiality constraints, the names of the generation units are anonymised for external publication.

5.3.1 Overview of the Portuguese system restoration

Restoration process timeline

- At 12:35(CEST) on 28/4/2025, REN requested the black-start mode start-up of HPP 1-Centro.
- At 12:43, REN requested the black-start mode start-up of CCGT 1-Norte. Following the restoration plan, the black-start operation mode was carried out using one generation unit in each of these power plants to establish electrical islands.
- At 12:45, HPP 1-Centro was operating in black-start mode. The events of the bottom-up restoration in the Centro island are described in detail in section 5.3.3.
- At 12:49, HPP 1-Centro tripped, after which the producer initiated several attempts to restart HPP 1-Centro, all of which failed due to problems starting up the auxiliary small hydro units.
- In the first few hours, CCGT 1-Norte was unsuccessful in closing the generator circuit breaker to energise the unit transformer. The events of the bottom-up restoration in the Norte island are described in detail in section 5.3.2
- At 15:40, HPP 1-Centro was restarted. It tripped again at 15:55 and was restarted at 16:13, re-establishing the first restoration area with the propagation of voltage to HPP 2-Centro and HPP 3-Centro at 150 kV.
- At 16:38, CCGT 1-Norte started. It propagated voltage to Canelas and Recarei SSs and progressively increased load supply until 17:26, when the unit tripped (see section 5.3.2.)
- At 17:26, CCGT 1-Norte started again, to create the second restoration area.

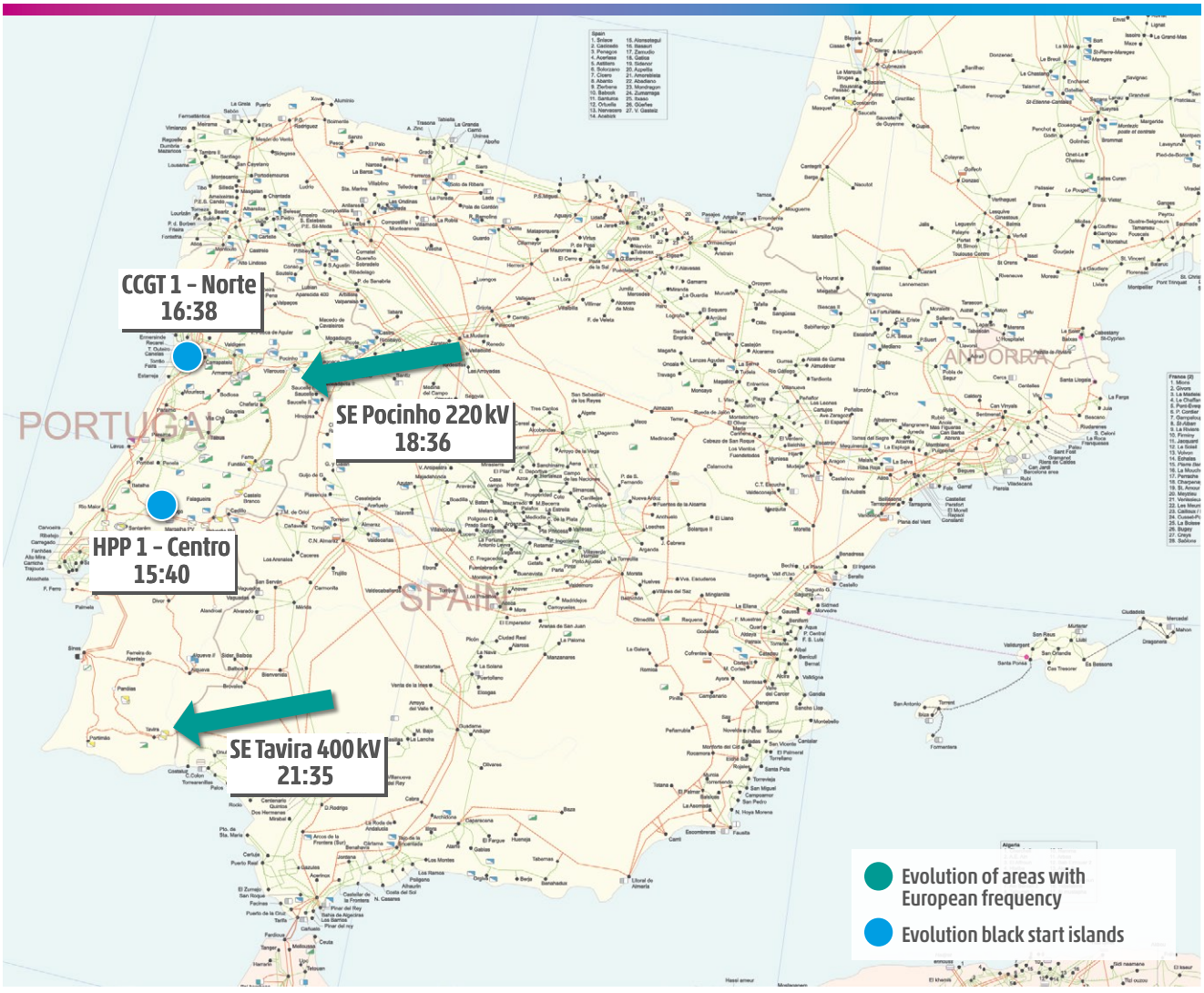


Figure 5-27: Starting times of power plants with black-start capability and later-stage support from interconnections with Spain – REN

» At 18:32, the Oporto region had voltage available in a relevant set of injection SSs in the distribution network, allowing E-Redes (the Portuguese DSO) to continue with the progressive load supply in this geographic area.

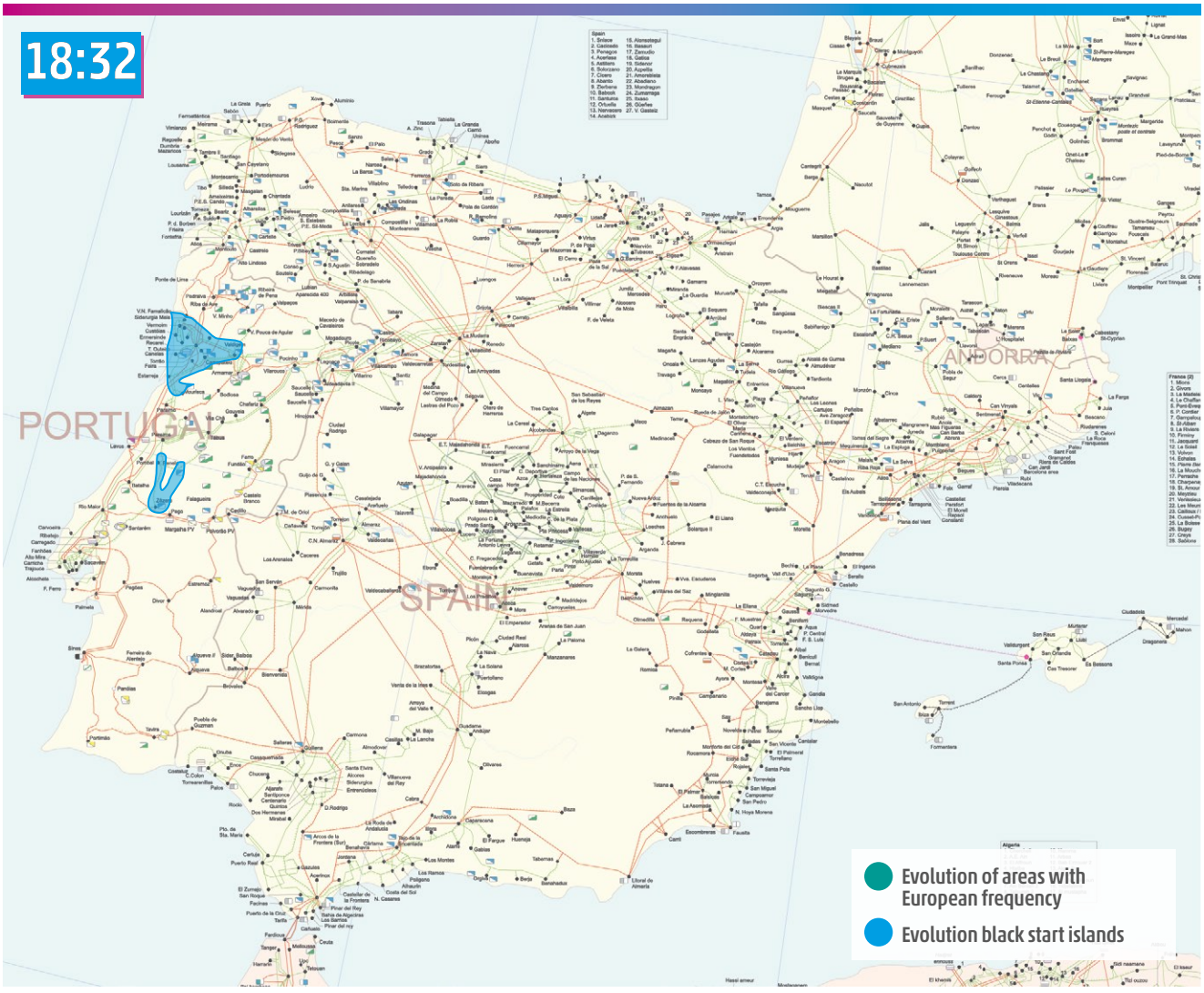


Figure 5-28: Evolution of the restoration in the black-start islands through to 18:32 (120 MW of load supplied) – REN

» At 18:36, voltage was propagated to the 220 kV busbar of Pocinho SS from the Spanish Aldeadávila SS (Pocinho–Aldeadávila 1 tie-line at 220 kV), leading to the creation of the third restoration area. This constituted the first synchronous connection with the continental European system through the Spanish electrical system.

At this time, the Portuguese system was being restored in three different electrical islands:

» First, by HPP 1-Centro

» Second, by CCGT 1-Norte

» Third, by interconnection with Spain (synchronous restoration)

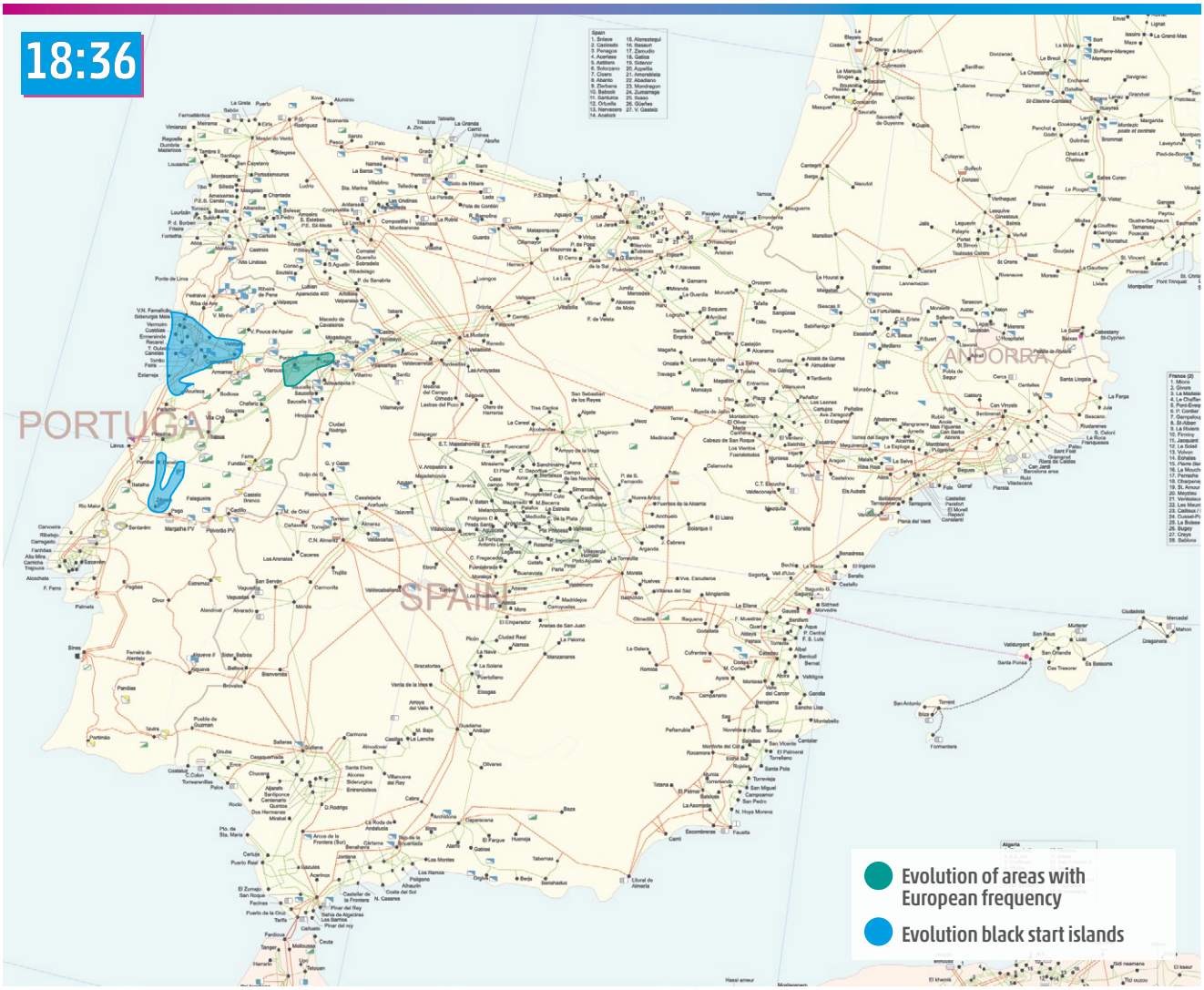


Figure 5-29: Evolution of the restoration in the transmission network and black-start islands through to 18:36 (120 MW of load supplied) – REN

» At 19:00, voltage was propagated, enabling progressive load supply and generation restoration along the corridors leading to the Zêzere SS, and thereby extending the third restoration area (synchronous with the Spanish/continental European region).

» At 19:58, the connection with the Spanish system was reinforced, connecting both the Pocinho–Aldeadávila 2 and Pocinho–Saucelle lines at 220 kV.

» At 20:01, the electric island powered by HPP 1-Centro was synchronised with the Spanish system, leading to the successful synchronisation of the first and third restoration areas with the continental European region. At this point, there were 18 HPP units and 1 TPP (CCGT Norte) connected to the grid: of these, ten were in the third restoration area, two in the first restoration area, and seven in the second restoration area.

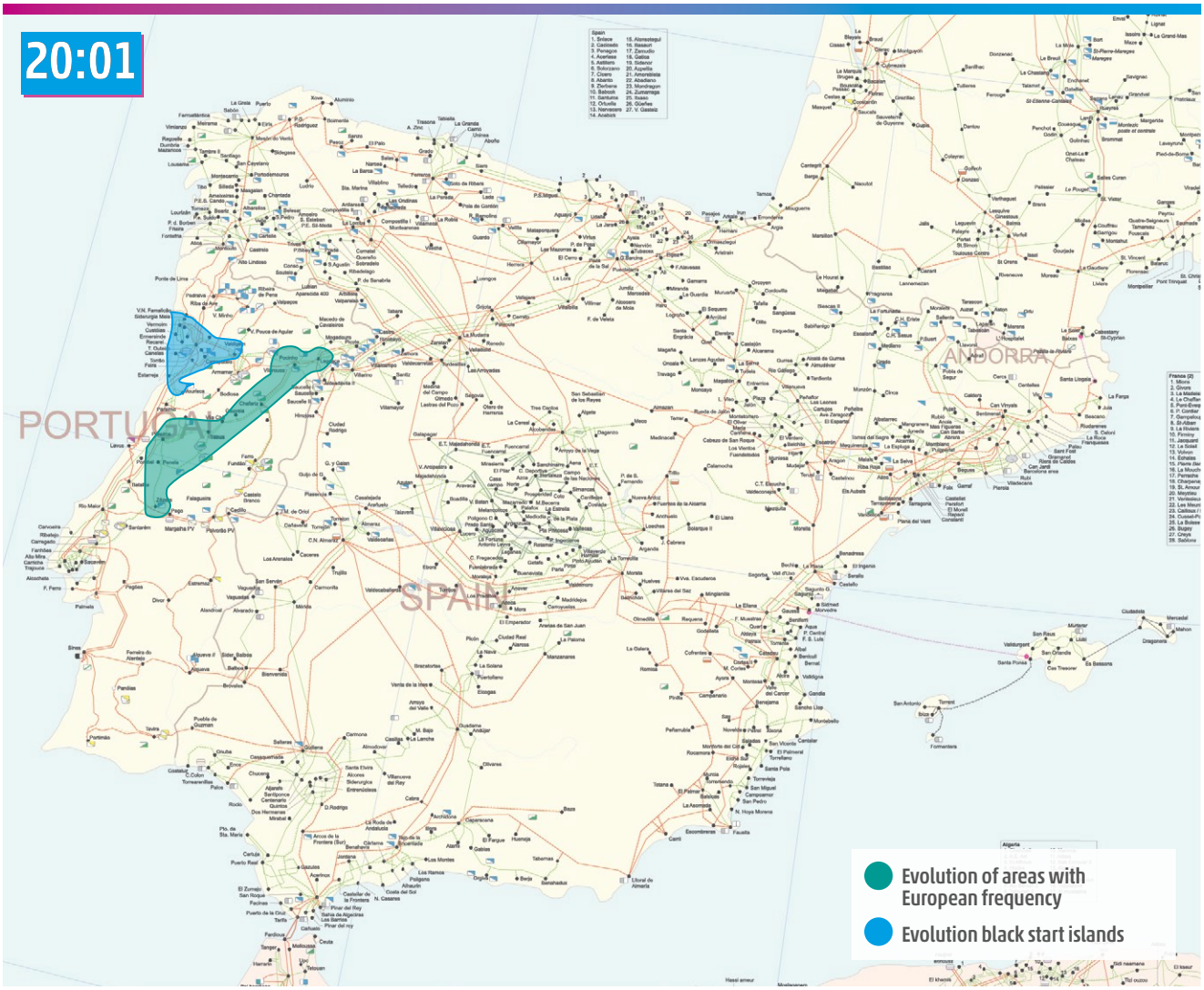


Figure 5-30: Evolution of the restoration in the transmission network and black-start islands, with the synchronisation of the electric island powered by HPP 1-Centro, through to 20:01 (376 MW of load supplied) – REN

» At 20:22, the Norte island was synchronised with the Spanish/continental European region, leading to the successful synchronisation of the second restoration area with the first and third areas, which were already in synchrony with the continental European region.

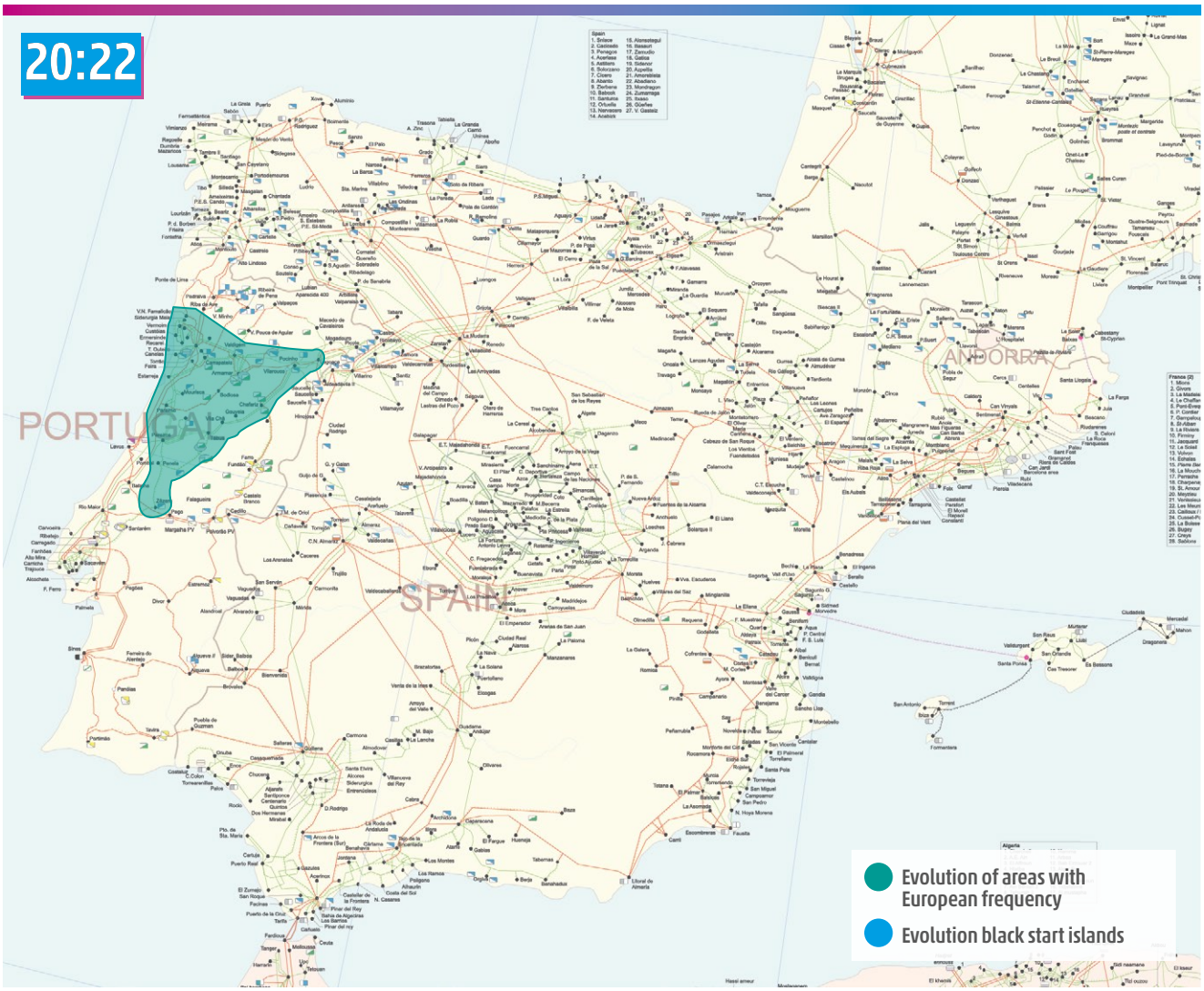


Figure 5-31: Evolution of the restoration in the transmission network, with the synchronisation of the electric island powered by CCGT 1-Norte, through to 20:22 (457 MW of load supplied) – REN

With all areas of the Portuguese system connected to the Spanish/European synchronous system, the restoration process for the transmission network accelerated substantially. The autonomy of the transmission network equipment ensured proper functioning of the voice communication system, telemetry, and remote control until the restoration process was completed. Permanent coordination and cooperation were maintained between REN, Red Eléctrica, E-REDES (the Portuguese DSO), and all national producers, namely the significant grid users (SGUs), during the Portuguese restoration process.

- At 20:25, automatic generation control (AGC) was activated, following an interconnection schedule programme of 0 MW (as agreed with Red Eléctrica), to balance the Portuguese system by automatically adjusting generation to match load.
- At 20:26, voltage propagation, progressive load supply, and generation restoration were occurring throughout the northern and central regions of Portugal.
- At 20:47, the interconnection with the Spanish grid was strengthened with the connection of the Lagoaça–Aldeadávila tie-line at 400 kV.
- At 20:57, the restoration of the main loads in the Lisbon region began, with voltage propagated to the Sacavém SS (REN's National Control Centre).

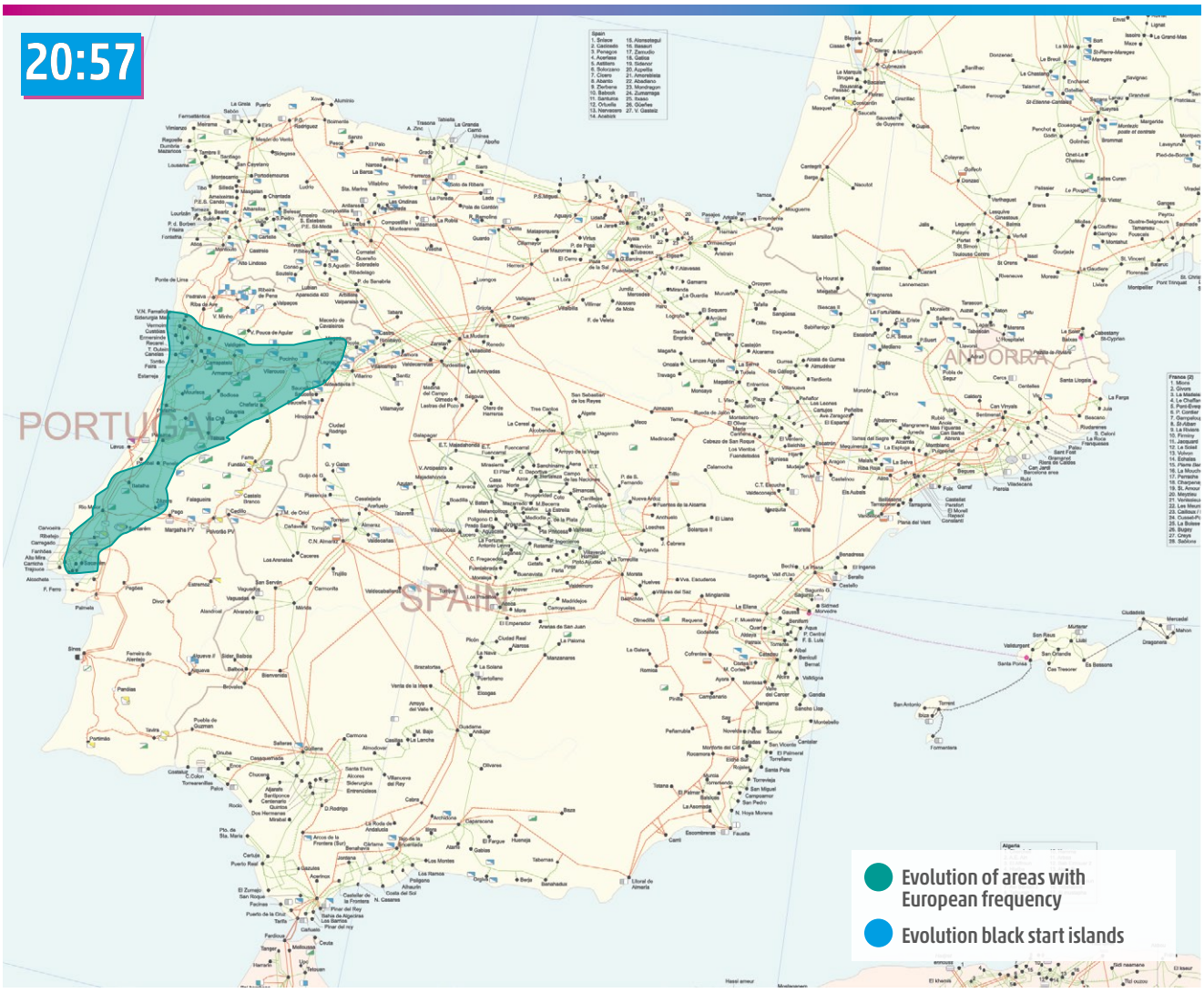


Figure 5-32: Evolution of the restoration in the transmission network through to 20:57 (715 MW of load supplied) – REN

» By 21:11, the Lisbon region had voltage available in a relevant set of injection SSs in the distribution network, allowing E-REDES to continue progressive load restoration in this geographic area.

» At 21:35, the 400 kV Tavira–Puebla de Guzmán tie-line was connected, initiating voltage propagation and progressive load supply in the Algarve and Alentejo areas, leading to the creation of the fourth restoration area through synchronous connection with the continental European region (via the Spanish system).

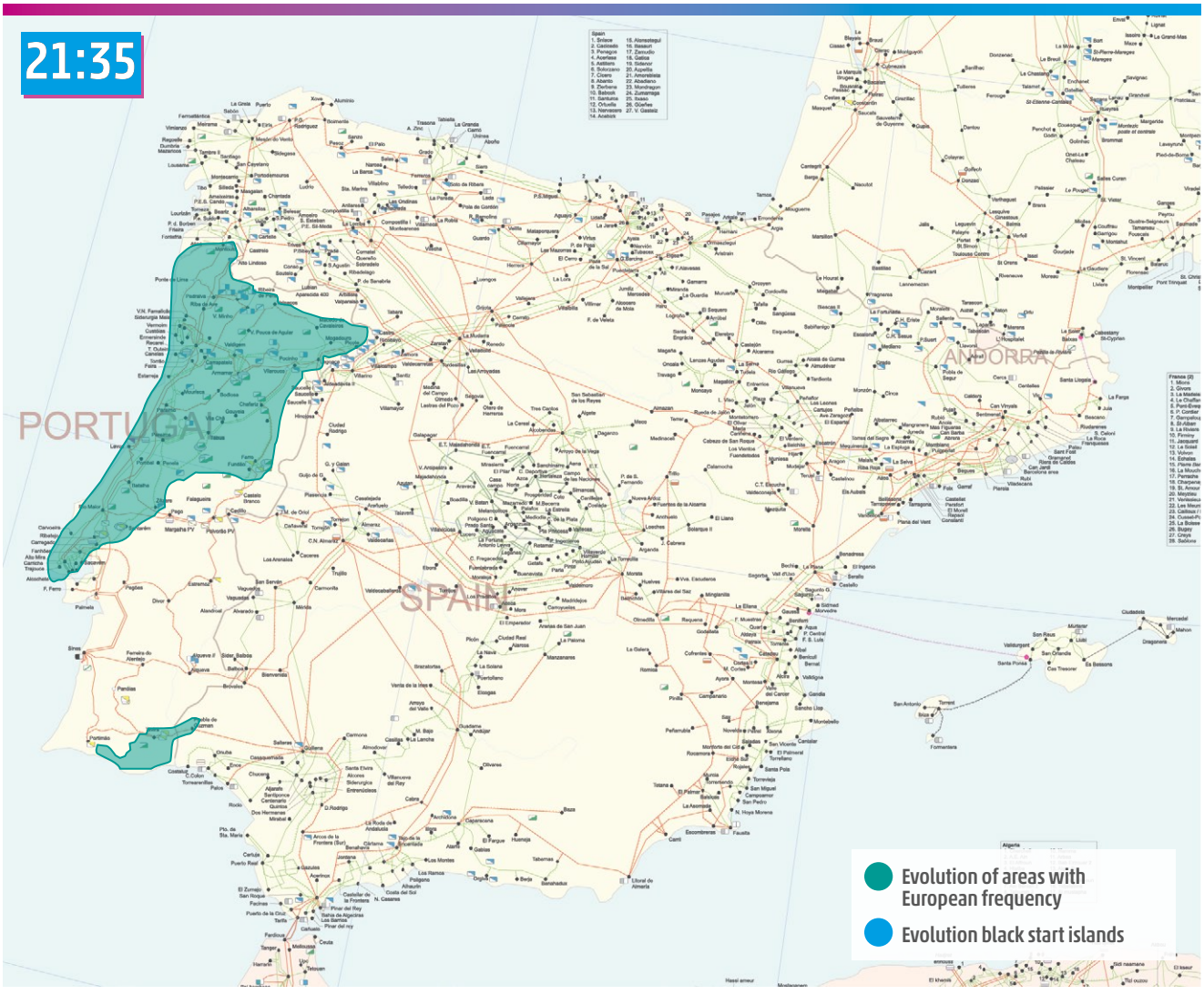


Figure 5-33: Evolution of the restoration in the transmission network through to 21:35 (1,244 MW of load supplied) – REN

» At 21:37 and 22:23, the interconnection with the Spanish system (synchronous with the continental European region) was strengthened with the connection of the 400 kV Alto Lindoso–Cartelle and Falagueira–Cedillo tie-lines, respectively.

» At 22:28, there was voltage propagation and progressive load supply in the Setúbal peninsula area from the extension of the 400 kV for the Lisbon region.

» At 23:19, the Algarve and Alentejo regions were connected with the rest of the network, leading to a successful merger of the main grid area with the fourth restoration area.

» At 00:22 on 29 April, the restoration of the TSO grid was completed through the connection of the last delivery point, the Divor SS.

» At 02:13, in coordination with Red Eléctrica, the Portuguese system state in the EAS was changed from restoration to emergency.

» At 14:37, in coordination with Red Eléctrica, the Portuguese system state in the EAS was changed from emergency state to alert state.

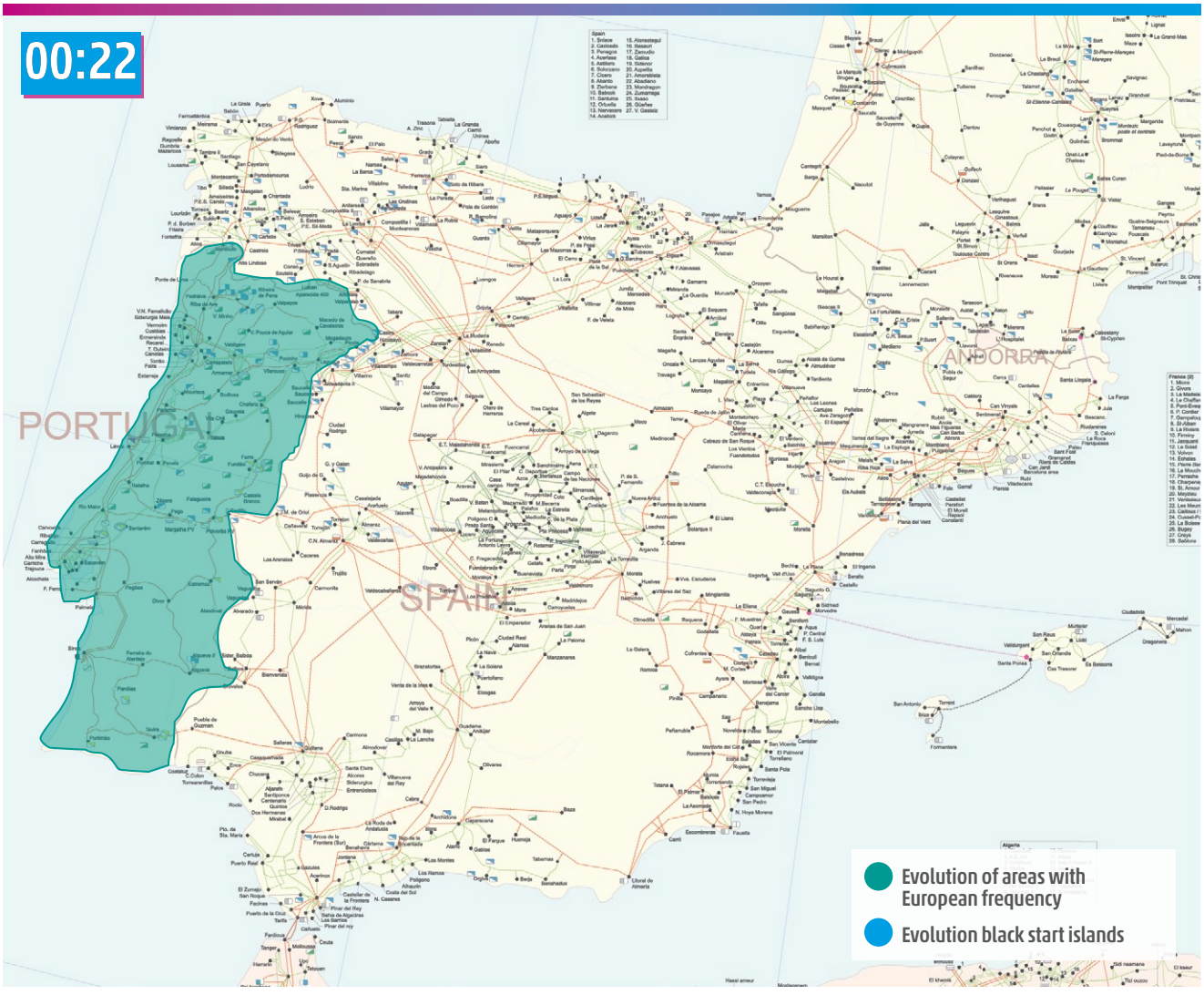


Figure 5-34: At 00:22, the restoration of the transmission network was completed (4,237 MW of load supplied) – REN

5.3.2 Bottom-up restoration of the Norte island in Portugal

- At 12:43 (CEST) on 28/4/2025, REN requested the black-start mode start-up of CCGT 1-Norte.
- In the first few hours, CCGT 1-Norte was unsuccessful in closing the generator circuit breaker to energise the unit transformer, due to unexpected problems in the control and command systems in the power plant substation. According to the producer, an interlock between excitation and the generator circuit breaker prevented the energisation of the generator step-up transformer. This interlock was identified and modified during this period.
- At 16:38, CCGT 1-Norte started, and voltage was propagated to the 220 kV busbar of the power plant SS.
- At 16:54, voltage was propagated to the Canelas and Recarei SSs, and there was a progressive load supply.
- At 17:23, a synchronous compensator was started using Unit 2 of HPP 1-Norte to support voltage control in the island, as provided for in the National Service Restoration Plan. For the reasons explained below, this caused the unwanted tripping of CCGT 1-Norte.
- During the start-up process of unit 2 of HPP 1-Norte as a synchronous condenser, its auxiliaries consumed approximately 17 MW in forcing the remaining water out of the hydro turbine and bringing the machine to synchronous speed. This additional load to the Norte island slowed down the CCGT 1-Norte generator and reduced the island frequency from 50 Hz to 49.7 Hz.
- The operator in charge at CCGT 1-Norte changed to load control in an attempt to increase the load set point and stop the speed drop. When changing to load control, the unit adopts the actual load as the load set-point.
- After 25 seconds, the auxiliary compensator's consumption dropped back to 0 MW. The sudden load drop after the manual change to load control was not anticipated by the operator. CCGT 1-Norte accelerated and was disconnected from the grid by its overspeed protection, as shown in Figure 5-35. CCGT 1-Norte did not trip, and continued on house load operation. Upon the opening of the HV circuit breaker, CCGT 1-Norte switched automatically from load control to speed control. A few minutes later, the switchyard was re-energised.

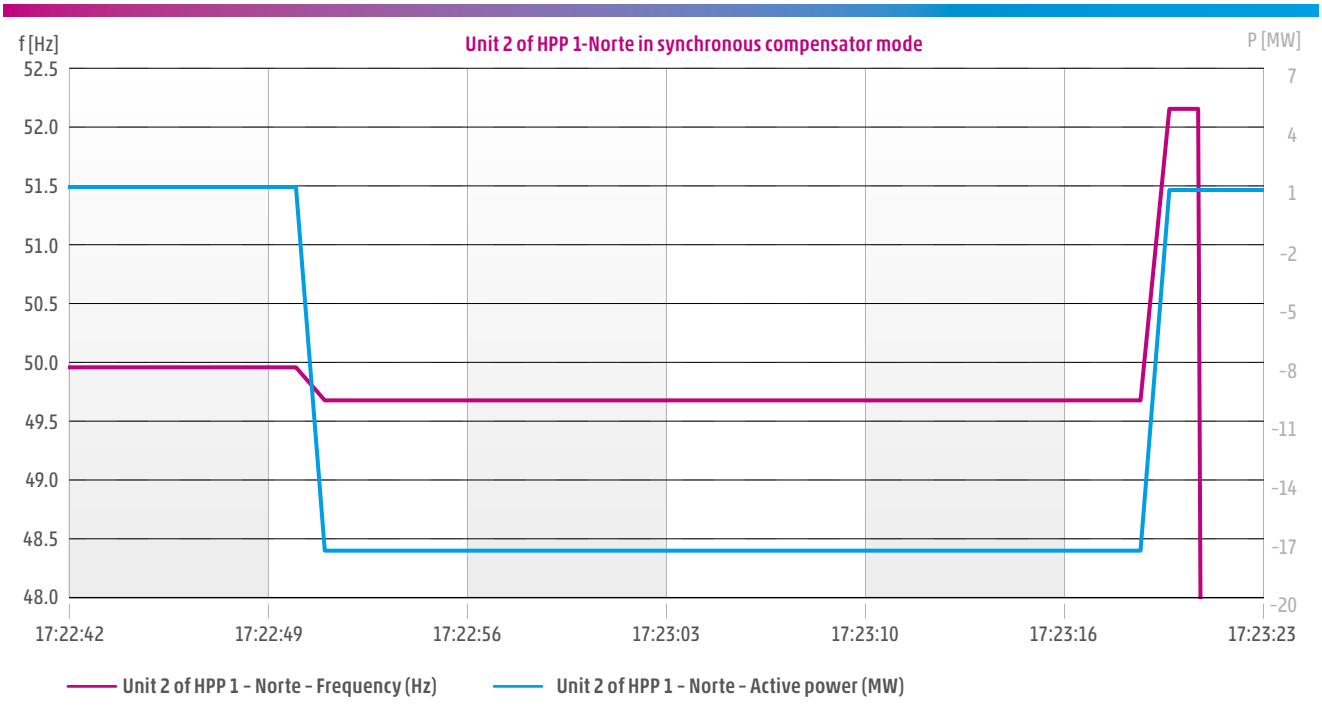


Figure 5-35: Impact of starting the synchronous compensator in HPP 1-Norte on the frequency in the Norte island

» Later in the restoration process, HPP1-Norte was started successfully in synchronous compensator mode. Moreover, REN confirmed that after April 28, the use of HPP 1-Norte was tested successfully in synchronous compensator mode in combination with CCGT 1-Norte.

» At 17:26, CCGT 1-Norte propagated voltage to the 220 kV busbar of power plant SS, leading to the creation of the second restoration area. Figure 5-36 shows the active power and the frequency (speed) of CCGT 1-Norte between 17:01 and 17:44.

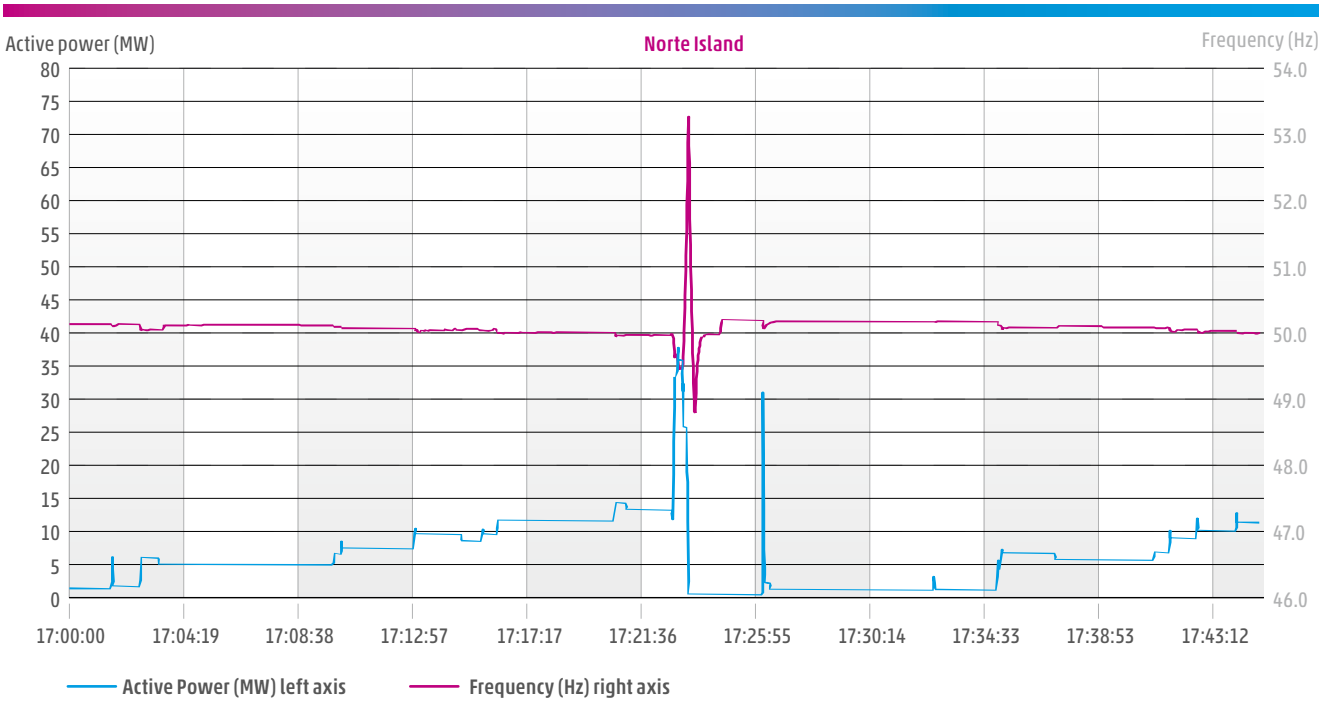


Figure 5-36: Active power and frequency in the Norte island between 17:01 and 17:44

» At 18:32, the Oporto region had voltage available in a relevant set of injection SSs in the distribution network, allowing E-Redes (the Portuguese DSO) to continue with the progressive load supply in this geographic area.

» At 20:22, successful synchronisation of the Norte island with the Spanish/continental European region was achieved. The synchronisation was accomplished without problems using a synchro-check device. Figure 5-37 shows the electrical parameters in the Norte island between 17:26 and 20:26, including the frequency difference between the Norte island and the Spain/European zone at synchronisation (20:22).

» The Expert Panel recommends that production and storage facilities involved in bottom-up restoration scenarios improve the accuracy of the measurements for key parameters, such as active power, reactive power, voltage and speed/frequency, to facilitate ex-post analysis.

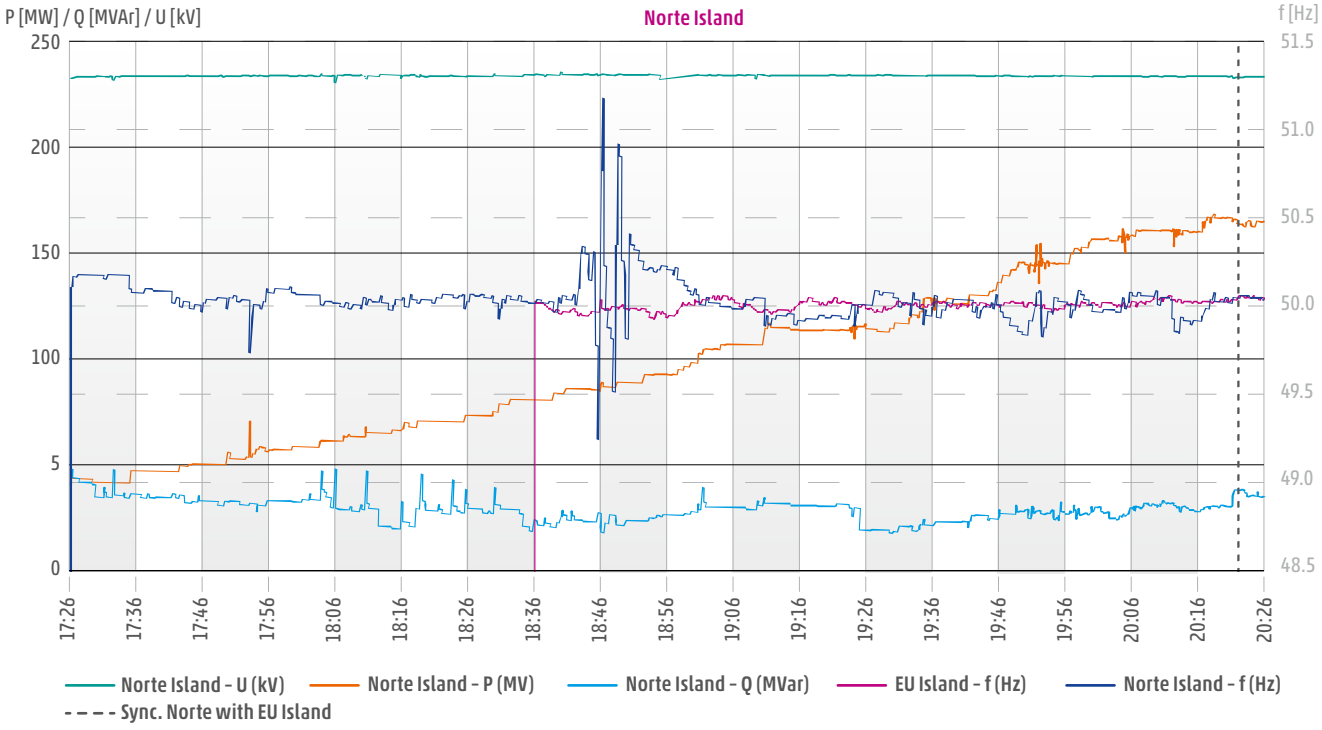


Figure 5-37: Electrical parameters in the Norte island between 17:26 and 20:26 (CEST)

5.3.3 Bottom-up restoration of the Centro island in Portugal

» At 12:35 (CEST) on 28/4/2025, REN requested the black-start mode start-up of HPP 1-Centro using unit 3.

» At 12:45, HPP 1-Centro was operating in black-start mode and propagating voltage to the 220 kV busbar of Zêzere SS.

» At 12:49, when connecting the 220/60 kV Transformer 4 of Zêzere SS with a rated power of 170 MVA, unit 3 of HPP 1-Centro tripped. This was due to the activation of the maximum voltage protection relay of the 400 V AC busbar (auxiliary system).

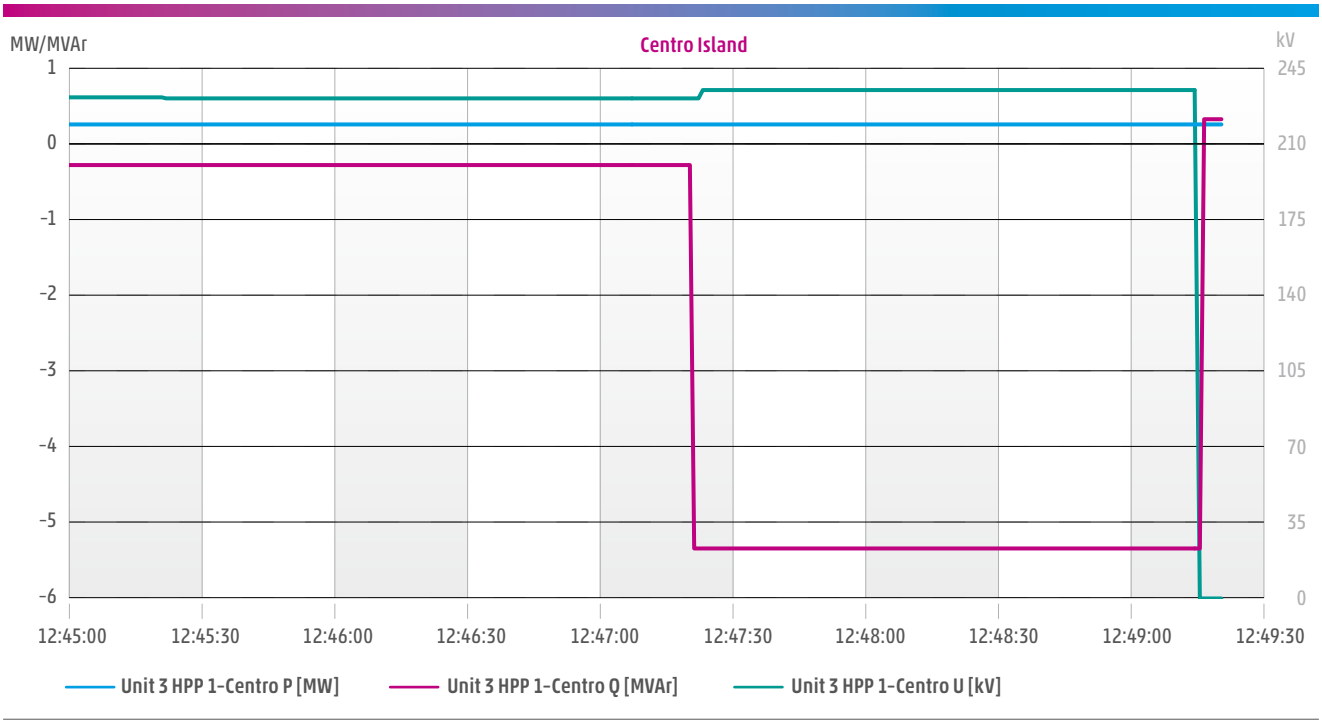


Figure 5-38: Active power, reactive power and voltage in the Centro island between 12:45 and 12:49

» HHP-Centro is composed of 3 main HPP generators and two small HPP generators that serve to energise the auxiliaries of the large HPPs. Following the activation of the maximum voltage protection relay of the 400 V auxiliary system, the small HPP generators could not stop in an appropriate way.

» The producer initiated several unsuccessful attempts to start the auxiliary services of HPP 1-Centro. The automation control function that triggered the urgent stop event in both the small HPP generators could not be reset until a diesel unit was connected to the low voltage busbar.

» After this, the low voltage busbar of the power plant was energised and the black start procedure of the main units was started.

» After the blackout, the producer installed additional over/under frequency protections and undervoltage protections on the small HPP generators, with the main objective of being able to stop them safely ready for a subsequent restart.

» At 15:40, unit 2 of HPP 1-Centro started and propagated voltage to the 220 kV busbar of Zêzere SS.

» At 15:51, the first load in the Zêzere SS was connected.

» At 15:55, unit 2 of HPP 1-Centro tripped with 5 MW, as shown in Figure 5-39. The trip was caused by a low oil level in the regulation accumulator, as the pumps were unable to restore the oil level within the required time, due to the period without energy in the power plant after the blackout.

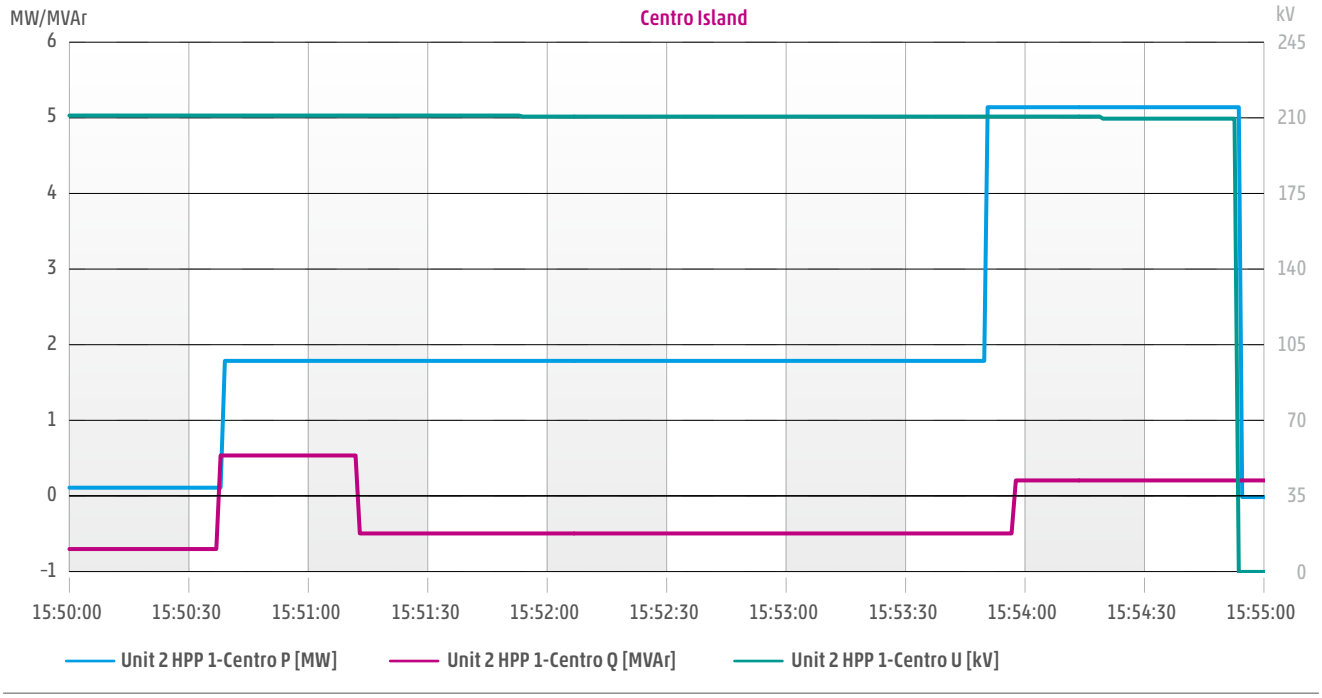


Figure 5-39: Active power, reactive power and voltage in the Centro island between 15:50 and 15:55

» At 16:13, unit 1 of HPP 1-Centro started, and propagated voltage to the 220 kV busbar of Zêzere SS, establishing the first restoration area.

» At 16:26, there was a progressive load restoration in Zêzere SS (60 kV) and propagation of voltage to two other hydropower plants, HPP 2-Centro and HPP 3-Centro, at 150 kV.

» At 20:01, the electric island powered by HPP 1-Centro was synchronised with the Spanish system. The synchronisation was accomplished without problems using a synchro-check device, which assesses the conditions for synchronisation: difference in voltage magnitude, difference in frequencies, and difference in phase angles. Figure 5-40 shows the electrical parameters in the Centro island between 16:11 and 20:11, including the frequency difference between the Centro island and the Spain/European zone at synchronisation (20:01).

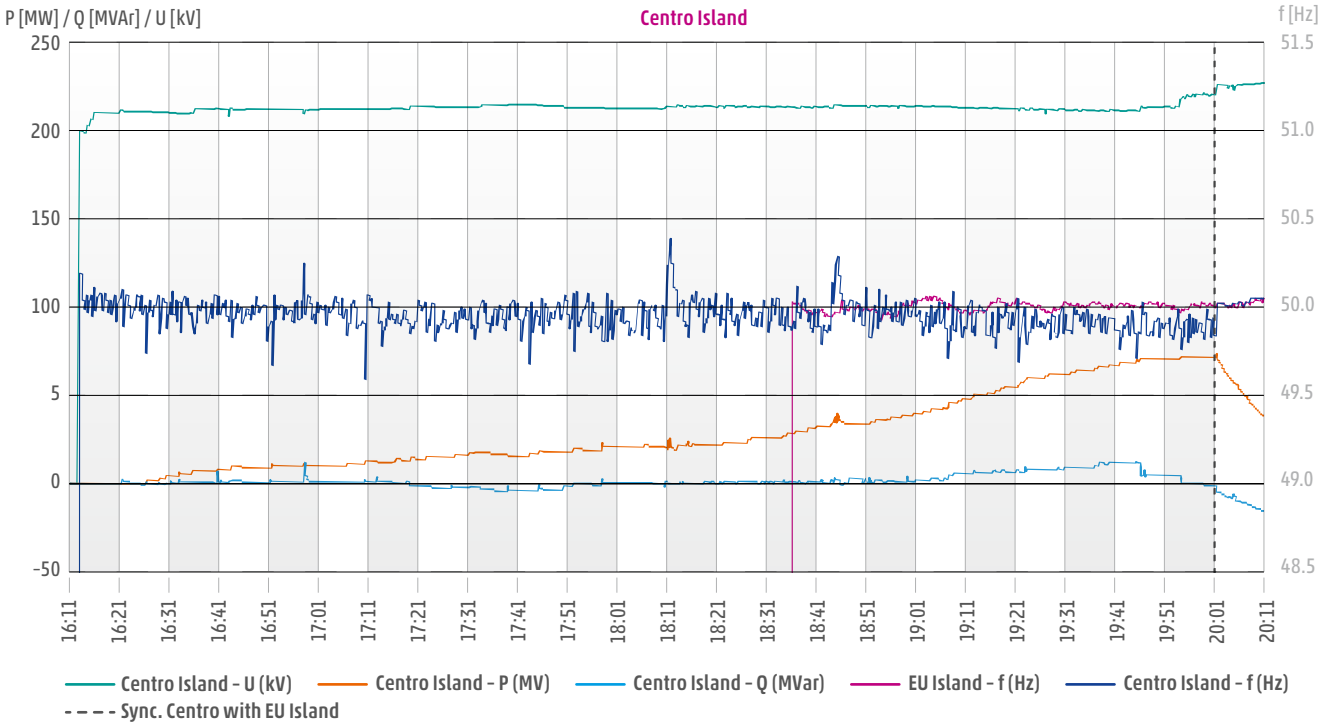


Figure 5-40: Electrical parameters in the Centro island between 16:11 and 20:11

» REN confirmed that the restoration procedure has been updated following the incident to stipulate that the lowest possible voltage setpoint should be applied on the main black-start source before the transformers are energised, so as to minimise inrush currents by keeping the operating point of the transformer within the linear part of the (current – magnetic flux) characteristic curve and avoiding the saturation zone.

» Alternatively, where technically possible, progressive voltage restoration (also known as "soft-start") is recommended to reduce transformer inrush phenomena. REN explained to the Expert Panel that it has increased the number of black-start facilities from two to four in Portugal. The two additional black-start facilities are hydropower plants with soft-start possibilities.

» The Expert Panel recommends that production and storage facilities involved in bottom-up restoration scenarios improve the accuracy of the measurements for key parameters, such as active power, reactive power, voltage and speed/frequency, to facilitate ex-post analysis.

5.4 French local area restoration by RTE

At 12:35 on 28/4/2025, operators from the Nantes control room checked that every shunt reactor available was in operation and asked every generation unit available to absorb maximum reactive power.

- At 12:38, three 400 kV overhead lines were opened to control voltage.
- At 12:45, two more 400 kV overhead lines were opened to control voltage.
- At 15:03, the HVDC unit at Baixas was energised.
- At 15:15, the HVDC unit at Baixas was absorbing enough reactive power to clear the remaining high voltage constraints, marking the end of the voltage-limit violation on the RTE network.

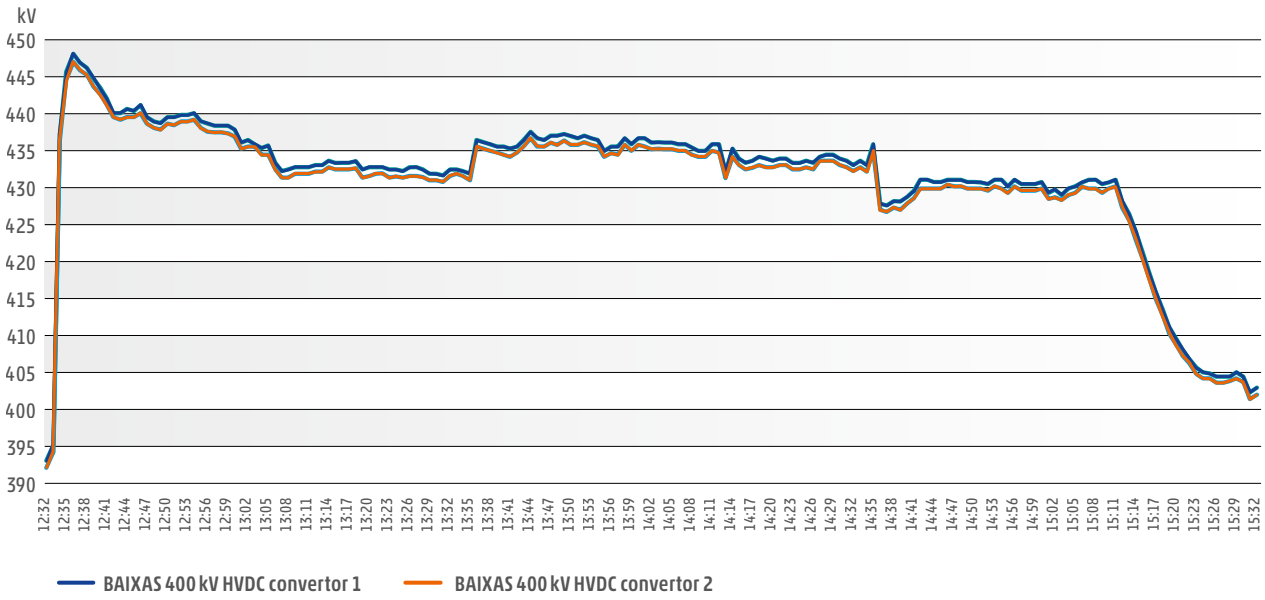
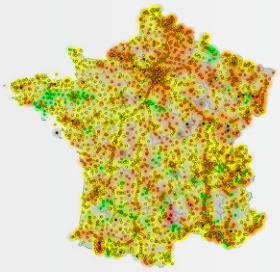


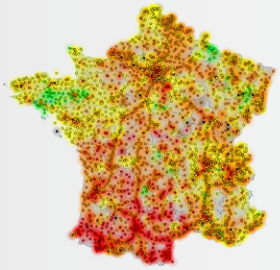
Figure 5-41: Voltages at the Baixas substation on 28 April 2025, at both HVDC 1 and 2 connection points at the 400 kV AC side.

Voltage evolution heat map



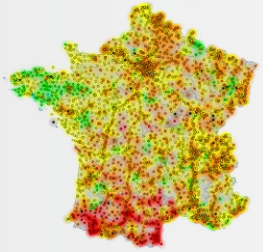
» **12:30**

Situation before the separation of the Iberian Peninsula from the rest of continental Europe.



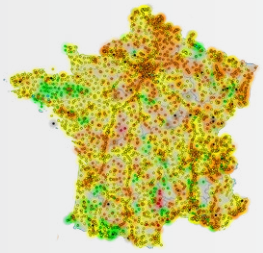
» **12:35**

After the tripping of the Golfech 1 power plant and the HVDC Baixas–Santa Llogaia, high voltages spread through all of southwest France.



» **12:40**

Seven minutes after the separation, and after several mitigation actions, the high-voltage episode was under control, except for close to the Spain–France border.



» **15:20**

After the connection of the HVDC in voltage control, voltage was back within normal range.

Legend

400 kV	225 kV	150 kV	90 kV	63 kV
396 < U ≤ 404	218 < U ≤ 227	147 < U ≤ 153	88 <U ≤ 91	62 < U ≤ 64
404 < U ≤ 412	227 < U ≤ 236	153 < U ≤ 159	91 < U ≤ 94	64 < U ≤ 66
412 < U ≤ 420	236 < U ≤ 245	159 < U ≤ 165	94 < U ≤ 97	66 < U ≤ 68
420 < U	245 < U	165 < U	97 < U	68 < U

Figure 5-42: Heat map of the voltage evolution in France

Power was available on the west part of the border via the 400 kV Argia–Hernani and 225 kV Argia–Arkale tie-lines. In coordination with Red Eléctrica, these tie-lines were brought into service at 12:43 and 12:54, respectively.

On the eastern part of the border, the 400 kV Baixas–Vic tie-line was energised at 13:35.

Restoration process timeline

- At 13:27, a maximum transit of 950 MW on the border was agreed upon with Red Eléctrica.
- At 14:41, a maximum transit of 1,200 MW on the border was allowed.
- At 15:59, HVDC Baixas–Sta. Llogaia link 1 was put into operation, followed by link 2 at 21:58.
- At 17:30, a maximum transit of 1,500 MW on the border was allowed.
- At 18:19, a maximum transit of 2,000 MW on the border was allowed until 23:59.
- At 22:33, an agreement was reached for a cross-border exchange from France to Spain of 1,400 MW, scheduled from 00:00 to 03:00.
- At 07:00 on 29 April, the maximum transit on the border was raised to 1,500 MW for 08:00.
- At 09:37, an agreement was reached to reduce the cross-border exchange from France to Spain to 800 MW from 10:00 to 11:00.
- At 10:05, an agreement was reached to reduce the cross-border exchange from France to Spain to 0 MW as of 11:00. This value was maintained until 23:00.
- At 22:00, an agreement was made to open the Spain-to-France intraday allocation (XBID) at 23:00.

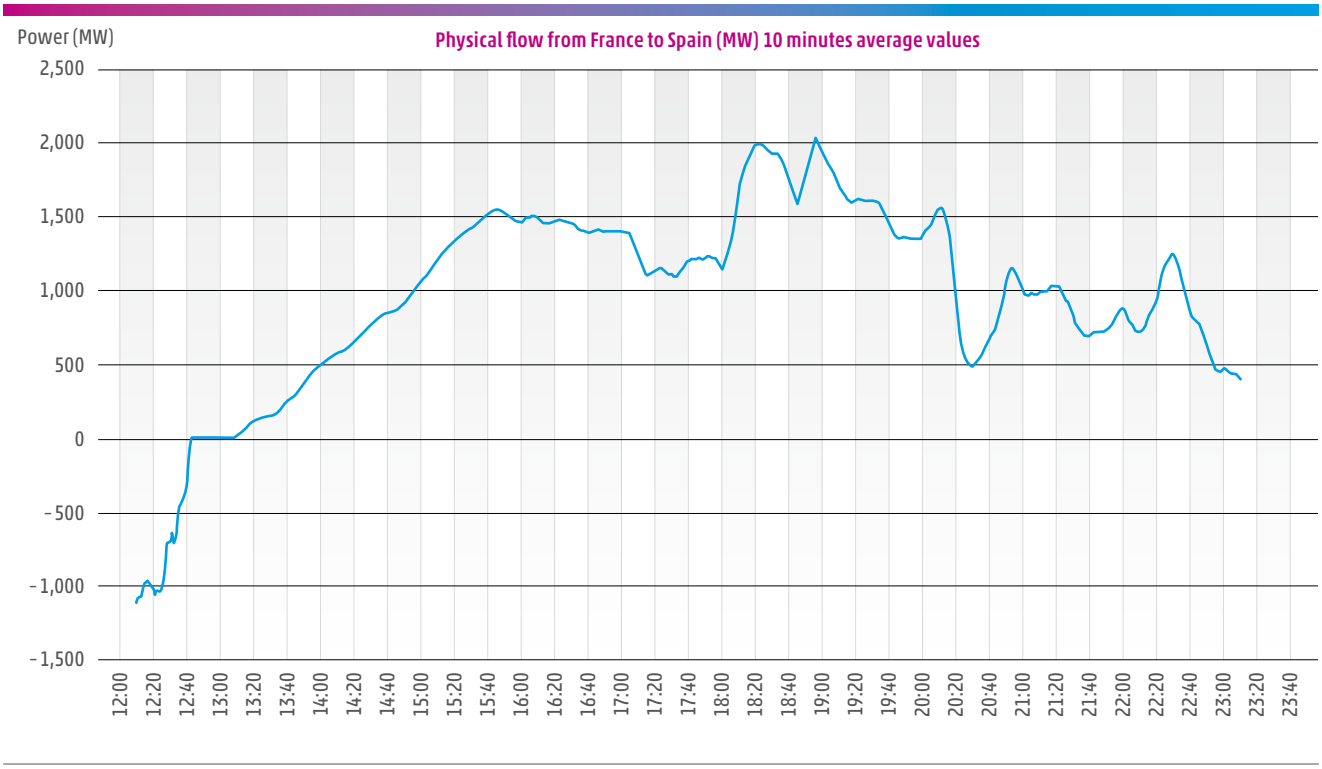


Figure 5-43: Physical flow from France to Spain (MW) in 10-minute average values

5.5 Generation and load recovery

The restoration of electrical grids after a total blackout is a complex and demanding coordinated process. TSOs worked collaboratively to re-establish electricity supply through a mixture of top-down and bottom-up approaches. Generation and load recovery occurred simultaneously, as the grid was being restored. This sub-chapter describes the generation and load recovery once system balance had been achieved.

5.5.1 Spain

The restoration of the Spanish transmission system from the Red Eléctrica side was concluded at 04:00 on 29 April. The restoration continued at the distribution system level, concluding at 07:00 on 29 April.

Evolution of generation and load during and after restoration

The restoration of the interrupted power supply began at 12:43 on 28 April with the support received through the Spain–France interconnection. Additionally, as noted in the previous sections, in the minutes following the incident, islands were created from hydroelectric plants with autonomous start-up across different electrical zones.

At 13:04, support was established through the Spain–Morocco interconnection. Subsequently, starting at 15:14, combined-cycle plants were progressively connected to contribute to the restoration of the power supply.

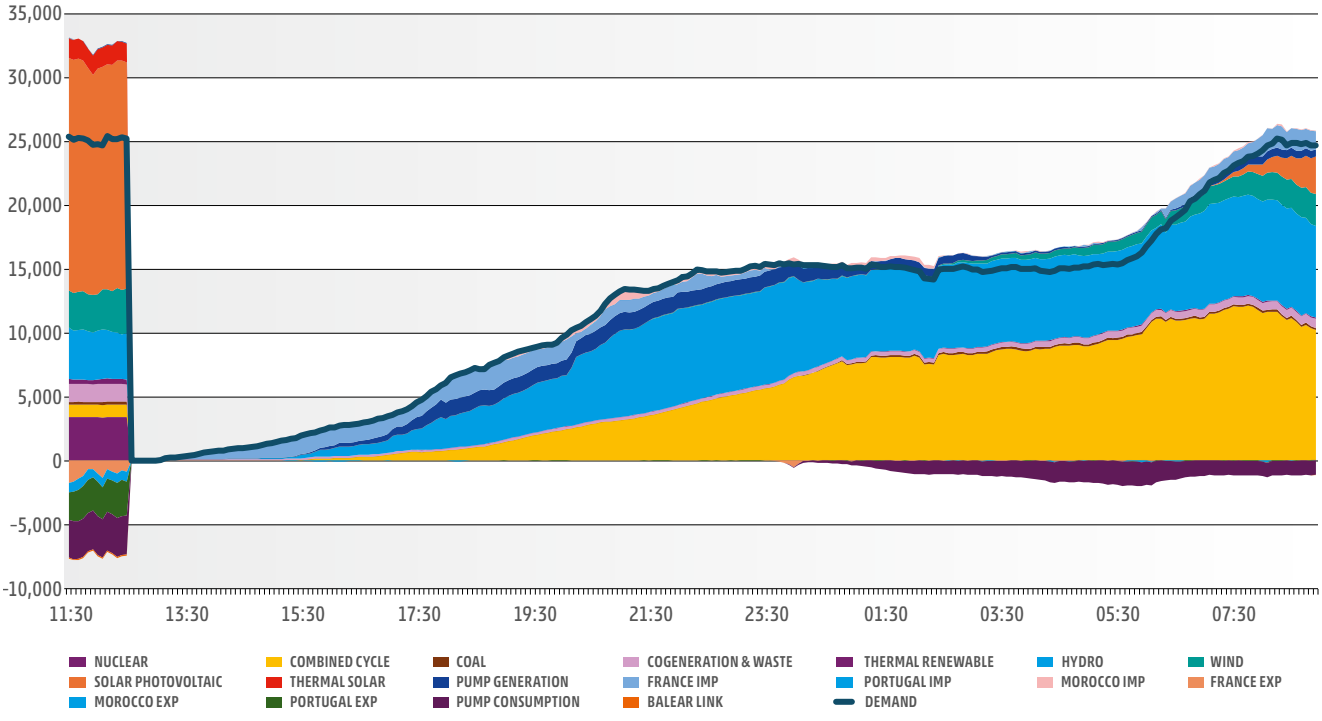


Figure 5-44: Evolution of the generation and load in the Spanish system during the restoration phase - Red Eléctrica

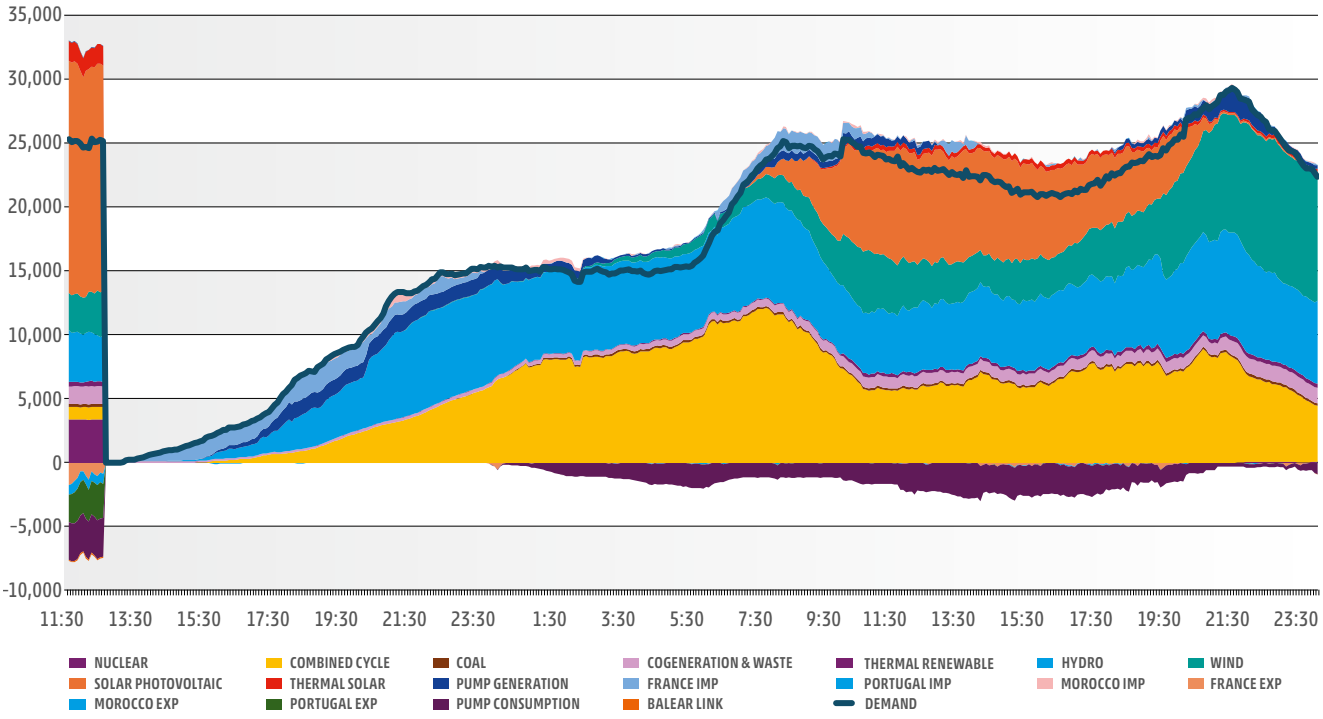


Figure 5-45: Evolution of the generation and load in the Spanish system through to the end of 29 April - Red Eléctrica

Evolution of renewable generation during restoration

Supervision and control of RCW in the Spanish electrical system is carried out by CECRE, which was commissioned in June 2006.

Under the Spanish regulation established in Royal Decree 413/2014, all single RCW facilities, or clusters sharing the same connection point with a total installed power greater than 1 MW, are required to send real-time telemetry of the active power they produce to CECRE. Additionally, all single RCW facilities, or clusters with a total installed capacity greater than 5 MW, receive active power setpoints from CECRE, with which they must comply.

CECRE, therefore, receives real-time telemetry from RCW facilities, and can issue real-time limits for maximum production in the RCW facilities connected to both the transmission and distribution network.

This real-time information is collected from the plants by the Renewable Energy Control Centres (RESCCs) and channelled to CECRE via the ICCP links connecting these control centres.

To minimise the number of points of contact with CECRE, the RESCCs serve as the sole real-time interface with the Spanish TSO. In other words, RESCCs act as intermediaries between CECRE and the RCW facilities. Each RCW facility is free to choose its preferred RESCC, regardless of the technology it uses or its connection point to the network.

Generation control is managed by CECRE, which sets individual limits for each facility through its corresponding RESCC.

Therefore, on the one hand, RESCCs collect real-time information from RCW facilities and send this to CECRE (supervision), while on the other hand, they collect the setpoints established by CECRE and send these to the facilities (control).

After the incident, at 13:38, CECRE sent a setpoint of 0 MW to all RCW facilities.

Once it was determined that RCW generation could be safely incorporated into the Spanish electrical system without affecting the restoration process, the setpoints were released as follows (it is important to note that the setpoints refer to the installed power of RCW generation):

Time	Setpoint for maximum production for RCW generation (MW)	% of total installed power
01:38	1,000	1.4
01:55	2,000	2.9
02:50	3,000	4.3
03:37	4,000	5.7
04:06	5,000	7.1
04:33	6,000	8.6
05:02	7,000	10
05:29	8,000	11.4
05:59	10,000	12.8
06:08	12,000	17.1
06:13	15,000	21.4
06:16	19,000	27.1
06:18	24,000	34.2
06:28	29,000	41.4
06:36	34,000	48.6
06:43	41,000	58.5
06:51	49,000	70.0
06:58	57,000	81.3
07:04	limitations released	100

Description of the evolution of balancing compensations from neighbouring TSOs

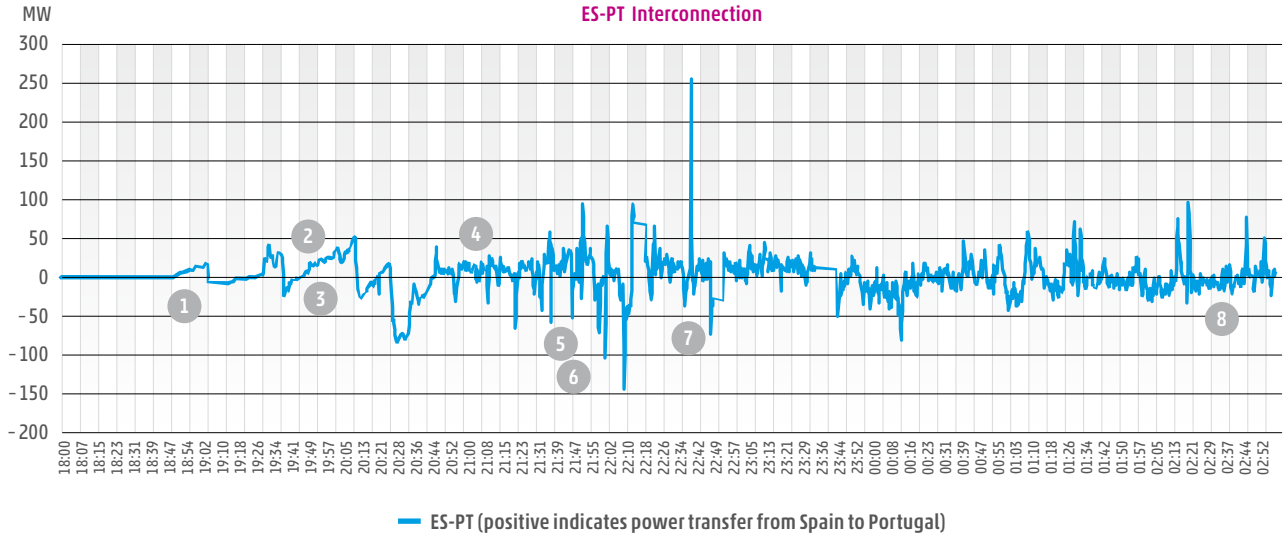


Figure 5-46: Active power exchange on the Spain–Portugal interconnection

The numbered sequences below relate to the numbers indicated in figure 5-46:

» 1. At 18:36, the L-220 kV Aldeadávila–Pocinho 1 interconnection was switched on, sending voltage to the Portuguese electrical system for the first time during the restoration process. From this point onwards, REN received the continental European frequency.

» 2. At 19:57, the L-220 kV Saucelle–Pocinho interconnection was switched on.

» 3. At 19:58, the L-220 kV Aldeadávila–Pocinho 2 interconnection was switched on.

» 4. At 20:47, the L-400 kV Aldeadávila–Lagoaça interconnection was switched on.

» 5. At 21:34, the L-400 kV Puebla de Guzman–Tavira interconnection was switched on.

» 6. At 21:37, the L-400 kV Cartelle–Lindoso 1 interconnection was switched on.

» 7. At 22:33, the L-400 kV Cedillo–Falagueira interconnection was switched on.

» 8. At 02:37 (29 April), the L-400 kV Cartelle–Lindoso 2 interconnection was switched on.

The interconnection between Spain and Portugal through the 400 kV Brovales–Alqueva line was unavailable due to maintenance work at the time of the incident. The return of the line was requested, ending the outage. This interconnection was switched on at 13:39 on 29 April.

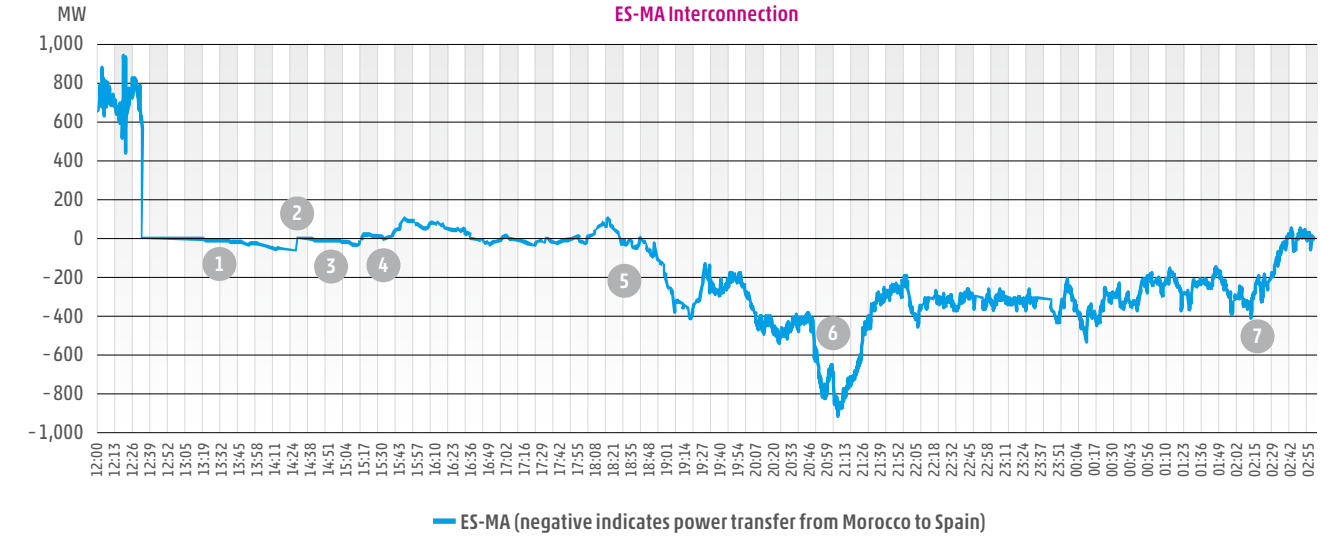


Figure 5-47: Active power exchange on the Spain–Morocco interconnection

The numbered sequences below relate to the numbers indicated in figure 5-47:

» 1. At 13:04 (April 28), support was received from Morocco through the L-400 kV Pto. de la Cruz–Mellousa 2 (ESMA 2) interconnection.

» 2. At 14:27, the interconnection with Morocco was disconnected due to an underfrequency relay trip, causing the disconnection of all the load supplied up to this moment.

» 3. At 14:34, voltage was resumed from Morocco through the L-400 kV Pto. de la Cruz–Mellousa 1 (ESMA 1) interconnection.

» 4. At 15:14, the first TTP was connected in the Southern zone.

» 5. At 18:16, the L-400 kV Bienvenida–Almaraz was switched on, synchronising the Southern zone with the Centro electrical island.

» 6. At 20:59, the L-400 kV Pto. de la Cruz–Mellousa 2 (ESMA 2) interconnection was switched on.

» 7. At 02:18, it was agreed with ONEE to discontinue emergency support from Morocco.

5.5.2 Portugal

Evolution of generation and load during and after restoration

As mentioned in the previous sections, the restoration of the supply began at 15:40 with the successful start-up of the first power plant in black-start mode. No support from the Spanish system was possible at that point.

At 18:36, support was established through the 220 kV Spain–Portugal interconnection to synchronise the two islands with the continental European power system. This significantly increased network stability, enabling accelerated restoration of the Portuguese system. No energy support was needed between the Spanish and Portuguese systems, and the interconnection capacity between Portugal and Spain was set at 0 MW in both the import and export direction.

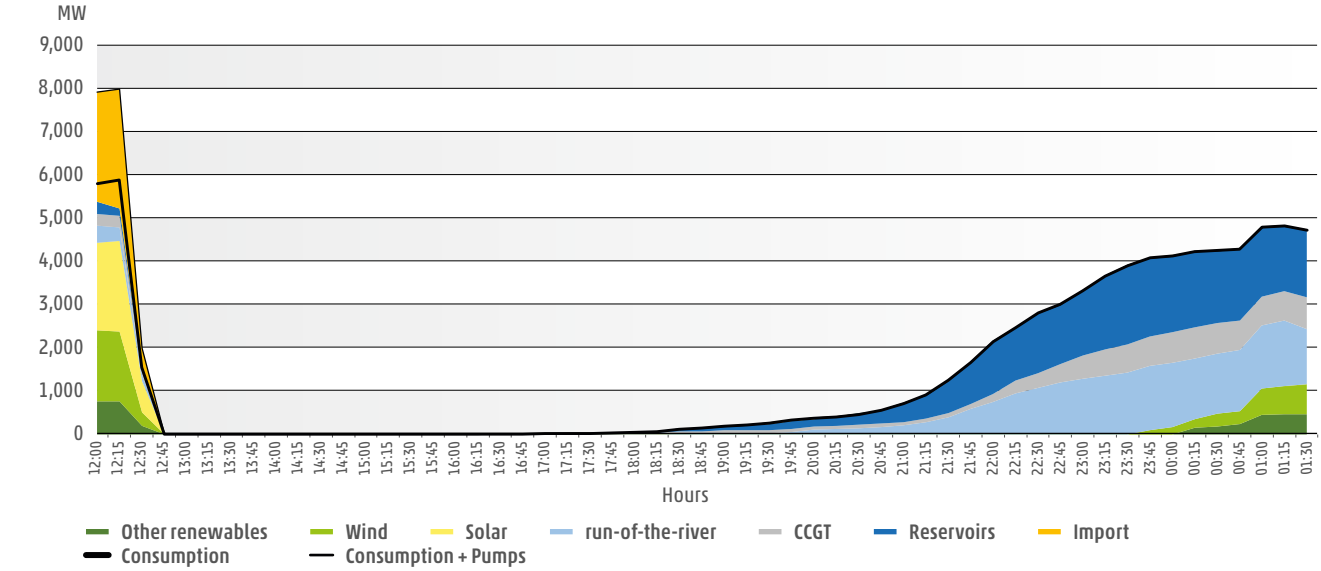


Figure 5-48: Evolution of the generation and load in the Portuguese system during the restoration phase – REN

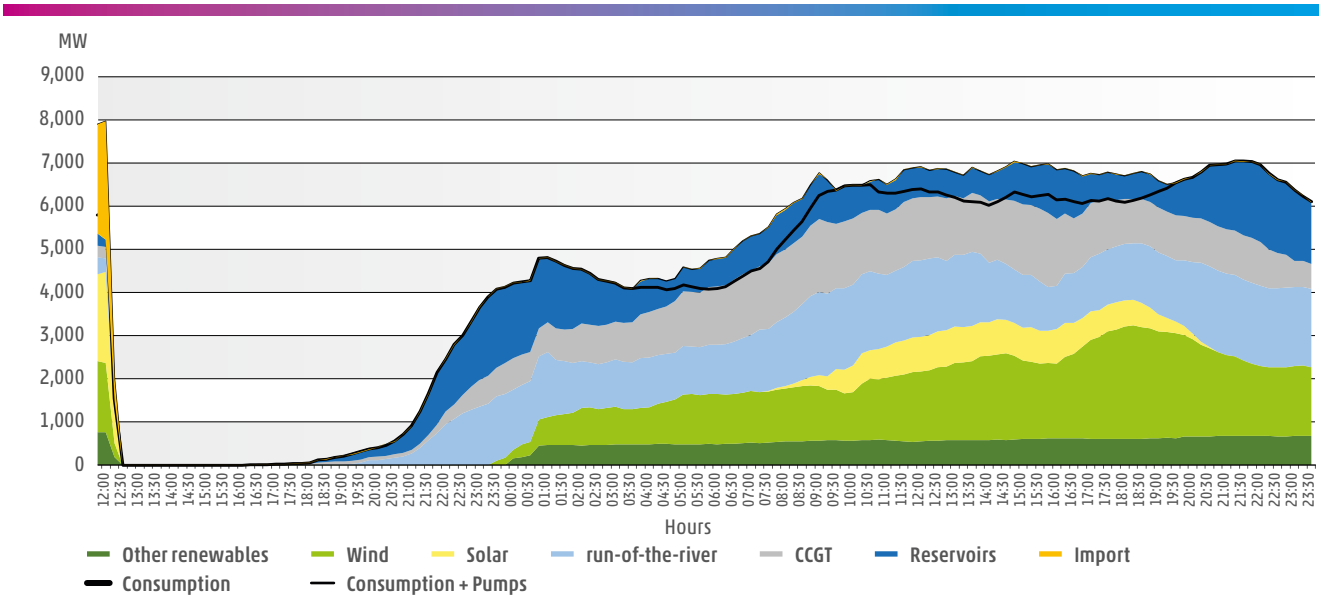


Figure 5-49: Evolution of the generation and load in the Portuguese system through to the end of 29 April – REN

The restoration of the Portuguese transmission system from the REN side was concluded at 00:22 on 29 April. At the distribution system level, the restoration continued, ending at 04:00 on 29 April.

Evolution of renewable generation (wind and solar) during restoration

According to the established Portuguese regulations, all single renewable power plants, or sets of power plants sharing the same connection point with a total installed power greater than 1 MW, should be observable and controllable (via setpoints) from REN's National Control Centre. However, no setpoint was sent to limit renewable production during the restoration phase.

According to the National Service Restoration Plan, the connection points of the renewable power plants (solar and wind generation) connected to the transmission system were not energised during the restoration of the transmission system. After the restoration, they were energised by REN throughout the day on 29 April, and instructions for production permissions were provided by phone.

For the wind generation connected to the transmission system (2,774 MW of installed capacity), production permissions were given from 16:00, while solar generation (1,767 MW of installed capacity) received permission from 21:00 on 29 April.

The renewable power plants connected to the distribution system began to be energised shortly before the end of the transmission system restoration, in coordination between REN and E-REDES. For the wind and solar generation connected to the distribution system (2,634 MW and 2,620 MW of installed capacity, respectively), production permissions were given from 23:00 on 28 April.

5.5.3 France

The D-1 commercial nomination on the Spain–France border for business day 28 August was scheduled from France to Spain (up to 1,688 MW) between 13:00 and 15:00, and then from Spain to France between 16:00 and 21:00 (1,800 MW from 16:00 to 18:00, then 3,000 MW).

As RTE supplied Red Eléctrica in the afternoon and evening of 28 April and in the morning of 29 April, RTE activated bids on the French balancing mechanism (up to 4,500 MW; total activated energy for 28 and 29 April: 51.9 GWh) and sought support from other TSOs (up to 500 MW, including 3,000 MWh from Swissgrid on 28 April and 8,850 MWh from Amprion on 28 and 29 April).

5.6 Steps after system restoration

This section describes the steps taken to ensure system balance after the network restoration.

5.6.1 Spain

The day-ahead market session for 29 April was conducted, but its results were not incorporated into system operations. The remaining electric market activities for 29 April in the Spanish system were suspended. They were re-established for 30 April. As the system restoration progressed, orders were given to synchronise all available hydraulic and thermal generators in the system. As they were synchronised, instructions were given to each of these generators by phone to adjust their generation to demand.

The master aFRR controller was connected at 00:06 on 29 April. The synchronised hydraulic and thermal generators with secondary regulation capacity were included in this system, thereby providing a significant volume of secondary regulation to manage variations in demand and generation.

From 01:38 to 07:00, the limitations on RCW generation were progressively removed to allow their inclusion in the system, as described in section 4.3.

Throughout 29 April, the synchronised hydraulic and thermal generators were instructed by phone to adjust their generation to demand and maintain adequate levels of secondary reserve.

From 00:00 to 24:00 on 29 April, export capacities with France, Portugal, and Morocco were reduced to 0 MW to enable all system demand to be supplied with the available generation.

From 06:30 to 11:00, due to the insufficient availability of synchronised generation to supply the demand, a support import programme was requested from France (800 MW from 06:30 to 08:00 and from 10:00 to 10:30, 1,500 MW from 08:00 to 10:00, and 400 MW from 10:30 to 11:00).

In the last hours of 29 April, hydraulic and thermal generators were instructed to adjust their generation to their market programmes, starting from 00:00 on 30 April.

5.6.2 Portugal

The day-ahead market session for 29 April was conducted, but the results were not incorporated into system operations. The remaining electric market activities for 29 April in the Portuguese system were suspended. They were re-established for 30 April.

As the system restoration progressed, HPPs and TPPs were given instructions by phone (non-market-based central dispatch). As they were synchronised, each of these generators received instructions by phone to adjust its generation to demand.

During the afternoon of 29 April, the production limitations on renewable generation (wind and solar) connected to the transmission grid were progressively removed to allow their inclusion in the system, as described in section 4.3.

Throughout 29 April, instructions were issued by phone to the synchronised hydro and thermal generators to adjust their generation to demand and maintain adequate secondary reserves. The AGC was put into service at 20:25 on 28 April, following an interconnection schedule programme at 0 MW, as agreed with Red Eléctrica.

5.7 Market restoration

This section describes the restoration of market activities in Spain and Portugal after the restoration and stabilisation of the systems.

5.7.1 Spain

On the morning of 29 April, the Iberian NEMO was informed by phone of the forecast for the completion of the system restoration and the resumption of market activities, starting with the clearing of the daily market for energy delivery on April 30.

At 11:39 on 29 April, in coordination with the Iberian NEMO, the Spanish TSO published a statement in the System Operator Information System (ESIOS) to all actors indicating that the restoration state would last until 23:59 that same day:

> *Following the incident in the peninsular electricity system that occurred at 12:33 p.m. yesterday, April 28, and until 11:59 p.m. today, April 29, market activities are suspended in accordance with Operating Procedure P.O.3.9. Therefore, no energy or capacity schedules or allocations made through market mechanisms for the periods within this time frame will be considered valid.*
>
> *The energy settlement will be carried out in accordance with sections 5 and 6 of the aforementioned procedure.*
>
> *The System Operator has already informed the NEMO that market activities will resume starting with the day-ahead market for energy delivery April 30.*

This information was also published on the privileged information platform of the System Operator.

> *Due to the incident in the Spanish Peninsular electrical system at 12:33 PM on April 28th, market activities for delivery April 29th are suspended, following both Article 35 of Commission Regulation (EU) 2017/2196 and Spanish Operational Procedure 3.9. Market activities for delivery April 30th have been restored.*

The following milestones in particular are highlighted:

- Participation in the single day-ahead coupling took place on 29 April at 12:00 for energy delivery on 30 April. For 30 April, the interconnection capacity between Spain and France, as well as between Spain and Portugal, was limited.
- The intraday market for 30 April was enabled at 23:00 on 29 April.
- Connection to the European platform for mFRR (MARI) at 00:15 on 30 April.
- Connection to the European platform for RR (TERRE) at 00:26 on 30 April.
- Connection to the imbalance netting platform (IGCC) at 11:55 on 30 April.

5.7.2 Portugal

On 29 April at 14:43 (CEST), REN validated the results of the daily market for 30 April.

On 29 April at 16:23, REN published a statement and an urgent market message indicating that the restoration and stabilisation of the system would last until 23:59 that same day:

Following the blackout situation that occurred on 28 April 2025, in the Portuguese electrical system and the subsequent, ongoing process of restoration and stabilisation that is taking place on 29 April, it is reported that:

- *The interconnection capacity between Portugal and Spain will be set on 30 April (CEST – Central European Summer Time), at 0 MW in the import and export direction.*
- *Transactions will resume in the day-ahead market, intraday market sessions, and the continuous intraday market for the delivery periods relating to 30 April (CEST)*
- *Contracts will resume, inclusive from 30 April (CEST), in the various markets and processes managed by REN – Rede Eléctrica Nacional, S.A., namely the processes for resolving technical restrictions, the replacement reserve (RR) market, the frequency restoration reserve market with manual activation (mFRR) and the secondary regulation band market*

Additionally, on 29 April at 18:26, a notification was sent to all market players indicating that the restoration state would last until 23:59 that same day.

REN reconnected to the balancing platforms as follows:

- IGCC: The reconnection occurred on 30 April at 19:08.
- TERRE: REN began submitting its requirements for RR for the delivery periods between 03:00 and 04:00 on 30 April.
- MARI: The reconnection occurred on 30 April for the quarter-hour from 19:30 to 19:45.

5.8 Behaviour of distributed energy resources and load during restoration

The Expert Panel asked the DSOs if they had observed any disturbing behaviour in DERs or any unexpected load behaviour (EVs, BESS, heat pumps, etc.) after re-energisation.

The DSOs replied that generation units < 1 MW connected to the DSO network currently have no obligation to provide real-time information to the DSO. DSOs do not, therefore, have visibility over DERs, their status, or their electrical parameters, meaning that it would have been difficult to detect any unwanted effects if they did occur.

One DSO explained that its choice of preferring urban areas at the start of the restoration was because this allowed the energised network to maintain a stable load with minimum impact from DERs.

Another DSO explained that the MV feeders that were switched on were selected before the blackout for having few DERs, ensuring a more stable load.

Prior to the initiation of the restoration process, all MV feeder switches were opened by the DSOs to ensure the complete disconnection of all loads and generators. Subsequently, a controlled reconnection sequence of loads and generation was carried out in coordination with the Spanish TSO.

5.9 Voice communication and remote-control problems during restoration

The Expert Panel asked the DSOs if they had experienced any issues with voice communication systems during the first 24 hours after the blackout.

A first contact between Red Eléctrica and one particular DSO control centre in Spain could not be established until 17:48, when it was made through the control centre of another DSO. Until that moment, there was no information about the condition of that particular DSO network following the incident.

Around 16:00, the South Control Centre of a DSO in Portugal experienced a failure in its primary communication system with the TSO Control System. As a contingency, satellite phones were used. By 18:00, the DSO had successfully implemented VOIP over the internal communications network, resolving the need for satellite phone use at the South Control Centre. The public earth-based and cell telecommunication networks failed. The DSO in Portugal resorted to private internal communication systems and satellite phones. The North Control Centre of that DSO can act as back-up for the South Control Centre to receive and execute instructions from the TSO.

A DSO in Galicia reported a clear degradation of its voice communication service during the first six hours after the start of the blackout, culminating in a complete service drop at 18:25. This was due to problems at the public telecom operator complicating the usual call routes. The DSO used redundant point-to-point lines and alternative communication centres to maintain communication with the TSO. The public operator's service was fully restored at around 01:15 on 29 April. Coordination and operational continuity were maintained throughout, with critical service continuity being prioritised.

The TSO confirmed major difficulties in contacting this DSO due to this drop in the voice communication system, with a significant impact on the restoration processes in Galicia. From 17:17 to 20:45, the TSO was not able to contact this DSO through the usual call routes, except on one occasion at 18:17.

The TSO explained that from 21:10 to 22:04 and from 22:27 to 00:02 the following day, it was unable to contact the DSO control centre in Sevilla due to communication issues at this DSO, although the TSO had no problems communicating by phone with other parties at that time. Several attempts were made to reach different phone numbers at their control centre, but none of these were successful. During these two periods, the restoration process for the demand in Andalucia was paused. At 00:02 on 29 April, communication was finally established using cloud-based communication software.

From 18:30 until 04:00 on 29 April, the TSO and a DSO control centre in Barcelona were only able to establish telephone communications occasionally due to communication issues at the DSO, although the TSO was able to communicate by phone with other parties without any problems at that time. This slowed down the restoration process in Cataluña, Galicia and León.

Another Spanish DSO reported that in all its DSO control centres, a priority phone line was available for communications with the TSO control centre, as requested in the applicable regulations. Nevertheless, between 18:00 and 19:00, voice communication issues were recorded with the TSO control centre, and linked to the latter's communication system. The TSO, however, rejects this hypothesis, as it was able to communicate by phone with other parties without any problems at that time, It reported that telephone calls made between the TSO and this DSO went through an automatic warning message that significantly delayed communications, since the call did not reach the DSO operator until the message had finished. In some cases, there were delays of more than 2 minutes before the load restoration instructions could be transmitted.

The TSO reported that at 18:17, a DSO informed them that some time earlier, they had lost remote control of their facilities in the Galicia and Madrid zone. This caused a significant, hours-long delay in the restoration of the Galicia and Madrid zone.

The TSO reported that one DSO informed them that from the start of the restoration process, they had neither remote control nor visualisation of many of their substations in the Andalucia distribution network. To supply demand from those substations as they received voltage from the transmission network, they had needed to dispatch personnel to the affected substations to recover remote control and telemetry signals. This situation delayed the restoration process. The DSO explained that some of these communication outages arose due to dependencies on the public telecommunications operators. Consequently, operational resources were deployed to the facilities in anticipation of a potential local intervention, and mobile power generators were placed on some communication repeaters. The public telecommunications operator's network lost coverage due to the failure of the autonomous power supply systems at its communication repeaters and switching centres. This loss of coverage varied depending on the location of the communication infrastructure and the timing of re-energisation.

Multiple DSOs explained that in areas where communication via public operators was lost, satellite terminals were used, for example, to enable communication between control centres and field personnel. The Spanish TSO reported that some generation companies had reported problems in communicating with their combined cycle power plants. This caused delays in the process of energising the auxiliary services of those plants, and consequently in their synchronisation.

5.10 Conclusion

The Expert Panel would like to recognise the dedication, commitment and good cooperation of the operators and employees of TSOs, DSOs, producers, renewable energy control centres, and other stakeholders. Their expertise and experience were instrumental in the successful restoration of the Spanish and Portuguese power systems following a blackout of unprecedented scope and magnitude.

The information gathered in the aftermath of the restoration has enabled the Expert Panel to identify key challenges and develop recommendations for further improving future restoration processes in the event of any similar incidents.

Restoration efforts proceeded most smoothly in areas that could be re-energised "top-down" from stable neighbouring transmission systems, which provided essential voltage support and system stability.

Regions farther away from healthy systems required parallel energisation from dedicated power plants – primarily hydropower facilities, but including one gas-fired plant serving Northern Portugal – equipped with black-start capabilities and following the established restoration procedures.

In Spain, renewable energy, cogen and waste facilities or clusters with an installed capacity greater than 5 MW were instructed to produce zero output during the first twelve hours of system restoration, to prevent uncontrolled energy injection. Once the system had stabilised, these production restrictions were gradually lifted, allowing these resources to be reintegrated.

The investigation found that some local restoration scenarios encountered challenges that led to delays in load recovery. These issues highlight the complexity of rebuilding an "islanded" section of the grid from a single source without broader system support. Nevertheless, certain areas energised through local black-start sources were able to successfully synchronise with the larger system.

Technical difficulties identified include issues with:

- power and frequency control of multiple power plants in the same island, as encountered in Extremadura 1 (section 5.2.2), Galicia (5.2.6), Cantabria (5.2.9), Extremadura 2 (5.2.10), and Norte island Portugal (5.3.2)
- synchronisation of the islands, as happened between Duero 1 and Duero 2 (section 5.2.3)
- machine protection settings, as encountered in the Centro island in Portugal (section 5.3.2) and the East Asturias island in Spain (5.2.5)
- inappropriate voltage control in certain generators leading to limits in their reactive absorption capability, as encountered in Galicia (section 5.2.6) and Levante (5.2.7)
- start-up and HV breaker closing with black-start plants, as encountered in Levante (section 5.2.7), Andalucía (5.2.13), Toledo (5.2.14) and both Portuguese black-start islands (5.3.2 and 5.3.3)
- impact from distributed energy resources, as encountered in Extremadura 1 (section 5.2.2)

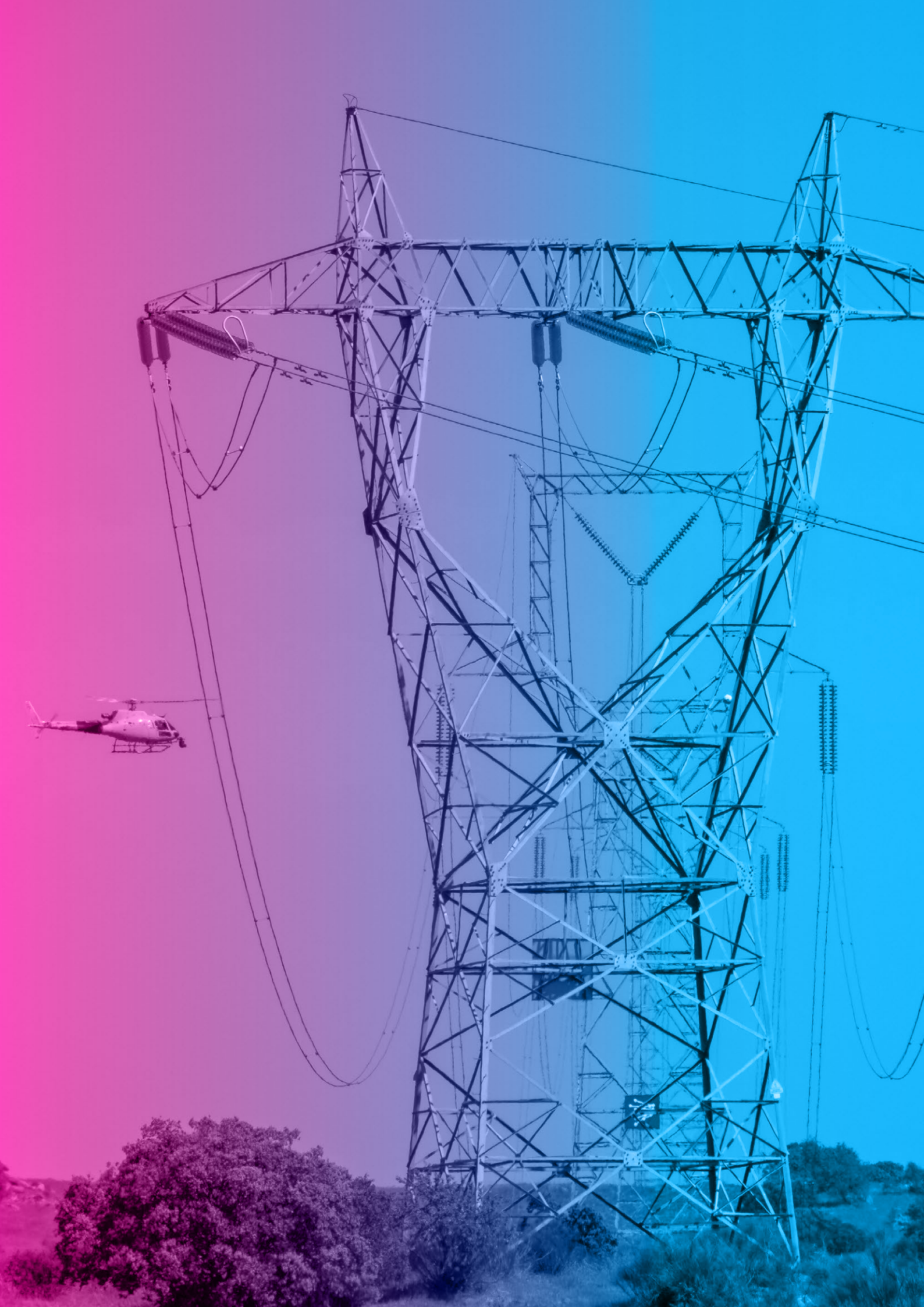
The incident underscored the critical need for reliable and redundant (voice) communication among key restoration actors, especially when public telecommunications are unavailable. In regions lacking such backup communication (Galicia, Sevilla, Barcelona and Madrid), load recovery was significantly delayed, as explained in section 5.9.

The Spanish system was fully restored within 16 hours, while the Portuguese system was back online within 12 hours after the blackout. Looking at restoration experiences following historic blackouts of similar magnitudes in other developed power systems, the restoration times in Spain and Portugal are within comparable limits, allowing for the specific complexities of grid restoration, and the limited interconnection of the Iberian Peninsula with Continental Europe, which reduced the possibilities for a top-down restoration strategy.

The restoration procedures and fallback strategies, combined with the full commitment of well-trained operators, enabled a rapid and effective restoration. Nevertheless, this experience highlights the ongoing importance of regular black start testing and joint operator training, frameworks for managing distributed energy resources during restoration, robust independent voice communication and remote control systems for DSOs and SGUs, and the value of high-resolution real-time measurements across both the transmission and distribution networks for the effective analysis of any future incidents.

Key recommendations to further improve future restoration processes in the event of similar incidents include:

- Regular live black start testing involving all relevant parties.
- Regular joint training sessions on restoration scenarios for all key stakeholders (TSOs, DSOs, producers).
- Ensuring that renewable energy sources in the distribution grids do not disturb the delicate restoration process.
- Ensuring that blackout-proof voice communication between key actors and critical remote control functions is available when public telecommunication networks are down.
- Making available high-resolution measurement data from SGUs in the transmission and distribution grids, to allow better after-incident analysis.



6 OPERATIONAL PLANNING

6.1 Introduction

This chapter addresses operational planning activities relevant to the incident under analysis, covering both regional-level processes performed by the Regional Coordination Centres (RCCs) and national operational planning procedures implemented by Red Eléctrica de España (RE) in its role as the Spanish transmission system operator (TSO). The chapter provides a comprehensive overview of the operational planning activities coordinated at the regional level and national levels.

With regard to regional coordination, the chapter presents the investigation and analysis performed by the RCCs in accordance with their responsibilities for Regional Incident Analysis and Reporting (RIAR). RIAR is a legally mandated task pursuant to Article 37(i) of Regulation (EU) 2019/943, requiring RCCs to carry out post-operation and post-disturbance analysis and reporting activities. The RCC Post-Operation and Post-Disturbances Analysis and Reporting Methodology was approved by ACER through decision No. 04/2022 and entered into force on 1 April 2022.

The RIAR task interfaces with the incident investigation process conducted by the ICS Expert Panel. In line with the applicable methodology, an RCC Investigation Subgroup is established within the ICS Expert Panel when the RCC investigation threshold is met. As confirmed in Chapter 8 of the Factual Report, this threshold has been met for the incident under consideration. Consequently, an RCC Investigation Subgroup was established and conducted a detailed investigation in accordance with Article 37 and Annex I of Regulation (EU) 2019/943. The conclusions of the RCC investigation are incorporated into this chapter of the ICS Final Report. This chapter also describes the national operational planning regulation in force in Spain on 28 April 2025, applied by RE and Spanish electricity undertakings. The explanations of the underlying procedures were provided by RE and are included to allow for the assessment of the implications of the RE operational planning environment on the incident. Accordingly, the following sections detail the RCC investigation, as well as the relevant operational planning procedures implemented by RE.

6.1.1 High-Level Description of RCC Tasks

In the given geographical area, RCC tasks are related to the coordinated operational planning phase and focus on steady state rather than dynamic stability analyses, which fall within the remit of the TSOs pursuant to Article 38 of the Commission Regulation 2017/1485.

Article 37.1 of the EU Regulation 2019/943 prescribes the following RCC tasks:

EU REGULATION 2019/943 ARTICLE 37.1	TASK
a	Coordinated capacity calculation (CCC)
b	Coordinated security analysis (CSA)
c	Common grid model (CGM)
d	Consistency assessment of defence and restoration plans
e	Short-term adequacy (STA)
f	Outage planning coordination (OPC)
g	Training and certification of staff working for RCCs (TCI)
h	Supporting the coordination and optimisation of regional restoration as requested by TSOs
i	Carrying out post-operation and post-disturbances analyses and reporting
j	Regional sizing of reserve capacity
k	Facilitating the regional procurement of balancing capacity
l	Supporting TSOs, at their request, in optimising inter-TSO settlements
m	Performing tasks related to the identification of regional electricity crisis scenarios if and to the extent they are delegated to the RCCs
n	Performing tasks related to seasonal adequacy assessments if and to the extent they are delegated to the RCCs
o	Calculating the maximum entry capacity available for participation of foreign capacity in capacity mechanisms
p	Performing tasks to support TSOs in identifying needs for new transmission capacity, upgrades to existing transmission capacity, or alternative solutions

Table 6-1: List of mandated RCC tasks.

Of these tasks, those carried out for the South West Europe (SWE) region are listed in 6.1.2, and those identified by the RCCs as relevant for this incident are outlined in 6.2.

6.1.2 Regional Coordination SWE

The SWE System Operation Region (SOR), established under Regulation (EU) 2019/943, comprises Portugal, Spain, and France. The RCC appointed for the region is Coreso.

The Portuguese and Spanish power systems are characterised by a high penetration of renewable energy sources (RES), particularly wind and solar generation. As a result, regional power flows are highly dependent on wind and solar generation patterns and forecasts. Owing to the variability of these sources, intraday (ID) system conditions may differ from those anticipated in the day-ahead (D-1) market time frame, notably during periods of high wind and/or solar output combined with low demand, when renewable reduction through market mechanisms may be required.

Under such conditions, the Portuguese TSO (REN) may identify tight downward regulation margins. This information is provided to the RCC for situational awareness. The situation is not a comparable limitation in the Spanish system, where a large share of renewable generation participates in balancing services. Relevant forecasts and system conditions are communicated to the RCC within the SWE day-ahead security analysis (SA DA) process and during regular coordination calls between TSOs.

In Spain, additional operational constraints arise from the geographical separation between major generation areas and load centres. During low-load conditions, long 400 kV transmission lines may exhibit capacitive behaviour, leading to high voltage levels. To manage these conditions, TSOs identify and implement additional operational measures (topological remedial actions (RAs)), such as the temporary disconnection of transmission lines. Consequently, D-1 grid models often include a higher number of lines out of service compared to the models stemming from the week-ahead (W-1) outage planning coordination (OPC) process. This difference is consistent with the respective objectives of the planning processes: W-1 coordination focuses on identifying potential incompatibilities among long-term planned outages, whereas D-1 and ID security analyses assess system security under expected operational conditions and determine the need for additional operational measures.

From an interconnection capacity calculation perspective, Coreso assesses the Portugal–Spain and Spain–France borders independently, as interactions between them remain limited due to the Spanish transmission network located between the two borders. Accordingly, capacity calculation in the SWE region is based on the net transfer capacity (NTC) methodology rather than a flow-based approach.

6.1.3 Geographical Scope of RCC SA in SWE

In line with the System Operation Guidelines (SO GL), RCC activities dealing with a representation of the grid focus on a defined subset of assets of the three TSOs' grids, defined on the basis of their cross-border impact, pursuant to the relevant regional methodologies. These processes, together with their applicable methodologies, are:

- Coordinated capacity calculation (CCC), in line with regional methodology pursuant to Article 21 of Capacity Allocation and Congestion Management (CACM), for the critical network elements (CNEs) and contingencies determined by applying the regional methodology and in line with Article 22 of CACM
- Coordinated security analysis (CSA), described in Articles 75 and 76 of the System Operation Guideline (SO GL) and its corresponding pan-European (PE) methodology for coordinating operational SA (CSAm) and regional methodology for regional operational security coordination (ROSC), which applies to the assets determined pursuant to Article 14 of CSAm
- OPC, described in Article 80 of the SO GL, which applies to the assets pursuant to the methodology for assessing the relevance of assets for outage coordination (RAOCm), implementing Article 84 of the SO GL

The RCC security assessment and coordination process applies to only a subset of elements of the national TSO network, assessed as cross-border relevant in line with the methodologies, and does not cover the entire control areas of the TSOs. In line with the SO GL, the network elements relevant for CSA and OPC tasks are defined as follows:

- Relevant for the CSA, cross-border relevant network elements (XNEs), defined as those above the influence threshold determined in the SWE region in line with the ranges specified in Annex I of CSAm
- For outage coordination, the relevant assets are defined as those above the influence threshold determined in the SWE region pursuant to RAOCm

The ROSC process that implements SWE regional mandated methodology pursuant to Article 26 of the SO GL is still under implementation (expected in Q3 2027). Currently, an interim regional security assessment is performed by Coreso as a coordinated approach agreed with REN, RE, and RTE. However, the current security assessment and outage coordination processes are supported by a legacy, non-automated optimisation tool that selects RAs in case a constraint or an outage incompatibility is detected. Coreso and the TSOs (RE, REN, and RTE) have agreed to limit the coordinated selection of RAs in the current SA and in OPC processes to cases where the simulated contingencies violate the limits of a subset of elements of cross-border relevance. This subset is referred to as the "Main area", reflecting the need for close coordination between the affected TSOs as per their agreement. Constraints related to less relevant XNEs are addressed by the concerned TSOs at the national level, with the potential to initiate bilateral coordination where and when necessary.

The Main area within Coreso's geographical scope is therefore the area in which, for every constraint and outage planning incompatibility (OPI) associated with an element of cross-border relevance, Coreso proposes RAs foreseen in the capacity calculation methodologies agreed by the involved TSO(s) and ensures coordination between all involved parties.

The Extended area, by contrast, includes network elements for which any event or modification (e.g. preventive topology modifications, trippings) may have consequences for the Main area.

Table 6-2 illustrates the number of substations included in the Main and Extended areas. Grid elements (namely lines and transformers) that connect to substations that belong to either area are considered part of that area as well.

The prescribed methodologies for capacity calculation, OPC, and CSA establish a formalised calculation of the elements that must be monitored in the three coordinated processes. Once ROSC is implemented, the Main and Extended areas are replaced by the formally recognised "Monitoring area".

	Main area		Extended area		
	400 kV Substations	**220 kV Lines**	**400 kV Substations**	**220 kV Substations**	**220 kV Lines**
RTE	8	7	11	16	0
RE	45 (*)	23	(All other 400 kV substations)(*)	0	2
REN	28	17	0	0	0
Impact[36]	Report constraints and find RAs when loading > 100 %		Report constraints when loading > 100 %. If loading is above 120 % for RTE or 115 % for RE, simulate the tripping, and if the cascading effect has consequences on the Main area, identify RAs		

(*) The legally established monitored area for the capacity calculation, CSA, and OPC processes includes 54 400 kV substations, 45 of which are considered in the Main area and the others in the Extended area.

Table 6-2: Number of assets included in the Main and Extended areas in the scope of SA and regional OPC.

36 Impact with regards to considering these elements in OPC and SA task

6.2 RCC Processes and Investigation

For this investigation, this report will focus on the tasks that are relevant for the specific hour of the event, which occurred on 28 April 2025 at around 12:33. These tasks are listed in Table 3 and further detailed in Sections 6.2.1–6.2.6. In all processes, no issues were forecasted in the concerned SWE region. Figure 1 presents a scheme of the time horizons associated with the relevant RCC tasks listed.

Task	RCC	Area	Time frame	Result	References to the chapters in this document
OPC	SeleneCC (SCC)	Europe	W-1 25.04.2025 13:00	OK	6.2.1
	Coreso	SWE (Regional)	W-1 25.04.2025 11:30		
STA	Nordic RCC (SCC)	Europe	W-1 (27.04.2025 09:50)	OK	6.2.2
CGM	Coreso, SCC	Europe, CGMES format	D-2, D-1, N/A	Not OK, No impact	6.2.1.4
	TSCNET, Baltic RCC	Europe, CGMES format	ID 28.04.2025 07:51 (Baltic) 11:12 (TSCNET)	OK	6.2.1.4
	Coreso	SWE, CGMES format	D-2, D-1 (26.04.2025 18:30, 27.04.2025 18:00)	OK	6.2.3.1
	Coreso	Continental Europe, UCTE DEF	D-1, ID (27.04.2025 18:30, 28.04.2025 03:30)	OK	6.2.3.2
CCC	Coreso	SWE (Regional)	D-2, D-1 (27.04.2025 08:00, 27.04.2025 22:00)	OK	6.2.4
CSA	Coreso	SWE (Regional)	D-1 (27.04.2025 23:00)	OK	6.2.5
E&R	Coreso	SWE region	Every 5 years, completed in October 2024	OK	6.2.6.1
Crisis Scenarios	Coreso	SWE region	Every 4 years, completed in September 2024	OK	6.2.6.3
RIAR	All RCCs	Europe	On demand	N/A	6.2.6.2

Table 6-3: Overview of the status of RCC tasks (according to the rotational calendar).

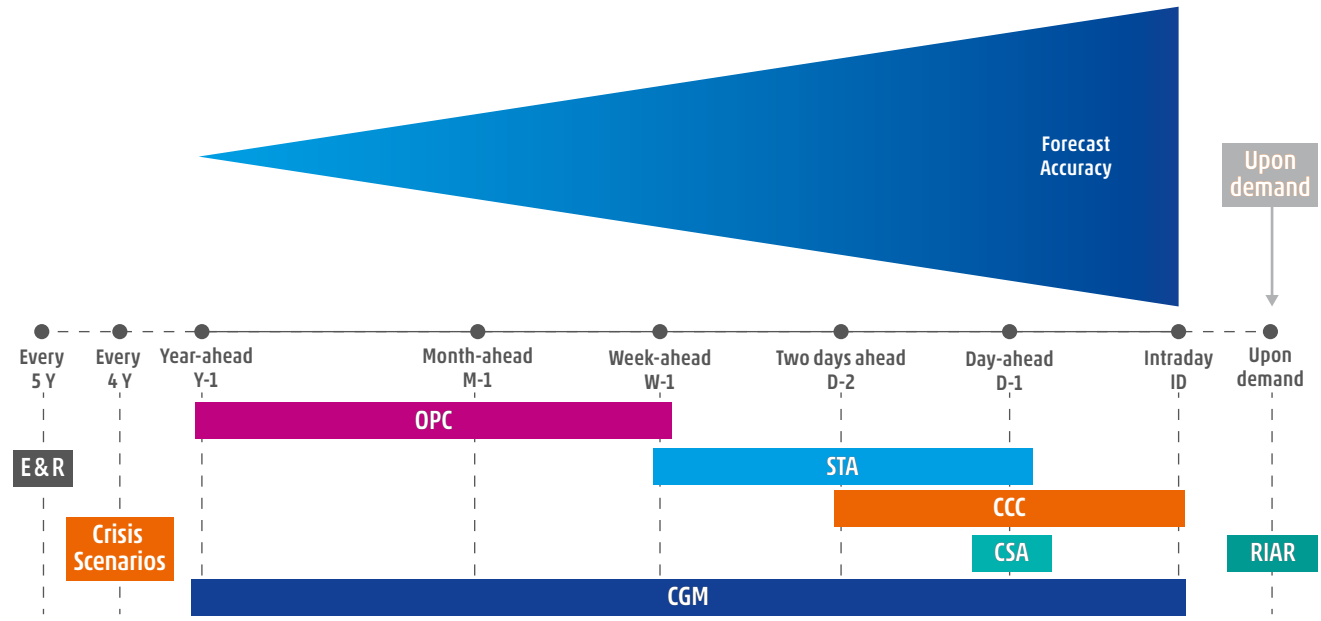


Figure 6-1: Relevant RCC tasks and their time horizons.

As shown above, RCCs perform two key PE tasks on a weekly basis following a rotational principle: OPC and short-term adequacy (STA). Coreso additionally performs the regional OPC service, which incorporates an SA, ensuring grid security through load-flow computations and SA.

6.2.1 OPC

The OPC process is designed to support the planning of grid element outages while ensuring the operational security of the interconnected European power system. This is achieved through procedures carried out at both the PE and regional levels.

Building on the PE OPC process, RCCs also conduct regional OPC processes, commonly referred to as regional OPI assessments. These aim to evaluate whether the outage plans of European TSOs are feasible from a grid security perspective. If potential congestions are identified, the process involves proposing RAs, verifying the feasibility of the coordinated unavailability plan (UAP) with respect to security limits, and issuing recommendations to address any detected incompatibilities in outage planning.

6.2.1.1 PE OPC Process

The PE OPC tool supports the coordination of outages by enabling the exchange of UAPs and maintaining a database of relevant assets. Relevant assets are grid elements that TSOs deem critical for regional coordination. A standardised procedure ensures data quality and consistency through the validation of planned statuses for cross-border lines managed by TSOs. All RCCs carry out the PE OPC process on a rotational basis. During the week of this incident, the RCC acting as the merge operator was SEleNe CC.

The W-1 OPC cycle covers Saturday to Friday, with TSOs required to submit preliminary disconnection plans by Wednesday at 12:00. After a series of validations and merges (first at 12:00, second at 16:00), RCCs perform security assessments. TSOs can adjust plans and request further analysis until Thursday at 16:00 (third merge). Final plans are reviewed in the weekly operational teleconference (WOPT) on Fridays at 10:25, for the CORE, Italy North (IN), and SWE regions, moderated by TSCNET. Final changes must be submitted by Friday at 13:00 (fourth merge), which defines the official plan. Outages submitted after that are deemed unplanned. RCCs use CGMs to evaluate outage impacts and propose RAs or cancellations if operational security limits are at risk.

The W-1 merges for Business Week 18 (BW18), which includes Business Day (BD) 28/04/2025, were performed successfully as follows:

- 1st Merge: 23 April 2025 12:00
- 2nd Merge: 23 April 2025 16:00
- 3rd Merge: 24 April 2025 16:00
- 4th Merge: 25 April 2025 13:00

The latest UAP updates were submitted by RE, REN, and RTE on 25 April 2025 after the fourth merge was performed. The remaining tie-line inconsistencies (TLIs) were not related to the event, as outage inconsistencies were due to end after 28 April.

The PE OPC tool currently considers only transmission assets in UAPs and not generation and demand assets. TSOs do not own these assets and therefore cannot submit outage details for them. In addition, further refining of the PE OPC tool would be required for them to be processed.

6.2.1.2 Regional OPI Assessment for SWE

As explained in 6.2.1.1, the PE OPC process merges individual UAPs four times. Once the second merge UAP is available, the regional OPI assessment starts at Coreso for all relevant regions.

The main steps performed by the OPC operator at Coreso include:

- Building the single OPC scenario to be analysed in detail for all regions. This is based on two main factors: the level of exchanges from reference cases provided by TSCNET and the volume of simultaneous relevant planned outages from TSOs. The operator selects one time stamp (TS) that reflects the most critical scenario – typically with the highest exchanges and most outages.
- Identifying any outage planning security constraints (OPSC) or OPIs at the regional level by running an SA using a contingency list supplied by TSOs (the same list used in the SA DA SWE). It should be noted that the SA considers static limits of power flows and voltages, rather than dynamic (transient) stability security limits. This approach is appropriate in the SWE case because:
  - Steady-state limits are reached before dynamic security limits in a statistically overwhelming majority of dispatching situations.
  - The subsequent SA process performed by TSOs, using an up-to-date scenario, captures potential dynamic stability risk in the transient regime associated with contingencies.
- Proposing solutions to TSOs when constraints are found, prioritising non-costly RAs (e.g. PST tap changes or outage rescheduling). Costly measures like generation redispatch or commercial exchange adjustments are considered only when necessary. The key distinction is that OPIs may involve costly measures and could even result in the cancellation of planned outages by the respective TSO.
- Sharing results during the WOPT and coordinating with other RCCs/TSOs when issues affect multiple regions, ensuring transparency and third-party awareness.
- Publishing Regional OPSC/OPI reports for each region. These reports summarise the analysed outages, cross-border physical flows, contingency outcomes, and proposed solutions. They are also shared internally to support upcoming capacity calculation and SA processes, especially for tracking postponed outages or agreed RAs.

The W-1 process is repeated weekly and remains flexible to accommodate last-minute changes, including urgent outage notifications requiring ad hoc analysis.

Focus on BD 28/04 TS 12:30

The W-1 Regional OPI assessment for Week 18 was carried out on Thursday and Friday, 24 – 25 April 2025. Results were distributed to all participating TSOs immediately after completion.

For Week 18, the RCC identified Monday BD 28/04 at TS 12:30 as the single (most critical, see the first bullet point in the list above) scenario for the CORE, IN, and SWE regions. This was based on the level of exchanges (considering those of BD 21/04, TS 18:30) and the amount of relevant planned outages across all three regions. All important parameters were within limits. The exchanges in the SWE region used during the regional OPI assessment process are displayed in Table 6-4, including details of the actual physical cross-border flows in real time for comparison purposes. The location of relevant outages is illustrated in Figure 6-2 below.

	OPC: commercial schedules (MW)	Real-Time physical flows (MW)
FR → ES	2,271	–960
ES → PT	1,243	2,453

Table 6- 4: Commercial exchanges in the SWE region in the grid model used during the OPC regional process. Real-time physical cross-border flows are displayed for comparison purposes. (Source: RTSN, BD 28/04 at TS 12:30)

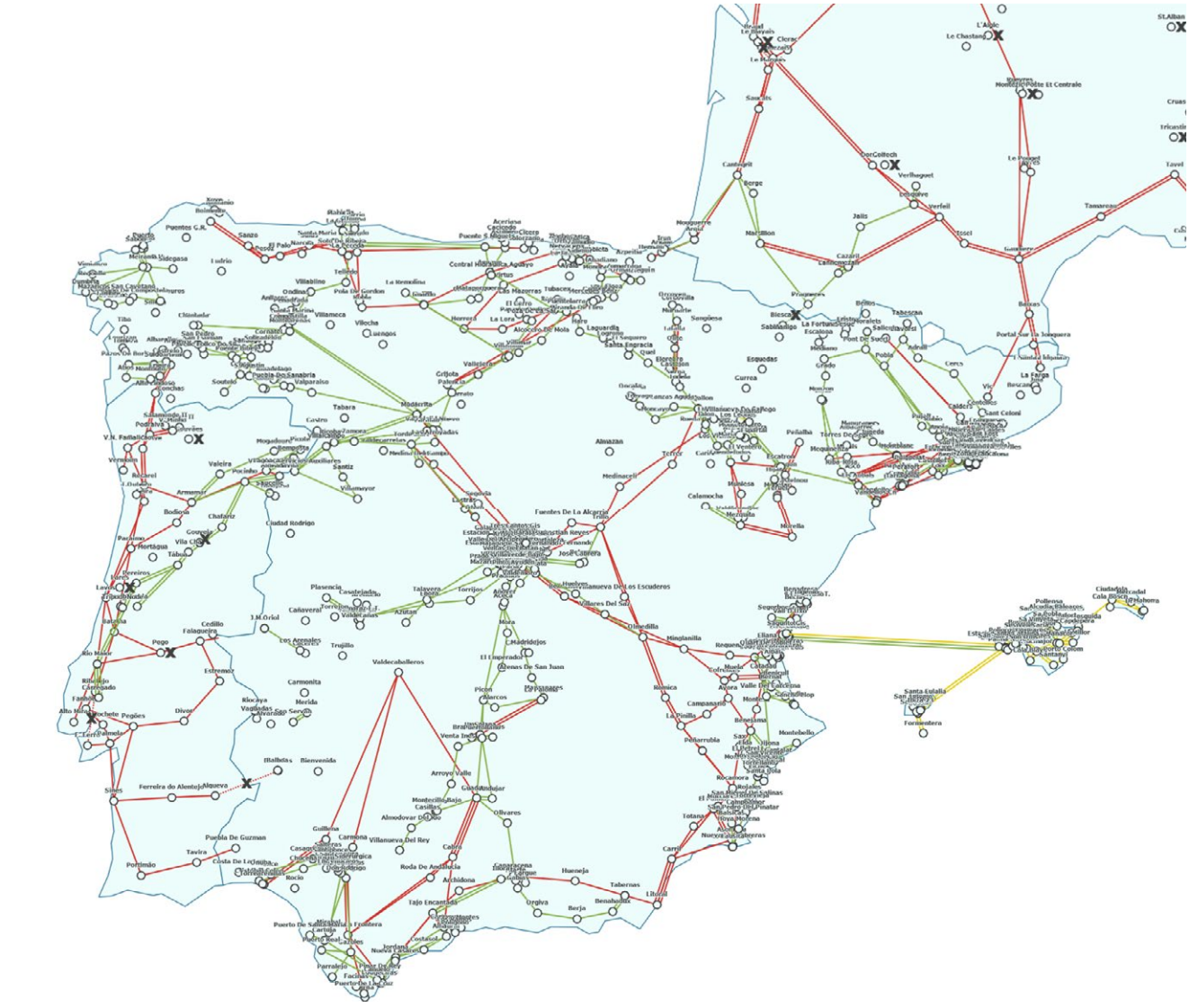


Figure 6-2: Visualisation of planned outages in the SWE region for BD 28/04. The lines are coloured based on their voltage level (red for 400 kV, green for 220 kV, yellow for voltages under 220 kV) and planned outages are signalled with a cross. (Source: PE OPC Tool)

The W-1 Regional OPI SA revealed no major violations arising from the planned outages requiring costly RAs, so the OPC report for the SWE region recorded no OPI.

6.2.1.3 Conclusion

The PE and regional OPC processes for Week 18 were executed in accordance with the defined procedures and timelines. All required W-1 merges were completed, relevant UAPs were exchanged, and the PE and regional security assessments were carried out as prescribed. The identification of BD 28/04 at TS 12:30 as the single scenario for detailed regional OPI assessment was based on the level of exchanges and the concentration of relevant planned outages across the CORE, IN, and SWE regions.

The regional OPI assessment for the SWE region did not identify any OPI requiring costly RAs or outage cancellations. One OPSC was detected and addressed through a non-costly RA (PST tap change), reducing the overload to within acceptable limits. Consequently, based on the planned transmission asset unavailabilities and the commercial exchanges used in the OPC grid model, the system was assessed as secure for the analysed scenario. Nevertheless, the underlying SAs assess power flows under steady-state conditions and do not cover dynamic or voltage-related operational security limits.

As the OPC was performed the week before, some structural limitations must be noted, such as the differences between the commercial exchanges considered and the actual real-time cross-border flows. In addition, the scope of the OPC process does not include generation or demand asset unavailabilities.

Accordingly, the OPC process, as designed and applied, was not intended to identify or prevent events arising from real-time operational conditions outside the scope of planned transmission outages. Within its defined remit, the OPC task was performed correctly and in line with existing procedures.

6.2.2 Short-Term Adequacy

The purpose of the STA service is to investigate whether the reliable available expected production capacity can meet the expected consumption using both upward and downward regulation at the regional level, pursuant to Article 81 of the SO GL. The calculation takes place daily, and the time horizon of the process is D-7 to D-1. The STA process is complementary to the adequacy analysis locally performed by TSOs pursuant to Article 104-107 of the SO GL. Although this incident was not related to any adequacy issues, a summary of the STA data and results is available in the following sections.

All SWE TSOs (RTE, RE, and REN) and RCCs participate in the PE (also known as cross-regional) STA process.

For the daily process, all TSOs are required to submit input data for the calculations. The process is then executed automatically and monitored by the main or backup RCC.

The regional STA process can also be triggered if the results of the deterministic calculation indicate inadequacy for a time horizon of D-3 or shorter. This situation rarely occurs. The Nordic RCC was the main adequacy assessment agent (AAA) for the PE process in the week when the incident occurred. No regional process was triggered on the days preceding the incident, and the focus of the analysis will be the process of the previous day. The process did not detect any adequacy issues for the day of the incident.

6.2.2.1 Input Data

All required input data (NTC files, W-1 generation, and W-1 load files) relevant for all affected and neighbouring TSOs were received before the STA calculations were performed on 27 April 2025 (the day before the incident). These files successfully passed validation and were used in the calculation.

The first file uploaded by RE for the W-1 generation was rejected, as it lacked information on installed generation for biomass, oil, pumped storage, water reservoir, other renewables, and waste. RE automatically sends two sets of input files for demand and generation forecasts.

The files are sent at 3 am and 8 am to cover the possibility of failure of the first set of files. During the day on 27 April, the second set of generation input files was automatically correctly uploaded.

To assess the accuracy of some of the input data provided by the affected TSOs, a comparison was made between the STA input data (forecast values) and the realised values (obtained from the ENTSO-E Transparency Platform (TP)) for solar and wind generation.

The comparison results for the affected regions are presented in Figure 6-3. The yellow area represents the forecasted probability range, where:

- The lower bound corresponds to the P05 value (5 % probability that the real value will be below this threshold)
- The higher bound corresponds to the P95 value (95 % probability that the actual value will be below this threshold)
- The centre line corresponds to the P50 (median forecast; 50 % probability of being above or below).

The forecasted values are used as input for the PE STA tool and form the basis for STA calculations. The observed/realised values (blue line) are based on data retrieved from the ENTSO-E TP and represent average power values for each corresponding hour, already including following markets and RAs. Due to Portugal's data being reported on an hourly basis, the final data point shown at 12:30 reflects the average for 12:00 – 13:00, which already includes the time frame during which the blackout occurred.

In general, the forecasted values for solar generation display narrow confidence ranges, indicating high certainty in the predictions that are not verified. In contrast, wind generation forecasts for Spain and France show broader confidence intervals, suggesting greater uncertainty. Across all TSOs and both forecast types, there is a discrepancy between observed/realised values and forecasted ranges.

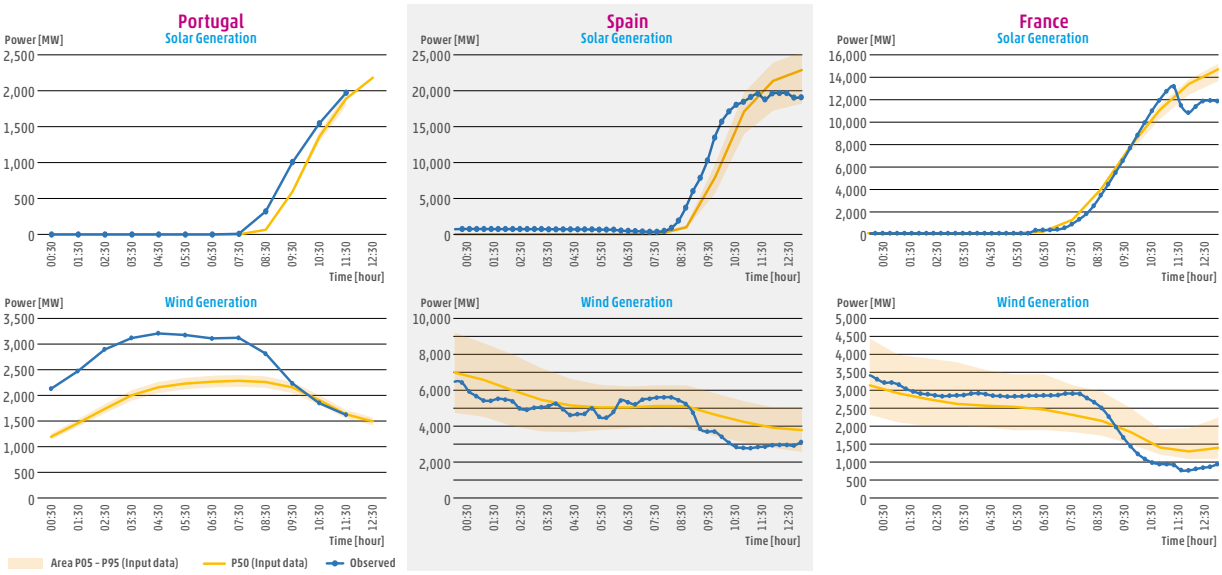


Figure 6-3: Graphs comparing the forecasted STA input data with the realised values, according to the ENTSO-E TP, for solar and wind generation for the affected regions (FR-PT-ES).

A 1-hour temporal offset is clearly visible in the RE input forecast results, particularly in the solar ramp profile between 07:30 and 10:00, where the forecasted ramp is systematically shifted. This deviation was detected after the CET-to-CEST time transition on 30 March. The issue was treated as an active case under investigation throughout April, with full correction implemented in May.

Besides this effect on the forecast, during the time frame 10:30 – 12:30, it can be observed that the solar values in Spain remain in the low part of the confidence range. STA files are sent at 8:00 on D-1. Therefore, the forecasts cannot account for possible RES limitations that are imposed later, after the STA process has been completed.

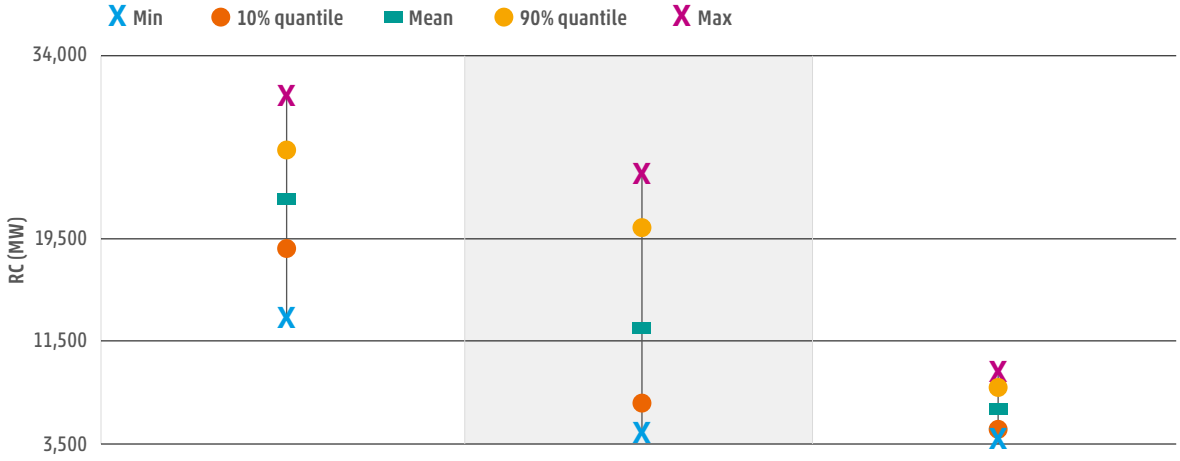
6.2.2.2 Calculation Results

STA calculation was completed within the standard time frame typical for the daily STA process. The automatic deterministic and probabilistic calculations were triggered at 09:00, finishing at 09:05 and 09:20, respectively.

The entire STA process was completed successfully, with no system failures or issues reported through the STA tool's ticketing system. The deterministic results for the period did not reveal any adequacy issues in affected and/or neighbouring TSOs after the STA calculation. Final exchanges between the affected TSOs and the remaining capacities (RCs) after the calculation are presented in Figure 6-4. As part of the daily procedures, the Nordic RCC operator on duty reviewed the deterministic results and emailed the rest of the RCCs along with the TSOs facing adequacy issues. RE, REN, and RTE were not among the countries identified with adequacy concerns.

Regarding the probabilistic calculations from the PE tool, no inadequacy was identified in the SWE region during the relevant time frame.

Furthermore, the results for all the SWE TSOs showed considerably high RCs, as shown in Figure 6-4. RC represents each TSO's ability to meet its load using available generation. These results indicate no adequacy issues and no reliance on imports from neighbouring areas.



	ES	FR	PT
Min	11,583	4,387	4,068
Q10	17,856	6,542	4,431
Mean	22,377	12,347	6,238
Q90	26,678	20,382	8,327
Max	30,559	24,873	9,431

Figure 6-4: RC statistic – final state per BDZ in SWE for the period 28/04/25 to 03/05/25.

A recently developed feature in the STA process allows the assessment of potential curtailment at the bidding zone level. This assessment is optional for TSOs, and only some currently use it. This analysis depends on input data classified as inflexible generation – generation that must run. The calculations then evaluate whether curtailment may be necessary and identify any surplus inflexible generation that cannot be consumed, while also considering the capacity to export this excess.

The must-run calculation results from RTE indicated no need for curtailment, as inflexible generation remained below the expected load. In the case of REN, the tool encountered a calculation error, so no meaningful results were available. For RE, no relevant data were provided for this feature; therefore, no calculation results are available. RE has already planned for the IT development to implement this capability in the STA process, with delivery planned in 2026.

6.2.2.3 Analysis

The outcomes of the STA calculation are dependent on the quality of the input data received from the TSOs. This data is crucial for defining the deterministic scenarios with the expected cross-border flows and serves as the basis for the probabilistic scenarios. Unrealistic input data could result in incorrect adequacy predictions or failure to identify inadequacies.

To determine whether the difference in forecast accuracy was an isolated event occurring only before the incident or a more systematic discrepancy, the scope of the analysis was extended to the entire month of April 2025. Several metrics were then used to evaluate the forecast quality, including accuracy and coverage of probable events.

The accuracy was evaluated by the difference between the realised values (from TP) and the forecasted input data (from TSOs) for all MTUs. The results, displayed in Table 6-5, show that there is no significant difference between the incident day and the observed values throughout the month. The difference is even smaller on the incident day, except for the wind forecast in Portugal. It can also be observed that the magnitude of the difference values is significantly reduced compared with the installed capacity of the generation, shown in Table 6-6.

Period		STA input data and TP realised values, averaged difference (MW)		
		PT	ES	FR
April 2025	Solar	126	2,244	490
	Wind onshore	400	1,012	1,106
Incident day	Solar	80	1,576	325
	Wind onshore	719	699	326

Table 6-5: Average difference between STA input data vs TP realised values.

	Installed capacity (MW)		
	PT	ES	FR
Solar	2,589	34,255	21,238
Wind onshore	5,366	30,932	24,606

Table 6-6: Spain, France, and Portugal installed capacities.

The coverage results, which evaluate the forecast intervals and are presented in Figure 6-5, show that the realised values are not usually between the forecast intervals, except for the wind onshore forecast for Spain and France, which displays more accurate results. Coverage analysis also shows that realised values frequently fall outside the 90 % forecast interval, particularly for solar, where RES limitations applied after D-1 08:00 cannot be captured. Wind forecasts for Spain and France show better alignment, though Portugal often exceeds the upper range, while France tends to fall below it.

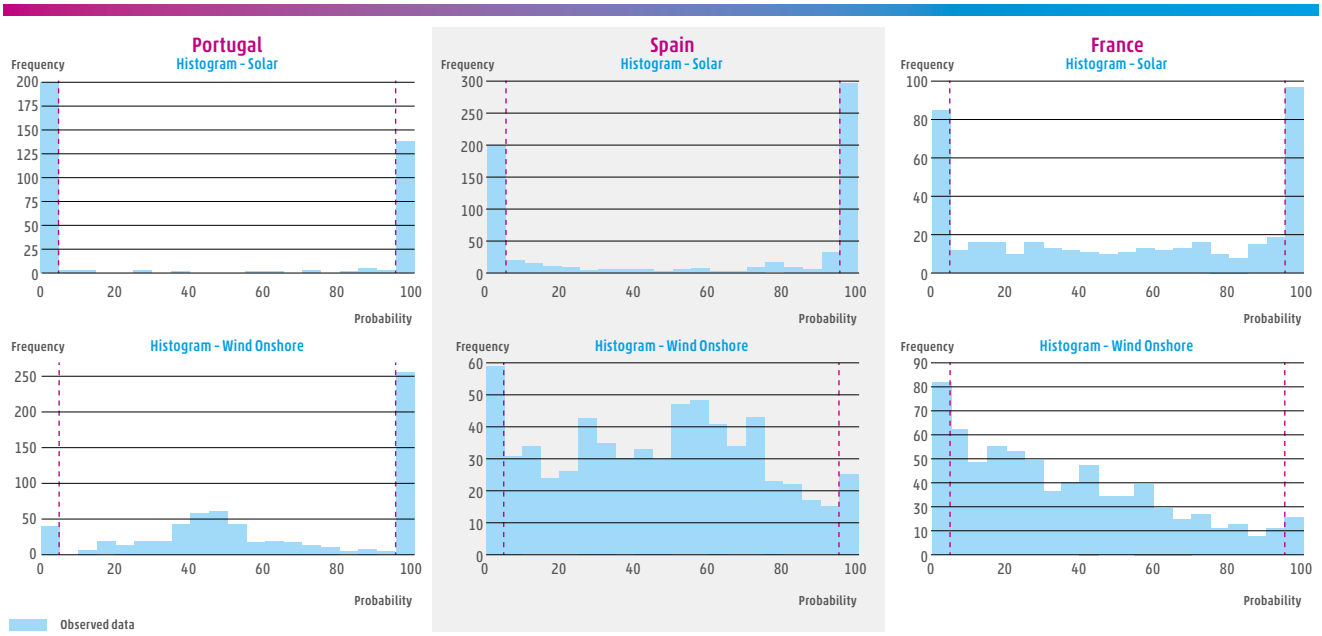


Figure 6-5: Probability range of the observed values compared with input data.

The most frequently observed values fall outside the 90 % probability range, indicating a significant deviation from the forecast. This is evident for the solar forecast, as the timing of the forecast prevents the inclusion of RES limitations. The wind forecast shows accurate alignment. There is also a high frequency of wind forecasts in the upper range for Portugal, showing that actual values often exceed predictions, while the opposite trend is observed for France.

The analysis of input data quality reveals significant discrepancies between the forecasted values provided by the TSOs and the actual observations, particularly for solar forecasts. Nevertheless, the results of the process would remain unchanged, without any adequacy issues.

It is worth mentioning that, given the timing for the forecast, limitations that affect RES (primarily solar in Spain) cannot be considered for the inputs to STA.

The second part of the analysis assesses the impact of those differences and forecast uncertainty on STA results, specifically whether the output would change if the input data were more accurate. A study was performed to assess the impact of input data on the results, where the forecast input data from the TSOs was replaced with the observed values for the day of the incident. The results indicate that cross-border flows are distinct, but the overall output of the STA process remains unchanged, with no issues in terms of adequacy and minimal impact on available capacity to meet demand.

6.2.2.4 Conclusion

The STA process for the period preceding the incident was performed in accordance with its defined methodology, scope, and timelines as set out in Article 81 of the SO GL. All required input data were received prior to the execution of the D-1 calculations, successfully passed validation checks, and were used in the STA assessment. The deterministic and probabilistic calculations were completed within the standard operational time frame, and no technical issues were reported during the process. The results were reviewed by the designated RCC operator, and no adequacy concerns were identified for the SWE region or the affected TSOs.

The STA results indicated sufficient RC for all relevant bidding zones, with no reliance on imports, and no identified risk to adequacy for the day of the incident. No regional STA process was triggered, and the outcome of the assessment was consistent with the absence of adequacy issues in real-time operation. Accordingly, the STA task was performed correctly and delivered results in line with its intended objective of assessing the balance between expected load and available generation capacity.

Regarding the incident under investigation, the STA process was not designed to identify or prevent events arising from operational security phenomena unrelated to adequacy. The STA process, within its defined scope, would not have been expected to identify or avert the incident.

Nevertheless, the investigation identified limitations related to the quality and representativeness of input data, particularly for solar generation forecasts. Differences between forecasted and realised values were observed, and the coverage of probabilistic forecast ranges was found to be limited. These limitations did not affect the final conclusions for the incident day; even when the STA input data was retrospectively replaced with realised generation values, the overall outcome of the adequacy assessment remained unchanged, with no inadequacy detected. Nevertheless, they highlight areas where the robustness of the STA process could be enhanced, particularly in data quality monitoring. The monitoring of STA process performance is a continuous activity, undertaken jointly by RCCs and TSOs in dedicated ENTSO-E Task Teams. Work is ongoing to improve data quality monitoring, forecast accuracy, and methodological alignment, in line with Articles 42 and 43 of the ACER decision on the amendment of the CSAm (13 May 2024).

Overall, the STA process fulfilled its intended function and was executed correctly. The results were reviewed, and no adequacy concerns were identified for the SWE region or the affected TSOs. While improvements in input data quality and forecast representativeness could further strengthen the accuracy of the assessment, they would not have altered the outcome of the STA assessment, nor enabled the identification or prevention of the incident under investigation.

6.2.3 Common Grid Model

As outlined in Table 6-3, the creation of common grid models (CGMs) is a responsibility assigned to RCCs under EU Regulation 2019/943, Article 37(c). This task must be carried out in accordance with the methodologies and procedures established under the SO GL, developed pursuant to Regulation (EC) No 714/2009, as well as CACM and forward capacity allocation (FCA). CGMs serve as the foundation for regional and PE processes such as CSA, OPC, and CCC, and are produced by merging the individual grid models (IGMs) provided by TSOs.

An IGM is a detailed model of a TSO's control area, containing information on network topology, load, generation, and relevant operational data. Each TSO is responsible for creating and maintaining its own IGM using a standardised format to ensure compatibility with other TSOs' models. Currently used standards include the Union for the Co-ordination of Transmission of Electricity Data Exchange Format (UCTE DEF)[37] and Common Grid Model Exchange Specification (CGMES).[38] UCTE DEF is a legacy standard created by the UCTE, which was the predecessor organisation to ENTSO-E. CGMES is an IEC (International Electrotechnical Commission) technical specification based on the IEC CIM (Common Information Model) family of standards.

In sum, a CGM is a consolidated network model produced by merging multiple IGMs from different TSOs. This provides a representation of a larger, interconnected portion of the power system, which is then utilised for system-wide processes. The CGM merging processes for different time horizons and regions are described in the following subsections.

6.2.3.1 PE CGM in CGMES

The PE CGM is created by merging the IGMs of European TSOs while considering reference data programmes (PEVF or CGMA files) and boundary sets (BDSs). It is created for four different time horizons: ID, D-1, Two-Day-Ahead (D-2), and Year-Ahead (YA).

In accordance with SOC Decision Number 11 dated 4 December 2019, the CGM is developed based on a rotational principle. Under this approach, CGM creation responsibilities are divided into groups based on time frames: Group 1 handles the D-1 and D-2 time frames, Group 2 is responsible for the ID time frame, and Group 3 manages the YA time frame. Each group operates with two roles: main and backup.

At any given time, one RCC assumes one role within a specific group and carries out the CGM creation accordingly. The rotation among RCCs occurs every four weeks – except for the YA time frame, which rotates annually – following a predefined schedule. The use of both main and backup roles ensures redundancy, as two RCCs work in parallel on the same time frame, ensuring that at least one CGM will always be available, even if the main RCC encounters technical or operational issues.

The merging process using the CGMES format during the period when the disturbance occurred was facing challenges. Convergence issues are often encountered, and certain CGMs are not published. During the development phase, one tactic used to identify problematic input files was to exclude them. This allowed the quality of the process to be assessed using the remaining files. To comply with the CGM methodology, RCCs jointly decided to discontinue this practice in the production environments as of Q3 2024. The end of this pre-emptive exclusion practice led to lower publication rates for CGMES CGMs. However, the published CGMs fully considered all available input data and were in line with the European Merging Function (EMF) requirements.

According to the CGMM and following the adopted timings in the CGM building process (implemented on 21 April 2025), the D-1 CGMES process begins at 18:00 – by which time all IGMs should be submitted – and concludes by 18:30, when validated CGMs for the relevant scenario become available via OPDE. Similarly, the D2CF CGMES process starts at 19:50 and ends at 20:00, with validated CGMs also made available through OPDE. For both processes, according to the rotation calendar, Coreso acted as the main RCC responsible for the merging, while SCC served as the backup RCC.

37 UCTE DEF: Union for the Co-ordination of Transmission of Electricity Data Exchange Format.

38 CGMES: Common Grid Model Exchange Specification.

The intraday congestion forecast (IDCF) process operates on an hourly basis, with validated CGMs released every hour via OPDE. TSCNET served as the main RCC for the IDCF CGMES process, and Baltic RCC acted as the backup.

For BD 28/04/25, all three SWE TSOs submitted valid D-2 and D-1 IGMs on time. However,11 TSOs from other regions did not provide D-2 IGMs, and five TSOs did not provide D-1 IGMs. Coreso's D-1 merge was successful for eight out of 24 TSs, while the remaining 16 – including the 12:30 TS – were non-convergent. The backup D-1 model from SCC for 12:30 was also non-convergent. For the D-2 process, neither Coreso's nor SCC's models converged for any TS.

For the ID merge, RTE delivered three batches (on 27 April 2025 at 20:41 UTC and 28 April 2025 at 04:40 and 12:40 UTC), REN delivered once, with 24 IGMs (on 27 April 2025 at 21:27 UTC), and RE delivered four batches (27 April 2025 at 20:47 and 23:08 UTC, and 28 April 2025 at 04:52 and 12:51 UTC). The transition to more frequent ID IGMS of these TSOs is pending. Seven other TSOs did not deliver any valid ID IGMs for this business day. TSCNET, as the main merge operator, performed hourly merges for the business day, which were all non-convergent. The latest merge before the incident took place at 11:12. For this latest merge, the IGMs from all but two TSOs were more than one hour old. As the backup merge operator, Baltic RCC was able to create a convergent model at 07:51 by substituting models based on additional validation rules not yet harmonised at the European level. This was the latest convergent PE CGM in CGMES before the incident.

Overall, the merging process in the CGMES format is still undergoing improvements, and the resulting models were of only limited use. Since all regional and cross-regional assessments are still based either on the legacy UCTE DEF merge or a regional CGMES merge, this had no impact on the incident under investigation. All security assessments regionally coordinated by Coreso, such as contingency assessments carried out within the capacity calculation process or the D-1 security assessment (legacy process), are performed using other models. The missing input data from other regions had no relevant impact on SWE.

In conclusion, while there were no available PE D-1 and D-2 CGMs in CGMES format at the time of the incident – only regional models for SWE, as explained below – this would not have made a difference.

6.2.3.2 Regional CGMES CGM

In addition to the PE CGMs, Coreso carries out a regional merge of the D-2 and D-1 CGMES IGMs from REN, RE, and RTE on a daily basis, producing CGMs that are mandatory inputs for the D-1 and ID capacity calculations, respectively.

For capacity calculation, the TSOs in SWE deliver:

- In D-2 (approximately 15:15 UCT), 24 models representing each hour of the day "D", used for capacity calculation in D-2
- In D-1 (approximately 15:30 UCT), 24 models representing each hour of the day "D", used for capacity calculation in the first IDCC.

For both capacity calculation processes, CGMs are produced each evening for six TSs (02:30, 05:30, 09:30, 12:30, 15:30, and 19:30) and shared with the TSOs as part of the process outputs.

The latest regional SWE CGMES merge was completed before 19:30 on Sunday. All required input data was valid and delivered on time. The merge and load flow simulations were successfully executed for both time horizons.

6.2.3.3 Continental Europe CGM in UCTE DEF

UCTE DEF is currently used for data exchange in various operational processes, including SA for SWE and other regions, and CCC in other regions. UCTE DEF is currently being replaced by CGMES. One major difference between UCTE DEF and the CGMES-based PE CGM implementation lies in the modelling approach: in UCTE DEF, models are built for all of Continental Europe, and each RCC additionally generates CGMs specifically for their own region. For example, for the day-ahead congestion forecast (DACF), Coreso produces a D-1 CGM for the SWE region (including applied model improvements), while SEleNe CC creates CGMs for southeast Europe. In this way, each RCC is responsible for developing CGMs tailored to its specific region.

The process of submitting and merging these individual models includes data validation. Detected issues are categorised by severity, which impacts the data's suitability for analysis:

- **Informational:** Indicates minor data conditions or deviations that do not prevent the use of the model.
- **Warning:** Signifies data inconsistencies or potential inaccuracies. Models with warnings may be used, but the associated data should be treated with caution.
- **Fatal Error:** Denotes a data issue that makes the model unusable for its intended purpose. Processing is typically halted until the error is rectified.

Currently, UCTE DEF-based IGMs are merged per synchronous area; in other words, the UCTE DEF CGM containing Spain, Portugal, and France will not include the Nordics.[39] In this subsection, the availability and validation of the affected TSO IGMs and the merge for the Continental Europe CGM in UCTE DEF are presented.

Since the IGMs are the starting point, the delivery of IGMs for the different time horizons is shown in Table 9. Each process has a different expected frequency of IGM creation. For D-1 and real-time snapshots (RTSN), one IGM per TS is expected, whereas for ID, it can go up to hourly delivery of IGMs before real time, e.g. after 12:30, new ID IGMs for 12:30 are not expected.

As shown in Table 6-7, most of the IGMs were valid, meaning they contained no fatal errors. The missing values are due to REN not providing ID or RTSN IGMs, and RE not providing RTSN IGMs. The provision of RTSN files in UCTE DEF is not mandatory, and both REN and RE already provide ID IGMs in CGMES format – one batch for REN and four for RE – for the timings referenced in 6.2.3.1. The valid UCTE DEF IGMs include informational and warning messages. The only invalid IGM is for the ID time horizon and was submitted by RE one hour before real time. The model for TS 12:30 was generated at 11:16, which was the last ID IGM submitted for TS 12:30, since RE does not provide IGM every hour. Concerning RTSN, REN and RE provided RTSN models generated on demand for the incident investigation.

TSO	D-1	ID	RTSN
REN	1/1	-	-
RE	1/1	4/5	-
RTE	1/1	19/19	1/1

Table 6-7: Delivery of valid IGMs for D-1, ID, and RT time horizons for BD 28/04/2025, TS 12:30, during the processes.

It is important to clarify that, for the SWE CCR, only D-1 UCTE CGMs are used by Coreso as input for the security assessment. A regular ID security assessment based on UCTE CGM merged models is not requested from Coreso by SWE TSOs. The Spanish and Portuguese IGMs are sent as part of the ID UCTE DEF-based merging process to facilitate data for security assessments by other TSOs or regions within the current legacy process (not the formal mandated ROSC process).

The RE ID IGM created at 11:16 for the 12:30 TS did not fulfil the quality check rules. RE ID IGM did not achieve convergence in RE's legacy power system analysis tool, and consequently also in the UCTE DEF IGM model, due to a malfunction in RE's tool responsible for modelling the status of reactors. High voltage shifts were provided, such as a 400 kV busbar operating at 280 kV. A substitution mechanism was applied to build up the corresponding ID UCTE CGM. Similar problems were observed on three separate days within the month: 4 April (one ID IGM), 14 April (two ID IGMs), and 28 April (five ID IGMs). On 28 April, out of the five affected models, four corresponded to the 10:30 TS and one to the 12:30 TS. The non-convergence of the ES IGM in RE's legacy power system analysis tool was not caused by grid conditions at that time, but rather by a bug in RE's reactor management script.

Concerning CGMs, the DA IGMs are utilised for the DACF process. Coreso is the RCC performing the SA for the SWE region, which is further described in 6.2.5. During the DACF process, four CGM versions were created: at 18:00, 18:27, 19:57, and 20:10.

Based on all the provided ID IGMs, CGMs were created. All CGMs created that day for the TS 12:30 were convergent. It should be noted that invalid IGMs were not included; instead, the latest provided IGM was used. For example, for the last ID CGM[40] created at 12:15, the ES IGM generated at 10:16 was utilised.

39 Although Baltic synchronisation took place in February 2025, Baltic TSOs do not provide IGM in UCTE DEF.

40 According to TSCNET merging.

The substitution mechanism applied to the non-converged IGM model delivered at 11:16 enabled the necessary load flow and ID security assessment processes for the regions and TSOs using UCTE CGM. In any case, the non-convergence of the UCTE IGM RE model delivered at 11:16 had no impact on the incident.

6.2.3.4 Conclusion

The CGM-related tasks relevant to the SWE region were performed in accordance with the applicable methodologies and operational practices, and any limitations observed in the PE CGMES processes did not affect system security or contribute to the incident under investigation.

Although convergent PE CGMES CGMs were not available for the D-1, D-2, and ID time horizons, this situation resulted from known and documented challenges in data completeness and model convergence at the European level. In line with the CGM methodology, the discontinuation of pre-emptive IGM exclusion ensured methodological compliance, even at the cost of lower publication rates. These constraints were external to the SWE region and did not compromise regional operational processes.

For SWE, all CGM inputs required for operational decision-making were available and fit for purpose. Regional grid models in CGMES format for D-1 and D-2 were successfully created by Coreso using complete and timely IGMs from REN, RE, and RTE, and were effectively used in the capacity calculation processes. In addition, D-1 security assessments relied on legacy UCTE DEF CGMs, which were fully convergent and produced in accordance with established procedures. Isolated issues affecting individual ID IGMs, notably the non-convergent RE model at 11:16, were appropriately managed through substitution mechanisms and did not affect the validity of the resulting CGMs used by other regions, other than SWE, for the ID operational security assessment. It seems of interest to clarify that the resulting CGM mentioned previously, where a substitution strategy was applied for the non-converging RE IGM, is not used as input for security assessment in SWE.

Consequently, the CGM creation tasks applicable to SWE were executed as intended, and the unavailability or limited usability of PE CGMES CGMs had no operational impact on SWE security assessments and no causal or contributory role in the incident.

6.2.4 Coordinated Capacity Calculation

The capacity calculation process described in this section refers to the regulated task that follows the CACM Guideline EU 1222/2015. As explained in 6.1.2, the SWE region has some specificities that define how cross-border capacities are calculated. Coreso performs the NTC-based capacity calculation using a coordinated methodology developed by SWE TSOs (REN, RE, and RTE) and approved by ACER and the NRAs. This task is currently performed for the following time frames:

- CCC Long-Term (LT) SWE: Currently performed on the TSO side. Under development at Coreso.
- CCC DA SWE: Performed two days ahead and calculates total transfer capacities (TTC) for six TSs for the DA market.
- CCC ID SWE: Two different runs apply:
  - 1st run: Performed one day ahead and calculates TTC for six TSs for the ID market.
  - 2nd run: Performed on the same day and calculates TTC for three TSs in the afternoon for the remaining ID market gates. This task is live as of 22 October 2025.

Based on a set of inputs from SWE TSOs (namely IGMs, GSKs, and CRACs, all in CGMES format), Coreso calculates TTC for a limited number of TSs (02:30, 05:30, 09:30, 12:30, 15:30, and 19:30) and for each border and direction independently. The calculated TTCs are sent as a proposal to the TSOs, who validate the proposal and allocate capacities for the 24-hour period by extrapolation. As part of the CCC DA SWE process, Coreso shall ensure that the proposed NTC fulfils the minimum capacity requirements according to Article 16(8) of the Electricity Regulation.

To achieve this, a threshold is defined over the maximum rating of the most limiting CNEC. Using calculated PTDFs and NTCs, the margin available for cross-zonal trade on the most limiting CNEC is assessed. If this margin is lower than 70 % of the rating, an additional NTC (ANTC) is added to the calculated NTC to obtain the proposed NTC. This virtual capacity is considered non-secure capacity and may be limited by a maximum available countertrading stated by TSOs within operational and security limits. Coreso then processes the TSOs' feedback to determine the final TTC for each border and direction.

To calculate the final output sent to markets, NTC, the following formula and parameters are applied:

- NTC = TTC – transmission reliability margin (TRM)
- TRM is 7.5 % of TTC for the Spain–France border, with a minimum of 200 MW.
- TRM is 10 % of TTC for the Portugal–Spain border, with a minimum of 100 MW.

Fallback values may be applied either if the TTC cannot be calculated or if the calculated TTC values are lower than the TRM values listed above. In such cases, the following fallback values are applied per process:

- For the CCC DA SWE process, the fallback values applied are TTC from the CCC LT SWE process.
- For CCC ID SWE processes, the fallback values applied are the available TTC from previous time frames. In the case of BD 28/04/2025, the fallback for CCC ID SWE (first run) is to use the TTC values from the CCC DA SWE process.

Although the task is well-established in the SWE region, a significant milestone was reached in February 2025 with the introduction of a new tool for the CCC D-1 process, replacing the previous technology.

In contrast, for the CCC ID process, the new tool was launched in March 2025.

6.2.4.1 Focus on BD 28/04, TS 12:30

Table 6-8 provides an overview of the proposed capacities for the market directions observed on 28 April.

Process	Direction	Calculated TTC (MW)	CNEC	Proposed TTC (MW) and reason
CCC DA	ES → FR	0	N-1 Braud–Preguillac 1 led to divergence	2,500 (LT TTC values applied)
CCC DA	ES → PT	3,900	N-2 contingency overloads	3,900
CCC ID	ES → FR	100	N-1 Braud–Preguillac 1 overloads Braud–Preguillac 2	2,500 (D2 TTC values applied due to TTC < TRM)
CCC ID	ES → PT	4,050	N-2 contingency overloads	4,050

Table 6-8: Overview of proposed TTC (MW) per process and direction. Only the real-time market directions are depicted. For all calculated capacities, refer to 6.2.4.2 and 6.2.4.3.

6.2.4.2 Long-term Capacity Calculation (CCC LT SWE)

The long-term capacity calculation process for yearly, quarterly, and monthly time frames is currently being developed. At the time of reporting, the go-live of this process is expected in Q3 2026.

Until then, the long-term values used as backup for the CCC DA SWE process are based on inputs received from RE through a bilateral monthly process among SWE TSOs (RE and REN for the ES–PT border; RE and RTE for the ES–FR border). Table 6-9 shows the values provided by RE for this BD for 12:30.

ES → FR	2,300 MW
ES → PT	3,500 MW
FR → ES	2,400 MW
PT → ES	2,200 MW

Table 6-9: Long-term NTC values provided for 28/04 TS 12:30. Source: RE

6.2.4.3 D-1 Capacity Calculation (CCC DA SWE)

The D-1 capacity calculation for BD 28/04 was performed on the evening of 26 April and concluded on 27 April at 08:00 and validated by TSOs. The results are shown in Table 6-10.

	TTC (MW)	Type of limitation
PT → ES	3,785	Voltage phase angle constraint
ES → PT	3,900	Thermal limit
FR → ES	2,500	Thermal limit
ES → FR	2,500	Divergent calculation LT TTC values applied

Table 6-10: Proposed TTC for TS 12:30 after the CCC DA SWE calculation for BD 28/04, with limiting reason.

Further analysis was conducted to examine the final direction of exchanges observed in real time on 28 April at 12:30. Regarding the ES → FR direction with 0 MW exchange, there was a violation in the model after N-1 Braud–Preguillac 1. Therefore, no TTC was computed, and the long-term TTC value (2,500 MW) was applied for this direction and TS by following the relation between NTC and TTC, as explained in 6.2.4. During the validation phase, TSOs did not decrease the proposed value, so it was confirmed as the final TTC value.

In the ES → PT direction, following the prescriptive N-2 contingency, the TTC was calculated to be 3,900 MW. During the validation phase, TSOs did not decrease the proposed value, so it was confirmed as the final TTC value.

Focusing on the capacities provided in the directions relevant to the ID market and to provide context, the final TTCs issued during CCC DA SWE were within normal values relative to those provided during 2025 so far, as illustrated in the figures below. In the Spain-to-France direction, the provided 2,500 MW belongs to the 16th percentile (see Figure 6-5), while for the Spain-to-Portugal direction, the provided 3,900 MW belongs to the 22th percentile (see Figure 6-6) of this year's distribution.

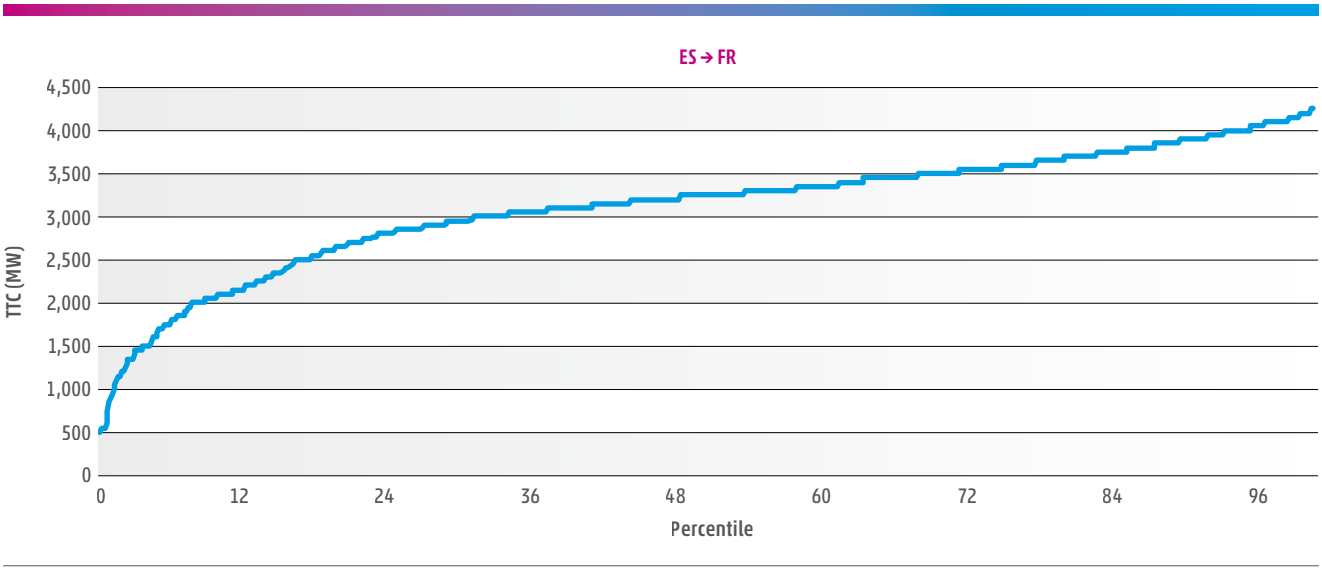


Figure 6-5: Proposed TTC values for the DA process and ES → FR direction since January 2025, sorted from lowest to highest.

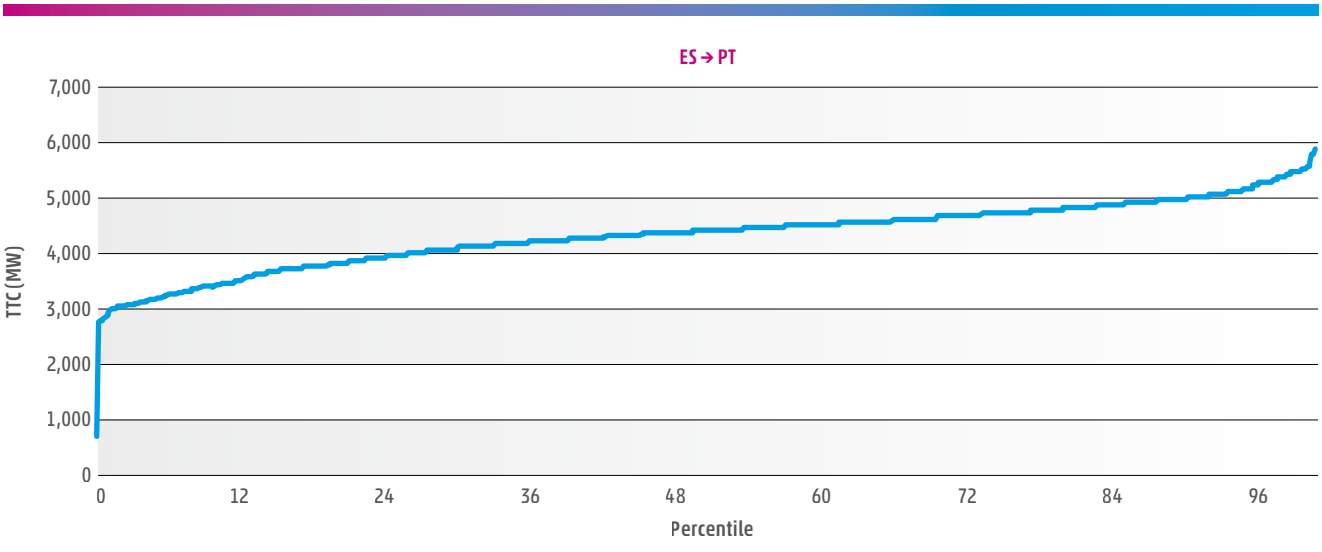


Figure 6-6: Proposed TTC values for the DA process and ES → PT direction since January 2025, sorted from lowest to highest.

6.2.4.4 Intraday Capacity Calculation (CCC ID SWE), First Run

The ID capacity calculation is performed in D-1, so for BD 28/04, the calculation was performed on the evening of 27 April, providing results by 22:00. The results are shown in Table 11.

	TTC (MW)	Limiting reason
PT → ES	3,050	Voltage phase angle constraint
ES → PT	4,050	Thermal limit
FR → ES	2,600	Thermal limit
ES → FR	2,500 (D2 TTC values applied due to TTC < TRM)	Base case divergent scenario D-2 TTC values applied

Table 6-11: Proposed TTC after CCC ID SWE calculation for BD 28/04, with limiting reason.

In the ES → FR direction, the computed TTC was 100 MW (i.e. lower than TRM) due to N-1 Braud–Preguillac 1 overloading Braud–Preguillac 2. Therefore, DA TTC values (2,500 MW) were applied. During the validation phase, TSOs did not decrease the proposed value, so it was confirmed as the final TTC value.

In the ES → PT direction, the prescriptive N-2 contingency resulted in a TTC value of 4,050 MW. During the validation phase, TSOs did not decrease the proposed value, so it was confirmed as the final TTC value.

6.2.4.5 Conclusion

The CCC task for the SWE region was performed in accordance with the approved methodologies and within the defined operational timelines for the relevant long-term, D-1, and ID time frames. All required inputs from the SWE TSOs were received, validated, and processed, and the CCC D-1 and ID calculations for BD 28/04 were completed successfully. Where TTC values could not be computed or fell below the applicable TRM, the predefined fallback procedures were correctly applied, and long-term or D-1 TTC values were used in line with the approved methodology. The proposed capacities were validated by the TSOs and confirmed as final values without adjustment.

For the directions relevant to the real-time exchanges observed at BD 28/04, TS 12:30, the TTC and resulting NTC values provided to the market were within the range of values observed historically in 2025. The application of fallback values, including for the ES → FR direction in both the D-1 and ID processes, followed the established rules and did not result in atypical capacity levels when compared to the distribution of past outcomes. No procedural deviations, calculation failures, or validation issues were identified in the execution of the CCC task.

With respect to the incident under investigation, the CCC process is designed to determine cross-border market capacities in compliance with operational security constraints, rather than to anticipate or prevent real-time system events. The capacities provided to the market were based on the contingency analyses and network conditions considered within the CCC framework and were confirmed by the TSOs during the validation phases. Accordingly, the CCC task, as designed and applied, would not have been expected to identify or avert the incident.

The investigation did not identify deficiencies in the execution of the CCC process.

Overall, the CCC task was executed correctly and in line with the approved methodology. Within its defined scope and objectives, it fulfilled its intended function and did not contribute to, nor was it capable of preventing, the incident under investigation.

The underlying SAs performed by Coreso assess the limits associated with power flows and voltage (permanent limits and voltage phase angle) related to operational security limits.

6.2.5 Coordinated Security Analysis

The implementation of the CSA methodology via the ROSC process, which will eventually replace the current SA service, is still under development, with an expected go-live of Q1 2027 for the SWE region. SA for the SWE region is conducted by Coreso daily using the DACF grid models in the D-1 time frame. This analysis is based on coordinated procedures and relies on the Continental Europe CGM in UCTE-DEF, as described in 6.2.3. Since the SA is performed on steady-state grid models, it only considers power flows and not dynamic security limits, while voltage limits are not assessed regularly due to data format and modelling limitations. Regarding its scope, SAs are run for the whole day, while RAs are proposed by Coreso operators for the worst TS for each constraint, in close coordination with the SWE TSOs. Currently, SAs in the ID time frame are not operational for the SWE region.

The SA DA process for the CORE, IN, and SWE regions unfolds in a tightly coordinated sequence beginning the day before16:00 and concluding by midnight.

It starts with the delivery of IGMs from the TSOs and the availability of market results, which serve as the foundational inputs for the process. By 18:00, the first version of the CGM (CGM V1) is generated. This version is then refined through several targeted improvements per region. For the SWE region, for example, Coreso adapts the tap position of the PSTs on the FR–ES border based on N-state loading and TSO operational guidelines to ensure the base case reflects a realistic scenario. At 18:15, a second, enhanced version of the CGM (CGM V2) is produced, incorporating these improvements. The process continues at around 18:30 – 18:45 with the launch of the SA for all regions and 24 TS, where the system's robustness is assessed against potential contingencies. Once the SA results are available, the operator interprets them and identifies the most relevant constraints within the Main and Extended area for the SWE region, as detailed in 6.1.2. Once selected, the worst-case TS for each constraint is analysed, addressing N-1 violations through the implementation of RAs. The applicability of these proposed actions is then verified through direct communication with the relevant TSOs. If confirmed, the actions are documented in the report; if not, the Coreso operator seeks and proposes alternative solutions.

By 21:00, Coreso, in collaboration with the TSOs, evaluates and coordinates RAs to address any identified issues. At 21:00, the Daily Operational Planning Teleconference (DOPT) is used to exchange and consolidate relevant cross-border data. The process then moves into the impact and deviation assessment phase at 22:00, which includes a specific study for the Central Eastern Europe (CEE) region.

A combined DACF CGM is built with the CGMs of Coreso and TSCNET, including all agreed preventive RAs from the study. At 22:30, the proposed RAs are confirmed with the TSOs, ensuring alignment and readiness. Finally, before midnight, the comprehensive D-1 Report (summarising the entire SA, including all assessments and confirmed actions) is published, marking the conclusion of the SA DA process.

As mentioned earlier, for the SWE region, no SA ID process is operational on the RCC side: instead, it is performed solely on the TSO side. This process is used to reassess the security situation with updated inputs, enabling more accurate results and solutions. Coreso provides SA ID services for the CORE and IN regions and supports TSOs from these regions during the close-to-real-time time frame as part of continuous monitoring and when requested. However, these services, which are voluntarily contracted by TSOs in other regions, are not provided for the SWE region, as the respective TSOs have not identified them as necessary or beneficial for this region.

For the 12:30 TS on 28 April 2025, the DA CGM used was the last one created around 18:00 on the previous day, following the standard procedure for SA DA activities. The study reveals that TS 12:30 was a non-stressed TS, meaning that no need for coordination of RA was forecasted in the D-1 time frame.

6.2.5.1 Overview of the Study for BD 28/04

The study concluded that critical constraints were detected on the France–Spain (FR–ES) border, requiring countertrading of up to 300 MW at 05:30, along with extreme curative tap positions in the Arkale PST to resolve the issue. Additional constraints were identified on the Spain–Portugal (ES–PT) border and within the internal Spanish grid. These were manageable through the usual topological RAs and internal redispatch.

Additional relevant information used during the SA DA SWE included the planned outage of interconnection tie-lines and the tight downwards generation margin announced by REN. There was no specific request to study N-2 contingencies, as the weather forecast was favourable.

TSO	Generation margins	Peak load (MW) and time
RE	Sufficient	30,800 at 21:30
REN	Tight–Downward	6,820 at 21:30
RTE	Sufficient	48,800 at 13:00

Table 6-12: Load and generation margin forecast declared by SWE TSOs at the start of the SA DA SWE process.

Day-Ahead Commercial Exchanges Overview

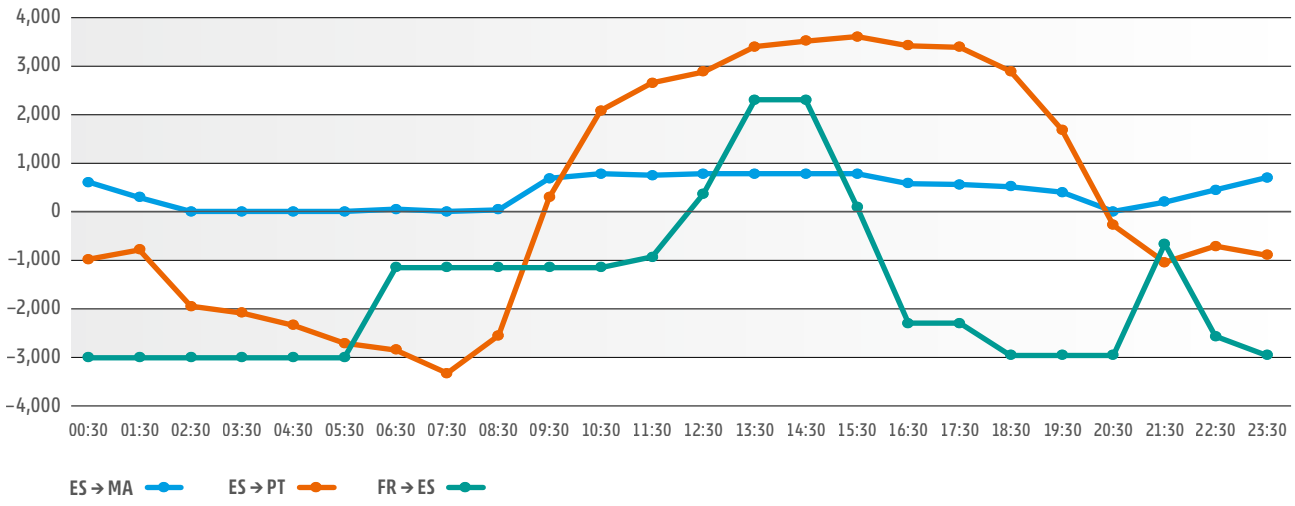


Figure 6-7: Commercial exchanges in the D-1 time frame for BD 28/04 (MW).

On 28 April, the commercial exchanges in D-1 showed Spain mostly exporting to France, except around midday, when it briefly imported up to 2,300 MW. Portugal exported to Spain overnight, but flows reversed in the morning, with Spain exporting up to 3,531 MW during the afternoon before Portugal resumed exports in the evening.

Some deviations between D-1 commercial flows and physical cross-border flows in the DA CGM were foreseen, particularly during the midday hours. On the Portugal–Spain border, Cedillo–Falagueira and Puebla de Guzman–Tavira were expected to have large flows around midday in the exporting direction to Portugal, while Aldeadavila–Lagoaca and Cartelle–Alto Lindoso were forecasted to handle the bulk of the export flows from Portugal during the night and early morning.

	00:30	01:30	02:30	03:30	04:30	05:30	06:30	07:30	08:30	09:30	10:30	11:30	12:30	13:30	14:30	15:30	16:30	17:30	18:30	19:30	20:30	21:30	22:30	23:30
ARKALE PST [1–33]	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	16	17	17	17
	-316	-309	-306	-301	-306	-319	-174	-180	-175	-132	-68	-66	50	220	214	-24	-161	-179	-262	-302	-265	-116	-284	-308
	56.2	54.9	53.8	53.1	54.1	56.6	34.3	34.5	33.6	27.9	20.7	20.4	14.1	37.3	36.5	14.9	30.2	33.6	47.5	54.5	48.9	28.9	52.8	55.7

Table 6-13: N-state overview for the Arkale PST, showing details for tap (top row), active power in MW (middle row), and loading in % (bottom row). Positive values of active power indicate flow from FR to ES. Yellow cells indicate loading over 50 %.

N-state Physical Flows

As explained earlier, the SWE region consists of two borders with a lower degree of interaction than in other regions, such as CORE. The main consequence is that, in general, the difference between commercial and physical flows is smaller. Consistent with the previous section, the highest N-state flows observed on the main corridors of the Spain–France border are all in the Spain-to-France direction, either in the early morning or evening. Since the capacity calculation process accounted for planned outages, the reported capacities did not require significant tap adaptation to manage flows.

The Spain–Portugal border has a high level of transfer capacity due to the nine tie-lines in operation. However, their distribution along the border is not balanced, with greater meshing in the north than in the south.

The lower meshing in the south part of the border means that the planned outages in the south impact both the N-state and N-1 loadings of tie-lines nearby. Additionally, it is worth mentioning the apparent loop in physical flows in the northern portion of the Spain–Portugal border in the morning, where some tie-lines export while others import. This behaviour is quite common in this part of the border due to the high level of meshing.

Since none of the high-loading values occurred near 12:30, the main conclusion is that this period was indeed less constrained in the D-1 time frame.

SA DA Study Results

Constraints were detected either in the morning or evening hours, mainly on the FR-ES border (Argia–Hernani, Arkale–Hernani, Hernani–Itxaso) and ES–PT border (Falagueira–Cedillo, Aldeadavila–Lagoaca). They were resolved using 300 MW countertrading (FR↑, ES↓), PST tap changes at Arkale (from 17 to 1 in steps), transformer disconnections at Argia, and internal redispatch by RE (400 MW) and REN (800 MW).

Coreso proposed RAs to bring all N-1 loadings below the 100 % threshold for each of the detected contingency constraints for BD 28/04.

6.2.5.2 Further Analysis

The CSA process relies on input data provided by TSOs. The list of contingencies specifies the cases to be simulated, while the list of monitored elements defines the elements to be monitored during the process. If a certain contingency is not simulated, the state of the grid under that scenario is not assessed. Similarly, if a monitored element is not included in the list, potential violations may go undetected.

Articles 76 and 77 of the SO Regulation constitute the legal basis for the regional proposal of CSA (ROSC methodology) and define several specific requirements that the methodology proposal should consider. SWE TSOs made a proposal in 2019 with an explanatory note.

On 3 December 2020, the SWE NRAs issued a decision, in which they agreed on 10 July 2020 to request amendments to the proposed methodology regarding:

- The definition of the relevant XNE
- The coordination between TSOs and the RCC to select a new set of RAs when a recommended RA is no longer needed
- The "on-request coordinated operational security assessment" process and updates to the TSOs' IGMs
- The implementation timeline

For the contingency scenarios:

According to Article 13 of the SWE ROSC methodology, which implements Article 76 of the SO Regulation and was approved by SWE NRAs in December 2020, SWE TSOs are required to send their contingency list, including both ordinary and exceptional contingencies, with each clearly classified.

Even under the same article and in accordance with the SO Regulation, contingencies that do not comply with the not comply with the N-1 criterion may be excluded, provided that such exclusions are properly communicated.

Limitations of the current SA DA SWE process:

The current SA DA process in the SWE region is implemented in line with the processes agreed between TSOs and Coreso. The geographical scope of the SA DA SWE is defined by network elements identified as cross-border relevant under the regional methodologies, ensuring a focused and proportionate assessment at the regional level. Internal network elements outside this scope are addressed through national processes.

The SA DA process is currently performed in the D-1 time frame and does not include an IDCF reassessment following ID market trading, such as changes in dispatch or voltage management actions. The assessment relies on predefined contingency lists and a defined set of assessed elements. For cross-border critical outages, one target scenario is assessed, and RAs are evaluated through established coordination procedures.

The current SA process performed by RCC is expected to evolve with the implementation of ROSC. ROSC will extend the temporal scope of CSAs to the ID time frame and support more frequent reassessments reflecting updated system conditions. It will also enable a broader and more systematic coverage of network elements for which RAs would be proposed, enhanced contingency assessment capabilities, and improved coordination of RAs, including voltage-related aspects, thereby strengthening the overall robustness and consistency of regional SA. The transition from UCTE DEF to CGMES will also enable appropriate consideration of voltage limits.

6.2.5.3 Conclusion

The UCTE DEF-based CSA in the D-1 time frame, currently implemented in the SWE region through the SA DA process, was performed in accordance with the agreed and coordinated operational procedures, not yet in line with the applicable regional methodology (ROSC methodology pursuant to Article 76 of SO GL). The SA DA process for BD 28/04 was executed using the latest available CGM generated on the previous day, following the standard timeline and incorporating validated TSOs' inputs, market results, planned outages, and declared generation margins. SAs were conducted for the full day, relevant contingencies were assessed, and preventive and curative RAs were proposed and coordinated with the concerned TSOs through established processes.

For BD 28/04 at TS 12:30, the SA DA identified this TS as non-stressed, with no forecasted need for coordinated RAs. Constraints detected in the SA DA process were limited to morning and evening periods and were addressed through countertrading, PST tap changes, transformer switching, and internal redispatch. None of the identified constraints or RAs was associated with the midday period corresponding to the incident. The CSA process, as currently implemented in the SWE region, is limited to the D-1 time frame and relies on predefined contingency lists, monitored elements, and the available information at that stage.

ID CSA is not operational at the RCC level for the SWE region and remains the responsibility of TSOs at the national level. The CSA framework is not designed to capture real-time operational dynamics or unforeseeable, rapidly evolving conditions occurring after the D-1 assessment. Accordingly, the CSA process, as designed and applied, would not have been expected to identify or prevent the incident under investigation. The SWE network is characterised by limited cross-border power flow interdependencies, enabling TSOs to adequately assess their networks, from a contingency assessment point of view, within their observability areas based on schedule information and to bilaterally coordinate any necessary RAs. Consequently, the added value of closer regional coordination performed by an RCC is more limited in the SWE region compared to other regions with higher structural interdependencies.

The UCTE DEF-based CSA performed by the RCC did not foresee the voltage collapse because it is not designed to, as elaborated in the previous section.

6.2.6 Other RCC Tasks

6.2.6.1 Emergency and Restoration Plans Consistency Check (NC ER)

In accordance with Article 6 of the NC ER, TSOs shall review their defence and restoration plans at least every five years, with a subsequent consistency assessment. RCCs are responsible for evaluating the coherence of specific measures, including inter-TSO assistance, frequency management, active power support, and top-down re-energisation. Following this consultation, the relevant RCC will prepare a technical report assessing the consistency of the measures implemented by the TSOs.

This task was last performed and completed in 2024, the second time since it was established. Coreso evaluated the defence and restoration plans for the SWE region and verified the existence of the necessary agreements outlining the procedures to be applied during emergency, blackout, or restoration states by RE, REN, and RTE. TSOs provided Coreso with references to the most recent versions of these documents, and no inconsistencies were identified between RTE and RE, or between RE and REN.

6.2.6.2 Regional Incident Analysis and Reporting

RIAR is the RCC task based on the requirement outlined in Article 37 (1) (i) of EU Regulation 2019/943 and ACER decision 04/2022. In this task, all RCCs jointly investigate the role of RCC tasks connected with incidents above a specific threshold. Since the establishment of the RCCs in 2022, the only incident before April 2025 that was above the threshold was the Scale 3 incident (blackout) in South-East Europe (SEE) on 21 June 2024. The system split on 24 July 2021 occurred before RCCs were established.

The investigation into the SEE blackout was completed on 25 February 2025 with the publication of the final report. With this report, the RCCs issued two recommendations for all ENTSO-E TSOs connected to the RCC tasks: one aiming at a more frequent update of the ID IGMs, the other addressing the voltage accuracy of the grid models. Both recommendations are also applicable to SWE, but are expected only in mid-2026. As of 28 April 2025, they were still pending, in line with the recommended timeline.

The TSOs included an additional recommendation concerning RCC tasks, aimed at resolving the tie-line rating inconsistencies in the SEE area. Generally, such inconsistencies are not limited to SEE but are also present in SWE. However, the recommendation only addressed the affected TSOs and did not include other TSOs. In contrast to the system split on 4 November 2006 in Germany, the tie-line rating inconsistencies did not contribute to the incident in SEE. Since RCCs generally consider the lower limits in regional processes, the impact of such inconsistencies is limited to local processes and not considered critical for RCC processes.

In addition to the mandatory RCC task of RIAR, Coreso provides a voluntary service to its shareholders: a "post-event analysis" service, which offers a basic investigation of incidents below the threshold. This may also involve the use of post-operational data, such as RTSN.

6.2.6.3 Crisis Scenarios

RCCs play a defined role in ENTSO-E's process of identifying regional crisis scenarios. This role is described in the **risk preparedness regulation** and further elaborated in the **methodology** for identifying regional electricity crisis scenarios, which was amended by ENTSO-E and RCCs in 2023 and submitted to ACER. The **amended methodology**, which addresses the role of RCCs at the different stages of the process for updating the regional crisis scenarios, was approved by ACER on 8 March 2024. The methodology describes four stages of updating the regional electricity crisis scenarios:

- Establishing a list of regional electricity crisis scenario candidates, starting with the previous list and providing updates
- Compiling a list of regional electricity crisis scenarios for evaluation
- Evaluation and ranking of electricity crisis scenarios
- Reporting on the most relevant regional electricity crisis scenarios

In all stages, the RCCs are supporting ENTSO-E, and TSOs are consulted or informed. ENTSO-E has the final responsibility for the report and the different process steps. This process delivers the most severe regional electrical crisis scenarios every fourth year. The TSOs are required to evaluate the inclusion of the regional electrical crisis scenarios in their risk preparedness plans.

In 2024, the regional electrical crisis scenarios were updated. This process resulted in the analysis of 23 scenarios, which can have different root causes, such as natural events (e.g. weather, earthquakes), anthropogenic causes (e.g. physical attacks), or other factors. So far, the investigation has identified no correlation between the incidents listed in the regional electricity crisis scenarios and the incident under investigation.

6.2.7 Outcome of the RCC Investigation

The RCC investigation reviewed the coordinated tasks applicable to the SWE region in relation to the event of BD 28/04. The review considered the objectives, scope, and execution of each task, considering the conditions observed before the incident.

Across all assessed tasks, the investigation found that the coordinated processes were carried out in line with the applicable methodologies, timelines, and roles defined at the regional level. The analyses performed did not identify abnormal conditions or elevated risks for the time frame corresponding to the event, and the outputs of the different tasks were mutually consistent within their respective horizons and assumptions. These analyses primarily focus on steady-state power flows and do not assess dynamic or voltage-related operational security limits.

The review also highlights that each coordinated task addresses specific aspects of system planning and operation, using forecasted inputs, predefined contingency sets, and fixed assessment horizons. As such, the RCC tasks provide a structured and forward-looking view of system conditions, but they are not designed to fully represent real-time operational dynamics or to anticipate rare, multifactorial events evolving over short time horizons.

In the context of the SWE region, the limited level of interdependence in terms of flows between borders means that many operational situations can be effectively managed at the national level, with regional coordination providing additional support where cross-border impacts are expected. This characteristic influences both the scope of RCC tasks and the extent to which regional coordination contributes to the management of events such as the one observed on BD 28/04.

During the incident under investigation, discussions highlighted several areas where the effectiveness and robustness of coordinated tasks could be further strengthened over time. These include improvements in data quality and forecast representativeness, closer continuity between planning and operational assessments, improvements in the modelling in the CGM used in CSA to assess voltage limits, and the extension of CSA into the ID time frame through the forthcoming implementation of ROSC. These developments are already underway within the existing regulatory framework and reflect the ongoing evolution of regional coordination.

Overall, the investigation concludes that the RCC tasks provided an appropriate assessment of system conditions within their defined scope, considering current modelling and process limitations. While the event of BD 28/04 unfolded outside the temporal and analytical reach of the coordinated tasks in place at the time, the findings of this investigation are in line with ongoing improvements in regional coordination and operational security arrangements.

6.3 Operational Planning in Spain

This section describes:

- The main regulatory framework relevant for the operational planning of the Spanish power system (6.3.1 to 6.3.5)
- As provided by RE upon request of the "Operational Planning" topic group established under the Expert Panel:
  - Technical solutions developed by RE to assess, plan, and operate the system in line with such regulatory framework (6.3.6 and 6.3.8)
  - A high-level explanation of the operational planning activity on 28 April 2025 (6.3.8)

When developing this description, the document seeks to answer specific questions raised by the Task Forces supporting the Expert Panel established to investigate the incident of 28 April 2025.

6.3.1 Spanish Regulation

The operational planning of the Spanish power system is highly regulated: the tasks of RE, as the system operator (SO), as well as the requirements for electricity undertakings, are described in detail in the following national regulations:

- Law 24/2013 (*Ley 24/2013 del Sector Eléctrico*)[41] and its amendments, complemented by its developments – e.g. RD 413/2014 (on matters relating to renewable generation, cogeneration, and waste) – establish the structure and responsibility for the electricity market.
- Methodologies, requirements, and procedures are adopted in line with national competences through Ministerial Orders, Resolutions of the Secretary of State of Energy, or CNMC Circulars, and are published in the national official gazette (B.O.E.).
- Operational Procedures/*Procedimientos de Operación* (P.O.) are a set of technical codes (about 40[42]) describing the roles, responsibilities, requirements, and criteria conditions for different processes, applied by RE as the SO of the Spanish power system and by other electricity undertakings, like system users, market participants, or network operators.
- Particularly relevant P.O.s for operational planning processes are:
  - **P.O. 1.1 and P.O. 1.4:** Security criteria and conditions for operating the transmission network
  - **P.O. 3.1:** General description of the operational planning process in Spain
  - **P.O. 3.2:** Establishes the technical restrictions process that addresses power system constraints in D-1, ID, and up to real time
  - **P.O. 7.4:** Specifics for voltage control in the transmission network

41 **BOE-A-2013-13645 Ley 24/2013, de 26 de diciembre, del Sector Eléctrico.**

42 The list of all up-to-date P.O. in force can be found at **https://www.ree.es/es/conocenos/marco-regulatorio/procedimientos-de-operacion**.

6.3.2 Operational Limits and Security Criteria in the Spanish Transmission Network

P.O. 1.1 sets the security criteria that are considered in the operational SA carried out by the SO. In particular, it establishes:

- The operational security limits to be respected in the normal state, in the "N situation" (base case), or under contingency conditions
- The set of ordinary and exceptional contingencies that shall be considered in the contingency analysis

The specific security limits set for the Spanish transmission system, in line with Article 25 of the SO GL, as well as the operational criteria applied by RE in the base case and under contingency, are summarised respectively in Table 6-14 and Table 6-15.

	Security Limits [43]
N situation	Voltages: 380 – 435 kV[44] (400 kV), 205 – 245 kV (220 kV)
Contingency (N-1) (line, trafo, PGM, reactance)	No permanent overload in lines (transitory ≤ 15 % for no more than 20 min) Transformers: Overload admitted up to 10 % in Winter (November–March), 0 % the rest of the year Voltages: 380 – 435 kV (400 kV), 205 – 245 kV (220 kV)
N-2 in specific conditions (double circuits, circuit + group)	Maximum overload in lines and transformers: 20 % in winter, 10 % in summer, 15 % in the rest Voltages: 375 – 435 kV (400 kV), 200 – 245 kV (220 kV)

Table 6-14: Security limits in the base case and under contingency in P.O. 1.1.

	Operational Criteria
N situation	Frequency and voltages within reference values[45] and prescribed limits No overloads
Contingency (N-1) (line, trafo, PGM, reactance)	No supply discontinuity
N-2 in specific conditions (double circuits, circuit + group)	No supply discontinuity No risk of voltage collapse

Table 6-15: Operational criteria in the base case and under contingency in P.O. 1.1.

In line with the requirements in P.O. 1.4, the SO aims to operate the network so as not to reach 420 kV. P.O. 1.1 sets the withstand capacity at 435 kV. This represents a deviation from the general rule allowed in Article 27(2) of the SO GL and constitutes a specific requirement for the Spanish system, which has long transmission axes and high variability in flows and line loading, and therefore covers the risk of voltages exceeding 420 kV under certain operational conditions.

6.3.3 Operational Planning Process

The Spanish operational planning process is primarily described in P.O.3.1.[46]

P.O.3.1, which was in force on 28 April, states the requirements for data exchange between RE as SO and market participants and introduces the processes (resolution of technical constraints, voltage control, balancing, etc.) that ensure that commercial trade schedules (internal and external) are compatible with operational security requirements, which are further regulated in other P.O.s (3.2, 7.4, and others).

The operational planning processes in P.O. 3.1 are summarised below and in Figure 6-15:

Before the D-1 market, RE publishes information related to the use of cross-border capacity as well as load and renewable generation forecasts and network unavailabilities.

43 The overvoltage limits are in line with Article 27(2) of the SO GL. Voltage limits in P.O. 1.1 have been in force since 1998

44 Under normal operating conditions, the voltage at the 400 kV level at the connection nodes will be between 390 and 420 kV. At the 220 kV level, the voltage will be between 205 and 245 kV. Eventually, maximum values of up to 435 kV and minimum values of up to 375 kV may occur at the 400 kV level. At the 220 kV level, voltages may eventually drop to 200 kV. Any installation directly connected to the transmission network must be able to withstand the above values without damage or disconnection.

45 Reference values in force on 28 April 2025 were 405 – 410 kV for assets connected at 400 kV and 220 – 225 kV for assets connected at 200 kV.

46 The one described here, which was in force on 28 April, was slightly modified in June 2025 to introduce new voltage control requirements (in line with the new voltage control operational procedure described in P.O. 7.4) and also in October, to integrate further temporary measures.

» At 13:00 of D-1 (the day before the day, D, of the energy delivery), the SO receives from OMIE (the NEMO for the Iberian Peninsula) the D-1 market results. Additionally, market participants deliver to the SO the disaggregation of their bids by physical units and the programme nominations. RE publishes a "daily basic programme" that is used as input in the D-1 technical constraints process (described in detail in P.O. 3.2).

» As a result of applying the D-1 technical constraints process in P.O.3.2, a "daily feasible programme" (PDVP) is published before 14h45 D-1. On some occasions, due to the complexity of SA in D-1 and the short time between the closure of D-1 market results and the first ID market auction (IDA1),[47] RE has had to inform NEMO that the TSO was not able to finish the technical constraints assessment before the IDA1. This was not the case on 28 April 2025.

» Market participants shall respect, when participating in ID auctions and in continuous ID markets, the limitations or restrictions requested in the previous technical constraints processes.

» ID market auctions are conducted in accordance with the EU-harmonised framework at 15:00 and 22:00 on D-1 and at 10:00 on D. For each auction, RE receives scheduling information from market participants from OMIE (NEMO), similar to the D-1 market, and publishes the "hourly programme" (PHC), 40 minutes after each auction.

» After each closure session of the ID continuous market, RE receives updated market schedules from OMIE, as well as generation unit unavailabilities and deviations communicated by market participants at the moment they occur, and also publishes an updated ID programme (PHFC).

» The continuous security assessment after the ID and continuous market takes place as part of what P.O. 3.2 describes as the "real-time technical constraints process". This process delivers a continuously updated feasible programme, called *P48*, that integrates, for the rest of the hours/MTUs of operational day (D) and all the hours/MTUs of the later day (assessment conducted in D-1 for the later D), the technically feasible schedule of all programming and physical units, including possible redispatches, limitations, and balancing.

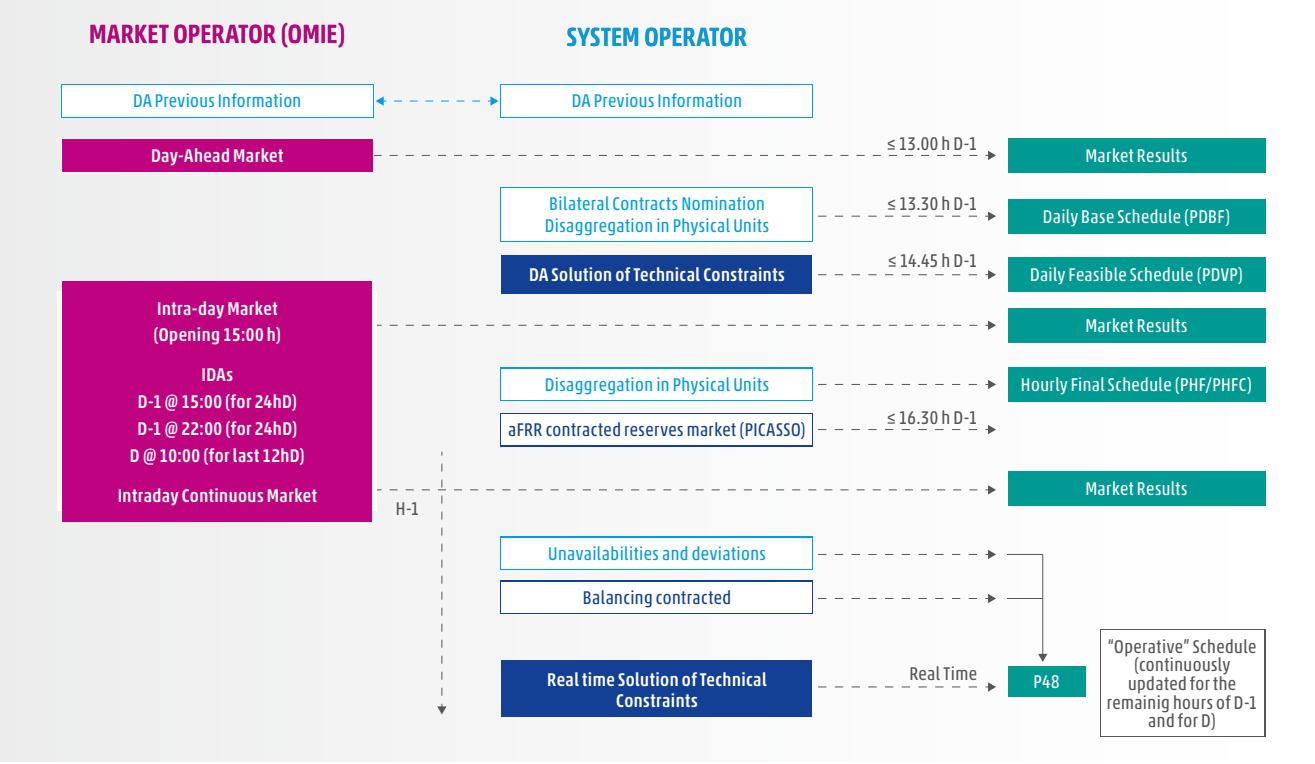


Figure 6-8: Flowchart summarising data exchanges and operational planning processes.[48]

47 Introduced pursuant to the implementation of Articles 37(5) and 55 of Regulation 2015/1222 or CACM GL, following ACER decision 01/2019.

48 Note that the timings in the flowchart were modified in June 2025 to introduce improvements for the reactive capacity market.

6.3.4 Technical Constraints Process

The technical constraints process of the Spanish power system is described in P.O.3.2. The one in force on 28 April[49] is described here.

In line with P.O. 3.2, a technical constraint is defined as "any circumstance or incident in the electrical system that, because of its impact on the electricity supply security, quality and reliability in line with the applicable P.O.s, require, pursuant to the SO technical assessment, the modification of the energy programs. In particular, Technical Constraints can be identified because of:

a) Non-compliance of the security conditions in steady-state or after contingency, as defined in the P.O. that establish the criteria for the operation and security of the power system.

b) Insufficient secondary and/or tertiary reserve.

c) Insufficient reserve capacity to ensure generation-demand adequacy.

d) Insufficient reserve capacity to control voltages in the transmission network.

e) Insufficient reserve capacity for system restoration."

The constraints solution process is applied to:

- The constraints in the transmission grid, which are assessed with respect to the requirements of Article 34 of the SO GL, and
- The constraints in the distribution grid,[50] communicated to RE by the distribution SOs.

As a result, RAs are ordered as needed, considering the activation time required.

6.3.5 Voltage Consideration Within the Technical Constraints Process (P.O. 7.4)

TSO operational security assessments monitor, for each element, overloads, short-circuit currents, angular differences, and voltage values, both in the base case and under contingency conditions.

Load flow calculations are performed as part of the operational planning processes described in Title 6.3.4, on D-1, ID, and continuously (at least every 10 minutes). These calculations verify that the thermal and voltage limits specified in P.O. 1.4 are respected for contingencies listed in Article 4.3.2 of P.O. 1.1, in line with Article 33 of the SO GL.

In accordance with Article 29(3) of the SO GL, to manage voltage variations inherent to the network evolution and dispatch conditions, measures are implemented already in the base case pursuant to P.O. 3.2 and P.O. 7.4, in force on 28 April 2025, requiring the dispatch of conventional generation units capable of providing real-time voltage control.

Resources to ensure voltage control are described in section 4.1.

49 It was modified in October 2025.

50 The Spanish technical constraints market is a local market that provides solutions for technical constraints in the distribution network when it involves the redispatch of generation units above 1 MW.

6.3.6 Criteria, Tools, and Data in the SAs Within the Technical Constraint Process

The volume of dispatched units is established by considering the programmed operational scenarios, which are characterised by network status parameters at both the global and zonal levels and informed by historical cases with similar network conditions and schedules. Consequently, units with real-time voltage control capability that are deemed necessary are programmed for scenarios typically characterised by:

- A low number of scheduled conventional power units in one or more regions (based on market schedules) and
- Low transmission network load (based on international schedules and renewable generation forecasts).

Additionally, assessments are regularly performed in the control room to check limits in contingency cases. When simulated contingencies show a case with voltages out of range, an evaluation is conducted and RAs are scheduled, including:

- Modification of line status (open/close)
- Shunt reactors switching/disconnection
- Dispatch of additional conventional power units to maintain voltage or to ensure it remains within acceptable limits in the event of contingencies

The input data and tools used to perform the security assessments in operational planning and real time, as referred to in the previous paragraph, are:

- Forecast cases (PSS/E format) built every hour at hh:15 using the GENES tool, covering each of the remaining hours of the current day and the 24 hours of the following day, with the following information:
  - International schedules (e.SIOS)
  - Production and storage unit schedules (available on the e.SIOS platform), including to disaggregation of programming units into physical units
  - Grid and production unit unavailabilities (DESC and e.SIOS databases)
  - TSO forecasts for demand, solar, and wind generation (Delfos forecast modules)
- Real-time (state estimation) cases built at least every 10 minutes. Data for the analysis are retrieved from SCADA, and load flow calculations are performed by the state estimator. Load flow values and contingency assessment results are available in the PCs of control room operators responsible for network assessment (transmission network operators and network operators in regional desks) in the different electrical regions. The information is presented:
  - In predesigned charts and screens tailored for each electrical region of the Spanish transmission network, allowing continuous monitoring of actual and historical voltage values at pilot nodes. Additional graphs and tables enable tracking of trends and other relevant variables.
  - In tables showing constrained element (thermal limits, voltage magnitudes, voltage phase angles, and short-circuit currents).
  - Real time/state estimation cases manually adjusted by the operator, for situations deemed necessary when electrical parameters indicate a potential risk of limit violation. In such cases, the operator can trigger an ad hoc manual contingency assessment.

Forecast scenarios (GENES) are used for the technical constraints process, which runs on D-1 and after the first ID auction. They are also used by operators when significant changes – like major outages, unavailabilities of generation units, or deviations from programming units – are communicated by market participants.

It is current practice for operators to continuously monitor the system and trigger specific assessments based on their network experience and judgement. This involves adjusting real-time cases from the state estimator with the forecast information when operators detect potentially risky trends in their regular monitoring of regional screens and graphs.

It is important to clarify that the simulations based on forecast scenarios, as well as the close-to-real-time security assessment performed by the TSO, must comply with regulations, consider contingencies in the contingency list, and account for voltage control capabilities in line with legally established requirements. Assessments are therefore performed using models and data delivered by system users for offline studies and SCADA representation.

6.3.7 Overview of the Security Assessment and Actions for BD 28/04

On 28 April 2025, none of security assessments performed by the SO detected the risk of the cascading overvoltage that occurred after 12:30.

The power system parameters were very different at 12:30 from those assessed in previous operational planning time frames. Nevertheless, the contingency assessment for the scenario around 12:30, based on structural data in line with operational procedures used by the TSO , indicated fulfilment of N-1 static security criteria. The cascading problem , as described in chapters 3 and 4 was unforeseeable by the applied security analysis.Below is an explanation of the most relevant decision timings and assessments, as well as the conditions considered on BD 28/04. They can be summarised as follows:

- At around 14:45 on 27 April 2025, D-1 technical restrictions were applied, with dispatching orders accounting for differences of approximately 2 GW in cross-border flows, as well as the typical uncertainty and variability of forecasts made many hours before real time.
- At around 22:30 on 27 April 2025, after the unavailability of Thermal Unit 4 was declared, IDAs 1 and 2 and the continuous market showed smaller variations in cross-border flows, while still representing 1.6 GW and assessing the night period as the most sensible from a voltage control perspective.
- At 9:00 on 28 April 2025, during the decision on whether to extend the limitation of Thermal Unit 5, a clear and increasing Spanish export tendency was observed, along with higher certainty in renewables programmes.
- At 10:17 on 28 April 2025, after a sustained tendency towards import reduction, the ID market schedules indicated a net exporting situation across all periods, with a downward trend in gradients.
- Schedules following IDA 3 confirm the exporting programmes.

A) Day-Ahead Studies: 27 April between 13:30 and 14:45

With regard to units providing voltage support in real time, in line with P.O. 7.4 in force, a total of 10 thermal units were initially scheduled as a result of the D-1 market and technical restrictions. These units were assessed as sufficient to provide the necessary voltage support.

In particular, in the South zone, one nuclear power plant had been scheduled in the market, and two additional thermal units were redispatched by the TSO,to support voltage variations associated with variability in internal and external schedules.

Those two units were selected in accordance with the technical and economic requirements and criteria set out in P.O. 3.2, as explained in 6.3.1.

It is relevant to detail the forecasted conditions at the time the needs were assessed during the D-1 technical restriction process, as well as their evolution in later ID assessments.

On 27 April during the D-1 technical restrictions process, the following conditions were observed:

- Strong gradients of photovoltaic production from 8:00 to 9:00 and 19:00 to 21:00.
- Changes in cross-border programmes: Spain exports most hours but imports from Portugal at night and from France from 13:00 to 16:00, resulting in gradient changes between 2:00 and 3:00 am and variability from 6:00 am until nearly the end of the day.
- Market prices (OMIE) – source of uncertainty in the production variation in Spain – decreasing from around 30 €/MWh to very low values (around 0 €/MWh) at 9:00 am, and then increasing from very low values at 20:00.

The Table 6-16 is an ex-post graphic representation of the Day Ahead assessment visualising the most challenging hours (hourly blocks in colours), from a voltage control perspective, associated with variations of renewable, importing/low transmission scenario and price changes.

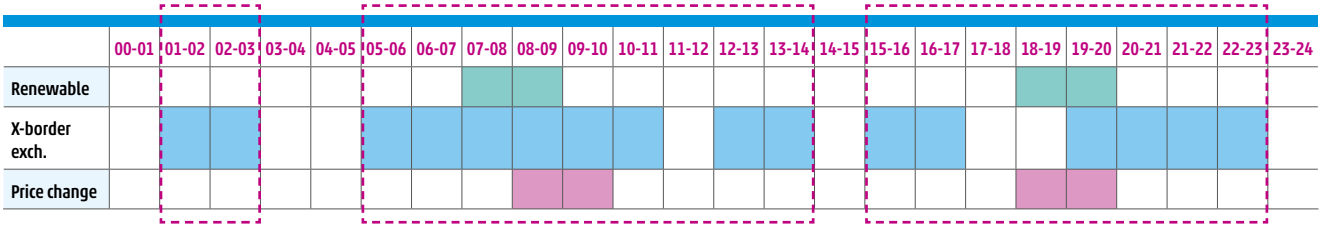


Table 6-16: Visualisation of the time periods requiring dynamic voltage control on 27 April around 15:00.

B) 27 April, ID Assessment After Unavailability Declared at 20:36

Around 20:00 on 27 April, the Thermal 4 generation unit scheduled through the D-1 technical restrictions process declared unavailability through 22:00 on 27 April. This was subsequently updated to the entire scheduling period for which it had been initially programmed (through 30 April).

At the time the unavailability was declared, the dispatching programme included other units that had been scheduled after the D-1 technical constraints process due to reserve requirements. These units also had the capability to provide real-time voltage control. In particular, in the South zone, Thermal Unit 5 (comprising a steam turbine and two gas turbines) had been dispatched until 0:00, and later from 16:45 on 28 April. Based on the unavailability of Thermal 4, a decision was made to extend the dispatch order of Thermal 5 until 9:00 to cover the voltage risk associated with the low-load night period. The assessment of a possible extension for the later period (9:00 –16:45) was postponed so that the decision could be made more efficiently using updated (less uncertain) information on market schedules and renewable production. During the night, as part of the real-time process (at 1:00), and taking into account the stability of cross-border programmes and the observed voltage profiles, operators adjusted the dispatch order for Thermal 5, shutting off one of the two gas turbines between 2:00 and 7:00.

C) 28 April, 8:30 – 9:00 through 12:00

Based on the available information on internal and international schedules for the period 9:00 to 17:00, and with reduced uncertainty regarding schedule changes compared to the D-1 situation, a re-evaluation performed close to 9:00 concluded that the dispatching programme from 09:00 to 16:45 was safe and efficient without extending the dispatch order of Thermal 5 between 9:00 and 16:45.

The trend in internal and external schedules was maintained during the subsequent hours (9:00 –12:00), with no indication of a risk of voltage slope due to schedule variations.

A few minutes after the third IDA, which closed at 10:30, the scenarios for 28 April from 12:00 onward were assessed, concluding that no new dispatch orders were required.

With Portugal, higher exports from Spain were observed, with no change in the flow direction (import/export).

Around 11:00, stable cross-border exchange and renewable production were expected for the 12:00 –13:00 scenarios.

For visual comparison with the conditions shown in Table 6-16, Table 6-17 illustrates the hours requiring real-time voltage control at the time when the decision to extend or schedule additional units for voltage control was made. The decision, made at around 9:00, was made based on updated (more accurate and less variable) information.

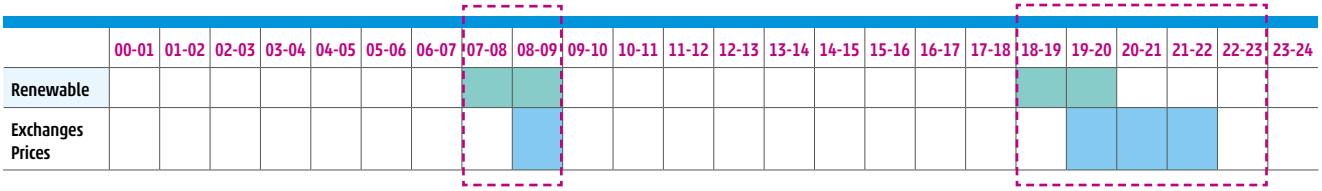


Table 6-17: Visualisation of the time periods requiring more real-time voltage control on 28 April at 9:00.[51]

51 This figure, like the previous one, is an ex-post graphic representation designed to illustrate the difference in the conditions assessed in D-1 (when Thermal Unit 5 was programmed for the later D day, 28 April) and those on D day around 9:00, at the time the decision was made whether to extend the dispatching order for Thermal Unit 5.

The security assessments associated with the technical constraints process in D-1, Intraday and real-time showed no violations of the operational security limits.

RE decisions in real time to solve oscillatory behaviour resulted in operational conditions significantly different from scenarios assessed in the operational planning.

D) 28 April, After 12:00

Presented below, and in Figure 6-9 and Figure 6-10 (depicting voltage in pilot nodes and topology actions in the period 12:00 – 12:32, is a summarised list of the events observed in the system and the real-time actions undertaken by operators to mitigate oscillatory phenomena and voltage issues:

- **12:03:** 0.6 Hz oscillation → voltage variations > 30 kV, undervoltage risk
  - Oscillation solved in < 5 minutes. Low damping observed until approximately 12:15
  - Actions associated with the oscillation and the resulting low damping situation: reduction (800 MW) of ES → FR exchange; pmode1 in HVDC, closing six circuits; switching off four REAs
- **12:16:** 0.6 Hz oscillation (low intensity and short time) → undervoltage risk
  - Action: switch off another REA
- **12:19:** Oscillation 0.2 → voltage variations > 20 kV, undervoltage risk
  - Oscillation solved in < 3.5 minutes
  - Actions: new reduction (500 MW) ES → FR exchange; closing an additional two circuits; switching off another REA
- **12:23:** Low damping, undervoltage in 400 kV SS of Almaraz and Carmona
  - Closing two circuits
  - Programming Thermal Unit 5 to reduce oscillatory risk (response time until 14h00)
  - Switch off two REAs
- **12:26:** Overvoltages
  - Action: connecting five REAs and changing the voltage setpoint of Baixas–Sta. Llogaia
- **12:30:** Voltage within allowed limits
- **12:32:** Voltages increase (approximately 15 kV in less than 1 minute)
  - An approximate reduction of 1 GW in the ES → FR exchange was observed, which, in isolation, was not considered sufficient to warrant a sharp 15 kV increase
- **12:32:55 – 12:33:18:** Trip of approximately 2 GW

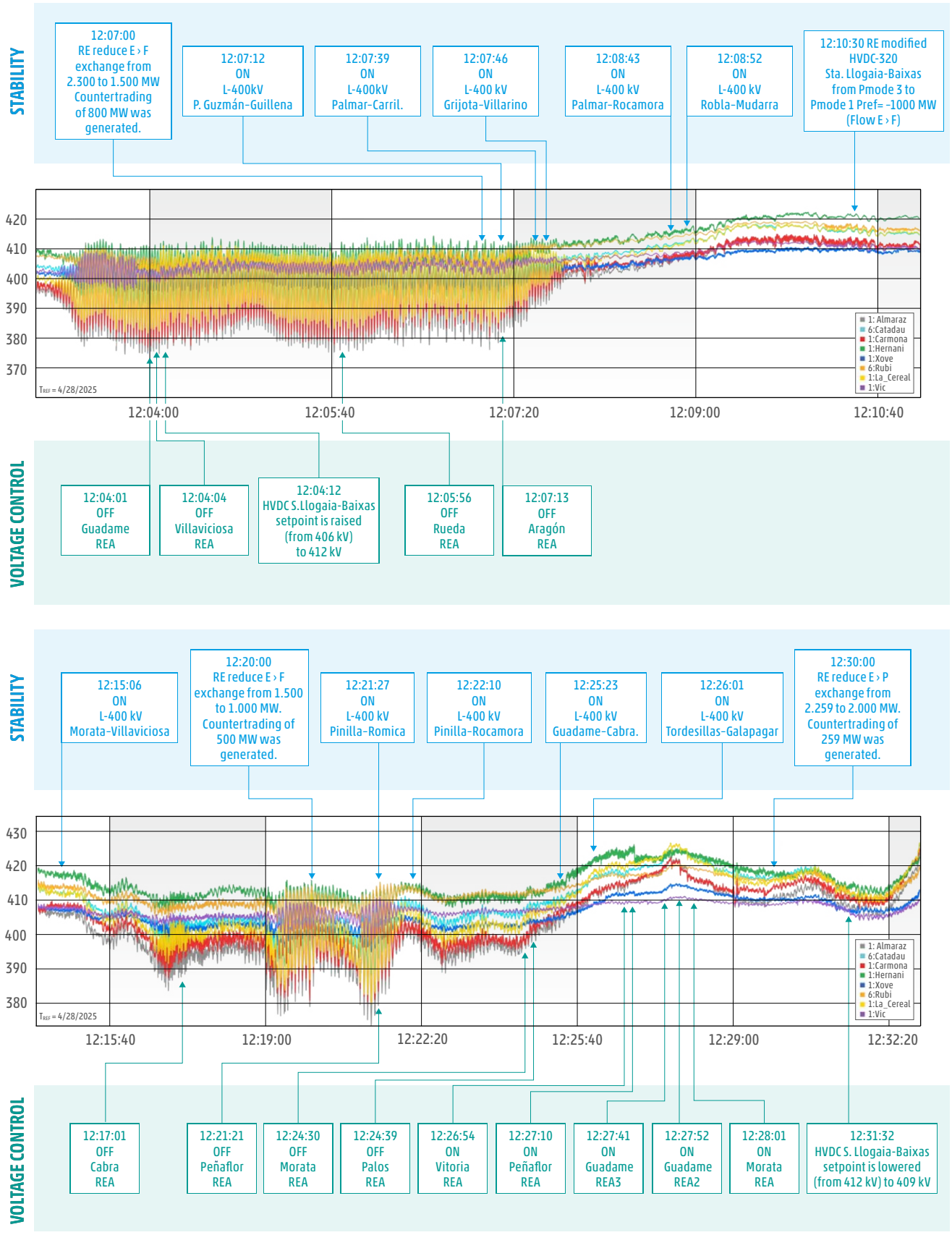


Figure 6-9 and Figure 6-10: Voltage in pilot nodes and topology actions in the period 12:00 -12:32.

6.3.8 Dynamic Stability Assessments

This section describes how Article 38 of the SO Regulation is currently implemented in Spain.

6.3.8.1 Offline Dynamic Stability Studies

Prospectively, as part of the analysis conducted to assess network development needs and the connection of new assets to the transmission network, offline dynamic stability studies are performed to assess the acceptability of dynamic conditions, in particular the maximum access capacity for generation per node and the fault clearing times These off-line dynamic stability assessments do consider transient thermal limits, voltage amplitude, rotor angle stability as well as frequency stability. In particular off-line dynamic voltage stability is assessed, under the perimeter of expected operational scenarios, to recommend a number of must-run of synchronous generation units, to assess the interest of dynamic voltage control network equipment capabilities, or to test the technical requirements to certain system users. Also, dynamic stability assessments are yearly conducted by Red Eléctrica to set the maximum value of the generation capacity that can be activated by the control room operator via the Active Power Reduction System (SRAP). Specific projects require ad hoc dynamic studies in close coordination with other TSOs, such as the new HVDC interconnection with France (Biscay Gulf) or the DSA performed in coordination with the neighbouring TSO (RTE) to assess the security of future higher cross-border exchange scenarios (8 GW).

Additionally, at the synchronous area level, dynamic frequency stability assessments are regularly performed using jointly developed dynamic grid models. These assessments are coordinated, including, for example, discussions on the adjustment of power system stabilisers (PSS) to mitigate inter-area oscillations.

6.3.8.2 DSA in Operation

The DSA tool available in the control centre performs stability assessments close to real time every 15 minutes. It simulates the dynamics for approximately 20 s, with a minimum calculation step of 1 ms, for 35 critical contingencies. Dynamic stability is evaluated with respect to frequency, voltage amplitude, phase angles, and thermal limits. Results are presented in the form of a semaphore indicating the main expected trends and risk probabilities.

The tool is undergoing improvements, with several parameters being refined to improve result accuracy. On the day of the incident, the tool outputs were not integrated with the operative process/alarming system. Ex-post verificationretrieval of the DSA tool results show that the tool did not indicate any stability risks for timestamps 11:50 and 12:25 (just before and after the oscillations).

6.4 Perspectives in forecasting oscillatory phenomena

The prediction of inter-area or forced oscillations in an electric power system is a challenging topic still under discussion in the technical literature and in the world of operation and planning.

The approach to the problem must inevitably consider uncertainty factors. First and foremost, the dynamic network models needed to correctly represent oscillatory phenomena. The process of building detailed supranational dynamic models for planning purposes is still ongoing and needs to be further improved and accelerated, as also suggested in the recommendations section of the report.

In addition to a thorough understanding of mathematical modeling, validation and measurement tools such as WAMS and on-site recordings and tests at the generators are required.

Having achieved correct system modeling, it is possible to evaluate the preventive measures: Power System Stabilisers, STATCOMs, Synchronous Condensers, critical grid configurations (i.e. lines opening for maintenance, generators displacements, ...), advanced system stabilization techniques (i.e. Wide Area Control Systems).

The planning and projecting stage can also define corrective interventions, such as Special Protection Schemes, advanced on line voltage control systems or Defence Plans improvements.

The availability of a correct modelling of the system jointly with a comparison with historical series of online monitoring of the system (i.e. WAMS) can help to deduce correlations between the triggering of oscillatory phenomena and particular grid topoligies or scenarios.

Some international research projects provide methodologies to forecast in advance with statistical approach the arising of an oscillation, but the level of uncertainty in the models (further complicated by the variability of renewables sources) illustrated thus far makes this stream still immature for systematic use, and real-time monitoring of the phenomena is more convenient in the immediate future.

Forecasting becomes even more challenging if, instead of predicting characteristic grid oscillations (e.g., inter-area), one wishes to predict phenomena related to interactions between inverters or between inverters and conventional generators. The degree of non-linearity and stochastic sequence of events that drive the generators to amplify an oscillation is really difficult to be intercepted in advance. In other words, the operational experience shows that in several situations theoretically prone to trigger oscillations, the phenomenon does not appear and, in some other cases, apparently stable, the oscillations arise.

That said, currently real-time modeling and monitoring remain the most effective tools available for studying and defining oscillations control strategies. The predictive approach, taking into account the limitations described and the efforts required to improve the tools, could be a valuable support for operations in the future.

It is important to explain that solving oscillations, moreover those of high amplitude, is an urgent priority[52] because of the high risk they entail for the power system. The rapidity of the actions is key. Actions shall be taken, of course, paying attention to the actual parameters of the system. The control room operators' sensitivity of the possible kV variations associated to the topological action is key (e.g. the operator is aware of the voltage levels in real time and of the few kV that the voltage would raise when closing a certain line). Additionally, operators have information on certain assets whose manoeuvre shall be prevented/limited.

52 The importance for the TSOs to act quick and avoid the high risk associated to oscillations is also explained and stated in several reports:
- Spanish oscillation 2016 **https://eepublicdownloads.entsoe.eu/clean-documents/SOC%20documents/Regional_Groups_Continental_Europe/2017/CE_inter-area_oscillations_Dec_1st_2016_PUBLIC_V7.pdf**,
- Italian oscillation 2017 **https://www.entsoe.eu/Documents/SOC%20documents/Regional_Groups_Continental_Europe/OSCILLATION_REPORT_SPD.pdf**

7 COMMUNICATION OF SYNCHRONOUS AREA MONITORS AND BETWEEN TSOs

This chapter outlines the key communications between synchronous area monitors (SAMs) and the affected Transmission System Operators (TSOs) during the incident, highlighting the exchanges that supported situational awareness and operational coordination. In addition to these interactions, it also includes messages exchanged via the ENTSO-E Awareness System (EAS) platform, as well as email correspondence between SAMs and the TSOs related to this event.

To maintain conciseness, the focus is on communications between TSOs; however, communications with significant grid users were also reviewed and considered during the analysis, despite not being detailed in this section.

7.1 Communication between TSOs

Within the Continental Europe Synchronous Area, control actions and reserves are organised in a hierarchical structure with scheduling areas, monitoring areas, load-frequency control (LFC) areas, LFC blocks, and the synchronous area with two coordination centres. The coordination centres facilitate communication and cooperation among TSOs, helping solve inter-TSO issues, and coordinate responses to potential disruptions or changes in power demand and generation. The coordination centres' role is carried out simultaneously by the TSOs Swissgrid and Amprion, with each responsible for a specific geographic portion of the synchronous area, ensuring that each TSO communicates with only one coordination centre. The TSOs affected by the incident on 28 April 2025 are all serviced by Swissgrid.

In the Continental Europe Synchronous Area, coordination centres also fulfil SAM obligations according to Article 133 of the System Operation Guidelines (SO GL). In this role, they monitor the synchronous area's frequency and other parameters and initiate international coordination, if needed. The SAM role is performed by Swissgrid in even-numbered months and by Amprion in odd-numbered months; therefore, Swissgrid was the active SAM during the event.

In the following, calls shown in blue are those with the SAMs designated for the Continental Europe Synchronous Area, namely Swissgrid or Amprion.

» **28 April, 09:08:**

RTE informed Swissgrid of notable frequency fluctuations observed in their network prior to the hourly change. No alarms were triggered at Swissgrid's control centre, and the oscillations were not clearly communicated by RTE, resulting in no immediate action.

» **28 April, 12:05:**

RE asked RTE to reduce cross-border exchange due to oscillations. Net transfer capacity (NTC) was set to 1,500 MW until 13:00.

» **28 April, 12:08:**

RE asked RTE to switch the HVDC interconnection Baixas (RTE)–Santa Llogaia (RE) system from PMode3 [ed. emulating an AC line] to PMode1 [ed. constant power set point] due to oscillations. RTE accepted.

» **28 April, 12:08:**

RE and REN coordinated on managing voltage oscillations and cross-border power flows. Initially, RE requested countertrading between 13:00 and 15:00 to reduce Portuguese imports, which REN accepted.

» **28 April, 12:11 – 12:13:**

RE contacted REN to request the cancellation of the previously agreed countertrading. Both parties agreed that REN would operate in TERRE (Trans European Replacement Reserves Exchange) while maintaining current import levels. REN then requested and received approval from RE to place 700 MW of replacement reserve needs in TERRE for the 13:00 – 14:00 period.

» **28 April, 12:14:**

RTE called RE to confirm the NTC value of 1,500 MW, which had not been updated via the tools.

» **28 April, 12:17:**

RE called REN to reinstate a countertrading request from 13:00 to 14:00 to reduce imports and alleviate power flows in Spain's 400 kV corridors. REN accepted.

» **28 April, 12:19:**

RE requested that RTE reduce NTC from 1,500 MW to 1,000 MW due to newly observed oscillations, with the reduction to remain in effect until 14:00. RTE accepted.

» **28 April, 12:20 – 12:27:**

RE requested that REN bring forward the countertrading originally scheduled for 13:00 – 14:00 [ed. final values activated were 13:00 – 13:15 with 507 MW, 13:15 – 13:30 with 590 MW, 13:30 – 13:45 with 531 MW, 13:45 – 14:00 with 538 MW] by adding quarter-hourly countertrading for the periods 12:30 – 12:45 with 247 MW and 12:45 – 13:00 with 88 MW to reduce Portuguese imports and alleviate power flows in Spain's 400 kV corridors. REN accepted both requests.

» **28 April, 12:34:**

RE informed RTE of a total blackout in the Iberian Peninsula and the initiation of system restoration. RE requested a gradual supply of 400 MW, which RTE agreed to, and both parties decided to set cross-border exchanges to 0 MW temporarily.

» **28 April, 12:34:**

Amprion reported a system split to Swissgrid and initiated discussions on coordinating with RE and REN, who were initially unable to communicate due to the blackout. It was noted that Swissgrid had attempted to reach the affected TSOs.

» **28 April, 12:34 – 12:36:**

RE called REN to confirm that both systems were in a blackout situation. RE and REN agreed to set cross-border exchange schedules to 0 MW and initiate restoration with black-start units and the assistance of the French system.

» **28 April, 12:37:**

RE requested RTE to energise the 400 kV TIE Baixas (RTE)—Vic (RE). RTE Toulouse was required to consult RTE Paris.

» **28 April, 12:39 – 12:40:**

RE requested RTE to restore voltage at the Hernani substation from the Argia substations. RTE successfully put the 400 kV TIE Argia (RTE)—Hernani (RE) into service.

» **28 April, 12:39:**

Amprion confirmed to Swissgrid that they, as SAM, would also issue the RG CE system split announcement within the EAS according to the system split procedure.

» **28 April, 12:41:**

Swissgrid contacted RTE for updates on the blackout; RTE confirmed plans to supply up to 400 MW to RE.

» **28 April, 12:41:**

RE asked RTE to initiate the agreed 400 MW exchange from France to Spain via AC, and both parties committed to expediting the process.

» **28 April, 12:47:**

TERNA called RTE to offer help.

» **28 April, 12:47:**

RE contacted ONEE and agreed to receive up to 100 MW of power from Morocco via one of the submarine cables connecting the two systems.

» **28 April, 12:47:**

RE informed Swissgrid of a blackout situation and shared initial restoration plans. No additional support was needed at that time.

» **28 April, 12:49:**

RE requested RTE to close the 400 kV TIE Baixas (RTE)—Vic (RE), but RTE clarified that the operation fell under the responsibility of their Paris control centre and not the control centre of Toulouse, which was called.

» **28 April, 12:49 –12:54:**

Swissgrid and Amprion confirmed the blackout in Spain and Portugal. They defined RE as the frequency leader for the Iberian Peninsula, Swissgrid for the rest of the Continental Europe Synchronous Area, and RTE as the resynchronisation leader according to the role definitions of the system split procedure.

» **28 April, 12:51:**

RTE informed RE that staff were being dispatched to Baixas to restore the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE), which could not be restarted automatically due to protection system constraints.

» **28 April, 12:57:**

Swissgrid confirmed to RTE the role assignments and that LFC controllers would remain unchanged.

» **28 April, 13:20:**

RE asked RTE about any issues or incidents on the tie-lines. RTE confirmed that no anomalies or incidents were observed. RE suggested that RTE send staff to close these tie-lines in case of remote-control failure.

» **28 April, 13:24:**

RTE informed Swissgrid that it had received too little information from Spain to assess the extent of the blackout. RTE reported Iberian frequencies around 50.3 – 50.4 Hz. It was still unclear to the RTE operator whether the French supply of power would be via AC or DC lines.

» **28 April, 13:27:**

RE asked RTE about the issue related to the lack of voltage transmission on the 400 kV TIE Baixas (RTE)—Vic (RE). RTE was seeing high voltages. It was agreed to energise the line. RE would adjust a load of 100 MW to help reduce the voltage. Additionally, it was agreed to increase NTC to 950 MW.

» **28 April, 13:36:**

RE provided an updated assessment, confirming a total blackout across the Iberian Peninsula, with all regions affected. The cause remained unknown.

» **28 April, 13:40:**

RTE contacted Swissgrid to request information on available capacity to support their system.

» **28 April, 13:40:**

RTE reported to RE that the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) could be started and voltage regulated, requesting a switch to PMode3 [ed. emulating an AC line]. However, RE indicated that the mode change was not feasible due to the network's weakened condition.

» **28 April, 13:41:**

REN enquired if RTE had already started the Spanish grid restoration. RTE confirmed that it was in progress.

» **28 April, 13:45:**

REN informed Swissgrid that Portugal was experiencing a nationwide blackout. REN attempted grid restoration using two black-start units, with no immediate support from Spain. One unit had failed multiple times. The cause of the blackout was still unknown.

» **28 April, 13:50:**

Swissgrid informed Amprion that RE was energised by RTE and REN was restoring bottom-up.

» **28 April, 14:02 –14:04:**

RE informed RTE of their ability to regulate voltage via the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) at Baixas and planned to activate the STATCOM function at Santa Llogaia. To support this, RE requested switching Link-2 from "Standby" to "Decoupled", which RTE approved.

» **28 April, 14:14:**

Swissgrid asked RTE about the grid situation in France, Spain, and Portugal. RTE reported a 600 MW exchange with Spain, with two interconnections [ed. 400 kV TIE Argia (RTE)—Hernani (RE) and 400 kV TIE Baixas (RTE)—Vic (RE)] already in service, and a third line expected to follow. RTE wanted to send 400 additional MW through the HVDC line.

» **28 April, 14:17:**

RTE asked RE if the 220 kV TIE Biescas 2 (RE)—Pragneres (RTE) could be put into service. RE answered that it would try to find out and then inform RTE.

» **28 April, 14:20:**

RE informed RTE about the closing of a switch at the converter and notified them of the move to "Stopped" and then "Decoupled" mode to regulate voltage as STATCOM.

» **28 April, 14:34:**

In response to the request made at 14:17, RE informed RTE that it was not possible to put the 220 kV TIE Biescas 2 (RE)—Pragneres (RTE) into service due to ongoing work.

» **28 April, 14:35:**

RE contacted ONEE and requested renewed voltage from the Moroccan electrical system.

» **28 April, 14:41:**

RE requested RTE to increase the cross-border exchange from 950 to 1,200 MW. RTE agreed with a 1,200 MW NTC.

» **28 April, 14:54:**

RE informed RTE that they could move Link-2 to "Stopped", which required a coordinated manoeuvre. RTE performed the manoeuvre first, followed by RE. RE and RTE then agreed to change to "Decoupled" mode without needing to coordinate operation in "STATCOM" mode.

» **28 April, 15:00:**

TERNA asked RTE about the situation at the Spanish border.

» **28 April, 15:01:**

RE asked if REN already had generation in service. REN reported that black-start capacity power plants were experiencing issues and that it was waiting for these power plants to be successfully started up and connected to the grid.

» **28 April, 15:06:**

Swissgrid informed Amprion that two tie-lines between France and Spain were back in operation and 600 MW were being exchanged. Amprion informed Swissgrid in its role as frequency leader that the Temelin power plant tripped in the Czech Republic, but that this was probably not linked to the blackout.

» **28 April, 15:13:**

Swissgrid enquired with RTE about a missing file with the scheduled exchange. There were misunderstandings between SG and RTE during the call. RTE confirmed that it would send the missing file.

» **28 April, 15:20:**

RTE reported to RE an exchange of 1,391 MW with an NTC of 1,200 MW. RE suggested increasing the cross-border exchange to 1,400 MW and NTC to 1,500 MW until 17:00.

» **28 April, 15:31:**

RTE informed RE that the 400 kV TIE Argia (RTE)—Hernani (RE) N-1 was causing an overload on the 220 kV TIE Argia (RTE)—Arkale (RE) due to the 220 kV PST Arkale (RE) being out of service.

» **28 April, 15:32:**

RE requested RTE to connect Link-1 from HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) for power transmission. Coordination was confirmed.

» **28 April, 15:35:**

RE informed RTE that the 220 kV PST Arkale (RE) [ed. and the 220 kV TIE Argia (RTE)—Arkale (RE)] would be put into service, establishing tap 17.

» **28 April, 15:37:**

REN reported issues with starting black-start plants to RE. RE could not assist REN at the time.

» **28 April, 15:39:**

RE informed RTE that they had found issues with docking the 220 kV PST Arkale (RE) and indicated that it may be necessary to open the line [ed. 220 kV TIE Argia (RTE)—Arkale (RE)].

» **28 April, 15:43:**

Swissgrid informed Amprion that three tie-lines between France and Spain [ed. 220 kV TIE Argia (RTE)—Arkale (RE), 400 kV TIE Baixas (RTE)—Vic (RE) and 400 kV TIE Argia (RTE)—Hernani (RE)] were back in operation.

» **28 April, 15:50:**

Amprion questioned RTE about a 2,500 MW demand on PICASSO. RTE mentioned a negative price and said that 500 MW would be coming back online soon. RTE noted that there would be an excess of power after 16:00. Amprion asked RTE about the situation in Spain, but RTE had no further information.

» **28 April, 15:52:**

RE requested power from RTE via the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) Link-1 in PMode3 [ed. the line was switched back to PMode1 after 1 min]. RE took the master control to make the changes.

» **28 April, 15:56:**

RE informed RTE that the 220 kV TIE Argia (RTE)—Arkale (RE) had to be decoupled to connect the PST.

» **28 April, 16:02:**

RTE reported to RE that the flow through Santa Llogaia was moving from Spain to France.

» **28 April, 16:04:**

RE informed RTE of the intention to change the flow direction at the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE). RTE confirmed and agreed to open the 220 kV TIE Argia (RTE)—Arkale (RE).

» **28 April, 16:09:**

Swissgrid asked RTE about the current grid situation and offered support. France did not require assistance at that time.

» **28 April, 16:13:**

RE informed RTE that the 220 kV TIE Argia (RTE)—Arkale (RE) and 220 kV PST Arkale (RE) were now operational. RE reported that power transmission had stopped due to an angular difference issue.

» **28 April, 16:13:**

Swissgrid asked RE about the current status of the system reposition and whether all transmission interconnection lines with France had been reconnected. RE confirmed that restoration efforts were ongoing and all interconnection lines with France were operational. The network was radial, and efforts were being made to create a mesh configuration. The exchange capacity stood at 1,500 MW. Swissgrid further enquired if any black-start power plants were in operation. RE said that one unit was active in the southern region, and more groups were expected to start up shortly. RE reported that an area had been connected to Morocco. Swissgrid asked about the situation at the border with Portugal, and RE conveyed that REN had reported issues with black-start plants. RE had several isolated zones, and once these were connected, RE would be able to provide voltage support to REN.

» **28 April, 16:39:**

Swissgrid enquired with RTE regarding the current grid situation, and there was no news to report.

» **28 April, 16:55:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 16:56:**

ONEE called RE and offered to provide up to 200 MW of support from its system. The offer was accepted on the same call.

» **28 April, 16:58:**

RTE informed RE their intention to couple busbars in ARGIA 220 kV. Approval was given.

» **28 April, 17:30:**

RTE informed RE that from 20:00, the cross-border exchange would proceed as scheduled before the incident, with 3,000 MW. RTE suggested informing its balance team to extend the 1,500 MW until midnight.

» **28 April, 17:35:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 18:06:**

RTE informed Swissgrid that as of 19:00, they would no longer be able to balance themselves due to a lack of energy from Spain and requested a countertrade of 500 MW.

» **28 April, 18:07:**

ONEE contacted RE to ask if the Spanish electricity system was divided into several electrical zones or if all electrical zones were connected. RE explained that it was divided, with the northern zone of the Spanish system connected to the European grid via the French grid, and the southern zone connected to the Moroccan grid.

» **28 April, 18:09:**

RTE reported a flow of 1,900 MW instead of 1,500 MW to RE. RE would regulate it.

» **28 April, 18:11:**

RTE informed RE that 1,700 MW was available until midnight.

» **28 April, 18:19:**

RTE reported to RE that the flow was 2,000 MW. RE would adjust it. RTE proposed increasing NTC to 2,000 MW.

» **28 April, 18:23:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 18:34:**

RE called REN and offered to energise the 220 kV TIE Aldeadávila I (RE)—Pocinho (REN), providing voltage support from its system (up to 20 MW of power support). The offer was accepted on the same call.

» **28 April, 19:08:**

ONEE contacted RE to request that no more load be taken from Morocco via ESMA1 due to low frequency. RE asked ONEE if it could continue taking 300 MW, and ONEE agreed, but instructed them not to exceed 300 MW.

» **28 April, 19:21:**

ONEE contacted RE to enquire about the progress of the restoration of the Spanish electricity system, specifically whether generation had been restored and whether the load on ESMA 1 would rise above 370 MW. RE advised that generation was gradually being restored and that the load on ESMA 1 was not expected to exceed that value.

» **28 April, 19:22:**

Swissgrid enquired with REN about the current grid situation. REN reported that there were three restoration areas, one of which was being supported by RE. At that time, 170 MW of load was being supplied.

» **28 April, 19:45:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 19:49:**

REN requested that RE reconnect the 220 kV tie-lines, specifically the 220 kV TIE Aldeadávila II (RE)—Pocinho (REN) or the 220 kV TIE Pocinho (REN)—Saucelle 1 (RE), with a preference for including both if feasible. RE suggested the reconnection of the 400 kV TIE Cedillo (RE)—Falagueira (REN). REN indicated that the reconnection of the 400 kV TIE Cedillo (RE)—Falagueira (REN) would occur at a later stage, and that they preferred to connect the 220 kV lines at that time. RE accepted.

» **28 April, 20:00 – 20:11:**

RTE requested emergency reserves from Amprion. It was agreed to deliver 400 MW from Germany to France from 21:00 until midnight.

» **28 April, 20:23:**

RE requested that RTE change the state of Link-2 from "Decoupled" to "Coupled" to enable power transmission. RE indicated a preference for receiving power through the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) rather than charging the 400 kV TIE Baixas (RTE)—Vic (RE). RTE agreed.

» **28 April, 20:31:**

RE reported to RTE that they had changed the power setpoint (PSP) from 600 MW to 300 MW for the HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) Link-1. RE confirmed no exchange variation, only the recovery of Link-2. RTE noted a 1,300 MW deviation [ed. Between the 2,000 MW schedule and the 700 MW physical flow] that may affect frequency.

» **28 April, 20:35:**

RE requested RTE to adjust the cross-border exchange to 800 MW from that moment on due to the existing deviation to maintain frequency stability. RTE approved the change.

» **28 April, 20:49:**

RE requested RTE to adjust the cross-border exchange to 1,400 MW due to the reduced imbalance. RTE approved.

» **28 April, 20:53:**

RE requested ONEE to reduce the cross-border exchange to the 300 MW agreed in the initial program. ONEE reported frequency issues as the reason for the higher flow and agreed to reduce generation.

» **28 April, 21:01:**

RE requested ONEE to close the second Spain–Morocco tie-line. RE energised the line and ONEE closed the line, reducing load on the other interconnection.

» **28 April, 21:02:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 21:08:**

ONEE informed RE that 900 MW was too much for their system and it was experiencing overloads.

» **28 April, 21:11:**

RE requested RTE to change the HVDC interconnection Baixas (RTE)–Santa Llogaia (RE) from "Decoupled" to "Stopped", waiting for 20 minutes, and then moving it to "Coupled" to put it into service. The manoeuvre was completed.

» **28 April, 21:13:**

ONEE asked RE about the interconnection between Spain and France. RE requested ONEE to regulate the 300 MW cross-border exchange, while ONEE asked RE to reduce production.

» **28 April, 21:14:**

Swissgrid contacted RTE for an assessment of the current situation. RTE reported that the frequency was stable and the situation at the border with Spain was under control. It confirmed that activation of the remaining power from Spain could continue.

» **28 April, 21:18:**

RTE informed RE that they planned to set the scheduled exchange to zero to regulate imbalances. They noted that increasing cross-border exchange could cause overloads when connecting two zones in a loop. RTE asked for cross-border exchange values for 29 April, suggesting 2,000 MW from France to Spain and 0 MW from Spain to France. RE confirmed these values.

» **28 April, 21:19:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 21:25:**

REN requested RE to connect the 400 kV TIE Puebla de Guzman (RE)–Tavira (REN) to begin supplying consumption in the Algarve (southern region of Portugal). RE accepted.

» **28 April, 21:26:**

ONEE asked RE about the deviation in the 300 MW exchange.

» **28 April, 21:31:**

RTE reported to Swissgrid that they had set the scheduled export value to Spain to 0 MW to better regulate with secondary control because otherwise, the scheduled cross-border values would change too frequently. The original scheduled exchange (without incident) with Spain was 3,000 MW from Spain to France, but with Spain now importing about 1,000 MW, RTE was buying 4,000 MW from Germany and Switzerland. Amprion and Swissgrid remained available to assist with any balancing issues.

» **28 April, 21:41:**

Swissgrid asked RE about nuclear plant issues or safe mode status. RE confirmed they were all in safe mode. Regarding the exchange with France, RE reported 1,400 MW. Swissgrid asked if RE was still in restoration mode and if assistance was needed. RE answered that the restoration was still ongoing and that no assistance was needed.

» **28 April, 21:41:**

RE requested RTE for information regarding the Link-2 manoeuvre. The waiting period had elapsed, and ready status could be applied. The manoeuvres were executed, and power transmission began. HVDC interconnection Baixas (RTE)—Santa Llogaia (RE) was normalised in PMode3 [ed. emulating an AC line], but PODP (power oscillation damping P) remained disabled.

» **28 April, 21:43:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 21:47:**

Swissgrid asked REN to confirm the existence of a synchronous connection from FR to PT.

» **28 April, 21:48:**

RE requested Coreso to validate the value provided by RTE, with an export of 0 MW and an import of 2,000 MW. Coreso confirmed.

» **28 April, 21:49:**

Swissgrid asked RE to verify if Portugal was connected to France through Spain and if Portugal was operating at the European frequency. The affected TSOs confirmed the existence of a synchronous connection from France to Portugal.

» **28 April, 21:50:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **28 April, 21:52:**

ONEE asked RE about the need for additional energy. RE responded that no further energy was currently required.

» **28 April, 22:03:**

ONEE asked RE about the connection between Spain and France. Information was provided, along with frequency measurements.

» **28 April, 22:10:**

REN informed RE that grid conditions were fulfilled for the connection of the 400 kV TIE Cedillo (RE)—Falagueira (REN) and requested the connection. RE accepted.

» **28 April, 22:28:**

RE informed REN that they would change their LFC to peninsular mode. REN agreed.

» **28 April, 22:33:**

RTE enquired with RE about cross-border exchange capacities for 29 April. RE confirmed that the capacity from Spain to France was 0 MW and reported readiness to reconnect secondary regulation. RTE requested information about the schedule to achieve balance. Both parties agreed on a 1,400 MW exchange from France to Spain between 00:00 and 03:00.

» **28 April, 22:34:**

RTE requested Amprion to extend and increase emergency power. It was agreed to continue with 400 MW from 00:00 until 01:00 on 29 April and then 650 MW from 01:00 to 10:00.

» **28 April, 22:50:**

RTE informed Swissgrid that the Spanish load frequency controller would soon be operational. Subsequently, the exchange program from France to Spain would be increased from 0 MW to 1,400 MW. However, RTE noted that the exchange program would remain unchanged for the next few hours to allow for better frequency regulation.

» **28 April, 22:55:**

RTE thanked Amprion for their support. RTE received a schedule and entered it into their system.

» **28 April, 23:14:**

Amprion confirmed to Swissgrid the delivery from Amprion to RTE for the next hours: 400 MW from 00:00 to 01:00 and 650 MW from 01:00 to 10:00.

» **28 April, 23:27:**

ONEE enquired with RE about the restoration process. RE stated that it was ongoing and could not provide a status update.

» **28 April, 23:41:**

RE informed RTE that the LFC would be reconnected with a 100 MW schedule and then revert to 1,400 MW. RTE required one hour's notice to adjust its system and did not support changes before 00:30. A 0 MW schedule from 0:00 was agreed, allowing the connection of the LFC.

» **28 April, 23:52:**

ONEE enquired with RE about the situation, and RE explained about the connections with other TSOs and noted that the restoration process had not yet been completed.

» **28 April, 23:55:**

REN and RE agreed to change their LFC mode to power and frequency control, with scheduled programs on the PT–ES border at 0 MW.

» **29 April, 00:03:**

RTE expressed concerns to Swissgrid about its ability to maintain balancing at the border with Spain. Swissgrid offered RTE the option of activating power via the Stage 2 frequency procedure in an emergency. RTE would respond promptly if necessary.

» **29 April, 00:08:**

RE informed RTE that the LFC in Spain has been successfully activated. RTE confirmed that frequency regulation in Spain was now being managed by the Spanish system. RE reported that it had recovered 70 % of the load. No information was available about the REN restoration.

» **29 April, 00:11:**

RTE announced to Swissgrid that the grid controller in Spain was in operation. The initial exchange program [ed. schedule agreed before the blackout] indicated 0 MW from Spain to France and 3,000 MW from France to Spain. RTE had no information about the LFC in Portugal.

» **29 April, 00:17:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **29 April, 00:47:**

REN asked RE to establish contact between the NCC management teams of REN and RE using the phone consoles in the control rooms. REN confirmed that it was already supplying all transmission substations with consumption and provided a summary of the generation profile (solar and wind connected to the transmission grid with restrictions). REN reported their intention to change the EAS state to emergency (red).

» **29 April, 01:54:**

REN informed RE that they were performing LFC regulation at 0 MW (schedule program) and asked for the match confirmation of the cross-border exchange schedule. RE confirmed that they would keep the scheduled programs (Spain to Portugal) at 0 MW on 29 April. RE reported that they would impose an NTC at 0 MW (Spain to Portugal) for 30 April. REN reported that they would continue limiting renewable (wind and solar) production connected to the transmission grid for better regulation. REN suggested changing the EAS status from the restoration to the emergency state, maintaining the suspension of energy markets. RE reported that they would remain in the restoration state for the time being.

» **29 April, 02:05:**

RE requested confirmation from REN regarding the scheduled exchanges from Portugal to Spain for 29 April. REN confirmed that the scheduled program would be 0 MW.

» **29 April, 02:16:**

RE informed ONEE that the supporting cross-border exchange was no longer required, and it was agreed to set it to 0 MW.

» **29 April, 02:19:**

Swissgrid asked REN for confirmation concerning the end of the restoration state of the Portuguese system. REN confirmed and reported that all transmission substations with consumptions were supplied.

» **29 April, 02:20:**

REN and RE called about an EAS malfunction. RE was seeing the Portuguese system in normal state when REN had switched to emergency state.

» **29 April, 02:22:**

Swissgrid informed RE that Portugal has updated the status in EAS to alert and confirmed that all loads were connected to the grid. Swissgrid requested information regarding the restoration process status in Spain. RE indicated that the restoration process was in its final stages. However, certain loads had not yet been recovered due to issues encountered by some DSOs when doing so. RE anticipated that all loads would likely be connected shortly.

» **29 April, 02:23:**

REN and RE called again about the EAS malfunction. RE was seeing the Portuguese system in normal state when REN had switched to emergency state.

» **29 April, 02:24:**

Swissgrid asked RTE for a general update. RTE confirmed ongoing stable border exchanges between Spain and France.

» **29 April, 02:25:**

REN and RE called again about an EAS malfunction. RE was viewing normal state of the Portuguese system when REN had changed to emergency.

» **29 April, 02:29:**

RTE asked RE about the status of the network restoration. RE reported that the transmission grid restoration was completed, and they were waiting for the restoration of loads at lower voltage levels. RE reported uncertainty about generation volume.

» **29 April, 02:29:**

REN and RE called again about the EAS malfunction. REN confirmed the end of the restoration state in the Portuguese system, switching to emergency state. REN confirmed that all transmission substations with consumptions were supplied.

» **29 April, 02:30:**

RE confirmed to REN that they were seeing the Portuguese system in emergency state in the EAS.

» **29 April, 02:44:**

Swissgrid provided Amprion with clarification regarding the current grid situation for the restoration of the Iberian Peninsula.

» **29 April, 03:11:**

Swissgrid asked RE about the EAS emergency status. RE confirmed all loads were connected, generation was stable, and there were no incidents. However, markets in the Iberian Peninsula remained suspended as a precaution. Swissgrid asked about next steps for balancing and exchange schedules. RE stated that the market was currently halted but expected normalisation by 30 April.

7.2 ENTSO-E Awareness System (EAS)

The EAS is activated in response to critical incidents, facilitating a coordinated and timely response among multiple TSOs within the ENTSO-E framework. This mechanism not only supports operational collaboration but also ensures effective communication of status assessments across Europe, enhancing situational awareness and enabling informed decision-making. Below is a detailed overview of the changes in system states as well as the information communicated through EAS for each affected TSO during the sequence of events.

» **28 April, 12:40:**
REN – Normal state to blackout state

» **28 April, 12:40:**
RE – Normal state to blackout state

» **28 April, 12:49:**
Swissgrid (as CC South) – Normal state to emergency state

» **28 April, 12:49:**
Amprion (as CC North) – Normal state to emergency state

» **28 April, 12:50:**
RTE – Normal state to emergency state

» **28 April, 13:08:**
RTE – Info: French synchronous leader for the split grid between France and Spain

» **28 April, 13:10:**
RE – Blackout state to restoration state

» **28 April, 13:20:**
Swissgrid (CC South) – Info: Swissgrid is the frequency leader of the main South Island, Spain is the frequency leader of their island, and France is the resynchronisation leader

» **28 April, 14:35:**
RTE – Emergency state to alert state

» **28 April, 17:05:**
REN – Blackout state to restoration state

» **29 April, 02:13:**
REN – Restoration state to emergency state

» **29 April, 03:00:**
RE – Restoration state to emergency state

» **29 April, 11:15:**
Swissgrid (as CC South) – Emergency state to normal state

» **29 April, 11:15:**
Amprion (as CC North) – Emergency state to normal state

» **29 April, 11:20:**
RTE – Alert state to normal state

» **29 April, 14:40:**
RE – Emergency state to alert state

» **29 April, 14:40:**
REN – Emergency state to alert state

» **30 April, 12:40:**
RE – Alert state to normal state

» **30 April, 12:50:**
REN – Alert state to normal state

7.3 Email Communications between SAM and the TSOs

This section includes the official email communications between SAM and the TSOs.

28 April, 12:53: From Swissgrid to all TSOs:

Dear Colleagues

We announce that a system split is occurred in RG CE. The "procedure in case of system separation and resynchronisation of separated grid areas" is started.

Details of the system split (number, borders, and TSOs of islands) and future information are shared via European Awareness System (EAS).

If the system split is inside of your control area or on your tie lines please set EAS to emergency state: critical event and separation from the grid (Flow chart: step 4).

Kind regards

28 April, 22:14 – From Swissgrid to all TSOs:

Update!

Dear Colleagues

RTE is still balancing the Iberian peninsula via aFRR/mFRR activations. RTE has to activate the difference to the scheduled values (up to 3,000 MW export Spain → France) as well. Currently the margins are sufficient.

In case there is a lack of balancing energy, Amprion and Swissgrid are ready to support RTE and stabilise the system frequency.

Kind regards

28 April, 23:20 – From Swissgrid to all TSOs:

Information!

Dear colleagues,

since 28.04.2025 23:17 (CET) we observed a permanent steady state frequency deviation with a fast-growing grid time deviation of +6 sec in the last 4 hours and a total grid time deviation of +4.27 sec.

The cause of the frequency deviation could be identified (reason: restoration state Spain and Portugal).

The Coordination Centres [ed. SAMs] will further monitor the situation.

If the situation is not getting worse no actions are necessary by uninvolved TSOs until further notice.

Thank you all for your collaboration.

Kind regards

29 April, 02:48 – From Swissgrid to all TSOs:

Information!

Dear colleagues,

from 28.04.2025 23:17 to 29.04.2025 02:45 (CET) we observed a permanent steady state frequency.

The grid time deviation in the last 4 hours is now: +1.93 sec

Total grid time deviation is now: +5.55 sec

Now the alarm is cancelled.

Reason for cancellation: grid time deviation is decreasing (≤2sec in the last 4 hours)

No further action is needed.

Thank you all for your collaboration.

Kind regards

8 CLASSIFICATION OF THE INCIDENT BASED ON THE ICS METHODOLOGY

The ICS methodology was originally developed in accordance with Regulation (EC) No 714/2009 of the European Parliament and of the Council of 13 July 2009, as repealed the by Regulation (EU) 2019/943 of the European Parliament and of the Council of 5 June 2019 on the internal market for electricity (recast), and updated to fulfil the objectives and the security indicator requirements laid out in Article 15 of Commission Regulation (EU) 2017/1485 of 2 August 2017 establishing a guideline on electricity transmission system operation (SOGL). The definitions and concepts in this methodology are in line with Articles 15 and 18 of the SOGL and further extended to describe the real-time situation of the TSO's system.

Scale 0 Noteworthy incident		Scale 1 Significant incident		Scale 2 Extensive incident		Scale 3 Major incident / 1TSO	
Priority / Short definition (Criterion short code)		**Priority / Short definition (Criterion short code)**		**Priority / Short definition (Criterion short code)**		**Priority / Short definition (Criterion short code)**	
#20	Incidents on load (L0)	#11	Incidents on load (L1)	#2	Incidents on load (L2)	#1	Blackout (OB3)
#21	Incidents leading to frequency degradation (F0)	#12	Incidents leading to frequency degradation (F1)	#3	Incidents leading to frequency degradation (F2)		
#22	Incidents on transmission network elements (T0)	#13	Incidents on transmission network elements (T1)	#4	Incidents on transmission network elements (T2)		
#23	Incidents on power generating facilities (G0)	#14	Incidents on power generating facilities (G1)	#5	Incidents on power generating facilities (G2)		
		#15	N-1 violation (ON1)	#6	N violation (ON2)		
#24	Separation from the grid (RS0)	#16	Separation from the grid (RS1)	#7	Separation from the grid (RS2)		
#25	Violation of standards on voltage (OV0)	#17	Violation of standards on voltage (OV1)	#8	Violation of standards on voltage (OV2)		
#26	Reduction of reserve capacity (RRC0)	#18	Reduction of reserve capacity (RRC1)	#9	Reduction of reserve capacity (RRC2)		
#27	Loss of tools and facilities (LT0)	#19	Loss of tools and facilities (LT1)	#10	Loss of tools and facilities (LT2)		

Figure 8-1: Incident classification scale

Figure 8-1 shows the criteria from the methodology and the corresponding scale, ordered by descending priority. An incident can comprise multiple events and meet various criteria. In this case, the highest criterion decides the scale of the incidents. In the case of a scale 2 or 3 event, an investigation is conducted by an Expert Panel. While only the highest priority criterion is relevant for deciding the scale, the other criteria are also assessed.

8.1 Scale of the Incident

The highest-violated ICS criterion during this incident was a scale 3 blackout (OB3). This criterion is met if there is a loss of more than 50 % of demand or a total absence of voltage for at least three minutes, assessed based on the TSO's control area. The lost load on 28 April 2025 was as follows:

- 25,638 MW for Spain (100 % of the demand in the Spanish Peninsula before the incident)
- 5,900 MW for Portugal (100 % of the demand in mainland Portugal before the incident)
- 7 MW for France (0.01 % of the demand before the incident).

Spain and Portugal also suffered a total absence of voltage. Therefore, both the loss of load and voltage criteria for OB3 were met.

For France, the lost load was below 100 MW and therefore no L criterion was met. France's highest criterion violation was an OV1 due to high voltage during the incident.

8.2 RCC Investigation Threshold

The RCC post-operation and post-disturbances analysis and reporting (RIAR) methodology has been developed to define the respective RCC task in accordance with Article 37 (1)(i) of the Regulation (EU) 2019/943 of the European Parliament and of the Council of 5 June 2019 on the internal market for electricity. This methodology foresees an RCC investigation in addition to the work of the Expert Panel if both of the following criteria are met:

a. a TSO has moved from a normal or alert system state to an emergency system state as a result of actions taken by another TSO being in emergency, blackout, or restoration system state, and
b. the incident has been confirmed as at least a scale 2 incident as defined by the ICS methodology.

Post-analysis confirmed that Spain and Portugal were in a blackout and restoration state during the incident, meaning that the incident met the RCC investigation threshold to initiate the investigation. The conclusions of the RCC investigation are added as a dedicated chapter in the final report.

8.3 Scale of all Violations Linked to the Incident

The OB3 was the most severe criterion violated during the incident in Spain and Portugal. Several other criteria were violated before the blackout occurred, which are presented in Table 8-1 and summarised below. When an incident meets several criteria, it is classified according to the criterion with the highest priority. Additionally, all sub-criteria information (i.e., ICS violations or events) is also collected. Figure 8-2 presents a timeline view of the ICS criteria violated during the incident. It should be noted that the violations of the criteria regarding loss of generation facilities (G) and loss of load (L) have been separated into smaller violations to better visualise the sequence of events and correlate the impact of these violations with the measured frequency and voltage during the incident. The following paragraphs consolidate the violations according to the ICS methodology and therefore do not fully align with the detailed timeline presented in Figure 8-2.

In Spain, there was a loss of generation facilities (G0) at 12:32, followed by a transformer trip at 12:32:57, which is classified as a scale 0 loss of network element (T0) violation, and a further 355 MW of generation tripped. Approximately 20 seconds later, at 12:33:16, there were successive trips of power generation facilities. These generation trips are classified as one G2 incident as they occurred within fifteen minutes and were greater than 3,000 MW. Further T incidents occurred with the loss of tie lines and transmission lines in Spain. The trip of the HVDC interconnector was classified as a T2 ICS violation due to the impact on a neighbouring TSO with OV (voltage) violations in France. The disconnections of all tie lines between France and Spain split the Iberian Peninsula from CE (involving more than one TSO with a load larger than 5 % of the total load of the synchronous area before the separation, thus violating the RS2 criterion). The loss of load (L2) criteria was met after the generation loss, as automatic TSO system defence measures were activated with the tripping of pumps and distribution load shedding.

There was high-voltage violations in Spain during the incident that contributed to the trip of generation, although they did not meet the time threshold for a voltage violation as defined in the ICS methodology. Voltage was above 1.1 per unit for 16 seconds, whereas the ICS methodology requires a time of greater than 30 seconds. The subsequent voltage collapse is captured under the OB3 (blackout) criteria.

Portugal experienced high voltage and – similarly to Spain – the duration did not reach the time thresholds specified in the ICS methodology, and therefore there were no OV incidents. There was a significant loss of load (L2) as pumps and industrial consumers were disconnected, as well as the distribution load. There were two T0 events with the loss of a transformer and an overhead line. Additionally, a G2 violation was recorded at the time of the blackout, as all generation facilities were disconnected.

The highest criteria experienced in France by RTE was scale 1 related to OV1. France also had G0 and T0 violations.

Frequency degradation (F) ICS criteria violation was a below scale (BS) violation (FBS). The reason is that the frequency deviation magnitude thresholds were met, but – similarly to the ICS voltage criteria (OV) – the duration of the deviations were not sufficiently long to be registered at a higher scale. Furthermore, the frequency deviations are registered at a synchronous area level, even if local anomalies might occur.

All oscillations that occurred prior to the blackout – as detailed in this report – did not cause any ICS criteria violations.

Criterion	Scale	Red Eléctrica	REN	RTE
OB	3	×	×	
L	2	×	×	
	1			
	0			
F	2			
	1			
	0			
	BS		×	
T	2	×		
	1			
	0	×	×	×
G	2	×	×	
	1			
	0			×
	BS			
ON	2	Not Violated		
	1			
RS	2	×	×	
	1			
	0			
OV	2	Not violated because the time threshold was not reached		
	1			×
	0			
	BS			
RRC	2	Not Violated		
	1			
	0			
LT	2	Not Violated		
	1			
	0			

Table 8-1: ICS criteria violations by TSO

Note: Each violated ICS criterion has an X in the cell. The scales shown are 0 – 3 and below scale (BS). There was no violation of the ICS ON, RRC, and LT criteria.

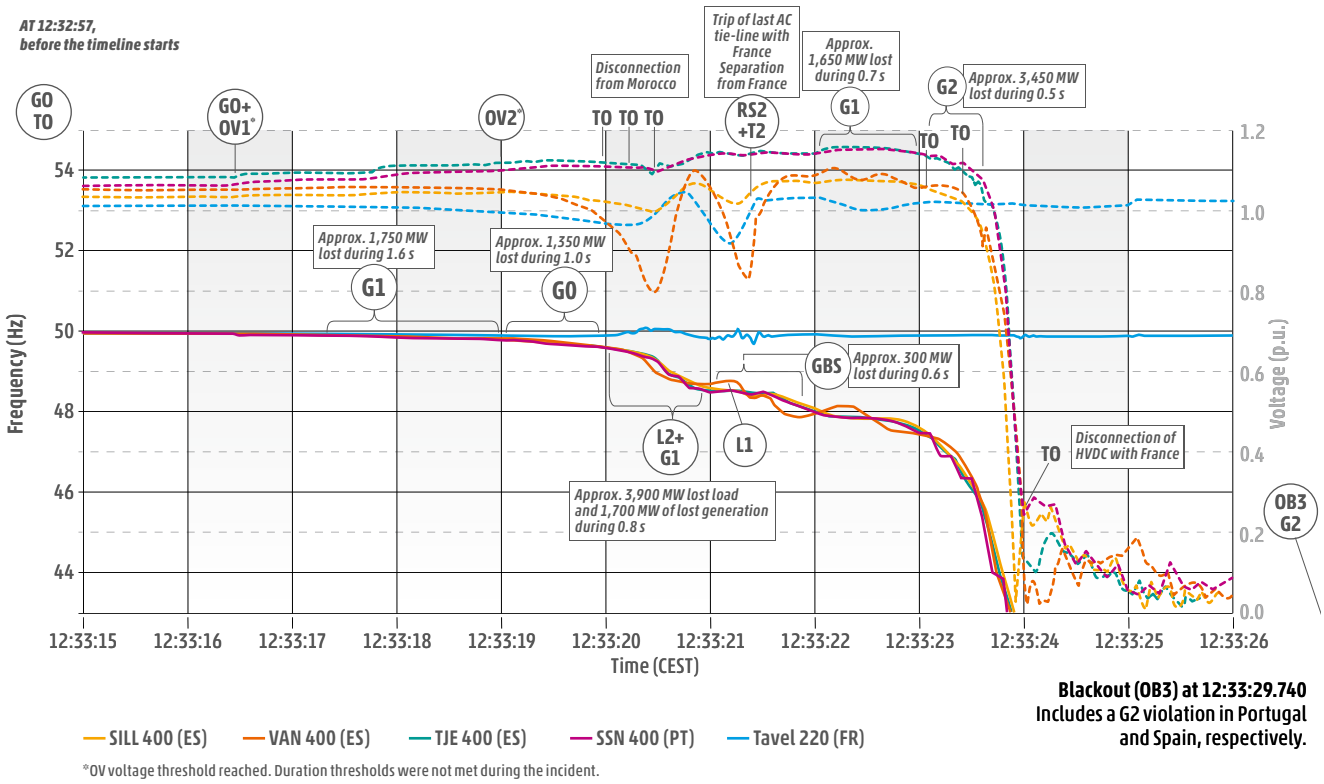


Figure 8-2: ICS criteria violations for Portugal and Spain during the incident
Note: All sub-ICS criteria violations for tripped generation facilities (G) and loss of load (L) are included in the graph but reported as a single G or L violation for Portugal and Spain. When the G and L criteria are regarded as a whole, Spain and Portugal each had a G2 violation, as the generation trips occurred within 15 minutes and amounted to more than 3,000 MW, and a blackout (OB3) due to the total loss of voltage in the control area. The L2 criterion was reached, but after more than 50 % of the load is lost in the control area, the L2 becomes an OB3 violation.

9 CONCLUSION AND RECOMMENDATIONS

9.1 Summary

Overall, the investigation concludes that the blackout that affected Spain and Portugal on 28 April 2025 was the result of multiple interacting factors. The incident evolved through a sequence involving a combination of voltage fluctuations and oscillatory phenomena, leading to widespread generation disconnections in Spain, particularly inverter-based resources, followed by a cascade of overvoltage disconnections and culminating in the loss of synchronism of the Iberian system with the Continental Europe Synchronous Area. The technical analyses indicate that the fast voltage increase was driven by a combination of interacting factors identified in the Root Cause Tree, which unfolded over very short timescales, leaving limited time for effective operational response. These include limitations in the availability and effectiveness of voltage control resources, the reactive power behaviour of connected assets, the protection settings of connected generators and the behaviour of small-embedded generation, compounded by limited observability.

The investigation relied on an extensive dataset collected from TSOs, DSOs, and SGUs, noting that collecting complete, high-quality, and timely data proved challenging and that some critical data remained missing, incomplete, or heterogeneous.

This experience underlines the importance of robust data collection processes, standardisation, and clear obligations to ensure that future investigations can be conducted with adequate accuracy and completeness. It also underscores the importance of all system users consistently fulfilling their obligations under the applicable requirements.

The Expert Panel's work has focused primarily on deriving clear, actionable recommendations to reduce the likelihood of similar events and strengthen the overall security and resilience of the European grid. These recommendations are the central outcome of the investigation and are structured into two categories:

(i) recommendations directly linked to the identified root causes, which are reflected graphically on the root cause tree (Figure 9-1); and

(ii) recommendations developed based on the investigation results, for which the Expert Panel considers that implementation may be beneficial for all system actors, but which are not related to the root causes of this incident.

Root cause tree

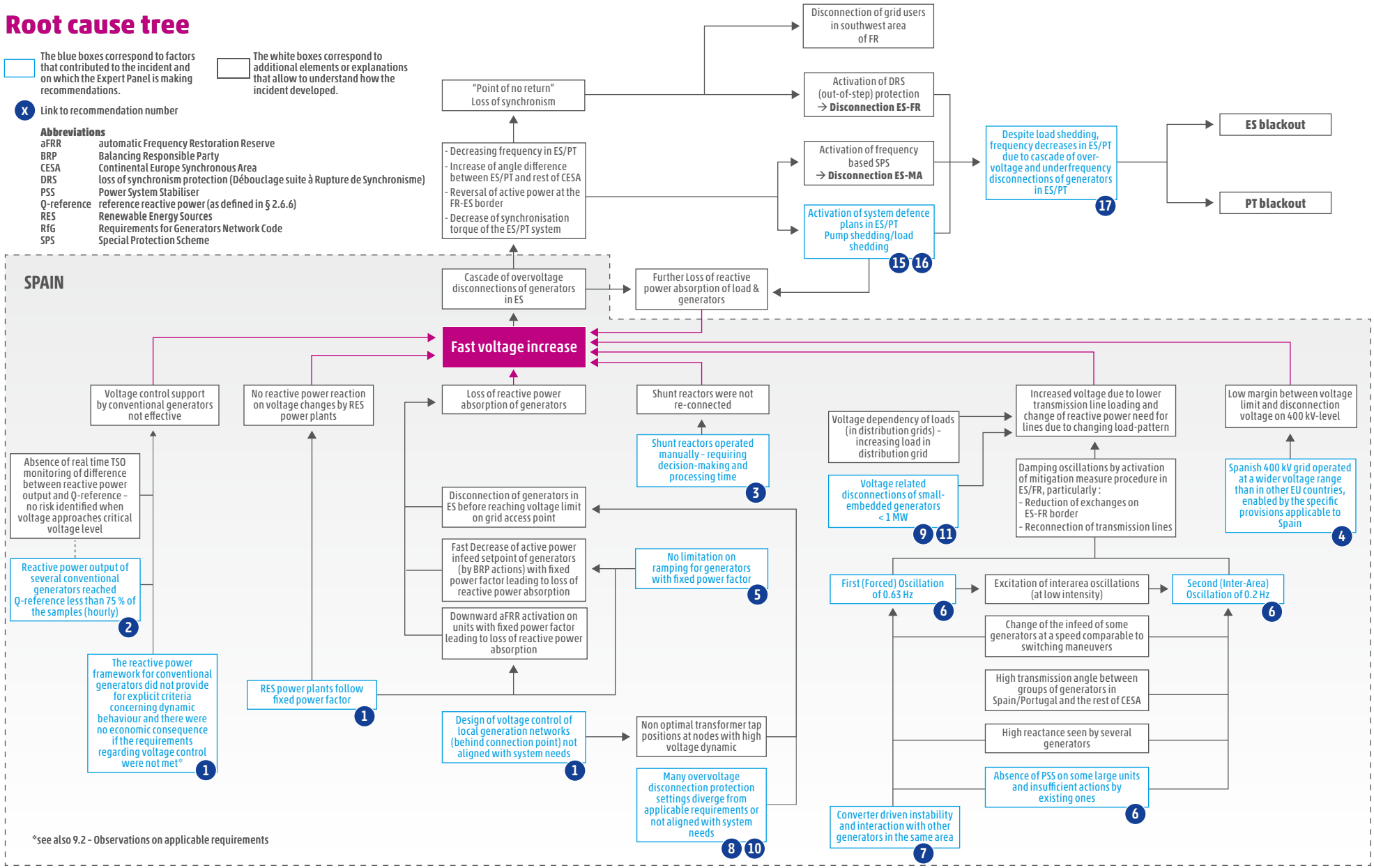


Figure 9-1: Root cause tree linked with recommendations.

The increasing penetration of variable renewable and distributed generation, further market integration, broader electrification, and evolving environmental and geopolitical risks place the European electricity system under increasingly challenging operational conditions, requiring higher levels of resilience. The recommendations address voltage control and reactive power management, oscillatory stability, and generation disconnection behaviour. They further aim to enhance the effectiveness of system defence mechanisms and the robustness and preparedness of restoration processes.

Ultimately, the recommendations outlined in this chapter represent a comprehensive set of measures designed to enhance operational robustness, improve cross-stakeholder information exchange, and help maintain a high level of security of supply across the European power system.

Where the recommendations in this chapter refer to "TSOs", "DSOs", "SGUs operating power-generating facilities", "the national competent authorities", etc., it is noted that the recommendations are targeted at all European TSOs, DSOs, SGUs operating power-generating facilities, national competent authorities, etc., and not only at those impacted by the investigated incident.

The monitoring of the implementation of the recommendations provided in this chapter does not fall within the mandate or remit of the Expert Panel. While voluntary, the recommendations are intended to prompt and support concrete action to help prevent the recurrence of similar events. Any responsibility for the consideration, follow-up and implementation of the recommendations lies with each addressee. However, none of the recommendations imposes an obligation on their addressees to act or to omit any actions. This is without prejudice to the requirements applicable to TSOs, in particular, to regularly monitor the appropriateness of the network operation tools required to maintain operational security and identify and implement improvements, if needed, taking into consideration the conclusions and recommendations stemming from the investigations into ICS incidents. Finally, the recommendations may serve as a basis for future regulatory development, inviting competent authorities to consider possible regulatory or implementation measures within their respective mandates, with a view to ensuring that the lessons learned from this incident are duly reflected in the applicable requirements and establishing a framework aimed at preventing similar incidents in the future.

9.2 Observations on the Applicable Requirements

The scope of this investigation and the mandate of the Expert Panel do not provide it with any authority or intention to interpret the applicable legislation or to determine liabilities for non-compliance with legal requirements. During the investigation, within the limits of its mandate, the Expert Panel made several observations regarding the applicable requirements. These observations are not to be interpreted as recommendations in the final report, and are included without prejudice to any inquiries and/or follow-up actions carried out by the competent national authorities.

More specifically, the Expert Panel observed certain divergences from the applicable requirements and, in some instances, a lack of alignment between the applicable legal and regulatory requirements and the current technical system needs and market conditions.

Moreover, during the investigation, the Expert Panel became aware that some parties had differing interpretations of the applicable requirements, based on the legislation in force at the time. This was particularly evident in the case of the definition of the upper voltage limits on the 400 kV grid and the voltage control obligations in Spain under Operational Procedures 1.4 and 7.4, where the applicable requirements were interpreted differently by two generators, the TSO, and the NRA. While acknowledging the differing interpretations on the matter, for the purpose of this report, the Expert Panel has followed the interpretation of the Spanish regulatory authority (CNMC).

Finally, the Expert Panel noted certain uncertainties regarding the allocation of decision-making responsibilities among national competent authorities in Spain (e.g., the national regulatory authority, the competent ministry, etc.), particularly in defining or further developing the applicable requirements. Such a lack of clarity in certain aspects has also been encountered regarding the respective roles and responsibilities of the different actors, including which party is responsible for which actions and in what sequence.

In general, the Expert Panel notes that the general principles of the applicable legislative procedure and correctly enforced mechanisms must apply to ensure legal clarity, compliance, and the enforceability of any applicable legislation. Measures that could contribute to that include, among others:

- **Issuing implementation guidance by the competent authorities for new legislation or for specific rules, either proactively or in response to requests for clarification submitted through official channels by multiple affected parties.** This approach is consistent with standard administrative law practice in the EU regulation. For example, the European Commission regularly publishes guidance, notices, and communications to clarify provisions in EU regulations and directives, thereby providing interpretations that increase legal certainty for businesses and national authorities.
- **Regular monitoring of the effectiveness of the applicable requirements and legislation by the competent authorities.** The energy sector is innovation-intensive, and technical rules may become outdated or ineffective over time. Such regular and proactive monitoring by the competent authorities should involve relevant market stakeholders and aim to align the applicable requirements with current (and, where possible, expected future) technical system needs and market conditions. Moreover, the competent authorities should regularly (re)consider whether the attribution of the roles and responsibilities is (still) effective and relevant in view of the evolving technical and market reality. Besides the systematic evaluation of the currently applicable requirements, regular monitoring should specifically focus on newly applicable legislation and its implementation, where the effects may be uncertain. Similarly, the relevant stakeholders should regularly assess whether the contractual framework in place between the parties remains fit for purpose and effective.
- **Making appropriate use of the existing/introducing new economic incentives, and enforcement mechanisms as appropriate, including economic penalties foreseen under the applicable law for breaches of the requirements.** Economic incentives, whether positive or negative, should be effective by ensuring the desired behaviour. Of course, in the design and application of both incentives and penalties, it is important to keep in mind that, in a system of systems as complex as the electrical system, with a multitude of system users, each party must perform its functions as established in the applicable regulations.

» **Consolidating and implementing institutional competencies pertaining to power system operation.** The competent national authorities should monitor and regularly evaluate the roles and responsibilities attributed by the regulatory framework to various actors and, where appropriate, (re)consider whether the attribution of those roles and responsibilities is (still) effective and relevant in view of evolving technical and market realities. Such an evaluation of roles and responsibilities should not exclude the competent authorities themselves, with changes introduced as appropriate. Moreover, the effectiveness of the overall governance concerning the system operation and design should also be regularly assessed and revised as necessary.

9.3 Recommendations Directly Linked to Root Causes

9.3.1 Voltage Control

Change needed (in magenta) for Recommendation 4 from the Grid Incident in South-East Europe of 2024 – Develop KPIs for Voltage Stability Risk and Risk of Rapid Voltage Change.

Description:

Analysis of the possibility to identify, and if possible, the development of a guideline for indicators to detect potential reduced voltage stability and risk of **rapid voltage change.**

Justification:

Any indication that the grid is in a weakened state can create a sense of urgency to act before any other incident occurs, and as such this might avoid **rapid voltage changes**, if actions can be taken sufficiently fast before the next incident.

Delivery owners:

Various specialists on system (voltage) stability within ENTSO-E, mainly from SPD. Specific TF to be set up.

Priority: Low

Linked to Subsection 4.1.3 Voltage Management.

Change needed (in magenta) for Recommendation 7 from the Grid Incident in South-East Europe of 2024 – Voltage and Reactive Power Assessment.

Description:

ENTSO-E to develop a guideline of good practice on voltage support means and studies on voltage stability.

TSOs to assess that a plan for the realisation of sufficient voltage support means is in place and remains adequate or is updated where necessary, and, where relevant add these means to the TYNDP and national grid development plans.

Justification:

Having sufficient MVAr support means will help avoid voltage collapse **or fast overvoltage-induced cascading disconnections** in a weakened grid.

Delivery owners:

TSOs and voltage experts in ENTSO-E StG Resilient Operation.

Priority: High

Linked to Subsection 4.1.2 Reactive power contribution by various assets.

R250428_1 – Voltage control mode

Description:

a) TSOs and SGUs operating power-generating facilities should ensure that generators use voltage control mode whenever possible, in line with the capabilities defined in the applicable requirements, e.g., Commission Regulation (EU) 2016/631 of 14 April 2016 establishing a network code on requirements for grid connection of generators (RfG).

b) National competent authorities should support this objective, e.g. by considering proposals from TSOs to extend voltage control requirements to existing power-generating facilities connected before the applicable requirements were in place (e.g. see Art. 4 RfG).

c) TSOs should explore the possibility of a centralised or zonal voltage regulation, where voltage or reactive power setpoints for generators are automatically calculated based on real-time voltage measurements in reference nodes.

d) ENTSO-E should provide comprehensive guidance to all TSOs, national competent authorities, and ACER on approaches to voltage control, encompassing all reactive power assets. For generators, the guidance should cover technical aspects (control mode), control aspects (reaction time, ramping requirements, measurement accuracy, resolution, service provision point), and implementation aspects (sampling, enforcement in the case of obligations or incentives, or penalties in the case of a service).

Justification:

Operating a significant capacity of generation resources with a fixed power factor control mode in Spain presented two issues:

1) for every (fast) active power ramp (e.g., schedule change), fixed-power-factor-mode resources also injected into the system a proportional fast reactive power ramp (and hence a voltage ramp);

2) fixed-power-factor-mode resources provided inadequate reactive power support that would compensate for voltage fluctuations.

The approach to reactive power control by generators in France and Portugal differed from that in Spain in several respects, and cross-country sharing of effective implementation practices was limited.

Delivery owners:

TSOs, SGUs operating power-generating facilities, national competent authorities, and ENTSO-E.

Priority: High

Linked to Subsection 4.1.2 Reactive power contribution by various assets.

R250428_2 – Reactive power capability visibility and adequacy

Description:

a) In the medium term (years ahead), TSOs should assess the system needs that will appear in real-time and technically design the system so that it can operate in the face of expected uncertainties through an appropriate combination of static and dynamic reactive power resources.

b) In operational planning processes and in real-time, TSOs should account for the occurrence of rapid voltage slopes (stemming from fast-changing active and reactive power flows) by ensuring sufficient and effective margins in dynamic reactive power resources and appropriate alarm settings regarding the performance monitoring of reactive power resources.

c) ENTSO-E should develop a guideline to facilitate the implementation of this recommendation.

Justification:

The investigation identified recurrent mismatches between expected reactive power provision and the actual real-time performance of some system users, reducing the effectiveness of voltage control during fast voltage variations. Such mismatches were not unique to 28 April but had occurred before in the Spanish system. Ensuring sufficient, visible, and effectively usable reactive power resources is therefore essential to manage rapid voltage changes and maintain system stability.

Delivery owners:

TSOs, DSOs, SGUs operating power-generating facilities, ENTSO-E, and national competent authorities.

Priority: Medium

Linked to Subsection 4.1.2 Reactive power contribution by various assets.

R250428_3 – Automatic activation of reactive power assets

Description:

TSOs should explore implementing automatic control of shunt reactors, as appropriate, using predefined control schemes to ensure voltage stability in the system. TSOs should consider more flexible devices (variable shunt reactors and capacitors, STATCOMs, etc.) when technically designing their systems in this regard.

Justification:

The investigation has identified that a substantial reactive power capacity from shunt reactors was available but not activated during the voltage rise preceding the blackout (mainly due to the low voltages a few minutes before and the manual action requiring more lead time). Approaches to such situations should be analysed in detail, and a common European approach should be considered where appropriate.

Delivery owners: TSOs.

Priority: Medium

Linked to Subsection 4.1.2 Reactive power contribution by various assets.

R250428_4 – Operating voltage range

Description:

TSOs should ensure that the harmonised operating voltage range foreseen at the European level is effectively applied across Europe, by avoiding or removing deviations that allow operation outside this range.

The harmonised voltage range (i.e., 380 – 420 kV, as defined in SO GL) is designed to ensure sufficient safety margins between normal operation and voltages where disconnections occur.

Furthermore, the harmonised range reduces the wear and tear of system components and ensures a level playing field.

Justification:

The Spanish Operational Procedure P.O. 1.4 (1998) and specific national provisions, together with references in European regulations applicable to Spain, enabled the Spanish TSO to operate the transmission grid in a transient mode at voltage levels up to 435 kV.

The decision to operate the system within an allowed voltage range of 380 – 435 kV, while generators remained connected at 435 kV or 440 kV (depending on commissioning date), left the voltage margin low or non-existent.

Delivery owners: TSOs, national competent authorities.

Priority: High

Applicable legislation: SO GL

Linked to Subsection 4.1.3 Voltage Management.

R250428_5 – Ensuring appropriate active and reactive power variation behaviour

Description:

Undesired rapid reactive power variations, particularly those occurring on time scales comparable to or faster than the system's voltage control response, should be avoided. In the case of a fixed power factor, until a more appropriate voltage control mode is implemented, reactive power ramps follow active power ramps, meaning that reactive power ramps can be temporarily addressed by ensuring that changes in active power are smoother.

This reduction in the rate of change can be ensured, e.g., by connection requirements and/or controller settings, and/or by incentivising a more gradual ramp between subsequent imbalance settlement periods (e.g., by correcting imbalances based on assumed ramping to incentivise balance responsible parties to follow that ramp).

The issue of reactive power generation on transmission lines stemming from rapid active power changes from generators should be further studied by TSOs.

Justification:

Rapid variations in power injected into the transmission/distribution grid by power plants, storage systems, and consumers (e.g., a rapid reduction in photovoltaic injection following negative prices) can lead to abrupt changes in power flow in the transmission grid. Such behaviours increase system stress and can contribute to greater TSO imbalances and defence plan activations, particularly when power factor is fixed.

Delivery owners:

TSOs, DSOs, market participants and national competent authorities.

Priority: Medium

Applicable legislation: SO GL

Linked to Subsection 4.1.2 Reactive power contribution by various assets.

9.3.2 Oscillations

R250428_6 – Continental Europe Synchronous Area Framework for inter-area oscillations mitigation

Description:

Establish a framework to improve the damping of inter-area oscillations in the Continental Europe Synchronous Area:

a) CE SA-TSOs should define the required level of damping for inter-area oscillations, collect damping measure roadmaps from TSOs, ensure their consistency, and verify that the combined measures are sufficient to meet the targeted minimum damping level.

b) CE SA-TSOs should establish a periodic and structured process for sharing validated Dynamic RMS models across TSOs, based on standardised CGMES dynamic profiles, including clear versioning, documentation of assumptions, and managed confidentiality arrangements, to support coordinated inter-area oscillation studies and post-event analyses.

c) In accordance with the applicable requirements (see Art. 38.4 of SO GL), ENTSO-E should conduct a coordinated study to assess the damping of inter-area modes in Continental Europe Synchronous Area regularly, based on a shared dynamic model of the Synchronous Area with two objectives:

- based on the above studies, tuning existing damping measures (PSS, POD) and the implementation of new ones will be proposed to reach a sufficient damping level;
- remedial actions for real-time management (e.g., topological, generation dispatch, or exchange reduction) of low-damping episodes will also be proposed if the minimum damping level cannot be reached.

d) The analysed damping measures in this report, which show a high impact on the damping of critical inter-area oscillations, such as POD-P/-Q on inverter-based devices, additional synchronous condenser or STATCOMs, should be prioritised in the context of system studies and implementation planning, without prejudice to the applicable regulatory frameworks.

Justification:

The analysis highlights that Continental Europe Synchronous Area has a low-damped inter-area mode that can directly or indirectly affect its operational security. Moreover, the evolution of the Continental Europe Synchronous Area (fewer synchronous generators, more inverters, extension with new countries' connections, etc.) could lead to new inter-area modes with low damping.

Delivery owners:

CE SA-TSOs, national competent authorities and ENTSO-E.

Priority: High

Applicable legislation: SO GL

Linked to Subsection 4.2.3 Validation of the Analytical Model.

R250428_7 – Enhance dynamic monitoring and operational oscillation detection

Description:

Improve and expand the monitoring and detection framework by efficiently using existing PMUs and oscillographs, or, where needed, installing additional PMUs, oscillographs, and power quality monitoring devices accessible to the TSO, particularly near interfaces with power plants, third-party-owned HV network sections, DSOs, and HV consumers. TSOs should analyse present and future system operations to determine the system needs for such measuring devices. Enable efficient PMU data exchange between TSOs to enhance wide-area visibility and coordinated assessment of system dynamics.

Implement automatic detection systems and effective early-warning tools to identify oscillations in a timely manner (including both inter-area and forced oscillations), locate their source in case of forced oscillation, and support prompt corrective actions, noting that automated corrective actions should be applied selectively.

Justification:

High-quality, real-time measurement data is primarily needed to reliably detect and localise oscillations and to tune, validate, and continuously monitor system-wide damping controls. Early and reliable detection constitutes the primary prerequisite for any further classification or response to oscillatory phenomena, for both inter-area and forced oscillations.

Delivery owners:

TSOs and national competent authorities.

Priority: High

Applicable legislation: SO GL

Linked to Subsection 4.2.3 Validation of the Analytical Model.

9.3.3 Disconnections

R250428_8 – Validated protection settings for generation units and generation evacuation grid connected to the TSO and DSO

Description:

TSOs and DSOs should assess the adequacy and consistency of the protection settings for generation units and evacuation grid elements (e.g., transformers below which several power-generating and storage modules are connected, including distribution-connected and distributed energy resources). This verification should ensure that withstand capabilities (high/low voltage, RoCoF) are aligned with the highest capabilities of grid users connected downstream, with explicit checks on over/undervoltage thresholds, and minimum time delays (commensurate with system dynamics and measurement uncertainties, e.g., ≥100 ms or even >1 s where applicable).

Compliance with the minimum timing requirements and voltage thresholds established by the applicable regulations and ensuring a sufficient margin in the adjusted thresholds to prevent potential measurement inaccuracies from causing unintended disconnections remains critically important. If this is not the case, appropriate corrective actions to achieve compliant protection settings should be identified and implemented.

If necessary, TSOs should assess, in coordination with national competent authorities, the possibility of applying the relevant requirements (e.g. Art. 4 RfG), where justified by system security needs.

Justification:

The analysis and expert inputs point to sensitivity to local protection settings (e.g., overvoltage thresholds measured away from the point of connection, instantaneous trips without delay) that can unnecessarily disconnect generation when voltages oscillate or rise transiently. In some cases, the protection appears to diverge from the applicable requirements.

Delivery owners:

TSOs, DSOs, and national competent authorities.

Priority: Medium

Applicable legislation: SO GL

Linked to Subsection 4.1.3 Voltage Management.

R250428_9 – High-voltage ride-through requirement for Type A power-generating modules

Description:

Type A power-generating modules should be capable of stable operation without disconnecting from the grid for a voltage-versus-time profile. ENTSO-E should evaluate and suggest relevant potential improvements to the applicable requirements under the RfG.

Justification:

The analysis in this report on small-scale PV generation trips, part of it of type A, highlights their sensitivity to high-voltage episodes, even when transmission-side voltage remains within acceptable limits.

Given the rapid expansion of PV installations across Europe in recent years – with a substantial share of systems being small-scale – it is critical to prevent unnecessary disconnection during high-voltage events. This is essential to maintain both grid stability and reliability.

Delivery owners: ENTSO-E.

Priority: High

Applicable legislation: RfG

Linked to Subsection 4.3.3 Analysis of Relevant Factors Influencing the TSO–DSO Flows.

R250428_10 – Regular controls on grid users' assessment of withstand capabilities (high/low voltage, RoCoF)

Description:

TSOs, DSOs, and private grid owners should put regular controls in place after any protection trip or unexpected disconnection to assess whether the observed behaviour of grid users during real events is consistent with the applicable withstand capability requirements (high- and low-voltage ride-through, RoCoF ride-through).

These assessments should be learning-oriented and aim to identify potential inadequacies in protection settings, control behaviour, measurement locations, or the applicability of existing requirements.

Justification:

The blackout analysis demonstrates that at least 10 disconnections were not in line with the applicable requirements. TSOs, DSOs, and private grid owners should take every opportunity to address this problem during everyday events that do not necessarily lead to serious consequences, to detect structural weaknesses early and avoid their manifestation during major system disturbances.

Delivery owners: TSOs, DSOs, and private grid owners.

Priority: Medium

Applicable legislation: SO GL

Linked to Subsection 4.1.3 Voltage Management.

R250428_11 – Investigation on behaviour of non-observable embedded generators

Description:

TSOs and DSOs, together with the electrical industry, should further investigate the disconnection and reconnection behaviour of inverters, in particular distribution-connected inverters (including rooftop PV systems and other embedded generation <1 MW). For this purpose, the Grid Connection and System Operation European Stakeholder Committees (GC ESC[53] and SO ESC[54]) should be used as a platform to exchange knowledge and experience.

Justification:

The data the Expert Panel received from PV system manufacturers during the investigation showed that many inverters tripped due to overvoltage during certain periods on 28 April. The Expert Panel also identified an important role of the reconnection time of the non-observable PV systems in the loss of generation.

53 GC ESC Website

54 SO ESC Website

However, the Expert Panel was unable to conduct an in-depth investigation and considers it beneficial to continue and intensify the exchange with DSOs and inverter manufacturers on this matter at the European level to better understand the voltage-related behaviour of non-observable embedded generators.

Delivery owners:

GC ESC and SO ESC (chaired by ACER), TSOs and DSOs.

Priority: Medium

Applicable legislation: RfG, SO GL

Linked to Subsection 4.3.3 Analysis of Relevant Factors Influencing the TSO–DSO Flows.

9.4 Recommendations Not Directly Linked to Root Causes

9.4.1 Voltage Control

R250428_12 – Reactive power support for inverter-connected installations in case of no active power transfer

Description:

TSOs should investigate options to ensure that HVDC systems and other comparable transmission-level inverter-based installations with voltage-support capability continue to provide reactive power support (functioning as a STATCOM) at all times, even in the immediate aftermath of a loss of DC connection.

Justification:

Following the blackout and loss of active power transfer over the HVDC, reactive power support at both ends was lost; however, subject to appropriate converter design and control strategies, HVDC converter stations could technically continue providing reactive power support.

Delivery owners:

TSOs, SGUs operating HVDC systems or other comparable transmission-level inverter-based installations.

Priority: Low

Applicable legislation: NC HVDC

Linked to Subsection 4.1.2 Reactive power contribution by various assets.

9.4.2 Oscillations

R250428_13 – Continental Europe Synchronous Area procedure to create snapshots common grid model

Description:

A common procedure should be established to create a snapshot common grid model of Continental Europe Synchronous Area promptly after a significant event, allowing accurate simulations of the system under similar conditions to those of the event.

The snapshot refers to a calculable network model derived after the event, based on available state-estimator results, topology, operating points, and relevant dynamic data. The procedure should define the minimum content and timing of the snapshot, while allowing TSOs to rely on their existing tools and vendor-specific implementations.

Justification:

Since the investigation of system oscillations relied on incomplete and non-harmonised post-event data across Continental Europe Synchronous Area, some simplifications had to be made for the simulation. Therefore, a new standard procedure for creating consistent post-event snapshots for dynamic analyses could be beneficial.

Delivery owners: CE SA-TSOs.

Priority: High

Applicable legislation: SO GL

Linked to Subsection 4.2.2 Construction of Analytical Framework for Oscillatory Analysis.

9.4.3 Disconnections

R250428_14 – Streamlined cross TSO/DSO/PGM investigation data governance

Description:

Establish a permanent, standardised framework for data provision during incident investigations or for broader system analyses to improve understanding of overall system behaviour. This framework should ensure fast, complete and uniform access to all required datasets from TSOs, DSOs (including distribution-connected generation data available to DSOs), SGUs operating power-generating facilities, SGUs operating large demand facilities, and SGUs operating HVDC or other inverter-based installations and other parties capable of providing relevant data on small-scale generation or storage facilities. The framework should be jointly developed by TSOs and DSOs, coordinated at the European level, and subject to oversight from the national competent authority. This framework should define:

- the exact data types and formats (event logs, dynamic recordings, high-resolution PMU/SCADA, DER status, protection activations);
- latency and completeness requirements, including defined escalation mechanisms;
- authorised data-sharing;
- predefined legal/contractual provisions removing the need for ad hoc negotiations.

This process should allow investigations to fully and reliably compare oscillation behaviour, disconnection patterns, and defence plan performance across all voltage levels, taking into account applicable SO GL Article 40(5) provisions and national RfG banding thresholds.

Where relevant for incident investigation purposes, the framework should define conditions under which selected operational data already available in real-time is preserved and made accessible for post-event analysis, without introducing additional real-time operational obligations.

In addition, regular data-provision readiness exercises should be conducted to verify and benchmark all partners' ability to deliver the required datasets within the specified timeframes and formats. This recommendation should apply to all installations (existing, new, restored or modified), ensuring uniform obligations and avoiding gaps in data availability.

This framework is intended to support incident investigation, system learning and analytical consistency and does not replace or modify existing regulatory compliance or enforcement processes.

Justification:

The investigation faced delays and analytical limitations due to gaps in the available data, particularly from SGUs operating power-generating facilities and from DSOs. This reduced the ability to reconstruct dynamic behaviour, identify oscillation sources, quantify DER disconnections, and understand protection responses. In future events, a structured, pre-agreed data governance framework would help ensure timely, consistent data provision and thereby enhance both the speed and the quality of incident analysis.

Delivery owners:

TSOs, DSOs, SGUs operating generation, demand or HVDC/inverter-based installations and national competent authorities, and other parties capable of providing relevant data on small-scale generation or storage facilities.

Priority: Medium

Applicable legislation: SO GL

Linked to Subsection 1.1.2 Data Collection.

9.4.4 System Defence Plan

R250428_15 – DER-aware dynamic LFDD

Description:

Develop and implement an adaptive LFDD scheme that accounts for high DER penetration and maintains effective load shedding at all times. This may include reverse-power blocking, feeder selection based on DER footprint, and other adaptive methods.

As this change affects the design and effectiveness of a regulated system defence measure, the updated defence plan should be submitted to the national competent authorities for information. Where specific elements require regulatory approval under national law, the relevant parts should be submitted accordingly.

Justification:

The defence analysis shows that the current LFDD often sheds DER <1 MW along with load due to missing reverse-power blocking and insufficient DER awareness. A DER-aware, adaptive LFDD directly addresses these gaps and improves the effectiveness of defence plans under high renewable penetration.

Delivery owners:

TSOs, DSOs and national competent authorities.

Priority: Low

Applicable legislation: NC ER

Linked to Subsection 4.4.4 System Defence Performance.

R250428_16 – Defence observability and readiness (data and drills)

Description:

Create a unified framework that provides real-time visibility at the feeder level, including power flows, directions, DER status, and LFDD activations. It should also ensure the delivery of a standardised post-event data package within 24 hours, containing LFDD logs, estimated DER losses, reverse-power blocking status, and a concise timeline of early actions. In addition, the framework must include regular joint drills between TSOs, DSOs, and producers to validate operational readiness, response times, and the quality of forensic data. This framework should be integrated into the defence plan and submitted to the national competent authorities for information. Where specific elements require regulatory approval under national law, the relevant parts should be submitted accordingly.

Justification:

The analysis highlights limited means to quantify DER <1 MW disconnections and evaluate defence performance. Consistent real-time observability guarantees forensic packages, and joint exercises directly close these gaps and demonstrate end-to-end readiness.

Delivery owners:

TSOs, DSOs and national competent authorities.

Priority: Low

Applicable legislation: NC ER

Linked to Subsection 4.4.4 System Defence Performance.

R250428_17 – Modernised system defence plan for fast voltage deviations

Description:

To reflect today's high RES and DER environment, TSOs should modernise the system defence plans to explicitly address fast voltage-related phenomena. This includes identifying fast-developing operational anomalies, such as rapid voltage changes, rapidly changing net positions, oscillatory behaviour, or voltage excursions beyond normal operating ranges.

The deliverable of this recommendation is a documented update of the system defence plan, defining:

- the scope of fast voltage-driven threats covered;
- the detection criteria and decision logic at the defence-level; and
- the governance for activating defence-level responses before existing measures become ineffective.
- The identification and implementation of concrete technical measures should be addressed through other relevant recommendations and as outcomes of the studies supporting this defence plan modernisation.

Justification:

The 28 April blackout showed that although current system defence plans include voltage-related elements, they were not sufficiently equipped to prevent a fast, overvoltage-induced cascading collapse. A streamlined, modernised defence framework, tailored to promptly identify and address operational anomalies under high DER conditions, would strengthen system resilience and help prevent similar cascading events in the future.

Delivery owners: TSOs.

Priority: Low

Applicable legislation: NC ER

Linked to Subsection 4.4.3 Voltage Stability.

9.4.5 Restoration

R250428_18 – Black-start and island-mode real tests

Description:

Make realistic black-start tests mandatory and periodic, preferably every 3 years or after major control or protection changes, in accordance with applicable requirements, covering power plant-related aspects such as auxiliaries, AVR/excitation, governor modes, and house load. Introduce a black-start readiness index and require TSO validation before certification renewal. Where possible, black-start tests should also consider the expected static and dynamic behaviour of system users during the planned island operation. To closely replicate real black-start scenarios while minimising impact on consumers, selected demand facilities (such as power-to-heat), BESS, and hydro pumps should be included in the tests.

Justification:

The investigation showed that black-start retries, misconfigured modes, and the oscillatory behaviour of islands impacted by non-observable RES can significantly delay the formation and preservation of sufficiently stable islands. Periodic, witnessed testing reduces first start failure risk, improves preparedness, and shortens restoration paths. Reliability of black-start facilities tested during testing should be considered in the design of restoration plans.

Delivery owners:

TSOs, Restoration System Providers (RSPs), DSOs and national competent authorities.

Priority: High

Applicable legislation: NC ER

Linked to Subsections 5.2 The restoration sequences in the Spanish system and 5.3 The restoration sequences in the Portuguese system.

R250428_19 – DER reconnection and DSO visibility

Description:

Define a framework for DER for controlled reconnection during restoration. This includes:

- a staged reconnection governance framework ensuring sufficient load before generation to integrate DER governance into restoration plans;
- improved real-time or near-real-time controllability and visibility for TSOs and DSOs of DER in distribution systems relevant for restoration operations;
- avoidance of early energisation of feeders with high DER penetration.

In addition, ensure adequate operational visibility for DSOs of production and storage facilities involved in bottom-up restoration scenarios of distribution systems, including connection status, availability for reconnection and blocking status during staged restoration.

High-resolution recordings and detailed post-event data required for incident analysis are addressed separately in Recommendation R14.

This framework should be integrated into restoration planning and submitted to the national competent authorities for information. Where specific elements require regulatory approval under national law, the relevant parts should be submitted accordingly.

Justification:

The Spanish experience demonstrates that controlled DER reconnection during restoration is feasible and effective when appropriate visibility and control frameworks are in place. However, the investigation also identified gaps for small-scale DER and in other systems, justifying the need to generalise such approaches.

Delivery owners:

TSOs, national competent authorities and DSOs.

Priority: Medium

Applicable legislation: NC ER, SO GL

Linked to Subsection 5.8 Behaviour of distributed energy resources and load during restoration.

R250428_20 – Enhance restoration common trainings

Description:

Promote common training across multiple involved entities in a restoration scenario (TSO, DSO, relevant SGUs, RSPs) at least every 3 years, to ensure that each entity's role and responsibilities are clear and regularly practiced by all involved operators. Training should focus at least on the following aspects:

- the role of the frequency master per island, with standard frequency control on the first black-start unit and with standard speed droop settings on additional sources running in parallel on the same island;
- controllable load and ensuring that the load is higher than the minimum stable active power limit of additional units before synchronising them to the same island;
- resynchronisation between islands;
- common restoration procedures defining agreed roles, decision authority, sequencing and time-bound operational steps for critical actions (e.g., generator synchronisation, load pick-up, island expansion and fallback actions).

Justification:

These trainings remove the main restoration blockers or control conflicts, such as overgeneration in small islands, reactive power balance fragility with risk of overvoltages, weak tie closures, and misunderstandings between TSOs, DSOs, relevant SGUs, or RSPs. This will reduce the time to first load and resynchronisation while lowering the risk of island re-collapse or trips during resynchronisation.

Delivery owners: TSOs, DSOs, relevant SGUs and RSPs.

Priority: Low

Applicable legislation: NC ER

Linked to Subsections 5.2 The restoration sequences in the Spanish system and 5.3 The restoration sequences in the Portuguese system.

R250428_21 – Ensure at least 24 h blackout-proof communication and critical tools

Description:

Ensure at least 24 h blackout-proof voice communication systems and critical tools (e.g., telemetry and remote-control signals) for DSOs, RSPs, and relevant SGUs, independent of public telecom providers.

These arrangements should be integrated into restoration and emergency procedures and submitted to national competent authorities for information. Where specific elements require regulatory approval under national law, the relevant parts should be submitted accordingly.

Justification:

During the restoration process, TSOs reported significant difficulties in contacting several DSOs and some SGUs due to failures in their voice communication systems, while TSOs' own communication systems remained operational. Several DSOs experienced issues receiving telemetry and remote-control signals at their facilities, which led to significant delays in restoring some areas.

Delivery owners:

TSOs, DSOs, RSPs, relevant SGUs and national competent authorities.

Priority: Low

Applicable legislation: NC ER

Linked to Subsection 5.9 Voice communication and remote-control problems during restoration.

LIST OF ABBREVIATIONS

AAA	Adequacy Assessment Agent
ACER	Agency for the Cooperation of Energy Regulators
ADC	Angle Difference Control
aFRR	automatic Frequency Restoration Reserves
AGC	Automatic Generation Control
AVR	Automatic Voltage Regulator
BRP	Balancing Responsible Party
BS	Below Scale
CACM GL	Capacity Allocation and Congestion Management Guideline
CC	Capacity Calculation
CCC	Coordinated Capacity Calculation
CCGT	Combined Cycle Gas Turbine Powerplant
CCR	Capacity Calculation Region
CE	Continental Europe
CE SA	Continental Europe Synchronous Area
CECRE	Renewable Energy Control Center
CEE	Central Eastern Europe
CEST	Central European Summer Time
CGM	Common Grid Model
CGMA	Common Grid Model Alignment
CGMES	Common Grid Model Exchange Standard
CGMM	Common Grid Model Methodology
CIM	Common Information Model
CPC	Constant Power Control
CRACs	Contingency List, Remedial Action and Additional Contraints
CSA	Coordinated Security Analysis
DA	Day-ahead
DA2CF	D-2 Congestion Forecast;
DACF	Day-ahead Congestion Forecast
DEF	Dissipative Energy Flow
DER	Distributed Energy Resource
DFR	Disturbance Fault Recording
DOPT	Daily Operational Planning Teleconference
DRS	Loss of synchronism protection (Débouclage suite à Rupture de Synchronisme)
DSA	Dynamic Stability Assessment
DSO	Distribution System Operator

EAS	ENTSO-E Awareness System
EMF	European Merging Function
EMT	Electromagnetic Transient
FACTS	Flexible AC Transmission Systems
FCA	Forward Capacity Calculation
FCIC	Fast Current Injection Controller
FFT	Fast Fourier Transform
FSSI	Fast Stochastic Subspace Identification
GCC	Generation Control Centre
GC ESC	Grid Connection European Stakeholder Committee
GS	Generation substation
HPP	Hydropower Plant
HV	High Voltages
IBR	Inverter-Based Resource
ICS	Incident Classification Scale
IDA	Intraday market Auction
ID	Intraday
IDCF	Intraday Congestion Forecast
IEC	International Electrotechnical Commission
IGCC	International Grid Control Cooperation
IGM	Individual Grid Model
KE	Kinetic Energy
LFC	Load-frequency Control
LFDD	Low Frequency Demand Disconnection
LT	Long term
LV	Low Voltage
MARI	Manually Activated Reserves Initiative
mFRR	manual Frequency Restoration Reserve
MMs	Modulation Modules
MV	Medium Voltage
NEMO	Nominated Electricity Market Operator
NC ER	Network Code on Emergency and Restoration
NPP	Nuclear Power Plant
NRA	National Regulatory Authority
NTC	Net transfer capacity
OHL	Overhead Line
OMIE	Operador del Mercado Ibérico de Energía

ONEE	L'Office National de l'Électricité et de l'Eau potable
OPC	Outage Planning Coordination
OPDE	Operational Planning Data Environment
OPI	Outage Planning incompatibility
OPR	Operating Power Range
OPSC	Outage Planning Security Constraints
OU	Omstein-Uhlenbeck
PCA	Principle Component Analysis
PDBF, PDVP, PHF, P48	Acronyms of the programs published before and after security processes on Day-Ahead and intraday.
PE	Pan-European
PEVF	Pan European Verification Function
PGM	Power Generation Module
PI	Proportional-Integral
PMU	Phasor Measurement Unit
POD	Power Oscillation Damping
PPC	Power Plant Controller
PSS	Power System Stabiliser
PST	Phase Shifting Transformer
PV	Photovoltaic
RA	Remedial Action
RC	Remaining capacity
RCC	Regional Coordination Centre
RCW	Renewable, cogeneration, and waste
RE	Red Eléctrica
REN	Redes Energéticas Nacionais
RES	Renewable Energy Sources
RG CE	ENTSO-E Regional Group Continental Europe
RIAR	Regional Incident Analysis and Reporting
RoCoF	Rate of Change of Frequency
ROSC	Regional Operational Security Coordination
RPCL	Reactive Power Control Loop
RR	Replacement reserve
RSC	Regional Security Coordinator
RSP	Restoration Service Provider
RTE	Réseau de Transport d'Electricité
RTSN	Real-time-snapshot

SA	Security Analysis
SAFA	Synchronous Area Framework Agreement
SAM	Synchronous Area Monitor
SC	Synchronous Condenser
SCADA	Supervisory Control And Data Acquisition
SEE	South-East-Europe
SGU	Significant Grid Users
SO	System Operator
SO ESC	System Operation European Stakeholder Committee
SOC	ENTSO-E System Operation Committee
SOGL	System Operation Guidelines
SPES	Spanish Peninsular Electric System
SS	Substation
STA	Short-term adequacy
STATCOM	Static Synchronous Compensator
SVR	Secondary Voltage Regulator
SWE	South West Europe
TCI	Training and certification of staff working for RCCs
TERRE	Trans European Replacement Reserves Exchange
TLIs	Tie-line Inconsistencies
TPP	Thermal Power Plant
TRM	Transmission Reliability Margin
TS	Transmission Substation
TS	Only for Chapter 6: Timestamp
TSO	Transmission System Operator
TTC	Total Transfer Capacity
UAP	Unavailability Asset Plan
UCTE DEF	Union for the Co-ordination of Transmission of Electricity Data Exchange Format
UKF	Unscented Kalman Filter
VSC	Voltage Source Converter
WOPT	Weekly Operational Teleconference
YA	Year-ahead

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